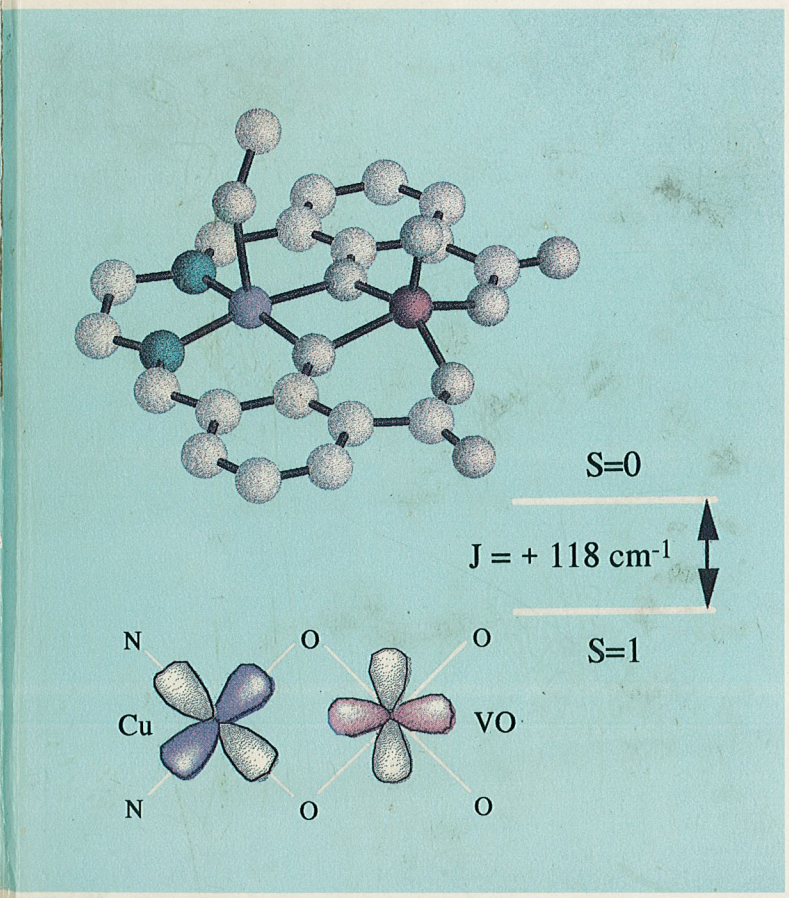
Olivier Kahn

Molecular Magnetism



VCH

Molecular Magnetism

Olivier Kahn

Olivier Kahn
Laboratoire de Chimie Inorganique
URA-CNRS 420
Université Paris-Sud
91405 Orsay
France

Sci
QD
940
K34
1993

This book is printed on acid-free paper.

Library of Congress Cataloging-in-Publication Data

Kahn, Olivier, 1942–
Molecular magnetism / Olivier Kahn.
p. cm.
Includes bibliographical references and index.
ISBN 1-56081-566-3 (alk. paper).
1. Molecular crystals—Magnetic properties. 2. Molecules--Magnetic properties. I. Title
QD940.K34 1993
541.3'78—dc20 93-10928
CIP

© 1993 VCH Publishers, Inc.

This work is subject to copyright

All rights are reserved, whether the whole or the part of the material is concerned, specifically those of translation, reprinting, re-use of illustrations, broadcasting, reproduction by photocopying or similar means, and storage in data banks.

Registered names, trademarks, etc., used in this book, even when not specifically marked as such, are not to be considered unprotected by law.

Printed in the United States of America

ISBN: 1-56081-566-3 VCH Publishers
ISBN: 3-527-89566-3 VCH Verlagsgesellschaft

Printing history:
10 9 8 7 6 5 4 3 2 1

Published jointly by

VCH Publishers, Inc.
220 East 23rd Street
New York N.Y. 10010-4606

VCH Verlagsgesellschaft mbH
P.O. Box 10 11 61
69451 Weinheim, Germany

VCH Publishers (UK) Ltd.
8 Wellington Court
Cambridge CB1 1HZ
United Kingdom

Preface

I have worked in the field of molecular magnetism for almost two decades and I have been more and more fascinated by the beauty of this area and by the diversity it presents. Even if this judgment is, to a large extent, subjective, I would like to explain why I hold such an impression.

Molecular magnetism is essentially multidisciplinary. First, it involves synthetic chemistry; one of its challenges is to design molecular systems that exhibit predictable magnetic properties. To achieve this aim, quite original compounds have been synthesized containing several kinds of metallic ions, organic radicals, and novel bridging networks. Second, it uses ideas from theoretical chemistry. To design compounds with expected magnetic behavior, it is necessary to use strategies that derive from an understanding of the underlying mechanism of the phenomena. Quite spectacular advances have been achieved in this way. Recently several microscopic and macroscopic approaches have been proposed that cover various aspects of molecular magnetism. These approaches actually derive directly from the basic concepts of quantum mechanics. Molecular magnetism may even be one of the most straightforward applications of the quantum mechanism. Molecular magnetism also allows us to stress the limits of theoretical models used in other fields. This is particularly the case for the molecular orbital model. The key concept, the interaction between two magnetic centers, cannot be properly understood within the framework of this model at the *self-consistent field* level. It is absolutely necessary to go further, and take into account the correlation. From a theoretical viewpoint, molecular magnetism is far from being a closed field, where all knowledge is well established. Many questions are still quite controversial. For instance, the interpretation of the phenomenon of interaction between magnetic centers uses several models that compete. Each model has some advantages and some disadvantages. All this makes the field very exciting.

Molecular magnetism is related to both molecular and solid state physics. Indeed, the pioneering approaches are often due to physicists. Some of them may be considered as the founding fathers of the field. Two famous names emerge in this respect, those of J.H. Van Vleck and P.W. Anderson, and I would like to express my admiration to them here.

It is also worth pointing out that molecular magnetism occupies a crossing point between two areas that would otherwise be incompatible. These are material science and study of the biological processes involving active sites containing metal ions. It

is, for instance, quite remarkable that the interaction between iron(III) and copper(II) (d^5 and d^9 ions) that occurs in cytochrome *c* oxidase (a metalloenzyme whose structure is still a mystery) is exactly of the same nature as the interaction between manganese(II) and copper(II) ions in a ferrimagnetic chain compound exhibiting a spontaneous magnetization at low temperature.

Last, molecular magnetism plays an important role in the emerging field of molecular electronics, i.e., the use of molecular systems in electronic circuits and devices. The spin transition phenomenon that occurs in some transition metal complexes represents the most spectacular example of molecular bistability.

At this stage, it is probably necessary to specify the scope of molecular magnetism. Molecular magnetism deals with the magnetic properties of isolated molecules and assemblies of molecules. These molecules may contain one or more magnetic centers. Assemblies of molecules are most often found in molecular crystals with very weak interactions between the molecular entities. They can also be found in extended systems, built from molecular precursors or "bricks," in a way that maximizes the interactions between the bricks and, hopefully, yields bulk magnetic properties. A few books have recently appeared that cover some aspects of this field, but none of them is really devoted entirely to molecular magnetism.

This book is organized as follows. The first chapter gives some generalities concerning magnetization and magnetic susceptibility, establishing the fundamental equations of molecular magnetism. The following four chapters concern molecules containing a unique magnetic center. In Chapter 2, the simplest situation is treated where this magnetic center has no first-order angular momentum. Chapter 3 deals with the significantly more complicated, and not yet fully resolved, case where the angular momentum is not totally quenched to first order. Chapter 4 is devoted to the phenomenon of spin transition and discusses its potential applications in molecular electronics. Chapter 5 deals with the difficult questions of intermediate-spin and spin-admixed states in transition metal compounds of low symmetry. The main part of this book, beginning with Chapter 6, concerns molecular systems containing several magnetic centers. The phenomena of isotropic, dipolar, anisotropic, and antisymmetric interactions in dinuclear compounds are first introduced and discussed in Chapters 6 and 7. The orbital theories appropriate for these phenomena are then presented in Chapters 8 and 9. Chapter 10 concerns molecular entities of higher nuclearity, i.e., trinuclear, tetranuclear species etc, and the concept of irregular spin state structure is introduced. Chapter 11 is devoted to the various types of magnetic chain compounds and describes the thermodynamic models appropriate to the quantitative interpretation of magnetic data. Chapter 12 concerns molecular magnetic materials and their models. The main emphasis is on the design of molecular-based magnets, and the first successes along these lines are outlined. Chapters 6 to 12 treat polynuclear compounds in which the unpaired electrons that lead to magnetic properties are localized in magnetic orbitals. The magnetic properties then arise from interaction between magnetic orbitals. In contrast, Chapter 13 is devoted to compounds where there is some synergy between electron delocalization and magnetic interaction, which leads to the concept of spin-dependent delocalization. Finally, some useful data and formulas are gathered in appendices at the end of the book.

Throughout this book, I have presented numerous examples. However, my purpose was never to write a comprehensive review of the field. The examples were selected out of a large choice, and, to a large extent, their choice is a reflection of my own subjectivity. Similarly, I did not attempt to reference all the papers dealing with molecular magnetism, or even all the good papers. Rather, I have quoted only a few to help illustrate the text or to provide additional information with regard to the text.

One of the difficulties in writing a scientific book is deciding on the prerequisite knowledge of the reader. My philosophy here was to assume that the reader is not acquainted with the field itself, but possesses a good background knowledge, particularly in symmetry, ligand field theory, and molecular orbital theory. I have also supposed that the reader is rather familiar with the operators and matrix notations used in quantum mechanics. This knowledge is provided during undergraduate studies in most, if not all, countries. A few textbooks are mentioned among the references, in which the reader could acquire the basic assumed knowledge. In another respect, this book is primarily written for chemists working in the field of molecular magnetism, or intending to do so. Therefore, the concepts have been described at the expense of mathematical technique. In other words, I have not attempted to prove the general formulas in the cases where this would have required tedious calculations. Instead, I preferred to consider specific cases first, and then generalize the results they yield.

I would like to thank heartily all my colleagues in my group at Orsay and in other groups who have collaborated with me on this subject. I have learned a lot from them. The writing of a book in addition to the normal research, teaching, and administrative duties is always a rather difficult task. I would like to express my deepest gratitude to those who have created an atmosphere around me that allowed me to achieve this work. I have in mind all my colleagues from the Laboratoire de Chimie Inorganique in Orsay, as well as my sons Fabrice and Sylvain, and my wife Joanna, who deserves special thanks. The crystalline beauty of her soul, the delicacy of her mind, the depth of her faith at all times have been an invaluable support for me.

Olivier Kahn
February 1992

Contents

List of the Main Symbols xiii

1. Magnetization and Magnetic Susceptibility 1

1.1 Definitions and Units 1
1.2 Diamagnetic and Paramagnetic Susceptibilities 2
1.3 Fundamental Equations in Molecular Magnetism 4
1.4 Van Vleck Formula 5
1.5 Temperature-Independent Paramagnetism 7
References 8

2. Molecules Containing a Unique Magnetic Center without First-Order Orbital Momentum 9

2.1 The Curie Law 9
2.2 Magnetization 10
2.3 Anisotropy of the *g*-Factor 12
2.4 Zero-Field Splitting: Qualitative Approach 14
2.5 Zero-Field Splitting: Quantitative Approach 17
2.6 Intermolecular Interactions 26
References 29

3. Molecules Containing a Unique Magnetic Center with a First-Order Orbital Momentum 31

3.1 First-Order Orbital Momentum and Spin-Orbit Coupling 31
3.2 Magnetism of a d^1 Ion in an Octahedral Field 32
3.3 Magnetism of a d^1 Ion in an Axially Distorted Octahedral Field 35
3.4 Magnetism of a Cobalt(II) Ion in Octahedral Surroundings 38
3.5 Magnetism of the Rare Earth-Containing Compounds: The Free-Ion Approximation 43
3.6 Magnetism of the Rare Earth-Containing Compounds: Some Examples 47
References 51

4. Low-Spin–High-Spin Transition 53

4.1 High-Spin Molar Fraction versus Temperature Curves for Spin Transition Compounds 54
4.2 Mechanism of the Spin Transition at the Molecular Scale 55
4.3 Spin Transition and Cooperativity 59
4.4 Regular Solution Model 60
4.5 Domain Model 67
4.6 Some Selected Examples 70
4.7 Spin Transition and Molecular Electronics 82
References 84

5. Intermediate-Spin and Spin-Admixed States 87

5.1 Intermediate Spin: One-Electron Approach 87
5.2 Spin-Admixed Ground State 89
5.3 Examples of Intermediate-Spin and Spin-Admixed States in Iron(III) Compounds 93
5.4 Examples of Intermediate-Spin and Spin-Admixed States in Iron(II) Compounds 96
5.5 Spin Transition and Spin Equilibrium Involving an Intermediate-Spin State 97
References 100

6. Isotropic Interaction in Dinuclear Compounds 103

6.1 Copper(II) Dinuclear Compounds 103
6.2 Other Symmetrical Dinuclear Compounds 112
6.3 Asymmetrical Dinuclear Compounds 120
6.4 Influence of the Local Anisotropy 125
6.5 Intermolecular Interactions between Dinuclear Units 131
References 132

7. Dipolar, Anisotropic, and Antisymmetric Interactions in Dinuclear Compounds 135

7.1 Dipolar and Anisotropic Interactions in Copper(II) Dinuclear Compounds 135
7.2 Antisymmetric Interaction in Copper(II) Dinuclear Compounds 138
7.3 Other Dinuclear Compounds 141
References 143

8. Orbital Models of the Isotropic Interaction: Case of Two Local Doublet States 145

8.1 Some Preliminary Remarks 145
8.2 Deficiency of the Molecular Orbital Approach at the Self-Consistent Field Level 146
8.3 Concept of Magnetic Orbitals: Orthogonalized and Natural Magnetic Orbitals 148
8.4 The Active-Electron Approximation 150
8.5 Beyond the Active-Electron Approximation: Ab Initio Calculation of the Singlet-Triplet Energy Gap 155
8.6 Theoretical Interpretation of Some Experimental Results 159
References 182

9. Orbital Models for Magnetic Interaction: Other Cases 185

9.1 Some Additional Comments on the Concept of Magnetic Orbital 185
9.2 The Active-Electron Approximation for Dinuclear Species with More Than Two Unpaired Electrons 186
9.3 Interaction between Pairs of Magnetic Orbitals 190
9.4 Ferromagnetic Interaction Through Crossed Interaction between Singly Occupied and Empty Orbitals 199
9.5 Mechanism of Anisotropic and Antisymmetric Interactions 202
9.6 Interaction Involving Ions with Unquenched Orbital Momentum 204
References 208

10. Trinuclear Compounds and Compounds of Higher Nuclearity 211

10.1 Symmetrical ABA Compounds 211
10.2 Nonsymmetrical Trinuclear Compounds 226
10.3 Tetranuclear Species with a Central Magnetic Ion 229

10.4 Cubane-Like Tetranuclear Species 236
10.5 Competing Interactions and Spin Frustration 241
10.6 High-Spin Molecules 245
References 248

11. Magnetic Chain Compounds 251

11.1 Chains of Equally Spaced Copper(II) Ions 251
11.2 Chains of Equally Spaced Magnetic Centers with Local Spins Larger Than 1/2 257
11.3 Ising and XY Models 261
11.4 Alternating Chains and Spin-Peierls Transition 263
11.5 Regular Ferrimagnetic Chains: Theory 269
11.6 Regular Ferrimagnetic Chains: Some Examples 275
11.7 Alternating Ferrimagnetic Chains 281
References 284

12. Magnetic Long-Range Ordering in Molecular Compounds: Design of Molecular-Based Magnets 287

12.1 Three-Dimensional Magnetic Ordering 287
12.2 Molecular-Based Magnets 288
12.3 Orbital Degeneracy and Ferromagnetic Interaction 290
12.4 Ferromagnetic Transitions in Decamethylferrocenium Tetracyanoethenide and Other Decamethylmetallocenium Charge-Transfer Salts 294
12.5 Intermolecular Ferromagnetic Interactions and Ferromagnetic Transition in Organic Radicals 300
12.6 Spin Polarization and Intermolecular Ferromagnetic Interactions 303
12.7 Topological Degeneracy and High-Spin Polycarbenes 309
12.8 Magnetic Ordering of Ferrimagnetic Chains 313
12.9 Spin Canting and Magnetic Ordering 321
12.10 Some Additional Results 325
References 328

13. Spin-Dependent Delocalization in Mixed-Valence Compounds 333

13.1 Mixed-Valence Dinuclear Compounds with a Unique Unpaired Electron 333
13.2 Spin-Dependent Delocalization in Mixed-Valence Dinuclear Compounds with Two Magnetic Sites 337

13.3 A Spin Hamiltonian Taking into Account the Spin-Dependent Delocalization 340
13.4 Role of the Nuclear Relaxation 343
13.5 Some Examples of Delocalized Mixed-Valence Pairs with Magnetic Cores 345
13.6 Mixed-Valence Compounds of Higher Nuclearity 350
References 352

Appendices 355

Appendix 1: Physical Constants and Units 355
Appendix 2: Action of $\mathbf{L}_x$, $\mathbf{L}_y$, and $\mathbf{L}_z$ on the *d* Orbitals 357
Appendix 3: Matrix Elements of $\mathbf{L}_x$, $\mathbf{L}_y$, and $\mathbf{L}_z$ Using the *d* Orbitals as a Basis Set 359
Appendix 4: Action of the Spin Operators $\mathbf{S}_x$, $\mathbf{S}_y$, and $\mathbf{S}_z$ on the Spin Functions 361
Appendix 5: Escalator Operators $\mathbf{S}_+$ and $\mathbf{S}_-$, and $\mathbf{L}_+$ and $\mathbf{L}_-$ 365
Appendix 6: Coupling of Two Spins S_A and S_B, and Clebsch–Gordon (or Wigner) Coefficients 367

Index 375

List of the Main Symbols

B	Electron delocalization parameter in a mixed-valence compound (Chapter 13)
$\mathbf{D}$	Zero-field splitting tensor
D	Axial zero-field splitting parameter
	Energy gap between magnetic orbitals (Chapter 9)
$\mathbf{d}$	Antisymmetric interaction vector
E	Energy
	Rhombic zero-field splitting parameter
$E^{(0)}$	Zero-field energy
$E^{(1)}$	First-order Zeeman coefficient
$E^{(2)}$	Second-order Zeeman coefficient
e	Molecular orbital energy
$\mathbf{F}$	Fock operator
$\mathbf{g}$	**g**-tensor
g	g-factor
g_e	g-factor for the free electron
g_u	Principal values of the **g**-tensor
G	Gibbs free energy (Chapter 4)
H	Magnetic field
	Enthalpy (Chapter 4)
$\mathbf{H}$	Hamiltonian
$\mathbf{J}, \mathbf{J}_z$	Polyelectronic angular momentum operator (Chapter 3)
J	Isotropic interaction parameter
	Angular momentum quantum number associated with **J** (Chapter 3)
j	Two-electron Coulomb integral
k	Boltzmann constant
	Two-electron exchange integral
	Force constant (Chapter 13)
$\mathbf{L}, \mathbf{L}_u$	Polyelectronic orbital operators
L	Orbital quantum number associated with $\mathbf{L}^2$
$\mathbf{l}, \mathbf{l}_u$	Monoelectronic orbital operators
l	Two-electron ionic integral
	Parameter related to the distortion in a mixed-valence compound (Chapter 13)

M Molar magnetization
Molecular weight (Chapter 1)
M_J Angular momentum quantum number associated with $\mathbf{J}_z$
M_L Orbital quantum number associated with $\mathbf{L}_z$
M_S Spin quantum number associated with $\mathbf{S}_z$
N Avogadro number
$\mathcal{N}$ Number of domains per mole (Chapter 4)
n Number of sites
Number of molecules per domain (Chapter 4)
R Molar gas constant (Chapter 4)
M_S Spin quantum number associated with $\mathbf{S}_z$
$\mathbf{S}$, $\mathbf{S}_u$ Polyelectronic spin operators
S Spin quantum number associated with $\mathbf{S}^2$
S Overlap integral
Entropy (Chapter 4)
$\mathbf{s}$, $\mathbf{s}_u$ Monoelectronic spin operators
T Temperature
U Energy gap between charge-transfer configuration and ground configuration
T_c Critical temperature
x High-spin molar fraction (Chapter 4)
Z Partition function (Chapters 1 and 2)
α Spin function for $m_s = 1/2$
One-site integral
canting angle (Chapters 7 and 12)
β Bohr magneton
Spin function for $m_s = -1/2$
Transfer integral
Γ Irreducible representation of a point group
Interaction Gibbs free energy accounting for cooperativity in an assembly of molecules (Chapter 4)
γ Interaction parameter accounting for cooperativity in an assembly of molecules (Chapter 4)
Δ Energy gap between states
κ Orbital reduction factor
λ Spin-orbit coupling parameter
Vibronic coupling parameter in a mixed-valence compound (Chapter 13)
μ Magnetic dipole
Ψ Wavefunction
φ Molecular orbital
θ Weiss constant
χ Molar magnetic susceptibility
χ_u Principal molar magnetic susceptibilities
ζ Spin-orbit coupling constant (Chapter 5)

Bold type is used for the quantum operators and the tensors throughout the book.

CHAPTER

1

Magnetization and Magnetic Susceptibility

1.1 Definitions and Units

In this first Chapter we establish some key equations in molecular magnetism, which will be used extensively throughout this book. To begin with, we consider a sample containing 1 mol of a molecular compound within an homogeneous magnetic field H. The sample acquires a molar magnetization M related to H through

$$\partial M/\partial H = \chi \tag{1.1.1}$$

where χ is the molar magnetic susceptibility. M, which is also called the molar magnetic moment, is a vector, H is an axial vector, and χ is a second rank tensor. It is always possible to choose the reference axes in order for χ to be diagonal with the χ_u ($u = x, y, z$) principal values. If the sample is magnetically isotropic, then χ becomes a scalar.

When the magnetic field is weak enough, χ is independent of H, such that one can write

$$M = \chi H \tag{1.1.2}$$

The problem of units deserves a few words [1.1]. The SI is the legal system, but legality is not science. Indeed, this system is particularly inappropriate in molecular magnetism and, like most of the researchers involved in this field, we prefer to use the cgsemu system. Strictly speaking the unit of magnetic field is the oersted. As many authors, however, we will express H in gauss which is the unit of magnetic induction created by the magnetic field H. In the vacuum B is related to H through $B = \mu_0 H$, and the permeability μ_0 in the cgsemu system is equal to 1. The volume magnetic susceptibility is a dimensionless quantity traditionally expressed in emu/cm^3, such that the dimension of emu is formally the cm^3. Therefore the molar magnetic susceptibility is expressed in cm^3 mol^{-1}. We will see that the quantity of interest is often χT, which is the product of the molar magnetic susceptibility by the temperature. χT is expressed in cm^3 K mol^{-1}. As for the molar magnetization M, it is expressed in cm^3 G mol^{-1}.

Alternatively, M may be expressed in $N\beta$ units, N being Avogadro's number and β the electronic Bohr magneton. The correspondence between the two units is

$$1\ N\beta = 5585\ \text{cm}^3\ \text{G mol}^{-1} \tag{1.1.3}$$

The other units used throughout this book are collected in Appendix 1. It should be stressed here that all energies will be expressed in cm^{-1}.

1.2 Diamagnetic and Paramagnetic Susceptibilities

Let us examine Eqs. (1.1.1) and (1.1.2). In principle χ is the algebraic sum of two contributions associated with different phenomena:

$$\chi = \chi^D + \chi^P \tag{1.2.1}$$

where χ^D and χ^P represent the diamagnetic and paramagnetic susceptibilities, respectively. The former is negative and the latter positive. When χ^D dominates, the sample is said to be diamagnetic; it is repelled by the magnetic field. When χ^P is the leading contribution, the sample is said to be paramagnetic; it is attracted by the applied field.

Diamagnetism is an underlying property of matter. It is always present, even when it is masked by the paramagnetism. The diamagnetism is due to the interaction of the magnetic field with the motion of the electrons in their orbits. The theory of this phenomenon has been well understood for several decades, and, for some small molecules, *ab initio* calculations of the molar diamagnetic susceptibility afford reasonably good results. This problem, however, is outside the scope of this book devoted to molecular paramagnetism. For our purpose it is sufficient to specify that χ^D is independent of the temperature and the strength of the applied field. Empirical formulas have been proposed to estimate its value. They are based on the idea that the diamagnetic susceptibility is essentially additive. χ^D can then be roughly calculated either from atomic susceptibilities and constitutive corrections, or from group susceptibilities (group meaning here ligand or counterion). The constitutive corrections are generally positive. They account for the fact that a molecule with multiple or conjugated bonds is less diamagnetic than a rather similar molecule with only single bonds. In the 1940s Pascal tabulated data allowing the application of these additive methods. Some of these data are gathered in Table 1.1. When only a very rough estimation of the diamagnetic susceptibility is required, we can use an equation of the type

$$\chi^D = kM \times 10^{-6}\ \text{cm}^3\ \text{mol}^{-1} \tag{1.2.2}$$

where M is the molecular weight of the compound and k is a factor varying between 0.4 and 0.5.

For paramagnetic compounds of low molecular weight the additive methods or even Eq. (1.2.2) may be sufficient to estimate the diamagnetic contribution χ^D. In such cases this contribution is indeed small and may even become negligible at low temperature. On the other hand, the situation is different when studying compounds of high molecular weight containing only a few paramagnetic centers. This often

Table 1.1. Diamagnetic Susceptibilities and Constitutive Corrections (in 10^{-6} cm^3 g atom^{-1} or 10^{-6} cm3 mol^{-1})[a]

Atoms			
H	−2.9	As(III)	−20.9
C	−6.0	As(V)	−43.0
N (ring)	−4.6	F	−6.3
N (open chain)	−5.6	Cl	−20.1
N (imide)	−2.1	Br	−30.6
O (ether or alcohol)	−4.6	I	−44.6
O (carbonyl)	−1.7	S	−15.0
P	−26.3	Se	−23.0
Constitutive corrections			
C=C	5.5	C=N	0.8
C≡C	0.8	N=N	1.8
C in aromatic ring	−0.25	N=O	1.7
C=N	8.1	C—Cl	3.1
Cations			
Li^+	−1.0	Ca^{2+}	−10.4
Na^+	−6.8	Sr^{2+}	−19.0
K^+	−14.9	Ba^{2+}	−26.5
Rb^+	−22.5	Zn^{2+}	−15.0
Cs^+	−35.0	Cd^{2+}	−24
NH_4^+	−13.3	Hg^{2+}	−40
Mg^{2+}	−5.0		
Transition metal cations			
$^*Ti^{3+}$	−9	$^*Fe^{2+}$	−13
Ti^{4+}	−5	$^*Fe^{3+}$	−10
$^*V^{2+}$	−15	$^*Co^{2+}$	−12
$^*V^{3+}$	−10	$^*Co^{3+}$	−10
$^*V^{4+}$	−7	$^*Ni^{2+}$	−10
V^{5+}	−1	Cu^+	−12
$^*Cr^{2+}$	−15	$^*Cu^{2+}$	−11
$^*Cr^{3+}$	−11	$^*Mo^{2+}$	−31
$^*Mn^{2+}$	−14	$^*Mo^{3+}$	−23
$^*Mn^{3+}$	−10	$^*Mo^{4+}$	−12
$^*Mn^{4+}$	−8	*Rare earth^{3+}	−20
Anions			
O^{2-}	−12.0	CN^-	−13.0
S^{2-}	−30	NO^{2-}	−10.0
F^-	−9.1	NO_3–	−18.9
Cl^-	−23.4	NCS^-	−31.0
Br^-	−34.6	CO_3^{2-}	−28
I^-	−50.6	ClO_4^-	−32.0
OH^-	−12.0	SO_4^{2-}	−40.1

(*Continued*)

Table 1.1. *(Continued)*

Some usual ligands	
H_2O	−13
NH_3	−18
CO	−10
CH_3COO^-	−30
$C_2O_4^{2-}$ oxalato	−25
$C_2H_8N_2$ ethylenediamine	−46
$C_5H_5^-$ cyclopentadienyl	−65
$C_5H_7O_2^-$ acetylacetonato	−52
C_5H_5N pyridine	−49
$C_4H_4N_2$ pyrazine	−50
$C_{10}H_8N_2$ bipyridine	−105
$C_{16}H_{14}N_2O_2^{-2}$ ethylenebis(salicylaminato)	−182
$C_{12}H_8N_2$ *o*-phenanthroline	−128

[a] Ions preceded by an asterisk are paramagnetic; only the underlying diamagnetism is indicated.

happens in biomagnetism. For instance, for a metalloprotein of molecular weight 50,000, χ^D is of the order of 2.5×10^{-2} cm^3 mol^{-1} according to (1.2.2). If the protein contains an high-spin iron(III) ion per molecular unit, the molar paramagnetic susceptibility is of the order of 1.5×10^{-2} cm^3 mol^{-1} at room temperature (vide infra), which does not compensate for χ^D. The accurate determination of χ^D is then crucial. Two methods may be used to solve this problem; the first consists in studying independently the diamagnetic susceptibility of the appropriate apoprotein, i.e., the derivative of the actual protein in which the paramagnetic ions have been eliminated, or replaced by diamagnetic ions. In a few cases such a method has been possible [1.2]. The second method consists of extracting mathematically a temperature-independent contribution from a temperature-dependent signal. Concerning the diamagnetic correction, the situation becomes even more dramatic when solutions, in particular biological solutions, are investigated. Diamagnetism of the solvent, and even the buffer, has to be taken into account and the temperature-dependent paramagnetic contribution may be only a very small part of the whole signal. High-sensitivity magnetometers like the superconducting quantum interference device (SQUID) are then absolutely necessary.

In this book we will always assume that the experimentally determined susceptibilities have been corrected for the diamagnetic contribution, and for the sake of simplicity we will use χ in the place of χ^P.

1.3 Fundamental Equations in Molecular Magnetism

The molar paramagnetic susceptibility (hereafter noted simply as molar magnetic susceptibility or even molar susceptibility) characterizes the way in which an applied

magnetic field H interacts with the angular momenta associated with the thermally populated states of a molecule. In classical mechanics, when a sample is perturbed by an external magnetic field, its magnetization is related to its energy variation through

$$M = -\partial E/\partial H \tag{1.3.1}$$

This equation may be easily translated into the language of quantum mechanics. We consider a molecule with an energy spectrum E_n (n = 1, 2, ...) in the presence of a magnetic field H. For each energy level we can define a microscopic magnetization μ_n as

$$\mu_n = -\partial E_n/\partial H \tag{1.3.2}$$

The macroscopic molar magnetization M is then obtained by summing the microscopic magnetizations weighted according to the Boltzmann distribution law, which leads to

$$M = \frac{N \Sigma_n(-\partial E_n/\partial H)\exp(-E_n/kT)}{\Sigma_n \exp(-E_n/kT)} \tag{1.3.3}$$

where T is the temperature and k is the Boltzmann constant. In the system of units used in this book k is equal to 0.695039 cm^{-1} K^{-1}.

This expression (1.3.3) may be considered as the fundamental expression in molecular magnetism. It does not lean on any approximation. The molar magnetic susceptibility is deduced from (1.3.3) through (1.1.1), or, if H/kT is not too large (vide infra), through (1.1.2).

Equation (1.3.3) may be transformed in a different way which will be used later on. For this we define the partition function Z as

$$Z = \Sigma_n \exp(-E_n/kT) \tag{1.3.4}$$

We can now write :

$$\partial \ln Z/\partial H = (1/kT)\frac{\Sigma_n(-\partial E_n/\partial H)\exp(-E_n/kT)}{\Sigma_n \exp(-E_n/kT)} \tag{1.3.5}$$

which leads to

$$M = NkT(\partial \ln Z/\partial H) \tag{1.3.6}$$

and

$$\chi = NkT(\partial^2 \ln Z/\partial H^2) \tag{1.3.7}$$

1.4 Van Vleck Formula

Although Eq. (1.3.3) is general, it is often difficult to apply. Indeed it requires knowledge on the $E_n = f(H)$ variations for all thermally populated states to calculate the $\partial E_n/\partial H$ derivatives. In 1932 Van Vleck proposed a simplification based on a few

approximations [1.3]. The first of them is that it is legitimate to expand the energies E_n according to the increasing powers of H :

$$E_n = E_n^{(0)} + EH_n^{(1)} + E_n^{(2)}H^2 + \cdots \tag{1.4.1}$$

where $E_n^{(0)}$ is the energy of level n in zero field. $E_n^{(1)}$ and $E_n^{(2)}$ are called first- and second-order Zeemann coefficients, respectively. From (1.4.1), μ_n in (1.3.2) becomes

$$\mu_n = -E_n^{(1)} - 2E_n^{(2)}H + \cdots \tag{1.4.2}$$

The second approximation is that H/kT is small with respect to unity. In other words, it is assumed that H is not too large and T not too small. The exponential in (1.3.3) may then be written as

$$\exp(-E_n/kT) = \exp(-E_n^{(0)}/kT)(1 - E_n^{(1)}H/kT) \tag{1.4.3}$$

From these two approximations, we obtain

$$M = \frac{N\Sigma_n(-E_n^{(1)} - 2E_n^{(2)}H)(1 - E_n^{(1)}H/kT)\exp(-E_n^{(0)}/kT)}{\Sigma_n (1 - E_n^{(1)}H/kT)\exp(-E_n^{(0)}/kT)} \tag{1.4.4}$$

In zero field, the magnetization vanishes, such that

$$\Sigma_n E_n^{(1)}\exp(-E_n^{(0)}/kT) = 0 \tag{1.4.5}$$

This condition (1.4.5) excludes molecular-based compounds exhibiting a spontaneous magnetization. Chapter 12 will be devoted to this subject.

Substituting (1.4.5) into (1.4.4) and retaining only terms linear in H results in

$$M = \frac{NH\Sigma_n(E_n^{(1)2}/kT - 2E_n^{(2)})\exp(-E_n^{(0)}/kT)}{\Sigma_n \exp(-E_n^{(0)}/kT)} \tag{1.4.6}$$

and finally

$$\chi = \frac{N\Sigma_n (E_n^{(1)2}/kT - 2E_n^{(2)})\exp(-E_n^{(0)}/kT)}{\Sigma_n \exp(-E\ n^{(0)}/kT)} \tag{1.4.7}$$

which is the Van Vleck formula. To apply this formula we only need to know the $E_n^{(0)}$, $E_n^{(1)}$, and $E_n^{(2)}$ quantities. It is no longer necessary to calculate the derivatives $\partial E_n/\partial H$. From a theoretical point of view, the Van Vleck formula can be utilized when the eigenvalues $E_n^{(0)}$ and eigenfunctions $|n>$ of the Hamiltonian in zero-field associated with the molecule under consideration are known. $E_n^{(1)}$ and $E_n^{(2)}$ are then calculated through perturbation theory as

$$E_n^{(1)} = <n \mid \mathbf{H}_{ZE} \mid n> \tag{1.4.8}$$

$$E_n^{(2)} = \Sigma'_m \frac{<n \mid \mathbf{H}_{ZE} \mid m>^2}{(E_n^{(0)} - E_m^{(0)})} \tag{1.4.9}$$

where Σ' means that the summation runs over the levels m with $E_m^{(0)} \neq E_n^{(0)}$. $\mathbf{H}_{ZE}$ is the Zeeman operator, which accounts for the interaction between the magnetic field and the electronic angular momenta:

$$\mathbf{H}_{ZE} = \beta\Sigma_i(\mathbf{l}_i + g_e\mathbf{s}_i) \cdot H \tag{1.4.10}$$

where $\mathbf{l}_i$ is the orbital momentum of electron i and $\mathbf{s}_i$ is the spin momentum of the same electron. g_e is the gyromagnetic factor, or the g-factor, of the free electron (equal to 2.0023), and β is the Bohr magneton already introduced in Section 1.1. In principle the summation in (1.4.10) runs over all electrons of the system. In a molecular orbital approach (without configuration interaction) only electrons occupying singly occupied orbitals contribute to the Zeeman perturbation. The expressions for $E_n^{(1)}$, in (1.4.8), and $E_n^{(2)}$, in (1.4.9), suppose implicitly that the Zeeman perturbation is diagonal using the functions $|n>$ as a basis set. In the following chapters we will see numerous examples of application of Van Vleck's formula. We would like to stress here that before using Van Vleck's equation, it is always necessary to check that the approximations on which it leans are valid. In particular, the Van Vleck equation gives the magnetic susceptibility only in the magnetic field range where the M versus H plot is linear. This point will be discussed again later on.

When all energies E_n are linear in H, the second-order Zeeman coefficients $E_n^{(2)}$ vanish and (1.4.7) becomes

$$\chi = \frac{N\Sigma_n E_n^{(1)2}\exp(-E_n^{(0)}/kT)}{kT\,\Sigma_n \exp(-E_n^{(0)}/kT)} \tag{1.4.11}$$

1.5 Temperature-Independent Paramagnetism

If the only thermally populated state of a molecule is a spin singlet without first-order angular momentum, then the paramagnetic susceptibility is intuitively expected to be zero, and the measured susceptibility will be negative. In a few cases a somewhat different situation is encountered, which the Van Vleck formula easily accounts for.

Let the energy $E_0^{(0)}$ of the ground state be the energy origin. Since this state has no angular momentum—it is said to be diamagnetic—$E_0^{(1)}$ is zero and (1.4.7) becomes

$$\chi = 2NE_0^{(2)} \tag{1.5.1}$$

or, from (1.4.9)

$$\chi = -2N\Sigma_{m\neq 0} \frac{<0 \mid \mathbf{H}_{ZE} \mid m>^2}{(E_0^{(0)} - E_m^{(0)})} \tag{1.5.2}$$

The diamagnetic ground state may couple with excited states through the Zeeman perturbation provided that the energy gaps are not too large. χ in (1.5.2) is positive since all denominators are negative, and temperature independent. This contribution is often called temperature-independent paramagnetism (TIP). The TIP is actually

rather small, often of the same order of magnitude as the diamagnetism, but of opposite sign. For example it is estimated to be about 60×10^{-6} cm^3 mol^{-1} for copper(II) mononuclear species, 100×10^{-6} cm^3 mol^{-1} for nickel(II) mononuclear species, and 200×10^{-6} cm^3 mol^{-1} for the octahedral complex $[Co(NH_3)_6]^{3+}$ possessing a $^1A_{1g}$ ground state.

TIP is not restricted to compounds with a diamagnetic ground state. The coupling between a magnetic ground state and nonthermally populated excited states may also give a weak temperature-independent contribution, which then superimposes to the dominant temperature-dependent contribution arising from the ground state. In the following we will assume that the measured susceptibilities have been corrected not only for diamagnetism, but also for TIP.

References

1.1. T.I. Quickenden, R.C. Marshall, *J. Chem. Ed.* 49 (1972), 114.

1.2. I. Morgenstern-Badarau, D. Cocco, A. Desideri, G. Rotilio, J. Jordanov, N. Dupré, *J. Am. Chem. Soc.* 108 (1986), 300.

1.3. J.H. Van Vleck, "The Theory of Electric and Magnetic Susceptibilities." Oxford University Press, Oxford, 1932.

CHAPTER

2

Molecules Containing a Unique Magnetic Center without First-Order Orbital Momentum

This chapter, and the three that follow, concern the magnetic properties of molecular species in that there is a unique magnetic center, generally a transition metal ion. The various situations that can be encountered will be treated in a progressive way from the simplest case, leading to the Curie law, through more complicated cases of orbital degeneracy, which are not yet perfectly understood.

2.1 The Curie Law

The simplest situation in molecular magnetism is that of molecules in which the $^{2S+1}\Gamma$ ground state has no first-order angular momentum and has a large separation in energy from the first excited states, such that any kind of coupling between ground and excited states may be neglected. In the absence of external magnetic field the $2S+1$ spin degeneracy is then retained. It is convenient to choose the energy of this $^{2S+1}\Gamma$ state as the energy origin. When the field is applied, the energies of the $2S+1$ Zeeman components are given by

$$E_n = M_S g \beta H \tag{2.1.1}$$

with M_s varying by an integer value from $-S$ to $+S$. Since the first excited states are assumed to be too high in energy to couple with the ground state, g is in principle isotropic and equal to $g_e = 2.0023$; the magnetic properties will be isotropic too. The energies E_n are linear in H, so it is possible to use the simplified form of the Van Vleck formula (1.4.11), provided that H/kT is small, with

$$E_n^{(0)} = 0 \tag{2.1.2}$$

Table 2.1 Values of $(4/g^2)\chi T$ as a Function of the Spin S

S	$(4/g^2)\chi T$ (cm^3 K mol^{-1})
1/2	0.375
1	1.000
3/2	1.876
2	3.001
5/2	4.377
3	6.002
7/2	7.878

$$E_n^{(1)} = M_S g\beta, \quad M_S = -S, -S+1, \ldots, S-1, S \tag{2.1.3}$$

which leads to

$$\chi = \frac{Ng^2\beta^2}{kT} \sum_{M_S=-S}^{+S} M_S^2/(2S+1) \tag{2.1.4}$$

and finally to

$$\chi = \frac{Ng^2\beta^2}{3kT} S(S+1) \tag{2.1.5}$$

The molar magnetic susceptibility varies as C/T, the constant C depending on the spin multiplicity of the ground state; this is the Curie law, which was proposed in 1910 from experimental data before the introduction of quantum mechanics. The most convenient way to establish experimentally that a compound obeys this Curie law is to obtain an horizontal straight line for the χT versus T plot. In the cgsemu unit system $N\beta^2/3k$ is equal to 0.12505 (very close to 1/8). Values of $(4/g^2)\chi T$ for different values of the spin S are given in Table 2.1.

Sometimes magnetic susceptibility data are given in the form of the temperature dependence of the so-called effective magnetic moment μ_{eff} defined as

$$\mu_{eff} = (3kT\chi/N\beta^2)^{1/2} \tag{2.1.6}$$

the unit being the Bohr magneton. We will avoid using this old-fashion notation, which is rather confusing. We prefer to reserve the Bohr magneton as a unit of magnetization.

2.2 Magnetization

It is important to keep in mind that the Curie law (2.1.5) is valid only when H/kT is small enough. The molar magnetization is then linear in H. When H/kT becomes large,

M must be calculated from the key Eqs. (1.3.3) or (1.3.6). It is actually more convenient to use the latter. The partition function Z is

$$Z = \sum_{M_S=-S}^{+S} \exp(-M_S g\beta H/kT) \tag{2.2.1}$$

$$= \sinh[(2S+1)x/2]/\sinh(x/2) \tag{2.2.2}$$

with

$$x = g\beta H/kT \tag{2.2.3}$$

$\partial \ln Z/\partial H$, which appears in (1.3.6), is given by

$$\partial \ln Z/\partial H = (g\beta/2kT)\{(2S+1)\coth[(2S+1)x/2] - \coth(x/2)\} \tag{2.2.4}$$

and may be rewritten as

$$\partial \ln Z/\partial H = (g\beta/2kT)\{(2S+1)\coth[(2S+1)y/2S] - \coth(y/2S)\} \tag{2.2.5}$$

with

$$y = g\beta SH/kT \tag{2.2.6}$$

The molar magnetization M is then

$$M = Ng\beta S B_S(y) \tag{2.2.7}$$

where $B_S(y)$ is the Brillouin function defined by

$$B_S(y) = \frac{2S+1}{2S}\coth\left(\frac{2S+1}{2S}y\right) - \frac{1}{2S}\coth\left(\frac{1}{2S}y\right) \tag{2.2.8}$$

We can check that for H/kT and y small, $B_S(y)$ may be replaced by

$$B_S(y) = y(S+1)/3S + \text{term in } y^3 + \cdots \tag{2.2.9}$$

such that, in this approximation, $\chi = M/H$ is effectively given by the Curie law (2.1.5). On the contrary, when H/kT becomes very large, $B_S(y)$ tends to unity and M tends to the saturation value M_S:

$$M_S = Ng\beta S \tag{2.2.10}$$

If the saturation magnetization is expressed in $N\beta$ units, its value is simply given by gS. The variations of M in $N\beta$ units for $g = 2$ and different values of S are shown in Figure 2.1.

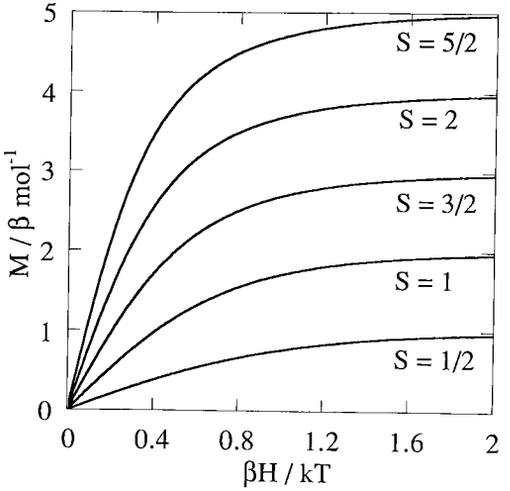


Figure 2.1. Magnetization M in $N\beta$ units versus $\beta H/kT$ plots for molecules with a $^{2S+1}\Gamma$ ground state. The g-factor is taken equal to 2.00.

2.3 Anisotropy of the *g*-Factor

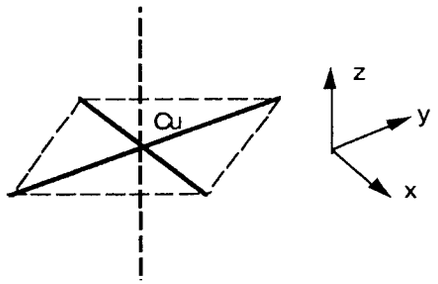
In a second step again we assume that the $^{2S+1}\Gamma$ ground state has no first-order angular momentum, but we no longer neglect the coupling between this state and some excited states arising from the same free-ion state through the spin-orbit coupling $\lambda\mathbf{L}\cdot\mathbf{S}$; $\mathbf{L}$ and $\mathbf{S}$ are the total electronic orbital and spin momentum operators, respectively, defined as

$$\mathbf{L} = \sum_i \mathbf{l}_i \tag{2.3.1}$$

$$\mathbf{S} = \sum_i \mathbf{s}_i \tag{2.3.2}$$

where the sums run over the electrons of the open shells. The excited states, however, are supposed to be high enough in energy to be totally depopulated in the temperature range of interest. This coupling may lead to two phenomena, namely the anisotropy of the g-factor, and, if the spin S associated with the ground state is larger than 1/2, the zero-field splitting. In this section we study the anisotropy of the g-factor, and since there is no zero-field splitting within a doublet state $^2\Gamma$, we take a compound with such a ground state as an example. More precisely we treat the case of a copper(II) mononuclear entity in which the metal ion is in elongated tetragonal surroundings with C_{4v} symmetry. There are actually a huge amount of complexes of this kind with copper(II) in square planar, square pyramidal, or elongated octahedral surroundings.

The relative energies of the d metal orbitals in such an environment, with their symmetry labels, are $e(d_{yz}, d_{zx}) < a_1(d_{z^2}) < b_2(d_{xy}) < b_1(d_{x^2-y^2})$ or $e(d_{yz}, d_{zx}) < b_2(d_{xy}) < a_1(d_{z^2}) < b_1(d_{x^2-y^2})$, the z axis being colinear with the fourfold rotation axis, and the x and y axes colinear with the projections of the short Cu–ligand bonds within the basal plane:



Since copper(II) has a d^9 configuration, the relative energies of the states are $^2B_1 < {}^2B_2 < {}^2A_1 < {}^2E$ or $^2B_1 < {}^2A_1 < {}^2B_2 < {}^2E$. The $^2B_1$ ground state has no first-order angular momentum. The matrix elements $< d_{x^2-y^2}|\mathbf{L}_u|d_{x^2-y^2} >$ are zero, $\mathbf{L}_u$ ($u = x, y, z$) being the components of the orbital momentum $\mathbf{L}$. On the other hand, $^2B_1$ may couple with $^2B_2$ through the component $\mathbf{L}_z$ of the orbital momentum, and with ^2E through the components $\mathbf{L}_x$ and $\mathbf{L}_y$. Indeed, we have the following non-zero matrix elements (see Appendicies 2 and 3):

$$< d_{xy} \mid \mathbf{L}_z \mid d_{x^2-y^2} > = 2i \tag{2.3.3}$$

$$< d_{yz} \mid \mathbf{L}_x \mid d_{x^2-y^2} > = < d_{zx} \mid \mathbf{L}_y \mid d_{x^2-y^2} > = -i \tag{2.3.4}$$

Taking these matrix elements together with those of the spin operators $\mathbf{S}_u$ ($u = x, y, z$) given in Appendix 4, the wavefunctions $\Psi\alpha$ and $\Psi\beta$ associated with the ground state at the first order are

$$\begin{aligned}
\Psi\alpha = {} & d_{x^2-y^2}\alpha - < d_{xy}\alpha \mid \lambda\mathbf{L}_z\mathbf{S}_z \mid d_{x^2-y^2}\alpha > d_{xy}\alpha/\Delta_1 \\
& - (< d_{yz}\beta \mid \lambda\mathbf{L}_x\mathbf{S}_x \mid d_{x^2-y^2}\alpha > d_{yz}\beta \\
& + < d_{zx}\beta \mid \lambda\mathbf{L}_y\mathbf{S}_y \mid d_{x^2-y^2}\alpha > d_{zx}\beta)/\Delta_2 \\
\Psi\beta = {} & d_{x^2-y^2}\beta - < d_{xy}\beta \mid \lambda\mathbf{L}_z\mathbf{S}_z \mid d_{x^2-y^2}\beta > d_{xy}\beta/\Delta_1 \\
& - (< d_{yz}\alpha \mid \lambda\mathbf{L}_x\mathbf{S}_x \mid d_{x^2-y^2}\beta > d_{yz}\alpha \\
& + < d_{zx}\alpha \mid \lambda\mathbf{L}_y\mathbf{S}_y \mid d_{x^2-y^2}\beta > d_{zx}\alpha)/\Delta_2
\end{aligned} \tag{2.3.5}$$

The closed shells have been ignored. Δ_1 and Δ_2 are the energy separations:

$$\Delta_1 = E(^2B_2) - E(^2B_1) \tag{2.3.6}$$

$$\Delta_2 = E(^2E) - E(^2B_1) \tag{2.3.7}$$

It follows

$$\Psi\alpha = d_{x^2-y^2}\alpha - i\lambda d_{xy}\alpha/\Delta_1 + \lambda(id_{yz}\beta - d_{zx}\beta)/2\Delta_2$$

$$\Psi\beta = d_{x^2-y^2}\beta + i\lambda d_{xy}\beta/\Delta_1 + \lambda(id_{yz}\alpha + d_{zx}\alpha)/2\Delta_2 \tag{2.3.8}$$

To calculate the components g_z and $g_x = g_y$ of the g-factor, we write

$$< \Psi\alpha \mid \mathbf{L}_z + g_e\mathbf{S}_z \mid \Psi\alpha > = < \Psi\alpha \mid g_z\mathbf{S}_z \mid \Psi\alpha > \tag{2.3.9}$$

and

$$< \Psi\alpha \mid \mathbf{L}_x + g_e\mathbf{S}_x \mid \Psi\beta > = < \Psi\alpha \mid g_x\mathbf{S}_x \mid \Psi\beta > \tag{2.3.10}$$

which affords

$$g_z = g_e - 8\lambda/\Delta_1 \tag{2.3.11}$$

$$g_x = g_y = g_e - 2\lambda/\Delta_2 \tag{2.3.12}$$

For transition metal ions with more than five d electrons the spin-orbit coupling parameter λ is negative such that g_z is larger than g_x and g_y, which in turn are larger than g_e. Typical values are $g_z = 2.20$ and $g_x = g_y = 2.08$.

The molar magnetic susceptibility χ_u for a magnetic field applied along the u direction is

$$\chi_u = Ng_u^2\beta^2/4kT \tag{2.3.13}$$

All the χ_u versus T plots are homothetic and the $\chi_u T$ versus T plots are horizontal straight lines. In fact EPR spectroscopy is much more appropriate than magnetic measurements to determine accurately the anisotropy of the g-factor. In most cases the magnetic data are recorded on polycrystalline samples and interpreted with an average value, g, of the g-factor defined by

$$g^2 = (g_x^2 + g_y^2 + g_z^2)/3 \tag{2.3.14}$$

In the rest of this book we will no longer consider this type of anisotropy.

2.4 Zero-Field Splitting: Qualitative Approach

When the ground state has a spin multiplicity larger than 2, its coupling with the excited states through the spin-orbit coupling may provoke a splitting of its Zeeman components in a zero applied magnetic field. This phenomenon, called zero-field splitting, also leads to an anisotropy of the magnetic properties. This anisotropy is more complicated and has more important consequences than that studied in the previous section.

To introduce qualitatively the zero-field splitting we consider a nickel(II) ion in octahedral surroundings. If the octahedron is regular, the symmetry is O_h but actually it is sufficient and more convenient to work with the group O of the 24 rotations retaining the octahedron invariant. The ground state arising from the $t_2^6 e^2$ configuration is $^3A_2$ and the spin triplet excited states arising from the $t_2^5 e^3$ configuration are $^3T_1$ and $^3T_2$. This latter is lower in energy than the former and, since we limit ourselves to a qualitative approach, we take into account only this $^3T_2$ excited state. First we suppose that the nickel(II) environment undergoes a trigonal distortion so that the appropriate symmetry group describing the system becomes D_3. The $O \rightarrow D_3$ symmetry lowering splits the $^3T_2$ excited state into $^3A_1 + {}^3E$, but does not affect on the ground state, which remains $^3A_2$. Second we look for the effect of the spin-orbit coupling, the molecular symmetry remaining that of a regular octahedron. The three spin functions associated with the spin triplet are a basis set for a representation $\Gamma_{S=1}$ of the group O, whose characters are given by

$$\chi(\phi) = \sin(3\phi/2)/\sin(\phi/2) \tag{2.4.1}$$

where ϕ is the rotation angle defining the operation. Applying (2.4.1) we find that $\Gamma_{S=1}$ is equal to T_1. It follows that the spin-orbit coupling splits the excited state $^3T_2$ into $T_1 \times T_2 = A_2 + E + T_1 + T_2$ components but does not affect the ground state. Indeed the direct product $T_1 \times A_2$ is T_2. The threefold degeneracy is retained. Therefore neither symmetry lowering alone nor spin-orbit coupling alone can affect the degeneracy of the ground state. On the other hand, if both perturbations are present, the result is different. If the distortion is applied first, $\Gamma_{S=1}$ transforms into $A_2 + E$ in D_3 so that the combined effect of trigonal distortion and spin-orbit coupling splits the ground state into $A_1 + E$ and the excited state into $A_1 + 2A_2 + 3E$. The same result is obviously obtained if the spin-orbit coupling is applied first. The T_1 and T_2 irreducible representations in the group O give $A_2 + E$ and $A_1 + E$ in the group D_3, respectively. What is important is that the components arising from the ground state are stabilized through the ground state–excited state interaction, but generally not by the same energy. Figure 2.2 summarizes what precedes. If the local symmetry is no longer axial but lower, the threefold degeneracy of the ground state is totally removed.

The discussion above can be generalized to all other spin multiplicities superior to 2. When the spin multiplicity is even (the number of electrons is then odd), the spin degeneracy cannot be totally removed in zero field. If the symmetry is low enough, all the components are twofold degenerate, and are called Kramers doublets. This is due to the fact that for an even spin multiplicity the spin functions no longer transform as the representations of a group isomorphous to the symmetry group Γ, but as the doubled-valued representations of a group isomorphous to the double group Γ'.

It is worth noting that the zero-field splitting does not always require the combined effect of distortion and spin-orbit coupling. In some cases it even occurs if the metal center is located in a rigorously cubic environment. This is the case for high-spin d^5 ions. The ground state arising from the $t_2^3 e^2$ configuration is $^6A_1$. The spin functions associated with the spin sextet transform as a representation $\Gamma_{S=5/2}$ of the double group O′, whose characters are given by

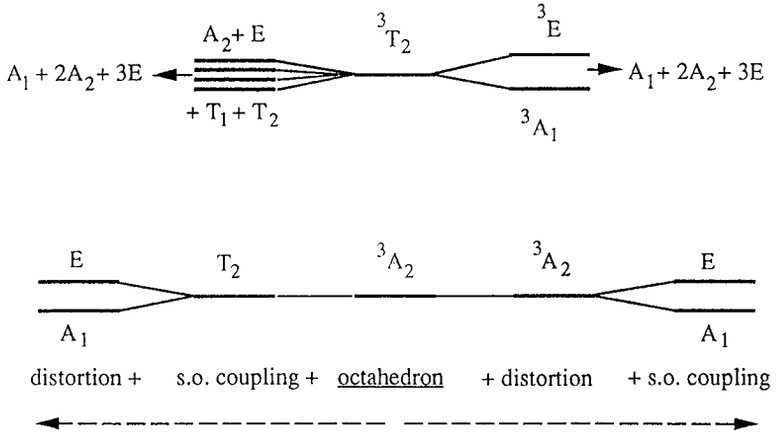


Figure 2.2. Zero-field splittings of the ground state and first excited spin triplet state for a Ni(II) ion in a trigonally distorted octahedral surroundings.

$$\chi(\phi) = \sin(3\phi)/\sin(\phi/2) \tag{2.4.2}$$

The characters of the doubled-valued irreducible representations of O′ as well as those of $\Gamma_{S=5/2}$ are given in Table 2.2. From Table 2.2 we can see that $\Gamma_{S=5/2}$ is equal to E_3 + G. Thus the spin-orbit coupling splits in zero-field the $^6A_1$ ground state into a twofold degenerate component E_3 and a fourfold degenerate component G. This zero-field splitting without distortion is actually a very weak phenomenon, at most of the order of 10^{-2} cm^{-1} for iron(III). In principle, it provokes a deviation of the magnetic susceptibility with respect to the Curie law, but of course does not create any magnetic anisotropy. Any symmetry lowering will split the G component further into two Kramers doublets and will lead to an anisotropy of the magnetic susceptibility.

Additional information on the symmetry considerations used in this section and throughout the book may be found in many textbooks; some of those, which we often used in our own work, are quoted in references [2.1–2.9].

Table 2.2. Doubled-Valued Representations of the double group O′ and Resentation $\Gamma_{S=5/2}$

			$4C_3$	$4C_3R$	$3C_2$	$3C_4$	$3C_4R$	$6C'_2$
O′	E	R	$4C_3^2R$	$4C_3^2$	$3C_2R$	$3C_4^3R$	$3C_4^3$	$6C'_2R$
E_2	2	–2	1	–1	0	$\sqrt{2}$	$-\sqrt{2}$	0
E_3	2	–2	1	–1	0	$-\sqrt{2}$	$\sqrt{2}$	0
G	4	–4	–1	1	0	0	0	0
$\Gamma_{S=5/2}$	6	–6	0	0	0	$-\sqrt{2}$	$\sqrt{2}$	0

2.5 Zero-Field Splitting: Quantitative Approach

The zero-field splitting within a $^{2S+1}\Gamma$ state without first-order angular momentum is expressed by the phenomenological Hamiltonian

$$\mathbf{H}_{ZFS} = \mathbf{S}\cdot\mathbf{D}\cdot\mathbf{S} \tag{2.5.1}$$

where **D** is a symmetric and traceless tensor. In matrix notation the total spin Hamiltonian taking into account the Zeeman perturbation is then

$$\mathbf{H} = \beta\mathbf{S}\cdot\mathbf{g}\cdot H + \mathbf{S}\cdot\mathbf{D}\cdot\mathbf{S} \tag{2.5.2}$$

g being the **g**-tensor. Assuming that the **D**- and **g**-tensors have the same principal axes, Eq. (2.5.2) may be rewritten as

$$\mathbf{H} = g_u\beta\mathbf{S}_u H_u + D[\mathbf{S}_z^2 - S(S+1)/3] + E(\mathbf{S}_x^2 - \mathbf{S}_y^2) \tag{2.5.3}$$

where the index u notes the direction of the applied magnetic field; g_u is the value of g and $\mathbf{s}_u$ is the component of **S** along this direction. D and E are the axial and rhombic zero-field splitting parameters, respectively. D and E are related to the principal values D_{uu} ($u = x, y, z$) of the **D**-tensor through

$$D = 3D_{zz}/2$$

$$E = |D_{xx} - D_{yy}|/2 \tag{2.5.4}$$

We will apply Eq. (2.5.3) to three examples, namely Ni(II), Cr(III), and Fe(III) ions in axially distorted surroundings. The rhombic parameter E is then zero.

2.5.1 Ni(II) Ion with d^8 Configuration in Axially Distorted Octahedral Surroundings

Let us begin with the case of a nickel(II) ion in trigonally distorted octahedral surroundings. When the field H_z is parallel to the unique axis, the energy matrix on the $|M_S\rangle$ basis set of the spin triplet is as follows:

$\|1\rangle$	$\|0\rangle$	$\|-1\rangle$
$g_z\beta H_z + D$	0	0
	0	0
		$-g_z\beta H_z + D$

(2.5.5)

In (2.5.5) the zero-field energy of the $|0\rangle$ component is taken as the energy origin. The energies are then

$$E_0 = 0$$

$$E_{1,2} = \pm g_z\beta H_z + D \tag{2.5.6}$$

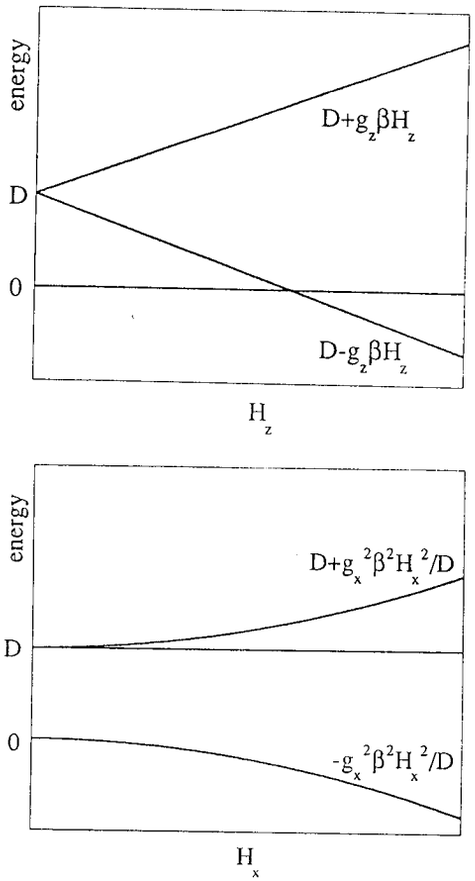


Figure 2.3. Field dependences of the energies of the Zeeman components arising from a spin triplet state with an axial anisotropy; (top) the field is along the distortion axis; (bottom) the field is perpendicular to the distortion axis.

When those energies are introduced in the Van Vleck formula, the parallel magnetic susceptibility χ_z is found as

$$\chi_z = \frac{2Ng_z^2\beta^2}{kT} \frac{\exp(-D/kT)}{1+2\exp(-D/kT)} \tag{2.5.7}$$

When the field H_x (or H_y) is perpendicular to the unique axis, the energy matrix is

|1>	|0>	|-1>
D	$\sqrt{2}g_x\beta H_x/2$	0
0		$\sqrt{2}g_x\beta H_x/2$
		D

(2.5.8)

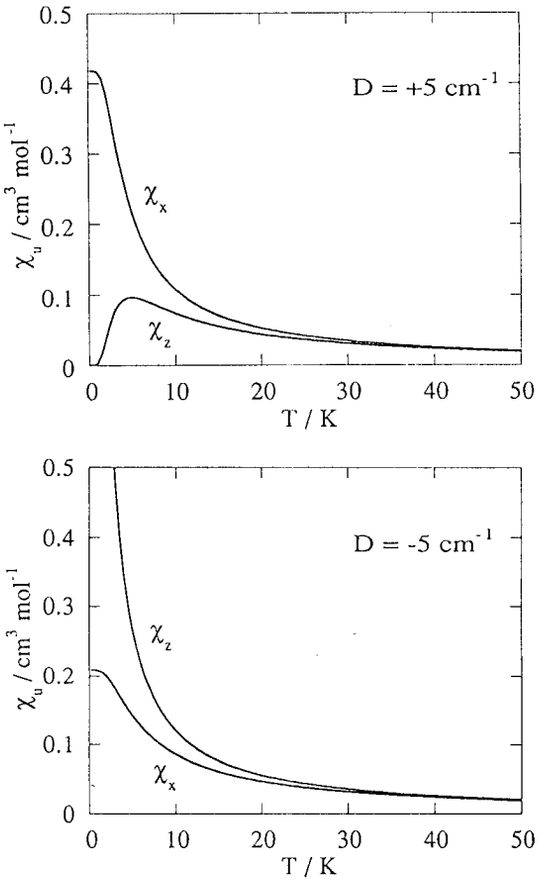


Figure 2.4. Principal susceptibilities χ_z and χ_x versus temperature plots for a spin triplet molecule with an axial anisotropy; (top) the axial zero-field splitting parameter D is equal to 5 cm^{-1}; (bottom) $D = -5$ cm^{-1}. Both g_z and g_x are taken equal to 2.00.

the eigenvalues of which are

$$E_1 = D$$

$$E_{2,3} = (\pm\sqrt{4g_x^2\beta^2H_x^2 + D^2} + D)/2 \tag{2.5.9}$$

To make use of the Van Vleck equation, we must assume that $|D|$ is much larger than $g_x\beta H_x$. The energies $E_{2,3}$ may then be approximated by

$$E_2 = -g_x^2\beta^2H_x^2/D$$

$$E_3 = g_x^2\beta^2H_x^2/D + D \tag{2.5.10}$$

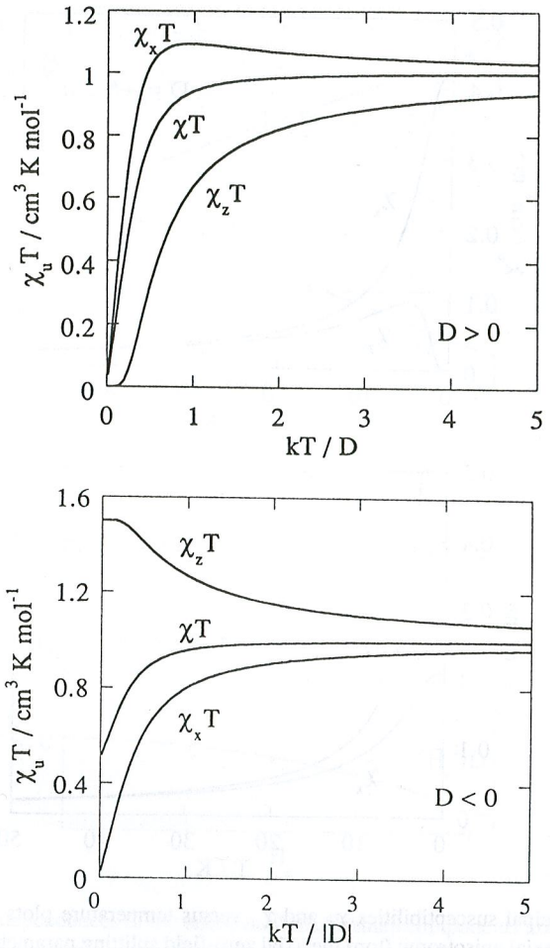


Figure 2.5. $\chi_u T$ ($u = z, x$) and χ versus $kT/|D|$ plots for a spin triplet molecule with an axial anisotropy; (top) the axial zero-field splitting D is positive; (bottom) $D < 0$. Both g_z and g_x are taken equal to 2.00.

We can note that the energies E_n (n = 1 to 3) have no term in H_x. The three $E_n^{(1)}$ coefficients are zero. The $E_n^{(0)}$ and $E_n^{(2)}$ coefficients may be introduced in the Van Vleck formula, which leads to the perpendicular magnetic susceptibility χ_x:

$$\chi_x = \frac{2Ng_x^2\beta^2}{D} \frac{1 - \exp(-D/kT)}{1 + 2\exp(-D/kT)} \tag{2.5.11}$$

The field dependences of the energies for both the parallel and perpendicular situations are plotted in Figure 2.3. The χ_z and χ_x versus T plots for $D = \pm 5$ cm^{-1} and $g_z = g_x = 2$ are shown in Figure 2.4. Whatever the sign of D, χ_x tends to a finite value

when T approaches zero. On the other hand, χ_z tends to zero for $D > 0$ and diverges for $D < 0$. We also give the $\chi_z T$ and $\chi_x T$ versus $kT/|D|$ curves for D positive and negative in Figure 2.5. For $D > 0$, both $\chi_z T$ and $\chi_x T$ tend to zero when $kT/|D|$ approaches zero; $\chi_z T$ continuously decreases while $\chi_x T$ passes through a maximum. For $D < 0$, $\chi_z T$ slightly increases on cooling and tends to the finite value $Ng_z^2\beta^2/k$ when $kT/|D|$ approaches zero. $\chi_x T$, on the other hand, tends to zero. In Figure 2.5 the curves χT versus $kT/|D|$ are also plotted, where χ is here the average magnetic susceptibility approximated by

$$\chi = (\chi_z + 2\chi_x)/3 \tag{2.5.12}$$

A more accurate value of the average magnetic susceptibility is given by

$$\chi = (4\pi)^{-1} \int_{\text{space}} \chi(\theta,\phi) \sin\theta \, d\theta \, d\phi \tag{2.5.13}$$

where $\chi(\theta,\phi)$ is the magnetic susceptibility in the direction noted by the polar angles θ and ϕ. It is important to note that the variations of χT are not very sensitive to the sign of D. This result is quite general. It is difficult, if not impossible, to determine unambiguously the sign of D from powder magnetic susceptibility measurements.

2.5.2 Cr(III) with d^3 Configuration in Axially Distorted Octahedral Surroundings

The second application of Eq. (2.5.3) will deal with the case of a Cr(III) ion in tetragonally distorted octahedral surroundings. The $^4A_2$ ground state is then split in zero field into two Kramers doublets. The energy matrix on the $|M_S>$ basis is diagonal when the field H_z is applied along the tetragonal axis, and the energies are

$$E_{\pm 1/2} = \pm g_z \beta H_z/2$$

$$E_{\pm 3/2} = 2D \pm 3g_z\beta H_z/2 \tag{2.5.14}$$

the zero-field energy of the $|\pm 1/2>$ components being taken as the energy origin. The parallel susceptibility is then

$$\chi_z = \frac{Ng_z^2\beta^2}{4kT} \frac{1 + 9\exp(-2D/kT)}{1 + \exp(-2D/kT)} \tag{2.5.15}$$

When the field H_x is perpendicular to the unique axis, the matrix energy is (see Appendix 4):

	$\|3/2>$	$\|-3/2>$	$\|1/2>$	$\|-1/2>$
$\|3/2>$	$2D$	0	$\sqrt{3}g_x\beta H_x/2$	0
$\|-3/2>$	0	$2D$	0	$\sqrt{3}g_x\beta H_x/2$
$\|1/2>$	—	—	0	$g_x\beta H_x$
$\|-1/2>$	—	—	—	0

(2.5.16)

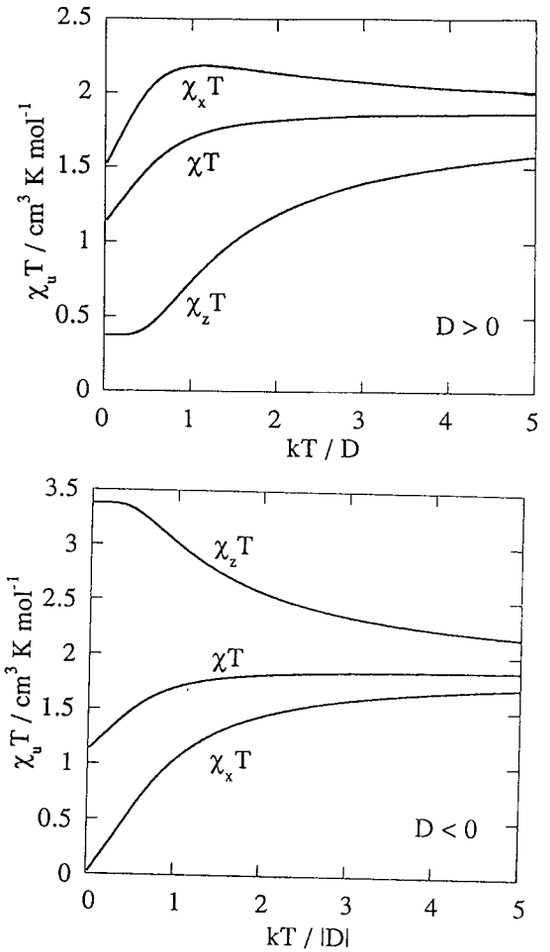


Figure 2.6. $\chi_u T$ ($u = z, x$) and χ versus $kT/|D|$ plots for a spin quartet molecule with an axial anisotropy; (top) the axial zero-field splitting D is positive; (bottom) $D < 0$. Both g_z and g_x are taken equal to 2.00.

Assuming that $|D|$ is much larger than $g_x\beta H_x$, the eigenvalues of the matrix (2.5.16) may be approximated by :

$$E_{1,2} = 2D + 3g_x^2\beta^2 H_x^2/8D$$

$$E_{3,4} = \pm g_x\beta H_x - 3g_x^2\beta^2 H_x^2/8D \qquad (2.5.17)$$

The $E_n^{(0)}$, $E_n^{(1)}$, and $E_n^{(2)}$ coefficients are

n	$E_n^{(0)}$	$E_n^{(1)}$	$E_n^{(2)}$
1	$2D$	0	$3g_x^2\beta^2/8D$
2	$2D$	0	$3g_x^2\beta^2/8D$
3	0	$g_x\beta$	$-3g_x^2\beta^2/8D$
4	0	$-g_x\beta$	$-3g_x^2\beta^2/8D$

(2.5.18)

Their insertion into the Van Vleck formula leads to

$$\chi_x = Ng_x^2\beta^2 \frac{(1/kT) + (3/4D) - (3/4D)\exp(-2D/kT)}{1 + \exp(-2D/kT)} \qquad (2.5.19)$$

or

$$\chi_x = \frac{Ng_x^2\beta^2}{kT} \frac{1 + (3/4x)[1 - \exp(-2x)]}{1 + \exp(-2x)} \qquad (2.5.20)$$

with

$$x = D/kT \qquad (2.5.21)$$

The variation of $\chi_z T$ and $\chi_x T$ versus $kT/|D|$ for both D positive and negative are plotted in Figure 2.6. For $D > 0$, $\chi_z T$ decreases on cooling and tends to $Ng_z^2\beta^2/4k$ when $kT/|D|$ approaches zero, and $\chi_x T$ passes through a maximum, then tends to $Ng_x^2\beta^2/k$. For $D < 0$, $\chi_z T$ increases on cooling and tends to a low-temperature limit of $9Ng_z^2\beta^2/4k$; $\chi_x T$ decreases continuously and tends to zero when $kT/|D|$ approaches zero. Finally, the χT versus $kT/|D|$ plots are also shown in Figure 2.6, χ being here the average magnetic susceptibility approximated through (2.5.12); χT is not at all sensitive to the sign of D.

2.5.3 d^5 Ion [Fe(III) or Mn(II)] in Axially Distorted Octahedral Surroundings

The third example of zero-field splitting concerns the high-spin d^5 ion in an axially distorted cubic field. This example again involves Kramers doublets. We assume that the spin Hamiltonian (2.5.3) with $E = 0$ is valid. The matrix energy on the $|M_S>$ basis set, for $H = H_z$ parallel to the unique axis, is diagonal with the following nonzero matrix elements

$$<\pm 5/2 \mid \mathbf{H} \mid \pm 5/2> = \pm 5g_z\beta H_z/2 + 25D/4$$

$$<\pm 3/2 \mid \mathbf{H} \mid \pm 3/2> = \pm 3g_z\beta H_z/2 + 9D/4 \qquad (2.5.22)$$

$$<\pm 1/2 \mid \mathbf{H} \mid \pm 1/2> = \pm g_z\beta H_z/2 + D/4$$

and the eigenvalues are

$$E_{1,2} = \pm g_z\beta H_z/2$$

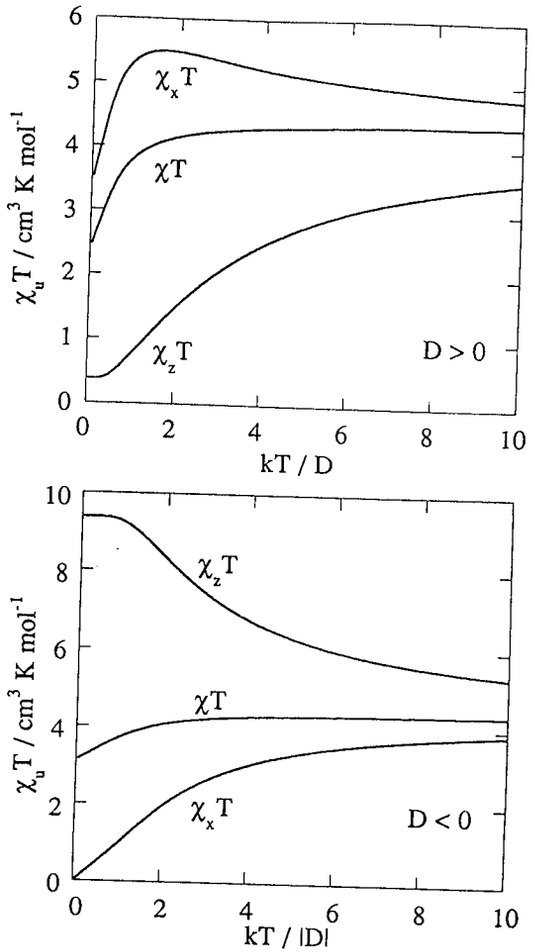


Figure 2.7. $\chi_u T$ ($u = z, x$) and χ versus $kT/|D|$ plots for a spin sextet molecule with an axial anisotropy; (top) the axial zero-field splitting D is positive; (bottom) $D < 0$. Both g_z and g_x are taken equal to 2.00.

$$E_{3,4} = 2D \pm 3g_z\beta H_z/2$$
$$E_{5,6} = 6D \pm 5g_z\beta H_z/2 \qquad (2.5.23)$$

where the zero-field energy of the $|\pm 1/2\rangle$ Kramers doublet has been taken as the energy origin. The insertion of the $E_n^{(0)}$ and $E_n^{(1)}$ coefficients deduced from (2.5.23) in the Van Vleck formula leads to

$$\chi_z = \frac{Ng_z^2\beta_2}{4kT}\,\frac{1 + 9\exp(-2D/kT) + 25\exp(-6D/kT)}{1 + \exp(-2D/kT) + \exp(-6D/kT)} \qquad (2.5.24)$$

For $H = H_x$ perpendicular to the unique axis, the elements of the matrix energy are calculated as (see Appendix 4)

|5/2>	|−5/2>	|3/2>	|−3/2>	|1/2>	|−1/2>
$6D$	0	$\sqrt{5}g_x\beta H_x/2$	0	0	0
	$6D$	0	$\sqrt{5}g_x\beta H_x/2$	0	0
		$2D$	0	$\sqrt{2}g_x\beta H_x$	0
			$2D$	0	$\sqrt{2}g_x\beta H_x$
				0	$3g_x\beta H_x/2$
					0

(2.5.25)

Assuming that $|D|$ is much larger than $g_x\beta H_x$, the eigenvalues of the matrix (2.5.25) may be approximated by

$$E_{1,2} = 6D + 5g_x^2\beta^2H_x^2/16D$$
$$E_{3,4} = 2D + 11g_x^2\beta^2H_x^2/16D$$
$$E_{5,6} = \pm 3g_x\beta H_x/2 - g_x^2\beta^2H_x^2/D \qquad (2.5.26)$$

The perpendicular magnetic susceptibility is then calculated from the $E_n^{(0)}$, $E_n^{(1)}$, and $E_n^{(2)}$ coefficients gathered below:

n	$E_n^{(0)}$	$E_n^{(1)}$	$E_n^{(2)}$
1	$6D$	0	$5g_x^2\beta^2/16D$
2	$6D$	0	$5g_x^2\beta^2/16D$
3	$2D$	0	$11g_x^2\beta^2/16D$
4	$2D$	0	$11g_x^2\beta^2/16D$
5	0	$3g_x\beta/2$	$-g_x^2\beta^2/D$
6	0	$-3g_x\beta/2$	$-g_x^2\beta^2/D$

(2.5.27)

which affords

$$\chi_x = \frac{Ng_x^2\beta^2}{4}\,\frac{(9/kT + 8/D) - 11\exp(-2D/kT)/2D - 5\exp(-6D/kT)/2D}{1 + \exp(-2D/kT) + \exp(-6D/kT)} \qquad (2.5.28)$$

or

$$\chi_x = \frac{Ng_x^2\beta^2}{4kT}\,\frac{9 + 8/x - 11e^{-2x}/2x - 5e^{-6x}/2x}{1 + e^{-2x} + e^{-6x}} \qquad (2.5.29)$$

with

$$x = D/kT \qquad (2.5.30)$$

The variation of $\chi_z T$ and $\chi_x T$ versus $kT/|D|$ for both D positive and negative are plotted in Figure 2.7. On cooling, for $D > 0$, $\chi_z T$ decreases and tends to $Ng_x^2\beta^2/4k$ when $kT/|D|$ approaches zero, whereas $\chi_x T$ passes through a maximum and tends to $9Ng_x^2\beta^2/4k$. For $D < 0$, $\chi_z T$ continuously increases and $\chi_x T$ continuously decreases on cooling. Figure 2.7 also shows the variation of χT, χ being the average magnetic susceptibility.

2.6 Intermolecular Interactions

Most magnetic measurements are carried out in the solid state and the molecular magnetic species, even if they are well separated from each other, are rarely perfectly isolated from a magnetic viewpoint. A large part of this book will be devoted to the phenomenon of interaction between magnetic centers (see chapters 6–13). Here, we restrict ourselves to showing how the Curie law in (2.1.5) has to be modified to account for weak intermolecular interactions. This problem may be treated in the molecular field approximation where a perturbation is added to the Zeeman term. This perturbation takes the form $-zJ\langle S_z\rangle\mathbf{S}_z$ where $\langle S_z\rangle$ is the mean value of the $\mathbf{S}_z$ component of the spin operator[2.10]. J is the interaction parameter between two nearest neighbor magnetic species and z is the number of nearest neighbors around a given magnetic molecule in the crystal lattice. According to whether J is positive or negative, the intermolecular interaction is said to be ferromagnetic or antiferromagnetic, respectively. In the former case, the neighboring spins tend to align in a parallel fashion, while in the latter, they tend to align in an antiparallel fashion. The total spin Hamiltonian is

$$\mathbf{H} = g\beta\mathbf{S}_z H - zJ\langle S_z\rangle\mathbf{S}_z \tag{2.6.1}$$

where the magnetic field is assumed to be along the z direction and the **g**-tensor to be isotropic. The eigenvalues $E(S,M_S)$ of (2.6.1) are given by

$$E(S,M_S) = M_S(g\beta H - zJ\langle S_z\rangle) \tag{2.6.2}$$

$\langle S_z\rangle$ is given by the Boltzmann distribution law

$$\langle S_z\rangle = \frac{\sum_{M_S=-S}^{S} M_S \exp[-E(S,M_S)/kT]}{\sum_{M_S=-S}^{S} exp[-E(S,M_S)/kT]} \tag{2.6.3}$$

From (2.6.2), (2.6.3) may be rewritten as

$$\langle S_z\rangle = \frac{\sum_{M_S=-S}^{S} M_S [1 - M_S(g\beta H - zJ\langle S_z\rangle)/kT]}{\sum_{M_S=-S}^{S}[1 - M_S(g\beta H - zJ\langle S_z\rangle)/kT]} \tag{2.6.4}$$

where the exponentials have been expanded up to the first order. It follows that

$$\langle S_z\rangle = -S(S+1)(g\beta H - zJ\langle S_z\rangle)/3kT \tag{2.6.5}$$

hence

$$\langle S_z\rangle = -\frac{S(S+1)g\beta H}{3kT - zJS(S+1)} \tag{2.6.6}$$

$\langle S_z\rangle$ is negative because the Zeeman components with negative M_S are more populated than those with positive M_S. The magnetic moment m of the electron with spin s is given by

$$m = -g_e\beta s \tag{2.6.7}$$

The minus sign comes from the negative charge of the electron. For a polyatomic system in the molecular field approximation, the magnetization M may be expressed as

$$M = -Ng\beta\langle S_z\rangle \tag{2.6.8}$$

such that the magnetic susceptibility χ defined as $\partial M/\partial H$ is

$$\chi = \frac{Ng^2\beta^2 S(S+1)}{3kT - zJS(S+1)} \tag{2.6.9}$$

χ may be rewritten as :

$$\chi = C/(T - \Theta) \tag{2.6.9}$$

which is known as the Curie–Weiss law. C is the Curie constant and Θ is the Weiss temperature (or the Weiss constant) defined by

$$\Theta = \frac{zJS(S+1)}{3k} \tag{2.6.10}$$

A plot of $\chi^{-1} = f(T)$ for a system obeying the Curie–Weiss law gives a straight line of which the slope is C^{-1}. The intercept with the T axis yields both the sign and the value of Θ (see Figure 2.8). In the framework of this model positive Θ indicates ferromagnetic intermolecular interactions and negative Θ indicates antiferromagnetic intermolecular interactions. In the former case ($\Theta > 0$) the Curie–Weiss law is obviously limited to the temperature range $T > \Theta$. If the magnetic data are represented in the form

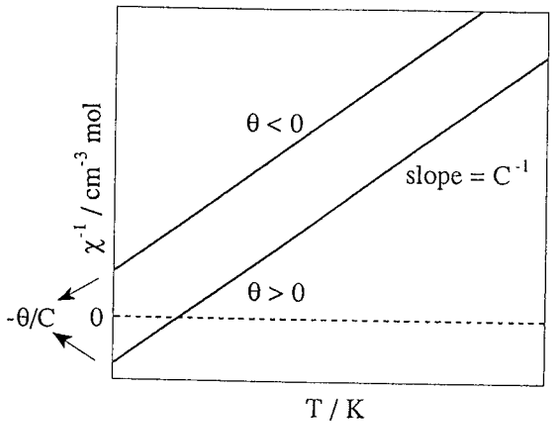


Figure 2.8. χ^{-1} versus temperature plot for an assembly of molecules obeying the Curie–Weiss law.

of a χT versus T plot, positive Θ leads to an increase and negative Θ to a decrease of $\chi_M T$ on cooling as shown in Figure 2.9. The lower the temperature, the more pronounced this effect. Here it is important to point out that a deviation with respect to the Curie law may have other origins than intermolecular interactions. Actually, we have seen in Section 2.5 that the zero-field splitting has quite similar effects on the average magnetic susceptibility.

From (2.6.6) and (2.6.8) the energies $E(S, M_S)$ in (2.6.2) may be expressed as

$$E(S,M_S) = M_S\, g\beta(H + H') \tag{2.6.11}$$

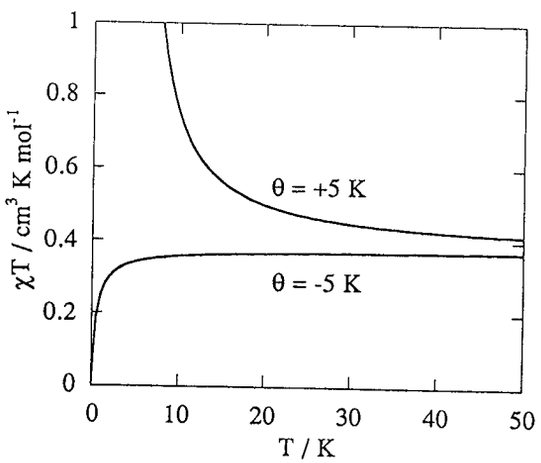


Figure 2.9. χT versus temperature plot for an assembly of molecules obeying the Curie–Weiss law with a Curie constant C = 0.375 cm^3 K mol^{-1} and the Weiss constant θ = 5 K (top) and –5 K (bottom).

where H' defined by

$$H' = zJ\chi_M H/N\beta^2 g^2 \tag{2.6.12}$$

is the molecular exchange field. For $J > 0$ the magnetic field experienced by the sample is larger than the applied magnetic field. For $J < 0$ the reverse situation holds.

References

2.1. E.P. Wigner, "Group Theory." Academic Press, New York, 1959.

2.2. C.J. Ballhausen, "Introduction to Ligand Field Theory." McGraw-Hill, New York, 1962.

2.3. R. McWeeny, "Symmetry, an Introduction to Group Theory and its Applications." Pergamon Press, Oxford, 1963.

2.4. M. Tinkham, "Group Theory and Quantum Mechanics." McGraw-Hill, New York, 1964.

2.5. V. Heine, "Group Theory in Quantum Mechanics." Pergamon Press, Oxford, 1966.

2.6. L.M. Falicov, "Group Theory and Its Physical Applications." The University of Chicago Press, Chicago, 1966.

2.7. D.M. Brink, G.R. Satchler, "Angular Momentum." Clarendon Press, Oxford, 1971.

2.8. J.S. Griffith, "The Theory of Transition-Metal Ions." Cambridge University Press, Cambridge, 1971.

2.9. R. Pauncz, "Spin Eigenfunctions; Construction and Use." Plenum Press, New York, 1979.

2.10. C.J. O'Connor, *Prog. Inorg. Chem.* 29 (1982), 203.

CHAPTER

3

Molecules Containing a Unique Magnetic Center with a First-Order Orbital Momentum

3.1 First-Order Orbital Momentum and Spin-Orbit Coupling

In the previous chapter we assumed that only the $^{2S+1}\Gamma$ ground state was thermally populated and that this state had no first-order orbital momentum. This means that the average values $\langle L_u \rangle$ ($u = x, y, z$) of the three components $\mathbf{L}_x$, $\mathbf{L}_y$, and $\mathbf{L}_z$ of the orbital momentum $\mathbf{L}$ are zero. Such a situation is observed when all the matrix elements $\langle \Psi_i \mid \mathbf{L}_u \mid \Psi_j \rangle$ are zero, Ψ_i being one of the wavefunctions associated with the $^{2S+1}\Gamma$ state. $\mathbf{L}_u$ has the same transformation properties as the rotation operator $\mathbf{R}_u$ in the group describing the symmetry properties of the molecule. It transforms as the irreducible representation Γ_u^{orb} of this group. The condition for $\langle L_u \rangle$ to be zero is that the direct product $\Gamma \times \Gamma$ does not contain Γ_u^{orb}. When this condition is not fulfilled for at least one of the directions $u = x$, y, or z, the first-order orbital momentum cannot be ignored. It couples with the spin momentum, which partially removes the degeneracy of the $^{2S+1}\Gamma$ state, and the magnetic properties may be different from what was described in Chapter 2. The magnetic anisotropy is often much more important, and the average susceptibility may not follow the Curie law, even at relatively high temperature. This chapter deals with this problem. We will not try to cover all situations where the magnetic properties are influenced at the first order by the orbital momentum. Rather, we will focus on two cases of transition ions of the first row, namely those of d^1 [for example, Ti(III)] and d^7 [for example, Co(II)] ions in octahedral surroundings, then we will speak about the magnetism of molecular compounds containing rare earths.

The Hamiltonian we need to consider when applying a magnetic field is

$$\mathbf{H} = \lambda \mathbf{L} \cdot \mathbf{S} + \beta(\mathbf{L} + g_e \mathbf{S}) \cdot H \tag{3.1.1}$$

where λ is the spin-orbit coupling parameter; λ is positive if the number of d electrons is smaller than five, and negative otherwise. The first term in the right-hand side of (3.1.1) corresponds to the spin-orbit coupling and the second to the Zeeman perturbation. This second term may be considered as a perturbation acting on the eigenstates of $\lambda\mathbf{L}\cdot\mathbf{S}$. In (3.1.1) the coupling between the thermally populated $^{2S+1}\Gamma$ ground state and the excited states is assumed to be negligible. Otherwise, $g_e = 2.0023$ should be replaced by an anisotropic g-factor (see section 2.3).

In principle, the determination of the eigenvalues of the Hamiltonian (3.1.1) requires the calculation of integrals like $\langle\Psi_i \mid \mathbf{L}_u \mid \Psi_j\rangle$. If the wavefunction Ψ_i is expressed as an antisymmetrized product of orbitals φ_u, then $\langle\Psi_i \mid \mathbf{L}_u \mid \Psi_j\rangle$ reduces to a sum of one-electron terms $\langle\varphi_\mu \mid \mathbf{l}_u \mid \varphi_\nu\rangle$. For a paramagnetic metal ion surrounded by ligands, φ_μ is generally expressed as a linear combination of atomic orbitals

$$\varphi_\mu = \sqrt{1 - x^2}\, d_\mu - x\phi_\mu \tag{3.1.2}$$

where d_μ notes a d metal orbital and ϕ_μ a linear combination of ligand orbitals. Neglecting small integrals like $\langle d_\mu \mid \mathbf{l}_u \mid \phi_\nu\rangle$ and $\langle\phi_\mu \mid \mathbf{l}_u \mid \phi_\nu\rangle$ results in

$$\langle\varphi_\mu \mid \mathbf{l}_u \mid \varphi_\nu\rangle = \kappa\langle d_\mu \mid \mathbf{l}_u \mid d_\nu\rangle \tag{3.1.3}$$

with

$$\kappa = 1 - x^2 \tag{3.1.4}$$

where κ is generally called the orbital reduction factor [3.1–3.3]. The delocalization of the unpaired electrons from the metal toward the ligands reduces the role of the orbital momentum in (3.1.1). To account for this effect, the Hamiltonian (3.1.1) may be replaced by

$$\mathbf{H} = \kappa\lambda\mathbf{L}\cdot\mathbf{S} + \beta(\kappa\mathbf{L} + g_e\mathbf{S})\cdot H \tag{3.1.5}$$

and the eigenfunctions are then expressed from pure d metal orbitals. The matrix elements $\langle d_\mu \mid \mathbf{l}_u \mid d_\nu\rangle$ are tabulated in Appendix 3.

3.2 Magnetism of a d^1 Ion in an Octahedral Field

A classical and simple application of the above is provided by the case of a d^1 ion situated at the center of a perfect octahedron. The ground state arising from the t_{2g}^1 configuration is $^2T_{2g}$. The components of the orbital momentum $\mathbf{L}$ transform as the T_{1g} irreducible representation of the O_h symmetry group, and the direct product $\Gamma \times \Gamma = T_{2g} \times T_{2g}$ $(= A_{1g} + E_g + T_{1g} + T_{2g})$ contains T_{1g}, such that the $^2T_{2g}$ ground state has a first-order orbital momentum. The degeneracy of this state is partially removed by the spin-orbit coupling. From the Appendices 3 and 4 we can easily build the matrix associated with the spin-orbit coupling operator using the six functions $d_{xy}\alpha$, $d_{yz}\alpha$, $d_{zx}\alpha$, $d_{xy}\beta$, $d_{yz}\beta$, and $d_{zx}\beta$ as a basis set. This matrix may be written in the block diagonalized form shown below:

$d_{xy}\alpha$	$d_{yz}\beta$	$d_{zx}\beta$	$d_{xy}\beta$	$d_{yz}\alpha$	$d_{zx}\alpha$
0	$\kappa\lambda/2$	$-i\kappa\lambda/2$	0	0	0
	0	$-i\kappa\lambda/2$	0	0	0
		0	0	0	0
			0	$-\kappa\lambda/2$	$-i\kappa\lambda/2$
				0	$i\kappa\lambda/2$
					0

(3.2.1)

The eigenvalues and eigenvectors of the matrix (3.2.1) are

$$E = -\kappa\lambda/2 \quad \Psi_1 = (\sqrt{6}/6)(2d_{xy}\alpha - d_{yz}\beta - id_{zx}\beta)$$
$$\Psi_2 = (\sqrt{6}/6)(2d_{xy}\beta + d_{yz}\alpha - id_{zx}\alpha)$$
$$\Psi_3 = (\sqrt{2}/2)(d_{yz}\alpha + id_{zx}\alpha)$$
$$\Psi_4 = (\sqrt{2}/2)(-d_{yz}\beta + id_{zx}\beta)$$

$$E = \kappa\lambda \quad \Psi_5 = (\sqrt{3}/3)(d_{xy}\alpha + d_{yz}\beta + id_{zx}\beta)$$
$$\Psi_6 = (\sqrt{3}/3)(d_{xy}\beta - d_{yz}\alpha + id_{zx}\alpha) \tag{3.2.2}$$

Ψ_1, Ψ_2, Ψ_3, and Ψ_4 transform as the irreducible representation Γ_8 of the double group O′, and Ψ_5 and Ψ_6 as the irreducible representation Γ_7. It follows that the spin-orbit coupling splits the $^2T_{2g}$ state into two levels separated in energy by $3\kappa\lambda/2$. The lower energy level is fourfold degenerate, whereas the higher energy level is twofold degenerate.

Using the Ψ_i functions diagonalizing the spin-orbit coupling as a basis set, the matrix associated with $\mathbf{H}$ defined in (3.1.5) is

Ψ_1	Ψ_2	Ψ_3	Ψ_4	Ψ_5	Ψ_6
$-\kappa\lambda/2 + (1-\kappa)\beta H/3$	0	0	0	$\sqrt{2}(2+\kappa)\beta H/3$	0
	$-\kappa\lambda/2 - (1-\kappa)\beta H/3$	0	0	0	$-\sqrt{2}(2+\kappa)\beta H/3$
		$-\kappa\lambda/2 + (1-\kappa)\beta H$	0	0	0
			$-\kappa\lambda/2 - (1-\kappa)\beta H$	0	0
				$\kappa\lambda - (1+2\kappa)\beta H/3$	0
					$\kappa\lambda + (1+2\kappa)\beta H/3$

(3.2.3)

The eigenvalues of this matrix (3.2.3) may be expanded according to the powers of H up to the quadratic term. The results are gathered in (3.2.4) in a form appropriate to the use of the Van Vleck formula, the energy of the Γ_8 level being taken as the origin:

n	$E_n^{(0)}$	$E_n^{(1)}/H$	$E_n^{(2)}/H^2$
1	0	$(1-\kappa)\beta/3$	$-4(2+\kappa)^2\beta^2/27\kappa\lambda$
2	0	$-(1-\kappa)\beta/3$	$-4(2+\kappa)^2\beta^2/27\kappa\lambda$
3	0	$(1-\kappa)\beta$	0
4	0	$-(1-\kappa)\beta$	0
5	$3\kappa\lambda/2$	$(1+2\kappa)\beta/3$	$4(2+\kappa)^2\beta^2/27\kappa\lambda$
6	$3\kappa\lambda/2$	$-(1+2\kappa)\beta/3$	$4(2+\kappa)^2\beta^2/27\kappa\lambda$

(3.2.4)

Introducing zero-field energies and Zeeman coefficients in the Van Vleck formula leads to :

$$\chi = \frac{N\beta^2}{27kT}\,\frac{[30(1-\kappa)^2x + 8(2+\kappa)^2] + [3(1+2\kappa)^2x - 8(2+\kappa)^2]\exp(-3x/2)}{x[2+\exp(-3x/2)]} \tag{3.2.5}$$

with

$$x = \kappa\lambda/kT \tag{3.2.6}$$

If the reduction of the orbital momentum operator $\mathbf{L}$ is neglected ($\kappa = 1$), then (3.2.5) becomes [3.4]

3.2.7

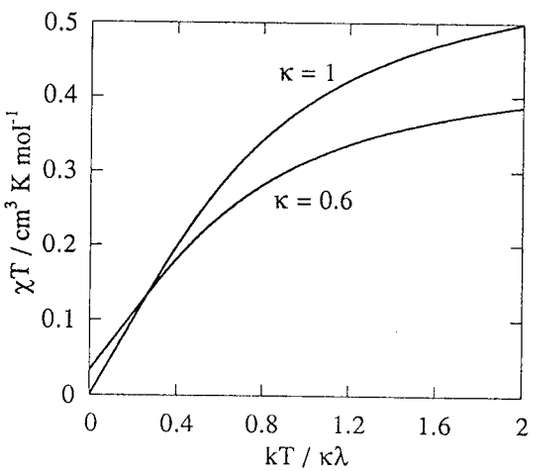


Figure 3.1. χT versus $kT/\kappa\lambda$ plots for a d^1 ion in a perfect octahedral environment. The curves are plotted for two values of the orbital reduction factor, namely $\kappa = 1$ and 0.6.

$$\chi = \frac{N\beta^2}{3kT}\,\frac{8 + (3x-8)\exp(-3x/2)}{x[2+\exp(-3x/2)]} \tag{3.2.7}$$

The χT versus $1/x = kT/\kappa\lambda$ plots for $\kappa = 1$ and 0.6 are shown in Figure 3.1. The magnetic susceptibility does not follow the Curie law. χT decreases as T is lowered and, for $\kappa = 1$, tends to zero for T approaching absolute zero. This rather surprising result is due to the fact that for this specific value of κ, the Γ_8 level is nonmagnetic at the first order; spin and angular momenta exactly cancel each other (in the approximation where g_e is taken equal to 2.00). For $\kappa \neq 1$ this situation is no longer true and the low-temperature limit of χT is $5N\beta^2(1-\kappa)^2/9k$.

It is worth noting that Eq. (3.2.5) is not valid for $\kappa = 0$ (or even close to 0). Indeed, Γ_7 and Γ_8 are then accidentally degenerate ($\kappa\lambda = 0$), and it is no longer legitimate to treat the Zeeman term as a perturbation with respect to the spin-orbit coupling.

3.3 Magnetism of a d^1 Ion in an Axially Distorted Octahedral Field

The derivation of Eqs. (3.2.5) and (3.2.7) is a pedagogically interesting exercise but is a little formal since it is rather unlikely for a d^1 ion to have a perfectly octahedral environment. Actually crystal packing and/or the Jahn–Teller effect are expected to remove the orbital degeneracy and to give an orbital singlet and an orbital doublet if the symmetry is axial, or three orbital singlets if the symmetry is lower. Let us assume that the actual symmetry is D_{4h} with a distortion along the z axis. The $^2T_{2g}$ state is then split into $^2B_{2g}$ and ^2E_g. Such a symmetry lowering may be accounted for by the effective operator $\mathbf{V}_{dis}$:

$$\mathbf{V}_{dis} = \Delta(2 - \mathbf{L}_z^2)/3 \tag{3.3.1}$$

which acts on the t_{2g} orbitals as (see Appendix 2):

d_{xy}	d_{yz}	d_{zx}
$-2\Delta/3$	0	0
	$\Delta/3$	0
		$\Delta/3$

(3.3.2)

such that the energy gap between the $^2B_{2g}$ and ^2E_g states is Δ. The Hamiltonian to consider is now

$$\mathbf{H} = \lambda\mathbf{L}\cdot\mathbf{S} + \mathbf{V}_{dis} + \beta(\mathbf{L} + g_e\mathbf{S})\cdot H \tag{3.3.3}$$

where, for the sake of simplicity, we have neglected the orbital reduction ($\kappa = 1$). The method used to solve the problem consists of diagonalizing the first two terms of (3.3.3) using the matrices (3.2.1) and (3.3.2), then of treating the Zeeman term as a

perturbation acting on the eigenstates of $\lambda \mathbf{L}\cdot\mathbf{S} + \mathbf{V}_{dis}$. This treatment must be carried out for two directions of the magnetic field, namely along the z axis and perpendicular to it, to find the parallel χ_z and perpendicular χ_x susceptibilities, respectively. The calculation is rather tedious but does not present any conceptual difficulty. It leads to the following zero-field energies [3.5, 3.6]:

$$E_1 = E_2 = (\lambda/2)[(\nu + 1/2) - (\nu^2 + \nu + 9/4)^{1/2}]$$

$$E_3 = E_4 = \lambda(\nu - 1/2)$$

$$E_5 = E_6 = (\lambda/2)[(\nu + 1/2) + (\nu^2 + \nu + 9/4)^{1/2}] \tag{3.3.4}$$

with

$$\nu = \Delta/\lambda \tag{3.3.5}$$

and to the following first- and second-order Zeeman coefficients:

H parallel to the z axis

$$E^{(1)}_{1,2} = \pm(2 - a^2)\beta/(1 + a^2)$$

$$E^{(1)}_{3,4} = \pm(2 - b^2)\beta/(1 + b^2)$$

$$E^{(1)}_{5,6} = 0 \tag{3.3.6}$$

$$E^{(2)}_{1,2} = -2(2 - ab)^2\beta^2/[\lambda(1 + a^2)(1 + b^2)c]$$

$$E^{(2)}_{3,4} = 2(2 - ab)^2\beta^2/[\lambda(1 + a^2)(1 + b^2)c]$$

$$E^{(2)}_{5,6} = 0 \tag{3.3.7}$$

H perpendicular to the z axis

$$E^{(1)}_{1,2} = \pm a(\sqrt{2} - a)\beta/(1 + a^2)$$

$$E^{(1)}_{3,4} = \pm b(\sqrt{2} - b)\beta/(1 + b^2)$$

$$E^{(1)}_{5,6} = 0 \tag{3.3.8}$$

$$E^{(2)}_{1,2} = -(2\beta^2/\lambda)\{[(a + b - ab\sqrt{2})^2/2(1 + a^2)(1 + b^2)c] + [(a - \sqrt{2})^2/(1 + a^2)(\nu - 3/2 + c)]\}$$

$$E^{(2)}_{3,4} = (2\beta^2/\lambda)\{[(a + b - ab\sqrt{2})^2/2(1 + a^2)(1 + b^2)c] - [(b - \sqrt{2})^2/(1 + b^2)(\nu - 3/2 - c)]\}$$

$$E^{(2)}_{5,6} = (2\beta^2/\lambda)\{[(a - \sqrt{2})^2/(1 + a^2)(\nu - 3/2 + c)]$$

$$+ [(b - \sqrt{2})^2/(1 + b^2)(\nu - 3/2 - c)]\} \tag{3.3.9}$$

with

$$a = (\sqrt{2}/2)[(\nu + 1/2) + (\nu^2 + \nu + 9/4)^{1/2}]$$

$$b = (\sqrt{2}/2)[(\nu + 1/2) - (\nu^2 + \nu + 9/4)^{1/2}]$$

$$c = (\nu^2 + \nu + 9/4)^{1/2} \tag{3.3.10}$$

Figure 3.2 shows the $\chi_z T$ and $\chi_x T$ versus kT/λ plots for $\nu = \pm 10$. When $\nu\,(>0)$ increases, the coupling between $^2B_{2g}$ ground state and ^2E_g excited state through $\lambda\mathbf{L}\cdot\mathbf{S}$ decreases and, since the $^2B_{2g}$ state has no first-order angular momentum, both the temperature dependences of $\chi_z T$ and $\chi_x T$ and the magnetic anisotropy (defined as $|\chi_z - \chi_x|\,T$) decrease. For $\nu < 0$, on the other hand, the ^2E_g state is the lowest. This state is split by the spin-orbit coupling into two Kramers doublets such that temperature dependences of $\chi_z T$ and $\chi_x T$, and magnetic anisotropy remain pronounced.

The most likely situation for a d^1 ion in octahedral surroundings is that where ν is positive with an orbital singlet ground state. It is especially the case if the distortion is due to the Jahn–Teller effect. In the temperature range where only this state is thermally populated, the magnetic properties may be approximated by considering a $S = 1/2$ spin state and using a Zeeman perturbation of the form

$$\mathbf{H}_{ZE} = \beta\mathbf{S}\cdot\mathbf{g}\cdot H \tag{3.3.11}$$

the principal values of the $\mathbf{g}$-tensor being

$$g_z = |\,2(2 - a^2)/(1 + a^2)\,|$$

$$g_x = g_y = |\,2a(\sqrt{2} - a)/(1 + a^2)\,| \tag{3.3.12}$$

As anticipated, for ν very large both g_z and g_x tend to 2. For $\nu = 10$, g_z and g_x are equal to 1.97 and 1.80, respectively. The principal magnetic susceptibilities in this approximation where only the $^2B_{2g}$ state is thermally populated are given by

$$\chi_{z(x)} = \frac{N g^2_{z(x)}\beta^2}{4kT} + C_{z(x)} \tag{3.3.13}$$

where $C_{z(x)}$ are positive temperature-independent contributions accounting for the coupling of the ground Kramers doublet with the excited levels through the Zeeman perturbation (see Section 1.5), which in this case may be rather large. Figure 3.2 confirms that $\chi_z T$ and $\chi_x T$ vary almost linearly versus temperature for $\nu = +10$.

The same kind of treatment may be carried out for any other case where the ground state has a first-order angular momentum, for instance for d^2, low-spin d^4, low-spin d^5, high-spin d^6, and high-spin d^7 ions in octahedral surroundings. The derivation of the magnetic susceptibilities is quite similar to what was performed above for a d^1 ion even if the calculation may be a bit more tedious since there are more populated states. Such an approach has been developed by Figgis et al. in the 1960s [3.6–3.9]. The important point emerging from the comparison between theoretical and experimental

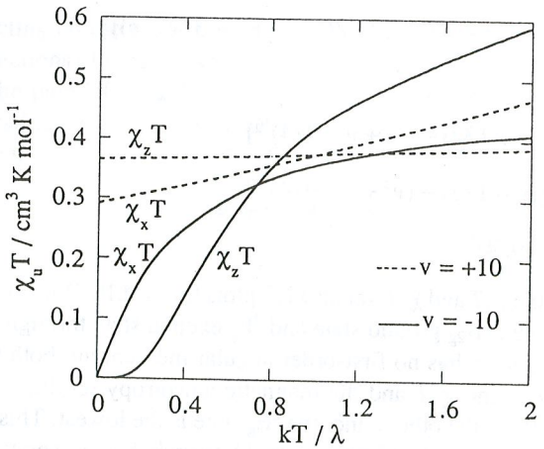


Figure 3.2. $\chi_z T$ and $\chi_x T$ plots for a d^1 ion in an axially distorted environment. The curves are plotted for $v = \Delta/\lambda = \pm 10$.

susceptibilities is that for octahedral complexes with an orbital triplet ground state the average magnetic susceptibility does not rigorously follow a Curie law. This is due to the second-order Zeeman coefficients. The deviation from the Curie law, however, is usually less pronounced than anticipated. This is due to two factors, namely the partial quenching of the orbital momentum resulting from the covalence ($\kappa < 1$), and the lowering of symmetry leading to an orbital singlet ground state. To explore further this situation, we will treat in detail the case of high-spin cobalt(II) complexes of octahedral symmetry in the next section.

3.4 Magnetism of a Cobalt(II) Ion in Octahedral Surroundings

The mononuclear compounds containing a high-spin cobalt(II) ion in octahedral surroundings present magnetic properties deviating substantially from a Curie law. If the symmetry was purely O_h, then the ground state would be $^4T_{1g}$. For real compounds, however, the symmetry is at best axial, generally with an orbital singlet ground state and an orbital doublet excited state. If the distortion is tetragonal (D_{4h}), these states may be labeled as $^4A_{2g}$ and ^4E_g. They are split by the spin-orbit coupling such that the spectrum is as shown in Figure 3.3 where the levels arising from the action of spin-orbit coupling are labeled with the irreducible representations of the double group D'_4. Γ_6 refers to $M_S = \pm 1/2$ and Γ_7 to $M_S = \pm 3/2$. If the symmetry were rhombic instead of axial, then the ^4E_g state would be split into two components and each would be split further into two Kramers doublets by spin-orbit coupling.

The derivation of magnetic properties for such a system may be carried out using the method described in the previous section. The basis set consists of the twelve

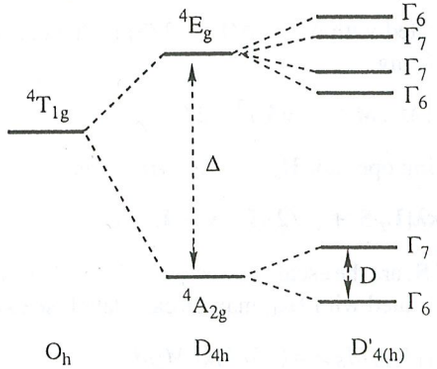


Figure 3.3. Energy spectrum for a high-spin Co(II) ion an axially distorted octahedral environment.

functions associated with the $^4T_{1g}$ state and the Hamiltonian is that given in (3.3.3). The spin-orbit coupling parameter λ for cobalt(II) is negative since there are more than five d electrons, and of the order of 170 cm^{-1}. The calculation requires the introduction of a new parameter describing the degree of mixing of the two $^4T_{1g}$ states arising from the ground ^4F and excited ^4P terms of the d^7 free ion. A convenient way of performing this calculation consists of using the analogy between the T_{1g} irreducible representation in O_h and the P irreducible representation in R_3, the rotation group of the sphere. The basis set is then made out of the functions $|M_L, M_S\rangle$ with $M_L = 0, \pm 1$, and $M_S = \pm 1/2, \pm 3/2$. The Hamiltonian is written down as

$$\mathbf{H} = \Delta(\mathbf{L}_z^2 - 2/3) - (3/2)\kappa\lambda\mathbf{L}\cdot\mathbf{S} + \beta[-(3/2)\kappa\mathbf{L}_u + g_e\mathbf{S}_u]\cdot H_u \qquad (3.4.1)$$

The factor –3/2 comes from the fact that the real angular momentum for the $^4T_{1g}$ state is equal to the angular momentum of the ^4P free-ion state multiplied by –3/2. As for the reduction factor κ, its meaning is a little bit more complex than in the case of the d^1 ion. It takes into account both the covalence of the Co–ligand bonds and the admixture of the $^4T_{1g}(^4P)$ excited state into the $^4T_{1g}$ ground state. Neglecting covalence, κ varies between 1 and 2/3. The higher value corresponds to the weak-field limit; there is no mixing of the two $^4T_{1g}$ states. The lower value corresponds to the strong-field limit. The $^4T_{1g}$ ground state arises from the $t_{2g}^5 e_g^2$ configuration and the wavefunction is of the form

$$\Psi[^4T_{1g}] = (2\sqrt{5}/5)\Psi[^4T_{1g}(^4F)] - (\sqrt{5}/5)\Psi[^4T_{1g}(^4P)] \qquad (3.4.2)$$

The method used to solve the problem again consists of diagonalizing spin-orbit coupling and distortion together, then calculating the Zeeman coefficients for the two directions parallel and perpendicular to the applied magnetic field. To help the reader who would like to become acquainted with this kind of calculation, we give some information on the way to follow.

The matrix associated with $\mathbf{V}_{dis} = \Delta(\mathbf{L}_z^2 - 2/3)$ is diagonal on the $|M_L, M_S\rangle$ basis, the matrix elements being

$$\langle M_L, M_S | \mathbf{V}_{dis} | M_L, M_S \rangle = \Delta(M_L^2 - 2/3) \tag{3.4.3}$$

The spin-orbit coupling operator $\mathbf{H}_{so}$ may be written as :

$$\mathbf{H}_{SO} = (-3/2)\kappa\lambda[\mathbf{L}_z \cdot \mathbf{S}_z + (1/2)(\mathbf{L}_+ \cdot \mathbf{S}_- + \mathbf{L}_- \cdot \mathbf{S}_+)] \tag{3.4.4}$$

where $\mathbf{L}_+$, $\mathbf{L}_-$, $\mathbf{S}_+$, and $\mathbf{S}_-$ are the escalator operators defined in Appendix 5. The nonzero matrix elements associated with $\mathbf{H}_{SO}$ may be calculated (see Appendix 5) as

$$\langle M_L, M_S | \mathbf{H}_{SO} | M_L, M_S \rangle = (-3/2)\kappa\lambda M_L M_S$$

$$\langle M_L, M_S | \mathbf{H}_{SO} | M_L + 1, M_S - 1 \rangle$$

$$= \langle M_L + 1, M_S - 1 | \mathbf{H}_{SO} | M_L, M_S \rangle$$

$$= (-3/4)\kappa\lambda[(2 + M_L)(1 - M_L)(5/2 - M_S)(3/2 + M_S)]^{1/2}$$

with $M_L \neq 1$, and $M_S \neq -3/2$

$$\langle M_L, M_S | \mathbf{H}_{SO} | M_L - 1, M_S + 1 \rangle$$

$$= \langle M_L - 1, M_S + 1 | \mathbf{H}_{SO} | M_L, M_S \rangle$$

$$= (-3/4)\kappa\lambda[(2 - M_L)(1 + M_L)(5/2 + M_S)(3/2 - M_S)]^{1/2}$$

with $M_L \neq -1$, and $M_S \neq 3/2$ (3.4.5)

The matrix associated with the Zeeman perturbation $\mathbf{H}_{ZE,z}$ for H parallel to the z axis is diagonal on the $|M_L, M_S\rangle$ basis, the matrix elements being

$$\langle M_L, M_S | \mathbf{H}_{ZE,z} | M_L, M_S \rangle = [(-3/2)\kappa M_L + g_e M_S]\beta H_z \tag{3.4.6}$$

The matrix associated with the Zeeman perturbation $\mathbf{H}_{ZE,x}$, for H perpendicular to the unique axis, can be determined from the matrices associated with $\mathbf{L}_x$ on the $|M_L\rangle$ basis, and with $\mathbf{S}_x$ on the $|M_S\rangle$ basis. These matrices are easily established from

$$\mathbf{L}_x = (\mathbf{L}_+ + \mathbf{L}_-)/2$$

$$\mathbf{S}_x = (\mathbf{S}_+ + \mathbf{S}_-)/2 \tag{3.4.7}$$

as

	|1>	|0>	|−1>
	0	$\sqrt{2}/2$	0
L_x =		0	$\sqrt{2}/2$
			0

	|3/2>	|1/2>	|−1/2>	|−3/2>
	0	$\sqrt{3}/2$	0	0
S_x =		0	1	0
			0	$\sqrt{3}/2$
				0

(3.4.8)

which leads to the nonzero matrix elements

$$\langle 1, M_S | \mathbf{H}_{ZE,x} | 0, M_S \rangle$$

$$= \langle 0, M_S | \mathbf{H}_{ZE,x} | -1, M_S \rangle = -3\sqrt{2}\kappa\beta H_x/4$$

$$\langle M_L, \pm 3/2 | \mathbf{H}_{ZE,x} | M_L, \pm 1/2 \rangle = \sqrt{3} g_e \beta H_z/2$$

$$\langle M_L, 1/2 | \mathbf{H}_{ZE,x} | M_L, -1/2 \rangle = g_e \beta H_x$$

with

$$\langle M_L, M_S | \mathbf{H}_{ZE,x} | M_{L'}, M_{S'} \rangle = \langle M_{L'}, M_{S'} | \mathbf{H}_{ZE,x} | M_L, M_S \rangle \tag{3.4.9}$$

The calculation of the parallel and perpendicular magnetic susceptibilities may then be easily performed on a microcomputer.

The χT versus $kT/|\lambda|$ curves for $v\ (= \Delta/|\lambda|) = 0$ and 10 are shown in Figure 3.4. In both cases κ is taken to be equal to 1 (no covalence and weak-field limit). Except at very low temperatures, χT is larger than expected for a $S = 3/2$ spin state with $g = g_e$ ($\chi_M T = 1.876$ cm^3 K mol^{-1}). The smaller v is, the more pronounced this difference.

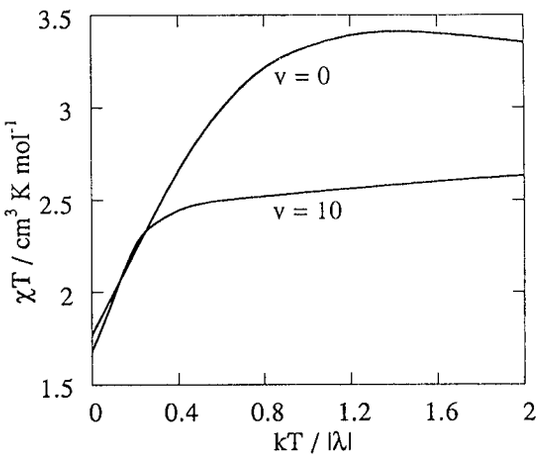


Figure 3.4. χT *versus* $kT/|\lambda|$ plots for a high-spin cobalt(II) ion in an axially distorted octahedral environment. The curves are plotted for $v = \Delta/|\lambda| = 0$ and 10.

The distortion quenches the orbital momentum provided that the lowest state has no first-order angular momentum. This approach, where the orbital momentum is explicitly taken into account, reproduces fairly well the temperature dependence of the average magnetic susceptibility, However, the situation is much less satisfying as far as the principal susceptibilities are concerned. All efforts to reproduce the details of the magnetic anisotropy for octahedral high-spin cobalt(II) compounds have more or less failed. This might be due to the fact that the model is oversimplified [3.10–3.12]. Let us mention two of these simplifications: (1) the mixing of the two $^4T_{1g}$ states is properly taken into account, but the coupling between the ground state and the other excited states, especially the $^4T_{2g}$ and $^4A_{2g}$ states arising from the ^4F free-ion state, through the spin-orbit coupling and the Zeeman perturbation are ignored; (2) the covalence factor involved in κ and the spin-orbit coupling parameter λ might not be isotropic. A more elaborate model, taking into account all these factors, can be constructed but is then overparameterized. It follows that the determination of the various parameters by fitting of the experimental magnetic data is almost meaningless. Therefore, it is worthwhile to look for more simple models where the magnetic properties are deduced from Hamiltonians involving only spin operators, and to examine their validity limits.

When Δ is large enough, only the two lowest Kramers doublets Γ_6 and Γ_7 arising from the $^4A_{2g}$ state are thermally populated. The energy separation D between Γ_7 and Γ_6 may then be considered as a zero-field splitting within the quartet state, and Eqs. (2.5.5) and (2.5.20) may be utilized to interpret the magnetic properties. In this case the Hamiltonian to consider is

$$\mathbf{H} = D(\mathbf{S}_z^2 - 5/4) + \beta \mathbf{S} \cdot \mathbf{g} \cdot H \tag{3.4.10}$$

The orbital momentum has disappeared. Its influence has been incorporated into D on the one hand, and the principal values g_z and g_x of the **g**-tensor on the other hand. D is of the order of several tens of wavenumbers, i.e., much larger than for a compound containing a metal ion with a spin-quartet ground state without first-order angular momentum and well separated in energy from the first excited states, for instance a Cr(III) ion. It follows that the magnetic anisotropy is also much more pronounced than for a Cr(III) compound.

A further approximation consists of assuming that only the lowest Kramers doublet, Γ_6, is thermally populated, which is valid if the magnetic data are limited to the very low temperature range. Since this Γ_6 level is split into two components by the Zeeman perturbation, it may be treated as an effective spin $S' = 1/2$. The magnetic properties then derive from the simple Hamiltonian

$$\mathbf{H} = \beta \mathbf{S}' \cdot \mathbf{g}' \cdot H \tag{3.4.11}$$

where $\mathbf{g}'$ is the Zeeman tensor associated with the effective spin $S' = 1/2$. The influence of the orbital momentum is incorporated into the principal values of $\mathbf{g}'$. For $\Delta > 0$ we have $g_z' < g'_x$. In the limit where Δ is very large, g'_z is close to 2 and g'_x is close to 4. For $\Delta = 0$, $\mathbf{g}'$ is evidently isotropic with the principal value $g' = 4.33$ for $\kappa = 1$. In a few cases Δ is found to be negative, the ground state being ^4E_g. If so, the lowest level

arising from ^4E_g again has the Γ_6 symmetry but the g′-tensor is such that $g'_z >> g'_x$ [3.13–3.16].

In this effective-spin approximation the principal susceptibilities may be written down as

$$\chi_{z(x)} = \frac{N g'^2_{z(x)} \beta^2}{4kT} + C_{z(x)} \tag{3.4.12}$$

where $C_{z(x)}$ are positive temperature-independent contributions accounting for the coupling of the Γ_6 populated level with the excited levels through the Zeeman perturbation (see Section 1.5).

In this section the same problem has been approached at three levels of approximation. The first treatment is, in principle, the most rigorous since it explicitly takes into account all the levels arising from the orbitally degenerate $^4T_{1g}$ state as well as the angular momentum in the Zeeman perturbation. The derivation of the principal susceptibilities is not difficult; it can be easily performed with a microcomputer. The size of the matrices to diagonalize to get the zero-field energies and the Zeeman coefficients is 12×12. For a polynuclear compound containing n cobalt(II) ions, on the other hand, the size of the matrices would be $12^n \times 12^n$, which would rapidly become intractable [3.17]. In the second approach the basis set is limited to four functions associated with the lowest Γ_6 and Γ_7 levels, and the orbital momentum is no longer considered explicitly. The spin, however, remains the actual spin of the problem, i.e., $S = 3/2$. Finally, the third approach is obviously the simplest one. The basis set is limited to the two spin functions associated with the effective spin $S' = 1/2$. Such an approach is of limited applicability; it is valid only at very low temperatures, say below 20 K if the $\Gamma_6 - \Gamma_7$ separation is of the order of 100 cm^{-1}. Its interest, however, is that it can be easily extended to polynuclear systems.

3.5 Magnetism of the Rare Earth-Containing Compounds: The Free-Ion Approximation

The electronic structure of the rare earth ions has been extensively investigated, essentially in relation to studies dealing with the EPR spectra [3.18, 3.19] or the optical properties [3.20] of these ions doped into diamagnetic host lattices. On the other hand, studies concerning the magnetic properties of rare earth-containing molecular compounds are not so numerous. This may be due to the fact that the number of well characterized compounds of this kind is not yet very large, and also to the fact that precise interpretation of the experimental data remains difficult in most cases [3.21, 3.22]. Some models have been developed for cubic or axial environments, but the actual symmetry is usually lower, especially for molecular compounds. It is, however, clear that the magnetism of rare earth containing species is a field where the role of the orbital momentum cannot be ignored. We are thus justified in approaching this problem in this chapter.

The ground configuration of the rare earth ions is $4f^n5s^25p^6$. These ions are generally trivalent; they can be divalent [Eu(II),Sm(II) and Yb(II)] or tetravalent [Ce(IV)] when the $4f^0$, $4f^7$, or $4f^{14}$ configurations are attained. The $4f$ orbitals, partly occupied by the magnetically active electrons, are very efficiently shielded by the fully occupied $5s$ and $5p$ orbitals. These $4f$ orbitals are almost uninvolved in the bonds between a rare earth ion and its nearest neighbors. It follows that the influence of the environment on the magnetic properties is much less pronounced for a rare earth compound than for a $3d$ ion compound. In a first approximation a rare earth ion in a molecular compound behaves as a free ion. In other respects the magnitude of the spin-orbit coupling is much larger than for a $3d$ ion, and increases from the left to the right of the Periodic Table. This spin-orbit coupling partially removes the degeneracy of the $^{2S+1}\Gamma$ ground term, where Γ is an irreducible representation of the R_3 group. This gives $^{2S+1}\Gamma_j$ states, with J varying by an integer value from $|L-S|$ *to* $L+S$. J is the quantum number associated with the total angular momentum $\mathbf{J}$ defined as

$$\mathbf{J} = \mathbf{L} + \mathbf{S} \tag{3.5.1}$$

The lowest state has the smallest J for the configurations $4f^1 - 4f^6$, and the highest J for the configurations $4f^8 - 4f^{13}$. For the configuration $4f^7$ there is no first-order angular momentum. The energy separation between the state of lowest energy and the first excited state varies from a few hundreds to several thousands of wavenumbers. When this energy separation is weak, the first excited state may be thermally populated; when it is large, only the ground state is thermally populated. The ground electronic configuration together with ground and first excited states, and their energy separation are given in Table 3.1.

Table 3.1 Ground Configuration, Ground and First Excited States, and Energy Separation between These States for the Rare Earth Ions

Ion	Configuration $4f^n$, n =	Ground state	First excited state	Energy state separation (cm^{-1})
Ce(III)	1	$^2F_{5/2}$	$^2F_{7/2}$	2200
Pr(III)	2	$^3H_4$	$^3H_5$	2100
Nd(III)	3	$^4I_{9/2}$	$^4I_{11/2}$	1900
Pm(III)	4	$^5I_4$	$^5I_5$	1600
Sm(III)	5	$^6H_{5/2}$	$^6H_{7/2}$	1000
Eu(III), Sm(II)	6	$^7F_0$	$^7F_1$	300
Gd(III), Eu(II)	7	$^8S_{7/2}$	$^6P_{7/2}$	30000
Tb(III)	8	$^7F_6$	$^7F_5$	2000
Dy(III)	9	$^6H_{15/2}$	$^6H_{13/2}$	–
Ho(III)	10	$^5I_8$	$^5I_7$	–
Er(III)	11	$^4I_{15/2}$	$^4I_{13/2}$	6500
Tm(III)	12	$^3H_6$	$^3H_5$	–
Yb(III)	13	$^2F_{7/2}$	$^2F_{5/2}$	10000

The free-ion states of the same J arising from different $^{2S+1}\Gamma$ terms may mix through spin-orbit coupling. The resulting states are split further and mixed by the ligand field, such that the spectrum of the low-lying states together with the magnetic properties may be rather complicated. A first approximation, however, consists of neglecting the ligand field effect as well as the mixing of the free-ion states through spin-orbit coupling, and of assuming that only the ground state is thermally populated. We are then faced with the magnetism of a system with a well isolated $^{2S+1}\Gamma_J$ ground state, and J, L, and S are then good quantum numbers. In the presence of a magnetic field H, the $^{2S+1}\Gamma_J$ state is split into $2J + 1$ components, the wavefunctions of which are noted $|J, M_J>$, with M_J varying by an integer value from $-J$ to $+J$. Let us now calculate the energies $E(J, M_J)$ of these components.

The magnetic momentum $\mathbf{M}$ is defined by

$$\mathbf{M} = \mathbf{L} + 2\mathbf{S} \tag{3.5.2}$$

$\mathbf{M}$ is not co-directional with the angular momentum $\mathbf{J}$ defined in (3.5.1) owing to the gyromagnetic factor $g_e = 2$ occurring in (3.5.2). This is illustrated in Figure 3.5. Because of the rapid precession of $\mathbf{M}$ about the direction of $\mathbf{J}$, it may be assumed that the component BC of $\mathbf{M}$ averages out to zero in any finite time, such that only the component AC of $\mathbf{M}$ along $\mathbf{J}$ needs to be considered (see Figure 3.5). Let α_L denote the angle between $\mathbf{L}$ and $\mathbf{J}$, and α_S the angle between $\mathbf{S}$ and $\mathbf{J}$. We can then write

$$\mathbf{S}^2 = \mathbf{J}^2 + \mathbf{L}^2 - 2\mathbf{L}\mathbf{J}\cos\alpha_L$$

$$\mathbf{L}^2 = \mathbf{J}^2 + \mathbf{S}^2 - 2\mathbf{S}\mathbf{J}\cos\alpha_S \tag{3.5.3}$$

The components $\mathbf{L}\cos\alpha_L$ of $\mathbf{L}$, and $\mathbf{S}\cos\alpha_S$ of $\mathbf{S}$ along $\mathbf{J}$ may then be expressed as

$$\mathbf{L}\cos\alpha_L = (\mathbf{J}^2 + \mathbf{L}^2 - \mathbf{S}^2)/2\mathbf{J}$$

$$\mathbf{S}\cos\alpha_S = (\mathbf{J}^2 + \mathbf{S}^2 - \mathbf{L}^2)/2\mathbf{J} \tag{3.5.4}$$

The Zeeman perturbation may then be written as

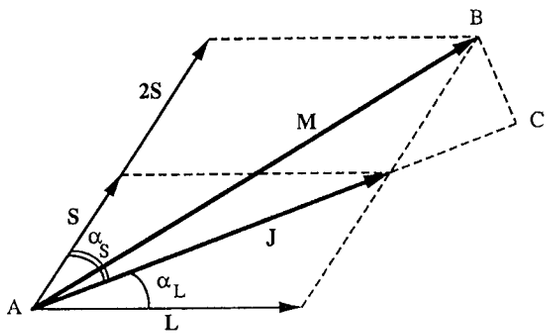


Figure 3.5. Magnetic momentum $\mathbf{M}$ and kinetic momentum $\mathbf{J}$ vectors (see text).

$$\mathbf{H}_{ZE} = \beta(\mathbf{L}\cos\alpha_L + 2\mathbf{S}\cos\alpha_S)\cdot H$$
$$= \beta[(\mathbf{J}^2 + \mathbf{L}^2 - \mathbf{S}^2)/\mathbf{J} + (\mathbf{J}^2 + \mathbf{S}^2 - \mathbf{L}^2)/\mathbf{J}]\cdot H$$
$$= \beta[3/2 + (\mathbf{S}^2 - \mathbf{L}^2)/2\mathbf{J}^2]\mathbf{J}\cdot H \qquad (3.5.5)$$

The eigenvalues of $\mathbf{L}^2$ and $\mathbf{S}^2$ being $L(L+1)$ and $S(S+1)$, respectively, $\mathbf{H}_{ZE}$ becomes

$$\mathbf{H}_{ZE} = \beta\left[\frac{3}{2} + \frac{S(S+1) - L(L+1)}{2J(J+1)}\right]\mathbf{J}\cdot H \qquad (3.5.6)$$

and the energy $E(J, M_J)$ of the $|J,M_J\rangle$ Zeeman component is

$$E(J,M_J) = \langle J,M_M|\,\mathbf{H}_{ZE}\,|J,M_J\rangle$$
$$= \beta g_J M_J H \qquad (3.5.7)$$

with

$$g_J = \frac{3}{2} + \frac{S(S+1) - L(L+1)}{2J(J+1)} \qquad (3.5.8)$$

As expected, for a pure spin ($L = 0$), $g_J = g_S = 2$, and for a pure orbital motion ($S = 0$), $g_J = g_L = 1$. For $J = 0$, Eq. (3.5.8) is indeterminate since L and S are then equal. However in that case we can put $J = S - L$, and write

$$g_0 = \frac{3}{2} + \frac{(S+L+1)(S-L)}{2(S-L)(S-L+1)}$$
$$= \frac{3}{2} + \frac{S+L+1}{2(S-L+1)}$$
$$= 2 + S \text{ (or } 2 + L) \qquad (3.5.9)$$

Of course a ${}^{2S+1}\Gamma_0$ state is not perturbed by the Zeeman perturbation at the first order. On the other hand, this state couples with the component $M_J = 0$ arising from the ${}^{2S+1}\Gamma_1$ state through the Zeeman perturbation, and the matrix element $\langle 0,0|\,\mathbf{H}_{ZE}\,|1,0\rangle$ depends on g_0 [see Eq. (3.5.12)].

The derivation of the magnetic susceptibility is rigorously parallel to the derivation in Section 2.1, J replacing S and g_J replacing g. This leads to

$$\chi = \frac{Ng_J^2\beta^2}{3kT}\,J(J+1) \qquad (3.5.10)$$

In this approximation the magnetic susceptibility obeys the Curie law. The values of g_J and χT calculated from (3.5.10) for the various rare earth ions are given in Table 3.2.

The presence of excited states not too far in energy from the ground state may add a significant temperature-independent contribution to the magnetic susceptibility calculated in (3.5.10). This contribution χ' is calculated as

Table 3.2. Values of g_J and χT for Rare Earth Ions in the Free-Ion Approximation

Ion	Configuration $4f^n$, n =	g_J	$\chi_M T$ (cm^3 K mol^{-1})
Ce(III)	1	6/7	0.80
Pr(III)	2	4/5	1.60
Nd(III)	3	8/11	1.64
Pm(III)	4	3/5	0.90
Sm(III)	5	2/7	0.09
Eu(III), Sm(II)	6	5	0.00
Gd(III), Eu(II)	7	2	7.88
Tb(III)	8	3/2	11.82
Dy(III)	9	4/3	14.17
Ho(III)	10	5/4	14.07
Er(III)	11	6/5	11.48
Tm(III)	12	7/6	7.15
Yb(III)	13	8/7	2.57

$$\chi' = -2N\sum_{M_J=-J}^{+J}\sum_{J'\neq J}^{+J'}\sum_{M_{J'}=-J'}^{+J'}\frac{\langle J,M_J|\,\beta(\mathbf{L}+2\mathbf{S})\,|J',M_{J'}\rangle^2}{[E(J,M_J) - E(J',M_{J'})]} \qquad (3.5.11)$$

and has been found as [3.23]

$$\chi' = \frac{2N\beta^2(g_J - 1)(g_J - 2)}{3\lambda} \qquad (3.5.12)$$

where λ is the spin-orbit coupling parameter occurring in the spin-orbit coupling operator $\lambda\mathbf{L}\cdot\mathbf{S}$ acting on the ${}^{2S+1}\Gamma$ term.

The magnetization is also given by an expression very close to (2.2.7), namely

$$M = Ng_J\beta JB_J(y) \qquad (3.5.13)$$

with

$$y = g_J\beta JH/kT \qquad (3.5.14)$$

3.6 Magnetism of Rare Earth-Containing Compounds: Some Examples

3.6.1 Gd(III) or Eu(II)

The simplest situation as far as the magnetism of the rare earth compounds is concerned is offered by Gd(III) and Eu(II) derivatives. The ground state arising from the $4f^7$ configuration is ${}^8S_{7/2}$. Since $L = 0$, there is no spin-orbit coupling. Furthermore the first excited state is located at some 30,000 cm^{-1} above the ground state. The

zero-field splitting within this ground state is very weak, of the order of 10^{-2} cm^{-1}, and not detectable in magnetism except in the temperature range of fractions of Kelvin. The magnetic susceptibility is almost perfectly isotropic, and follows the Curie law $\chi T = 21N\beta^2/k$ (= 7.88 cm^3 K mol^{-1}).

3.6.2 Eu(III) or Sm(II)

For many rare earth compounds the average value of χT at room temperature is close to what is predicted in the free-ion approximation, even if the χT versus T plot does not exactly follow the Curie law. For Sm(III), Sm(II), and Eu(III), on the other hand, the situation is quite different owing to the presence of thermally populated excited states. First let us consider the Eu(III) [or Sm(II)] case. The ^{7}F ground term is split by the spin-orbit coupling into seven states ^7F_J with J taking the integer values from 0 to 6. The spin-orbit coupling operator is:

$$\mathbf{H}_{SO} = \lambda \mathbf{L}\cdot\mathbf{S} \tag{3.6.1}$$

where λ is the spin-orbit coupling parameter. This operator may also be expressed as

$$\mathbf{H}_{SO} = \lambda(\mathbf{J}^2 - \mathbf{L}^2 - \mathbf{S}^2)/2 \tag{3.6.2}$$

and the energies of the states are

$$E(J) = \lambda J(J+1)/2 \tag{3.6.3}$$

where the energy of the $^7F_0$ ground state is taken as the origin. Since λ is small enough for the first excited states to be thermally populated, of the order of 300 cm^{-1}, the magnetic susceptibility may be expressed as

$$\chi = \frac{\sum_{J=0}^{6} (2J+1)\chi(J)\exp[-\lambda J(J+1)/2kT]}{\sum_{J=0}^{6} (2J+1)\exp[-\lambda J(J+1)/2kT]} \tag{3.6.4}$$

where from (3.5.10) and (3.5.12) $\chi(J)$ is

$$\chi(J) = \frac{Ng_J^2\beta^2 J(J+1)}{3kT} + \frac{2N\beta^2(g_J-1)(g_J-2)}{3\lambda} \tag{3.6.5}$$

In the present case all the g_J factors are equal to 3/2, except g_0 which is equal to 5 from (3.5.9). It follows that χ can be expanded as [3.24]

$$\begin{aligned}\chi = (N\beta^2/3kTx)[&24 + (27x/2 - 3/2)e^{-x} + (135x/2 - 5/2)e^{-3x} \\ &+ (189x - 7/2)e^{-6x} + (405x - 9/2)e^{-10x} + (1485x/2 - 11/2)e^{-15x} \\ &+ (2457x/2 - 13/2)e^{-21x}]/[1 + 3e^{-x} + 5e^{-3x} \\ &+ 7e^{-6x} + 9e^{-10x} + 11e^{-15x} + 13e^{-21x}]\end{aligned} \tag{3.6.6}$$

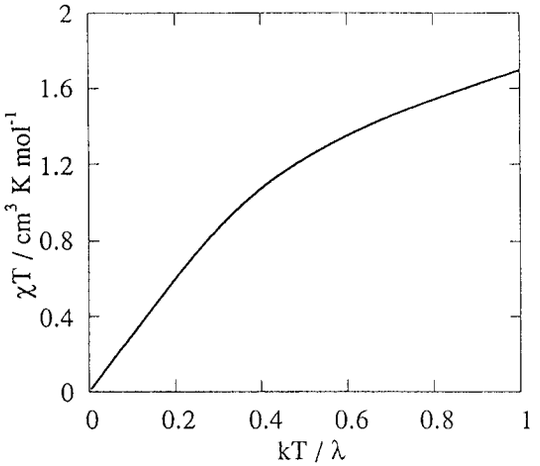


Figure 3.6. χT versus kT/λ plot for an Eu(III) compound.

with

$$x = \lambda/kT \tag{3.6.7}$$

The high-temperature limit $(\chi T)_{HT}$, for $kT >> \lambda$, is obtained by assuming that the magnetic susceptibility of the ^{7}F term is the sum of an orbital contribution and a spin contribution, according to

$$(\chi T)_{HT} = (N\beta^2/3k)[g_L^2 L(L+1) + g_S^2 S(S+1)] \tag{3.6.8}$$

with $g_L = 1$, and $g_S = 2$. This high-temperature limit would then be equal to $12N\beta^2/k$, namely 4.50 cm^3 K mol^{-1}. Actually this limit cannot be reached. Indeed, owing to the value of λ, only the first three low-lying states with the energies 0, λ, and 3λ can be significantly populated. As for the low-temperature limit of χT, it is zero because the $^7F_0$ ground state is nonmagnetic. The low-temperature limit of the susceptibility χ is finite and not zero owing to the term $\chi'(0)$ arising from the coupling between the $^7F_0$ and $^7F_1$ states through the Zeeman perturbation. The χT versus kT/λ plot is shown in Figure 3.6. As expected, χT decreases continuously as T is lowered down to absolute zero. The curve of Figure 3.5 is limited to the $0 \le kT/\lambda \le 1$ range. The upper limit $kT/\lambda = 1$ corresponds to T ≈ 430 K.

3.6.3 Sm(III)

The ^{6}H ground term for Sm(III) is split by spin-orbit coupling into six levels. The energies, $E(J)$, increase from $^6H_{5/2}$ to $^6H_{15/2}$. These energies are :

$$E(J) = \lambda[J(J+1) - 35/4]/2 \tag{3.6.9}$$

The energy of the ground state is again taken as the origin. The spin-orbit coupling parameter is of the order of 200 cm^{-1}, such that the first excited state $^6H_{7/2}$ can be

populated at room temperature and above. The expression of the magnetic susceptibility taking into account the six states arising from ^6H is

$$\chi = \frac{\sum_{J=5/2}^{15/2} (2J+1)\chi(J)\exp[-E(J)/kT]}{\sum_{J=5/2}^{15/2} (2J+1)\exp[-E(J)/kT]} \qquad (3.6.10)$$

$\chi(J)$ being given by (3.6.5), and $E(J)$ by (3.6.9). χ may be expanded as

$$x = (N\beta^2/3kTx)[a_1x + b_1 + (a_2x + b_2)e^{-7x/2} + (a_3x + b_3)e^{-8x} + (a_4x + b_4)e^{-27x/2} + (a_5x + b_5)e^{-20x} + (a_6x + b_6)e^{-55x/2}]/[3 + 4e^{-7x/2} + 5e^{-8x} + 6e^{-27x/2} + 7e^{-20x} + 8e^{-55x/2}] \qquad (3.6.11)$$

with

$a_1 = 2.143$	$b_1 = 7.347$
$a_2 = 42.92$	$b_2 = 1.641$
$a_3 = 283.7$	$b_3 = -0.6571$
$a_4 = 620.6$	$b_4 = -1.9400$
$a_5 = 1122$	$b_5 = -2.835$
$a_6 = 1813$	$b_6 = -3.556$

and x defined as in (3.6.7). As was the case for Eu(III), E(J) and $\chi(J)T$ decrease with J. Therefore χT decreases as T decreases, and tends to the value 0.089 cm^3 K mol^{-1} as T approaches absolute zero. This magnetic behavior is rather unusual in the sense that Eq. (3.6.11) predicts that the χ versus T curve should exhibit a rounded minimum for a T value defined by $kT/\lambda = 0.956$. This susceptibility minimum is shown in Figure 3.7, where λ has been set equal to 200 cm^{-1}.

3.6.4 A Few Additional Remarks

We have already mentioned that with the exception of Sm(III), Sm(II), and Eu(III), the measured χT value at room temperature is rather close to that tabulated in the third column of Table 3.2. This does not mean that the free-ion approximation allows to interpret all the details of the magnetic properties. In most cases the situation is much more complicated. The $^{2S+1}\Gamma_J$ free-ion ground state is partially split by the ligand field. If this ligand field has a rhombic symmetry, which is the most likely for molecular compounds, the number of components is $2J + 1$ for integer J, and $J + 1/2$ for half-integer J. In the latter case, the components are Kramers doublets. The total multiplet width of such splitting may vary from a few tens to a few hundreds of wavenumbers. If this width is small enough, all the components arising from the

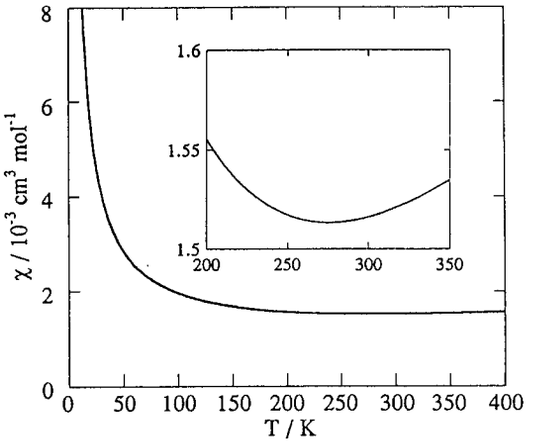


Figure 3.7. χ versus T plot for an Sm(III) compound. The minimum of the curve is emphasized in the inset.

$^{2S+1}\Gamma_J$ ground state are statistically populated at room temperature, and the free-ion approximation applies. As T is lowered, however, the components of higher energy are successively depopulated, and the free-ion approximation becomes less and less valid. This may have two consequences: first, the magnetic susceptibility does not follow the Curie law; second, the system becomes more and more anisotropic on cooling. When the multiplet width is larger, the components of higher energy arising from the free-ion ground state are not statistically populated, even at room temperature, and the compound may present a significant magnetic anisotropy. In such a case the fact that the average value of χT is close to the value predicted in the free-ion approximation may be somewhat fortuitous. The magnetic anisotropy always increases as the temperature is lowered. The simulation of the magnetic properties is in principle possible. The first requirement is to describe the effect of the ligand field in a satisfying fashion. For rare earth compounds the electrostatic model seems to be well suitable. The diagonalization of the spin-orbit coupling, the ligand field and the Zeeman perturbations using all the kets associated with the $^{2S+1}\Gamma_J$ free-ion states arising from the $4f^n$ configuration as basis set is now tractable, thanks to the storage capacity and the speed of the vectorial computers. In such an approach, however, the chemical intuition tends to vanish. For this reason we will not pursue this subject.

References

3.1. K.W.H. Stevens, *Proc. Roy. Soc. (London) A Ser.* 219 (1953), 542.

3.2. J.E. Griffiths, J. Owen, I.M. Ward, *Proc. Roy. Soc. (London) A Ser.* 226 (1953), 96.

3.3. J. Owen, J.H.M. Thornley, *Rep. Prog. Phys.* 29 (1966), 675.

3.4. M. Kotani, *J. Phys. Soc. Jpn.* 4 (1949), 293.

3.5. F.E. Mabbs, D.J. Machin, "Magnetism and Transition Metal Complexes." Chapman and Hall, London, 1973.

3.6. B.N. Figgis, *Trans. Faraday Soc.* 57 (1961), 198.

3.7. B.N. Figgis, J. Lewis, F.E. Mabbs, G.A. Webb, *J. Chem. Soc. A* (1966), 1411.

3.8. B.N. Figgis, J. Lewis, F.E. Mabbs, G.A. Webb, *J. Chem. Soc. A* (1967), 442.

3.9. B.N. Figgis, M. Gerloch, J. Lewis, F.E. Mabbs, G.A. Webb, *J. Chem. Soc. A* (1968), 2086.

3.10. M. Gerloch, P.N. Quested, *J. Chem. Soc. A* (1971), 3729.

3.11. M. Gerloch, P.N. Quested, R.C. Slade, *J. Chem. Soc. A* (1971), 3741.

3.12. R.B. Bentley, M. Gerloch, J. Lewis, P.N. Quested, *J. Chem. Soc. A* (1971), 3751.

3.13. A. Abragam, M.H.L. Pryce, *Proc. Roy. Soc. (London) Ser. A* 206 (1951), 173.

3.14. K. Kambe, S. Koide, Y. Usui, *Prog. Theor. Phys.* 7 (1952), 15.

3.15. C.A. Bates, P.H. Wood, *Contemp. Phys.* 16 (1975), 547.

3.16. L. Banci, A. Bencini, C. Benelli, D. Gatteschi, C. Zanchini, *Struct. Bonding (Berlin)* 52 (1982), 37.

3.17. M.E. Lines, *J. Chem. Phys.* 55 (1971), 2977.

3.18. A. Abragam, B. Bleaney, "Electron Paramagnetic Resonance of Transition Ions," Oxford University Press, Oxford, 1970.

3.19. J.W. Orton, "Electron Paramagnetic Resonance." Iliffe, London, 1968.

3.20. G.H. Dieke, "Spectra and Energy Levels of Rare Earth Ions in Crystals." Interscience, New York 1968.

3.21. T.H. Siddall III, in "Theory and Applications of Molecular Paramagnetism," E.A. Boudreaux, L.N. Mulay eds., p. 257. Interscience, New York, 1976.

3.22. A.T. Casey, S. Mitra, in "Theory and Applications of Molecular Paramagnetism," E.A. Boudreaux, L.N. Mulay eds., p. 271. Interscience, New York 1976.

3.23. A.T. Casey, in "Theory and Applications of Molecular Paramagnetism," E.A. Boudreaux, L.N. Mulay, eds., p. 27. Interscience, New York, 1976.

3.24. J.H. Van Vleck, A. Frank, *Phys. Rev.* 34 (1929), 1494; 34 (1929), 1625.

CHAPTER

4

Low-Spin–High-Spin Transition

When a transition metal ion of configuration d^n, $n = 4$ to 7, is in octahedral surroundings, its ground state may be low spin (LS) or high spin (HS), depending to a first approximation on the magnitude of the Δ energy gap between the e_g and t_{2g} metal orbitals relative to the mean spin pairing energy P. More precisely, for $\Delta >> P$, the ground state arises from the configuration where the d electrons occupy first the t_{2g} orbitals of lowest energy, and then, if there are more than six electrons, the e_g orbitals of highest energy. The ground state is then LS. For $\Delta << P$, on the other hand, Hund's rule is obeyed. The HS ground state has the same spin multiplicity as the free metal ion. The nature of the ground state for d^n ions, $n = 4$ to 7, in O_h symmetry is specified in Table 4.1. When the symmetry is lowered with respect to O_h, the orbital degeneracy is partially or totally removed but the spin multiplicity is retained.

When the conditions $\Delta <<$ or $>> P$ are no longer fulfilled, a LS $\leftrightarrow$ HS transition may occur, which leads to quite spectacular physical properties [4.1–4.5]. The condition for a spin transition to occur is often defined by $|\Delta\text{-}P| \approx kT$. Such a formulation is confusing and, in our opinion, must be avoided. As a matter of fact, Δ depends on the electronic state; for a spin transition compound Δ(LS) and Δ(HS) are clearly not the same. Similarly, the mean spin pairing energy P is defined in a rather ambiguous way. The occurrence of spin transition is related to LS and HS potential energy curves as will be specified in Section 4.2. A spin transition may be induced either by temperature or by pressure. The temperature-induced spin transitions are by far the most widely studied and we will focus on them. This chapter is organized as follows:

Table 4.1. Ground State of the d^n Ions, $n = 4$ to 7, in Octahedral Environment

	$\Delta >> P$	$\Delta << P$
d^4	$^3T_{1g}$	^5E_g
d^5	$^2T_{2g}$	$^6A_{1g}$
d^6	$^1A_{1g}$	$^5T_{2g}$
d^7	^2E_g	$^4T_{1g}$

first, we will discuss the various types of spin transition and the techniques used to characterize them. Second, we will study the mechanism of spin transition at the molecular level. We will show that in many cases this purely molecular approach does not allow the correct interpretation of experimental data; it is necessary to take into account the cooperativity. The main models including intermolecular interactions within an assembly of molecules are presented. A few selected examples are discussed. Finally, the spin transition phenomenon is discussed within the more general framework of molecular electronics.

4.1 High-Spin Molar Fraction versus Temperature Curves for Spin Transition Compounds

A thermally induced LS $\leftrightarrow$ HS transition is characterized by an $x = f(T)$ curve, where x is the molar fraction of HS molecules and $(1 - x)$ that of LS molecules. Several techniques are currently used to obtain such curves. The simplest method consists of measuring the temperature dependence of the molar magnetic susceptibility. In most cases the orbital degeneracy in a strictly octahedral environment is totally removed by the symmetry lowering. If so, χT is constant in the temperature range where all the molecules are in the same spin state, with $(\chi T)_{LS}$ and $(\chi T)_{HS}$ values, respectively. Therefore, the $x = f(T)$ curve can be deduced in a straightforward manner from the experimental $\chi T = f(T)$ curve according to:

$$x = \frac{\chi T - (\chi T)_{LS}}{(\chi T)_{HS} - (\chi T)_{LS}} \qquad (4.1.1)$$

Any other technique that provides different responses according to whether the state is LS or HS may also be used to determine the high-spin molar fraction x. Distinct signals are obtained in principle when the spin flipping frequency is slower than the technique frequency scale. Otherwise, a single average signal is observed. Among these techniques, Mössbauer spectroscopy has been particularly useful in the case of iron compounds. The Mössbauer spectra then show two well resolved quadrupole doublets with variable intensity as a function of temperature, provided that the coexisting spin isomers have lifetimes larger than 10^{-7} s. One of these doublets is associated with LS molecules and the other with HS molecules. The high-spin molar fraction x may then be deduced from the relative intensities of these quadrupole doublets [4.6].

Temperature-dependent infrared studies also allow the study of spin transition. The metal–ligand stretching vibrations are systematically higher in energy in the LS state than in the HS state. Other stretching vibrations, such as the ν_{N-C} of the thiocyanate groups, may be strongly affected by a spin transition. An example of spin transition for which the temperature dependence of infrared spectra was the only available investigation method is given in Section 4.6.

For a few iron(III) and cobalt(II) compounds the x versus T plot has been calculated from the temperature dependence of the intensities of EPR signals [4.7]. For iron(II)

compounds which are EPR silent, EPR may be used in another way, by doping the lattice with an EPR active ion such as copper(II). In the HS phase, owing to the magnetic concentration, the spectrum is poorly resolved. On the other hand, in the LS phase, the host lattice is diamagnetic and the EPR spectrum may be beautifully resolved with fine, hyperfine, and superhyperfine structures [4.8]. If the LS $\leftrightarrow$ HS transition is abrupt, the switching from one kind of spectrum to the other one may be quite spectacular.

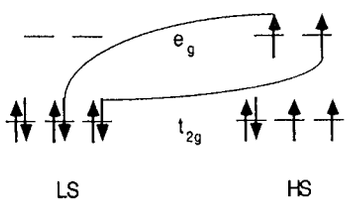
Let us return to the $x = f(T)$ curves, irrespective of the techniques used to plot them. Various types of transition have been observed, which can be summed up as follows:

- The transition may be abrupt, occurring within a few Kelvin, or smooth, occurring within a large temperature range.
- The transition may be complete both at low temperature ($x = 0$) and at high temperature ($x = 1$), or incomplete. The incompleteness may occur at low temperature ($x \neq 0$) and/or at high temperature ($x \neq 1$). The incompleteness at low temperature is more often observed.
- The transition may occur in two steps in exceptional cases.
- The $x = f(T)$ curves may be strictly identical in the cooling and heating modes, or exhibit a hysteresis effect. In the latter case the critical temperature $T_c\downarrow$ on cooling is lower than the critical temperature $T_c\uparrow$ upon heating.

4.2 Mechanism of the Spin Transition at the Molecular Scale

In agreement with the general spirit of this book, we discuss the mechanism of the spin transition by focusing on a specific example, that of octahedral iron(II) complexes. These compounds are the most intensively investigated in the field.

The LS and HS states for iron(II) compounds may be schematized as



At the molecular scale the spin transition corresponds to an intraionic electron transfer with spin flip of the transferred electrons. The word "intraionic" means that the electrons are transferred from the t_{2g} to the e_g orbitals but remain in the immediate vicinity of the metal ion. For an iron(II) compound this transfer involves two electrons and the spin variation is $\Delta S = 2$. For a d^5 ion two electrons are also transferred. For d^4 and d^7 ions a single electron is transferred and the spin variation is $\Delta S = 1$.

The occupancy of the e_g orbitals is higher in the HS state than in the LS state; these orbitals are more antibonding than the t_{2g}s. It follows that the average metal–ligand

bond length is longer in the HS state than in the LS state. This difference is in the range 0.14–0.24 Å for iron(II) compounds, and 0.11–0.15 Å for iron(III) compounds, both with $\Delta S = 2$. It is in the range 0.09–0.11 Å for cobalt(II) with $\Delta S = 1$.

At the first view it could be thought that a spin transition is more likely when the energies of the LS and HS states are closer. Actually the situation is a bit more complicated. The quantity we need to consider in order to define what state is the lowest is the Gibbs free energy G which takes into account the entropy factor S in addition to the enthalpy factor H. At constant pressure the Gibbs free energy variation ΔG accompanying the transformation LS $\leftrightarrow$ HS is

$$\Delta G = \Delta H - T\Delta S \tag{4.2.1}$$

ΔG (= $G_{HS} - G_{LS}$), ΔH (= $H_{HS} - H_{LS}$) and ΔS (= $S_{HS} - S_{LS}$) refer to an assembly of, say, N molecules. The critical temperature T_c for which there is the same amount of LS and HS molecules is defined by $\Delta G = 0$, hence

$$T_c = \Delta H / \Delta S \tag{4.2.2}$$

In order for T_c to be positive, ΔH and ΔS must have the same sign. The entropy variation, ΔS, neglecting the intermolecular interactions, may be written as the sum of electronic ΔS_{el} and vibrational ΔS_{vib} contributions:

$$\Delta S = \Delta S_{el} + \Delta S_{vib} \tag{4.2.3}$$

ΔS_{el} is related to the ratio of the degeneracies Ω_{HS}/Ω_{LS} between the HS and LS states according to

$$\Delta S_{el} = Nk \ln(\Omega_{HS}/\Omega_{LS}) \tag{4.2.4}$$

and is therefore always positive. For a perfectly octahedral iron(II) compound Ω_{HS}/Ω_{LS} is equal to 15 and ΔS_{el} is equal to 1.882 cm^{-1} K^{-1}. If the symmetry is sufficiently low, such that the orbital degeneracy is completely removed, Ω_{HS}/Ω_{LS} is equal to 5 and ΔS_{el} to 1.119 cm^{-1} K^{-1}. As for ΔS_{vib}, it is also positive. Indeed, the vibrational disorder is more pronounced in the HS state than in the LS state owing to the longer metal–ligand bond lengths. Hence, both ΔS and ΔH are positive. Therefore, in order for a spin transition to be observed, the minimum of the LS potential energy curve must be slightly lower than the minimum of the HS potential energy curve, as shown in Figure 4.1 where the nuclear coordinate Q in the abscissa refers to the breathing vibration of the coordination sphere. At low temperatures, below T_c, the enthalpy factor dominates; ΔG is positive and the LS species is the most stable. At high temperatures, above T_c, the entropy factor dominates. ΔG is negative and the HS species is the most stable.

In Figure 4.1 we have drawn a crossing between the LS and HS curves due to the fact that these states do not have the same symmetry. This crossing, however, may be avoided if we take into account the spin-orbit coupling which mixes LS and HS states in the crossing region. This situation has been investigated by Bacci [4.9, 4.10]. In other respects, ΔH in Figure 4.1 is defined as the energy difference between the minima of the LS and HS potential energy curves. Strictly speaking, ΔH is the energy difference

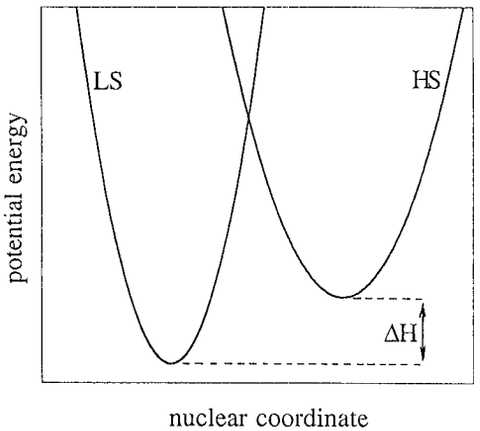
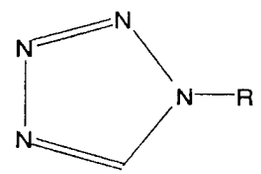


Figure 4.1. LS and HS potential energy curves for a spin transition compound.

between the lowest vibrational states associated with the LS and HS electronic states, respectively.

The existence of two minima in the potential energy curves, the LS minimum being lower in energy than the HS minimum, has allowed the observation of a very interesting phenomenon called light-induced excited spin state trapping (LIESST) by Decurtins et al. [4.4, 4.11–4.14]. These authors first recorded the single crystal electronic spectra of $Fe(ptz)_6(BF_4)_2$ with ptz = 1-propyltetrazole



R = -$CH_2CH_2CH_3$

at room temperature and at 10 K. This compound was known to exhibit a spin transition at 135 K with a spectacular change of color. $Fe(ptz)_6(BF_4)_2$ is purple in the LS phase and white in the HS phase. The spectrum at room temperature is actually typical of a $^5T_{2g}$ ground state with a $^5T_{2g} \rightarrow {}^5E_g$ spin-allowed transition at 12,250 cm^{-1}, and that at 10 K of a $^1A_{1g}$ ground state with $^1A_{1g} \rightarrow {}^1T_{1g}$ and $^1A_{1g} \rightarrow {}^1T_{2g}$ spin-allowed transitions at 18,400 and 26,700 cm^{-1}, respectively. Then, Decurtins et al. irradiated the sample at 10 K with a 15 W xenon arc lamp. Within a few minutes the sample turned white and showed all the spectroscopic (electronic and Mössbauer spectra) and magnetic properties of the high-temperature phase. The spin state conversion is quantitative and the trapped HS state does not decay within several hours or days when maintaining the temperature around 10 K. The thermal energy kT is too low for the

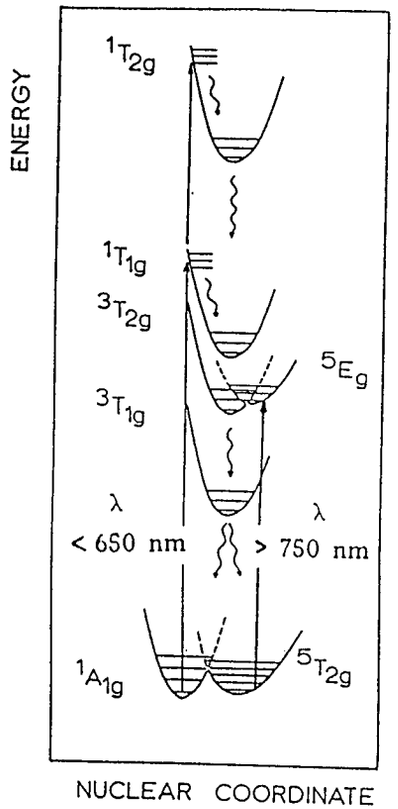


Figure 4.2. Potential curves for the ground and excited states of an iron(II) spin transition compound. Arrows indicate the mechanism for LIESST and reverse LIESST (from ref. [4.4]).

system to clear the energy barrier between the HS and LS states. Above 50 K the HS state relaxes back to the stable LS state. The same result is obtained when the LS phase is irradiated at 10 K into the $^1A_{1g} \rightarrow {}^1T_{1g}$ d-d absorption band with the 514.5 nm (19,436 cm^{-1}) beam of an Ar^+ laser.

The interpretation of LIESST is visualized in Figure 4.2 showing the potential energy curves for the states involved in the phenomenon. The irradiation with the intense xenon arc lamp occurs in the absorption region of the LS spectrum and leads to a population of the excited spin singlet states. These states are very short lived and mostly decay back to the LS ground state. However, an alternative decay path is possible, due to spin-orbit coupling. This decay leads to the lowest spin triplet state $^3T_{1g}$, and from there either to the LS ground state, or to the metastable (at 10 K) HS state. The phenomenon is reversible. When irradiating a sample trapped in the HS state at 10 K, using a light corresponding to the $^5T_{2g} \rightarrow {}^5E_g$ transition, a pumping back to the normal LS state is observed [4.4, 4.13].

4.3 Spin Transition and Cooperativity

In this section we consider an assembly of N molecules exhibiting a LS ↔ HS transition and we define the molar fraction of HS molecules as x and that of LS molecules as $1 - x$.

Even if we assume that there is no intermolecular interaction, i.e., that each molecule ignores its neighboring molecules, we must introduce an additional entropy term, the mixing entropy, to the Gibbs free energy G. This term accounts for the fact that there are many ways of distributing xN HS molecules and $(1 - x)$N LS molecules within the assembly of N molecules. This mixing entropy is given by

$$S_{mix} = k[N \ln N - xN \ln xN - (1 - x)N \ln (1 - x)N] \tag{4.3.1}$$

which may be written as

$$S_{mix} = -R[x \ln x + (1 - x) \ln (1 - x)] \tag{4.3.2}$$

where $R = Nk$ is the gas constant. S_{mix} is maximum for $x = 0.5$ and vanishes for $x = 0$ and 1. Taking into account S_{mix}, the molar Gibbs free energy is

$$G = xG_{HS} + (1 - x)G_{LS} - TS_{mix} \tag{4.3.3}$$

where G_{HS} and G_{LS} are the molar Gibbs free energy for the HS and LS species, respectively. The partial derivative of G with respect to x is

$$\partial G/\partial x = \Delta G + RT \ln\left(\frac{x}{1 - x}\right) \tag{4.3.4}$$

and at any temperature and pressure the equilibrium condition for the tranformation LS → HS is defined by

$$(\partial G/\partial x)_{T,P} = 0 \tag{4.3.5}$$

hence

$$\ln\left(\frac{1 - x}{x}\right) = \frac{\Delta G}{RT} = \frac{\Delta H}{RT} - \frac{\Delta S}{R} \tag{4.3.6}$$

Taking into account (4.2.2), (4.3.6) may be transformed as

$$x = \frac{1}{1 + \exp[(\Delta H/R)(1/T - 1/T_c)]} \tag{4.3.7}$$

An example of x versus T plot is shown in Figure 4.3, where ΔH has been taken equal to 600 cm^{-1} and T_c to 150 K. At $T = 0$ K, x is zero. All the molecules are LS. On the other hand, at high temperature, the transformation is incomplete; the high-spin molar fraction x tends to $[1 + \exp(-\Delta H/RT_c)]^{-1}$ and not to unity. Moreover, the transformation is very smooth, covering a large temperature range. Such smooth transitions are normally observed in solution, but are rather exceptional in the solid state. Therefore, it is not legitimate to ignore intermolecular interactions. If the

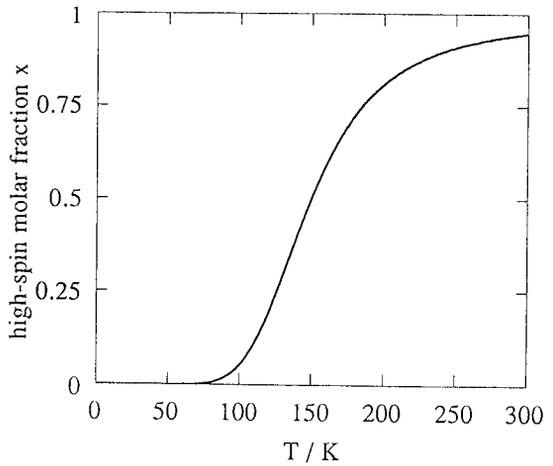


Figure 4.3. High-spin molar fraction x versus T plot for a spin transition compound in the absence of intermolecular interaction. ΔH is taken equal to 600 cm–1 and Tc to 150 K.

origin of the spin transition phenomenon is essentially molecular, it is due to a subtle balance between enthalpy and entropy factors, the manifestation of the phenomenon is largely influenced by the cooperativity within the assembly of molecules.

So far two types of thermodynamic models have been proposed to account for this cooperative character. One of these models is based on the hypothesis of regular solutions, and the other on the idea that there is formation of domains within the molecular assembly. We will discuss both approaches successively and we will look for their ranges of validity and their limits.

4.4 Regular Solution Model

This model was proposed in 1972 by Slichter and Drickamer, who added an interaction term $\Gamma(x, T, P)$ to the Gibbs free energy defined in (4.3.3) [4.15, 4.16]. These authors expressed $\Gamma(x, T, P)$ in a simple and symmetrical fashion as

$$\Gamma(x, T, P) = \gamma x(1 - x) \tag{4.4.1}$$

where only the x dependence is taken into account; γ is an interaction parameter assumed not to depend on T in a first approximation. Although this model was first applied to pressure induced transitions, we assume that we work at constant pressure, which is by far the most frequent case. Taking G_{LS} as the origin of the energies, we can write

$$G = x\,\Delta H + \gamma x(1 - x) + T\{R[x \ln x + (1 - x) \ln (1 - x)] - x\,\Delta S\} \tag{4.4.2}$$

which allows us to plot the $G = f(x)$ curves for various values of ΔH, ΔS, and γ, and at different temperatures T. The x values corresponding to extrema of G are defined by the condition $(\partial G/\partial x)_{T,P} = 0$, which leads to the implicit equation

$$\ln\left(\frac{1 - x}{x}\right) = \frac{\Delta H + \gamma(1 - 2x)}{RT} - \frac{\Delta S}{R} \tag{4.4.3}$$

One way to study this implicit equation is to plot point by point the explicit equation $T = f(x)$ deduced from (4.4.3). When γ is smaller than $2RT_c$, the derivative dT/dx is positive in the whole $0 \le x \le 1$ range, and for any T value there is a corresponding x value. The transition is relatively smooth and occurs without a hysteresis effect. When γ is equal to $2RT_c$, dT/dx is zero for $x = 0.5$ and positive otherwise. The transition is abrupt around $T = T_c$, but again without hysteresis (see Figure 4.4). When γ is larger than $2RT_c$, dT/dx is zero for two x values, denoted x_A and x_B, the corresponding temperatures being denoted T_A and T_B, respectively; x_A is smaller and x_B larger than 0.5. The derivative dT/dx is positive for $x < x_A$ and $x > x_B$, and negative in the $x_A < x < x_B$ range. The T versus x plot then has the shape of a double fold with the external branches XA and BY corresponding to minima of G, and the central branch AB corresponding to maxima of G (see Figure 4.4). For any T value between T_A and T_B, there are two x values minimizing and one maximizing the Gibbs free energy. A new situation is encountered, which may be explained as follows (see Figure 4.4): When the temperature increases up to T_A, x increases up to x_A (branch XA). At $T = T_A$, x jumps from x_A to $x_{A'}$ (horizontal line AA′) and when warming further above T_A, x increases again and tends to unity (branch A′Y). Now, when T decreases from high temperatures down to T_B, x decreases down to x_B (branch YB). At $T = T_B$, x jumps from x_B to $x_{B'}$ (horizontal line BB′) and when cooling further below T_B, x decreases again and tends to zero. The warming and cooling routes are no longer similar. The spin transition occurs in an abrupt fashion with a hysteresis effect.

In their original paper published in 1972 Slichter and Drickamer proposed another method to solve the implicit equation (4.4.3). The solutions of (4.4.3) are provided by the intersects of the logarithmic curve

$$y = \ln\left(\frac{1 - x}{x}\right) \tag{4.4.4}$$

and the straight line

$$y = \frac{\Delta H + \gamma(1 - 2x)}{RT} - \frac{\Delta S}{R} \tag{4.4.5}$$

which at any temperature passes through the point P defined by $x = 1/2 + \Delta H/2\gamma$ and $y = -\Delta S/R$ (see Figure 4.5).

For $\gamma < 2RT_c$ there is a unique intersect corresponding to a minimum of G, whatever the temperature may be. The transition is continuous without hysteresis. For $\gamma = 2RT_c$ the straight line is tangent to the logarithmic curve at $T = T_c$. Again a unique solution is found, although there is a jump of x at T_c due to the vertical tangent of the $x = f(T)$ curve. Finally, for $\gamma > 2RT_c$, there is a temperature range $T_B < T < T_A$ in which there

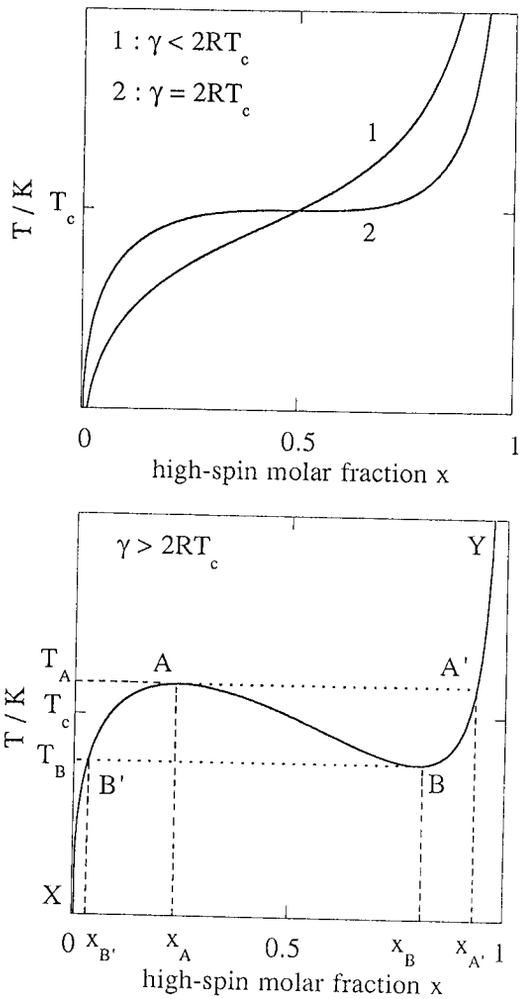


Figure 4.4. Temperature T versus high-spin molar fraction x plots for a spin transition compound in the presence of intermolecular interaction. Top: $\gamma \leq 2RT_c$; bottom: $\gamma > 2RT_c$.

are three solutions, the two extreme ones corresponding to minima of G and the middle one corresponding to a maximum of G. An abrupt change in x occurs at T_A when warming up, and at T_B when cooling down. The transition is discontinuous with hysteresis.

In order to visualize the ideas presented above, we consider two concrete cases, both with $\Delta H = 600$ cm^{-1} and $\Delta S = 4$ cm^{-1} K^{-1}. The critical temperature is then equal to 150 K. In the first case we take $\gamma = 208.5$ cm^{-1}, i.e., exactly $2RT_c$. Indeed R is equal to 0.69504 cm^{-1} K^{-1}. The $G = f(x)$ curves are plotted in Figure 4.6 for several temperatures around 150 K. At $T = 140$ K, G has a minimum for an x value close to zero. The regular solution is rich in LS molecules. The position of the unique minimum

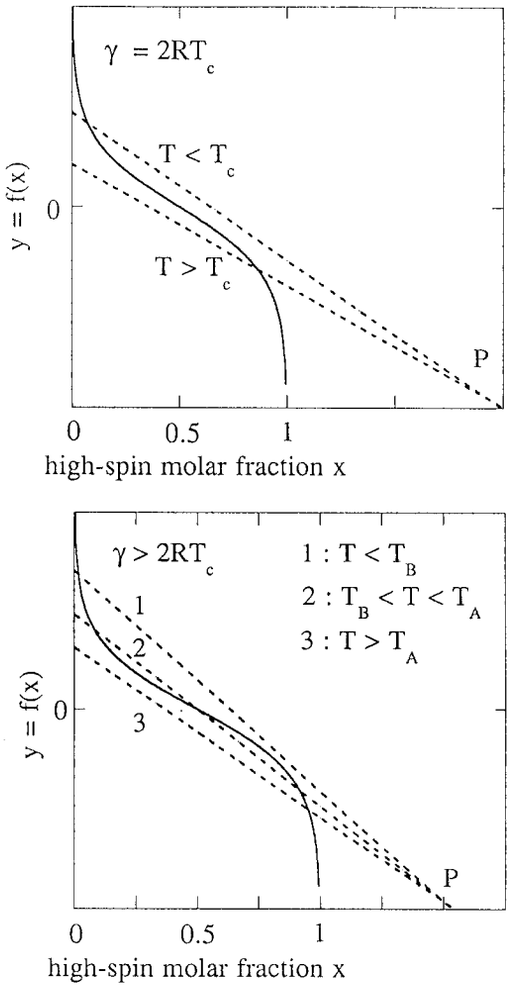


Figure 4.5. $y = \ln[(1-x)/x]$ logarithmic curve (solid line) and $y = [\Delta H + \gamma(1-2x)]/RT - \Delta S/R]$ (dotted line). The intersects give the x values corresponding to the extrema of the Gibbs free energy G. Top: $\gamma = 2RT_c$; bottom: $\gamma > 2RT_c$.

moves to the right as T increases, reaches 0.5 for $T = T_c = 150$ K, and rapidly becomes close to unity above T_c. The regular solution is then rich in HS molecules. The $x = f(T)$ curve deduced from (4.4.3) is represented in Figure 4.7. It exhibits a rather abrupt transition with a vertical tangent for $x = 0.5$, but without hysteresis.

In the second case we take $\gamma = 300$ cm^{-1}, hence larger than $2RT_c$. The $G = f(x)$ curves are plotted in Figure 4.8 for several temperatures between 138.3 and 157.5 K. Outside this temperature range G only shows one minimum. In this 138.3–157.5 K range G exhibits two minima, one for x close to zero and the other for x close to unity. For $T <$

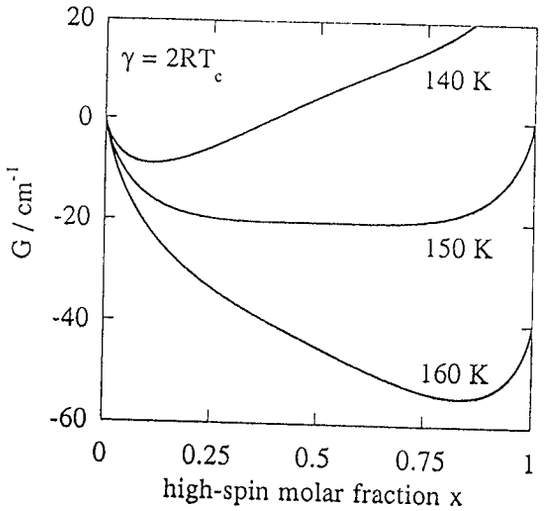


Figure 4.6. G versus x curves for $\Delta H = 600\ \text{cm}^{-1}$, $\Delta S = 4\ \text{cm}^{-1}\ \text{K}^{-1}$, and $\gamma = 208.5\ \text{cm}^{-1}$ in the regular solution model. γ is then equal to $2RT_c$.

T_c the minimum close to $x = 0$ is lower in energy than that close to $x = 1$. The former corresponds to a stable state, and the latter to a metastable state. For $T > T_c$ the reverse situation holds. At $T = T_c$ the two minima are symmetrically situated with respect to $x = 0.5$ and have the same Gibbs free energy. Between 138.3 and 157.5 K hysteresis can occur.

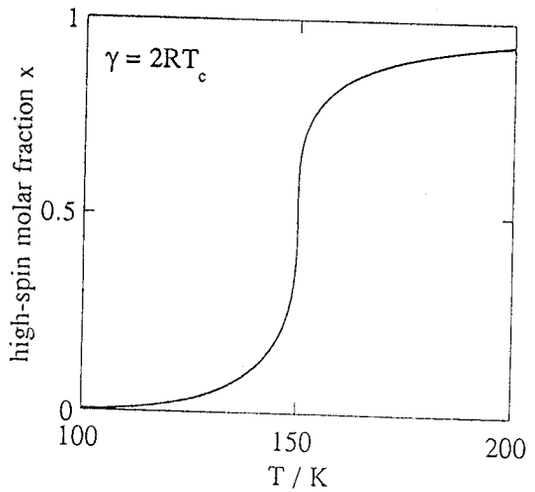


Figure 4.7. High-spin molar fraction x versus T plot for $\Delta H = 600\ \text{cm}^{-1}$, $\Delta S = 4\ \text{cm}^{-1}\ \text{K}^{-1}$, and $\gamma = 208.5\ \text{cm}^{-1}$ in the regular solution model. γ is then equal to $2RT_c$; the transition is abrupt but without hysteresis.

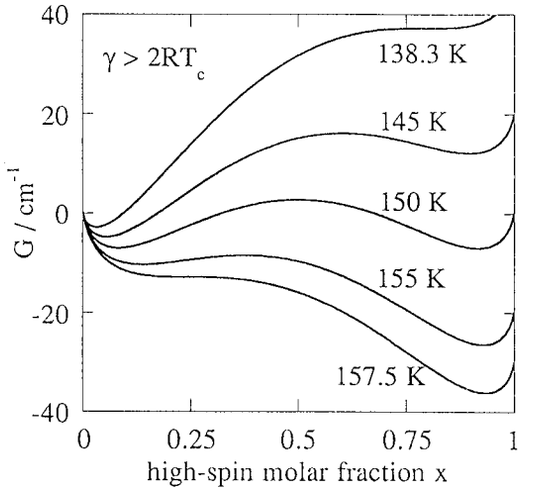


Figure 4.8. G versus x curves for $\Delta H = 600\ \text{cm}^{-1}$, $\Delta S = 4\ \text{cm}^{-1}\ \text{K}^{-1}$, and $\gamma = 300\ \text{cm}^{-1}$ in the regular solution model. γ is then larger than $2RT_c$.

The $x = f(T)$ curve associated with the minima of G is shown in Figure 4.9. This curve shows an hysteresis loop around 150 K with a maximum magnitude of 19.2 K.

To understand the microscopic origin of the parameter γ introduced by Slichter and Drickamer, we can write a general expression of the x dependence of $\Gamma(x, T, P)$ as

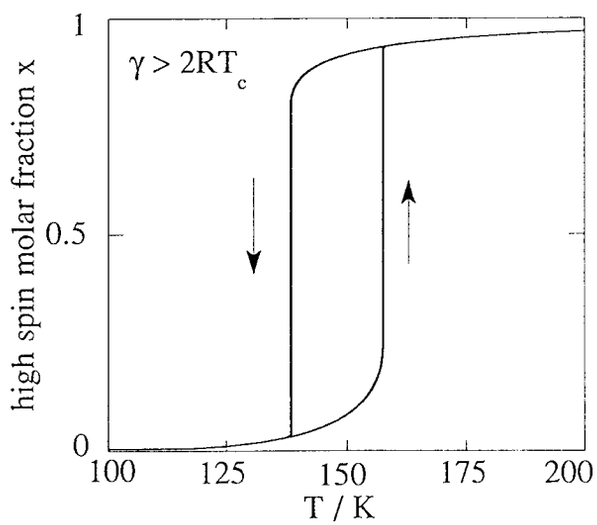


Figure 4.9. High-spin molar fraction x versus T plot for $\Delta H = 600\ \text{cm}^{-1}$, $\Delta S = 4\ \text{cm}^{-1}\ \text{K}^{-1}$, and $\gamma = 300\ \text{cm}^{-1}$ in the regular solution model. γ is then larger than $2RT_c$; the transition is abrupt, with hysteresis, the maximum width of which is 19.2 K.

$$\Gamma(x,T,P) = \gamma_{LL}(1-x)^2 + 2\gamma_{LH}x(1-x) + \gamma_{HH}x^2 \quad (4.4.6)$$

where γ_{LL} is the interaction parameter between LS molecules, γ_{LH} between LS and HS molecules, and γ_{HH} between HS molecules. The probability of having a pair of LS molecules is $(1-x)^2$; similarly the probability of having a pair of HS molecules is x^2, and the probability of having a pair with one LS and one HS molecule is $2x(1-x)$. Eq. (4.4.6) may be rewritten as

$$\Gamma(x) = \gamma_0 + \gamma_1 x + \gamma_2 x^2 \quad (4.4.7)$$

with

$$\gamma_0 = \gamma_{LL} \quad (4.4.8)$$

$$\gamma_1 = 2(\gamma_{LH} - \gamma_{LL}) \quad (4.4.9)$$

$$\gamma_2 = \gamma_{LL} + \gamma_{HH} - 2\gamma_{LH} \quad (4.4.10)$$

In the absence of intermolecular interactions the Gibbs free energy already contains an enthalpy term independent of x and an enthalpy term linear in x [see Eq. (4.3.3)]. The intermolecular interaction introduces a term quadratic in x, the coefficient of which is γ_2. In Slichter and Drickamer's approximation the interaction coefficient γ may be expressed as

$$\gamma = 2\gamma_{LH} - \gamma_{LL} - \gamma_{HH} \quad (4.4.11)$$

and represents the interaction energy between molecules of different spin states with respect to the average interaction energy between molecules of the same spin state. This is a measure of the tendency for the molecules of one type to be surrounded by like rather than unlike molecules [4.17, 4.18].

Several modifications of Slichter and Drickamer's model have been proposed. None of them changes its basic philisophy. Rao et al. worked with the complete expression (4.4.6) [4.19, 4.20]. Zimmermann and König assumed that the intermolecular interactions were of the Ising type (see Chapter 12). If the two spin states are paramagnetic (d^4, d^5, and d^7 ions), the full expression (4.4.6) must be used. On the other hand, if the LS state is diamagnetic (d^6 ions), γ_{LL} and γ_{LH} in (4.4.6) as well as γ_0 and γ_1 in (4.4.7) are zero. Actually, these authors focused on iron(II) species so that the only interaction parameter used was between spin-quintet molecules [4.21].

Concerning the temperature dependence of γ, we can mention the work of Purcell who used an expression of the type

$$\gamma = \gamma_a + \gamma_b T \quad (4.4.12)$$

to interpret the magnetic data obtained with the $Fe(phen)_2(NCR)_2$ compounds with R = BH_3, BPh_3, S, and Se [4.22].

The interaction Gibbs free energy $\Gamma(x, T, P)$ in what precedes has been introduced in a phenomenological way. At the microscopic level the real mechanism of cooperativity in an assembly of spin transition molecules is still badly understood, even if several hypotheses have been proposed. Our opinion is that the available experimental data cannot determine which hypothesis is most suitable, even if each author is obviously convinced that his own model is by far the most appropriate. Among these hypotheses let us mention an electron–phonon coupling [4.23], an intermolecular coupling of the molecular distortions of the Jahn–Teller type [4.24–4.26], and an elastic interaction between molecules of different spins [4.27–4.29].

The main merit of the regular solution model arises from the fact that it can be applied to continuous as well as discontinuous transitions. Moreover, it can account for hysteresis, which has been observed in quite a few compounds, particularly in iron(II) chemistry. The problem of hysteresis deserves some additional comments. The condition for observing hysteresis has been found as

$$\gamma > 2RT_c \quad (4.4.13)$$

This condition has to be considered as necessary but not sufficient. When (4.4.13) is satisfied, hysteresis will occur provided that the system can be trapped in the secondary free energy minimum. However, tunneling through the barrier between the two minima due to a demixing process can reduce or even suppress hysteresis. Thus, this model permits a prediction of only the maximum amplitude of the hysteresis, for instance 19.2 K in the second example treated above (γ = 300 cm^{-1}). The demixing process depends on many factors such as grain size, frontier domains, rate of nucleation, etc, which are not easy to quantify, and to our knowledge have not yet been introduced into models describing the spin transition [4.30].

4.5 Domain Model

This model was developed for the first time by Sorai and Seki [4.31]. These authors considered that LS and HS molecules were not distributed in a random fashion within the assembly, but formed domains of like spin. These domains are assumed to have a uniform size around the critical temperature T_c. The number of domains per mole is $\mathcal{N}$, such that the number of molecules per domain is $n = N/\mathcal{N}$.

The mixing entropy is now given by

$$\mathcal{S}_{mix} = k[\mathcal{N}\ln\mathcal{N} - x\mathcal{N}\ln x\mathcal{N} - (1-x)\mathcal{N}\ln(1-x)\mathcal{N}] \quad (4.5.1)$$

or

$$\mathcal{S}_{mix} = -(R/n)[x\ln x + (1-x)\ln(1-x)] \quad (4.5.2)$$

Replacing $\mathcal{S}_{mix}$ as expressed in the model of the regular solutions in (4.4.2) by (4.5.2) and imposing $(\partial G/\partial x)_{T,P} = 0$ results in

$$\ln\left(\frac{1-x}{x}\right) = \frac{n\Delta G}{RT} = \frac{n\Delta H}{RT} - \frac{n\Delta S}{R} \quad (4.5.3)$$

thus

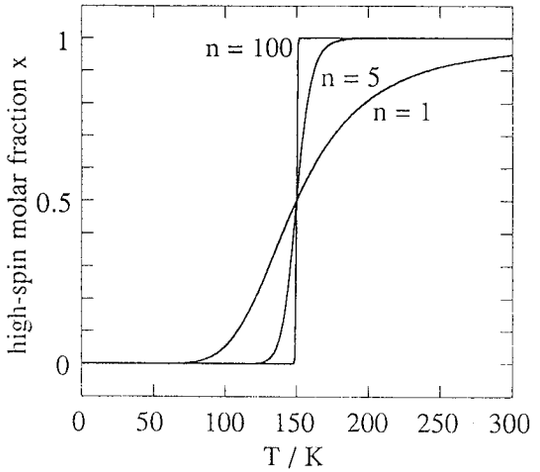


Figure 4.10. High-spin molar fraction x versus temperature T plots for $\Delta H = 600\ \text{cm}^{-1}$, $\Delta S = 4\ \chi\mu^{-1}\ \text{K}^{-1}$, and $n = 1$, 5, and 100 in the domain model.

$$x = \frac{1}{1 + \exp[n\Delta H/R)(1/T - 1/T_c)]} \tag{4.5.4}$$

For given values of ΔH and T_c the abruptness of the transition, deduced from the slope of the $x = f(T)$ curve around T_c, increases with the number n of molecules per domain. Figure 4.10 shows the x versus T plots for $\Delta H = 600\ \text{cm}^{-1}$, $T_c = 150$ K and $n = 1$, 5 and 100.

If we know ΔH from thermodynamic measurements as described in Section 4.6, then n may be determined from the slope of the straight line $\ln[(1 - x)/x] = f(1/\text{T})$.

In this model it also possible to calculate the molar heat capacity at constant pressure C_p, defined as

$$C_p = \partial H/\partial T = \partial[(1 - x)H_{LS} + xH_{HS}]/\partial T \tag{4.5.5}$$

$$= (1 - x)C_{LS} + xC_{HS} + \Delta H\partial x/\partial T \tag{4.5.6}$$

C_{LS} and C_{HS} (which stand for $C_{p,LS}$ and $C_{p,HS}$, respectively) are the molar heat capacities for the LS and HS species, respectively. Calculating $\partial x/\partial T$ from (4.5.4) and inserting into (4.5.6) leads to

$$C_p = C_{LS} + \frac{C_{HS} - C_{LS}}{1 + A} + \frac{n\Delta H^2 A}{RT^2(1 + A)^2} \tag{4.5.7}$$

with

$$A = \exp[(n\Delta H/R)(1/T - 1/T_c)] \tag{4.5.8}$$

At T = T_c, C_p is maximum and takes the value $C_p(T_c)$ given by

$$C_p(T_c) = \frac{C_{LS}(T_c) + C_{HS}(T_c)}{2} + \frac{n\Delta H^2}{4RT_c^2} \tag{4.5.9}$$

The number n of molecules per domain may be deduced in principle from the $C_p = f(T)$ curve, using

$$n = \frac{4RT_c}{\Delta H^2}\left[C_p(T_c) - \frac{C_{LS}(T_c) + C_{HS}(T_c)}{2}\right] \tag{4.5.10}$$

Indeed, this curve presents a λ-peak with a maximum $C_p(T_c)$ at T_c. The integration of the peak gives ΔH. $C_{LS}(T_c)$ and $C_{HS}(T_c)$ can be estimated by extrapolating up to T_c the C_p values obtained far from T_c in the low- and high-temperature regions, respectively, as is schematized in Figure 4.11.

It is worth noting here that it is impossible to derive an expression for C_p in regular solution model since x cannot be expressed explicitly versus T. However, a numerical calculation would be feasible.

The physical bases of the domain model are probably easier to understand, and the calculations are particularly simple. However, in its present form, this model has a severe limitation. It cannot account for hysteresis. Hysteresis could appear if the sizes of the domains were different in the cooling and warming modes.

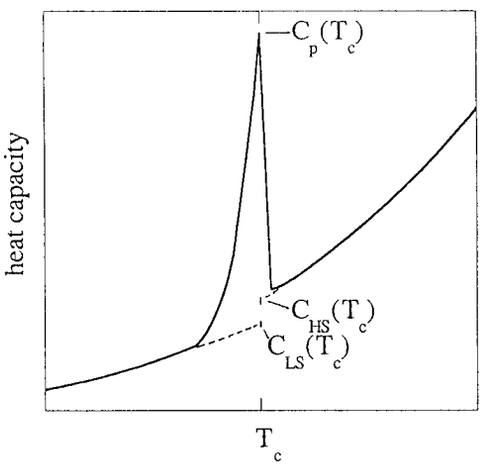
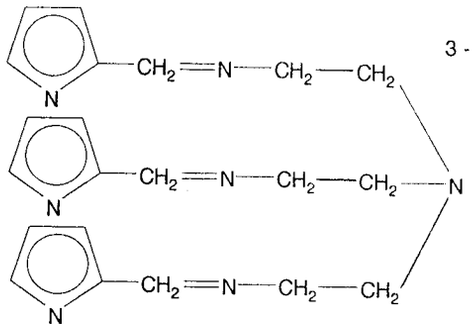


Figure 4.11. Heat capacity C_p versus temperature T curve showing the meaning of the symbols $C_p(T_c)$, $C_{LS}(T_c)$ and $C_{HS}(T_c)$.

4.6 Some Selected Examples

4.6.1 d^4 Ions

There is only one report on spin transition in a d^4 compound [4.32]. This report concerns Mn[(pyrol)$_3$tren] where a Mn(III) ion is sixfold coordinated by the trianionic Schiff base (pyrol)$_3$tren:



The structure of the molecule is presented in Figure 4.12. The MnN_6 core comprises the nitrogen atoms of the pyrol and imine groups. The coordination is distorted about the threefold axis joining the metal center and the amine nitrogen from octahedral toward trigonal-prismatic symmetry. The χT versus T plot displays an abrupt discon-

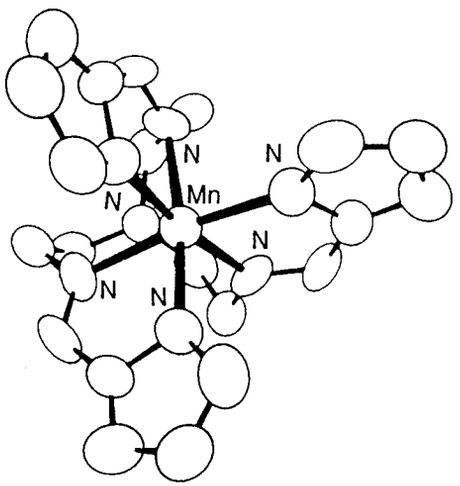
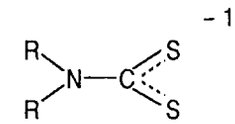


Figure 4.12. Molecular structure of Mn[(pyrol)$_3$tren] (from ref. [4.32]). In this figure, as in all the other figures of molecular structures contained in this book, the carbon atoms are not labeled.

tinuity around 48 K, which is characteristic of an LS ($S = 1$) $\leftrightarrow$ HS ($S = 2$) transition. No other information on this transition has been reported so far.

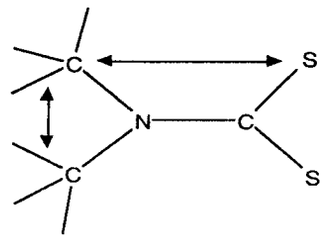
4.6.2 d^5 Ions

The spin transition phenomenon was observed for the first time by Cambi in the 1930s for tris(dithiocarbamato)iron(III) compounds [4.33–4.35]. The ligands dithiocarbamato(–) are of the type

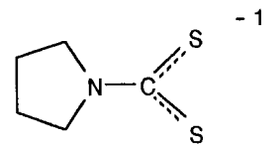


and the complex has a FeS_6 core with a pseudo-threefold axis [4.36, 4.37], as shown in Figure 4.13.

The actual magnetic behavior of these molecules strongly depends on the nature of the substituents R, and the solvent molecules which may be present in the crystal lattice. The substituents R interfere sterically with the sulfur atoms of the FeS_6 core and thereby influence the ligand bite as schematized below, the arrows indicating the steric interactions:



The most bulky R substituents, e.g., isopropyl groups, give strong steric interaction and small ligand bite resulting in a high T_c critical temperature. T_c is defined as the temperature for which the molar fractions of LS and HS molecules are both equal to 0.5. Even at room temperature χT is rather close to what is expected for an isolated doublet ground state. Less bulky substituents, like R = methyl, result in spin crossover LS ($S = 1/2$) $\leftrightarrow$ HS ($S = 5/2$) with T_c around 200 K. Finally, small steric interference, e.g., with the ligand tetramethylene dithiocarbamato



results in HS behavior in the whole temperature range [4.38].

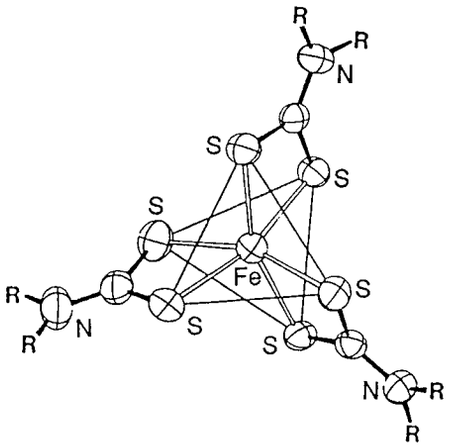
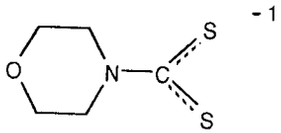


Figure 4.13. Molecular structure of a tris(dithiocarbamato)iron(III) compound $Fe(S_2CNR_2)_3$ (from ref. [4.37]).

A rationalization of the role of solvent molecules has been proposed in the case of tris(4-morpholinecarbodithioato)iron(III) compounds with the following ligand:



- 1

An LS behavior is obtained with nonpolar solvents, e.g., benzene and nitrobenzene. On the other hand, a spin transition or an HS behavior is obtained with polar solvents, e.g., dichloromethane, chloroform, or water [4.39].

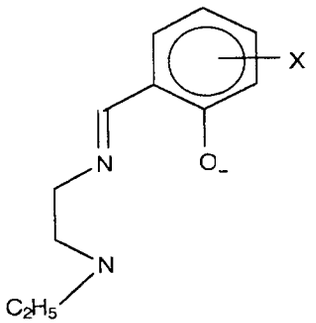
In the family of tris(dithiocarbamato)iron(III) compounds, if there is a spin transition in the solid state, it is always very smooth, without detectable hysteresis. We are typically in a situation of noncooperativity. The $\ln [(1-x)/x] = f(1/T)$ curve for tris(dimethyldithiocarbamato)iron(III) has similar slopes in the solid state and in solution. The enthalpy variation ΔH is in the range 500–750 cm^{-1} (6–9 kJ mol^{-1}). This indicates that the interaction parameter γ, in the regular solution model, is close to zero, or the domain size n, in the domain model, is close to unity [4.40].

The ^{57}Fe Mössbauer spectra of the tris(dithicarbamato)iron(III) spin transition compounds do not show distinct LS and HS signals, but rather an average signal suggesting that the spin flipping frequency is faster than the technique frequency scale [4.41].

From magnetic data limited to the 80–300 K range it had been reported that some tris(dithiocarbamato)iron(III) compounds exhibited an $S = 3/2 \leftrightarrow S = 5/2$ spin transition. It is now clear that this is never the case. On the other hand, several halogenato-bis(dithiocabamato)iron(III) derivatives have an $S = 3/2$ ground state, well

separated in energy from the first excited states. The magnetic properties of these compounds are discussed in Section 5.3.

Other interesting examples of spin transition in iron(III) chemistry have been reported by Hendrickson et al. These concern the series of complexes noted [Fe(X-saleten)$_2$]Y, where X-saleten(-) is a monoanion shown below [4.42–4.44]:



This monoanion arises from the condensation of salicylaldehyde (X = H) or 3(or 5)-methoxysalicylaldehyde (X = OCH_3) with *N*-ethylethylenediamine. Y^- is a counteranion like NO_3^-, PF_6^-, or BPh_4^-. These complexes are assumed to have a pseudo-octahedral geometry with a FeN_4O_2 core. All exhibit an LS ($S = 1/2$) $\leftrightarrow$ HS ($S = 5/2$) transition in the solid state, which is characterized by magnetic susceptibility, Mössbauer spectroscopy, and EPR. In no case is an intermediate spin state $S = 3/2$ observed. The Mössbauer spectroscopy sees either a superposition of $S = 1/2$ and $S = 5/2$ spin states, or only one of them. For most of the compounds the transition is gradual, and often incomplete at low temperature. For one of them, [Fe(3-OCH_3saleten)$_2$]PF_6, it is abrupt, occurring within 2 K around 159 K, and complete, at both high and low temperatures.

Two observations have been made on the [Fe(X-saleten)$_2$]Y compounds, which give important insights about the mechanism of the phenomenon. First, grinding one of the microcrystalline samples makes the transition incomplete, or more incomplete, at both high and low temperatures. The more efficient the grinding is, the more pronounced this effect. The modifications resulting from the grinding are reversible. As a matter of fact, dissolving a ground sample in methanol and precipitating the compound again gives a new sample which behaves in the same way as the original unground sample. The second observation deals with the abruptness of the transition. As the degree of grinding increases, the transition becomes more gradual. A similar observation has been made for iron(III) active species doped into isostructural cobalt(III) and chromium(III) hosts. As the percentage of iron(III) decreases, the transition again becomes smoother.

Hendrickson et al. have explained these observations by the nucleation and growth mechanism of phase transitions in solids. The transformation from HS to LS requires the formation of domains of LS molecules in the HS phase and the subsequent growth of these domains upon cooling. Before a domain can persist, and then grow, it has to

attain a critical size. The rate of growth of critical-sized domains of LS molecules determines the nature of the spin transition on cooling. Similarly, the rate of growth of critical-size domains of HS molecules would be the key factor on warming from the low temperatures. All the imperfections in the solid such as defects, grain boundaries or vacancies are preferred sites for nucleation. Grinding a sample leads to an increase in defects, and hence to an increase in nucleation sites. This explains the presence of LS molecules at high temperature. The incompleteness of the transition at low temperature is attributed to an increase in the activation barrier for the transition when the sample is ground. This higher activation barrier would be due to the fact that there is impingement between growing domains which are more abundant in the ground sample. In a doped sample the host molecules play the same role as the defects in a ground sample.

4.6.3 d^6 Ions

We have already mentioned that the iron(II) spin transition complexes were by far the most extensively studied. An exhaustive review article on this subject was published in 1981 by Gütlich [4.2]. Among these iron(II) compounds, two of them may be considered as archetypes of spin transition systems, namely $Fe(NCS)_2(bipy)_2$ and $Fe(NCS)_2(phen)_2$ with bipy = 2,2′-bipyridine and phen = 1,10-phenanthroline. Both exhibit an abrupt spin transition with T_c = 212 K and 176 K, respectively. The hysteresis is very small, 0.15 K for the former compound and 0.40 K for the latter.

The crystal structures of $Fe(NCS)_2(bipy)_2$ and $Fe(NCS)_2(phen)_2$ have been solved, both above and below T_c, the former by König et al. [4.45], the latter by Gallois et al.

Table 4.2. Comparison of the Main Structural Data for $Fe(NCS)_2(phen)_2$ at 293 K and 130 K

	293 K	130 K
Space group	*Pbcn*	*Pbcn*
Lattice constants (Å)		
a	13.1612 (18)	12.7699 (21)
b	10.1633 (11)	10.0904 (25)
c	17.4806 (19)	17.2218 (30)
Bond lengths (Å)		
Fe—N(1)	2.199 (3)	2.014 (4)
Fe—N(2)	2.213 (3)	2.005 (4)
Fe—N(20)	2.057 (4)	1.958 (4)
Bond angles (°)		
N(1)—Fe—N(2)	76.1	81.8
N(1)—Fe—N(20)	103.2	95.3
N(2)—Fe—N(20)	89.6	89.1

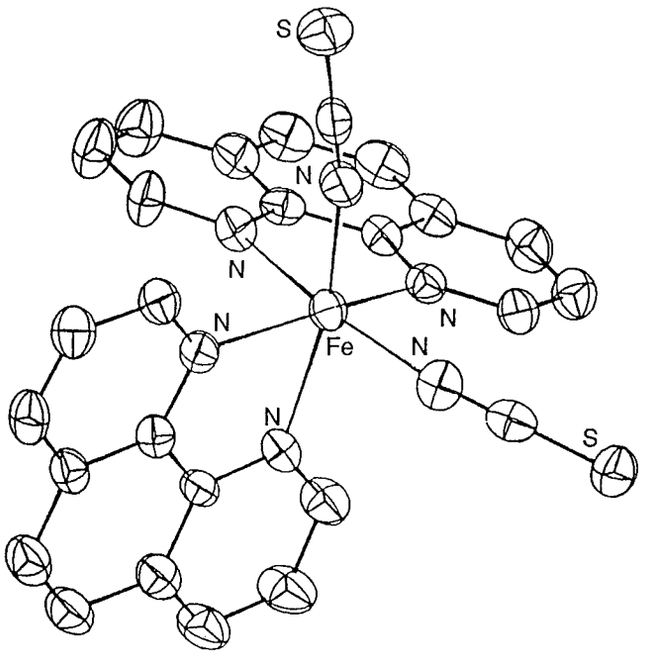


Figure 4.14. Molecular structure of $Fe(phen)_2(NCS)_2$ (from ref. [4.46]).

[4.46]. The coordination of the thiocyanato groups is cis. In both cases there is no structural phase transition. The main structural data for $Fe(NCS)_2(phen)_2$, the structure of which was determined more accurately, are collected in Table 4.2. The labeling of the atoms refers to Figure 4.14 where the structure of the compound is presented. The high- and low-temperature structures are superimposed in Figure 4.15.

The transition is accompanied by a large reorganization of the iron(II) coordination sphere. Passing from the HS to the LS phase results in an important shortening of the Fe—N(phen) and Fe—N(NCS) distances (0.20 and 0.10 Å, respectively), and a noticeable variation of the N—Fe—N angles, leading to a less distorted FeN_6 octahedron. The transitions have been followed by ^{57}Fe Mössbauer spectroscopy, which detects only one phase, LS at low temperature and HS at high temperature, far from T_c, and the coexistence of the two phases in the critical temperature region. The fact that distinct signals are seen indicates that the rate at which each molecule interchanges from LS to HS or vice versa is less than the inverse of the ^{57}Fe Mössbauer spectroscopy time scale, namely about 10^7 s^{-1} [4.47]. Variable temperature infrared spectra also detect the presence of two spin isomers around T_c. Other LS and HS crystal structures have been compared by König [4.48].

An elegant work was carried out by Sorai and Seki who measured very accurately the temperature dependence of the heat capacity C_p for $Fe(NCS)_2(phen)_2$ in the 13–375 K temperature range [4.31]. This curve is shown in Figure 4.16. The maximum of the

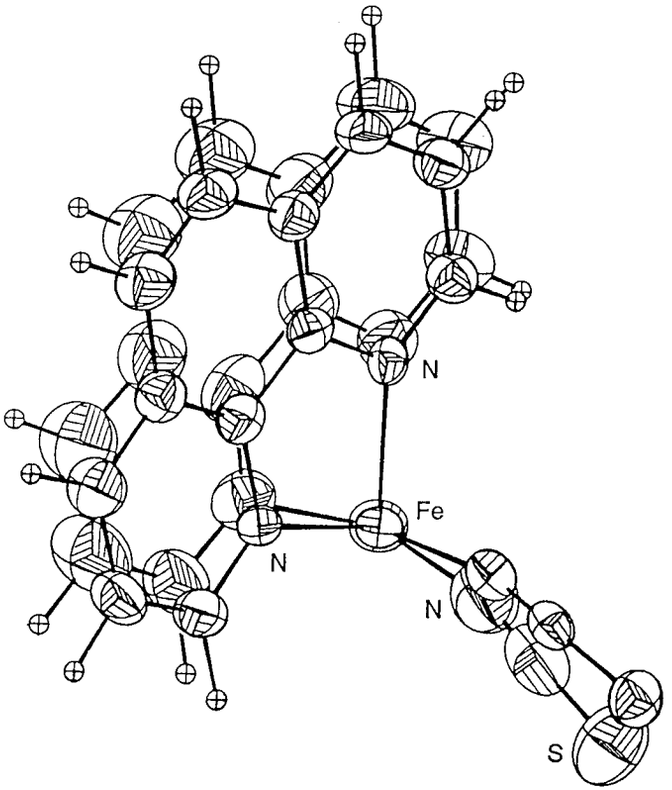


Figure 4.15. Superposition of the high- and low-temperature structures of $Fe(phen)_2(NCS)_2$. The LS structure appears in the front of the figure and the HS structure in the back. The molecule has a twofold rotation axis so that only half of the molecule is represented (from ref. [4.46]).

C_p anomaly due to spin transition appears at 176.3 K. Enthalpy and entropy variations are found as $\Delta H = 718(12)$ cm^{-1} and $\Delta S = 4.07(6)$ cm^{-1} K^{-1}. As for $C_{HS}(T_c) - C_{LS}(T_c)$ (see Section 4.5), it is estimated as 1.56 cm^{-1} K^{-1}. The key point arising from this study is that the magnetic entropy variation $R\ln(\Omega_{HS}/\Omega_{LS})$ due to the change of spin multiplicity accounts for only 27% of the total entropy variation. Therefore, 73% of ΔS is of vibrational nature. According to Sorai and Seki most of the vibrational contribution is intramolecular. It arises mainly from the changes in Fe—N stretching frequencies and N—Fe—N deformation frequencies. From the C_p versus T curve Sorai and Seki determined the average number n of molecules per domain (see Section 4.5) and found that n is equal to 100.

Interesting metal dilution studies have been reported [4.49, 4.50], in particular for $Fe_xM_{1-x}(NCS)_2(phen)_2$ systems with M = Mn, Co, Ni, and Zn. The x versus T plots were deduced from ^{57}Fe Mössbauer spectroscopy. Two effects were found. First, the transition becomes more gradual when x decreases whatever the nature of M, which confirms the role of cooperativity in the abruptness of the transition. On dilution the cooperative forces may be considerably weakened. The second effect is more subtle.

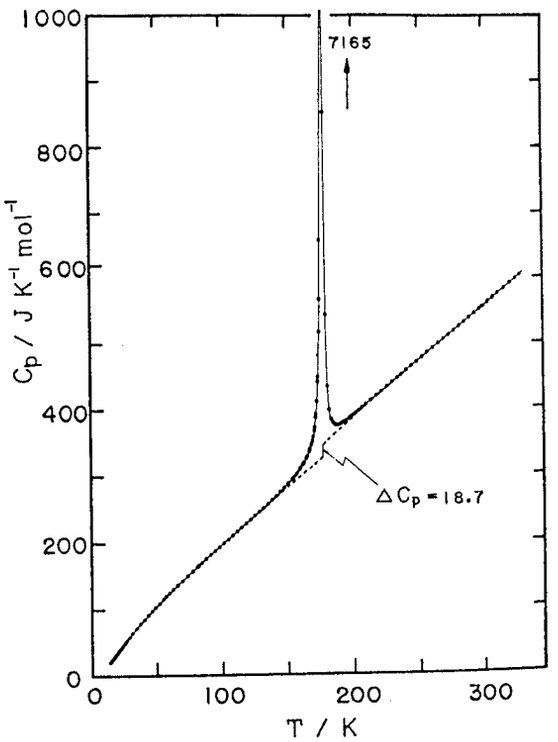


Figure 4.16. Heat capacity C_p versus temperature T curve for $Fe(phen)_2(NCS)_2$ (from ref. [4.31]).

When the ionic radius of M(II) is larger than that of Fe(II), dilution increases the stability of the HS phase. T_c is shifted toward lower temperatures and the amount of HS molecules at low temperature ($T << T_c$) increases with metal dilution. When the ionic radius of M(II) is smaller than that of Fe(II), the LS phase seems to be stabilized. In particular, the amount of LS molecules at high temperature ($T >> T_c$) increases with metal dilution. In the former case the volume of the unit cell increases on dilution, which may be considered as applying a *negative* lattice pressure on the remaining iron(II) sites. Therefore, the Fe—N bond may be longer in the dilute $Fe_xM_{1-x}(NCS)_2(phen)_2$ phase, which favors the HS state. In the latter case the M(II) ions exert a *positive* lattice pressure on the iron(II) sites. This may allow the Fe—N bonds to be shorter, which favors the LS state. This role of the ionic radii would deserve to be reinvestigated with other doped systems; it is not obvious that the above interpretation is valid in all situations. The case where the atomic radius of M(II) is assumed to be smaller than that of Fe(II), in particular, might be more complicated. Only HS Fe(II) was taken into account (the radius is 0.92 Å) whereas, below T_c, Fe(II) is in the LS form with a much smaller ionic radius (0.75 Å).

Many other iron(II) compounds with an FeN_6 core and two NCS^- ligands in cis positions exhibit more or less abrupt spin transitions. On the other hand, to date, only

one iron(II) compound with two NCS^- groups in trans positions has been reported as a spin transition system. This compound is trans-bis(thiocyanato)bis(4,4′-bis-1,2,4,-triazole)iron(III) hydrate. The transition is complete and abrupt. Moreover, in contrast with the cis-compounds, it presents an hysteresis of 21 K with $T_c\downarrow = 123.5$ K when cooling the sample and $T_c\uparrow = 144.5$ K when heating it. Let us also mention that when this compound loses its noncoordinated water molecule, the spin transition completely disappears, the HS phase being stable in the whole temperature range [4.51]. Crystal structure and magnetic properties of trans-bis(thiocyanato)bis(4,4′-bis-1,2,4,-triazole)iron(III) hydrate are shown in Figure 4.17.

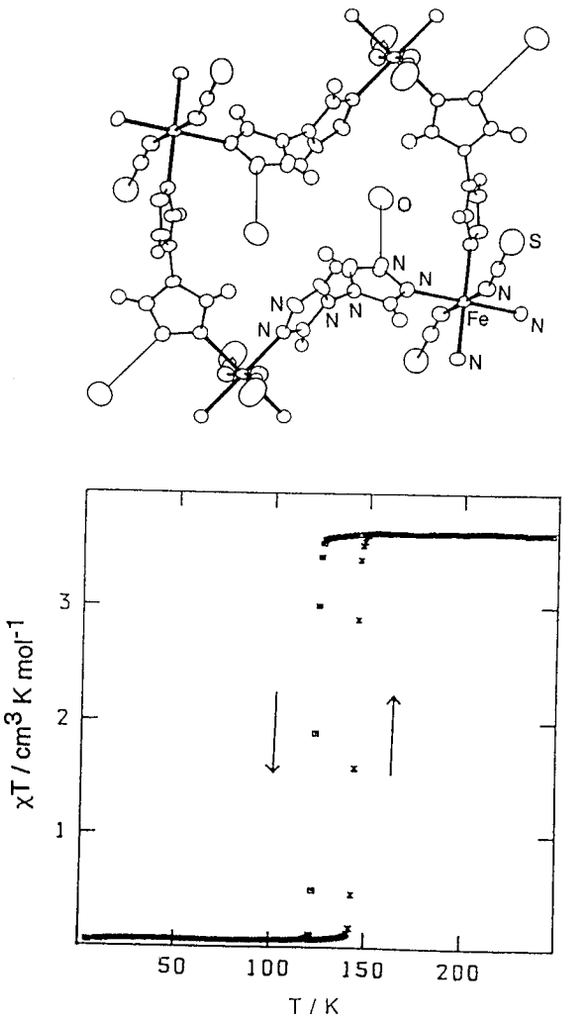
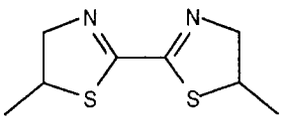


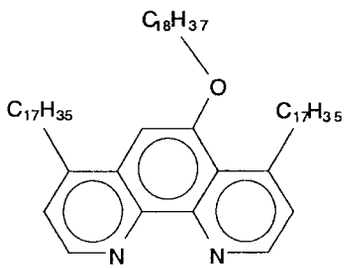
Figure 4.17. Crystal structure and χT versus T curves for [trans-bis(thiocyanato)bis(4,4′-bis-1,2,4-triazole)iron(III) hydrate (from ref. [4.51]).

König et al. have used high-resolution X-ray powder diffraction as a function of temperature and the analysis of the resulting peak profiles to study the spin transition in several iron(II) compounds, some of them exhibiting a gradual transition, and some an abrupt transition [4.3]. For gradual transitions diffraction peaks of a single phase are obtained, the individual peaks showing a systematic shift of their position versus temperature. This situation is encountered, for instance, for $Fe(bts)_2(NCS)_2$, bts being the ligand 2,2′-bis(5-methyl-2-thiazoline):



The transition occurs within ca. 100 K, with $T_c = 219$ K [4.52]. In X-ray powder diffraction the system behaves as a solid solution of two spin isomers within the same lattice. For abrupt transitions diffraction peaks belonging to both the LS and HS phases are observed. The peaks of one phase replace those of the other, as the transition progresses in either direction. Among many other compounds, $Fe(phen)_2(NCS)_2$ and $Fe(bipy)_2(NCS)_2$ present such a behavior. Concerning $Fe(phen)_2(NCS)_2$, only small changes of the diffraction patterns are apparent, which is consistent with the fact that the LS and HS isomers crystallize in the same space group.

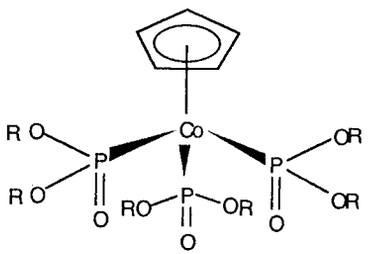
A spin transition iron(II) compound has been incorporated in a Langmuir–Blodgett (LB) film, namely $Fe(X_2Yphen)_2(NCS)_2$. X_2Yphen is the trisubstituted 1,10-phenanthroline:



containing three hydrophobic long chains. Two hundred layers were transferred onto a CaF_2 slide. The spin transition in the LB film was studied by infrared spectroscopy, focusing on the 2000–2150 cm^{-1} frequency range, which corresponds to the C—N stretching vibrations of the NCS groups. At room temperature a doublet corresponding to HS molecules was observed as well as a doublet of higher frequency corresponding to LS molecules, with a much weaker intensity. Upon cooling to about 200 K the intensity of the HS doublet continuously decreases in favor of the LS doublet. The $x = f(T)$ curve was deduced from the relative intensities of the two doublets. The transition is far from being abrupt. It takes place over a temperature interval of ca. 75 K. Nevertheless, it is less smooth than a simple Boltzmann distribution law between $S = 0$ and $S = 2$ states separated by a Δ energy gap would suggest. Some cooperativity is clearly retained in the two-dimensional network [4.53].

We have already covered the LIESST discovered with $[Fe(ptz)_6](BF_4)_2$. Subsequent results on $Fe(mtz)_6(BF_4)_2$ with mtz = 1-methyltetrazole bring new insights on this phenomenon [4.54]. $Fe(mtz)_6(BF_4)_2$ has two different iron(II) sites A and B. Sites A undergo a LS $\leftrightarrow$ HS transition at 160 K whereas sites B are HS in the whole temperature range. At 20 K sites A can be converted into the metastable HS state by irradiation with a Xe arc lamp, which is the normal LIESST effect. At the same temperature sites B can be converted into the metastable LS state by irradiation with red light ($\lambda > 700$ nm). This is the first example of a successful light induced formation of a metastable LS state for iron(II) HS species.

A unique family of cobalt(III) compounds undergoing a spin transition was described by Kläui et al. The general formula of the compounds is $(CoL_2)PF_6$, where L is a tridentate ligand of the type:

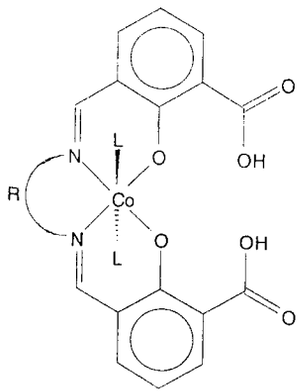


– 1

The transition has been studied for various R substituents in the solid state through magnetic susceptibility measurements, and in solution through NMR measurements [4.55–4.57]. In both cases the transition is very gradual. Interestingly, NMR studies have been performed using both ^{31}P and ^{59}Co NMR chemical shifts.

4.6.4 d^7 Ions

d^7 spin transition compounds are relatively rare. The most thoroughly investigated series is probably $Co[H_2(fsa)_2R]L_2$, the basic structure of which is



$H_2(fsa)_2R$ is the planar Schiff base resulting from the condensation of NH_2—R—NH_2 on 3-formylsalicylic acid. L is a ligand in an apical position [4.58–4.61].

For R = CH_2—CH_2 and L = H_2O a rather abrupt spin transition is observed with $T_c\downarrow = 81.5$ K and $T_c\uparrow = 84.6$ K. ΔH, ΔS, and γ in the regular solution model have been estimated from the $x = f(x)$ plot as $\Delta H = 161$ cm^{-1}, $\Delta S = 1.94$ cm^{-1} K^{-1} and $\gamma = 130$ cm^{-1}. In the domain model the domain size n is estimated as about 25, which is intermediate between the value for $Fe(phen)_2(NCS)_2$, $n = 100$, and that for tris(dimethyldithiocarbamato)iron(III), $n = 1$. The EPR spectrum of the LS phase is typical of an $S = 1/2$ cobalt(II) ion with a strong axial elongation, on which a rhombic distortion is superimposed. The actual symmetry is, indeed, C_{2v}. Interestingly, when warming $Co[H_2(fsa)_2en](H_2O)_2$ above 70°C, the two apical water molecules are lost, so that the cobalt(II) ion is in square planar surroundings and a LS behavior is observed in the whole temperature range.

For R = CH_2—CH_2 and L = py the transition is again rather abrupt with a more pronounced hysteresis. For a sample consisting of well shaped small crystals the critical temperatures are found as $T_c\downarrow = 115$ K and $T_c\uparrow = 127$ K. If a pressure of 10 kbar is exerted and released on this sample, then the transition becomes more gradual, the temperature at which there is 50% of LS and HS molecules is lower and the transition is incomplete at low temperatures. These observations are similar to those reported by Hendrickson et al. for $[Fe(X\text{-saleten})_2]Y$ compounds, except that the

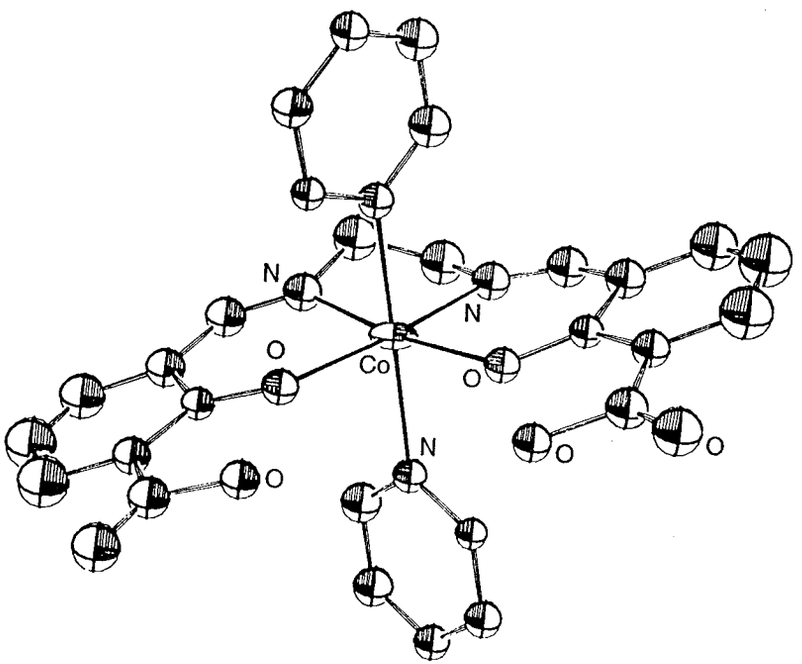


Figure 4.18. Molecular structure of $CoH_2(fsa)_2en(py)_2$ (from ref. [4.61]).

transition for the sample subjected to 10 kbar pressure seems complete at high temperature. Finally, a new observation is made, namely a significant broadening of the hysteresis loop for the perturbed sample (33 K instead of 12 K). The crystal structure of $Co[H_2(fsa)_2en](py)_2$ has been solved at room temperature [4.62]. A view of the molecule is shown in Figure 4.18.

One of the most abrupt transitions in this series of compounds was observed for R = CH_2—CH_2 and L = 4-*t*-butylpyridine with $T_c\downarrow$ = 138 K and $T_c\uparrow$ = 154 K. In contrast, for L = 4-methylpyridine, the transition is of the continuous type.

4.7 Spin Transition and Molecular Electronics

One of the most exciting perspectives in molecular chemistry is the utilization of isolated molecules, or of assemblies of molecules, in electronic circuits and devices [4.63–4.65]. It is probably a rather long-term perspective and the challenge is not to replace classical electronics, which may be defined as silicon electronics, but to use molecular systems to perform functions that are not possible for silicon. Some ideas have already put forward along this line and some preliminary advances have been achieved in several directions, namely in switching, amplification, information storage, and signal processing. In spite of the different end points, a pervasive concept appears in most of these fields; they require the use of molecular systems able to exist in two different electronic states. Such a property may happen for either an isolated molecule, or only an assembly of weakly interacting molecules. The spin transition phenomenon might well be the most spectacular example of transition between two electronic states in molecular chemistry [4.66], and devices for display and data recording incorporating spin transition molecular materials have been decribed [4.67].

In this section we will generalize the concept of transition in molecular chemistry, in relation with the possible emergence of molecular electronics.

To be appealing in molecular electronics, a molecular system must be able to evolve from a stable (or metastable) electronic state to another stable (or metastable) electronic state in a reversible and detectable fashion when applying an appropriate and controllable perturbation.

Almost every word of this definition deserves a comment. "Detectable" means that it is possible to measure a response function R for any value of the applied perturbation P and that the $R = f(P)$ curve exhibits a discontinuity for a critical value of the perturbation P_c. In fact, the situation of genuine discontinuity with a vertical tangent at $P = P_c$ is rather exceptional. More generally, the two states, stable below and above P_c, may couple, leading to a softening of the transition. The thermal population of the excited state in a given range of perturbation gives the same softening effect. There is of course no precise limit to the concept of transition. When the softening of the $R = f(P)$ plot becomes too pronounced, the concept tends to vanish.

"Reversible" means that the initial state of the system, where the perturbation varies in a given way, is identical to the final state where the perturbation varies in the opposite way. This does not require the routes to be identical. The case where the response versus perturbation plots for P increasing (P↑) and P decreasing (P↓) are different

corresponds to the hysteresis phenomenon. It is then possible to define two critical perturbations, $P_c\uparrow$ and $P_c\downarrow$. For any value of P between $P_c\uparrow$ and $P_c\downarrow$ there is a coexistence of stable and metastable states. In a certain sense, between these two limits, the system remembers its history. This is a particularly interesting situation since it opens the door to information storage.

Finally, the perturbation must be appropriate and controllable. Appropriate refers to the fact that the concept of transition includes the system as well as the perturbation. The couple molecular system–perturbation is indissociable. Controllable points out that it must be possible to fix the value of the perturbation, either below P_c, or above P_c, or also between $P_c\downarrow$ and $P_c\uparrow$ in the case of hysteresis.

The key concept is the existence of double minimum energy curves. An important distinction has to be made based on whether the phenomenon is purely molecular, or it is associated with an assembly of interacting molecules. In the former case we consider the potential energy variation versus one or several nuclear coordinates characterizing the molecular geometry [$E = f(Q)$]. Such a double well potential energy curve is represented in Figure 4.1. In the latter case the notion of nuclear coordinate becomes meaningless because, at a given instant, all molecules are not in the same state. Here we consider the Gibbs free energy variation versus the molar fraction of one of the constituents [$G = f(x)$]. Such curves are represented in Figures 4.6 and 4.8.

Although the two kinds of curves $E = f(Q)$ and $G = f(x)$ may appear rather similar, they do not correspond to the same physical processes. Thus, in the $E = f(Q)$ diagram, there are at least three different ways to move from one well to another: (1) either by climbing the activation barrier: this is the thermal process; (2) or by passing horizontally through the barrier: this is the quantum mechanical tunnelling effect; (3) or, finally, by passing through an excited state in a Franck–Condon vertical process followed by vibrational relaxation. This is the optical process. On the contrary, in the $G = f(x)$ diagrams, the system always evolves along the minimum energy path and can be trapped in a secondary minimum, which leads to the hysteresis effect.

Let us explain this behavior on the $G = f(x)$ curves of Figure 4.8 where the perturbation is the temperature. In the cooling mode, down to 150 K, the system is described by the lowest minimum corresponding to a regular solution rich in HS molecules. At 150 K the two minima have the same energy but the system is trapped in the one of the right-hand side, which is rich in HS molecules. Between 150 and 138.3 K the system remains trapped in the secondary minimum. Finally, at $T_c\downarrow$ = 138.3 K this secondary minimum disappears. The transition occurs and the system is described by the minimum which is rich in LS molecules. In the heating mode the system is described by the left-hand side minimum which is rich in LS molecules up to 157.5 K. The transition occurs at $T_c\uparrow$ = 157.5 K when this minimum vanishes.

Concerning the perturbation used to modify the relative positions of the two energy minima, it is important to point out that the temperature may be the appropriate and controllable perturbation mentioned in our definition only in the case of an assembly of molecules. Indeed, T is then involved in the entropy component of the Gibbs free energy. For an isolated molecule the only role of T is to change the population of the LS and HS states and not to displace a potential energy curve with respect to another.

For spin transition compounds it is well established that the occurrence of hysteresis is due to intermolecular interactions within the assembly of molecules, even if the microscopic origin of these interactions is not yet perfectly understood. The hysteresis observed for molecular-based magnets discussed in Chapter 12 also results from cooperativity within a crystal lattice. An important question may then be raised: can hysteresis be observed for an assembly of noninteracting molecules, and if so, what conditions must be fulfilled? Several groups are working on this problem and although the definitive answer is not known yet, it seems that the occurrence of hysteresis at the purely molecular level cannot be excluded [4.68, 4.69].

References

4.1. H.A. Goodwin, *Coord. Chem. Rev.* 18 (1976), 293.

4.2. P. Gütlich, *Struct. Bonding(Berlin)* 44 (1981), 83.

4.3. E. König, G. Ritter, S.K. Kulshreshtha, *Chem. Rev.* 85 (1985), 219.

4.4. P. Gütlich, A. Hauser, *Coord. Chem. Rev.* 97 (1990), 1.

4.5. J.K. Beattie, *Adv. Inorg. Chem.* 32 (1988), 1.

4.6. P. Gütlich, in "Mössbauer Spectroscopy Applied to Inorganic Chemistry," G.J. Long, ed., Vol. 1, Modern Inorganic Chemistry Series. Plenum Press, New York, 1984.

4.7. J. Zarembowitch, R. Claude, P. Thuéry, *Nouv. J. Chim.* 9 (1985), 467.

4.8. W. Vreugdenhil, J.G. Haasnoot, O. Kahn, P. Thuéry, J. Reedijk, *J. Am. Chem. Soc.* 109 (1987), 5272.

4.9. M. Bacci, *Inorg. Chem.* 25 (1986), 2322.

4.10. M. Bacci, *Coord. Chem. Rev.* 86 (1988), 245.

4.11. S. Decurtins, P. Gütlich, C.P. Köhler, H. Spiering, A. Hauser, *Chem. Phys. Lett.* 105 (1984), 1.

4.12. S. Decurtins, P. Gütlich, K.M. Hasselbach, A. Hauser, H. Spiering, *Inorg. Chem.* 24 (1985), 2174.

4.13. A. Hauser, *Chem. Phys. Lett.* 124 (1986), 543.

4.14. S. Decurtins, P. Gütlich, C.P. Köhler, H. Spiering, *J. Chem. Soc., Chem. Commun.* (1985), 430.

4.15. C.P. Slichter, H.G. Drickamer, *J. Chem. Phys.* 56 (1972), 2142.

4.16. H.G. Drickamer, C.W. Franck, in "Electronic Transitions and the High Pressure Chemistry and Physics of Solids," p. 44. Chapman and Hall, London, 1973.

4.17. R.A. Swalin, "Thermodynamics of Solids." John Wiley, New York, 1962.

4.18. J.H. Hildebrand, R.L. Scott, "Regular Solutions." Prentice-Hall, Englewood Cliffs, NJ, 1962.

4.19. C.N.R. Rao, *Int. Rev. Phys. Chem.* 4 (1985), 19.

4.20. S. Usha, K. Srinivasan, C.N.R. Rao, *Chem. Phys.* 100 (1985), 447.

4.21. R. Zimmermann, E. König, *J. Phys. Chem. Solids* 38 (1977), 779.

4.22. K. F. Purcell, M.F. Edwards, *Inorg. Chem.* 23 (1984), 2620.

4.23. R. Zimmermann, *J. Phys. Chem. Solids*, 44 (1983), 151.

4.24. T. Kambara, *J. Phys. Soc. Jpn. 49 (1980), 1806.*

4.25. N. Sasaki, T. Kambara, *J. Chem. Phys.* 74 (1981), 3472.

4.26. T. Kambara, *J. Chem. Phys.* 74 (1981), 4557.

4.27. S. Ohnishi, S. Sugano, *J. Phys.* C14 (1981), 39.

4.28. H. Spiering, E. Meissner, H. Köppen, E.W. Müller, P. Gütlich, *Chem. Phys.* 68 (1982), 65.

4.29. P. Adler, L. Wiehl, E. Meissner, C.P. Köhler, H. Spiering, P. Gütlich, *J. Phys. Chem. Solids*, 48 (1987), 517.

4.30. O. Kahn, J.P. Launay, *Chemtronics* 3 (1988), 140.

4.31. M. Sorai, S. Seki, *J. Phys. Chem. Solids*, 35 (1974), 555.

4.32. P.G. Sim, E. Sinn, *J. Am. Chem. Soc.* 103 (1981), 241.

4.33. L. Cambi, A. Gagnasso, *Atti. Accad. Naz. Lincei* 13 (1931), 809.

4.34. L. Cambi, L. Szegö, *Ber. Dtch. Chem. Ges.* 64 (1931), 2591; 66 (1933), 656.

4.35. L. Cambi, L. Szegö, A. Cagnasso, *Atti. Accad. Naz. Lincei* 15 (1932a), 266; 15 (1932b), 329.

4.36. A. H. Ewald, R.L. Martin, E. Sinn, A.H. White, *Inorg. Chem.* 8 (1969), 1837.

4.37. K. Stahl, I. Ymén, *Acta Chem. Scand.* A37 (1983), 729.

4.38. K. Stahl, *Inorg. Chim. Acta*, 75 (1983), 85.

4.39. E.J. Cucauskas, B.S. Deaver Jr., E. Sinn, *Inorg. Nucl. Chem. Lett.* 13 (1977), 283.

4.40. J. Albertsson, A. Oskarsson, K. Stahl, *Acta Chem. Scand.* A36 (1982), 783.

4.41. P.B. Merrithew, P.G. Rasmussen, *Inorg. Chem.* 11 (1972), 325.

4.42. M.S. Haddad, W.D. Federer, M.W. Lynch, D.N. Hendrickson, *J. Am. Chem. Soc.* 102 (1980), 1468.

4.43. M.S. Hadddad, M.W. Lynch, W.D. Federer, D.N. Hendrickson, *Inorg. Chem.* 20 (1981), 123.

4.44. M.S. Haddad, W.D. Federer, M.W. Lynch, D.N. Hendrickson, *Inorg. Chem.* 20 (1981), 131.

4.45. E. König, K.J. Watson, *Chem. Phys. Lett.* 6 (1970), 457.

4.46. B. Gallois, J.A. Real, C. Hauw, J. Zarembowitch, *Inorg. Chem.* 29 (1990), 1152.

4.47. P. Ganguli, P. Gütlich, E.W. Müller, W. Irler, *J. Chem. Soc., Dalton Trans.* (1981), 441.

4.48. E. König, *Prog. Inorg. Chem.* 35 (1987), 527. *Struct. Bonding(Berlin)* 76 (1991), 51.

4.49. P. Ganguli, P. Gütlich, E.W. Müller, *Inorg. Chem.* 21 (1982), 3439.

4.50. I. Sanner, E. Meissner, H. Köppen, H. Spiering, P. Gütlich, *Chem. Phys.* 86 (1984), 227.

4.51. W. Vreugdenhil, J.H. van Diemen, R.A.G. de Graaff, J.G. Haasnoot, J. Reedijk, A.M. van der Kraan, O. Kahn, J. Zarembowitch, *Polyhedron* 9 (1990), 2971.

4.52. G. Bradley, V. MacKee, S. M. Nelson, J. Nelson, *J. Chem. Soc., Dalton Trans.* (1978), 522.

4.53. P. Coronel, A. Barraud, R. Claude, O. Kahn, A. Ruaudel-Teixier, J. Zarembowitch, *J. Chem. Soc., Chem. Commun.* (1989), 193.

4.54. P. Poganiuch, S. Decurtins, P. Gütlich, *J. Am. Chem. Soc.* 112 (1990), 3270.

4.55. W. Kläui, *J. Chem. Soc.* (1979), 700.

4.56. P. Gütlich, B.R.MacGarvey, W. Kläui, *Inorg. Chem.* 19 (1980), 3704.

4.57. G. Navon, W. Kläui, *Inorg. Chem.* 23 (1984), 2722.

4.58. J. Zarembowitch, O. Kahn, *Inorg. Chem.* 23 (1984), 589.

4.59. J. Zarembowitch, R. Claude, O. Kahn, *Inorg. Chem.* 24 (1985), 1576.

4.60. P. Thuéry, J. Zarembowitch, *Inorg. Chem.* 25 (1986), 2001. P. Thuéry, J. Zarembowitch, A. Michalowicz, O. Kahn, *Inorg. Chem.* 26 (1987), 851.

4.61. J. Zarembowitch, *New. J. Chem.* 16 (1992), 255.

4.62. P. Charpin, M. Nierlich, D. Vigner, M. Lance, P. Thuéry, J. Zarembowitch, F. d'Yvoire, *J. Crystallogr. Spectrosc. Res.* 18 (1988), 429.

4.63. F.L. Carter, ed., "Molecular Electronic Devices." Marcel Dekker, New York 1982. "Molecular Electronic Devices," F.L. Carter, ed. Marcel Dekker, New York, 1987.

4.64. E.T.T. Jones, O.M.Chayan, M.S. Wrighton, *J. Am. Chem. Soc.* 109 (1987), 5526.

4.65. A. Aviram, *J. Am. Chem. Soc.* 110 (1988), 5687.

4.66. J. Zarembowitch, O. Kahn, *New J. Chem.* 15 (1991), 181.

4.67. O. Kahn, J. Kroeber, C. Jay, *Adv. Mater.* 4 (1992), 718.

4.68. M. Sano, H. Taube, *J. Am. Chem. Soc.* 113 (1991), 2327.

4.69. H. Bolvin, B. Vekhter, O. Kahn, *New. J. Chem.* 15 (1991), 889.

CHAPTER

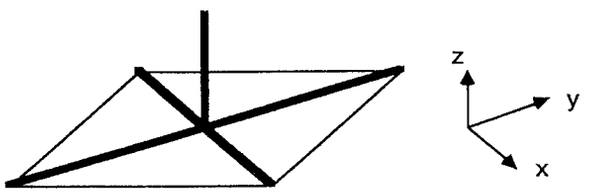
5

Intermediate-Spin and Spin-Admixed States

5.1 Intermediate Spin: One-Electron Approach

In the preceding chapter we saw that for d^4, d^5, d^6 and d^7 metal ions in octahedral environments, the nature of the ground state depends on the strength of the ligand field. For d^4 and d^7 ions, the spin variation between the two possible ground states is $\Delta S = 1$. For d^5 and d^6 ions, this spin variation is $\Delta S = 2$, so that we may wonder whether a ground state with an intermediate spin state can exist. Such an intermediate spin would be $S = 3/2$ for d^5 ions like iron(III), and $S = 1$ for d^6 ions like iron(II). As long as the environment of the metal ion remains roughly octahedral, even strongly distorted, the answer is negative. On the other hand, for five- or four-coordinate species such intermediate spin states have actually been observed.

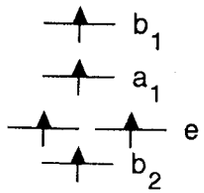
A first approach to this problem consists in considering a five-coordinate d^5 compound with a square pyramidal environment and C_{4v} symmetry.



The d orbitals in C_{4v} symmetry transform as $a_1(d_{z^2})$, $b_1(d_{x^2-y^2})$, $b_2(d_{xy})$, and $e(d_{yz}$ and $d_{zx})$. Their relative energies can be deduced from the t_{2g} and e_g subsets valid in octahedral symmetry. One of the ligands in the apical position is moved away along the z axis toward infinity, such that the a_1 orbital pointing toward this ligand is stabilized with respect to the b_1 orbital. As for the orbitals arising from the t_{2g} subset, their relative energies depend on the in-plane and out-of-plane π interactions between the metal ion and the ligands in the basal plane. If this basal plane is a tetracoordinating conjugated macrocycle like porphyrins or phthalocyanins, then the b_2

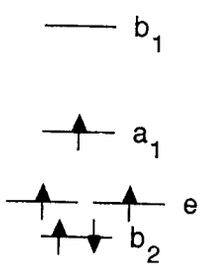
orbital is slightly stabilized with respect to the e orbitals. The relative energies are then $b_2 < e < a_1 < b_1$.

If the ligand in the apical position creates a rather weak field, then the most antibonding $b_1(x^2 - y^2)$ orbital is not so high in energy with respect to the other four orbitals. The spin-pairing energies are larger than the various energy gaps between orbitals and the ground configuration is



The ground state arising from this configuration is high-spin with the symmetry label $^6A_1$. If this state is well separated in energy from the first excited states, the magnetic susceptibility follows the Curie law, except in the very low-temperature range where a weak zero-field splitting within the spin sextet may influence the magnetic data. The Curie constant is $\chi T = 35Ng^2\beta^2/12k$, and the Zeeman factor is almost isotropic and close to 2.00.

Conversely, if the apical ligand creates a strong field, the $e_g - t_{2g}$ splitting in hypothetical six-coordinate species is large. The stabilization of the $a_1(z^2)$ orbital in the actual five-coordinate species is large, such that the $b_1 - a_1$ energy gap may be greater than the spin-pairing energy. The ground configuration is then



with an $S = 3/2$ intermediate ground state. Its symmetry label is $^4A_2$. If this state is the only thermally populated one, then the magnetic behavior is that of a spin-quartet state without first-order angular momentum (see Section 2.5). The high-temperature limit of χT is $Ng^2\beta^2/4k$. The zero-field splitting, which gives rise to two Kramers doublets $M_S = \pm 1/2$ and $\pm 3/2$, may be important, such that the magnetic anisotropy may be pronounced, even at relatively high temperature.

In principle a third situation might be anticipated, namely a low-spin ground state ^2E arising from the $(b_2)^2(e)^3$ configuration. However, up to now, such a low-spin state has never been observed for a genuine five-coordinate d^5 ion. This is due to the fact that the stabilization of the $a_1(z^2)$ orbital with respect to the hypothetical six-coordinate case is always sufficiently pronounced for the $a_1 - e$ energy gap to be smaller than the

spin pairing energy. On the other hand, low-spin $S = 1/2$ ground state is well documented for six-coordinate compounds.

Although the existence of an $S = 3/2$ ground state was postulated by Pauling et al. [5.1, 5.2] as early as 1936, it has been ignored or rejected by most of the authors from the 1940s up to the beginning of the 1970s. Some authors even stated from simple theoretical considerations that an $S = 3/2$ state could not have the lowest energy [5.3]. Actually, the $S = 3/2$ intermediate spin state is not easy to distinguish from the thermal equilibrium between $S = 1/2$ low-spin and $S = 5/2$ high-spin states if the magnetic data are not accurately recorded down to the lowest temperatures. Until the pioneering work of Maltempo and Moss [5.4] in 1976, most of the cases where the χT values were intermediate between what is expected for low spin ($Ng^2\beta^2/4k$) and high spin ($35Ng^2\beta^2/12k$) have been attributed to such an $S = 1/2 - S = 5/2$ thermal equilibrium.

The above discussion may be easily extended to the case of d^6 ions. The high-spin ground state is then $^5B_2$. It arises from the $(b_2)^2(e)^2(a_1)^1(b_1)^1$ configuration. The intermediate-spin ground state ^3E arises from the $(b_2)^2(e)^2(a_1)^1(b_1)^0$ configuration.

5.2 Spin-Admixed Ground State

Let us return to the case of a d^5 ion in square pyramidal C_{4v} symmetry. We have shown that according to the strength of the ligand occupying the apical position, the ground state may be $^6A_1$ or $^4A_2$. In fact, in some compounds, the situation may be even more complicated and can no longer be visualized in a purely one-electron orbital approach. If the two states $^6A_1$ or $^4A_2$ are close enough in energy, they can couple through the spin-orbit coupling to give a so-called spin-admixed ground state. Group theory allows a qualitative approach to this phenomenon. The C'_{4v} double group is as

C'_{4v}	E	R	$2C_4$ $2C_4^3R$	$2C_4^3$ $2C_4R$	C_2 C_2R	$2\sigma_v$ $2\sigma_vR$	$2\sigma_d$ $2\sigma_dR$
A_1	1	1	1	1	1	1	1
A_2	1	1	1	1	1	−1	−1
B_1	1	1	−1	−1	1	1	−1
B_2	1	1	−1	−1	1	−1	1
E_1	2	2	0	0	−2	0	0
E_2	2	−2	$\sqrt{2}$	$-\sqrt{2}$	0	0	0
E_3	2	−2	$-\sqrt{2}$	$\sqrt{2}$	0	0	0

(5.2.1)

The spin functions associated with $S = 5/2$ span the $E_2 + 2E_3$ irreducible representations, and those associated with $S = 3/2$ span the $E_2 + E_3$ irreducible representations of C'_{4v}. For the unperturbed $^6A_1$ state the $E_2 + 2E_3$ Kramers doublets are accidentally degenerate since there is no first-order angular momentum. In the same

way, for the unperturbed $^4A_2$ state, the $E_2 + E_3$ Kramers doublets are accidentally degenerate. When the $^6A_1$ and $^4A_2$ states interact, the E_2 zero-field split states on the one hand, and the E_3 zero-field split states on the other hand can mix. The magnitude of this mixing depends on the separation Δ between the unperturbed $^6A_1$ and $^4A_2$ states, and the spin-orbit coupling constant ξ.

For practical purposes it is more appropriate to label the Kramers doublets arising from the unperturbed $^6A_1$ and $^4A_2$ states with the $|^6A_1, \pm M_S\rangle$ (M_S = 1/2, 3/2, and 5/2), and $|^4A_2,\pm M_S\rangle$ (M_S = 1/2 and 3/2) notations. This is possible because the symmetry is axial, and the zero-field splitting operator acting within the unperturbed states is of the form DS_z^2. $|^6A_1, \pm 1/2\rangle$ and $|^4A_2, \pm 1/2\rangle$ correspond to the E_2 irreducible representation, and the other Kramers doublets correspond to E_3.

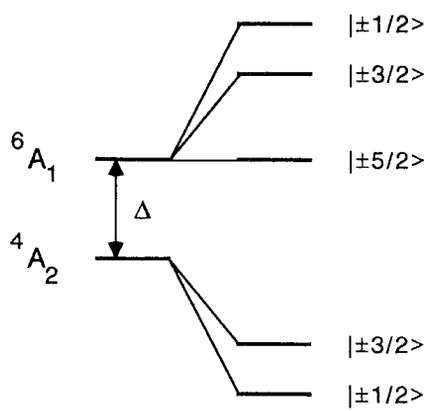
The Hamiltonian appropriate to describe the energy levels in zero field is

$$\boldsymbol{H} = \mathbf{V} + \xi\Sigma_i \mathbf{l}_i \cdot \mathbf{s}_i \tag{5.2.2}$$

where $\mathbf{V}$ is the ligand field provoking the Δ energy gap between unperturbed $^6A_1$ and $^4A_2$ states, and $\xi\Sigma_i \mathbf{l}_i \cdot \mathbf{s}_i$ is the spin-orbit coupling operator, ξ being the spin-orbit coupling constant for the five electrons occupying the d orbitals. Since the states $^4A_2$ and $^6A_1$ in which we are interested do not arise from the same free-ion state, the spin-orbit coupling operator may not be simplified in $\lambda\, \mathbf{L}\cdot\mathbf{S}$. The matrix elements of this operator have been calculated by Maltempo and Moss [5.4]. The only nonzero matrix elements are

$$\begin{aligned}
&\langle {}^6A_1,\pm 1/2 \,|\, \xi\Sigma_i \mathbf{l}_i \cdot \mathbf{s}_i \,|\, {}^4A_2,\pm 1/2 \rangle = \langle {}^4A_2,\pm 1/2 \,|\, \xi\Sigma_i \mathbf{l}_i \cdot \mathbf{s}_i \,|\, {}^6A_1,\pm 1/2\rangle \\
&\qquad = -(6/5)^{1/2}\xi \\
&\langle {}^6A_1,\pm 3/2 \,|\, \xi\Sigma_i \mathbf{l}_i \cdot \mathbf{s}_i \,|\, {}^4A_2,\pm 3/2 \rangle = \langle {}^4A_2,\pm 3/2 \,|\, \xi\Sigma_i \mathbf{l}_i \cdot \mathbf{s}_i \,|\, {}^6A_1,\pm 3/2\rangle \\
&\qquad = -(4/5)^{1/2}\xi
\end{aligned} \tag{5.2.3}$$

It follows that the energy levels are



where the unperturbed $^4A_2$ state is assumed to be the lowest. When Δ is large with respect to ξ, S = 3/2 and 5/2 remain good quantum numbers. The $^6A_1 - {}^4A_2$ interaction leads to zero-field splittings within the almost pure spin states. The zero-field splitting parameter D is

$$D = \xi^2/5\Delta \tag{5.2.4}$$

Conversely, when Δ and ξ are of the same order of magnitude, the mixing of the states is important. S = 3/2 and 5/2 are no longer good quantum numbers. On the other hand, $M_S = \pm 1/2, \pm 3/2, \pm 5/2$ remain good quantum numbers in zero field. The wavefunctions associated with each Kramers doublet, except $|^6A_1, \pm 5/2\rangle$, are of the form

$$|\pm M_S\rangle = a\,|{}^6A_1,\pm M_S\rangle + b\,|{}^4A_2,\pm M_S\rangle \tag{5.2.5}$$

where the mixing coefficients a and b are determined by diagonalizing the energy matrices :

$\lvert {}^6A_1,\pm 1/2\rangle$	$\lvert {}^4A_2,\pm 1/2\rangle$
Δ	$-(6/5)^{1/2}\xi$
	0

$\lvert {}^6A_1,\pm 3/2\rangle$	$\lvert {}^4A_2,\pm 3/2\rangle$
Δ	$-(4/5)^{1/2}\xi$
	0

(5.2.6)

As expected, for $\Delta/\xi = 0$, there is an equal mixture of high-spin and intermediate-spin states for each of the four Kramers doublets with $M_S = \pm 1/2$ and $\pm 3/2$. For large values of $|\Delta/\xi|$, the composition of the wavefunctions for the ground Kramers doublet approaches pure intermediate-spin or pure high-spin according to whether Δ is positive or negative. From the matrices above we can also determine the energies of the five Kramers doublets as

$$\begin{aligned}
E_1 &= 0 \\
E_2 &= [(25x^2 + 120)^{1/2} - (25x^2 + 80)^{1/2}]/10 \\
E_3 &= [5x + (25x^2 + 120)^{1/2}]/10 \\
E_4 &= [(25x^2 + 120)^{1/2} + (25x^2 + 80)^{1/2}]/10 \\
E_5 &= (25x^2 + 120)^{1/2}/5
\end{aligned} \tag{5.2.7}$$

with

$$x = \Delta/\xi \tag{5.2.8}$$

where the zero energy is assigned to the M_S = 1/2 lowest Kramers doublet. The variation of these energies versus Δ/ξ is shown in Figure 5.1.

The calculation of the parallel and perpendicular magnetic susceptibilities does not present any conceptual difficulty. Using the appropriate eigenfunctions for the five Kramers doublets as a basis set, the Zeeman perturbation $g\beta\mathbf{S}\cdot H$ is diagonalized, and the first- and second-order Zeeman coefficients are calculated for both the parallel (z) and perpendicular (x,y) directions of the applied magnetic field, and finally introduced

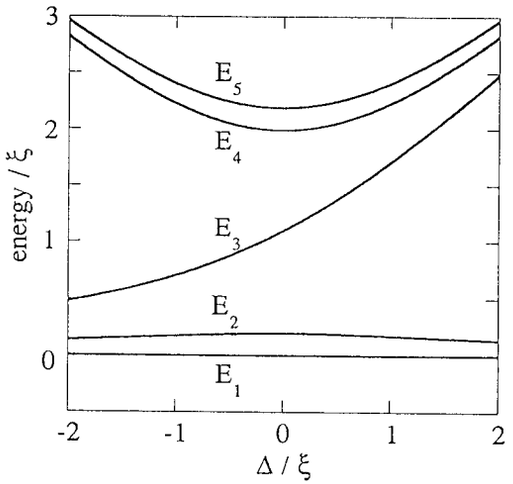


Figure 5.1. Energy (in ξ units) versus Δ/ξ curves for the five low-lying Kramers doublets in the case of $^6A_1 - {}^4A_2$ spin-admixed state. The energy of the lowest doublet is taken as the energy origin. Δ is the $^6A_1 - {}^4A_2$ energy gap for unperturbed states, and ξ the spin-orbit coupling constant.

into the Van Vleck formula (1.4.7). We can note here that, owing to the orthogonality of the orbital functions, the Zeeman perturbation does not couple unperturbed states. Typical χT versus kT/ξ curves for $g = 2$, and $\Delta/\lambda = 0$ and ± 2 are shown in Figure 5.2, where χ is the average magnetic susceptibility.

Interestingly, the degree of mixing in principle may also be deduced from EPR spectroscopy. Resonance within the $M_S = \pm 1/2$ Kramers doublet is observed at $g_z = 2$

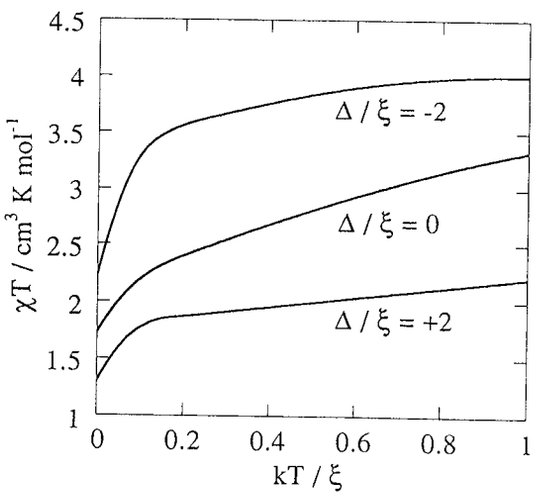


Figure 5.2. χT versus kT/ξ plots in the case of $^6A_1 - {}^4A_2$ spin-admixed state for $\Delta/\xi = 0$ and ± 2.

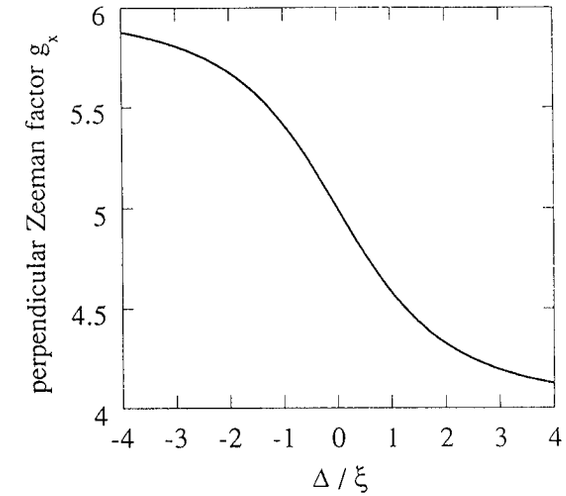


Figure 5.3. Variation of the perpendicular Zeeman factor g_x associated with the lowest Kramers doublet as a function of Δ/ξ in the case of $^6A_1 - {}^4A_2$ spin-admixed state.

and $g_x = g_y = 6$ for a pure $^6A_1$ state in axial symmetry, and at $g_z = 2$ and $g_x = g_y = 4$ for a pure $^4A_2$ state in axial symmetry. For the ground Kramers doublet of a spin-admixed species, the value of g_z is independent of Δ/ξ, and equal to 2. As for the value of g_x, it varies as shown in Figure 5.3 from a minimum of 4 for large positive values of Δ/ξ to a maximum of 6 for large negative values of Δ/ξ, according to [5.5]:

$$\Delta/\lambda = \pm\left[-\frac{24(g_x-5)^2}{5(g_x-6)(g_x-4)}\right]^{1/2} \tag{5.2.9}$$

where the plus sign stands for $g_x < 5$, and minus for $g_x > 5$.

5.3 Examples of Intermediate-Spin and Spin-Admixed States in Iron(III) Compounds

The first well-documented examples of iron(III) compounds with intermediate $S = 3/2$ states are provided by halogenobis(dialkyldithiocarbamato)iron(III). Single crystal X-ray structures for several derivatives of this family are available [5.6–5.9]. In all cases the iron atom has a highly distorted square pyramidal geometry, the halogen atom being at the apex of the pyramid and the iron atom at its center, about 0.6 Å above the mean basal plane of the four sulfur atoms belonging to the dithiocarbamato ligands. The overall symmetry is close to C_2. The structure of chlorobis(diethyldithiocarbamato)iron(III) is sketched in Figure 5.4. Down to around 10 K the average magnetic susceptibility [5.9–5.11] for this compound follows the Curie law anticipated for an $S = 3/2$ ground state with $\chi T = 1.90$ cm^3 K mol^{-1}. Magnetic anisotropy measurements

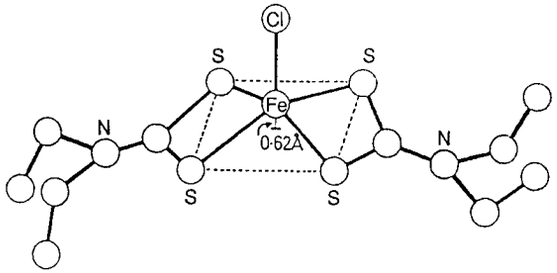


Figure 5.4. Molecular structure of chlorobis(diethyldithiocarbamato) iron(III) (from ref. [5.7]).

have permitted an accurate determination of the energy gap δ between the two Kramers doublets, related to the axial D and rhombic E zero-field splitting parameters appearing in the spin Hamiltonian in zero-field:

$$\mathbf{H} = D(\mathbf{S}_z^2 - 5/4) + E(\mathbf{S}_x^2 - \mathbf{S}_y^2) \tag{5.3.1}$$

through

$$\delta = 2(D^2 + 3E^2)^{1/2} \tag{5.3.2}$$

δ has been found to be equal to 3.4 cm^{-1} [5.12, 5.13]. Interestingly, δ has also been determined directly by far infrared spectroscopy [5.14, 5.15] and found to be equal to 3.85 cm^{-1}. It should be noted, however, that this spectroscopic determination has been carried out at 2.2 K, i.e., in the three-dimensional ferromagnetic state. Indeed, surprisingly, the compound orders ferromagnetically at 2.5 K, such that the two values of δ are not exactly comparable. Mössbauer data confirm that only the $S = 3/2$ ground state is thermally populated throughout the 4.2–300 K temperature range [5.9, 5.10, 5.16].

Low-spin, high-spin, intermediate-spin, and spin-admixed states in iron(III) porphyrins have been extensively investigated, in particular by Scheidt and Reed [5.17]. These compounds are invariably found to have one or two axial ligands so that the overall geometry is tetragonal, or nearly so. The spin state is essentially controlled by the nature and the number of axial ligands. More precisely, the coordination of strong field ligands leads to six-coordinate compounds with an $S = 1/2$ low-spin ground state. This is the case, for instance, for $[Fe(TPP)(CN)_2]^+$ or $[Fe(TPP)(im)_2]^+$ with TPP = meso-tetraphenylporphyrinato and im = imidazole [5.18, 5.19]. The iron atom is then located within the basal plane, and the Fe–N bond lengths in this plane are on the order of 1.99 ± 0.02 Å. Weaker field ligands lead to either six- or to five-coordinate compounds with an $S = 5/2$ high-spin ground state. Six-coordinate geometry is encountered with neutral ligands like H_2O or dimethylformamide (DMF), while five-coordinate geometry is observed for weak field monoanionic ligands like halogeno. For high-spin six-coordinate derivatives, for instance $[Fe(TPP)(H_2O)_2]^+$, the iron atom remains within the basal plane, but the Fe—N bond lengths are about 0.06 Å larger than in the low-spin case [5.20]. For high-spin five-coordinate derivatives,

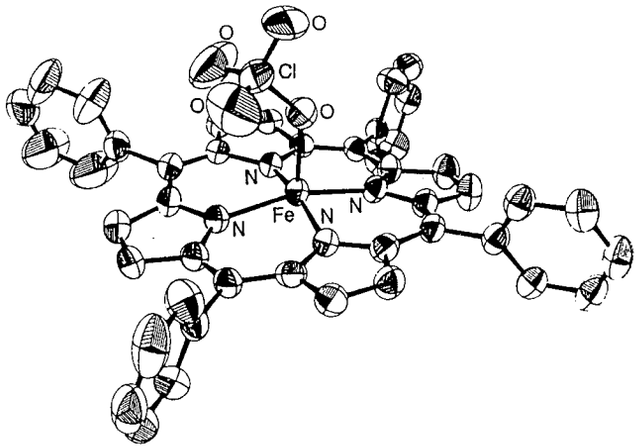


Figure 5.5. Molecular structure of $Fe(TPP)(ClO_4)$ with TPP = meso-tetraphenylporphyrinato (from ref. [5.22]).

for instance [Fe(TPP)Cl], the iron atom is extruded out of the porphyrinato plane in the direction of the apical ligand, the distance between the metal atom and the center of the basal plane being of the order on 0.5 ± 0.1 Å. The Fe—N bond lengths involving the porphyrinato nitrogen atoms are around 2.07 ± 0.02 Å[5.21]. Finally, moderate field ligands may lead to five-coordinate compounds with either a pure $S = 3/2$ ground state, or, more frequently, an $S = 3/2$, 5/2 spin-admixed state. The displacement of iron from the basal plane is then on the order of 0.3 Å.

A well-documented case of a spin-admixed derivative is offered by $[Fe(TPP)(ClO_4)]$, the structure of which is shown in Figure 5.5. Perchlorato is linked to iron by one of its four oxygen atoms with an unusually short Fe—O bond length (2.029 Å) [5.23, 5.23]. The Fe—N bond lengths (mean value: 2.001 Å) are shorter than in high-spin iron(III) porphyrins as is the displacement of the iron atom above the mean porphyrinato plane (0.28 Å). At room temperature χT for $[Fe(TPP)(ClO_4)]$ is equal to 3.18 cm^3 K mol^{-1}, which is significantly higher than anticipated for a pure $S = 3/2$ state (≈ 1.88 cm^3 K mol^{-1}), but lower than anticipated for a pure $S = 5/2$ state (≈ 4.38 cm^3 K mol^{-1}). On cooling, χT first smoothly decreases down to ca. 30 K, and then more rapidly below 30 K. At 4.2 K, χT is equal to 1.71 cm^3 K mol^{-1}. These magnetic properties are fairly well reproduced by Maltempo's model; the parameters are $\Delta = 108$ cm^{-1}, $\xi = 188$ cm^{-1}, and the g-values associated with the lowest Kramers doublet are calculated as $g_z = 2$ and $g_x = 4.74$ [5.24–5.26]. These values correspond to wavefunctions for the ground state containing 65% of $^4A_2$ and 35% of $^6A_1$. The experimental EPR spectrum gives $g_z = 2$ and $g_x = 4.75$. In spite of this apparent agreement, the Δ and ξ values must be considered at best as indicative. Indeed, the electronic structure of $[Fe(TPP)(ClO_4)]$ is certainly more complicated than is suggested by Maltempo's model in which only two low-lying states are taken into account. Thus, this model does not allow a proper interpretation of the magnetization data, even if the Δ and ξ values are different from those given above. Mössbauer spectroscopy

confirms that only one state is thermally populated. This spectrum shows a well-defined quadrupole doublet with a large quadrupole splitting that remains unchanged throughout the whole temperature range [5.26]. This unique populated state might arise from the mixing of more than two spin states.

In addition to six-coordinate low-spin and five-coordinate high-spin species, intermediate-spin and spin-admixed species have also been characterized in iron(III) phthalocyanine chemistry [5.27]. Thus, [Fe(Pc)Br] with Pc = phthalocyaninato shows an essentially pure $S = 3/2$ state. χT remains equal to 2.0 cm^3 K mol^{-1} in the 30–300 K range. Below 30 K, χT decreases owing to the zero-field splitting within the quartet ground state. In contrast, [Fe(Pc)Cl] shows a significant degree of $S = 3/2$, 5/2 mixing. χT is equal to 2.55 cm^3 K mol^{-1} at 300 K and decreases down to 1.32 cm^3 K mol^{-1} at 4.2 K. The g_x value of 4.67 is consistent with a $^4A_2 - {}^6A_1$ mixing. For both [Fe(Pc)Br] and [Fe(Pc)Cl], Mössbauer spectra confirm that there is no equilibrium between two spin states. The spectra remain essentially unchanged throughout the temperature range.

5.4 Examples of Intermediate-Spin and Spin-Admixed States in Iron(II) Compounds

The $S = 1$ intermediate spin state has also been observed in iron(II) porphyrins as well as the $S = 0$ low-spin and $S = 2$ high-spin states. The situation as a whole may be summed up as follows [5.17]: strong-field ligands lead to five-coordinate or six-coordinate low-spin compounds. In five-coordinate low-spin compounds the iron atom is pushed away from the basal plane by ca. 0.2 Å whereas in six-coordinate high-spin compounds it remains located within this basal plane. The Fe—N bond lengths involving porphyrinato nitrogen atoms are on the order of 1.99 ± 0.02 Å. [Fe(TPP)(NO)] is an example of a five-coordinate [5.28], and [Fe(TPP)(CO)(py)] or [Fe(TPP)(NO)(1-meim)] of a six-coordinate [5.29, 5.30] compound with an $S = 0$ ground state (1-meim = 1-methylimidazole). Weak-field ligands lead to high-spin compounds, which may be five- or six-coordinate. The Fe—N bond lengths are larger (ca. 2.08 ± 0.02 Å) as is the extrusion of the metal from the porphyrinato plane for five-coordinate derivatives. [Fe(TPP)(2-meim)] is an example of a five-coordinate [5.31] and [Fe(TPP)$(THF)_2$] of a six-coordinate [5.32] high-spin compound (2-meim = 2-methylimidazole; THF = tetrahydrofuran). Finally, $S = 1$ intermediate-spin state has been reported for the planar compound, without an apical ligand, Fe(TPP) [5.33].

The best analysis of the magnetic behavior of Fe(TPP) has been proposed by Mitra et al. [5.34]. At room temperature χT is equal to 2.20 cm^3 K mol^{-1}, which is twice as much as would be expected for a triplet state without orbital contribution. When T is lowered, χT decreases, first very smoothly, and then more and more rapidly below ca. 80 K, and eventually tends to zero when T approaches absolute zero. This low-temperature behavior is indicative of large zero-field splitting within the triplet state with a nonmagnetic level lying the lowest. The magnetic anisotropy curve $(\chi_x - \chi_z) = f(T)$ has also been recorded. χ_x is larger than χ_z in the whole temperature range, and the anisotropy increases as T is lowered. The important point is that the

principal magnetic susceptibilities χ_x and χ_z do not follow Eqs. (2.5.15) and (2.5.19) valid for an $S = 1$ orbital singlet split in zero-field. Mitra et al. reached a satisfactory explanation of the magnetic data from the following hypotheses: At least three $S = 1$ states strongly couple through spin-orbit coupling. These states along with the configuration from which they arise are $^3A_2(a_1^2b_2^2e^2)$, $^3E(a_1b_2^2e^3)$ and $^3B_2(a_1b_2e^4)$ in ascending order of energy, the order of the one-electron orbitals being $a_1(z^2) < e(yz, zx) < b_2(xy) << b(x^2 - y^2)$. The strong stabilization of the a_1 orbital is obviously due to the fact that there is no apical ligand. It is worth noting that the pronounced mixing of the $^3A_2$ ground state with the closely lying ^3E state, which explains the large orbital contribution to the magnetic susceptibility, has also been deduced from a proton magnetic resonance study [5.35–5.37].

5.5 Spin Transition and Spin Equilibrium Involving an Intermediate-Spin State

For all the intermediate-spin or spin-admixed state compounds mentioned so far in this chapter, a single spin state was thermally populated in the temperature range of interest, i.e., up to 300 K. In addition to those compounds there are a few species in which a spin transition or a spin equilibrium between a low-spin (or a high-spin) and an intermediate-spin state was characterized. The first report of this kind concerns [Fe(salen)(NO)] with salen = *N*,*N*′-ethylenebis(salicylideneiminato) [5.38, 5.39]. The molecular structure of [Fe(salen)(NO)] was determined both at 296 and 98 K [5.40]. The results are shown in Figure 5.6. There are only marginally significant structural changes between the two structures although the compound presents a spin transition at 175 ± 3 K (vide infra). The iron(III) ion is in square pyramidal environment with the oxygen and nitrogen coordinating atoms of salen in the basal plane, and the NO group in the apical position. The Fe—N—O linkage is bent, with an angle of 141° above the transition and 127° below. The temperature dependence of χT for [Fe(salen)(NO)] shows a rather abrupt transition around $T_c = 175$ K occurring over a temperature interval of ca. 8 K (see Figure 5.7); χT is equal to 1.7 cm^3 K mol^{-1} above T_c and to 0.5 cm^3 K mol^{-1} below T_c. These values indicate that the transition occurs between $S = 1/2$ and $S = 3/2$ states [5.41]. The variation of the EPR spectra versus temperature recorded on a powder sample is consistent with such a spin transition. At room temperature the spectrum shows a broad signal covering the $g = 2$–4 range. This signal is the envelope of the two transitions at $g = 2$ and $g = 4$, respectively, expected for the $M_S = \pm 1/2$ Kramers doublet arising from the zero-field splitting within the $S = 3/2$ state. The other Kramers doublet, $M_S = \pm 3/2$, is nonresonant. When cooling down and passing through the transition, the component at $g = 4$ vanishes, and at temperatures below 170 K only a sharp signal at $g = 2$ is observed, corresponding to the $S = 1/2$ state. At 77 K the anisotropy of the **g**-tensor associated with the $S = 1/2$ state is resolved with $g_z = 2.133$ and $g_x = 2.036$. The spin transition can also be followed by Mössbauer spectroscopy. Around T_c a superposition of two spectra belonging to

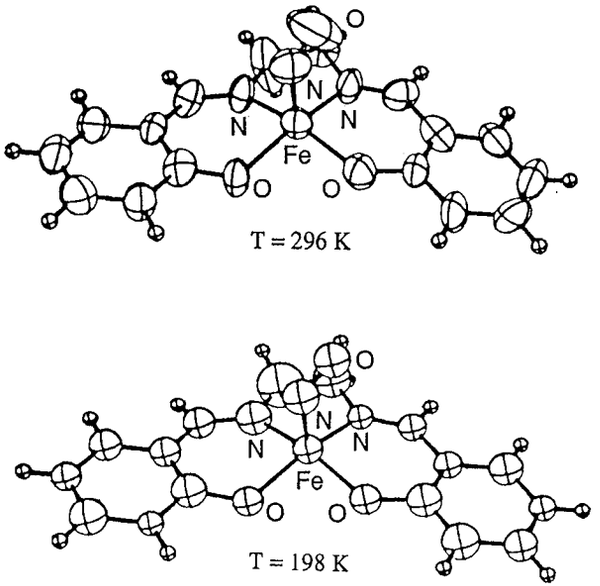


Figure 5.6. Molecular structure of [Fe(salen)(NO)] with salen = *N,N'*-ethylenebis(salicylideneiminato) at 296 and 198 K, i.e., above and below the S = 3/2 ↔ *S* = 1/2 spin transition, respectively. The position of the oxygen atom of NO is disordered; only one of the two possible positions is indicated (from ref. [5.40]).

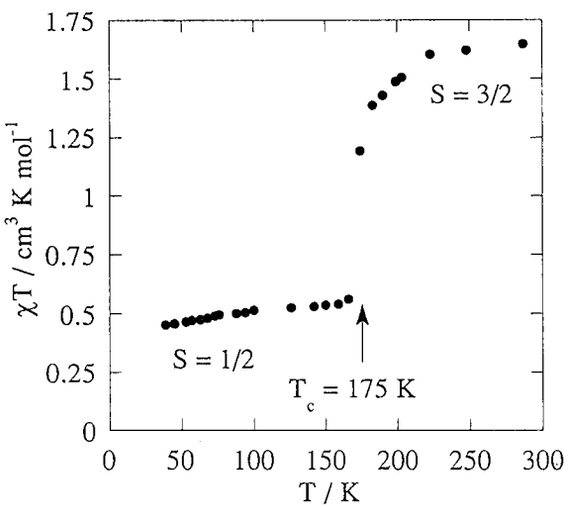
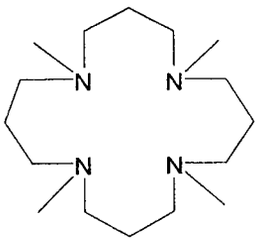


Figure 5.7. Experimental χT *versus* T variation for [Fe(salen)(NO)] with salen = *N,N'*-ethylenebis(salicylideneiminato) (from ref. [5.41]).

the $S = 1/2$ and $S = 3/2$ states, respectively, is seen, which indicates that fluctuation rates are slow on the Mössbauer time scale.

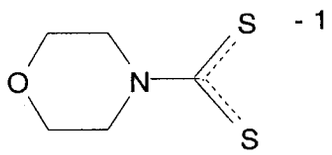
Several other NO complexes with substituted salen rings were also investigated. For [Fe(4-Cl-salen)(NO)] the high- and low-temperature limit values of χT are much the same as for the unsubstituted species, but the decrease of χT when T is lowered is gradual. This behavior is consistent with a spin equilibrium between a ground $S = 1/2$ state and a thermally populated excited $S = 3/2$ state. If so, the population of the two low-lying states varies with temperature according to the Boltzmann law. For [Fe(5-Cl-salen)(NO)], on the other hand, only the $S = 3/2$ state is thermally populated throughout the temperature range.

Another case of spin equilibrium between $S = 1/2$ and $S = 3/2$ states has been characterized [5.42]. It deals with the compound [Fe(TMC)(NO)]$(BF_4)_2$ (TMC stands for 1,4,8,11-tetramethyl-1,4,8,11-tetraazacyclotetradecane). This macrocyclic tetramine shown below binds the iron(III) ion in such a way that the four *N*-methyl substituents lie on the same side of the plane formed by the four nitrogen atoms, which favors the formation of five-coordinate species. The NO group occupies the apical



position. An investigation of the magnetic properties of [Fe(TMC)(NO)]$(BF_4)_2$ shows a continuous decrease of χT as the temperature is lowered, with $\chi T = 1.7$ cm^3 K mol^{-1} at 286 K and 0.9 cm^3 K mol^{-1} at 4.2 K. This behavior is actually very similar to that observed for [Fe(4-Cl-salen)(NO)] and is attributed to an $S = 1/2 - S = 3/2$ spin equilibrium. The Mössbauer spectra at various temperatures show a single quadrupole-split doublet. Both the isomer shift and the quadrupole splitting are temperature dependent, confirming that more than one spin state is thermally populated.

At least one convincing example of an $S = 3/2 - S = 5/2$ spin equilibrium has been reported [5.43]. It concerns the dichlorodimethane monosolvate of tris(4-morpholinecarbodithioato-S,S')iron(III). 4-morpholinecarbodithioato is a bidentate monoanion:



In this complex the iron(III) ion is in a distorted octahedral environment of six sulfur atoms. The study of the magnetic properties reveals a continuous decrease of χT as T is lowered from $\chi T = 3.7$ cm^3 K mol^{-1} at 300 K down to 1.75 cm^3 K mol^{-1} at 5 K. This latter value is much too high for an $S = 1/2$ state, but is close to what is expected for an $S = 3/2$ state. Temperature dependences of EPR and Mössbauer spectra are consistent with such a spin equilibrium involving quartet and sextet spin states [5.44]. Interestingly, the unsolvated derivative exhibits a much more common $S = 1/2 - S = 5/2$ spin equilibrium.

References

5.1. L. Pauling, C.D. Coryell, *Proc. Natl. Acad. Sci. U.S.A.* 22 (1936), 210.

5.2. C.D. Coryell, F. Stitt, L. Pauling, *J. Am. Chem. Soc.* 59 (1937), 633.

5.3. J.S. Griffith, *Proc. R. Soc. Ser. A* 235 (1956), 23.

5.4. M.M. Maltempo, T.H. Moss, *Q. Rev. Biophys.* 9 (1976), 181.

5.5. W.R. Scheidt, S.R. Osvath, Y.J. Lee, C.A. Reed, B. Shaevitz, G.P. Gupta, *Inorg. Chem.* 28 (1989), 1591.

5.6. B.F. Hoskins, R.L. Martin, A.H. White, *Nature (London)* 211 (1966), 627.

5.7. B.F. Hoskins, A.H. White, *J. Chem. Soc. A* (1970), 1668.

5.8. P.C. Healy, A.H. White, B.F. Hoskins, *J. Chem. Soc. Dalton Trans.* (1972), 1369.

5.9. G.E. Chapps, S.W. McCann, H.H. Wickman, R.C. Sherwood, *J. Chem. Phys.* 60 (1974), 990.

5.10. H.H. Wickman, A.M. Trozzolo, H.J. Williams, G.W. Hull, F.R. Merritt, *Phys. Rev.* 155 (1967), 563.

5.11. R.L. Martin, A.H. White, *Inorg. Chem.* 6 (1967), 712.

5.12. P. Ganguli, V.R. Marathe, S. Mitra, R.L. Martin, *Chem. Phys. Lett.* 26 (1974), 529.

5.13. P. Ganguli, R. Marathe, S. Mitra, *Inorg. Chem.* 14 (1975), 970.

5.14. G.C. Brackett, P.L. Richards, H.H. Wickman, *Chem. Phys. Lett.* 6 (1970), 75.

5.15. G.C. Brackett, P.L. Richards, W.S. Caughy, *J. Chem. Phys.* 54 (1971), 4383.

5.16. H.H. Wickman, *J. Chem. Phys.* 56 (1972), 976.

5.17. W.R. Scheidt, C.A. Reed, *Chem. Rev.* 81 (1981), 543.

5.18. W.R. Scheidt, K.J. Haller, K. Hatano, *J. Am. Chem. Soc.* 102 (1980), 3017.

5.19. D.M. Collins, R. Countryman, J.L. Hoard, *J. Am. Chem. Soc.* 94 (1972), 2066.

5.20. W.R. Scheidt, I.A. Cohen, M.E. Kastner, *Biochemistry* 18 (1979), 3546.

5.21. J.L. Hoard, G.H. Cohen, M.D. Glick, *J. Am. Chem. Soc.* 89 (1967), 1992.

5.22. C.A. Reed, T. Mashiko, S.P. Bentley, M.E. Kastner, W.R. Scheidt, K. Spartalian, G. Lang, *J. Am. Chem. Soc.* 101 (1979), 2948.

5.23. H. Masuda, T. Taga, K. Osaki, H. Sugimoto, Z. Yoshida, H. Ogoshi, *Inorg. Chem.* 19 (1980), 950.

5.24. S. Mitra, V.R. Marathe, R. Birdy, *Chem. Phys. Lett.* 96 (1983), 103.

5.25. R. Birdy, D.V. Behere, S. Mitra, *J. Chem. Phys.* 78 (1983), 1453.

5.26. K. Spartalian, G. Lang, C.A. Reed, *J. Chem. Phys.* 71 (1979), 1832.

5.27. B.J. Kennedy, K.S. Murray, P.R. Zwack, H. Homborg, W. Kalz, *Inorg. Chem.* 25 (1986), 2539.

5.28. W.R. Scheidt, M.E. Frisse, *J. Am. Chem. Soc.* 97 (1975), 17.

5.29. W.R. Scheidt, P.L. Piciulo, *J. Am. Chem. Soc.* 98 (1976), 1913.

5.30. S.M. Peng, J.A. Ibers, *J. Am. Chem. Soc.* 98 (1976), 8032.

5.31. J.P. Collman, N. Kim, J.L. Hoard, G. Lang, L.J. Radonovitch, C.A. Reed, "Abstracts of Papers." 167th National Meeting of the American Chemical Society, Los Angeles, CA, April 1974.

5.32. C.A. Reed, T. Mashiko, W.R. Scheidt, K. Spartalian, G. Lang, *J. Am. Chem. Soc.* 102 (1980), 2302.

5.33. J.P. Collman, J.L. Hoard, N. Kim, G. Lang, C.A. Reed, *J. Am. Chem. Soc.* 97 (1975), 2676.

5.34. P.D.W. Boyd, D.A. Buckingham, R.F. McMeeking, S. Mitra, *Inorg. Chem.* 18 (1979), 3585.

5.35. H. Goff, G.N. LaMar, C.A. Reed, *J. Am. Chem. Soc.* 99 (1977), 3641.

5.36. J. Mispelter, M. Momenteau, J.M. Lhoste, *Mol. Phys.* 33 (1977), 1715.

5.37. J. Mispelter, M. Momenteau, J.M. Lhoste, *J. Chem. Phys.* 72 (1980), 1003.

5.38. A. Earnshaw, E.A. King, L.F. Larkworthy, *Chem. Commun.* (1965), 180.

5.39. A. Earnshaw, E.A. King, L.F. Larkworthy, *J. Chem. Soc. A* (1969), 2459.

5.40. K.J. Haller, P.L. Johnson, R.D. Feltham, J.H. Enemark, J.R. Ferraro, L.J. Basile, *Inorg. Chim. Acta* 33 (1979), 119.

5.41. F.V. Wells, S.W. McCann, H.H. Wickman, S.L. Kessel, D.N. Hendrickson, R.D. Feltham, *Inorg. Chem.* 21 (1982), 2306.

5.42. K.D. Hodges, R.G. Wollmann, S.L. Kessel, D.N. Hendrickson, D. G. Van Derveer, E.K. Barefield, *J. Am. Chem. Soc.* 101 (1979), 906.

5.43. R.J. Butcher, E. Sinn, *J. Am. Chem. Soc.* 98 (1976), 2440.

5.44. D.P. Rininger, N.V. Duffy, R.C. Weir, E. Gelerinter, J. Stanford, D.L. Uhrich, *Chem. Phys. Lett.* 52 (1977), 102.

CHAPTER

6

Isotropic Interaction in Dinuclear Compounds

There is no doubt that the most important developments in molecular magnetism in the last two decades have concerned compounds where several magnetic centers interact. These magnetic moment carriers may be transition metal ions, rare earths, or organic radicals. This and all the following chapters will be devoted to the various problems related to the interaction between magnetic centers within molecular units. We will first study the physics of polymetallic systems in this chapter and in Chapter 7, and then we will treat the orbital mechanism of the interaction between magnetic centers. This chapter deals with molecular entities containing only two magnetic centers and focuses on the dominant phenomenon, namely the isotropic interaction.

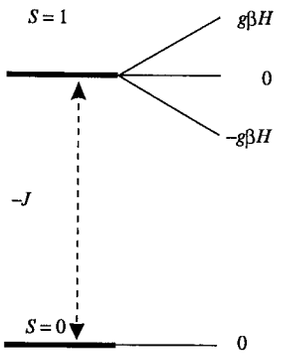
6.1 Copper(II) Dinuclear Compounds

The most thoroughly investigated dinuclear compounds by far are those involving copper(II) ions, where the interaction occurs between two local doublet states. Let us consider a compound of this nature, in which two copper(II) ions, noted A and B, in the same molecular entity are bridged by a diamagnetic ligand capable of transmitting the electronic effects between A and B. If the two metal ions interact through the bridge, then the local spins $S_A = S_B = 1/2$ are not good quantum numbers. The good spin quantum numbers are $S = 0$ and 1. Let $E(S = 0)$ and $E(S = 1)$ be the energies of the two pair states. Due to electrostatic reasons studied in detail in chapter 8, $E(S = 0)$ and $E(S = 1)$ are in general not equal, but separated by a J energy gap defined as

$$J = E(S = 0) - E(S = 1) \tag{6.1.1}$$

J is often referred to as the isotropic interaction parameter. When the state $S = 0$ is the ground state, the interaction is said to be antiferromagnetic; J is then negative. When the state $S = 1$ is the ground state, the interaction is said to be ferromagnetic and J is positive. This latter situation is much less frequent than the former. About 95% of copper(II) dinuclear compounds present an antiferromagnetic interaction.

In the absence of any other phenomenon, such as those studied in Section 6.5 and in Chapter 7, the calculation of the magnetic susceptibility is straightforward. The Zeeman perturbation does not affect the singlet state and splits the components of the triplet state as



If the energy of the triplet state is taken as the origin, the $E_n^{(0)}$ and $E_n^{(1)}$ coefficients intervening in Eq. (1.4.11) are

n	$E_n^{(0)}$	$E_n^{(1)}$
1	0	$-g\beta$
2	0	0
3	0	$+g\beta$
4	J	0

and the magnetic susceptibility is given by

$$\chi = \frac{2Ng^2\beta^2}{kT[3 + \exp(-J/kT)]} \qquad (6.1.3)$$

This expression (6.1.3) was derived for the first time by Bleaney and Bowers [6.1]. The χ versus T plot is shown in Figure 6.1, for $J = 0$ and $\pm$ 100 cm^{-1}. For $J < 0$, the magnetic susceptibility presents a maximum, and then tends to zero when T approaches zero. Indeed, at low temperature, only the diamagnetic ground state is thermally populated. This maximum in χ is a signature of antiferromagnetic interaction. The temperature T_{max} for which it occurs is related to J through

$$|J|/kT_{max} = 1.599 \qquad (6.1.4)$$

For instance, for $J = -100$ cm^{-1}, T_{max} is observed at 90 K since k is equal to 0.695 cm^{-1} K^{-1} (see Appendix 1). For $J > 0$, on cooling, χ increases faster than in C/T. We notice that the difference between the $\chi = f(T)$ curves for $J = 0$ and $J = 100$ cm^{-1} is weakly pronounced. The best way to reveal a ferromagnetic interaction is to plot χT

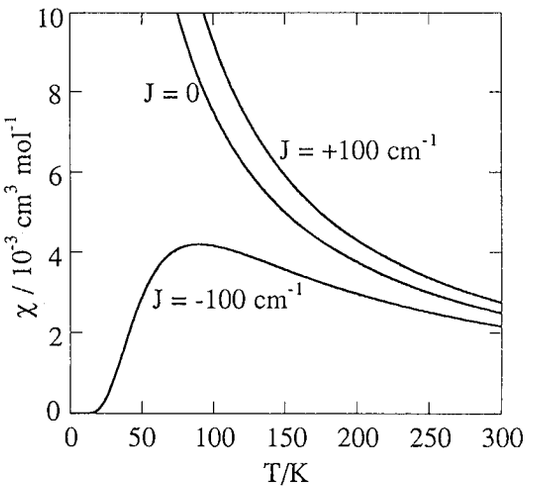


Figure 6.1. χ versus T curves for a copper(II) dinuclear compound with $J = 0$ and ± 100 cm^{-1}. For the sake of simplicity the Zeeman factor is taken to be equal to 2.

versus T. For $J = 0$, χT is constant and equal to $Ng^2\beta^2/2k$. For $J > 0$, χT is close to $Ng^2\beta^2/2k$ when $kT >> J$. On cooling, χT increases, due to the depopulation of the diamagnetic excited state in favor of the triplet ground state, and tends to a plateau with $\chi T = 2Ng^2\beta^2/3k$ corresponding to the temperature range where the excited singlet state is fully depopulated. The ratio $(\chi T)_{LT}/(\chi T)_{HT}$ between the low- and high-temperature limits is equal to 4/3. Of course, for $J < 0$, χT continuously decreases upon cooling. The χT versus $kT/|J|$ plots for J positive, negative or zero are compared in Figure 6.2.

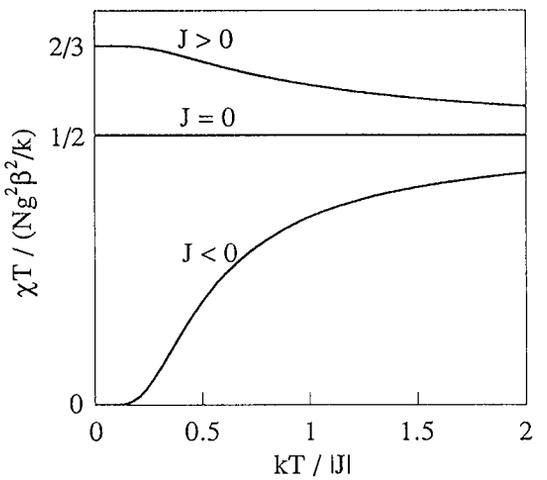


Figure 6.2. χT (in $Ng^2\beta^2/k$ units) versus $kT/|J|$ curves for a copper(II) dinuclear compound with J positive, negative, or zero.

The isotropic interaction phenomenon is purely electrostatic in nature, as discussed in Chapter 8. However, it is often formally described by a coupling between the local spin operators $\mathbf{S}_A$ and $\mathbf{S}_B$. The phenomenological Hamiltonian accounting for this may be written as

$$\mathbf{H} = -J\,\mathbf{S}_A \cdot \mathbf{S}_B \tag{6.1.5}$$

Such a phenomenological description of the isotropic interaction was introduced first by Heisenberg, then discussed by Dirac and Van Vleck. **H** in (6.1.5) is generally called the Heisenberg–Dirac–Van Vleck (HDVV) Hamiltonian. Since

$$\mathbf{S} = \mathbf{S}_A + \mathbf{S}_B \tag{6.1.6}$$

hence

$$\mathbf{S}^2 = \mathbf{S}_A^2 + \mathbf{S}_B^2 + 2\mathbf{S}_A \cdot \mathbf{S}_B \tag{6.1.7}$$

H may be rewritten as

$$\mathbf{H} = -\frac{J}{2}(\mathbf{S}^2 - \mathbf{S}_A^2 - \mathbf{S}_B^2) \tag{6.1.8}$$

the eigenvalues of which are

$$E(S, S_A, S_B) = -\frac{J}{2}[S(S+1) - S_A(S_A+1) - S_B(S_B+1)] \tag{6.1.9}$$

which, after a change of origin, can be rewritten

$$E(S) = -\frac{J}{2}S(S+1) \tag{6.1.10}$$

such that Eq. (6.1.1) is obeyed. It is important to stress that the HDVV Hamiltonian is easy to handle. Moreover, as we will see in the next section, it can be generalized. However, it is purely phenomenological, and does not provide any information on the real mechanism of the isotropic interaction.

An enormous amount of work has been devoted to copper(II) dinuclear compounds and our goal is not to write a review here. Several review papers have been published to which the reader can refer [6.2–6.7]. We limit ourselves to presenting a couple of examples. Other important examples will be discussed in Chapter 8, which is devoted to the orbital mechanism of the isotropic interaction.

The magnetic interaction phenomenon within a molecule was discovered in 1951 by Guha [6.8], then by Bleaney and Bowers [6.1], on a compound known at that time as copper(II) acetate monohydrate. Guha found that the magnetic susceptibility exhibits a maximum as a function of temperature; Bleaney and Bowers observed that the EPR spectra resemble those of triplet states rather than doublet states expected for noninteracting copper(II) ions. Furthermore, the intensity of the signal also passes through a maximum around 270 K and rapidly decreases as the temperature is lowered. The molecule is actually dimeric, as shown in Figure 6.3. The copper(II) ions are in square pyramidal environments, with oxygen atoms from acetato groups in the basal

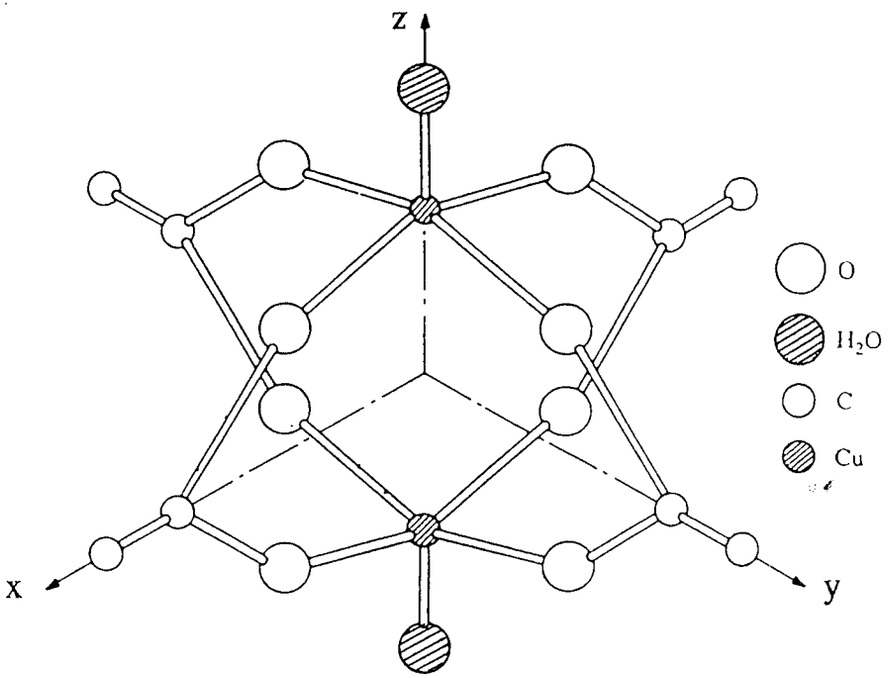


Figure 6.3. Molecular structure of diaquatetra-μ-acetato-dicopper(II) (from ref. [6.3]).

planes and water molecules occupying the apical positions. The metal atoms are bridged by four acetato groups, the two basal planes being parallel, so the molecular symmetry is very close to D_{4h}. The Cu...Cu separation is equal to 2.64 Å. The χ versus T plot for a sample of diaquatetra-μ-acetato-dicopper(II) $Cu_2(CH_3COO)_4(H_2O)_2$, recrystallized twice, is shown in Figure 6.4. As described, χ presents a broad maximum around 265 K and becomes almost negligible below 100 K. On cooling further, χ increases again. This effect, which is often observed for polymetallic compounds with a singlet ground state, is not intrinsic to the compound but is due to the presence of a very small amount of noncoupled species in the sample. To account for this effect, Eq. (6.1.3) may be corrected as follows:

$$\chi = \frac{2Ng^2\beta^2}{kT[3 + \exp(-J/kT)]}(1-\rho) + \frac{Ng^2\beta^2}{2kT}\rho \tag{6.1.11}$$

in which ρ is the molar fraction of noncoupled species. In (6.1.11) it is assumed that the "impurity" follows the Curie law and has the same molecular weight and the same g-factor as the actual compound. Least-squares fitting of the data of Figure 6.4 with the corrected equation (6.1.11) leads to $J = -296\ \text{cm}^{-1}$, $g = 2.09$, and $\rho = 0.0085$. The singlet-triplet energy gap J has been measured directly by inelastic neutron scattering

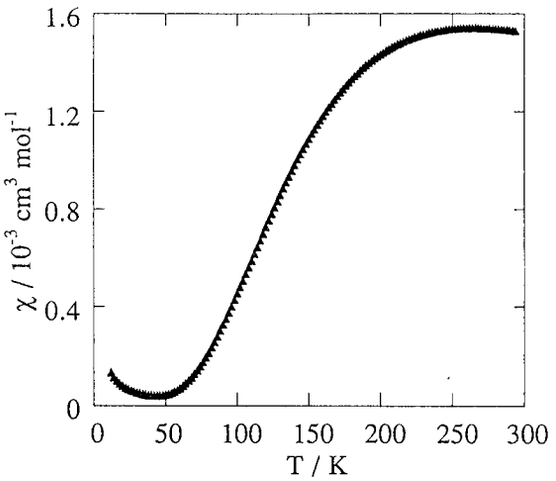


Figure 6.4. χ versus T plot for diaquatetra-μ-acetato-dicopper(II).

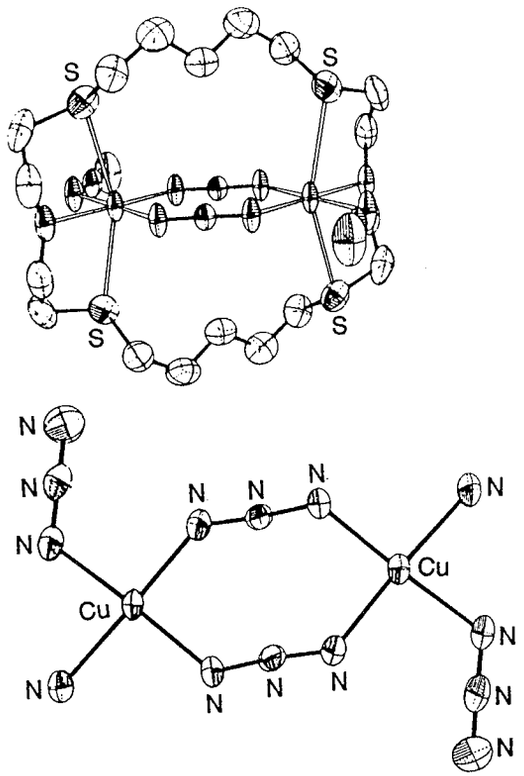
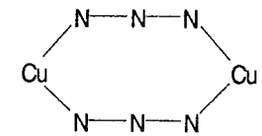


Figure 6.5. Molecular structure of the inclusion complex $[Cu(N_3)_2(C_{18}H_{38}N_2S_4)]$ with two bridging and two terminal azido groups (from ref. [6.13]).

on a deuterated sample $Cu_2(CD_3COO)_4(D_2O)_2$ by Güdel et al. and found to be equal to $-298(4)$ cm^{-1} [6.9, 6.10].

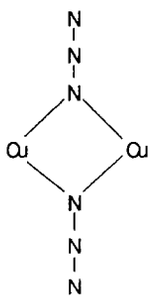
Our second example deals with azido-bridged copper(II) dinuclear compounds, whose magnetic properties present a rather intriguing versatility [6.11, 6.12]. We will consider three compounds, successively, with different types of bridging networks. The first one is represented in Figure 6.5. The planar network



is contained within a cryptate entity. The two metal ions are separated by 5.145 Å. In spite of this large separation the metal ions are so strongly coupled in an antiferromagnetic fashion that the compound is diamagnetic. Even at room temperature the triplet excited state is totally depopulated. Its energy with respect to the singlet ground state should be larger than 10^3 cm^{-1}. All the copper(II) dinuclear compounds with symmetrical end-to-end azido bridges present such a strong antiferromagnetic interaction [6.13–6.15].

The second azido bridged compound is shown in Figure 6.6. Each copper(II) ion is surrounded by five nitrogen atoms, two from the terminal ligand tmen = N,N,N′,N′-tetramethylethylenediamine and three from azido groups, which occupy the vertices of a slightly distorted square-based pyramid. Two azido groups bridge in an end-to-end fashion between the apical positions of one copper atom and the basal plane of the adjacent copper atom. The short and long Cu—N bond lengths in the bridges are equal to 1.979 and 2.456 Å, respectively, and the Cu...Cu separation is 5.004 Å. The magnetic and EPR properties of this compound have not detected any interaction between the copper(II) ions. In the 4.2–300 K temperature range the magnetic susceptibility follows the Curie law expected for two noncoupled copper(II) ions with $\chi T = 0.85(2)$ cm^3 K mol^{-1}. In the same way, the EPR spectrum is typical of a noncoupled copper(II) ion in an elongated tetragonal environment [6.16].

The third compound is $[Cu_2(t\text{-bupy})_4(N_3)_2](ClO_4)_2$ where t-bupy stands for 4-ter-butylpyridine. The dinuclear cation shown in Figure 6.7 contains the planar network:



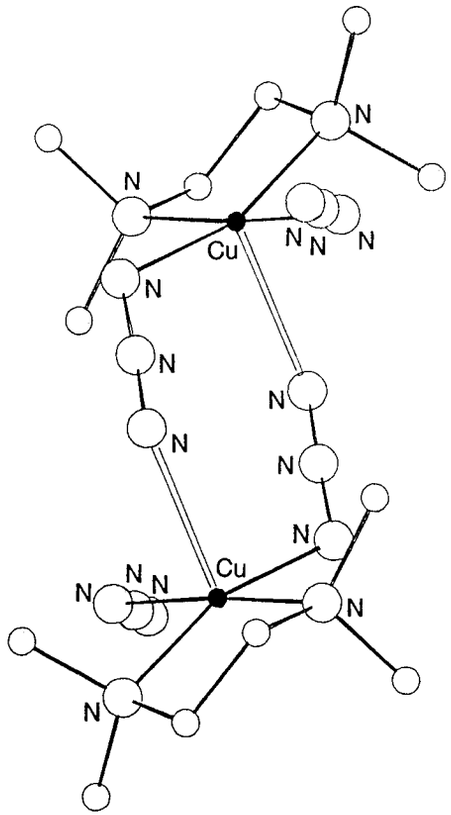


Figure 6.6. Molecular structure of di-μ-azido-bis[[*N*,*N*,*N*′,*N*′-tetramethylethylenediamine-azido-copper(II)] (from reference [6.16]).

with two end-on azido bridges. The Cu...Cu separation is 3.045 Å. The magnetic and EPR properties of this compound are typical of ferromagnetically coupled copper(II) species. The χT versus T plot is given in Figure 6.8. At room temperature χT is equal to 0.92 cm^3 K mol^{-1}, a value that is already higher than expected for two noncoupled copper(II) ions; χT increases on cooling and reaches a plateau below 60 K with χT = 1.09(2) cm^3 K mol^{-1}. The ground triplet-excited singlet separation as deduced from the magnetic susceptibility data and Eq. (6.1.3) is found to be equal to J = 110(20) cm^{-1} [6.17]. We must point out here that the determination of J from the magnetic data is much more accurate for $J < 0$ than for $J > 0$. When $J < 0$, χ exhibits a maximum at a temperature directly related to the J value (see Table 6.1) whereas, when $J > 0$, both χ and χT vary monotonically as a function of T. Moreover, the derivative of Eq. (6.1.3) with respect to J,

$$d\chi/dJ = \frac{2Ng^2\beta^2}{k^2T^2}\frac{\exp(-J/kT)}{[3+\exp(-J/kT)]^2} \qquad (6.1.12)$$

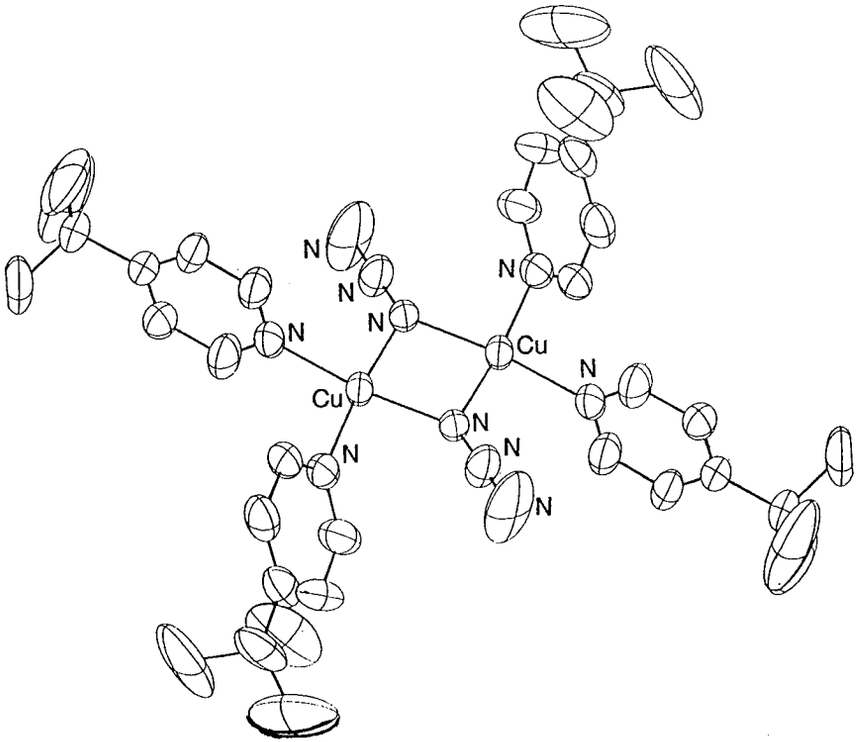


Figure 6.7. Molecular structure of the dication di-μ-azido-tetrakis(4-ter-butylpyridine)dicopper(II) (from ref. [6.17]).

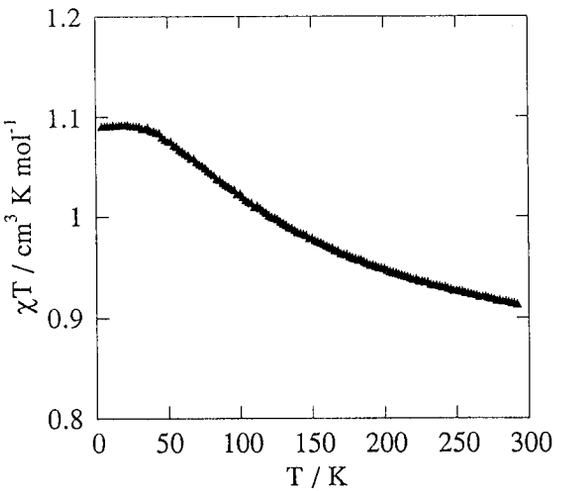


Figure 6.8. χT versus T plot for di-μ-azido-tetrakis(4-ter-butylpyridine)dicopper(II) perchlorate.

Table 6.1. Ratio $|J|/kT_{max}$ Relating the Interaction Parameter J and the Temperature T_{max} for Which a Maximum of χ is Observed for Pairs of Local Spins S_A Antiferromagnetically Coupled

S_A	$\lvert J\rvert/kT_{max}$
1/2	1.599
1	0.976
3/2	0.648
2	0.462
5/2	0.347

is maximum for a negative J value ($J/kT = -1.099$). Both the shape and the temperature dependence of the EPR spectra of $[Cu_2(t\text{-bupy})_4(N_3)_2](ClO_4)_2$ confirm that the ground state is the triplet state [6.18].

6.2 Other Symmetrical Dinuclear Compounds

The HDVV Hamiltonian (6.1.5) describing the isotropic interaction is valid for any pair of interacting magnetic centers with local spins S_A and S_B, provided that the local states have no first-order angular momentum. In this section we consider that the two magnetic centers are identical, with $S_A = S_B$, and symmetry related within the dinuclear unit. The total spin Hamiltonian taking into account the Zeeman perturbation is then

$$\mathbf{H} = -J\mathbf{S}_A\cdot\mathbf{S}_B + \beta(\mathbf{S}_A\cdot\mathbf{g}_A + \mathbf{S}_B\cdot\mathbf{g}_B)\cdot H \tag{6.2.1}$$

where $\mathbf{g}_A$ and $\mathbf{g}_B$ are the local $\mathbf{g}$-tensors, which, in our hypothesis, are equal. Moreover, we assume that they are isotropic with the g principal value. In (6.2.1) we neglect the zero-field splitting (local anisotropy) of interacting ions with $S_A > 1/2$ as well as smaller effects, such as anisotropic interaction, which will be treated in Chapter 7. The pair states are defined by S varying by an integer value from 0 to $2S_A$. The relative energies of these pair states are given by Eq. (6.1.10). The general expression for magnetic susceptibility is then

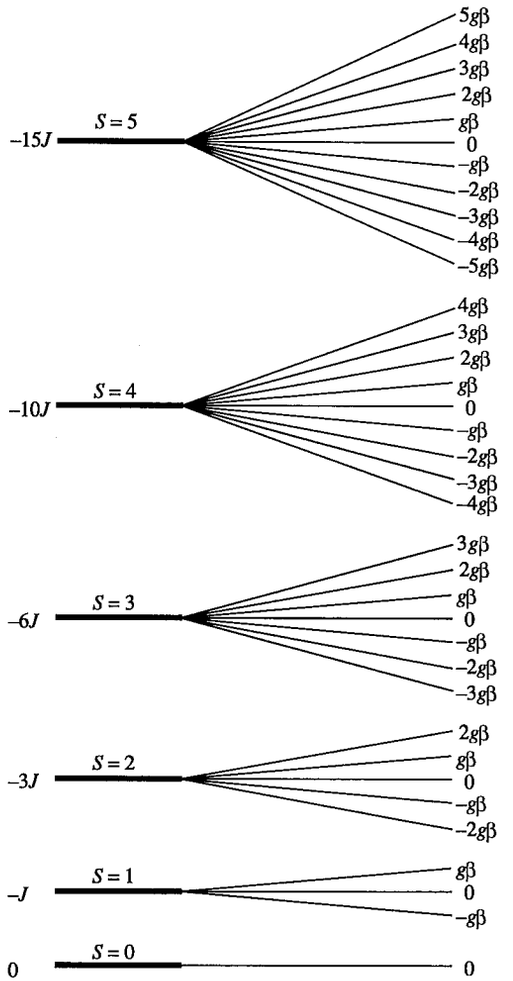
$$\chi = \frac{Ng^2\beta^2}{kT}\frac{\sum_S \sum_{M_S=-S}^{+S} M_S^2 \exp[-E(S)/kT]}{\sum_S (2S+1)\exp[-E(S)/kT]} \tag{6.2.2}$$

which can be rewritten as

$$\chi = \frac{Ng^2\beta^2}{3kT}\frac{\sum_S S(S+1)(2S+1)\exp[-E(S)/kT]}{\sum_S (2S+1)\exp[-E(S)/kT]} \tag{6.2.3}$$

The variations of χ and χT as a function of T are qualitatively similar to those obtained with copper(II) dinuclear species. When J is negative, the ground state is diamagnetic and χ presents a maximum at a finite temperature T_{max}. The ratio $|J|/kT_{max}$ as a function of local spin S_A (= S_B) is given in Table 6.1.

Let us treat explicitly the case of a pair of local spins $S_A = S_B = 5/2$ [high-spin iron(III) or manganese(II)]. The spectrum of the low-lying states and the first order Zeeman coefficients associated with the Zeeman perturbation are as follows:



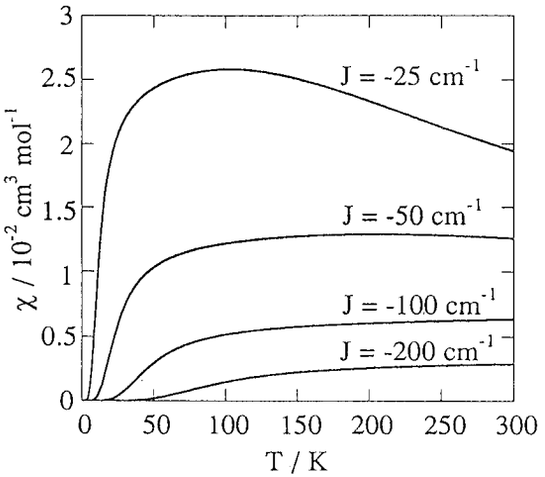


Figure 6.9. χ versus T curves for a high-spin iron(III) dinuclear compound with various values of $J < 0$. The Zeeman factor is taken to be equal to 2.

The molar magnetic susceptibility χ is calculated as

$$\chi = \frac{2Ng^2\beta^2}{kT} \frac{e^x + 5e^{3x} + 14e^{6x} + 30e^{10x} + 55e^{15x}}{1 + 3e^x + 5e^{3x} + 7e^{6x} + 9e^{10x} + 11e^{15x}} \tag{6.2.4}$$

with

$$x = J/kT \tag{6.2.5}$$

The χ versus T curves for various values of $J < 0$ are represented in Figure 6.9. For local spins $S_A = S_B = 2$, the last term in both the numerator ($55e^{15x}$) and the denominator ($11e^{15x}$) of (6.2.4) has to be suppressed. For local spins $S_A = S_B = 3/2$, an additional term at the right-hand side of the numerator and the denominator has to be suppressed, and so on.

We focus on high-spin iron(III) compounds as examples of symmetrical dinuclear units with more than two unpaired electrons. Many systems of this kind have been reported. For examples, we can mention the μ-oxo derivatives with the Fe—O—Fe linkage [6.19, 6.20]. The structure of one of these, $[Fe(salen)]_2O$, is shown in Figure 6.10. The Fe—O—Fe bridging angle is significantly smaller than 180°, as in most of the μ-oxo iron(III) compounds, being equal to 145° [6.21]. The magnetic susceptibility of $[Fe(salen)]_2O$, corrected for the uncoupled impurity, continuously decreases on cooling from room temperature, characterizing a strong antiferromagnetic interaction. The interaction parameter J deduced from the magnetic data and the theoretical Eq. (6.2.4) was found to be equal to -178 cm^{-1} [6.22]. All the μ-oxo iron(III) compounds display rather similar magnetic properties with $-J$ values in the range

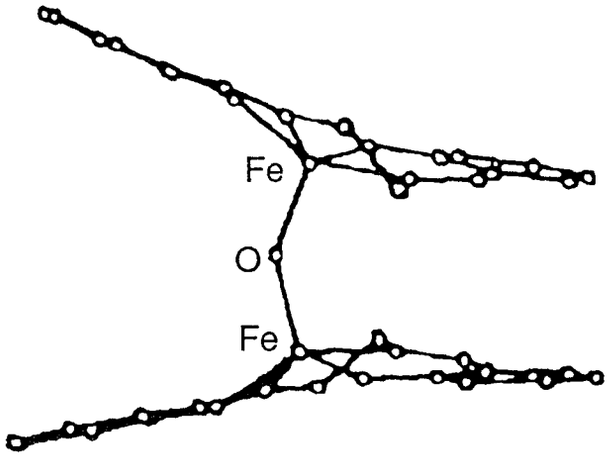


Figure 6.10. Molecular structure of $[Fe(salen)]_2O$ with salen = *N,N'*-ethylenebis(salicylideneiminato) (from ref. [6.21]).

170–230cm^{-1}. In contrast with other series of related compounds (see Section 8.5) J does not seem to be very sensitive to small changes in bridging angle values. In the μ-oxo derivatives the Fe—O—Fe angle may take values between 135° and 175° [6.19, 6.23].

The reaction of $[Fe(salen)]_2O$ with $(Me_3Si)_2S$ in DMF affords the μ-sulfido compound $[Fe(salen)]_2S$. The Fe—S—Fe angle, equal to 121.8°, is substantially smaller than the Fe—O—Fe angle (see Figure 6.11). At equal bridging angles the antiferromagnetic interaction would be expected to be more pronounced in the μ-sulfido than in the μ-oxo derivatives, due to a stronger delocalization of the spin density toward the bridging atom in the sulfur-containing compound. If the bridge X remained the same, then a decrease of the Fe—X—Fe bridging angle should lead to a weaker antiferromagnetic interaction (see Section 8.5). Apparently, in the present case, the two factors almost exactly compensate each other. Indeed J in $[Fe(salen)]_2S$ was reported as -172 cm^{-1} [6.24].

Quite an interesting example of iron dinuclear unit is provided by hemerythrin, an iron-containing respiratory protein found in some marine phila. In the deoxy form both iron centers are in the +2 oxidation state, and the protein may reversibly bind 1 mol of oxygen to form oxyhemerythrin [6.25]. The most stable form of the protein is methemerythrin. The structure of the active site has been determined by X-ray diffraction for both methemerythrin and the azido derivative, and consists of two triply bridged iron(III) ions. One of the bridges is a μ-oxo and the other two are carboxylato groups of amino acid residues, a glutamate and an aspartate. The peripheral coordination sites are occupied by three histidine groups around one of the iron atoms, and only two histidine groups around the other, such that the dinuclear unit is not rigorously symmetrical as shown in Figure 6.12. In azidomethemerythrin N_3^- occupies the vacant

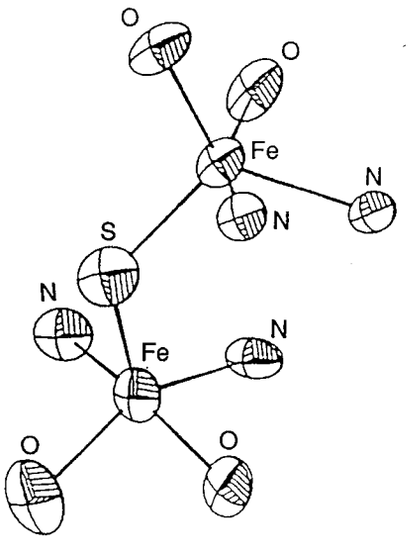
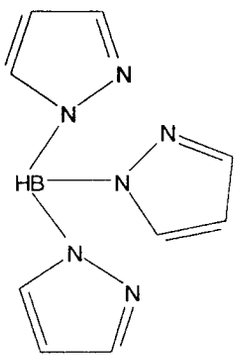


Figure 6.11. Structure of the coordination spheres in $[Fe(salen)]_2S$ (from ref. [6.24]).

site around the second metal ion [6.26]. The magnetic properties of methemerythrin were very carefully studied with an ultrasensitive magnetometer, which revealed a strong antiferromagnetic interaction characterized by $J = -268$ cm^{-1}[6.27]. Two synthetic models of methemerythrin have been synthesized and structurally characterized. One of these is $[(HBpz_3)FeO(CH_3COO)_2Fe(HBpz_3)]$, where $HBpz_3^-$ is the tridentate tris-1-pyrazolylborate anion [6.28, 6.29]:



The structure of this compound is shown in Figure 6.13. The other synthetic model is the dication $[LFeO(CH_3COO)_2FeL]^{2+}$ where L is 1,4,7-triazacyclononane (or 1,4,7-trimethyl-1,4,7-triazacyclononane) [6.30, 6.31]:

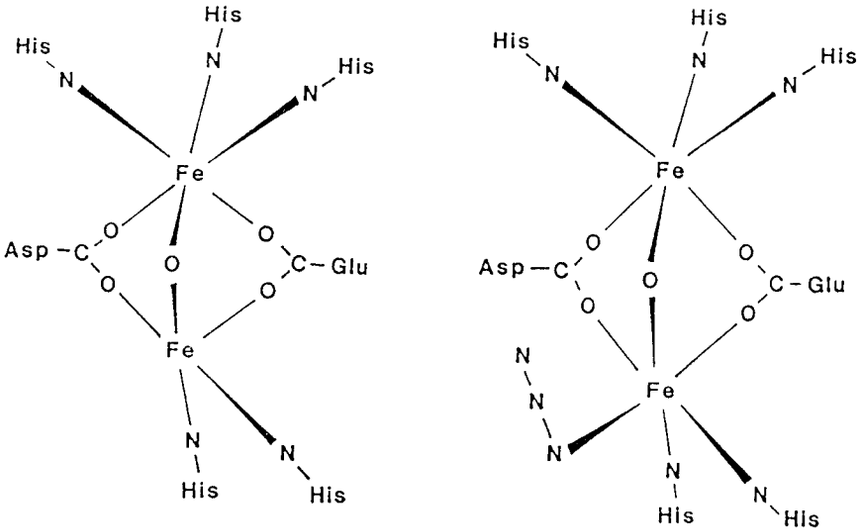
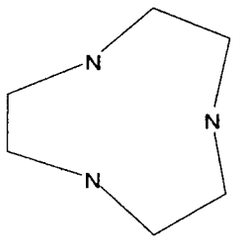


Figure 6.12. Structure of the dinuclear unit in methemerythrine (left) and azidomethemerythrine (right) (from ref. [6.26]).



In both compounds the iron(III) ions are bridged by an oxo and two acetato groups with a $FeO(CH_3COO)_2Fe$ core very similar to that found in azidomethemerythrin. For instance, the average length of the Fe—O bond involving the oxo bridge is 1.784 Å and the Fe—O—Fe angle is 124.6° in $[(HBpz_3)FeO(CH_3COO)_2Fe(HBpz_3)]$. These values are 1.79 Å and 132°, respectively, in azidomethemerytrin. The magnetic properties of both compounds are quite similar. They reveal an antiferromagnetic interaction characterized by $J = -240(2)$ cm^{-1}. In addition to the similarity of their magnetic properties with those of the biological species, both model compounds exhibit ligand-field bands characteristic of all methemerythrin derivatives. In contrast with hemerythrin, however, the two iron(III) coordination spheres in each of the synthetic models are similar; both iron atoms are six-coordinate. The synthetic compounds do not present the specific reactions of the protein and, in particular, they do not reversibly bind oxygen.

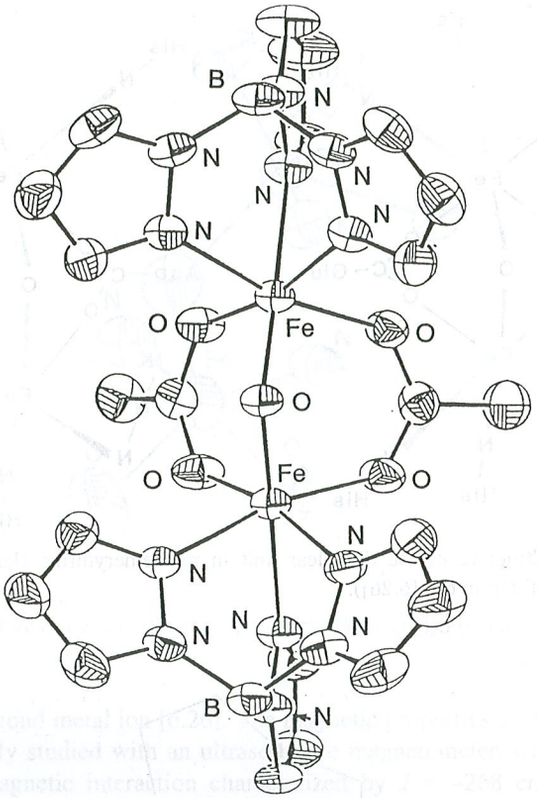
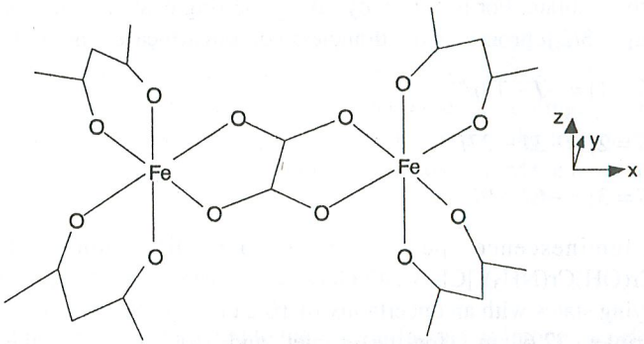


Figure 6.13. Molecular structure of $[(HBpz_3)FeO(CH_3COO)_2Fe(HBpz_3)]$ where $HBpz_3^-$ is the tridentate tris-1-pyrazolylborate anion (from ref. [6.28]).

It is worth mentioning that $[(HBpz_3)FeO(CH_3COO)_2Fe(HBpz_3)]$ can be protonated to afford the μ-hydroxo-di-μ-acetato cation $[(HBpz_3)Fe(OH)(CH_3COO)_2Fe(HBpz_3)]^+$, the structure of which has been solved. The bond lengths in the Fe—O—Fe linkage of this compound are much longer than in the μ-oxo precursor. The average values are 1.954 and 1.784 Å, respectively, whereas the Fe—O—Fe angles are virtually identical. In contrast, the Fe—O bond length involving acetato bridges are much less modified. They are, on average, 0.05 Å shorter in the μ-oxo derivative. The magnitude of the antiferromagnetic interaction is considerably diminished in the μ-hydroxo compound; J is found to be equal to -34 cm^{-1}. The comparison between the two compounds suggests that the interaction between the iron(III) ions is essentially, if not exclusively, mediated by the O^{2-} or OH^- bridges. In the μ-hydroxo compound, owing to the lengthening of the Fe—O bonds in the Fe—O—Fe linkage, the spin density delocalized from the metal towards the bridge is strongly reduced and the antiferromagnetic interaction is much less pronounced [6.32].

The fact that the interaction in μ-oxo-di-μ-acetato iron(III) species is most likely transmitted by the monoatomic bridge O^{2-} does not mean that an extended bridge is unable to transmit a substantial interaction between two high-spin iron(III) ions. Thus, in the μ-oxalato-tetrakis(acetylacetonato) diiron(III) compound



the Fe...Fe separation is larger than 5 Å. However, the χ versus T plot exhibits a maximum around 30 K, corresponding to an interaction parameter $J = -7.22$ cm^{-1} [6.33]. We will study the mechanism of the interaction through extended bridges in Section 8.5. We can state here, however, that bisbidentate ligands like oxalato have a specific ability to transmit the electronic effects between magnetic centers relatively far apart from each other, provided that the *xy*-type interaction pathway is operative (see referential axes above). For high-spin d^5 ions all the *d*-type orbitals are singly occupied, including that of *xy*-symmetry.

In the examples discussed in this section so far the magnetic data have been interpreted by neglecting the zero-field splittings within the excited magnetic states. Such an approximation is quite valid so long as the isotropic interaction is antiferromagnetic and the magnitude of the zero-field splitting is much smaller than that of the isotropic interaction. The EPR spectrum of μ-oxalato-tetrakis(acetylacetonato)diiron(III) dissolved in a 1/1 toluene–chloroform mixture reveals a zero-field splitting of 0.10(6) cm^{-1} between the $M_S = \pm 1$ and the $M_S = 0$ components of the triplet state located at 7.22 cm^{-1} above the singlet ground state. The influence of this small effect on the χ versus T plot is within the experimental uncertainties. Any zero-field splitting effect is obviously without influence on the magnetic susceptibility curve when $-J$ is of the order of 200 cm^{-1}.

To fit the magnetic data it was also assumed that the low-lying states rigorously respect the Lande interval rule:

$$E(S-1) - E(S) = JS \tag{6.2.6}$$

deduced from (6.1.10). This assumption can be not strictly valid for a few compounds. To account for this discrepancy, which is weak in any case, a biquadratic term of the form $j(\mathbf{S_A}\cdot\mathbf{S_B})^2$ may be added to the bilinear term $-J\mathbf{S_A}\cdot\mathbf{S_B}$, j being here a biquadratic interaction parameter. The eigenvalues of the $(\mathbf{S_A}\cdot\mathbf{S_B})^2$ operator are the square of those

of the $\mathbf{S_A}\cdot\mathbf{S_B}$ operator, so that the eigenvalues of the isotropic interaction Hamiltonian including this biquadratic term

$$\mathbf{H} = -J\mathbf{S_A}\cdot\mathbf{S_B} + j(\mathbf{S_A}\cdot\mathbf{S_B})^2 \tag{6.2.7}$$

are easy to calculate. For instance, by taking the singlet state as the energy origin, in the $S_A = S_B = 3/2$ [chromium(III) dinuclear compound] case, one finds

$$E(S=1) = -J - 13j/2$$

$$E(S=2) = -3J - 27j/2$$

$$E(S=3) = -6J - 9j \tag{6.2.8}$$

From luminescence spectroscopy on a modification of the compound $[(NH_3)_5Cr(OH)Cr(NH_3)_5]Cl_5\cdot 3H_2O$ Güdel et al. determined the relative energies of the low-lying states with an uncertainty of ± 0.2 cm^{-1}. These authors found 27.5 cm^{-1} for the triplet, 72.6 cm^{-1} for the quintet, and 151.5 cm^{-1} for the septet states. Least-squares fitting leads to the interaction parameters $J = -26.5$ cm^{-1} and $j = -0.15$ cm^{-1}. In this case the observed energy splitting is very close to a perfect Lande pattern [6.34]. It is clear that such a small j value cannot be determined from magnetic susceptibility data, even if these data are very accurately measured.

6.3 Asymmetrical Dinuclear Compounds

We now suppose that the A and B magnetic centers are not symmetry related. This occurs in heterodinuclear compounds with different S_A and S_B local spins. This may also occur in bimetallic compounds containing metal ions of the same nature but in different environments, or in species containing both a transition ion and an organic radical. As in the preceding section, we neglect local anisotropy and anisotropic interactions, which are treated in Section 6.4 and Chapter 7, respectively. The spin Hamiltonian (6.2.1) is still valid and the pair states are defined by S varying by an integer value from $|S_A - S_B|$ to $S_A + S_B$ with the relative energies given by (6.1.10).

We can note that, whatever S_A and S_B may be, the spin S varies monotonically versus the energy of the pair states. The system is said to present a regular spin state structure. This regularity leads to a simple correspondence between the nature of the interaction and the shape of the χT versus T plot. The high-temperature limit of χT, for $kT >> |J|$, is the sum of what is expected for each of the magnetic centers, i.e.,

$$(\chi T)_{HT} = \frac{N\beta^2}{3k}\left[g_A{}^2 S_A(S_A+1) + g_B{}^2 S_B(S_B+1)\right] \tag{6.3.1}$$

where g_A and g_B denote the principal values of the local **g**-tensors assumed to be isotropic. In the absence of interaction ($J = 0$), χT remains constant over the whole temperature range with a value given in (6.3.1); this is the Curie law. If the interaction is antiferromagnetic ($J < 0$), then the ground state has the smallest spin $|S_A - S_B|$ and the most excited state the highest spin $S_A + S_B$. Between these two limits S increases

with increasing energy. χT continuously decreases on cooling and tends toward the low-temperature limit:

$$(\chi T)_{LT} = \frac{Ng_S^2\beta^2}{3k}\left[(S_A - S_B)^2 + |S_A - S_B|\right] \tag{6.3.2}$$

where g_S ($S = |S_A - S_B|$) is related to the local **g**-factors as indicated below. Only in the $S_A = S_B$ case, is the ground state diamagnetic, and the χ versus T plot exhibits the characteristic maximum. Otherwise, although χT decreases on cooling, χ continuously increases. If the interaction is ferromagnetic, the spectrum of the low-lying states is reversed. χT continuously increases on cooling and tends toward the low-temperature limit:

$$(\chi T)_{LT} = \frac{Ng_S^2\beta^2}{3k}\left[(S_A + S_B)(S_A + S_B + 1)\right] \tag{6.3.3}$$

where g_S ($S = S_A + S_B$) is again related to the local g-factors as indicated below and in Table 6.2.

For asymmetrical dinuclear compounds the $\mathbf{g_A}$ and $\mathbf{g_B}$ local **g**-tensors have no reason to be equal; each $\mathbf{g_S}$ tensor associated with the pair state of spin S has to be calculated as a function of $\mathbf{g_A}$ and $\mathbf{g_B}$. Before presenting the general formula relating the $\mathbf{g_S}$, $\mathbf{g_A}$, and $\mathbf{g_B}$ tensors, it seems worthwhile to detail a specific case, that of a Cu(II)Ni(II) pair with $S_{Cu} = 1/2$ and $S_{Ni} = 1$ local spins. The spin Hamiltonian (6.2.1) becomes here

$$\mathbf{H} = -J\mathbf{S_{Cu}}\cdot\mathbf{S_{Ni}} + \beta(\mathbf{S_{Cu}}\cdot\mathbf{g_{Cu}} + \mathbf{S_{Ni}}\cdot\mathbf{g_{Ni}})\cdot H \tag{6.3.4}$$

For the sake of simplicity, we assume that $\mathbf{g_{Cu}}$ and $\mathbf{g_{Ni}}$ are isotropic with g_{Cu} and g_{Ni} principal values. The isotropic interaction between the two magnetic centers gives rise to two pair states characterized by $S = 1/2$ and $S = 3/2$, and separated by $3J/2$. The local spin functions $|M_S>$ for copper(II) are $|\pm 1/2>$ and for nickel(II) $|\pm 1>$ and $|0>$. The spin functions $|S,M_S>$ for the doublet and quartet pair states are easily obtained from the coupling coefficients given in Appendix 6. They may be written down as linear combinations of the $|M_S(\mathrm{Cu}), M_S(\mathrm{Ni})>$ product functions:

$$|1/2,1/2> = (\sqrt{3}/3)(\sqrt{2}\,|-1/2,1> - |1/2,0>)$$

$$|1/2,-1/2> = (\sqrt{3}/3)(|-1/2,0> - \sqrt{2}\,|1/2,-1>) \tag{6.3.5}$$

and:

$$|3/2,3/2 = |1/2,1>$$

$$|3/2,1/2> = (\sqrt{3}/3)(|-1/2,1> + \sqrt{2}\,|1/2,0>)$$

$$|3/,-1/2> = (\sqrt{3}/3)(\sqrt{2}\,|-1/2,0> + |1/2,-1>)$$

$$|3/2,-3/2> = |-1/2,-1> \tag{6.3.6}$$

Since g_S is expressed through

Table 6.2. Relationships between $\mathbf{g}_S$, $\mathbf{g}_A$, and $\mathbf{g}_B$ Tensors in Asymmetrical Dinuclear Compounds with A and B Magnetic Centers[a]

S_A	S_B	S	$\mathbf{g}_A$	$\mathbf{g}_B$
1/2	1	1/2	−1/3	4/3
		3/2	1/3	2/3
1/2	3/2	1	−1/4	5/4
		2	1/4	3/4
1/2	2	3/2	−1/5	6/5
		5/2	1/5	4/5
1/2	5/2	2	−1/6	7/6
		3	1/6	5/6
1	3/2	1/2	−2/3	5/3
		3/2	4/15	11/15
		5/2	2/5	3/5
1	2	1	−1/2	3/2
		2	1/6	5/6
		3	1/3	2/3
1	5/2	3/2	−2/5	7/5
		5/2	4/35	31/35
		7/2	2/7	5/7
3/2	2	1/2	−1	2
		3/2	1/5	4/5
		5/2	13/35	22/35
		7/2	3/7	4/7
3/2	5/2	1	−3/4	7/4
		2	1/12	11/12
		3	7/24	17/24
		4	3/8	5/8
2	5/2	1/2	−4/3	7/3
		3/2	2/15	13/15
		5/2	12/35	23/35
		7/2	26/63	37/63
		9/2	4/9	5/9

[a]This table must be read as follows (see the first line): for the $S = 1/2$ pair state arising from the interaction between $S_A = 1/2$ and $S_B = 1$ local states, we have $\mathbf{g}_1 = -\mathbf{g}_A/3 + 4\mathbf{g}_B/3$.

$$g_S = M_S < S,M_S \mid g_{Cu}\mathbf{S}_{Cu} + g_{Ni}\mathbf{S}_{Ni} \mid S,M_S> \tag{6.3.7}$$

it results that

$$g_{1/2} = (4g_{Ni} - g_{Cu})/3$$

$$g_{3/2} = (2g_{Ni} - g_{Cu})/3 \tag{6.3.8}$$

It can be also seen that the $M_S = \pm 1/2$ components arising from the doublet and quartet pair states couple through the Zeeman perturbation. Indeed, we have

$$<1/2,\pm 1/2 \mid g_{Cu}\mathbf{S}_{Cu} + g_{Ni}\mathbf{S}_{Ni} \mid 3/2,\pm 1/2> = -\delta\sqrt{2} \tag{6.3.9}$$

with

$$\delta = (g_{Cu} - g_{Ni})/3 \tag{6.3.10}$$

We can now calculate the magnetic susceptibility. Owing to the off-diagonal terms in (6.3.9) the second-order Zeeman coefficients are not all zero. The $E_n^{(0)}$, $E_n^{(1)}$, and $E_n^{(2)}$ coefficients are gathered below:

n	$E_n^{(0)}$	$E_n^{(1)}$	$E_n^{(2)}$
1	0	$-g_{1/2}\beta/2$	$4\delta^2\beta^2/3J$
2	0	$g_{1/2}\beta/2$	$4\delta^2\beta^2/3J$
3	$-3J/2$	$-3g_{3/2}\beta/2$	0
4	$-3J/2$	$-g_{3/2}\beta/2$	$-4\delta^2\beta^2/3J$
5	$-3J/2$	$g_{3/2}\beta/2$	$-4\delta^2\beta^2/3J$
6	$-3J/2$	$3g_{3/2}\beta/2$	0

(6.3.11)

and χ may be written as

$$\chi = \frac{N\beta^2}{4kT}\frac{g_{1/2}^2 + 10g_{3/2}^2\exp(3J/2kT)}{1 + 2\exp(3J/2kT)} - \frac{8\delta^2\beta^2}{3J}\frac{1 - \exp(3J/2kT)}{1 + 2\exp(3J/2kT)} \tag{6.3.12}$$

δ is of the order of 10^{-1} and the latter term on the right-hand side of (6.3.12) is extremely small and actually may be reasonably neglected, except when the doublet and quartet pair states are very close to each other. However, in such a case, the local anisotropy of nickel(II) becomes preponderant and strongly mixes the pair states such that S is no longer a good quantum number.

Using the Wigner–Eckart theorem, Chao [6.35], Bullugiu [6.36–6.37], then Scaringe et al. [6.38] in a more complete manner generalized the relations between local and pair tensors and derived the following expression:

$$\mathbf{g}_S = (1 + c)\mathbf{g}_A/2 + (1 - c)\mathbf{g}_B/2 \tag{6.3.13}$$

with

$$c = \frac{S_A(S_A + 1) - S_B(S_B + 1)}{S(S + 1)} \tag{6.3.14}$$

Since Eq. (6.3.13) is of general application, the relationships between $\mathbf{g}_S$, $\mathbf{g}_A$, and $\mathbf{g}_B$ tensors are tabulated in Table 6.2 for different couples S_A and S_B ($\neq S_A$). For $S_A = S_B$, we have $\mathbf{g}_S = (\mathbf{g}_A + \mathbf{g}_B)/2$ whatever S.

The Cu(II)Ni(II) compounds represent the simplest examples of heterobimetallic systems. Indeed, since one of the interacting ions is a local doublet, the isotropic interaction leads to only two pair states. When the interaction is antiferromagnetic, the ground state is not split in zero-field so that in the temperature range where only this ground doublet state is thermally populated the magnetic susceptibility should follow

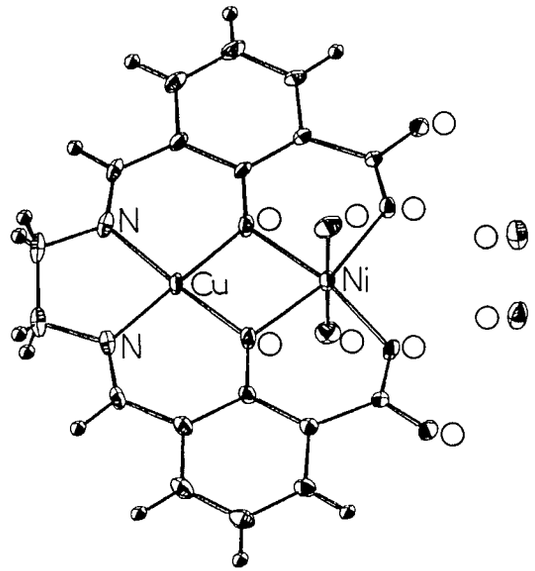


Figure 6.14. Molecular structure of $CuNi(fsa)_2en(H_2O)\cdot H_2O$ with $H_4(fsa)_2en$ = *N,N'*-(2-hydroxy-3-carboxybenzilidene)ethylenediamine (from ref. [6.41]).

the Curie law with $\chi T = Ng_{1/2}{}^2\beta^2/4k$. This allows an accurate determination of the average value $g_{1/2}$. Furthermore, if the doublet-quartet energy gap $3J/2$ is large enough with respect to the zero-field splitting within the excited quartet state, this zero-field splitting will have no influence on the magnetic properties, and Eq. (6.3.12) (with $\delta = 0$) will be valid. One of the most extensively studied Cu(II)Ni(II) compounds is shown in Figure 6.14. Its formula is $CuNi(fsa)_2en(H_2O)_2\cdot H_2O$, where $(fsa)_2en^{4-}$ stands for the dinucleating ligand derived from the Schiff base *N,N'*-(2-hydroxy-3-carboxybenzilidene)ethylenediamine. The copper atom is in a strictly planar environment and the nickel atom is in a pseudo-octahedral environment with two water molecules on each side of the molecular plane [6.39–6.42]. The magnetic data represented in Figure 6.15 are characteristic of an antiferromagnetic interaction. Below 60 K only the doublet ground state is thermally populated, and in the 60–10 K range χT is constant and equal to 0.52 cm^3 K mol^{-1}, corresponding to $g_{1/2} = 2.35$. The slight decrease of χT below 10 K might be due to intermolecular interactions (see Section 6.5). On heating above 60 K population of the excited quartet state begins and χT progressively increases. Least-squares fitting of the magnetic data leads to a doublet-quartet energy gap of $3J/2 = -213$ cm^{-1}, $g_{3/2}$ being equal to 2.23. The EPR spectrum of the compound at low temperature is typical of a doublet state with $g_z = 2.23$ and $g_x = g_y = 2.30$. It is interesting to remark that this doublet pair state cannot be confused with the local doublet state of copper(II). Indeed, for a copper(II) ion in a planar environment, g_z is always larger than g_x and g_y (see Section 6.3). On the contrary, for the Cu(II)Ni(II) pair, g_z is the smallest principal value of $\mathbf{g}_{1/2}$. This difference can be understood by examining the relationship between $\mathbf{g}_{1/2}$, $\mathbf{g}_{Cu}$, and $\mathbf{g}_{Ni}$ in (6.3.8) or Table 6.2. Since the

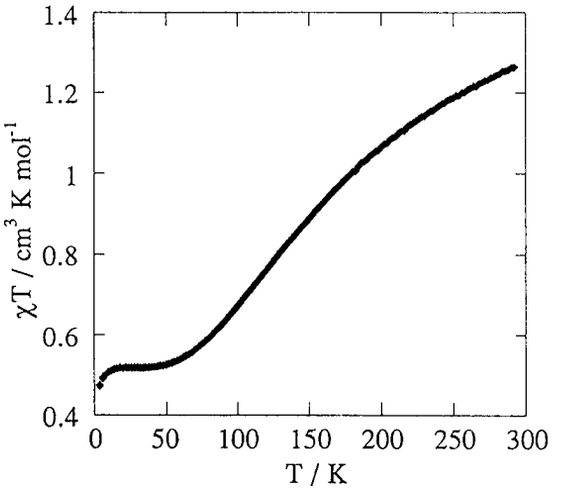


Figure 6.15. χT versus T plot for $CuNi(fsa)_2en(H_2O)_2.H_2O$.

nickel(II) ion in $CuNi(fsa)_2en(H_2O)_2\cdot H_2O$ is in weakly distorted octahedral surroundings, $\mathbf{g}_{Ni}$ may be assumed to be almost isotropic and the direction where $\mathbf{g}_{Cu}$ is maximum is that where $\mathbf{g}_{1/2}$ is minimum.

6.4 Influence of the Local Anisotropy

In this chapter we have not yet accounted for the possible local anisotropy of the interacting ions. This anisotropy may be present when the local spin S_A (and/or S_B) is larger than 1/2. The phenomenological Hamiltonian appropriate for describing the low-lying levels in AB, taking into account the Zeeman perturbation, is then

$$\mathbf{H} = -J\mathbf{S}_A\cdot\mathbf{S}_B + \mathbf{S}_A\cdot\mathbf{D}_A\cdot\mathbf{S}_A + \mathbf{S}_B\cdot\mathbf{D}_B\cdot\mathbf{S}_B + \beta(\mathbf{S}_A\cdot\mathbf{g}_A + \mathbf{S}_B\cdot\mathbf{g}_B)\cdot H \qquad (6.4.1)$$

where $\mathbf{D}_A$ and $\mathbf{D}_B$ are the local anisotropy tensors introduced in Section 2.4. When the isotropic interaction is the dominant term in (6.4.1), the total spin quantum number S remains a good quantum number for the AB pair and it is possible to associate an effective spin Hamiltonian $\mathbf{H}_S$ with each of the states defined by S. This Hamiltonian is

$$\mathbf{H}_S = \mathbf{S}\cdot\mathbf{D}_S\cdot\mathbf{S} + \beta\mathbf{S}\cdot\mathbf{g}_S\cdot H \qquad (6.4.2)$$

In Section 6.3 we indicated how the $\mathbf{g}_S$, $\mathbf{g}_A$ and $\mathbf{g}_B$ tensors were related. Using again the Wigner–Eckart theorem, it is possible to relate the $\mathbf{D}_S$, $\mathbf{D}_A$, and $\mathbf{D}_B$ tensors in a similar way [6.35–6.38]. The relationship is the following:

$$\mathbf{D}_S = (c_1 + c_2)\mathbf{D}_A/2 + (c_1 - c_2)\mathbf{D}_B/2 \qquad (6.4.3)$$

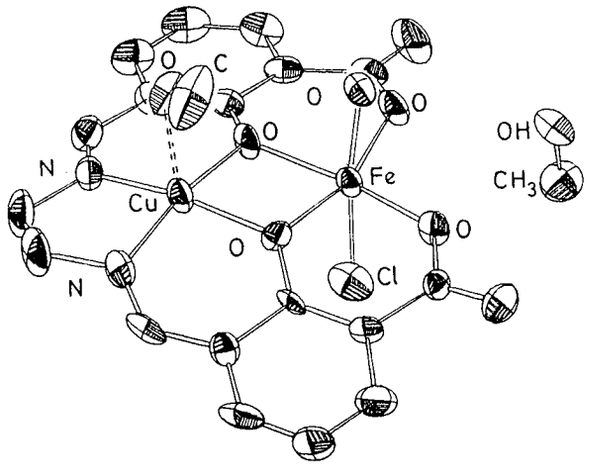


Figure 6.16. Molecular structure of $FeCu(fsa)_2en(CH_3OH)(Cl)\cdot CH_3OH$ (from ref. [6.43]).

with

$$c_1 = \frac{3[S_A(S_A+1) - S_B(S_B+1)]^2 + S(S+1)[3S(S+1) - 3 - 2S_A(S_A+1) - 2S_B(S_B+1)]}{(2S+3)(2S-1)S(S+1)}$$

$$c_2 = \frac{[4S(S+1) - 3][S_A(S_A+1) - S_B(S_B+1)]}{(2S+3)(2S-1)S(S+1)} \qquad (6.4.4)$$

The relationships between $\mathbf{D}_S$, $\mathbf{D}_A$, and $\mathbf{D}_B$ tensors are tabulated in Table 7.2 in Chapter 7. In fact, in the hypothesis where J is the leading term, the pair states are well separated in energy from each other and the zero-field splittings within the excited states do not play any significant role in the magnetic properties. It is sufficient, therefore, to take into account the zero-field splitting within the ground state, which may influence the magnetic data in the low-temperature range. It follows that when the ground state is a spin singlet ($S_A = S_B$ and $J < 0$) or a spin doublet ($|S_A - S_B| = 1/2$ and $J < 0$), the local anisotropy may be neglected, provided that $|J|$ is large enough.

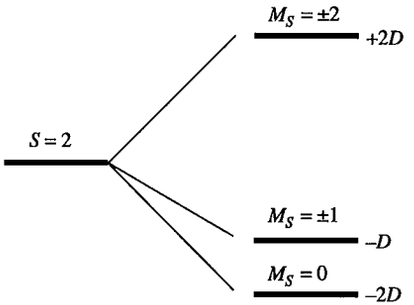
Let us apply these ideas to a real case, that of an antiferromagnetically coupled Cu(II)Fe(III) dinuclear compound with $S_{Cu} = 1/2$ and $S_{Fe} = 5/2$. The structure of such a complex, namely $FeCu(fsa)_2en(CH_3OH)(Cl)\cdot CH_3OH$, is shown in Figure 6.16 [6.43]. The low-lying states are a quintet and a septet separated by $3J$. The $\mathbf{g}_2$ and $\mathbf{g}_3$ tensors are related to the local $\mathbf{g}_{Cu}$ and $\mathbf{g}_{Fe}$ tensors through (see Table 6.2):

$$\mathbf{g}_2 = (7\mathbf{g}_{Fe} - \mathbf{g}_{Cu})/6$$
$$\mathbf{g}_3 = (5\mathbf{g}_{Fe} + \mathbf{g}_{Cu})/6 \qquad (6.4.5)$$

The χT versus T curve for this compound indicates that J is negative (vide infra). The $S = 2$ state is then the lowest and the $\mathbf{D}_2$ tensor describing the splitting in zero-field within the ground quintet is related to $\mathbf{D}_{Fe}$ through

$$\mathbf{D}_2 = 4\mathbf{D}_{Fe}/3 \qquad (6.4.6)$$

Assuming that $\mathbf{D}_2$ is axial we define an axial zero-field splitting parameter $D = 3D_{zz}/2$. The quintet ground state is then split as



The principal susceptibilities are determined as follows: We add the susceptibility arising from the thermal population of the excited septet state to the parallel and perpendicular magnetic susceptibilities for the ground quintet state axially split in zero field. These quintet state susceptibilities are calculated according to the method described in Section 2.5. We obtain

$$\chi_z = (2N\beta^2/kT)\{g_2^2[\exp(D/kT) + 4\exp(-2D/kT)] + 14g_3^2\exp(3J/kT)\}/$$
$$[\exp(2D/kT) + 2\exp(D/kT) + 2\exp(-2D/kT) + 7\exp(3J/kT)] \qquad (6.4.7)$$

$$\chi_x = 2N\beta^2\{(g_2^2/3D)[9\exp(2D/kT) - 7\exp(D/kT) - 2\exp(-2D/kT)]$$
$$+ (14g_3^2/kT)\exp(3J/kT)\}/[\exp(2D/kT) + 2\exp(D/kT)$$
$$+ 2\exp(-2D/kT) + 7\exp(3J/kT)] \qquad (6.4.8)$$

The average magnetic susceptibility for $FeCu(fsa)_2en(CH_3OH)(Cl)\cdot CH_3OH$ is displayed in Figure 6.17. On cooling from room temperature χT decreases, which indicates that the septet excited state is depopulated in favor of the quintet ground state. The rather abrupt decrease of χT below ca. 30 K is due to the zero-field splitting within the $S = 2$ state. The fitting of the magnetic data leads to $|D| = 8\ cm^{-1}$ and a quintet-septet separation $3J = -240\ cm^{-1}$. As we emphasized in Section 2.5, the average magnetic measurements do not allow the determination of the sign of D.

It is worth mentioning that Eqs. (6.3.13) and (6.4.3), and Table 6.2 and 7.2 are valid only when $|J|$ is much larger than the local anisotropies. Let us develop this point in the case of a Cu(II)Ni(II) pair [6.44–6.46]. Assuming that the local anisotropy around

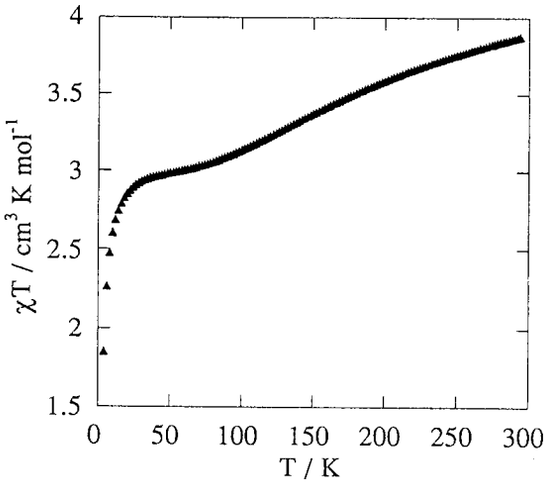


Figure 6.17. χT versus T plot for $FeCu(fsa)_2en(CH_3OH)(Cl)\cdot CH_3OH$.

the nickel(II) ion is axial and that the $\mathbf{g}_{Cu}$ and $\mathbf{g}_{Ni}$ tensors are isotropic with g_{Cu} and g_{Ni} principal values, the spin Hamiltonian may be written as

$$\mathbf{H} = -J\mathbf{S}_{Cu}\cdot\mathbf{S}_{Ni} + D(\mathbf{S}_{Ni}^2 - 2/3) + \beta(g_{Cu}\mathbf{S}_{Cu} + g_{Ni}\mathbf{S}_{Ni})\cdot H \qquad (6.4.9)$$

where D is the energy gap between the $M_S = \pm 1$ and $M_S = 0$ components arising from the local triplet state for the Ni(II) ion. When the magnetic field H is parallel to the unique axis of the nickel(II) ion, the nonzero matrix elements for the spin Hamiltonian (6.4.9) using the $|S,M_S\rangle$ functions (6.3.5) and (6.36) as a basis set are

$$\begin{aligned}
&\langle 3/2,3/2|\mathbf{H}|3/2,3/2\rangle = -J/2 + D/3 + 3g_{3/2}\beta H/2\\
&\langle 3/2,1/2|\mathbf{H}|3/2,1/2\rangle = -J/2 - D/3 + g_{3/2}\beta H/2\\
&\langle 3/2,-1/2|\mathbf{H}|3/2,-1/2\rangle = -J/2 - D/3 - g_{3/2}\beta H/2\\
&\langle 3/2,-3/2|\mathbf{H}|3/2,-3/2\rangle = -J/2 + D/3 - 3g_{3/2}\beta H/2\\
&\langle 1/2,1/2|\mathbf{H}|1/2,1/2\rangle = J + g_{1/2}\beta H/2\\
&\langle 1/2,1/2|\mathbf{H}|1/2,1/2\rangle = J - g_{1/2}\beta H/2\\
&\langle 3/2,1/2|\mathbf{H}|1/2,1/2\rangle = \langle 1/2,1/2|\mathbf{H}|3/2,1/2\rangle = -\sqrt{2}\,\delta\beta H + \sqrt{2}D/3\\
&\langle 3/2,-1/2|\mathbf{H}|1/2,-1/2\rangle = \langle 1/2,-1/2|\mathbf{H}|3/2,-1/2\rangle = -\sqrt{2}\,\delta\beta H - \sqrt{2}D/3
\end{aligned} \qquad (6.4.10)$$

where $g_{3/2}$ and $g_{1/2}$ have been defined in (6.3.8) and δ in (6.3.10). The last two matrix elements of (6.4.10) couple the doublet and quartet pair states. In the hypothesis where

$|D|$ is smaller than $|J|$ but where $|D/J|$ is no longer negligible, the energies of the states may be expanded according to increasing powers of the field H, which leads to

$$\begin{aligned}
&E(3/2,\pm 3/2) = -J/2 + D/3 \pm 3g_{3/2}\beta H/2\\
&E(3/2,\pm 1/2) = -J/2 - D/3 + a \pm g'_{3/2}\beta H/2 + b\delta^2\beta^2 H^2\\
&E(1/2,\pm 1/2) = J - a \pm g'_{1/2}\beta H/2 - b\delta^2\beta^2 H^2
\end{aligned} \qquad (6.4.11)$$

with

$$\begin{aligned}
&a = -4D^2(27J + 6D)\\
&b = -12/(9J + 2D)
\end{aligned} \qquad (6.4.12)$$

and

$$\begin{aligned}
&g'_{3/2} = [2g_{Ni}(1 - 2x) + g_{Cu}(1 + 4x)]/3\\
&g'_{1/2} = [4g_{Ni}(1 + x) - g_{Cu}(1 + 4x)]/3
\end{aligned} \qquad (6.4.13)$$

x being defined by

$$x = \frac{4D}{9J + 2D} \qquad (6.4.14)$$

We see that the $g'_{3/2}$ and $g'_{1/2}$ Zeeman factors within the $|3/2,\pm 1/2\rangle$ and $|1/2,\pm 1/2\rangle$ Kramers doublets, respectively, deviate from the $g_{3/2}$ and $g_{1/2}$ values given in (6.3.8). As a matter of fact, for $|J|/D = 10$ with J negative and D positive, we have

$$\begin{aligned}
&g'_{3/2} = (2.182g_{Ni} + 0.818g_{Cu})/3\\
&g'_{1/2} = (3.818g_{Ni} - 0.818g_{Cu})/3
\end{aligned} \qquad (6.4.15)$$

values which differ significantly from those obtained in the limit $D = 0$ although $|J|$ is still one order of magnitude larger than D. If the local anisotropy around Ni(II) was rhombic instead of axial, then the g-factor within the $|3/2,\pm 3/2$ Kramers doublet would also deviates from the $g_{3/2}$ value.

From the energies in (6.4.11) it is possible to determine the parallel magnetic susceptibility using the Van Vleck formula.

The two heterodinuclear compounds we discussed with Cu(II)Ni(II) or Cu(II)Fe(III) pairs have only two pair states since one of the interacting ions is a spin doublet. We would now like to mention a final example of heterodinuclear species with a Fe(III)Ni(II) pair. Its formula is $[FeNi(bpmp)(C_3H_7O)](BPh_4)_2$ where bpmp is the anion of 2.6-bis[(bis(2-pyridyl-methyl)amino]-4-methylphenol [6.47]. The structure of the complex cation is shown in Figure 6.18. Both Fe(III) and Ni(II) ions are in distorted octahedral surroundings; they are bridged by the phenolic oxygen atom and two propionato groups. Fe(III) is high spin, and the local spins are $S_{Fe} = 5/2$ and $S_{Ni} = 1$. The interaction gives rise to $S = 3/2$, 5/2, and 7/2 pair states with the relative energies

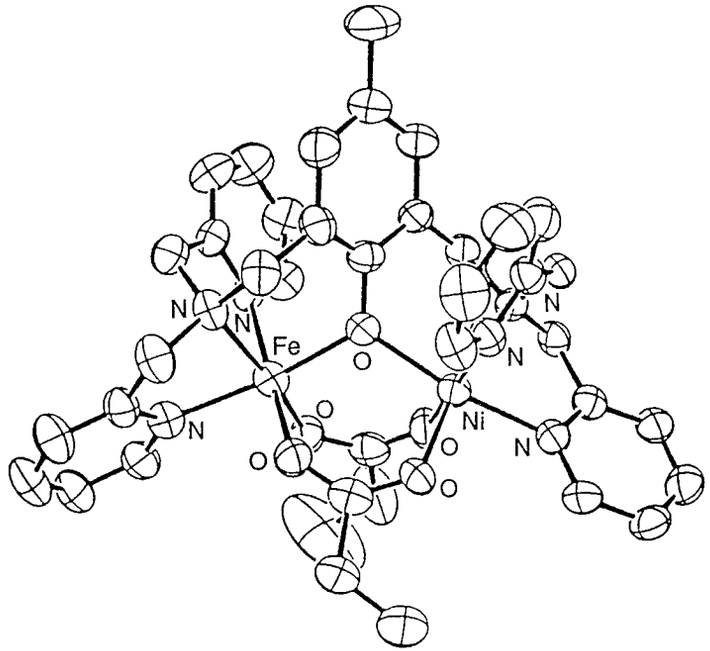


Figure 6.18. Molecular structure of $[FeNi(bpmp)(C_3H_7O)]^{2+}$ where bpmp is the anion of 2.6-bis[(bis(2-pyridyl-methyl)amino]-4-methylphenol (from ref. [6.47]).

0, $-5J/2$, and $-6J$. These pair states are split further into Kramers doublets due to the local anisotropies of the interacting ions. If J is negative and large enough with respect to the local anisotropies, only the zero-field splitting within the quartet ground state affects the magnetic properties. The zero-field splitting tensor $\mathbf{D}_{3/2}$ for the $S = 3/2$ state is related to the local tensors $\mathbf{D}_{Fe}$ and $\mathbf{D}_{Ni}$ through (see Table 7.2):

$$\mathbf{D}_{3/2} = (28\mathbf{D}_{Fe} + \mathbf{D}_{Ni})/15 \tag{6.4.16}$$

which shows that the splitting within the ground state is governed essentially by the Fe(III) anisotropy. The Zeeman tensors $\mathbf{g}_{3/2}$, $\mathbf{g}_{Fe}$, and $\mathbf{g}_{Ni}$ are related through (see Table 6.2):

$$\mathbf{g}_{3/2} = (7\mathbf{g}_{Fe} - 2\mathbf{g}_{Ni})/5 \tag{6.4.17}$$

Least-squares fitting of the magnetic susceptibility curve leads to $J = -23$ cm^{-1}, and to an energy separation δ between the two Kramers doublets arising from the $S = 3/2$ state of 0.7 cm^{-1}. The J value has been confirmed by analysis of the temperature dependence of the $g = 4.2$ signal observed in the EPR spectrum which belongs to the excited $S = 5/2$ state. Similarly, the zero-field splitting pattern within the ground state has been confirmed by the analysis of Mössbauer spectra.

6.5 Intermolecular Interactions between Dinuclear Units

In Section 2.6 we demonstrated that intermolecular interactions between entities containing a unique magnetic center lead to the Curie–Weiss law $\chi = C/(T - \theta)$, the Weiss constant θ characterizing the nature and the magnitude of these interactions. Here we treat the case where weak intermolecular interactions superimpose on a dominant intramolecular isotropic interaction in an assembly of dinuclear units [6.48]. The calculation closely follows that of Section 2.6. The spin Hamiltonian takes the form

$$\mathbf{H} = -J\mathbf{S}_A\cdot\mathbf{S}_B + \mathbf{g}\beta\mathbf{S}_z H - zJ' \langle S_z\rangle \mathbf{S}_z \tag{6.5.1}$$

where the third term on the right-hand side describes the intermolecular interactions in the molecular-field approximation. zJ' plays the role attributed to zJ in Section 2.6, and we have $|J| >> |zJ'|$. The eigenvalues $E(S,M_S)$ of $\mathbf{H}$ in (6.5.1) are given by

$$E(S,M_S) = E_0(S,M_S) + M_S(g\beta H - zJ'\langle S_z\rangle) \tag{6.5.2}$$

$E_0(S,M_S)$ being the eigenvalues of $-J\mathbf{S}_A\cdot\mathbf{S}_B$. $\langle S_z\rangle$ is calculated as

$$\langle S_z\rangle = \frac{\sum_S \sum_{M_S} M_S \exp[-E(S,M_S)/kT]}{\sum_S \sum_{M_S} \exp[-E(S,M_S)/kT]} \tag{6.5.3}$$

which may be approximated by

$$\langle S_z\rangle = \frac{\sum_S \sum_{M_S} M_S \exp[-E_0(S,M_S)/kT][1 - M_S(g\beta H - zJ'\langle S_z\rangle)/kT]}{\sum_S \sum_{M_S} \exp[-E_0(S,M_S)/kT]} \tag{6.5.4}$$

Since

$$\sum_S \sum_{M_S} M_S \exp[-E_0(S,M_S)/kT] = 0 \tag{6.5.5}$$

(6.5.4) becomes

$$\langle S_z\rangle = -\frac{\sum_S \sum_{M_S} M_S^2 \exp[-E_0(S,M_S)/kT](g\beta H - zJ'\langle S_z\rangle)/kT}{\sum_S \sum_{M_S} \exp[-E_0(S,M_S)/kT]} \tag{6.5.6}$$

which gives

$$\langle S_z \rangle = -\frac{g\beta HF(J,T)}{kT - zJ'F(J,T)} \tag{6.5.7}$$

with

$$F(J,T) = \frac{\sum_S \sum_{M_S} M_S^2 \exp[-E_0(S,M_S)/kT]}{\sum_S \sum_{M_S} \exp[-E_0(S,M_S)/kT]} \tag{6.5.8}$$

From the expression of the magnetization given in (2.6.8) we obtain the following expression for the magnetic susceptibility:

$$\chi = \frac{Ng^2\beta^2F(J,T)}{kT - zJ'F(J,T)} \tag{6.5.9}$$

This expression has some similarity with the Curie–Weiss law in the sense that the temperature is corrected by a term θ

$$\theta = zJ'F(J,T)/k \tag{6.5.10}$$

This term θ, however, depends on the temperature. For a copper(II) dinuclear compound, we have

$$F(J,T) = 2[3 + \exp(-J/kT)]^{-1} \tag{6.5.11}$$

and χ may be written down as

$$\chi = 2Ng^2\beta^2[kT - \frac{2zJ'}{3 + \exp(-J/kT)}]^{-1}\,[3 + \exp(-J/kT)]^{-1} \tag{6.5.12}$$

The effect of intermolecular interactions may be important when the ground state is magnetic. For instance, a few copper(II) dinuclear compounds with a triplet ground state ($J > 0$) have been unambiguously characterized, and in some of these the triplet molecular states couple on the scale of the lattice in an antiferromagnetic fashion ($zJ' < 0$). If this is the case, a maximum in the χT versus T curve at low temperature is observed instead of the plateau expected in the temperature range where only the triplet ground state is thermally populated.

References

6.1. B. Bleaney, K.D. Bowers, *Proc. Roy. Soc. (London) Ser. A* 214 (1952), 451.

6.2. M. Kato, H.B. Jonassen, J.C. Fanning, *Chem. Rev.* 64 (1963), 99.

6.3. R.L. Martin, in "New Pathways in Inorganic Chemistry," E.A.V. Ebsworth, A. Maddock, A.G. Sharpe, eds. Cambridge University Press, Cambridge, 1968.

6.4. A.P. Ginsberg, *Inorg. Chim. Acta Rev.* 5 (1971), 45.

6.5. R.J. Doedens, *Prog. Inorg. Chem.* 19 (1975), 173.

6.6. O. Kahn, *Angew. Chem. Int. Ed. Engl.* 24 (1985), 834.

6.7. M. Kato, Y. Muto, *Coord. Chem. Rev.* 92 (1988), 45.

6.8. B.C. Guha, *Proc. Roy. Soc. (London) Ser. A* 206 (1951), 353.

6.9. H.U. Güdel, A. Stebler, A. Fürrer, *Inorg. Chem.* 18 (1979), 1021.

6.10. H.U. Güdel, in "Magneto-Structural Correlation in Exchange Coupled Systems," R.D. Willett, D. Gatteschi, O. Kahn, eds., NATO ASI Series, p. 329. Reidel, Dordrecht, 1985.

6.11. O. Kahn, M.L. Boillot, in "Biological and Inorganic Copper Chemistry." K.D. Karlin, J. Zubieta, eds., Vol. I, p. 187. Adenine Press, Guilderland, NY, 1985.

6.12. M.F. Charlot, O. Kahn, M. Chaillet, C. Larrieu, *J. Am. Chem. Soc.* 108 (1986), 2574.

6.13. Y. Agnus, R. Louis, J.P. Gisselbrecht, R. Weiss, *J. Am. Chem. Soc.* 106 (1984), 93.

6.14. J. Comarmond, P. Pluméré, J.M. Lehn, Y. Agnus, R. Louis, R. Weiss, O. Kahn, I. Morgenstern-Badarau, *J. Am. Chem. Soc.* 104 (1982), 6330.

6.15. V. McKee, J.V. Dadgegian, R. Bau, C.A. Reed, *J. Am. Chem. Soc.* 103 (1981), 7000.

6.16. I. Bkouche-Waksman, S. Sikorav, O. Kahn, *J. Crystallogr. Spectrosc. Res.* 13 (1983), 303.

6.17. S. Sikorav, I. Bkouche-Waksman, O. Kahn, *Inorg. Chem.* 23 (1984), 490.

6.18. M.L. Boillot, Y. Journaux, A. Bencini, D. Gatteschi, O. Kahn, *Inorg. Chem.* 24 (1985), 263.

6.19. K.S. Murray, *Coord. Chem. Rev.* 12 (1974), 1.

6.20. S.J. Lippard, *Angew. Chem. Int. Ed. Engl.* 27 (1988), 344.

6.21. J.E. Davies, B.M. Gatehouse, *Acta Crystallogr. Sect. B* B29 (1973), 1934.

6.22. R.G. Wollman, D.N. Hendrickson, *Inorg. Chem.* 16 (1977), 723.

6.23. R.E. Norman, R.C. Holz, S. Menage, C.J. O'Connor, J.H. Zhang, L. Que, Jr *Inorg. Chem.* 20 (1990), 4629.

6.24. J.R. Dorfman, J.J. Girerd, E.D. Simhon, T.D.P. Stack, R.H. Holm, *Inorg. Chem.* 23 (1984), 4407.

6.25. P.C. Wilkins, R.G. Wilkins, *Coord. Chem. Rev.* 79 (1987), 195.

6.26. R.E. Stenkamp, L.C. Sieker, L.H. Jensen, *J. Am. Chem. Soc.* 106 (1984), 618. M.A. Holmes, R.E. Stenkamp, *J. Mol. Biol.* 220 (1991), 723.

6.27. J. W. Dawson, H.B. Gray, H.E. Hoenig, G.R. Rossman, J.M. Schredder, R.H. Wan, *Biochemistry* 11 (1972), 461.

6.28. W.H. Armstrong, S.J. Lippard, *J. Am. Chem. Soc.* 105 (1983), 4837.

6.29. W.H. Armstrong, A. Spool, G.C. Papaefthymiou, R.B. Frankel, S.J. Lippard, *J. Am. Chem. Soc.* 106 (1984), 3653.

6.30. K. Wieghardt, K. Pohl, W. Gebert, *Angew. Chem. Int. Ed. Engl.* 22 (1983), 727.

6.31. J.A.R. Hartman, R. L. Rardin, P. Chandhuri, K. Paul, K. Wieghardt, B. Nuber, J. Weiss, G.C. Papaefthymiou, R.B. Frankel, S.J. Lippard, *J. Am. Chem. Soc.* 109 (1987), 7387.

6.32. W.H. Armstrong, S.J. Stephen, *J. Am. Chem. Soc.* 106 (1984), 4632.

6.33. M. Julve, O. Kahn, *Inorg. Chim. Acta* 76 (1983), L39.

6.34. H.U. Güdel, in "Magneto-Structural Correlation in Exchange Coupled Systems," R.D. Willett, D. Gatteschi, O. Kahn, eds., NATO ASI Series, p. 297. Reidel, Dordrecht, 1985.

6.35. C.C. Chao, *J. Mag. Reson.* 10 (1973), 1.

6.36. E. Buluggiu, A. Vera, *Z. Naturforsh.* 31a (1976), 911.

6.37. E . Buluggiu, *J. Phys. Chem. Solids* 41 (1980), 1175.

6.38. R.P. Scaringe, D. Hodgson, W.E. Hatfield, *Mol. Phys.* 35 (1978), 701.

6.39. H. Okawa, Y. Nishida, M. Tanaka, S. Kida, *Bull. Chem. Soc. Jpn.* 50 (1977), 127.

6.40. N. Torihara, H. Okawa, S. Kida, *Inorg. Chim. Acta* 26 (1976), 97.

6.41. I. Morgenstern-Badarau, M. Rerat, O. Kahn, J. Jaud, J. Galy, *Inorg. Chem.* 21 (1982), 3050.

6.42. S. Desjardins, I. Morgenstern-Badarau, O. Kahn, *Inorg. Chem.* 23 (1984), 3833.

6.43. Y. Journaux, O. Kahn, J. Zarembowitch, J. Galy, J. Jaud, *J. Am. Chem. Soc.* 105(1983), 7585.

6.44. A. Bencini, D. Gatteschi, *Mol. Phys.* 54 (1985), 969.

6.45. J. Hulliger, Ph.D. Thesis, University of Zürich, Zürich, Switzerland, 1984.

6.46. Y. Journaux, O. Kahn, I. Morgenstern-Badarau, J. Galy, J. Jaud, A. Bencini, D. Gatteschi, *J. Am. Chem. Soc.* 107 (1985), 6305.

6.47. T.R. Holman, C. Juarez-Garcia, M.P. Hendrich, L. Que, Jr., E. Münck, *J. Am. Chem. Soc.* 112 (1990), 7611.

6.48. C.J. O'Connor, *Prog. Inorg. Chem.* 29(1982), 203.

CHAPTER

7

Dipolar, Anisotropic, and Antisymmetric Interactions in Dinuclear Compounds

In Chapter 6 we studied the isotropic interaction in dinuclear compounds, which leads to a splitting between pair states. In most actual cases this interaction, characterized in sign and magnitude by the J parameter, is by far the dominant phenomenon. It is, however, not the only phenomenon arising from the presence of two magnetic centers in the molecular unit. In this chapter we will consider the other effects, namely the dipolar, anisotropic, and antisymmetric interactions. We will focus on the phenomenological aspects, the orbital mechanisms being discussed in Chapter 9. For the sake of simplicity we will treat first the copper(II) dinuclear species, then we will generalize the results to other dinuclear systems involving similar or different metal ions.

7.1 Dipolar and Anisotropic Interactions in Copper(II) Dinuclear Compounds

In addition to the singlet-triplet splitting the interaction between two copper(II) ions within a symmetrical dinuclear entity leads to a zero-field splitting within the triplet state. This effect has two origins, the dipolar and anisotropic interactions.

The dipolar interaction arises from the influence of the magnetic field created by one of the magnetic ions on the magnetic moment due to the other. In classical mechanics the interaction energy between two magnetic dipoles μ_A and μ_B is

$$E = (1/r^3)\left[\mu_A \cdot \mu_B - \frac{3(\mu_A \cdot \mathbf{r})(\mu_B \cdot \mathbf{r})}{r^2}\right] \tag{7.1.1}$$

where $\mathbf{r}$ is the vector joining the two dipoles and r its length. If the magnetic dipoles are due to electron spins, we have

$$\mu_A = -\beta \mathbf{g}_A \cdot \mathbf{S}_A \tag{7.1.2}$$

and a similar expression for μ_B; $\mathbf{g}_A$ and $\mathbf{g}_B$ are the local **g**-tensors. The expression of the spin Hamiltonian when introducing (7.1.2) into (7.1.1) is in general rather complicated but can always been written as $\mathbf{S}_A \cdot \mathbf{D} \cdot \mathbf{S}_B$, where **D** is a symmetric and traceless tensor, provided that A and B are symmetry related [7.1–7.4]. In the simplest case where the local **g**-tensors are isotropic, the elements D_{uv} of **D** are

$$D_{uv} = g^2\beta^2\langle\frac{r^2\delta_{uv} - 3uv}{r^5}\rangle \tag{7.1.3}$$

with the brackets indicating the integration over the spatial part of the wavefunction; δ_{uv} is the Kronecker symbol. The components of **D** arising from the dipolar interaction may be calculated in principle if the spin density map is known. The simplest method, valid when A and B are rather far apart from each other, consists of assuming that the spins S_A and S_B are fully localized on the magnetic centers A and B. This is the point charge approximation, in which the zero-field splitting within the triplet state varies as r^{-3} [7.4, 7.5]. A more elaborate approach takes into account the delocalization of the local spin densities. This can be made from the magnetic orbitals a_i and b_j occupied by the unpaired electrons (see Chapter 8).

The anisotropic interaction, sometimes called pseudo-dipolar interaction, results from the combined effect of the local spin-orbit coupling and the interaction between the magnetic centers [7.6, 7.7]. Its mechanism will be discussed more thoroughly in Chapter 9. It is sufficient here to say that it can be phenomenologically described by the same spin Hamiltonian as the dipolar interaction, such that the **D** tensor is actually the sum of dipolar $\mathbf{D}^{dip}$ and anisotropic $\mathbf{D}^{ani}$ contributions.

The spin Hamiltonian including the Zeeman perturbation appropriate to describing the low-lying states of a symmetrical copper(II) pair takes the form

$$\mathbf{H} = -J\mathbf{S}_A \cdot \mathbf{S}_B + \mathbf{S}_A \cdot \mathbf{D} \cdot \mathbf{S}_B + \beta(\mathbf{S}_A \cdot \mathbf{g}_A + \mathbf{S}_B \cdot \mathbf{g}_B) \cdot H \tag{7.1.4}$$

The first term on the right-hand side describes the isotropic interaction, the second describes both the dipolar and the anisotropic interactions, and the last term describes the Zeeman perturbation. The matrix elements of **H** in (7.1.4) using the $|S,M_S\rangle$ functions as a basis set may be extracted from (7.2.2) by putting $d_x = d_y = d_z = 0$. $\mathbf{S}_A \cdot \mathbf{D} \cdot \mathbf{S}_B$ splits in zero field the triplet state. This splitting may be described by the effective spin Hamiltonian $\mathbf{H}_1$ for the $S = 1$ state:

$$\mathbf{H}_1 = \mathbf{S} \cdot \mathbf{D}_1 \cdot \mathbf{S} \tag{7.1.5}$$

the **D** and $\mathbf{D}_1$ tensors being related by

$$\mathbf{D}_1 = \mathbf{D}/2 \tag{7.1.6}$$

A generalization of this result will be given in Section 7.3. This zero-field splitting within the triplet state may influence the magnetic properties of the system in the low-temperature range if the triplet state is the lowest, i.e., if the interaction is ferromagnetic. In the case where the symmetry is axial, the parallel and perpendicular magnetic susceptibilities are easily calculated as

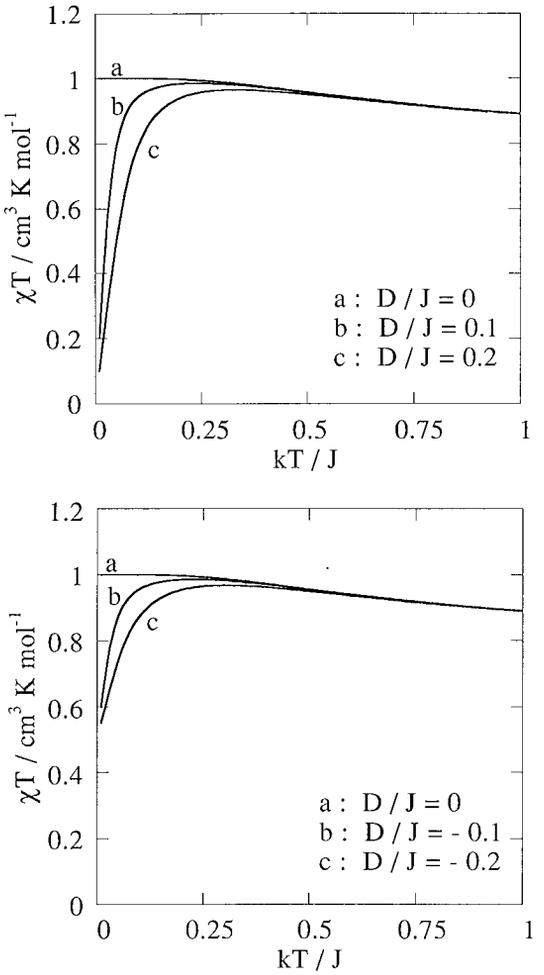


Figure 7.1. χT versus kT/J curves for a ferromagnetically coupled copper(II) pair with a triplet ground state axially split in zero field. D is the axial zero-field splitting parameter within the triplet state, and J the singlet-triplet energy gap. Top: $D \geq 0$; bottom: $D \leq 0$. For the sake of simplicity g_x (=g_y) and g_z are taken equal to 2.00.

$$\chi_z = \frac{2Ng_z^2\beta^2}{kT}\frac{\exp(-D/3kT)}{\exp(2D/3kT) + 2\exp(-D/3kT) + \exp(-J/kT)} \tag{7.1.7}$$

$$\chi_x = \frac{2Ng_x^2\beta^2}{D}\frac{\exp(2D/3kT) - \exp(-D/3kT)}{\exp(2D/3kT) + 2\exp(-D/3kT) + \exp(-J/kT)} \tag{7.1.8}$$

where the axial zero-field splitting parameter D is related to the matrix element D_{zz} through

$$D = 3D_{zz}/4 \tag{7.1.9}$$

g_z in (7.1.7) and g_x in (7.1.8) are the principal values of the $\mathbf{g}_1$ tensor associated with the triplet state. In order to establish Eq. (7.1.8) it was assumed that $|D|$ is much larger than $g_x\beta H$. Otherwise, the Van Vleck formula cannot be used, and χ_x cannot be expressed in an algebraic form. If the magnetic measurements are carried out on a polycrystalline sample, then the plateau expected in the absence of zero-field splitting for the χT versus T curve in the temperature range where only the triplet ground state is thermally populated (see Section 6.1) is no longer observed. It is replaced by a maximum and a decrease of χT when the sample is cooled further. Figure 7.1 shows the χT versus kT/J curves for various values of the D/J ratio in the case of ferromagnetic interaction ($J > 0$). Actually, for copper(II) dinuclear compounds, $|D|$ is of the order of 1 cm^{-1}, at the most [7.5–7.8], and its influence on the magnetic susceptibility curve is very weak. Moreover the effect of D is often hidden by intermolecular interactions (see Section 6.5). The best technique to determine accurately the principal values and directions of the **D** tensor is evidently EPR. Several chapters of books and review articles are devoted to this subject [7.1–7.4].

7.2 Antisymmetric Interaction in Copper(II) Dinuclear Compounds

When the dinuclear entity is no longer symmetrical, but rather of low symmetry, another phenomenon may be operative, namely the antisymmetric interaction. This contributes to the zero-field splitting within the triplet state and, in addition, couples the $S = 0$ and $S = 1$ pair states. The origin of the antisymmetric interaction is the same as that of the anisotropic interaction, i.e., the synergistic effect of local spin-orbit coupling and interaction between magnetic centers (see Chapter 9). This effect vanishes not only when the dinuclear unit is centrosymmetric but also when its molecular symmetry is C_{nv} ($n \geq 2$) with the n-fold axis joining the interacting centers, or higher. Therefore, for most compounds, the antisymmetric interaction is actually zero.

The most general form for the spin Hamiltonian including the Zeeman perturbation appropriate to describing the low-lying states of a asymmetrical copper(II) dinuclear compound is

$$\mathbf{H} = -J\mathbf{S}_A\cdot\mathbf{S}_B + \mathbf{S}_A\cdot\mathbf{D}\cdot\mathbf{S}_B + \mathbf{d}\mathbf{S}_A\wedge\mathbf{S}_B + \beta(\mathbf{S}_A\cdot\mathbf{g}_A + \mathbf{S}_B\cdot\mathbf{g}_B)\cdot H \tag{7.2.1}$$

where the new term appearing in (7.2.1) with respect to (7.1.4) describes the antisymmetric interaction. This term was first proposed by Dzialoshinski [7.9], and then analyzed by Moriya [7.10, 7.11], although the phenomenon had been pointed out before by Stevens [7.5]. In choosing the reference axes in such a way that diagonalizes

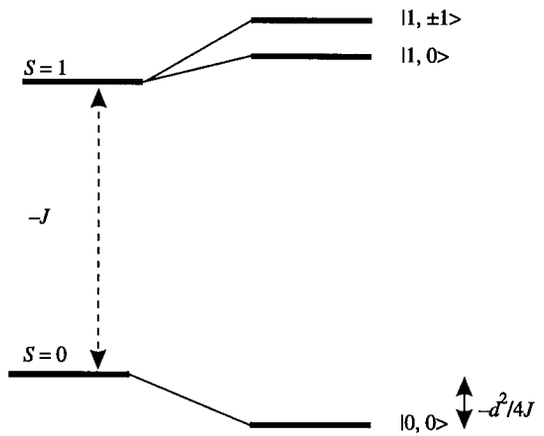
D, assuming that **D**, $\mathbf{g}_A$, and $\mathbf{g}_B$ are coaxial and using the $|S,M_S\rangle$ functions as a basis set, we find the matrix associated with **H** in (7.2.1) as

$\|1, 1\rangle$	$\|1, 0\rangle$	$\|1, -1\rangle$	$\|0, 0\rangle$
$-J/4 + D_{zz}/4 + \beta g_z H_z$	$(\sqrt{2}/2)\beta(g_xH_x - ig_yH_y)$	$(D_{xx}-D_{yy})/4$	$(\sqrt{2}/4)(id_x+d_y) - (\beta/\sqrt{2}/4)(\delta_xH_x+i\delta_yH_y)$
$(\sqrt{2}/2)\beta(g_xH_x+ig_yH_y)$	$-J/4 - D_{zz}/2$	$(\sqrt{2}/2)\beta(g_xH_x-ig_yH_y)$	$-id_z/2 + \beta\delta_zH_z/2$
$(D_{xx}-D_{yy})/4$	$(\sqrt{2}/2)\beta(g_xH_x+ig_yH_y)$	$-J/4 + D_{zz}/4 - \beta g_zH_z$	$(\sqrt{2}/4)(-id_x+d_y) + (\beta/\sqrt{2}/4)(\delta_xH_x-i\delta_yH_y)$
$(\sqrt{2}/4)(-id_x+d_y) - (\beta/\sqrt{2}/4)(\delta_xH_x-i\delta_yH_y)$	$id_z/2 + + \beta\delta_zH_z/2$	$(\sqrt{2}/4)(-id_x+d_y) + (\beta/\sqrt{2}/4)(\delta_xH_x+i\delta_yH_y)$	$3J/4$

(7.2.2)

where D_{uu} ($u = x, y, z$) are the principal values of the **D** tensor and d_u the components of the **d** vector; δ_u stands for $g_{A(u)} - g_{B(u)}$.

If we neglect **D** in (7.2.2) and take into account the antisymmetric interaction characterized by the **d** vector, then we obtain the spectrum of the low-lying states in zero-field



where J is assumed to be negative and large in absolute value with respect to the components of d. The newest aspect of the antisymmetric interaction is the singlet-triplet coupling. The zero-field energies $E(S,M_S)$ of the levels deduced from (7.2.2) are

$$E(0,0) = 3J/4 - d^2/4J$$

$$E(1,\pm1) = -J/4 + (d_x^2 + d_y^2)/8J$$

$$E(1,0) = -J/4 + d_z^2/4J \qquad (7.2.3)$$

with

$$d^2 = d_x^2 + d_y^2 + d_z^2 \qquad (7.2.4)$$

such that the D parameter which measures the zero-field splitting within the triplet state is

$$D = (d_x^2 + d_y^2 - 2d_z^2)/8J \qquad (7.2.5)$$

This zero-field splitting is added to that arising from dipolar and anisotropic interactions. In the case of antiferromagnetic interaction the wavefunction Ψ_0 for the ground state is

$$\Psi_0 = [1 - (d^2/4J^2)]^{-1/2}\{ \mid 0,0> + (1/2J)[(\sqrt{2}/2)(id_x + d_y) \mid 1,1> - id_z \mid 1,0>$$

$$- (\sqrt{2}/2)(id_x - d_y) \mid 1,-1>]\} \qquad (7.2.6)$$

This ground state is no longer rigorously diamagnetic but contains a small proportion of triplet state [7.12, 7.13]. The average value $<S^2>$ of $\mathbf{S}^2$ defined by

$$<S^2> = <\Psi_0|\mathbf{S}^2|\Psi_0> \qquad (7.2.7)$$

is easily obtained as

$$<S^2> = d^2/2J^2 \qquad (7.2.8)$$

The average values of $\mathbf{S}_u$ ($u = x, y, z$), on the other hand, are all zero.

Due to the singlet-triplet mixing, the average magnetic susceptibility defined as

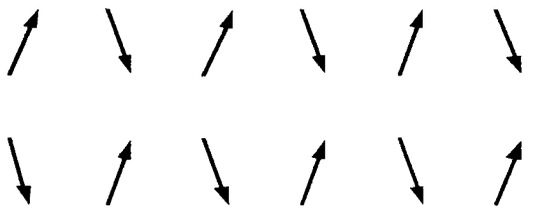
$$\chi = (\chi_x + \chi_y + \chi_z)/3 \qquad (7.2.9)$$

does not tend to zero when T approaches zero but to a low-temperature limit χ_{LT} given by

$$\chi_{LT} = -Ng^2\beta^2d^2/3J^3 \qquad (7.2.10)$$

in the approximation where $g_x = g_y = g_z$. Since J is negative, χ_{LT} is positive. In the temperature range where only the ground state is thermally populated, the magnetic susceptibility is constant and given by (7.2.10). This effect is rather similar to the TIP we discussed in Section 1.5, and may be very difficult to detect. To our knowledge the presence of antisymmetric interaction in antiferromagnetically coupled copper(II) pairs has not yet been demonstrated, although some dinuclear species of very low symmetry have been synthesized [7.14]. On the other hand, in extended lattices of low symmetry, the antisymmetric interaction is at the origin of weak ferromagnetism. This phenomenon may be visualized as follows: The antisymmetric interaction tends to orient the neighboring spins perpendicular to each other while the isotropic interaction tends to orient them in either parallel or antiparallel directions. In the antiferromagnetic

case the antisymmetric interaction leads to a canting between the spins in the state that otherwise would be diamagnetic, and to the onset of a magnetic ordering with a weak spontaneous magnetization. This is schematized below in the form of a two-dimensional array of spins. It should be noted, however, that in most cases the onset of a spontaneous magnetization results from a three-dimensional ordering. In no case, can it result from a one-dimensional ordering (see Chapter 12).



Some examples of weak ferromagnetism for molecular compounds will be discussed in Section 12.9.

7.3 Other Dinuclear Compounds

The spin Hamiltonian including the Zeeman perturbation and describing the low-lying states for any pair of interacting magnetic centers may be written as

$$\mathbf{H} = -J\mathbf{S}_A\cdot\mathbf{S}_B + \mathbf{S}_A\cdot\mathbf{D}\cdot\mathbf{S}_B + d\mathbf{S}_A\wedge\mathbf{S}_B + \mathbf{S}_A\cdot\mathbf{D}_A\cdot\mathbf{S}_A + \mathbf{S}_B\cdot\mathbf{D}_B\cdot\mathbf{S}_B$$

$$+ \beta(\mathbf{S}_A\cdot\mathbf{g}_A + \mathbf{S}_B\cdot\mathbf{g}_B)\cdot H \qquad (7.3.1)$$

provided that the local states have no first-order orbital momentum. This Hamiltonian (7.3.1) differs from (7.2.1) in that it takes into account the local anisotropies characterized by the $\mathbf{D}_A$ and $\mathbf{D}_B$ tensors. If the local spin S_A (or S_B) is equal to 1/2, then evidently $\mathbf{D}_A$ (or $\mathbf{D}_B$) vanishes. To date, to our knowledge, there is no report of dinuclear species of low symmetry where the antisymmetric interaction has been characterized and we will no longer consider this contribution. We recall, however, that in extended lattices the antisymmetric interaction can give rise to the very interesting phenomenon of weak ferromagnetism.

If the isotropic interaction $-J\mathbf{S}_A\cdot\mathbf{S}_B$ is the dominant term in (7.3.1), as is most often the case, the quantum number S characterizing the spin of the pair states and varying by an integer value from $|S_A - S_B|$ to $S_A + S_B$ is a good quantum number. It is then possible to associate an effective Hamiltonian with each of the spin states as indicated in Section 6.4. This effective Hamiltonian may be written as

$$\mathbf{H}_S = \mathbf{S}\cdot\mathbf{D}_S\cdot\mathbf{S} + \beta\mathbf{S}\cdot\mathbf{g}_S\cdot H \qquad (7.3.2)$$

$\mathbf{g}_S$ is related to $\mathbf{g}_A$ and $\mathbf{g}_B$ through the Eqs. (6.3.13), and (6.3.14), and Table 6.2. As for $\mathbf{D}_S$, it is related to $\mathbf{D}_A$, $\mathbf{D}_B$, and $\mathbf{D}$ through [7.15]

Table 7.1. Relationships between $\mathbf{D}_S$, $\mathbf{D}_A$, $\mathbf{D}_B$, and **D** Tensors in Dinuclear Compounds with A and B Magnetic Centers[a]

S_A	S_B	S	$\mathbf{D}_A$	$\mathbf{D}_B$	**D**
1/2	1/2	1	—	—	1/2
1/2	1	3/2	—	1/3	1/3
1/2	3/2	1	—	3/2	−1/4
		2	—	1/2	1/4
1/2	2	3/2	—	7.5	−1/5
		5/2	—	3/5	1/5
1/2	5/2	2	—	4/3	−1/6
		3	—	2/3	1/6
1	1	1	−1/2	−1/2	1
		2	1/6	1/6	1/3
1	3/2	3/2	−4/15	1/5	8/15
		5/2	1/10	3/10	3/10
1	2	1	1	21/10	−3/5
		2	−1/6	1/2	1/3
		3	1/15	2/5	4/15
1	5/2	3/2	1/15	28/15	−7/15
		5/2	−4/35	23/35	8/35
		7/2	1/21	10/21	5/21
3/2	3/2	1	−6/5	−6/5	17/10
		2	0	0	1/2
		3	1/5	1/5	3/10
3/2	2	3/2	−3/5	0	4/5
		5/2	−3/70	3/14	29/70
		7/2	1/7	2/7	2/7
3/2	5/2	1	3/10	14/5	−21/20
		2	−5/14	10/21	37/84
		3	−1/20	11/30	41/120
		4	3/28	5/14	15/56
2	2	1	−21/10	−21/10	13/5
		2	−3/14	−3/14	5/7
		3	1/10	1/10	2/5
		4	3/14	3/14	2/7
2	5/2	3/2	−1/10	−4/15	17/15
		5/2	−3/14	1/10	39/70
		7/2	1/21	2/9	23/63
		9/2	1/6	5/18	5/18
5/2	5/2	1	−16/5	−16/5	37/10
		2	−10/21	−10/21	41/42
		3	−1/45	−1/45	47/90
		4	1/7	1/7	5/14
		5	2/9	2/9	5/18

[a]This table must be read as follows (see the second line): for the $S = 3/2$ pair state arising from the interaction between $S_A = 1/2$ and $S_B = 1$ local states, we have $\mathbf{D}_{3/2} = \mathbf{D}_B/3 + \mathbf{D}/3$.

$$\mathbf{D}_S = \frac{c_1 + c_2}{2}\mathbf{D}_A + \frac{c_1 - c_2}{2}\mathbf{D}_B + \frac{1 - c_1}{2}\mathbf{D} \tag{7.3.3}$$

Equation (7.3.3) completes Eq. (6.4.3) where the anisotropic interaction tensor **D** was ignored. The coefficients c_1 and c_2 are given by (6.4.4). The relationships between $\mathbf{D}_S$, $\mathbf{D}_A$, $\mathbf{D}_B$, and **D** tensors are tabulated in Table 7.1 for the different couples S_A and S_B. It is likely worthwhile repeating here that **D** consists of two contributions, the mechanisms of which are different, namely the dipolar and anisotropic interactions. In fact, what is experimentally determined, either from EPR spectra or from magnetic susceptibility data, is the zero-field splitting tensor $\mathbf{D}_S$ associated with the pair state S. There is no trivial way to analyze quantitatively the different contributions. One can state, however, that generally the local anisotropy $\mathbf{D}_A$ (and/or $\mathbf{D}_B$) seems to dominate the interaction term **D**. This is, for instance, the case for the Cu(II)Fe(III) compound shown in Figure 6.16.

References

7.1. A. Abragam, B. Bleaney, "Electron Paramagnetic Resonance of Transitions Ions." Dover, New York, 1986.

7.2. J. Owen, E.A. Harris, in "Electron Paramagnetic Resonance," S. Geschwind, ed. Plenum, New York, 1972.

7.3. N. M. Atherton, "Electron Spin Resonance." John Wiley, New York, 1973.

7.4. A. Bencini, D. Gatteschi, "EPR of Exchange Coupled Systems." Springer-Verlag, Berlin, 1990.

7.5. L. Banci, A. Bencini, D. Gatteschi, *J. Am. Chem. Soc.* 105 (1983), 761.

7.6. K.W.H. Stevens, *Rev. Mod. Phys.* 25 (1953), 166.

7.7. J. Kanamori, in "Magnetism," G.T. Rado, H. Suhl, eds., Vol. 1, p. 127, Academic Press, New York, 1963.

7.8. D. Gatteschi, A. Bencini, in "Magneto-Structural Correlations in Exchange Coupled Systems," R.D. Willett, D. Gatteschi, O. Kahn, eds., p. 241. Reidel, Dordrecht, 1985.

7.9. I. Dzialoshinsky, *J. Phys. Chem. Solids* 4 (1958), 241.

7.10. T. Moriya, *Phys. Rev.* 120 (1960), 91.

7.11. T. Moriya, in "Magnetism," G.T. Rado, H. Suhl, eds., Vol. 1, p. 86. Academic Press, New York, 1963.

7.12. P. Erdös, *J. Phys. Chem. Solids* 27 (1966), 1705.

7.13. P. Erdös, *Helv. Phys. Acta* 39 (1966), 155.

7.14. M. Julve, M. Verdaguer, A. Gleizes, M. Philoche-Levisalles, O. Kahn, *Inorg. Chem.* 23 (1984), 3808.

7.15. S.P. Scaringe, D.J. Hodgson, W.E. Hatfield, *Mol. Phys.* 35 (1978), 701.

CHAPTER

8

Orbital Models of the Isotropic Interaction: Case of Two Local Doublet States

8.1 Some Preliminary Remarks

Few problems in chemistry, particularly in inorganic chemistry, have attracted the attention of the theoreticians more than that of the electronic structure of magnetic coupled systems. A large number of studies devoted to this subject have been published and numerous models have been proposed [8.1]. At present, it is rather difficult for the profane to find his route among the maze of theoretical approaches devoted to this problem.

Chemists interested in the problem of chemical bonding in closed-shell molecules know that they have an efficient tool at their disposal, namely the molecular orbital model. There are, of course, several levels of sophistication in this model, but the criteria of molecular topology, overlap between atomic orbitals, and electronegativity often allow a qualitative interpretation of the phenomena. This explains and justifies the astonishing success of rustic approaches like Hückel or extended Hückel methods. In molecular magnetism the situation is different. No model dominates and there are still a number of controversies about the advantages and limits of the various approaches. One point, however, has emerged in the last few years; the phenomenon of magnetic interaction cannot be understood properly in the framework of one "classical" molecular orbital model without configuration interaction (CI). In fact, this situation is perhaps the best illustration of the limits of the molecular orbital model at the self-consistent field (SCF) level.

In this chapter we do not intend to review all the models proposed so far. This would be fastidious, and probably not very useful for the reader who is not a specialist in this field. Instead, we would like to stress a few key concepts, discuss their merits and limits, and show how they can be applied to some selected examples. This chapter is restricted to the case of isotropic interaction between two local doublet states. More complicated situations will be treated in the following chapter, and in Chapter 12, which deals with the design of molecular-based magnets.

We will explain why the molecular orbital approach at the SCF level is inappropriate, and then we will introduce the concept of magnetic orbitals. We will emphasize that there are two ways to define these magnetic orbitals. We will then consider the active-electron approximation, in which only the two unpaired electrons are explicitly taken into account and we will derive mathematical expressions for J. Then, we will go beyond this approximation. Finally, we will discuss some experimental results.

8.2 Deficiency of the Molecular Orbital Approach at the Self-Consistent Field Level

Let us consider an AXB centrosymmetrical model system, where A and B are two magnetic centers with one unpaired electron each, and X is a closed-shell diamagnetic bridge. The physics of this kind of system tells us that normally the two low-lying states, a spin singlet $^1\Gamma_g$ and a spin triplet $^3\Gamma_u$, are close enough in energy to both be populated at room temperature. The singlet–triplet (S–T) energy gap is noted J.

The problem here is to examine whether it is possible to analyze properly the nature of the $^1\Gamma_g$ and $^3\Gamma_u$ states by considering a four-electron three-orbital problem. The three atomic orbitals involved are d_A and d_B centered on A and B, respectively, and ϕ_X centered on the bridge X; ϕ_X is assumed to have a g symmetry with an energy lower than that of d_A and d_B, as is generally the case for transition metal compounds.

From d_A, d_B, and ϕ_X, we can construct the three molecular orbitals of increasing energy:

$$\varphi_0 \approx \phi_X + \varepsilon(d_A + d_B) \qquad (\varepsilon << 1)$$

$$\varphi_1 \approx d_A - d_B$$

$$\varphi_2 \approx (d_A + d_B) - \varepsilon'\phi_X \qquad (\varepsilon' << 1) \tag{8.2.1}$$

φ_0, of g symmetry, is bonding with preponderant ligand character. φ_1, of u symmetry, is nonbonding provided that the direct A–B interaction is negligible. φ_2, again of g symmetry, has the highest energy with preponderant metal character, as schematized in Figure 8.1. The corresponding energy diagram is also shown in Figure 8.1.

Let us now try to describe the $^1\Gamma_g$ and $^3\Gamma_u$ low-lying states at the SCF level, with monodeterminantal functions. In this framework the singlet, noted S_1, is

$$S_1 = |\varphi_0 \overline{\varphi}_0 \varphi_1 \overline{\varphi}_1| \tag{8.2.2}$$

and the $M_S = 1$ component of the triplet, noted T, is

$$T = |\varphi_0 \overline{\varphi}_0 \varphi_1 \varphi_2| \tag{8.2.3}$$

where the bar notes $m_S = -1/2$. The quality of this description is not the same for both states and depends on the energy of ϕ_X with respect to that of d_A and d_B. More precisely, the lower in energy ϕ_X is, the better the description of $^3\Gamma_u$, and the less satisfying that of $^1\Gamma_g$. Indeed, when ϕ_X is very low in energy, both ε and ε' in (8.2.1) are close to zero. The $e_2 - e_1$ gap between the energies of the φ_2 and φ_1 molecular orbitals is small. These

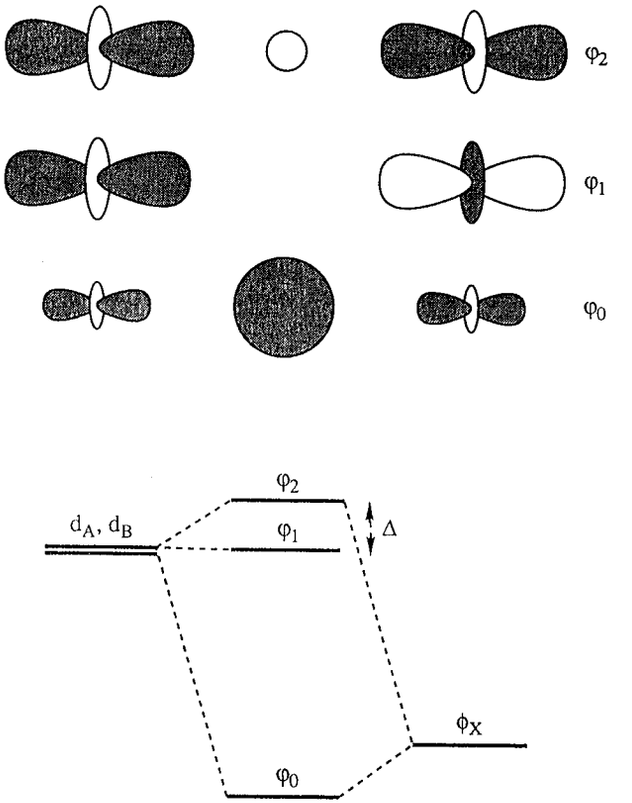


Figure 8.1. Schematic representation and energy diagram of the molecular orbitals for the AXB model system.

two molecular orbitals are occupied by two electrons, which leads to the following configurations and states:

$$\varphi_0^2\varphi_1^2 \quad {}^1\Gamma_g$$

$$\varphi_0^2\varphi_2^2 \quad {}^1\Gamma_g$$

$$\varphi_0^2\varphi_1\varphi_2 \quad {}^3\Gamma_u + {}^1\Gamma_u$$

If the molecular orbitals are determined by an SCF open-shell calculation on the triplet state, this state is well described by the single determinant T. In contrast, the low-lying singlet state is described by mixing the two determinants S_1 defined above and S_2 defined as

$$S_2 = |\varphi_0 \overline{\varphi}_0 \varphi_2 \overline{\varphi}_2| \tag{8.2.4}$$

In the limit where ϕ_X is very low in energy with respect to d_A and d_B, S_1 and S_2 have almost the same weight in the description of the lowest singlet state. This result

emphasizes the absolute necessity of going beyond the SCF level in order to understand the magnetic properties of the AXB system. More generally, in a molecular orbital approach, the configuration interaction is an inevitable step in any attempt to calculate the spectrum of the low-lying states for a magnetic polymetallic compound.

The S–T energy gap J is related to the difference in electronic correlation between those two states. It is possible in principle to calculate the energy of each of these states separately and to obtain J as the difference. This method, however, presents the difficult problem of evaluating a small quantity as a difference between two large numbers. Moreover, a direct calculation of the correlation energy requires an extended basis set along with a large number of perturbating configurations. Such a method is still difficult for real polymetallic compounds. Furthermore, owing to its intrinsic complexity, this multiconfigurational approach retains the character of a "black box." In other words, it appears tedious, if not impossible, to extract some rules allowing a qualitative understanding of the isotropic interaction phenomenon.

8.3 Concept of Magnetic Orbitals: Orthogonalized and Natural Magnetic Orbitals

The molecular orbital plus configuration interaction treatment is not only difficult but is also very far from the local approach used by experimentalists to interpret magnetic data. Indeed, the HDVV Hamiltonian for the isotropic interaction (see Chapter 6) may be regarded as the phenomenological translation of the electrostatic interaction occurring between the two unpaired electrons of A and B magnetic centers. This interaction is weak with respect to the normal two-electron chemical bond since it leads to singlet and triplet states close enough in energy to both be thermally populated. Moreover, when the interaction vanishes, the borderline situation corresponds to two uncorrelated electrons, one around A, and the other around B. These requirements are compatible with a Heitler–London approach using two semilocalized orbitals a and b as a basis set; a is singly occupied by the unpaired electron arising from A and b by the unpaired electron arising from B. Two definitions of the magnetic orbitals have been proposed to date and used in a heuristic way. One definition concerns orthogonalized magnetic orbitals, and the other concerns natural magnetic orbitals [8.1]. We present both definitions here.

8.3.1 Orthogonalized Magnetic Orbitals

a and b may be deduced from the molecular orbitals φ_1 and φ_2 through [8.1–8.3]

$$a = (\varphi_1 + \varphi_2)/\sqrt{2}$$

$$b = (\varphi_1 - \varphi_2)/\sqrt{2} \qquad (8.3.1)$$

a and b are obviously orthogonal but not perfectly localized; a has a nonvanishing contribution on B and b has a nonvanishing contribution on A. This tail of a on B (and

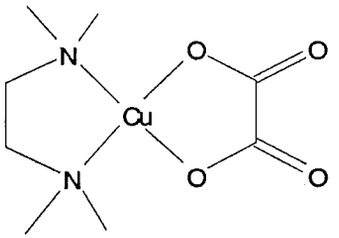
b on A) becomes more important as the $e_2 - e_1$ energy separation between the φ_1 and φ_2 molecular orbitals becomes larger and the isotropic interaction in AXB becomes more pronounced. In other respects, a and b are related through an inversion center and are degenerate such that they are singly occupied in the ground configuration.

The localization criterion (8.3.1) is valid only when the A and B centers are symmetry related, either through an inversion center, or through a twofold axis. Otherwise, the localization criteria are no longer unique. The concept of orthogonalized magnetic orbital has never been extended to asymmetrical systems, for instance copper(II) dinuclear compounds in which the surroundings of the metal ions are different, except in the very peculiar case of $CuVO(fsa)_2en{\cdot}CH_3OH$, which will be discussed in Section 8.6.

8.3.2 Natural Magnetic Orbitals

Due to their defect of localization, orthogonalized magnetic orbitals are difficult to visualize and even more difficult to utilize for a qualitative prediction of the magnitude of the interaction. We have already pointed out that the exchange interaction in AXB, in a certain sense, may be viewed as the borderline case of a very weak chemical bond, and for more than half a century the chemists have enjoyed describing the chemical bond in terms of overlap between localized orbitals. This suggests that another approach might be worthwhile, at least for a qualitative description of the magnetic interaction phenomenon, in which the magnetic orbitals would be better localized and in general nonorthogonal. This leads to the concept of natural magnetic orbitals.

The natural magnetic orbital a in AXB is defined as the singly occupied molecular orbital for the AX fragment in its local ground state; b is defined in the same way with respect to the XB fragment. It follows that a and b are now eigenfunctions of local Hamiltonians. The cutting of the AXB entity into two AX and XB fragments with a common bridging region is in general somewhat arbitrary, which is certainly the weak point of this approach. The local states and the natural magnetic orbitals are better defined when the AX and XB fragments do actually exist. This happens in quite a few cases, for instance for the oxalato-bridged copper(II) compound $[tmen(H_2O)Cu(ox)Cu(H_2O)tmen]^{2+}$ shown in Figure 8.11. In addition to this dinuclear complex, it is possible to synthesize the mononuclear species Cu(tmen)(ox):



Cu(tmen)(ox) may be considered as a mononuclear fragment, in which the copper(II) ion has almost the same nearest neighbors as in $[tmen(H_2O)Cu(ox)Cu(H_2O)tmen]^{2+}$. Indeed, in the dinuclear species, the H_2O molecule in the apical position interacts only very weakly with the metal center. Consequently, the natural magnetic orbital a in $[tmen(H_2O)Cu(ox)Cu(H_2O)tmen]^{2+}$ may be visualized as the highest singly occupied molecular orbital in Cu(tmen)(ox). This orbital a has obviously no delocalization tail on the second copper atom (on which it is not centered) and, in principle, is not orthogonal to the symmetry related magnetic orbital b. Let us remark here that the concept of natural magnetic orbital is straightforward when the interaction is direct instead of through a bridge. This occurs, for instance, in the face-to-face $(TCNQ^-)_2$ dimer. Each natural magnetic orbital is then the singly occupied π^* molecular orbital of the $TCNQ^-$ unit.

Another way of determining the natural magnetic orbital a in AXB consists of contracting the atomic orbitals of B in such a way that prevents any orbital interaction between the magnetic center B and its surroundings. The orbital a is then the singly occupied molecular orbital of highest energy for the dissymmetrized AXB molecule. Similarly, b may be determined by contracting the atomic orbitals of A.

The concept of natural magnetic orbital can be easily extended to the case where AXB is not centrosymmetric. The AX and XB fragments are then different and the singly occupied molecular orbitals in these fragments are no longer symmetry related. The atomic orbital contraction method is also well adapted for defining the natural magnetic orbitals in an asymmetrical dinuclear compound [8.4].

8.4 The Active-Electron Approximation

For numerous compounds the highest occupied molecular orbitals of the bridging ligands that interact with the metal ions are low in energy compared to the d orbitals. They are also well separated from other metal or ligand occupied orbitals. This is true, for example, for oxo- or fluoro-bridged compounds. In such cases an extensively used approximation consists of taking into account only the two unpaired electrons occupying the magnetic orbitals a and b; all other electrons are considered as passive. This active-electron approximation may be developed using either orthogonalized, or natural magnetic orbitals. We will present both of these approaches, and then we will attempt to compare them.

8.4.1 Active-Electron Approximation and Orthogonalized Magnetic Orbitals

In order to calculate the S–T energy gap J in AXB, we may write down the effective electrostatic Hamiltonian [8.5]

$$\mathbf{H} = \mathbf{h}(1) + \mathbf{h}(2) + 1/r_{12} \tag{8.4.1}$$

where all electrons except the two unpaired electrons are considered as passive. In (8.4.1) $\mathbf{h}(i)$ is the one-electron Hamiltonian for electron i taking into account its kinetic energy and its interaction with the nuclei and the passive electrons, and r_{12} is the interelectronic distance. In the following we need the two one-electron and the four two-electron integrals given below:

$$\alpha = \langle a(1) \,|\, \mathbf{h}(1) \,|\, a(1) \rangle$$

$$\beta = \langle a(1) \,|\, \mathbf{h}(1) \,|\, b(1) \rangle$$

$$j^0 = \langle a(1)a(2) \,|\, r_{12}^{-1} \,|\, a(1)a(2) \rangle$$

$$j = \langle a(1)b(2) \,|\, r_{12}^{-1} \,|\, a(1)b(2) \rangle$$

$$k = \langle a(1)b(2) \,|\, r_{12}^{-1} \,|\, a(2)b(1) \rangle$$

$$l = \langle a(1)b(2) \,|\, r_{12}^{-1} \,|\, b(1)b(2) \rangle \tag{8.4.2}$$

There are two states that arise from the ab ground configuration, $^1\Gamma_g(GC)$ and $^3\Gamma_u(GC)$, and two additional states that arise from the a^2 (or b^2) excited charge-transfer configuration, $^1\Gamma_g(CTC)$ and $^1\Gamma_u(CTC)$. The orbital parts of the corresponding wavefunctions are $(2)^{-1/2}[a(1)b(2) \pm a(2)b(1)]$ where the positive sign holds for $^1\Gamma_g$ (GC) and the negative sign for $^3\Gamma_u$ (GC), and $(2)^{-1/2}[a(1)a(2) \pm b(1)b(2)]$ where the positive sign holds for $^1\Gamma_g(CTC)$ and the negative sign for $^1\Gamma_u(CTC)$.

Neglecting the GC–CTC interaction, the energies of the low-lying singlet and triplet states are

$$E[^1\Gamma_g(GC)] = 2\alpha + j + k$$

$$E[^3\Gamma_u(GC)] = 2\alpha + j - k \tag{8.4.3}$$

such that the triplet state is found stabilized by $J_F = 2k$ with respect to the singlet state. Let us take into account the GC–CTC interaction, which stabilizes $^1\Gamma_g(GC)$ and destabilizes $^1\Gamma_g(CTC)$. The energy difference U between the two $^1\Gamma_g$ states is

$$U = j^0 - j \tag{8.4.4}$$

where we assumed that due to the relaxation of the inner shells, the electroneutrality of the magnetic centers is retained. If, on the contrary, the A and B ions separated by a distance R have both a charge q in the GC and $q+1$ and $q-1$, respectively, in the CTC, then U becomes

$$U = j^0 - j - 1/R \tag{8.4.5}$$

Whatever the expression for U, the stabilization J_{AF} of the singlet state of lowest energy due to the GC–CTC interaction is

$$J_{AF} = -\frac{4(\beta + l)^2}{U} \tag{8.4.6}$$

J may then be expressed as the sum of a positive contribution J_F stabilizing the triplet and a negative contribution J_{AF} stabilizing the singlet:

$$J = J_F + J_{AF} = 2k - \frac{4(\beta + l)^2}{U} \tag{8.4.7}$$

The ionic integral l is often neglected without any explicit reason, such that J_{AF} reduces to $-4\beta^2/U$.

8.4.2 Active-Electron Approximation and Natural Magnetic Orbitals

If we use natural magnetic orbitals, an additional one-electron integral is necessary, namely the overlap integral $\mathcal{S}$:

$$\mathcal{S} = \langle a(1) \mid b(1) \rangle \tag{8.4.8}$$

The normalization coefficients of the orbital parts of the wavefunctions become $[2(1 \pm \mathcal{S}^2)]^{-1/2}$ where the positive sign holds for the two $^1\Gamma_g$ states and the negative sign for $^3\Gamma_u$ and $^1\Gamma_u$.

Neglecting the GC–CTC interaction, the energies of the lowest states are

$$E[^1\Gamma_g(\mathrm{GC})] = \frac{2(\alpha + \beta\mathcal{S}) + j + k}{1 + \mathcal{S}^2}$$

$$E[^3\Gamma_u(\mathrm{GC})] = \frac{2(\alpha - \beta\mathcal{S}) + j - k}{1 - \mathcal{S}^2} \tag{8.4.9}$$

such that, at this level of approximation, the S–T energy gap J is given by

$$J = 2k + 4\beta\mathcal{S} - 2\mathcal{S}^2(2\alpha + j) \tag{8.4.10}$$

where, after having performed all calculations, $\mathcal{S}^2$ was neglected compared to unity. The first term $2k$ in (8.4.10) is always positive and favors the ferromagnetic interaction. For $\mathcal{S}$ small enough β and $\mathcal{S}$ are of opposite sign and $4\beta\mathcal{S}$ is negative and favors the antiferromagnetic interaction. As for the third term in $\mathcal{S}^2$, its sign cannot be easily predicted since α and j are of opposite sign. For $\mathcal{S}$ small enough this quadratic term may reasonably be assumed to be weak with respect to the two others and actually is often neglected. If so, J appears again as the sum of a ferromagnetic contribution $J_F = 2k$ and an antiferromagnetic contribution $J_{AF} = 4\beta\mathcal{S}$.

If we consider the interaction between the two $^1\Gamma_g$ states separated in energy by U now defined as

$$U = \frac{j^0 - j}{1 - \mathcal{S}^2} \tag{8.4.11}$$

the energy of the lowest singlet state becomes

$$E(^1\Gamma_g) = \frac{2(\alpha + \beta\mathcal{S}) + j + k}{1 + \mathcal{S}^2} - \frac{4[(\beta - \alpha\mathcal{S})(1 - \mathcal{S}^2) + l(1 + \mathcal{S}^2) - (j + k)\mathcal{S}]^2}{U(1 + \mathcal{S}^2)^4} \tag{8.4.12}$$

and J is found as

$$J = 2k + 4\beta\mathcal{S} - 2\mathcal{S}^2(2\alpha + j) - \frac{4[\beta + l - (\alpha + j + k)\mathcal{S}]^2}{U} \tag{8.4.13}$$

where $\mathcal{S}^2$ was again neglected compared to unity. The term in $1/U$ appearing in (8.4.13) is obviously negative and gives an additional antiferromagnetic contribution. The problem when we examine Eq. (8.4.13) is how to recognize whether the main contribution in the stabilization of the singlet state is $4\beta\mathcal{S}$ or the term in $1/U$. There is no definitive answer yet. However, it has been suggested that in transition metal dinuclear compounds the metal–metal charge transfer configuration is too high in energy to couple significantly with the ground configuration. For instance, in copper(II) dimers, the levels arising from the copper(III)–copper(I) excited configuration have never been unambiguously detected by electronic spectroscopy. U is probably beyond the commonly accessible energy range, and the stabilization of the singlet state arising from the GC–CTC interaction would be a second-order effect compared to $4\beta\mathcal{S}$. The situation might be different in face-to-face radical dimers like $(TCNQ^-)_2$. The magnetic orbital is then a π^* molecular orbital delocalized on the whole radical and the self-repulsion j^0 between two electrons occupying such an orbital may be strongly reduced. U can then be weak enough to contribute significantly to the stabilization of the singlet state. In $(TCNQ^-)_2$ U is equal to ca. 12,000 cm^{-1}. The interaction between ground state and charge-transfer excited states also plays an essential role in the magnetic properties of mixed valence compounds. This question will be treated in Chapter 13.

Setting $\mathcal{S} = 0$ in Eq. (8.4.13) results in Eq. (8.4.7) as expected. However, it is important to stress that the integrals α, β, j, k, and l do not have the same values according to whether the orthogonalized or natural magnetic orbitals are used as a basis set. The comparison of the two expressions for J in (8.4.7) and (8.4.13) is interesting in so far as it allows the emphasis of the two alternative ways of describing the isotropic interaction in the active-electron approximation. When using orthogonalized magnetic orbitals, the antiferromagnetic interaction appears to be due entirely to the GC–CTC interaction. For very large U the antiferromagnetic contribution in $1/U$ completely vanishes and J reduces to $2k$. In contrast, when natural magnetic orbitals are used, the antiferromagnetic contribution also arises from the GC and is driven by the overlap $\mathcal{S}$ between the magnetic orbitals. When $\mathcal{S}$ is zero, or close to zero, the dominant term is $2k$ and the interaction is expected to be ferromagnetic. The role played by the overlap $\mathcal{S}$ makes this latter approach particularly appropriate in predicting qualitatively the nature of the interaction in a novel compound. Orthogonalized and natural magnetic orbital approaches are compared in Figure 8.2.

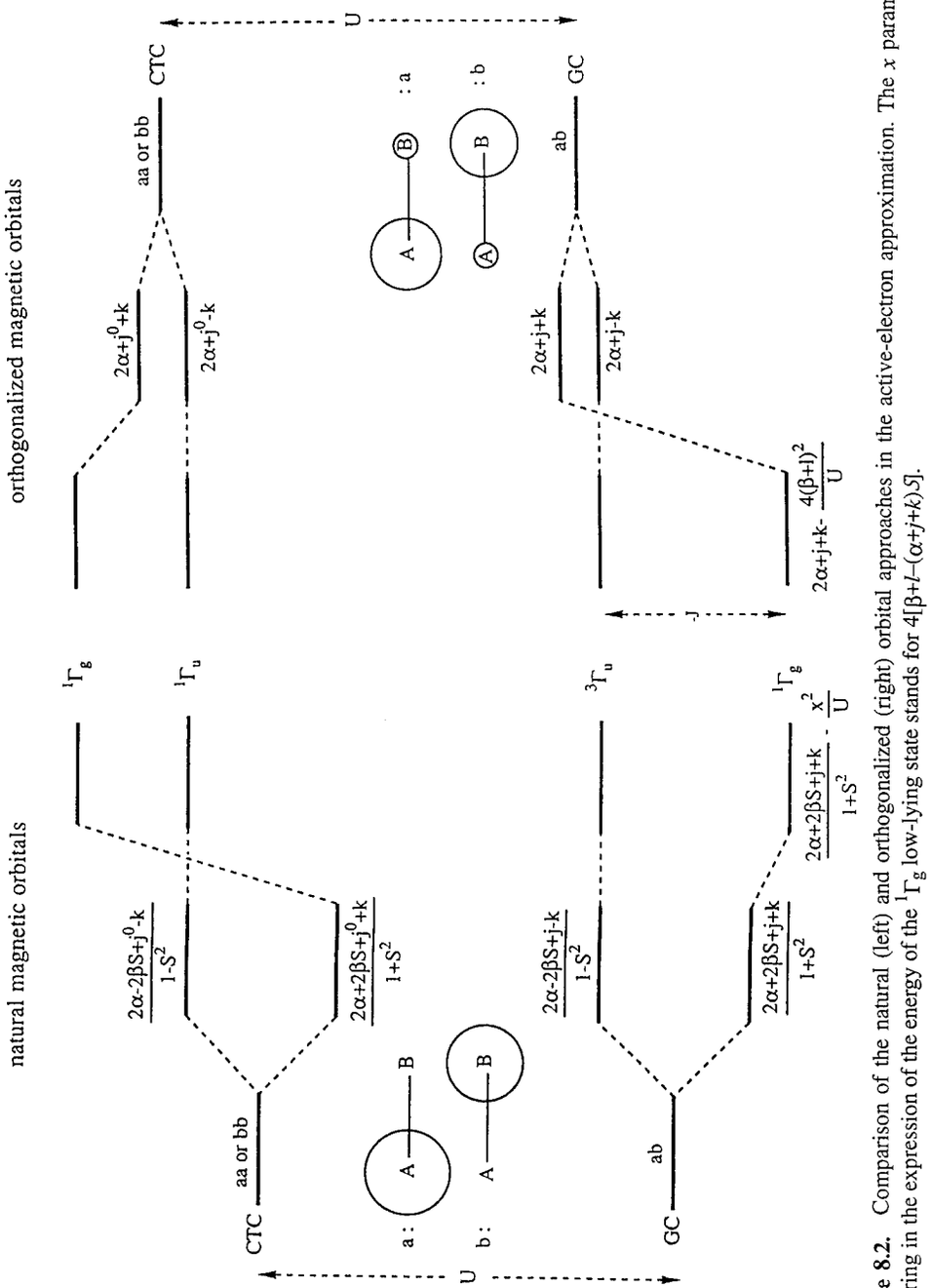


Figure 8.2. Comparison of the natural (left) and orthogonalized (right) orbital approaches in the active-electron approximation. The x parameter appearing in the expression of the energy of the $^1\Gamma_g$ low-lying state stands for $4[\beta+J-(\alpha+j+k).S]$.

8.5 Beyond the Active-Electron Approximation: Ab Initio Calculation of the Singlet-Triplet Energy Gap

In this section we do not intend to review all the calculations that go beyond the active-electron approximation, but rather to focus on one of these that, to us, seems to be particularly important. This calculation uses a perturbational approach to the configuration interaction. The pioneering work along this line was carried out by de Loth et al. [8.6], following an idea developed by Malrieu who showed that it was not necessary to calculate the total correlation energy for both the singlet and triplet states when using ordinary Rayleigh–Schrödinger perturbation theory [8.7]. Instead, it was possible to derive direct expressions for the energy gap J between these two states. We present the main steps and results of this approach.

First, an open-shell calculation is carried out with the main Fock operator, according to Nesbet [8.8]:

$$\mathbf{F} = \mathbf{h} + \sum_i (2\mathbf{J}_i + \mathbf{K}_i) + (\mathbf{J}_1 - \mathbf{K}_1/2) + (\mathbf{J}_2 - \mathbf{K}_2/2) \tag{8.5.1}$$

where, as usual, $\mathbf{h}$ is the monoelectronic part of the Hamiltonian, $\mathbf{J}$ and $\mathbf{K}$ the two-electron Coulomb and exchange operators, respectively, defined according to

$$\mathbf{J}_k\varphi_l(1) = [\int \varphi_k^2(2)(1/r_{12})\, d\tau]\varphi_l(1)$$

$$\mathbf{K}_k\varphi_l(1) = [\int \varphi_k(2)\varphi_l(2)(1/r_{12})\, d\tau]\varphi_k(1) \tag{8.5.2}$$

The subscript i in (8.5.1) refers to closed-shell molecular orbitals, and φ_1 and φ_2 to the singly occupied molecular orbitals in the triplet state; a and b are the magnetic orbitals defined in (8.3.1). The calculation is of the spin-restricted Hartree–Fock type, which means that the spin orbitals i and $\bar{i}$ have the same energy. A more thorough comparison between spin-restricted and spin-unrestricted methods will appear in Chapter 12.

To zeroth order, the wavefunctions associated with the low-lying states are

$$\Psi_S = (|a\bar{b}| + |b\bar{a}|)/\sqrt{2}$$

$$\Psi_T = (|a\bar{b}| - |b\bar{a}|)/\sqrt{2} \tag{8.5.3}$$

where the closed shells have been omitted. If $\mathbf{H}$ stands for the electrostatic Hamiltonian of the system including the perturbative correlation energy, the zeroth-order S–T splitting $J^{(0)}$ is

$$J^{(0)} = \langle\Psi_S^{(0)}|\mathbf{H}|\Psi_S^{(0)}\rangle - \langle\Psi_T^{(0)}|\mathbf{H}|\Psi_T^{(0)}\rangle = 2k \tag{8.5.4}$$

k being the two-electron exchange integral defined in (8.4.2). $J^{(0)}$ is equivalent to the ferromagnetic contribution in the active-electron approximation. This zeroth-order contribution has been called the potential exchange by Anderson [8.3].

Second-order corrections on the singlet and triplet states are

Table 8.1. Second-Order Contributions and Corrections in the Ab Initio Calculation of the Singlet–Triplet Energy Gap

Contribution	|I>	Correction
Kinetic exchange	$\|a\bar{a}\|$	$\frac{4F_{ab}^2}{\Delta E_{a\to b}}$
Particle polarization	$\|a\bar{i^*}\|$	$2\Sigma_{i^*}\left[\frac{F_{ai^*}(ab,i^*b)}{\Delta E_{a\to i^*}}+\frac{F_{bi^*}(ba,i^*a)}{\Delta E_{b\to i^*}}\right]$
Hole polarization	$\|\bar{b}ia\bar{b}\|$	$2\Sigma_i\left[\frac{F_{ai^*}(ab,ib)}{\Delta E_{i\to a}}+\frac{F_{bi}(ba,ia)}{\Delta E_{i\to b}}\right]$
Double-spin polarization	$\|\bar{i^*}\bar{i}ab\|$	$-4\sum_i\sum_{i^*}\frac{(ia,i^*a)(ib,i^*b)}{\Delta E_{i\to i^*}}$
Ligand→metal charge transfer	$\|\bar{b}ij\bar{a}\bar{b}\|$	$2\sum_i\sum_j\frac{(ai,bj)(aj,bi)}{\Delta E_{i\to a}+\Delta E_{j\to b}}$
Metal→ligand charge transfer	$\|i^*\bar{j^*}\|$	$2\sum_{i^*}\sum_{j^*}\frac{(ai^*,bj^*)(aj^*,bi^*)}{\Delta E_{a\to i^*}+\Delta E_{b\to j^*}}$
Kinetic exchange + polarization	$\|\bar{i^*}i\bar{b}\bar{b}\|$	$\sum_i\sum_{i^*}\{\frac{4(ab,ii^*)}{\Delta E_{a\to b}+\Delta E_{i\to i^*}}\times 2[(ab,ii^*)-(ai,bi^*)-(ai^*,bi)]\}$

$$E_S^{(2)}=\sum_{I\neq\Psi_S^{(0)}}<\Psi_S^{(0)}|\mathbf{H}|I>^2/(E^{(0)}-E_I^{(0)}) \tag{8.5.5}$$

$$E_T^{(2)}=\sum_{I\neq\Psi_T^{(0)}}<\Psi_T^{(0)}|\mathbf{H}|I>^2/(E^{(0)}-E_I^{(0)}) \tag{8.5.6}$$

$E^{(0)}$ is the energy associated with the determinant $|a\bar{b}|$ (or $|b\bar{a}|$), and $E_I^{(0)}$ is the unperturbed energy of the excited configuration represented by a single determinant $|I>$. The second-order correction $J^{(2)}$ to the S–T splitting is then

$$J^{(2)}=E_S^{(2)}-E_T^{(2)} \tag{8.5.7}$$

$$=2\sum_{I\neq\Psi_S^{(0)},\Psi_T^{(0)}}<|a\bar{b}|\,|\mathbf{H}|I><I|\mathbf{H}|\,|b\bar{a}>/(E^{(0)}-E_I^{(0)}) \tag{8.5.8}$$

The determinants $|I>$ must interact with both $|a\bar{b}|$ and $|\bar{b}a|$. There are thus fewer configurations playing a role in the S–T energy gap than in the energies of the singlet and triplet states. These configurations can be determined by diagrammatic methods or Slater rules. The determinants $|I>$ must differ, at most, by two spin-orbitals from

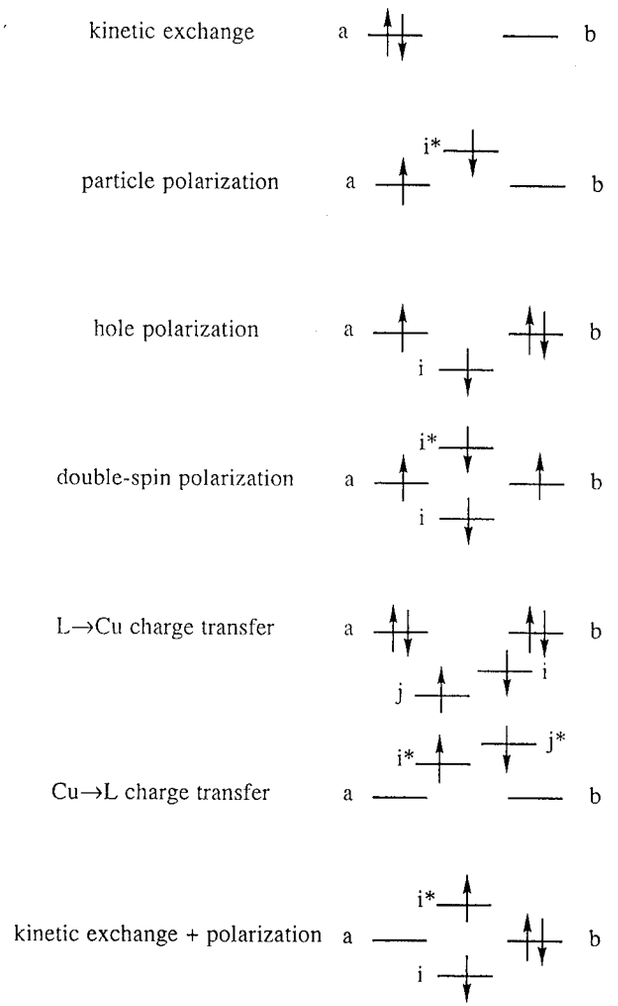


Figure 8.3. Schematic representation of the $|I>$ determinants that interact with both $|a\bar{b}|$ and $|b\bar{a}|$ in the second-order correction to the S–T splitting.

both $|a\bar{b}|$ and $|b\bar{a}|$ to give a nonzero numerator in Eq. (8.5.8). Table 8.1 and Figure 8.3 give a schematic representation of all such determinants $<I|$, along with the name of the contribution and the expression of the second-order correction. In Table 8.1 $i,j\ldots$ note doubly occupied molecular orbitals, and $i^*,j^*\ldots$ vacant molecular orbitals in the ground state. The integral (ij,kl) stands for $<i(1)j(2)\,|r_{12}^{-1}|\,k(1)l(2)>$. The energy differences $E^{(0)}-E_I^{(0)}$ in (8.5.8) reduce one to negative transitions $\Delta E_{p\to q}$, and are calculated as the difference between eigenvalues of the SCF Hamiltonian:

$$\Delta E_{p\to q}=e_p-e_q \tag{8.5.9}$$

except for $\Delta E_{a\to b}$ which would be equal to zero. $\Delta E_{a\to b}$ is actually calculated as the difference between the mean values of the exact Hamiltonian:

$$\Delta E_{a\to b} = <|a\bar{b}|\,|\mathbf{H}|\,|a\bar{b}|> - <|a\bar{a}|\,|\mathbf{H}|\,|a\bar{a}|> \tag{8.5.10}$$

which may be written as

$$\Delta E_{a\to b} = -U = -(j^0 - j) \tag{8.5.11}$$

using the notations of Section 8.4.

The so-called kinetic exchange term generally dominates among the second-order perturbation corrections. This is true, in particular, when the doubly occupied molecular orbitals i are low in energy with respect to φ_1 and φ_2. This term is strictly equivalent to the antiferromagnetic contribution J_{AF}, which was found in the active-electron approximation, provided that closed-shell molecular orbitals of low energy noted i are neglected. Indeed, one has

$$\begin{aligned} a\mathbf{J}_1 b &= l + k \\ a\mathbf{J}_2 b &= l - k \\ a\mathbf{K}_1 b &= l + j \\ a\mathbf{K}_2 b &= l - j \end{aligned} \tag{8.5.12}$$

so that

$$F_{ab} = <a|\mathbf{F}|b> = \beta + l \tag{8.5.13}$$

F_{ab} may also be written as

$$F_{ab} = [<\varphi_1|\mathbf{F}|\varphi_1> - <\varphi_2|\mathbf{F}|\varphi_2>]/2 = (e_1 - e_2)/2 \tag{8.5.14}$$

where e_1 and e_2 are the SCF energies of φ_1 and φ_2, respectively. These are determined by an open-shell process. The kinetic exchange (KE) term, or antiferromagnetic contribution, then becomes

$$\Delta E_{KE}^{(2)} = -\frac{(e_1 - e_2)^2}{j^0 - j} \tag{8.5.15}$$

Equation (8.5.15) was first derived by Hay et al. [8.9], and used for qualitative correlations between magnetic properties and structures. Examples of its use will be discussed in the next section.

In a few cases fourth-order terms have also been considered. Their expressions are rather complicated and we prefer not to give them here.

This *ab initio* perturbational approach has been applied to several symmetrical copper(II) dinuclear compounds [8.10–8.13]. Examples will be presented in Section 8.6. Its extension to other symmetrical pairs does not present any conceptual difficulty provided that the whole symmetry of the system is high enough for all singly occupied molecular orbitals to transform as different irreducible representations of the molecular symmetry group. This happens, for instance, for nickel(II) dinuclear compounds resulting from the sharing of an edge between two octahedra. Otherwise, cross-terms appear at the first order, which might make the treatment very tedious or even intractable.

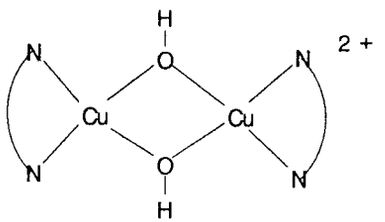
To finish this brief discussion, we will stress that the weak point of this perturbational approach is convergence. As a matter of fact, for most compounds J is found experimentally to be negative whereas the zeroth-order term in the perturbational expansion of the configuration interaction, $2k$, is obviously positive. This is not a fully satisfying situation. Moreover, it is not perfectly clear whether it is legitimate to stop the expansion of J at the second-order terms. Actually, the status of the fourth- and higher-order terms is not yet well understood.

8.6 Theoretical Interpretation of Some Experimental Results

In this section we would like to review and discuss some of the experimental studies concerning the isotropic interaction between two local doublet states within a molecular unit. These studies do not all have the same status; some are only qualitative whereas some correspond to *ab initio* calculations. The examples have been chosen in a way which introduces a new idea with each new example.

8.6.1 Influence of Geometry in Bibridged Copper(II) Dimers

The first magnetostructural correlation dealing with copper(II) dinuclear compounds was proposed by Hatfield and Hodgson [8.14]. This correlation concerns planar hydroxo-bridged copper(II) dimers of the type



in which the terminal groups are bidentate nitrogen-containing ligands. Twelve compounds of this kind have been structurally characterized to date, and magnetically investigated. These compounds differ by the nature of the terminal ligands and/or of the counteranions. Hatfield and Hodgson have shown that the Cu—N and Cu—O bond lengths along the series are almost constant [1.98(2) Å for Cu—N and 1.92(2) Å for Cu—O]. The main structural changes concern the bridging angle CuOCu = α and the Cu...Cu separation. This α angle varies from 95.6° to 104.1° along the series, and the S–T energy gap J is found to vary in parallel from +172 to –509 cm^{-1}. From the

Table 8.2. Bridging Angle and S–T Energy Gap in Planar Hydroxo-bridged Copper(II) Dimers

Compound[a]	CuoCu (degrees)	J (cm^{-1})
$[Cu(bipy)OH]_2(NO_3)_2$	95.5	+ 172
$[Cu(bipy)OH]_2(ClO_4)_2$	96.6	+ 93
$[Cu(bipy)OH]_2(SO_4)\cdot 5H_2O$	97	+ 49
β-$[Cu(dmaep)OH]_2(ClO_4)_2$	98.4	− 2.3
$[Cu(eaep)OH]_2(ClO_4)_2$	98.8 and 99.5	−130
$[Cu(2miz)OH]_2(ClO_4)_2\cdot 2H_2O$	99.4	− 175
α-$[Cu(dmaep)OH]_2(ClO_4)_2$	100.4	− 200
$[Cu(tmen)OH]_2(ClO_4)_2$	102.3	− 306
$[Cu(tmen)OH]_2(NO_3)_2$	101.9	− 367
α-$[Cu(teen)OH]_2(ClO_4)_2$	103.0	− 410
β–$[Cu(teen)OH]_2(ClO_4)_2$	103.7	− 469
$[Cu(tmen)OH]_2Br_2$	104.1	− 509

Source: From ref. [8.15].

[a]bipy, 2,2′-bipyridine; dmaep, 2-(2-dimethylaminoethyl)pyridine; eaep, 2-(2-ethylaminoethyl)pyridine; 2miz, 2-methylimidazole; tmen, *N,N,N′,N′*-tetramethylethylenediamine; teen = *N,N,N′,N′*-tetraethylethylenediamine.

experimental data collected in Table 8.2 Hatfield and Hodgson deduced a linear correlation between J and α [8.14]. This correlation

$$J(cm^{-1}) = -74\ \alpha(\text{degrees}) + 7270 \tag{8.6.1}$$

predicts that J is equal to zero for $\alpha = 97.5°$. For $\alpha > 97.5°$ the ground state is a singlet and for $\alpha < 97.5°$ the ground state is a triplet. It appears from Eq. (8.6.1) that a change of one degree in bridging angle results in a variation of about 74 cm^{-1} in S–T separation.

A semiquantitative interpretation of these data may be obtained in the framework of the active-electron approximation [8.9, 8.16]. This approximation is assumed to be valid since the $2p$ valence orbitals of the bridging oxygen atoms are low in energy with respect to the d copper orbitals. Replacing the bridging oxygen atoms by less electronegative atoms like sulfur could make this approximation no longer valid. In the present case J is expressed as

$$J = 2k - \frac{(e_1 - e_2)^2}{U} \tag{8.6.2}$$

$U = j^0 - j$ is not thought to change significantly when α varies, such that the magnitude of the antiferromagnetic contribution is governed by $(e_1 - e_2)^2$, e_1 and e_2 being the energies of the two singly occupied molecular orbitals in the triplet state. Referring to the D_{2h} molecular symmetry one of the orbitals transforms as b_{1g}, and the other as b_{2u}, as schematized in Figure 8.4. The energy variation of these molecular orbitals when

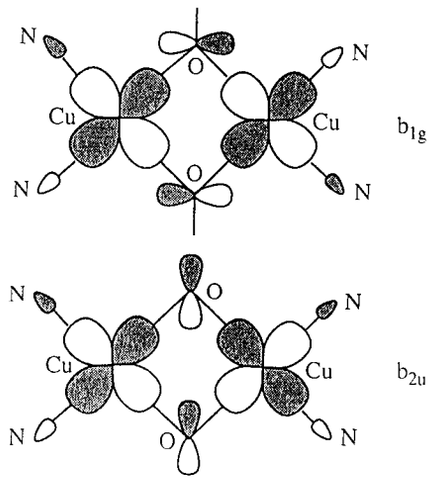


Figure 8.4. b_{1g} and b_{2u} molecular orbitals for di-μ-hydroxo copper(II) dimers. Those two molecular orbitals are singly occupied in the low-lying triplet state.

α varies is essentially governed by the $3d_{xy}$(copper)–$2p$(oxygen) antibonding overlaps. For α_0 value of α close to 90° the overlaps $\langle d_{xy} | p_x \rangle$ in b_{1g} and $\langle d_{xy} | p_y \rangle$ in b_{2u} are equal and negative, and the two molecular orbitals have the same energy; α_0 is slightly larger than 90° owing to the $2s$–$2p$ mixing. When α increases, $\langle d_{xy} | p_x \rangle$ becomes more negative and $\langle d_{xy} | p_y \rangle$ less negative. Therefore, the energy difference $e(b_{1g}) - e(b_{2u})$ increases together with the stabilization of the singlet state. For a peculiar value of α ($> \alpha_0$) the antiferromagnetic contribution $-(e_1 - e_2)^2/U$ is exactly compensated by the ferromagnetic contribution $2k$, and the S–T energy gap is zero. This peculiar value is found equal to 97.5° in the series of compounds of Table 8.2. When α decreases from this value, $2k$ predominates and the triplet state becomes the ground state. The variations of $e(b_{1g})$ and $e(b_{2u})$ versus α are schematized in Figure 8.5. In this interpretation it is implicitly assumed that $2k$ is weakly sensitive to small structural changes. It is worth noting that this approach suggests that, for α values much smaller than 90°, the singlet state would be again stabilized. Experimentally, compounds of this sort are probably inaccessible due to the Cu(II)–Cu(II) electrostatic repulsion.

The situation corresponding to $\alpha = 97.5°$ is interesting. The S–T energy gap is then zero. This does not mean that there is no interaction between the two local doublets. Actually, for this peculiar geometry ferro- and antiferromagnetic contributions compensate each other and the singlet and triplet states arising from this interaction are accidentally degenerate.

The same type of qualitative approach may be used to understand how the S–T energy gap would vary if the molecule was bent along the direction joining the two bridging oxygen atoms [8.16, 8.17]. When the dihedral angle δ between the two CuO_2 planes is no longer equal to 180°, the molecular symmetry becomes C_{2v} and the molecular orbitals, b_{1g} and b_{2u} in D_{2h}, transform as a_2 and b_1, respectively. When δ

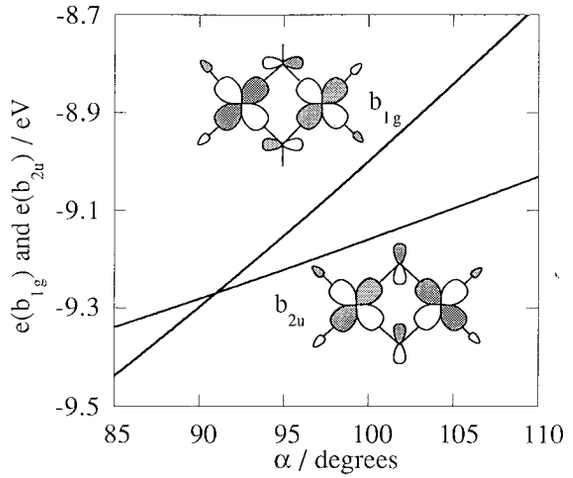


Figure 8.5. Variations versus the bridging angle α of the energies of the b_{1g} and b_{2u} molecular orbitals for di-μ-hydroxo copper(II) dimers. The energies result from an Extended Hückel calculation.

decreases with respect to 180°, the $<d_{xy}|p_x>$ overlap in b_1 is unchanged whereas the $<d_{xy}|p_y>$ overlap in a_2 becomes less negative. It follows that the energy $e(b_1)$ remains almost constant and the energy $e(a_2)$ decreases as shown in Figure 8.6. The $e(a_2) - e(b_1)$ gap, therefore, decreases when δ decreases and the interaction becomes less antiferromagnetic. Experimental data have confirmed these predictions.

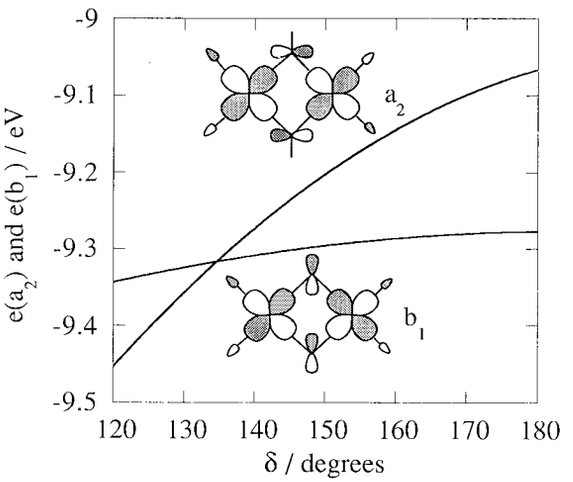


Figure 8.6. Variations versus the dihedral angle δ of the energies of the a_2 and b_1 molecular orbitals for di-μ-hydroxo copper(II) dimers. δ is the dihedral angle between the two CuOO planes, the distance O—O remaining constant. The energies result from an Extended Hückel calculation.

Table 8.3. *Ab Initio* Calculation of the S–T Splitting J for $[Cu(tmen)OH]_2Br_2$ with the Actual Geometry (Left Column) and with α = 95° (Right Column)

	Value (cm^{-1})	
Contribution	α = 104.1°	α = 95°
Potential exchange	+ 1380	+ 1275
Kinetic exchange	− 1036	− 431
Double-spin polarization	+ 11	+ 40
Ligand→metal charge transfer	− 477	− 492
Metal→ligand charge transfer	− 35	− 35
Kinetic exchange + polarization	− 236	− 260
Calculated value	− 393	+ 97
Experimental values	− 509	+ 161[a]

[a]This value is deduced from Eq. (8.6.1.) (from ref. 8.10).

Daudey et al. attempted to justify the correlation $J = f(\alpha)$ in (8.6.1) using the *ab initio* perturbational approach described in Section 8.5 [8.10]. They considered two compounds: $[Cu(tmen)OH]Br_2$, which has the largest bridging angle and presents the strongest antiferromagnetic interaction, and a hypothetical compound deduced from the actual compound by modification of the value of the bridging angle to α = 95°. The values of the different contributions to J for both compounds are compared in Table 8.3. It clearly emerges from this table that the variation of the so-called kinetic exchange term $-(e_1 - e_2)^2/U$ is actually the main cause of the J variation. It also emerges that the other second-order terms are far from negligible, even if they are less sensitive to small structural variations. Thus, without them, J would be found positive for α = 104.1° whereas it is experimentally strongly negative. Table 8.3 provides interesting information on the possibilities and the limits of the active-electron approximation. We should note that in the calculation particle and hole polarization terms were not considered.

The same rationalization of Hatfield and Hodgson's correlation may be obtained in a natural magnetic orbital approach [8.18]. The stabilization of the singlet state is then attributed to $4\beta S$ where S is the overlap integral between two magnetic orbitals of the type schematized in Figure 8.7. It is straightforward to observe that for $\alpha = \alpha_0$ close to 90°, S is equal to zero. The two magnetic orbitals are then said to be accidentally orthogonal. The word "accidentally" here means that the orthogonality is not due to the fact that these magnetic orbitals have different symmetries. They are obviously symmetry related; both transform as the b_1 irreducible representation of the C_{2v} site group. For $\alpha = \alpha_0$ positive and negative zones of the overlap density ρ(1) between the magnetic orbitals a and b, defined as

$$\rho(1) = a(1)b(1) \qquad (8.6.3)$$

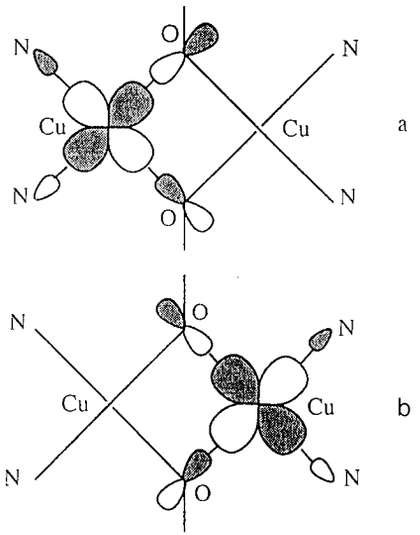


Figure 8.7. Magnetic orbitals *a* and *b* localized in the left and right fragments, respectively, for a di-μ-hydroxo copper(II) dimer.

compensate each other such that

$$S = \int_{\text{space}} \rho(1) d\tau(1) = 0 \qquad (8.6.4)$$

We will study the case where the magnetic orbitals are not accidentally but strictly orthogonal in section 8.6.4.

8.6.2 Orbital Complementarity and Countercomplementarity

Up to now we have focused on copper(II) dinuclear species with two identical bridges. The situation is obviously more complicated when the two bridges are no longer identical. This problem was tackled by Nishida et al. [8.19], and McKee et al. [8.20, 8.21], who demonstrated that in the active-electron approximation the two bridges may either add or counterbalance their effects.

To introduce this concept we consider a planar copper(II) dinuclear compound with an hydroxo (or alkoxo) bridge on one side and a carboxylato bridge on the other. The bridging angle on the hydroxo (or alkoxo) side is much larger than 90°. The symmetry of the system is then C_s with a mirror-plane perpendicular to the molecular plane as a unique symmetry element. We assume that only the highest occupied molecular orbital (HOMO) of the carboxylato is involved in the problem. This HOMO is symmetric with respect to the mirror-plane. The singly occupied molecular orbitals in the triplet state, transforming as a' and a'', are schematized in Figure 8.8. In the absence of the carboxylato group the molecular orbital a'' would be higher in energy, due to the large value of the CuOCu bridging angle. However, the HOMO of the carboxylato raises

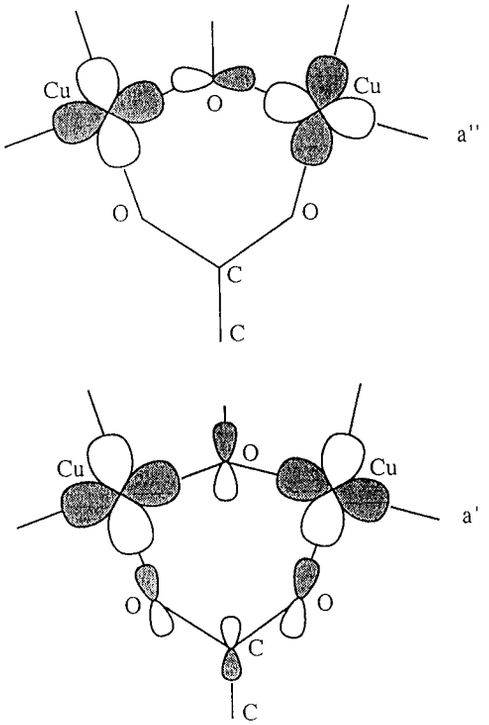
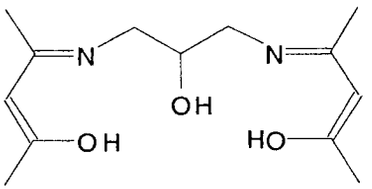


Figure 8.8. a' and a'' molecular orbitals for a μ-hydroxo-μ-carboxylato copper(II) dinuclear compound. The energies of these molecular orbitals are close to each other because of the anticomplementarity of the two bridging ligands.

the energy of a', which in turn diminishes the $e(a'') - e(a')$ energy gap. It follows that the carboxylato exerts a countercomplementarity effect and the magnitude of the antiferromagnetic interaction is expected to be small in spite of the large CuOCu angle. This concept of countercomplementarity has been invoked for the compound $[Cu_2L(CH_3OO)]$, the structure of which is shown in Figure 8.9. L^{3-} is the binucleating ligand derived from the Schiff base obtained by condensation of one molecule of 1,3-diaminopropan-2-ol with two molecules of acetylacetone:



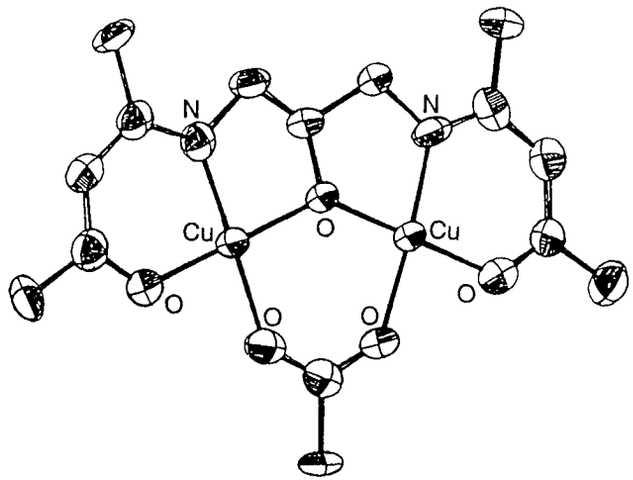


Figure 8.9. Molecular structure of $[Cu_2L(CH_3COO)]$. H_3L is the Schiff base derived from the condensation of one molecule of 1,3-diaminopropan-2-ol with two molecules of acetylacetone (from ref. [8.19]).

The CuOCu angle is very large, at 133.3°, and the S–T splitting is reported as $J = -165$ cm^{-1}. A rather similar compound but without carboxylato bridge presents an S–T splitting $J = -635$ cm^{-1}. Its CuOCu bridging angle is equal to 137.7°.

If carboxylato is replaced by another bridge, the HOMO of which combines with the a'' molecular orbital and not with the a', the $e(a'') - e(a')$ energy gap is enhanced

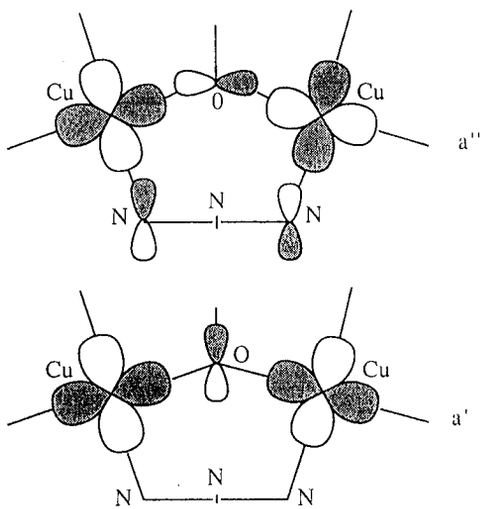


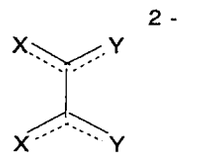
Figure 8.10. a' and a'' molecular orbitals for a μ-hydroxo-μ-1,3-azido copper(II) dinuclear compound. The energy of a'' is much higher than that of a' because of the complementarity of the two bridging ligands.

together with the stabilization of the singlet state. This happens, for instance, for N_3^- bridging in an end-to-end fashion. The HOMO of N_3^- then has π_u symmetry, referring to the $D_{\infty h}$ group, as schematized in Figure 8.10. In a μ-alkoxo-μ-1,3-azido copper(II) dinuclear unit, the singlet state has been found to be stabilized by more than 1000 cm^{-1} with respect to the triplet state [8.21].

8.6.3 Interaction between Magnetic Centers Far Apart from Each Other Through Extended Bridges

The key factor determining the magnitude of the antiferromagnetic interaction in a symmetrical copper(II) dinuclear compound is the energy gap $|e_1 - e_2|$ between singly occupied molecular orbitals. This energy gap may be large even if the Cu...Cu separation is significant, provided that certain requirements are fulfilled. The possibility of achieving strong interactions between magnetic centers far away from each other through extended bridges actually appears as a specificity of molecular chemistry.

The dinuclear copper(II) compounds with bisbidentate bridging ligands of the type



X = Y = O: oxalato
X = Y = S: tetrathiooxalato
X = O, Y = NR: oxamido
X = S, Y = NR: dithiooxamido

have been more particularly investigated, both experimentally and theoretically [8.22–8.24]. We first examine the dinuclear cation $[tmen(H_2O)Cu(ox)Cu(H_2O)tmen]^{2+}$, which was already discussed in Section 8.3. The Cu...Cu separation through the oxalato group is equal to 5.14 Å, but nevertheless the copper(II) ions strongly interact; J is equal to -384.5 cm^{-1}. The dinuclear unit shown in Figure 8.11 is centrosymmetric with a quasi-planar Cu(ox)Cu bridging network. The environment of each copper(II) ion is square pyramidal with a water molecule weakly bonded to the metal in the apical position. Let us also note that the C–C bond length is equal to 1.54 Å, which corresponds to a single bond. The C–O bonds, on the other hand, have some double-bond character since their average length is 1.24 Å.

The concept of natural magnetic orbital is convenient to understand qualitatively why the S–T splitting is large in spite of the Cu...Cu distance. Indeed, as mentioned in Section 8.3, the mononuclear species Cu(tmen)(ox) exists, and the natural magnetic orbital a may be defined as the highest singly occupied molecular orbital for this compound. Because of the square pyramidal environment of the metal, and to the

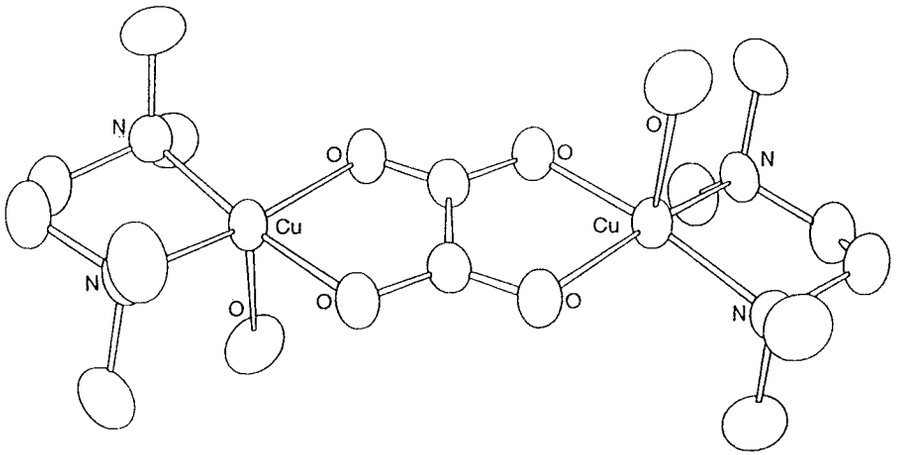


Figure 8.11. Molecular structure of the dinuclear cation $[tmen(H_2O)Cu(ox)Cu(H_2O)tmen]^{2+}$ with tmen = N,N,N′,N′-tetramethylethylenediamine and ox = oxalato (from ref. [8.24]).

quasi-coplanarity of the basal plane and the oxalato group, a is built from the d_{xy} metal orbital. It is delocalized toward the oxygen atoms of the bridge as shown in Figure 8.12. In the dinuclear unit the two magnetic orbitals are related through the inversion center. They are properly oriented to interact efficiently in the region of the bridge.

The interpretation above is certainly oversimplified and does not account for all the subtlety of the interaction through the oxalato bridge. Nevertheless, it contains at least two key ideas. One of these ideas is topology; the two magnetic orbitals are located in the same plane as the bridging ligand. The second idea deals with the energy; each magnetic orbital is strongly delocalized toward the oxygen atoms of the bridge owing to the relatively weak energy gap between the HOMOs of $C_2O_4^{2-}$ and the d metal

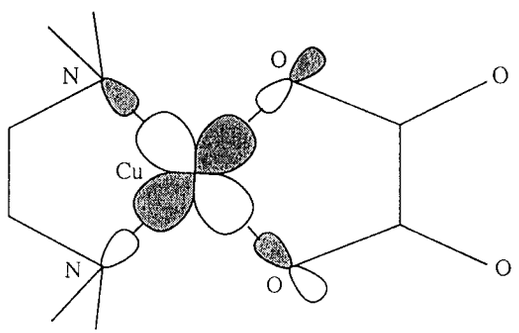


Figure 8.12. Highest singly occupied molecular orbital for Cu(tmen)(ox) with tmen = N,N,N′,N′-tetramethylethylenediamine and ox = oxalato. This molecular orbital may be viewed as one of the two magnetic orbitals for the dinuclear cation $[tmen(H_2O)Cu(ox)Cu(H_2O)tmen]^{2+}$.

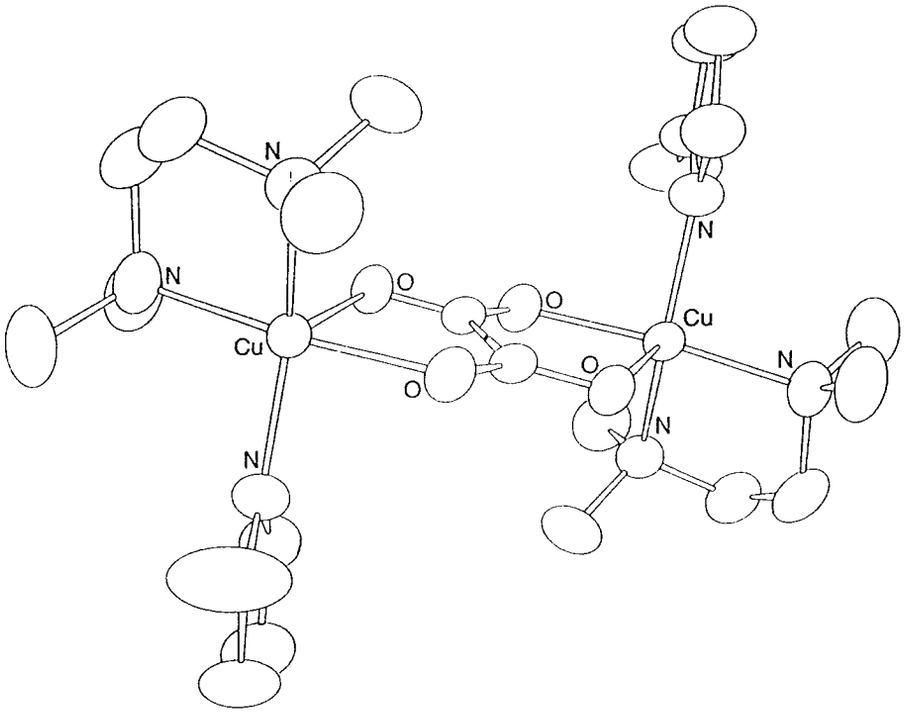


Figure 8.13. Molecular structure of $[tmen(2Meim)Cu(ox)Cu(2Meim)tmen]^{2+}$ with tmen = N,N,N′,N′-tetramethylethylenediamine, 2meim = 2-methylimidazole and ox = oxalato (from ref. [8.24]).

orbitals. The former idea, topology, is substantiated by the following result: the two water molecules in $[tmen(H_2O)Cu(ox)Cu(H_2O)tmen]^{2+}$ can be replaced by some nitrogen-containing ligands like 2Meim = 2-methylimidazole. The structure of $[tmen(2Meim)Cu(ox)Cu(2Meim)tmen]^{2+}$ is represented in Figure 8.13. The inversion center is retained, but the environment of the metal is strongly modified. The four nearest neighbors of copper(II) are the nitrogen atoms of tmen and 2Meim and only one of the oxygen atoms of the bridge. In such an environment, each magnetic orbital undergoes a reversal with respect to the situation encountered in $[tmen(H_2O)Cu(ox)Cu(H_2O)tmen]^{2+}$. Instead of being located in the same plane as the bridge, it is located in a plane perpendicular to this bridge. The two symmetry-related magnetic orbitals are now unfavorably oriented to interact, as shown in Figure 8.14, and the S–T energy gap is dramatically reduced; it is equal to $-13.8\ cm^{-1}$.

The role of the energy factor is emphasized by the value of the S–T splitting in a dinuclear species where oxalato is replaced by tetrathiooxalato $C_2S_4^{2-}$. This compound was described by Vicente et al. [8.25] and contains a practically planar $Cu(C_2S_4)Cu$ network. The Cu...Cu separation is equal to 6.06 Å, and the S–T splitting is so large

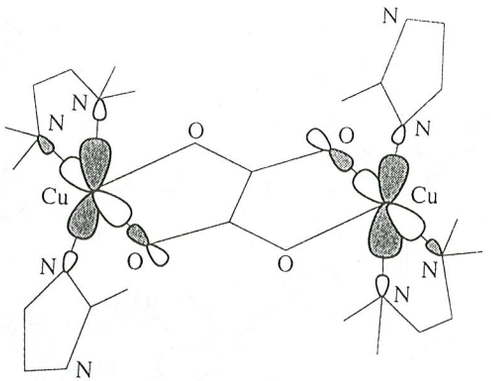


Figure 8.14. Schematic representation of the two magnetic orbitals in [tmen(2Meim)Cu(ox)-Cu(2Meim)tmen]$^{2+}$.

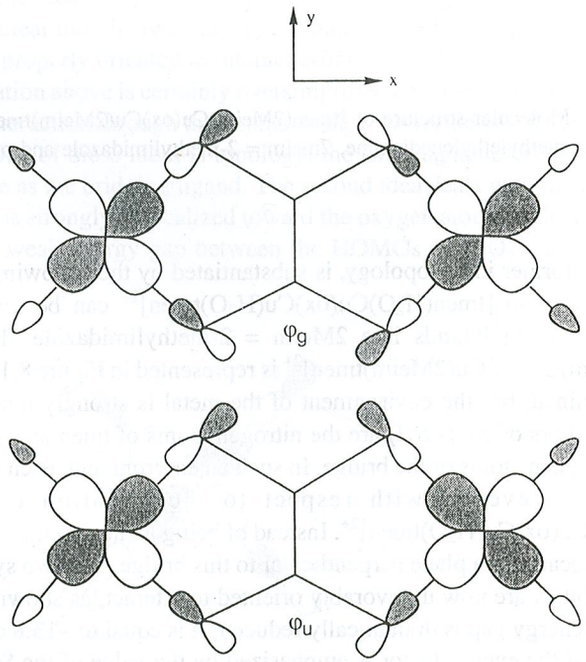


Figure 8.15. φ_g and φ_u molecular orbitals for a planar μ-oxalato copper(II) dinuclear species. Those two molecular orbitals are singly occupied in the low-lying triplet state.

that even at room temperature the excited triplet state appears to be fully depopulated; $-J$ is above 800 cm^{-1}. The enhancement of the interaction, compared to the oxalato-bridged unit of Figure 8.11, may be attributed to the fact that the sulfur $3s$ and $3p$ valence orbitals are higher in energy than the oxygen $2s$ and $2p$ orbitals. This results in a reduction of the energy gap between the HOMOs of $C_2S_4^{2-}$ and the metal d orbitals. The delocalization of each magnetic orbital toward the sulfur atoms of $C_2S_4^{2-}$ is more pronounced than toward the oxygen atoms of $C_2O_4^{2-}$.

Basically, the same situation occurs when oxalato is replaced by oxamido, and then by dithiooxamido. The energies of the valence orbitals vary as $2p_O < 2p_N < 3p_S$ and the delocalization of the magnetic orbitals toward the bisbidentate bridges varies accordingly. The magnitude of the antiferromagnetic interaction thus follows the sequence oxalato < oxamido < dithiooxamido, provided that the magnetic orbitals remain localized in the plane of the bridging network [8.26, 8.27]. A dithiooxamido copper(II) derivative has been reported with a Cu...Cu separation of 5.67 Å and an S–T splitting J of –730 cm^{-1} [8.28].

An *ab initio* calculation of J in a unit modelling the cation of Figure 8.11 has been performed. Each tmen ligand is replaced by two ammonia molecules. The SCF open-shell calculation uses pseudopotentials to describe the inner-shell electrons and a basis of Slater-type atomic orbitals developed on Gaussian-type functions to describe the valence electrons. The two singly occupied molecular orbitals schematized in Figure 8.15 are symmetric and antisymmetric with regard to inversion; they are noted φ_g and φ_u, respectively. The overlap between $2p$ orbitals centered on oxygen atoms bonded to the same carbon atom is antibonding in φ_g and bonding in φ_u such that φ_g is higher in energy than φ_u. From φ_g and φ_u, and using Eq. (8.3.1), we can build the two orthogonalized magnetic orbitals. One of these is shown in Figure 8.16 in the form of electronic isodensity curves a^2 = constant, drawn in the mean molecular plane. As expected, a is constructed from the copper d_{xy} orbital and is delocalized toward the oxygen atoms of the bridge. The delocalization toward the peripheral nitrogen atoms is less pronounced. A consequence of the orthogonality of the magnetic orbitals a and b is the presence of delocalization tails that appear in a on copper atom B and its close surroundings. The values of the different contributions to the S–T splitting are given in Table 8.4. The calculation was performed without particle and hole polarization, but with inclusion of fourth-order terms, which were not discussed in Section 8.5. The theory–experiment agreement is far from perfect. It would probably be improved if the particle and hole polarization terms were taken into account, but this is not the most important point. This calculation again indicates the limits of the active-electron approximation. As for $[Cu(tmen)OH]_2Br_2$, the potential exchange $2k$ dominates the kinetic exchange $-[e(\varphi_g) - e(\varphi_u)]^2/(j^0 - j)$ while experimentally the singlet state is strongly stabilized with respect to the triplet state. It is necessary to go further than this active-electron approximation and to consider the other second-order terms in order to obtain a negative J value.

Other compounds have been described in which two local doublet states interact efficiently, in spite of a large separation between the magnetic centers. We would like to mention two spectacular results along this line. The former, reported by Chaudhuri et al. [8.29], deals with the compound $[LCu(H_2O)(terephthalato)(H_2O)CuL](ClO_4)_2$,

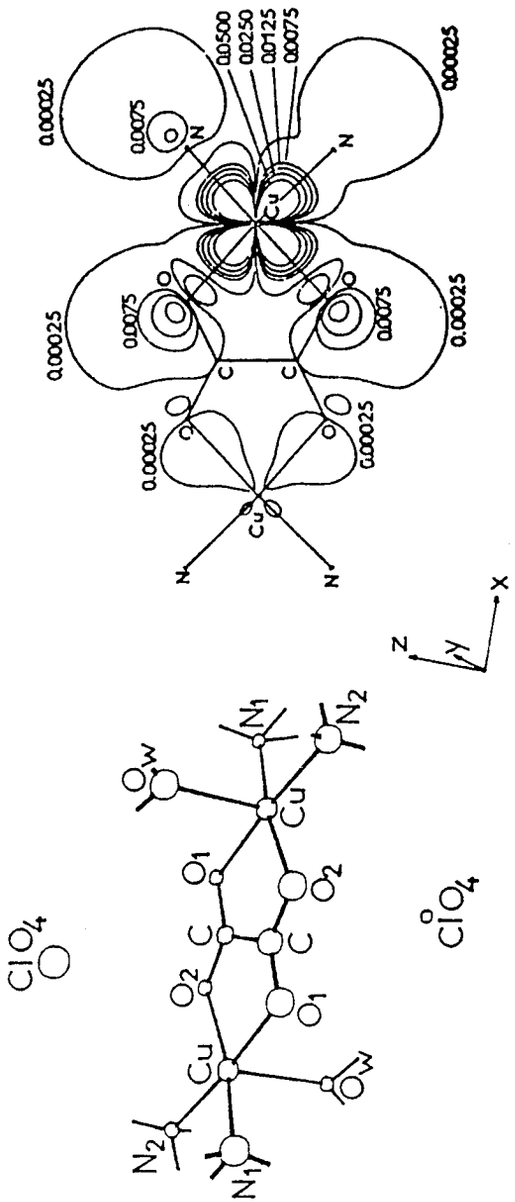


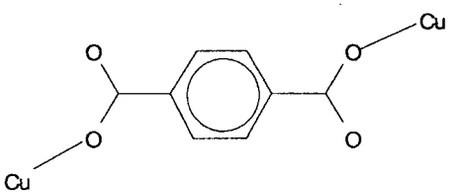
Figure 8.16. Modelized μ-oxalato copper(II) dinuclear species and electronic isodensity curves for one of the two magnetic orbitals (from ref. [8.22]).

Table 8.4. *Ab Initio* Calculation of the S–T Splitting J for $[(NH_3)_2(H_2O)Cu(ox)Cu(H_2O)(NH_3)_2]^{2+}$

Contribution	Value (cm^{-1})
j^0	152,000
j	25,400
$e(\varphi_g) - e(\varphi_u)$	7,572
Potential exchange	720
Kinetic exchange	–453
Double spin polarization	–38
Ligand→metal charge transfer	–146
Metal→ligand charge transfer	–9
Kinetic exchange + polarization	–177
Fourth-order terms	–192
Calculated value	–295
Experimental value	–385

Source: From ref. [8.23].

where L is the cyclic amine 1,4,7,-trimethyl-1,4,7-triazacyclononane. The bridging network is



The dinuclear dication has an inversion center located at the center of the bridge benzene ring. Each copper(II) has square pyramidal surroundings with an oxygen atom of the terephtalato, a water molecule and two nitrogen atoms of the triamine in the basal plane, the third nitrogen atom of L occupying the apical position. The two basal planes are strictly coplanar with the terephtalato ligand. The Cu...Cu distance is equal to 11.25 Å. The S–T splitting is reported as $J = -70\ cm^{-1}$, which is quite a surprisingly large value. The basic mechanism of the interaction is probably the same as for the systems with bisbidentate bridges discussed above. The two magnetic orbitals located in the same plane as the bridge are favorably oriented to interact.

The second result concerns the compound $Rh_2(CF_3COO)_4(tempo)_2$ in which two tempo (= 2,2,6,6-tetramethylpyridinyl-1-oxy) organic radicals interact through the singly bonded $Rh_2(CF_3COO)_4$ diamagnetic core [8.30]. The structure is shown in Figure 8.17. The spin density for each tempo group is shared between the oxygen and nitrogen atoms. The distance between the two axial nitroxyl oxygen atoms is 6.85 Å, which indicates that the separation between the barycenters of the local spin densities is larger than this distance. The S–T splitting is found equal to $J = -478\ cm^{-1}$. Since the unpaired electron of the nitroxide radical occupies a π^* orbital, the mechanism of

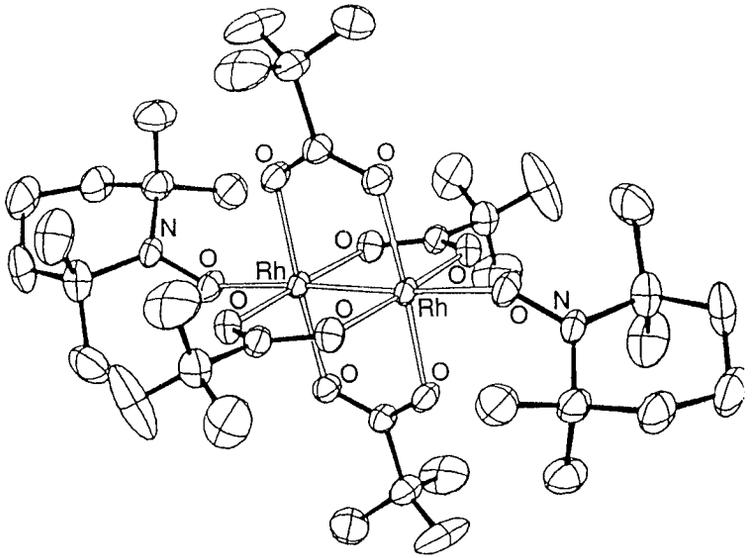


Figure 8.17. Molecular structure of $Rh_2(CF_3COO)(tempo)_2$ with tempo = 2,2,6,6-tetramethylpyridinyl-1-oxy) (from ref. [8.30]).

the interaction is not the same as in the terephthalato-bridged compound described above. This interaction is certainly propagated along the N—O—Rh—Rh—O—N linkage via π overlaps.

The results reported and discussed in this section tell us that the isotropic interaction may be propagated over very large distances. The question is then whether there is a limit beyond which the interaction completely vanishes. From a statistical analysis of data available up to 1979 Coffman and Buettner have proposed an equation relating the limit of J, noted J_{limit}, and the separation r between two interacting magnetic centers with local spins 1/2 [8.31]. This equation is

$$J_{limit}\,(cm^{-1}) = -1.35 \times 10^7 \exp[-1.80r\,(Å)] \qquad (8.6.5)$$

It is clear that for the last two examples as well as the dithiooxamido derivative mentioned before the actual S–T splitting exceeds the J_{limit} value deduced from Eq. (8.6.5). Therefore, the limit, if any, is far beyond what Eq. (8.6.5) suggests.

8.6.4 Strict Orthogonality of the Magnetic Orbitals and Ferromagnetic Interaction

In some compounds the orthogonality of the magnetic orbitals is not accidental but strict in the sense that the orbitals a and b transform as different irreducible representations of the symmetry group adapted to the system. This strict orthogonality is achieved more easily when the two interacting magnetic centers are not identical. We

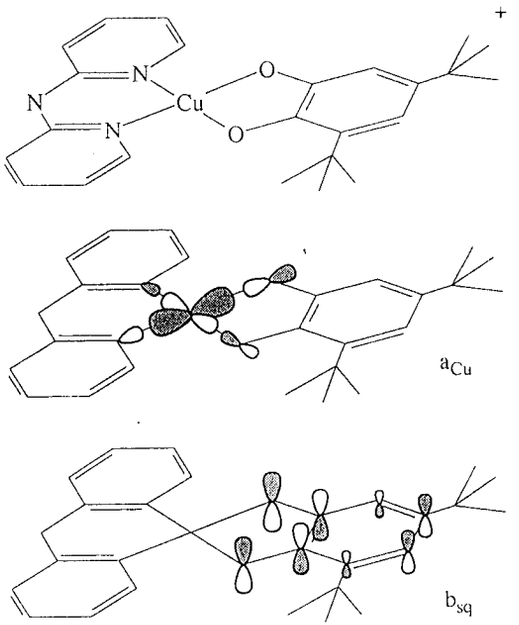


Figure 8.18. Molecular structure of $[Cu(NHpyr_2)(DTBSQ)]^+$ with $NHpyr_2$ = di-2-pyridylamine and DTBSQ = 3,5-di-*tert*-butyl-*o*-semiquinone (top), and a_{Cu} and b_{sq} magnetic orbitals centered on the copper(II) ion and the organic radical, respectively.

assume first that the active-electron approximation is valid. If so, both orthogonalized and natural magnetic orbital approaches predict that the triplet state will be the lowest. Indeed, in the orthogonalized orbital approach a and b do not transform as the same irreducible representation of the symmetry group, and the β and l integrals in Eq. (8.4.7) are zero. The S–T energy gap reduces to its ferromagnetic contribution $2k$. In the natural orbital approach S is taken to be equal to 0 in Eq. (8.4.13), and J reduces again to $2k$. Let us notice here that the two-electron exchange integrals k in the two approaches are not exactly identical, but they are both always positive.

We present now two compounds in which the strict orthogonality of the magnetic orbitals is achieved. For one of these, no thorough theoretical study has yet been carried out, whereas, for the other, an *ab initio* calculation has been performed.

The first compound contains the $[Cu(NHpy_2)(DTBSQ)]^+$ cation with $NHpy_2$ = di-2-pyridylamine and DTBSQ = 3,5-di-*tert*-butyl-*o*-semiquinone. Its structure is shown in Figure 8.18 [8.32]. The copper(II) ion is in a 4 + 2 distorted octahedral environment with two nitrogen atoms of $NHpy_2$ and the two oxygen atoms of DTBSQ in the basal plane. Two oxygen atoms of perchlorate groups occupy apical positions with apical bond lengths above 2.4 Å. In addition, the copper(II) ion is slightly displaced from the basal plane toward one of the apical sites, such that the molecular skeleton is not rigorously planar. If we neglect this weak bending of the magnetic

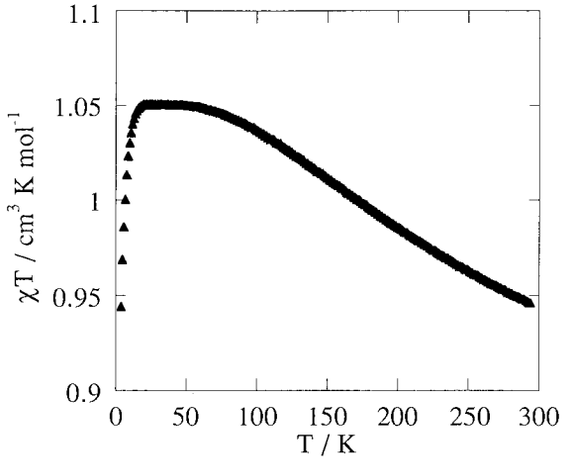


Figure 8.19. χT versus T plot for [Cu(NHpyr$_2$)(DTBSQ)](ClO$_4$) (from ref. [8.32]).

cation and ignore the *tert*-butyl groups, the symmetry is C_{2v}. When considering this bending, the symmetry is C_s with a mirror-plane σ perpendicular to the pseudo-molecular plane.

The unpaired electron from the copper(II) ion is described by a magnetic orbital a_{Cu} pointing along the Cu—N and Cu—O bonds in the basal plane; a_{Cu} is antisymmetric with regard to the mirror-plane σ and transforms as b_1 in C_{2v} and a'' in C_s. The unpaired electron of the orthosemiquinone radical is found from elementary molecular

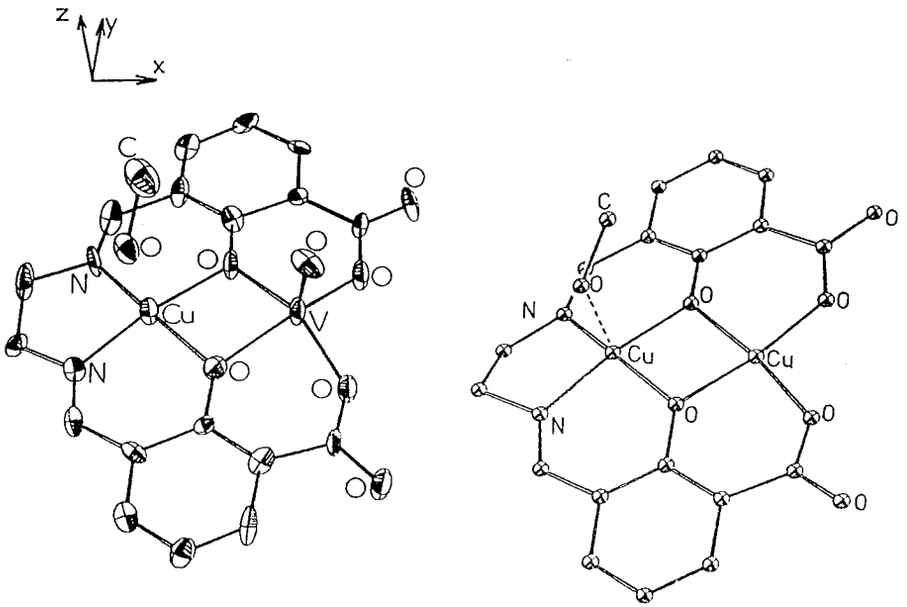


Figure 8.20. Molecular structures of CuVO(fsa)$_2$en·CH$_3$OH and Cu$_2$(fsa)$_2$en·CH$_3$OH with H$_4$(fsa)$_2$en = *N*,*N*′-(2-hydroxy-3-carboxybenzilidene)ethylenediamine (from ref. [8.34]).

orbital theory to occupy the π^* orbital noted b_{sq}. This orbital transforms as b_2 in C_{2v} and a' in C_s and is symmetric with regard to the mirror-plane σ. Both a_{Cu} and b_{sq} are schematized in Figure 8.18. Even taking into account the slight deviation of the [Cu(NHpy$_2$)(DTBSQ)]$^+$ unit from planarity, a_{Cu} and b_{SQ} are strictly orthogonal and the interaction is expected to be purely ferromagnetic with a triplet ground state. The experimental χT versus T plot shown in Figure 8.19 agrees with this prediction. At room temperature χT is equal to 0.92 cm^3 K mol^{-1}; it smoothly increases on cooling, reaches an extended plateau between 150 and 20 K and, finally, slightly decreases below 20 K. This behavior is quite characteristic of a strong intramolecular ferromagnetic interaction. The plateau corresponds to the temperature range where only the triplet ground state is thermally populated. The magnetic susceptibility then follows the Curie law $\chi T = 2Ng^2\beta^2/3k$. Below 20 K intermolecular interactions are probably operative. Quantitative fitting of the experimental data leads to J larger than 200 cm^{-1}. A similar case of strict orthogonality of magnetic orbitals in a Ni(II)(DTBSQ) compound has been reported by Benelli et al. [8.33].

The second example concerns the compound CuVO(fsa)$_2$en·CH$_3$OH, the structure of which is represented in Figure 8.20 along with that of Cu$_2$(fsa)$_2$en·CH$_3$OH [8.34]. The two metal ions, copper(II) and vanadium(IV), as well as the oxygen atoms of the vanadyl group and of the methanol molecule weakly bound to copper are located in a pseudo mirror-plane perpendicular to the plane of the macrocycle. "Pseudo" here means that this plane is not a crystallographic plane, but the molecular symmetry is very close to C_s. From elementary ligand-field considerations we find that the unpaired electron around copper(II) is described by an *xy*-type magnetic orbital transforming as a'' and the unpaired electron around vanadium(IV) is described by a x^2–y^2-type magnetic orbital transforming as a'. Thus, strict orthogonality is achieved. The triplet state has effectively been found to be the lowest with a ground triplet-excited singlet separation of 118 cm^{-1}. The $\chi T = f(T)$ curve is shown in Figure 8.21. χT first increases

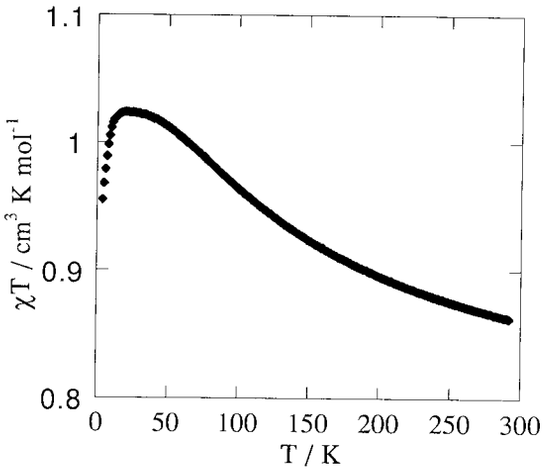


Figure 8.21. χT versus T plot for CuVO(fsa)$_2$en·CH$_3$OH (from ref. [8.34]).

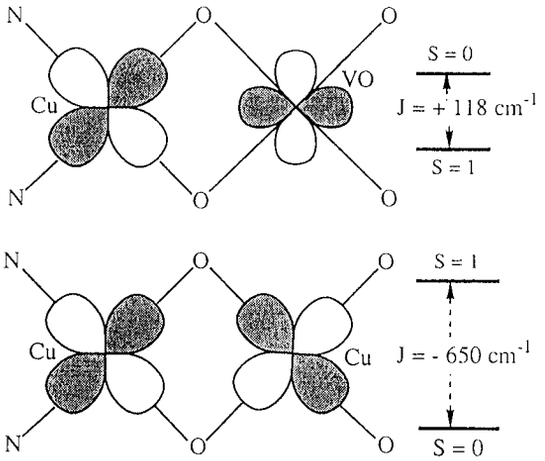


Figure 8.22. Symmetries of the magnetic orbitals and nature of the interaction in $CuVO(fsa)_2en{\cdot}CH_3OH$ and $Cu_2(fsa)_2en{\cdot}CH_3OH$ (from ref. [8.34]).

on cooling down, and then reaches a plateau below 60 K, which corresponds to the temperature range where the excited singlet state is totally depopulated. It is worth comparing $CuVO(fsa)_2en{\cdot}CH_3OH$ with $Cu_2(fsa)_2en{\cdot}CH_3OH$. The two compounds have similar molecular structures (see Figure 8.20). In the latter, a copper(II) ion replaces the vanadyl group in the external coordination site. This ion is in square planar surroundings and the two magnetic orbitals have now the same symmetry a''. The CuOCu bridging angle is much larger than 90° and the antiferromagnetic contribution dominates. The S–T splitting J for $Cu_2(fsa)_2en{\cdot}CH_3OH$ is found equal to $-650\ cm^{-1}$. The key role of the symmetry in determining the nature of the interaction is emphasized in Figure 8.22.

The compound $CuVO(fsa)_2en{\cdot}CH_3OH$ is particularly important because it provides clear proof that the nature of the interaction is related to the symmetry of the molecular species and the interacting magnetic orbitals as a whole, and not to the local symmetry along each metal–bridge–metal linkage. As a matter of fact, the magnetic orbitals around copper(II) and vanadyl(II) are delocalized toward the bridges, in an antibonding σ fashion for the copper(II) one, and in an antibonding π fashion for the vanadium(IV) one, as schematized in Figure 8.23. Along each Cu—O—V linkage the magnetic orbitals overlap, and neglecting the phase relation between the two bridges, we could claim that the interaction is antiferromagnetic. In fact, the overlap density as defined in (8.6.3) presents a positive zone around one of the bridges and a negative zone around the other, and the interaction is strongly ferromagnetic.

The concept of strict orthogonality lies implicitly on a qualitative model where the interaction parameter J can be decomposed into ferromagnetic J_F and antiferromagnetic J_{AF} contributions, this latter term vanishing when all magnetic orbitals are orthogonal. However, in the perturbational development of J given in Section 8.5, we saw that other terms may contribute significantly to the S–T energy gap. In all

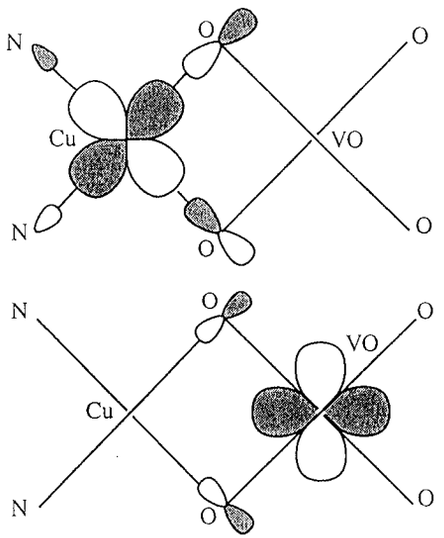


Figure 8.23. Schematic representation of the magnetic orbitals centered on copper(II) (top) and vanadyl(II) (bottom) for $CuVO(fsa)_2en{\cdot}CH_3OH$.

copper(II) dinuclear compounds studied so far the balance of these additional terms is negative, i.e., in favor of the antiferromagnetic interaction, even if some of these may be positive. Moreover, in some cases, this negative balance prevails over the potential exchange term $2k$. Therefore, it is important to see what the contribution of these terms is with regard to the potential exchange term in the case of $CuVO(fsa)_2en{\cdot}CH_3OH$.

The calculation was carried out as follows [8.35]: a model complex retaining the main characteristics of the actual complex, namely the CuO_2VO bridging network and the peripheral nitrogen and oxygen atoms bound to the metal centers, was built. This model complex has exactly C_S symmetry, with the mirror-plane containing copper(II) and vanadyl(II). The calculation is restricted to the valence electrons by using pseudopotential operators to replace all core electrons. The open-shell SCF process with the Nesbet operator gives directly the magnetic orbitals. The lowest magnetic orbital is well localized on copper, and the highest on vanadium, as shown in Figure 8.24. The former has essentially xy-character and is antisymmetric with regard to the mirror plane; the latter has essentially $x^2–y^2$ character with some admixture of z^2 and is symmetric with regard to the mirror plane.

The magnetic orbitals are strictly orthogonal such that the kinetic exchange term is exactly zero. The potential exchange term $2k$ is found to be equal to $623\ cm^{-1}$. This rather large contribution is understood when we consider the overlap density $\rho(1)$ between the two magnetic orbitals, as defined in Eq. (8.6.3). A map of isooverlap density in the plane containing the copper and the bridging atoms is shown in Figure 8.25. The overlap density $\rho(1)$ is, of course, antisymmetric with regard to the mirror plane. It presents nonnegligible contributions around the bridging oxygen atoms as

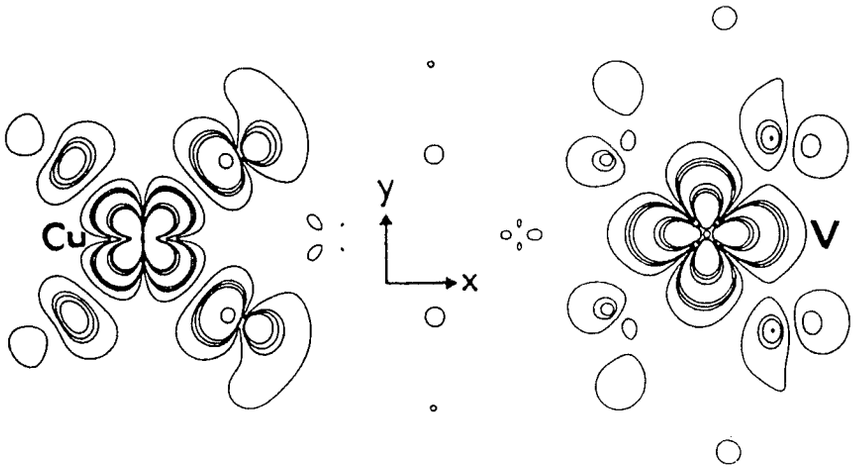


Figure 8.24. Electronic isodensity curves for the two magnetic orbitals centered around copper(II) and vanadyl(II) drawn in the CuO_2V mean plane of $CuVO(fsa)_2en{\cdot}CH_3OH$ (from ref. [8.35]).

well as the metal atoms and the peripheral nitrogen and oxygen atoms. The two-electron exchange integral k may be written as

$$k = \int_{\text{space}} \rho(1)\rho(2) r_{12}^{-1}\, d\tau \qquad (8.6.6)$$

which points out that the magnitude of k is governed by the extrema of the overlap density. Another ferromagnetic contribution, of the second order, is provided by the

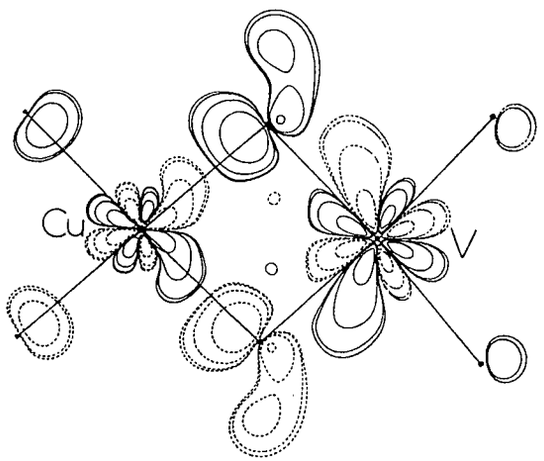


Figure 8.25. Isooverlap density curves drawn in the CuO_2V mean plane of $CuVO(fsa)_2en{\cdot}CH_3OH$ (from reference [8.35]).

Table 8.5. Different Types of Contribution Obtained at Zeroth and Second Order and Numerical Values for the Copper(II)–Vanadyl(II) Model Complex Mimicking $CuVO(fsa)_2en{\cdot}CH_3OH$

Type of contribution	Value (cm^{-1})
Potential exchange	623
Kinetic exchange	0
Double-spin polarization	79
Ligand→metal charge transfer	–114
Metal→ligand charge transfer	–38
Kinetic exchange + polarization	–113
Particle polarization	–33
Hole polarization	–60
Total value after second order	343

Source: From ref. [8.35].

double spin polarization process. All other second-order contributions are negative. Globally, the balance of the second-order terms is –280 cm^{-1}, which does not compensate for the strong potential exchange term $2k$. The results of the perturbational development of J are gathered in Table 8.5.

The calculated J value (343 cm^{-1}) has the correct sign but is more positive than the reported experimental one (120 cm^{-1}). The important point arising from this calculation is that the terms appearing beyond the active-electron approximation do not modify the qualitative predictions made in the framework of this approximation, regarding the nature of the ground state.

8.6.5 Some Other Theoretical Approaches

As far as quantitative calculations of S–T splitting are concerned, we have emphasized the approach where the configuration interaction problem is treated by a perturbational expansion. Other approaches have also been used where the energies of the singlet and triplet pair states are calculated separately. However, those approaches pose the difficult problem of evaluating small quantities (< 0.1 eV) as differences of very large numbers, and have been applied only to very simple model systems like $(Be_2N_3)^+$ [8.36] or $(Cu_2Cl_2X_2)^{2-}$, where X^- notes a pure negative charge [8.37].

Another approach is based on the Xα approximation scheme initially proposed by Johnson [8.38]. The determination of the interaction parameter J does not require the explicit calculation of the energies of each of the low-lying states, but only that of the state of the highest spin multiplicity noted $E(S_{max})$. The relative energies of the states

are calculated using the broken-symmetry formalism [8.39–8.42]. A broken-symmetry state with an energy E_B is defined as follows: the dinuclear species is considered as consisting of two identical monomeric species with opposite spins. This state is represented by a single determinant constructed by lowering the space symmetry of the dinuclear species and removing any symmetry element connecting the two halves of the molecule. A mirror symmetry is imposed in the spin up and spin down potentials. It has been shown that E_B is a weighted average of the energies of the low-lying states arising from the interaction of the local states. The isotropic interaction parameter is then given by

$$J = 2[E_B - E(S_{max})]/S_{max}^2 \tag{8.6.7}$$

which, in the case of interaction between local doublets, becomes

$$J = 2[E_B - E(S = 1)] \tag{8.6.8}$$

This method has been applied satisfactorily to the case of the $(Cu_2Cl_4)^{2-}$ dianion in which the environment of the metal atom may vary from planar to pseudotetrahedral according to the nature of the countercation. The experimentally observed variation of J with the degree of tetrahedral distortion is quantitatively reproduced [8.42].

References

8.1. R.D. Willett, D. Gatteschi, O. Kahn, eds., "Magneto-Structural Correlations in Exchange Coupled Systems," NATO ASI Series. Reidel, Dordrecht, 1985.

8.2. P.W. Anderson, *Phys. Rev.* 115 (1959), 2.

8.3. P.W. Anderson, in "Magnetism," G.T. Rado, H. Suhl, eds., Vol. 1, p. 25. Academic Press, New York, 1963.

8.4. B. Gillon, C. Cavata, P. Schweiss, Y. Journaux, O. Kahn, D. Schneider, *J. Am. Chem. Soc.* 111 (1989), 7124.

8.5. J.J. Girerd, Y. Journaux, O. Kahn, *Chem. Phys. Lett.* 82 (1981), 534.

8.6. P. de Loth, P. Cassoux, J.P. Daudey, J.P. Malrieu, *J. Am. Chem. Soc.* 103 (1981), 4007.

8.7. J.P. Malrieu, *J. Chem. Phys.* 47 (1967), 4555.

8.8. R. K. Nesbet, *Proc. R. Soc. London, Ser. A* 230 (1955), 312.

8.9. P.J. Hay, J.C. Thibeault, R.H. Hoffmann, *J. Am. Chem. Soc.* 97 (1975), 4884.

8.10. J.P. Daudey, P. de Loth, J.P. Malrieu, in ref. 1, p. 87.

8.11. P. de Loth, J.P. Daudey, H. Astheimer, L. Walz, W. Haase, *J. Chem. Phys.* 82 (1985), 5048.

8.12. F. Nepveu, W. Haase, H. Astheimer, *J. Chem. Soc., Faraday II* 82 (1986), 551.

8.13. H. Astheimer, W. Haase, *J. Chem. Phys.* 85 (1986), 1424.

8.14. W. H. Crawford, H.W. Richardson, J.R. Wasson, D.J. Hodgson, W.E. Hatfield, *Inorg. Chem.* 15 (1976), 2107.

8.15. W.E. Hatfield, in ref. 1, p. 555.

8.16. M.F. Charlot, S. Jeannin, Y. Jeannin, O. Kahn, J. Lucrèce-Abaul, J. Martin-Frère, *Inorg. Chem.* 18 (1979), 1675.

8.17. M.F. Charlot, O. Kahn, S. Jeannin, Y. Jeannin, *Inorg. Chem.* 19 (1980), 1410.

8.18. M.F. Charlot, O. Kahn, *Nouv. J. Chim.* 4 (1980), 567.

8.19. Y. Nishida, S. Kida, *J. Chem. Soc. Dalton Trans.* (1986), 2633.

8.20. V. McKee, M. Zvagulis, C.A. Reed, *Inorg. Chem.* 24 (1985), 2914.

8.21. V. McKee, M. Zwagulis, J.V. Dagdigian, M.G. Patch, C.A. Reed, *J. Am. Chem. Soc.* 106 (1984), 4765.

8.22. O. Kahn, Angew. *Chem. Int. Ed. Engl.* 24 (1985), 834.

8.23. M.F. Charlot, M. Verdaguer, Y. Journaux, P. de Loth, J.P. Daudey, *Inorg. Chem.* 23 (1984), 3802.

8.24. M. Julve, M. Verdaguer, A. Gleizes, M. Philoche-Levisalles, O. Kahn, *Inorg. Chem.* 23 (1984), 3808.

8.25. R. Vicente, J. Ribas, S. Alvarez, A. Segui, X. Solanz, M. Verdaguer, *Inorg. Chem.* 26 (1987), 4004.

8.26. M. Verdaguer, O. Kahn, M. Julve, A. Gleizes, *Nouv J. Chim.* 9 (1985), 325.

8.27. J.J. Girerd, Y. Jeannin, S. Jeannin, O. Kahn, *Inorg. Chem.* 18 (1979), 3034.

8.28. R. Veit, J.J. Girerd, O. Kahn, F. Robert, Y. Jeannin, *Inorg. Chem.* 25 (1986), 4175.

8.29. P. Chandhuri, K. Oder, K. Wieghardt, S. Gehring, W. Haase, B. Nuber, J. Weiss, *J. Am. Chem. Soc.* 110 (1988), 3657.

8.30. T.R. Felthouse, T.Y. Dong, D.N. Hendrickson, H.S. Shieh, M.R. Thompson, *J. Am. Chem. Soc.* 108 (1986), 8201.

8.31. R.E Coffman, G.R. Buettner, *J. Phys. Chem.* 83 (1979), 2387.

8.32. O. Kahn, R. Prins, J. Reedijk, J.S. Thompson, *Inorg. Chem.* 26 (1987), 3557.

8.33. C. Benelli, A. Dei, D. Gatteschi, L. Pardi, *Inorg. Chem.* 27 (1988), 2831.

8.34. O. Kahn, J. Galy, Y. Journaux, I. Morgenstern-Badarau, *J. Am. Chem. Soc.* 104 (1982), 2165.

8.35. P. de Loth, P. Karafiloglou, J.P. Daudey, O. Kahn, *J. Am. Chem. Soc.* 110 (1988), 5676.

8.36. M.F. Charlot, O. Kahn, M. Chaillet, C. Larrieu, *J. Am. Chem. Soc.* 108 (1986), 2574.

8.37. R. Broer, W.J.A. Maaskant, *Chem. Phys.* 102 (1986), 103.

8.38. K.H. Johnson, *Adv. Quantum Chem.* 7 (1973), 143.

8.39. A.P. Ginsberg, *J. Am. Chem. Soc.* 102 (1980), 111.

8.40. L. Noodleman, *J. Chem. Phys.* 74 (1981), 5737.

8.41. L. Noodleman, E.R. Davidson, *Chem. Phys.* 109 (1986), 131.

8.42. A. Bencini, D. Gatteschi, *J. Am. Chem. Soc.* 108 (1986), 5763.

CHAPTER

9

Orbital Models for Magnetic Interaction: Other Cases

9.1 Some Additional Comments on the Concept of Magnetic Orbital

Whatever the number N of magnetic centers within a molecule may be, the isotropic interaction Hamiltonian in the HDVV model is expressed as a sum of terms involving pairs of local spins [9.1]:

$$\mathbf{H} = -\sum_{k=1}^{N} \sum_{l>k}^{N} J_{kl} \mathbf{S}_k \cdot \mathbf{S}_l \tag{9.1.1}$$

It is therefore legitimate to restrict our discussion to dinuclear compounds noted AB, where, in contrast with the previous chapter, we do not explicitly mention the bridge X. In Sections 9.2–9.4 of this chapter, we will treat the case where at least one of the local spins is larger than 1/2. Moreover, we will again assume that the local ground states $^{2S_A+1}\Gamma_A$ and $^{2S_B+1}\Gamma_B$ have no first-order angular momentum. The HDVV Hamiltonian (6.1.5) is then appropriate for the description of the isotropic interaction and the problem we face is relating the interaction parameter J to the molecular integrals involved in the actual mechanism of the phenomenon.

In this chapter we could try to follow the same plan as in Chapter 8. However, some of the results concerning the interaction between two local doublets remain valid, but some of the others have not yet been extended to systems with more than two magnetic electrons. So, our remarks in Chapter 8 on the deficiency of the molecular orbital approach at the SCF level also apply here, as well as the concept of magnetic orbitals. On the other hand, at present there is no report of an extension of the *ab initio* calculation of J to AB systems with more than two unpaired electrons. We can thus eliminate this question in this chapter.

Let us add, however, some comments on the key concept of magnetic orbital. We assume that in AB there are $n_A = 2S_A$ unpaired electrons centered on A, occupying n_A orbitals denoted a_μ. In the same way, the $n_B = 2S_B$ unpaired electrons centered on B occupy n_B orbitals denoted b_ν. These magnetic orbitals a_μ and b_ν may be either

orthogonalized or natural. When they are orthogonalized, they are obtained through localization of the $n_A + n_B$ singly occupied molecular orbitals for the AB compound in its state of highest spin multiplicity. The procedure of localization is well established for the symmetrical systems of high symmetry in which the magnetic orbitals are pairwise symmetry related. The situation is much more complicated for the asymmetrical systems, or even for symmetrical systems of low symmetry in which the magnetic orbitals do not all transform as different irreducible representations of the site symmetry group. In the case of natural magnetic orbitals, the a_μ functions are the singly occupied molecular orbitals for the A fragment in its $^{2S_A+1}\Gamma_A$ local group state, and the b_ν functions are the singly-occupied molecular orbitals for the B fragment in its $^{2S_B+1}\Gamma_B$ local group state. These natural magnetic orbitals can be determined whatever the symmetry of the AB species may be; the case where the A and B fragments are different does not present any specific difficulty. As mentioned in Section 8.3, it is sometimes possible to give these orbitals a more chemical meaning. A novel example of this concerns the Cu(II)Ni(II) compound whose structure is represented in Figure 6.14. We have $n_A = 1$ unpaired electron around copper(II) in planar surroundings and $n_B = 2$ unpaired electrons around nickel(II) in octahedral surroundings. Two other compounds of the same series are known, with Cu(II)Mg(II) and Ni(II)Ni(II) pairs, respectively [9.2, 9.3]; they are isostructural with the Cu(II)Ni(II) compounds shown in Figure 6.14. In the Cu(II)Mg(II) compound there is just one unpaired electron that may be considered as occupying the a_1 magnetic orbital; in the Ni(II)Ni(II) compound, the Ni(II) ion in the inside site is also diamagnetic while the other Ni(II) ion possesses two unpaired electrons that may be considered as occupying the b_1 and b_2 magnetic orbitals.

It is worth mentioning here that the atomic orbital contraction method (see Section 8.3) is well adapted to define a_1, b_1, and b_2 analytically. In this method a_1 is determined by contracting the nickel atomic orbitals in such a way that prevents any orbital interaction between the nickel(II) ion and its surroundings. In a similar manner, b_1 and b_2 are determined by contracting the copper atomic orbitals in order to prevent any interaction between the copper(II) ion and its surroundings. The magnetic orbitals then strictly describe the unpaired electrons in the absence of any interaction. The Heitler–London wavefunctions built from these magnetic orbitals correspond well to zeroth-order functions in an approach where the interaction is weak enough to be treated as a perturbation.

9.2 The Active-Electron Approximation for Dinuclear Species with More Than Two Unpaired Electrons

Only the $n = n_A + n_B$ unpaired electrons are explicitly taken into account in the active-electron approximation. The electrostatic Hamiltonian may be written as

$$\mathbf{H} = \sum_{i=1}^{n} \mathbf{h}(i) + \sum_{i=1}^{n} \sum_{j>i}^{n} 1/r_{ij} \qquad (9.2.1)$$

$\mathbf{h}(i)$ takes into account the kinetic energy of the electron i and the one-electron potential acting on it due to the nuclei and the passive electrons; r_{ij} is the interelectronic distance. As we pointed out previously, the isotropic interaction is purely orbital in nature and, in a certain sense, corresponds to a very weak bond between the A and B fragments. To describe such a situation, it is appropriate to use Heitler–London wavefunctions constructed from the magnetic orbitals. It is sufficient to work in the subspace $M_S = 0$ if n is even or $M_S = 1/2$ if n is odd since all the pair states have a component with $M_S = 0$ or 1/2. The basis set is then made of all Heitler–London functions with $M_S = 0$ (or 1/2) constructed from the n_A orbitals a_μ and n_B orbitals b_ν. Once again, two approaches have to be considered according to whether we use orthogonalized or natural magnetic orbitals.

9.2.1 Orthogonalized Magnetic Orbital Approach

Orthogonalized magnetic orbitals result from the localization of molecular orbitals transforming as the irreducible representations of the symmetry group appropriate for the AB species [9.4, 9.5]. In general, they can be determined unambiguously only when there is at least one symmetry operation that interchanges the A and B fragments. It follows that this approach cannot be used for heterodimetallic compounds. There is, however, an exception to this limitation, namely the case where any pair (a_μ, b_ν) of orbitals involves functions transforming as different irreducible representations of the site group; all the $\beta_{\mu\nu}$ integrals defined in (9.2.4) are then zero. The singly occupied molecular orbitals for AB are then the magnetic orbitals of the problem; they are directly localized as described in the case of $CuVO(fsa)_2en{\cdot}CH_3OH$ (see Section 8.6). Ignoring this very peculiar situation, we assume that AB is symmetrical. The magnetic orbitals centered on A and B, respectively, are then two-to-two symmetry related. Let us write the isotropic interaction Hamiltonian as:

$$\mathbf{H} = -\sum_{\mu=1}^{n_A} \sum_{\nu=1}^{n_B} J_{\mu\nu} \mathbf{s}_\mu \cdot \mathbf{s}_\nu \qquad (9.2.1)$$

where $\mathbf{s}_\mu$ and $\mathbf{s}_\nu$ are monoelectronic spin operators for the a_μ and b_ν orbitals, respectively, and the $J_{\mu\nu}$'s are contributions involving pairs of magnetic orbitals. Eq. (9.2.1) is just a generalization of the HDVV Hamiltonian to the case where each interacting center has more than one unpaired electron. Since $\mathbf{s}_\mu = \mathbf{S}_A/2S_A$ and $\mathbf{s}_\mu = \mathbf{S}_B/2S_B$ for all μ and ν, (9.2.1) may be written as [9.1]:

$$\mathbf{H} = -J\mathbf{S}_A \cdot \mathbf{S}_B \qquad (9.2.2)$$

with

$$J = \frac{1}{n_A n_B} \sum_{\mu=1}^{n_A} \sum_{\nu=1}^{n_B} J_{\mu\nu} \qquad (9.2.3)$$

Using the orthogonalized orbitals to build the Heitler–London wavefunctions, it is possible to express each $J_{\mu\nu}$ contribution as:

$$J_{\mu\nu} = 2k_{\mu\nu} - 2(2\beta_{\mu\nu} + l_{\mu\nu} + l_{\nu\mu})^2[(j_{\mu\mu}{}^0 - j_{\mu\nu})^{-1} + (j_{\nu\nu}{}^0 - j_{\mu\nu})^{-1}] \tag{9.2.4}$$

the integrals appearing in (9.2.4) being defined as:

$$\beta_{\mu\nu} = \langle a_\mu(1)|\mathbf{h}(1)|b_\nu(1)\rangle$$

$$j_{\mu\mu}{}^0 = \langle a_\mu(1)a_\mu(2)|r_{12}{}^{-1}|a_\mu(1)a_\mu(2)\rangle$$

$$j_{\mu\nu} = \langle a_\mu(1)b_\nu(2)|r_{12}{}^{-1}|a_\mu(1)b_\nu(2)\rangle$$

$$k_{\mu\nu} = \langle a_\mu(1)b_\nu(2)|r_{12}{}^{-1}|a_\mu(2)b_\nu(1)\rangle$$

$$l_{\mu\nu} = \langle a_\mu(1)b_\nu(2)|r_{12}{}^{-1}|b_\nu(1)b_\nu(2)\rangle \tag{9.2.5}$$

As we mentioned previously, the ionic integrals $l_{\mu\nu}$ and $l_{\nu\mu}$ in (9.2.3) are often ignored, generally without explicit justification. When the site symmetry is high enough for the orbitals a_μ to transform as different irreducible representations of the site symmetry group, then the $\beta_{\mu\nu}$, $l_{\mu\nu}$, and $l_{\nu\mu}$ integrals for $\mu \neq \nu$ vanish, and (9.2.3) becomes

$$J_{\mu\nu} = 2k_{\mu\nu} - \frac{4\delta_{\mu\nu}(\beta_{\mu\mu} + l_{\mu\mu})^2}{j_{\mu\mu}{}^0 - j_{\mu\mu}} \tag{9.2.6}$$

where $\delta_{\mu\nu}$ is the Kronecker symbol. Equation (9.2.5) is then just an extension of what we described in Section 8.4. Hay et al. have shown that (9.2.5) may be rewritten as [9.8]

$$J_{\mu\nu} = 2k_{\mu\nu} - \frac{\Delta_\mu{}^2\delta_{\mu\nu}}{j_{\mu\mu}{}^0 - j_{\mu\mu}} \tag{9.2.7}$$

where Δ_μ is the energy difference between the two molecular orbitals used to obtain the two localized orbitals a_μ and b_μ.

9.2.2 Natural Magnetic Orbital Approach

The natural magnetic orbital approach again consists of expressing the wavefunctions associated with the low-lying states in AB as Heitler–London functions constructed from the magnetic orbitals. However, two magnetic orbitals a_μ and b_ν centered on A and B, respectively, now overlap. Therefore, we have to introduce the overlap integrals

$$\mathcal{S}_{\mu\nu} = \langle a_\mu(1) \,|\, b_\nu(1)\rangle \tag{9.2.7}$$

We will not develop the calculation here; it is rather tedious but without conceptual difficulty [9.7]. The method consists of expanding the energies of the low-lying pair states according to increasing powers of the overlap integrals $\mathcal{S}_{\mu\nu}$. If all the $\mathcal{S}_{\mu\nu}{}^2$ terms are neglected with respect to 1, then the relative energies of the pair states are found

to follow the Lande interval rule (6.2.6) derived from the HDVV Hamiltonian. J can be developed as indicated in (9.2.2). The expression of $J_{\mu\nu}$ limited to the zeroth- and first-order in $\mathcal{S}_{\mu\nu}$ is

$$J_{\mu\nu} = 2k_{\mu\nu} + 4t_{\mu\nu}\,\mathcal{S}_{\mu\nu} + 4\{\sum_{\substack{\omega=1\\ \neq\mu}}^{n_A}[(a_\omega a_\omega \,|\, a_\mu b_\nu) - (a_\omega a_\mu \,|\, a_\omega b_\nu)]$$

$$+ \sum_{\substack{\rho=1\\ \neq\nu}}^{n_B}[(a_\mu b_\nu \,|\, b_\rho b_\rho) - (a_\mu b_\rho \,|\, b_\rho b_\nu)]\}\mathcal{S}_{\mu\nu} \tag{9.2.8}$$

$k_{\mu\nu}$ is the two-electron exchange integral defined in (9.2.4), and $t_{\mu\nu}$ a one-electron term defined as

$$t_{\mu\nu} = \langle a_\mu(1) \,|\, h(1) - \frac{\alpha_{\mu A} + \alpha_{\nu B}}{2} \,|\, b_\nu(1)\rangle \tag{9.2.9}$$

$$a_{\mu A} = \langle a_\mu \,|\, \mathbf{h}(1) \,|\, a_\mu(1)\rangle \tag{9.2.10}$$

$(a_\mu b_\nu \,|\, c_\omega d_\rho)$ is a simplified notation for the two-electron integral $\langle a_\mu(1)c_a(2)\,|\Gamma_{12}{}^{-1}\,|\,b_\nu(1)d_\rho(2)\rangle$. The first term, $2k_{\mu\nu}$, in the right-hand side of (9.2.8) is the positive ferromagnetic contribution. The second term, $4t_{\mu\nu}\mathcal{S}_{\mu\nu}$, is negative for $\mathcal{S}_{\mu\nu}$ small, and favors an antiferromagnetic interaction. It varies roughly as $-\mathcal{S}_{\mu\nu}{}^2$. It is difficult to estimate the magnitude, or even the sign, of the third term involving the two-electron exchange integrals. The integrals of the type $(a_\omega a_\omega \,|\, a_\mu b_\nu)$ appear to be larger in absolute value than those of the type $(a_\omega a_\mu \,|\, a_\omega b_\nu)$. They also have the same sign as $\mathcal{S}_{\mu\nu}$. Thus, this term would be positive.

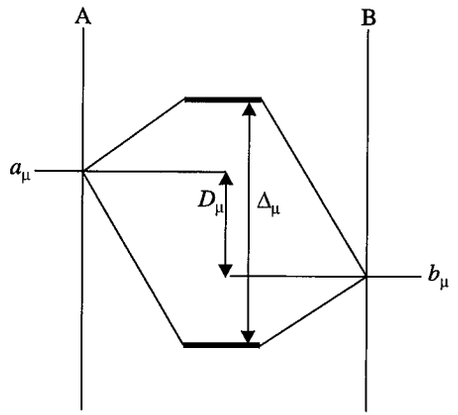
If the A and B fragments have the same site symmetry and that around B with $n_B \geq n_A$ there is never more than one magnetic orbital b_μ giving a nonzero overlap with the magnetic orbital a_μ, then (9.2.8) becomes

$$J_{\mu\nu} = 2k_{\mu\nu} + 4t_{\mu\mu}\mathcal{S}_{\mu\mu}\delta_{\mu\nu} + 4\{\sum_{\substack{\omega=1\\ \neq\mu}}^{n_A}[(a_\omega a_\omega \,|\, a_\mu b_\mu) - (a_\omega a_\mu \,|\, a_\omega b_\mu)]$$

$$+ \sum_{\substack{\rho=1\\ \neq\mu}}^{n_B}[(a_\mu b_\mu \,|\, b_\rho b_\rho) - (a_\mu b_\rho \,|\, b_\rho b_\mu)]\}\mathcal{S}_{\mu\mu}\delta_{\mu\nu} \tag{9.2.11}$$

In (9.2.11) μ denotes the irreducible representation spanned by a_μ and b_μ. From each pair of orbitals a_μ and b_μ localized on the A and B fragments, respectively, we can construct two molecular orbitals delocalized on AB. Each of these molecular orbitals is singly occupied in the pair state of highest spin multiplicity. If Δ_μ denotes the energy

gap between these two molecular orbitals for this state and D_μ the energy gap between the magnetic orbitals as shown



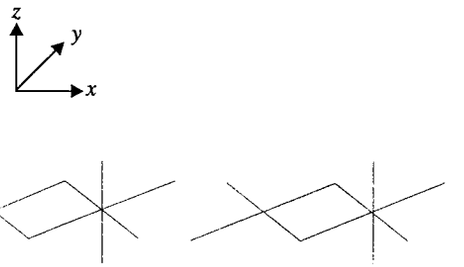
it is then possible to reexpress (9.2.11) as [9.8]

$$J_{\mu\nu} = 2k_{\mu\nu} - 2(\Delta_\mu^2 - D_\mu^2)^{1/2} S_{\mu\mu}\delta_{\mu\nu} \qquad (9.2.12)$$

In the right-hand side of (9.2.12) we have again two terms that favor the ferromagnetic and the antiferromagnetic interactions, respectively. This expression (9.2.12) clearly demonstrates that the antiferromagnetic contribution corresponds to the tendency for two electrons that were originally present in the magnetic orbitals of the same symmetry, to pair in a bonding molecular orbital. This trend is governed by the overlap integral between the magnetic orbitals. When this overlap is zero, the parallel-spin situation is favored, as expected from Hund's rule.

9.3 Interaction between Pairs of Magnetic Orbitals

Qualitative rules allowing the prediction of the nature of the interaction between two A and B spin carriers, according to the symmetry of the AB pair, were proposed in the sixties by Goodenough and Kanamori [9.9–9.11]. A clear presentation of these rules was given by Ginsberg [9.12]. What we intend to do in this section is to show how the concepts of natural orbital, overlap integral and overlap density introduced in Chapter 8 allow the generalization and extension of Goodenough–Kanamori's rules. We will treat in detail the case where the structure may be described as resulting from the sharing of an edge between two square planes, two octahedra or a square plane and an octahedron. Quite a few compounds of this kind have been investigated. The basic structures and the reference axes we use are



If the bridging network is planar, then the site symmetry is C_{2v} and the symmetry of the molecule as a whole is D_{2h} or C_{2v}. The value of the bridging angle α is always larger than 90° due to the electrostatic repulsion between the metal ions. In the following we will assume that this angle may vary from 90° up to about 110°. Our purpose in this section is to discuss qualitatively the nature and, if possible, the order of magnitude of the $J_{\mu\nu}$ terms involving pairs of magnetic orbitals.

The five *d*-type metal orbitals span the $a_1(d_{x^2-y^2}$ and $d_{z^2})$, $a_2(d_{yz})$, $b_1(d_{xy})$, and $b_2(d_{zx})$ irreducible representations of the C_{2v} group. These orbitals are delocalized toward the bridges as shown in Figure 9.1. The d_{z^2}-type orbital is denoted a_1' to distinguish it from the $d_{x^2-y^2}$-type orbital denoted a_1. First let us consider the diagonal terms $J_{\mu\mu}$ with $\mu = a_1, a_1', a_2, b_1$, and b_2. The two orbitals concerned are not strictly orthogonal, and $J_{\mu\mu}$ contains both ferromagnetic (F) and antiferromagnetic (AF) contributions. $J_{a_1a_1}$ involves orbitals pointing toward each other along the x axis and weakly delocalized toward the bridges in a π-fashion. The direct overlap of the $d_{x^2-y^2}$ orbitals favors the AF interaction. When the bridging angle α is smaller, the metal–metal separation becomes shorter and, in turn, this AF contribution becomes more pronounced. As for the role of the bridges, it can be F if α is close to 90° or AF if α is much larger. In any case, the role of the bridges is limited by the fact that the delocalization of the a_1 orbitals is weak. It turns out that $J_{a_1a_1}$ is expected to be an AF contribution of medium magnitude.

$J_{a_1'a_1'}$ deals with orbitals weakly delocalized in the molecular plane, so this contribution is anticipated to be rather weak whatever its sign. The metal–bridge overlap is of the σ-type. When α is close to 90°, the situation of accidental orthogonality may be achieved (see Section 8.6) and the contribution may be F. When α is large, the contribution becomes AF.

$J_{b_1b_1}$ has already been investigated in Section 8.6. The b_1 orbitals point along the metal–bridge directions and may be strongly delocalized toward the bridges. Accidental orthogonality is achieved for an α value close to 90°. This contribution may then be strongly F. When α increases, $J_{b_1b_1}$ becomes AF and may take strongly negative values for α larger than about 100°.

Both $J_{a_2a_2}$ and $J_{b_2b_2}$ concern orbitals for which the molecular plane is a nodal plane. These orbitals are weakly delocalized toward the bridges through π-type metal–bridge overlaps. There is no possibility of accidental orthogonality and the contributions are expected to be very weakly AF.

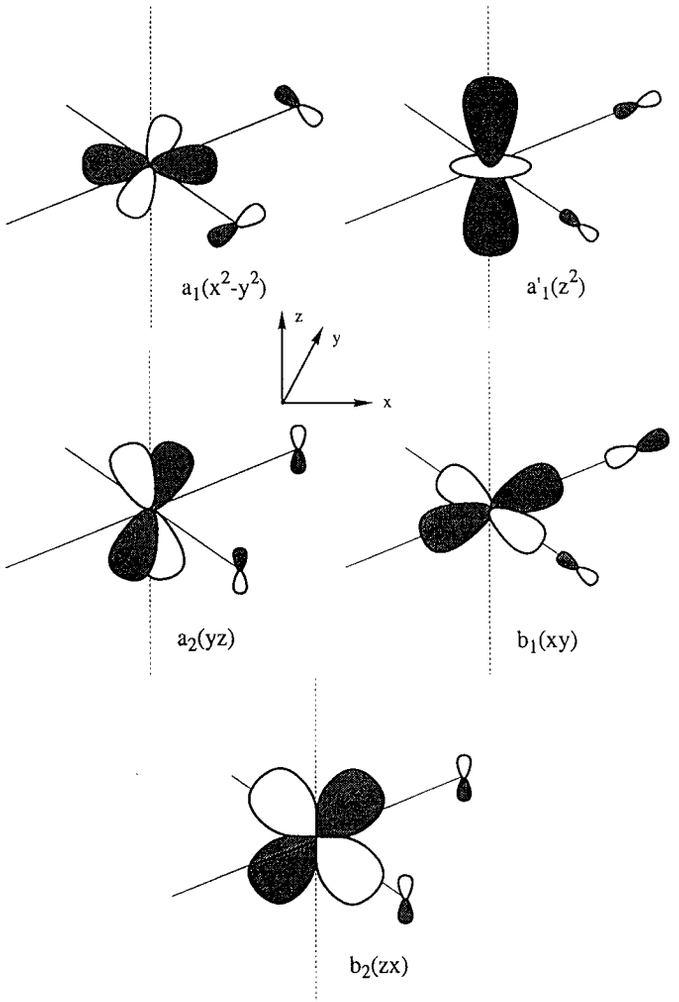


Figure 9.1. Schematic representation of the *d*-type orbitals for a metal site of C_{2v} symmetry.

All off-diagonal terms $J_{\mu\nu}$ involve orthogonal orbitals except $J_{a_1a_1'}$. The a_1 and a_1' orbitals are delocalized toward the bridges in such a way that they may overlap efficiently; the metal–bridge overlaps are of the π-type for a_1 and σ-type for a_1'. There is no possibility of accidental orthogonality and $J_{a_1a_1'}$ is expected to be significantly AF.

All other $J_{\mu\nu}$ terms with $\mu \neq \upsilon$ are strictly positive. Among them, only $J_{a_1b_1}$ is expected to be important as discussed in detail in Section 8.6. The a_1 and b_1 orbitals are both situated in the molecular plane, and are orientated in such a way that the overlap density $\rho_{a_1b_1}(1) = a_1(1)b_1(1)$ displays zones of large magnitude around the

Table 9.1. Prediction of the Nature and of the Order of Magnitude for Contributions $J_{\mu\nu}$ Involving Pairs of Magnetic Orbitals[a]

$J_{\mu\nu}$	a_1	a_1'	a_2	b_1	b_2
a_1	AF_m	AF_m	F_w	F_S	F_w
a_1'		From F_w to A F_w	F_w	F_w	F_w
a_2			AF_w	F_w	F_w
b_1				From F_S to AF_S	F_w
b_2					AF_w

[a]F, ferromagnetic; AF, antiferromagnetic. The subscripts are w, weak; m, medium; s, strong.

bridges. These qualitative predictions are gathered together in Table 9.1, in the/form of a symmetric matrix.

Let us use the results in Table 9.1 to either analyze real situations or predict the magnetic behavior of hypothetical compounds. We exclude cases where at least one of the local ground states possesses an unquenched orbital momentum from our discussion; this occurs for the configurations d^1, d^2, low-spin d^4 and d^5, and high-spin d^6 and d^7. The HDVV Hamiltonian is then inappropriate to describe properly the isotropic interaction (see Section 9.6). Before examining the various cases successively, a point needs to be stressed. When there is a competition between AF and F contributions, most often the former takes over. In other words, a notation AF_S in Table 9.1 suggests a larger $J_{\mu\nu}$ term in absolute value than a notation F_S.

We first consider symmetrical compounds, then asymmetrical ones.

$d^3 - d^3$ in octahedral environments. The magnetic orbitals arising from the $(t_{2g})^3$ configuration are a_1, a_2, and b_2. J is given by

$$J = (1/9)(J_{a_1a_1} + J_{a_2a_2} + J_{b_2b_2} + 2J_{a_1a_2} + 2J_{a_1b_2} + 2J_{a_2b_2}) \tag{9.3.1}$$

The leading term is $J_{a_1a_1}$, and J is expected to be negative (AF). Since $J_{a_1a_1}$ is dominated by the direct $d_{x^2-y^2} - d_{x^2-y^2}$ overlap, we may anticipate that the smaller the bridging angle, the more pronounced the antiferromagnetic interaction. Such a correlation has been observed for a series of $(Cr_2O_{10})^{14-}$ dimeric units resulting from the sharing of an edge between two CrO_6 octahedra [9.13]. Other chromium(III) dinuclear compounds relevant with this discussion were investigated by Hodgson [9.14] and Glerup et al. [9.15].

$d^4 - d^4$ in distorted octahedral environments. The site geometry is most likely elongated along the z direction due to Jahn–Teller effect. Therefore, the fourth magnetic orbital is a_1'. In the expression of J

$$J = (1/16) \sum_{\mu=a_1,a_1',a_2,b_2} \sum_{\nu=a_1,a_1',a_2,b_2} J_{\mu\nu} \tag{9.3.2}$$

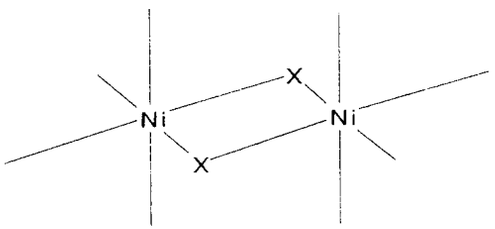
the dominant term is again $J_{a_1a_1}$, although the number of positive terms becomes important. J may be expected to be weakly negative (AF).

High-spin $d^5 - d^5$ in octahedral environments. The five d-type orbitals on each site are occupied. The sign of J results from a competition between a number of strictly positive terms, most of them weak, and a few diagonal terms, the most important being $J_{a_1a_1}$ and $J_{b_1b_1}$ For a bridging angle α close to 90°, the AF contribution arising from $J_{a_1a_1}$ might dominate the F contribution arising from $J_{b_1b_1}$. For large values of α, $J_{b_1b_1}$ is strongly AF. The resulting AF contribution which arises from the diagonal terms usually dominates the F contributions due to off-diagonal terms, and J is negative (AF). The opposite situation where the off-diagonal terms take over, however, may not be completely ruled out, particularly for small bridging angles.

$d^8 - d^8$ in octahedral environments. The two magnetic orbitals arising from the $(e_g)^2$ configuration are a_1' and b_1. J is given by

$$J = (1/4)(J_{a_1'a_1'} + J_{b_1b_1} + 2J_{a_1'b_1}) \tag{9.3.3}$$

The sign of the first two terms in the right-hand side of (9.3.3) depends on the bridging angle α. The third term is weakly positive. It follows that the sign of J will depend on α in a simple way. J is expected to be positive (F) for α values close to 90°, and negative when α becomes large enough. All the available data concerning nickel(II) dimeric species are consistent with those predictions [9.16–9.22]. Thus, quite a few nickel(II) dimeric compounds containing the bridging network



where X is a bridging monoanion like Cl^-, Br^-, or NCS^- have a spin quintet ground state.

$d^9 - d^9$ in elongated environments. When the Jahn–Teller effect operates, each site can be either square planar, or present a 4 + 1 or 4 + 2 coordination. In all cases the single magnetic orbital is b_1 and $J = J_{b_1b_1}$ depends on α as we discussed in Section 8.6. J is positive (F) for α close to 90°, and becomes negative (AF) when α increases. All magnetostructural correlations concerning bibridged copper(II) dimers are in line with these predictions (see Table 8.2).

d^3 in octahedral environment–d^4 in elongated environment. J is given by

$$J = (1/12) \sum_{\mu=a_1,a_2,b_2} \sum_{\nu=a_1,a_1',a_2,b_2} J_{\mu\nu} \tag{9.3.4}$$

The dominant terms are $J_{a_1a_1}$ and $J_{a_1a_1'}$. They are both negative, and thus J should be negative (AF). Since the local spins are not equivalent, the antiferromagnetic interaction gives rise to a nondiamagnetic ground state characterized by $S = 1/2$.

$d^3 - d^5$ in octahedral environments. The dominant terms in the expansion of J are $J_{a_1a_1}$, $J_{a_1a_1'}$, and $J_{a_1b_1}$. The first two may dominate, giving a negative J (AF), and the ground state is then characterized by $S = 1$. A Cr(III)Fe(III) compound with an $S = 1$ ground state has been reported but its structure is not exactly that of two octahedra sharing an edge [9.23].

$d^3 - d^8$ in octahedral environments. This, the following and the $d^4 - d^9$ situations are particularly interesting. The magnetic orbitals a_1, a_2, and b_2 for the d^3 ion arise from the $(t_{2g})^3$ configuration, and the magnetic orbitals a_1' and b_1 for the d^8 ion arise from the $(e_g)^2$ configuration. If the site symmetries were O_h, all the orbitals involved in the interaction would be strictly orthogonal. However, the site symmetries are not O_h but C_{2v}. J may be expanded as

$$J = (1/6)(J_{a_1a_1'} + J_{a_1b_1} + J_{a_2a_1'} + J_{a_2b_1} + J_{b_2a_1'} + J_{b_2b_1}) \tag{9.3.5}$$

All the $J_{\mu\nu}$ terms are strictly positive except $J_{a_1a_1'}$. J may be expected to be positive, due in particular to the strongly positive $J_{a_1b_1}$ term. The molecular spin associated with the ground state is then $S = 5/2$. There is no report about $d^3 - d^8$ dinuclear compounds. On the other hand, a Cr(III)Ni(II)$_3$ cluster has been described, of formula $\{Cr[(ox)Ni(Me_6\text{-}[14]ane\text{-}N_4)]_3\}(ClO_4)_3$ with ox = oxalato and Me_6-[14]ane-N_4 = ±5,7,7,12,14,14-hexamethyl-1,4,8,11 -tetraazacyclotetradecane [9.24]. The skeleton of the tetrametallic cation is shown in Figure 9.2. A central Cr(III) ion in distorted

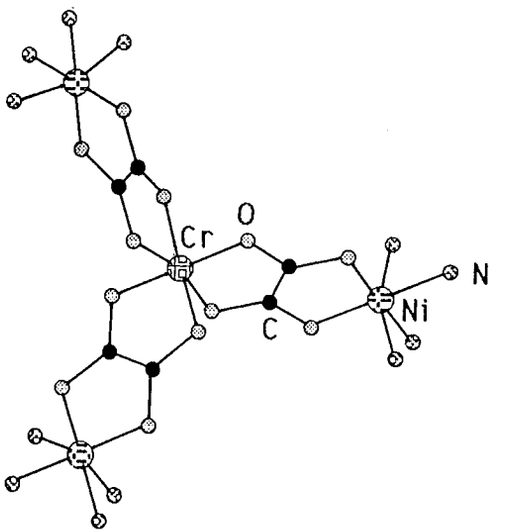


Figure 9.2. Skeleton of the tetranuclear cation $\{Cr[(ox)Ni(Me_6\text{-}[14]ane\text{-}N_4)]_3\}^{3+}$ with ox = oxalato and Me_6-[14]ane-N_4 = ±5,7,7,12,14,14-hexamethyl-1,4,8,11-tetraazacyclotetradecane (from ref. [9.24]).

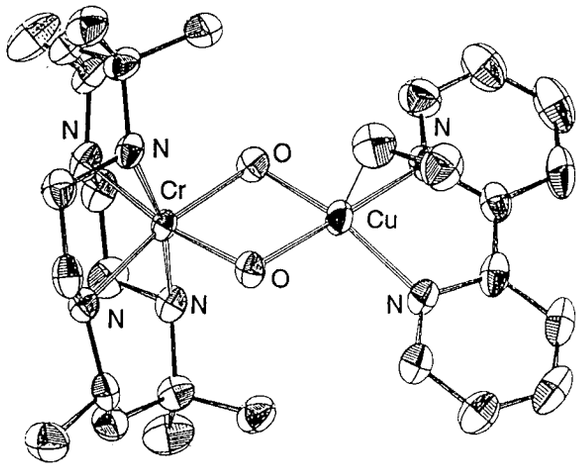


Figure 9.3. Molecular structure of $[Cr(Me_6\text{-}[14]ane\text{-}N_4)(OH)_2Cu(bpy(CH_3OH)]^{3+}$ with bpy = 2,2′-bipyridine. Due to the orthogonality of the magnetic orbital centered on the Cr(III) and Cu(II) ions, the interaction is ferromagnetic (from ref. [9.27]).

octahedral surroundings is linked through oxalato bridges to three Ni(II) ions also in distorted octahedral surroundings. The magnetic susceptibility and magnetization data have confirmed that the Cr(III)–Ni(II) interactions are ferromagnetic, and the spin associated with the ground state is $S = S_{Cr} + 3S_{Ni} = 9/2$. The interaction parameter J has been found equal to 5.3 cm^{-1}.

d^3 in octahedral environment–d^9 in elongated environment. J is given by

$$J = (1/3)(J_{a_1b_1} + J_{a_2b_1} + J_{b_2b_1}) \tag{9.3.6}$$

containing only positive terms, one of which, $J_{a_1b_1}$, being important. This is a remarkable example of strict orthogonality of the magnetic orbitals. The interaction is certainly ferromagnetic with an $S = 2$ ground state well separated in energy from an $S = 1$ excited state. Several Cu(II)Cr(III) dinuclear compounds have been described [9.25–9.27]. One of them is $[CuCr(fsa)_2en(H_2O)_2]Cl$, $(fsa)_2en^{4-}$ being the binucleating ligand derived from the Schiff base N,N′-(2-hydroxy-3-carboxybenzilidene)ethylenediamine. The structure of this compound is rather similar to that of the Cu(II)Fe(III) species shown in Figure 6.13. The two apical sites around Cr(III) are water molecules. In this compound the $S = 2$ state has the lowest energy. The quintet–triplet energy gap has been determined as $2J$ = 210 cm^{-1}. The structure of another Cu(II)Cr(III) unit is displayed in Figure 9.3. Its formula is $[Cr(Me_6\text{-}[14]ane\text{-}N_4)(OH)_2Cu(bpy)(CH_3OH)]^{3+}$, with bpy = 2,2′-bipyridine. The ground quintet–excited triplet energy gap is $2J$ = 100 cm^{-1}.

d^4 in elongated environment–d^5 in octahedral environment. In such a case we may write

$$J = (1/20)[25J(d^5\text{–}d^5) - J_{b_1a_1} - J_{b_1a_1'} - J_{b_1a_2} - J_{b_1b_1} - J_{b_1b_2}] \tag{9.3.7}$$

where $J(d^5\text{–}d^5)$ refers to a $d^5 - d^5$ compound, the structure of which would be similar to that of the $d^4 - d^5$ compound of interest. In (9.3.7) the two main $J_{\mu\nu}$ terms are $J_{b_1a_1}$ and $J_{b_1b_1}$. The term $J_{b_1a_1}$ is always positive, whereas the sign of $J_{b_1b_1}$ depends on the bridging angle α. For α close to 90°, $J_{b_1b_1}$ is also positive, and the interaction will be more antiferromagnetic than in the related $d^5 - d^5$ species. For large values of α $J_{b_1b_1}$ is strongly negative, and the interaction is predicted to be less antiferromagnetic than for the related $d^5 - d^5$ species. A weak overall ferromagnetic interaction cannot be excluded.

d^4 in elongated environment–d^8 in octahedral environment. J may be expanded as

$$J = (1/8)(J_{a_1a_1'} + J_{a_1b_1} + J_{a_1'a_1'} + J_{a_1'b_1} + J_{a_2a_1'} + J_{a_2b_1} + J_{b_2a_1'} + J_{b_2b_1}) \tag{9.3.8}$$

and the two main terms are $J_{a_1a_1'}$ and $J_{a_1b_1}$. The interaction could be weakly ferromagnetic.

$d^4 - d^9$ in elongated environments. This is another case of strict orthogonality of the magnetic orbitals. Indeed, J is given by

$$J = (1/4)(J_{a_1b_1} + J_{a_1'b_1} + J_{a_2b_1} + J_{b_2b_1}) \tag{9.3.9}$$

and only contains positive terms, one of them being important. The ground state is predicted to be that of highest spin $S = 5/2$, with a rather large separation between this state and the excited state $S = 3/2$.

$d^5 - d^8$ in octahedral environments. The expansion of J contains eight terms $J_{\mu\nu}$ with $\mu = a_1, a_1', a_2, b_1$, and b_2, and $\nu = a_1'$ and b_1. Again, the nature of the interaction may be governed by the bridging angle α. For α close to 90°, both $J_{a_1'a_1'}$ and $J_{b_1b_1}$ are positive, and the only negative term is $J_{a_1a_1'}$. The interaction may then be ferromagnetic. For larger values of α both $J_{a_1'a_1'}$ and $J_{b_1b_1}$ become negative and may dominate the positive off-diagonal terms. The interaction is then predicted to be antiferromagnetic with an $S = 3/2$ ground state. Such a situation has been reported for a Fe(III)Ni(II) compound the structure of which is somewhat different to that considered here [9.28] (see Figure 6.18).

d^5 in octahedral environment–d^9 in elongated environment. The sign of J is governed by the sign of $J_{b_1b_1}$. J is positive for α values close to 90°, and negative otherwise. For the Cu(II)Fe(III) compound presented in Section 6.4, the quintet–septet energy gap was found to be $3J$ = –240 cm^{-1}, which is consistent with the fact that the mean value of the CuOFe bridging angle is 100.1° [9.27].

d^8 in octahedral environment–d^9 in elongated environment. J may be written as

$$J = (1/2)(J_{a_1'b_1} + J_{b_1b_1}) \tag{9.3.10}$$

and is dominated by the $J_{b_1b_1}$ term. As in several other cases, the interaction may be ferromagnetic for α close to 90°, and antiferromagnetic otherwise. For all the Cu(II)Ni(II) dinuclear compounds of known structure that are relevant with this discussion, the CuONi bridging angles are close to 100° and, accordingly, the interaction is antiferromagnetic with a doublet ground state and a quartet excited state [9.2, 9.29]. For instance, for $CuNi(fsa)_2en(H_2O)_2{\cdot}H_2O$, whose structure is shown in Figure 6.14, the mean value of CuONi is 98.9°, and the doublet-quartet energy gap is found

Table 9.2. Prediction of the Nature of the Isotropic Interaction in Various Dinuclear Compounds with a Structure Resulting from the Sharing of an Edge between Two Square Planes, Two Octahedra, or a Square Plane and an Octahedron[a]

	d^3	d^4 elongated	d^5	d^8	d^9 elongated
d^3	AF ← AF	AF	AF	F	F
d^4 elongated	AF	AF ← AF	F	F	F
d^5			AF (or F) → AF	F → AF	F → AF
d^8				F → AF	F → AF
d^9 elongated				F → AF	

[a]F, ferromagnetic; AF, antiferromagnetic. → means that the interaction is expected to be more antiferromagnetic when the bridging angle increases from 90°. ← means that the interaction is expected to be more antiferromagnetic when the bridging angle decreases toward 90°

as $3J/2 = -213\ cm^{-1}$ [9.2]. A unique Cu(II)Ni(II) compound with a quartet ground state has been reported [9.30]. Its structure, however, does not correspond to that considered in this discussion. Copper(II) is in tetrahedral rather than square planar surroundings.

The results of this discussion are collected in Table 9.2.

Before we close this section, it is worthwhile stressing that the predictions of Table 9.2 are based on the active-electron approximation. This approximation is hoped to give results that are qualitatively correct when the HOMOs of the bridging ligands are low in energy with respect to the magnetic orbitals. Otherwise, it may become nonvalid, and the predictions to which it leads may be erroneous. It is also important to stress that it is rather exceptional to have bridging angles close to 90°, due to the electrostatic repulsion between the metallic cations. Therefore, a notation like F → AF in Table 9.2 means that the interaction will usually be antiferromagnetic, particularly if the metal ions are trivalent.

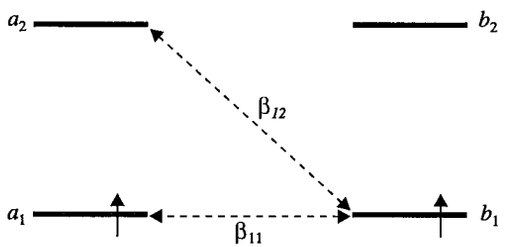
In this section we focused on one type of symmetry, but obviously the same approach could be used for other symmetries, and provides the theoretical background to rationalize magnetostructural correlations which have been experimentally observed. We already mentioned the correlation concerning the bibridged copper(II) dimers. Another interesting correlation concerns the dinuclear iron(III) centers bridged by a ligand oxygen atom like oxo, hydroxo, or alkoxo, and at least one other bridging ligand, in general carboxylate, but also sulfate or phosphate [9.31]. In all these cases the interaction is antiferromagnetic. The J interaction parameter has been found to correlate with a P parameter defined as half of the shortest interaction pathway between two iron(III) ions, i.e., (Fe_A—O + Fe_B—O)/2. When multiple-atom bridges like carboxylato are present, only the Fe—O(bridge) distances are taken into account to calculate the average length of the interaction pathway. The correlation established from 36 sets of experimental data is

$$J(cm^{-1}) = -1.753\times10^{10}\exp[-12.663P(Å)] \qquad (9.3.11)$$

which shows that the magnitude of the antiferromagnetic interaction rapidly falls as the interaction pathway increases.

9.4 Ferromagnetic Interaction Through Crossed Interaction between Singly Occupied and Empty Orbitals

Up to now we have seen a unique mechanism favoring the ferromagnetic interaction in the active-electron approximation. This mechanism is governed by the two-electron exchange integral $k_{\mu\nu}$. Another mechanism has been proposed by Goodenough [9.10], which we will discuss in this section. This novel mechanism is based on the interaction between the ground configuration and an excited configuration where an electron has been transferred from a singly occupied orbital centered on a site into an empty orbital centered on the other site, or, which is strictly equivalent, from a doubly occupied orbital centered on a site into a singly occupied orbital centered on the other site. First let us present the qualitative arguments in favor of this mechanism. We consider the a_1 and a_2 orbitals centered on A, and b_1 and b_2 orbitals centered on B, with one electron around each site occupying the orbital of lower energy:



The transfer of the unpaired electron from b_1 to a_2 (or from a_1 to b_2) leads to the charge transfer excited configuration a_1a_2 (or b_1b_2). The triplet state arising from this configuration is lower in energy than the singlet state, due to Hund's rule. Consequently, the interaction between the two triplet states arising from ground and excited configurations, respectively, may be expected to be more important than the interaction between the two singlet states. Thus, the low-lying triplet state will be stabilized with respect to the low-lying singlet.

Quantitatively, the situation is a little less clear. We assume again that AB is centrosymmetric. It follows that the states arising from the ground configuration a_1b_1 are $^1\Gamma_g$(GC) and $^3\Gamma_u$(GC) with the orbital parts of the wavefunctions given in Section 8.4 (adding here the subscripts 1). Two singlet states, $^1\Gamma_g$(CTC) and $^1\Gamma_u$(CTC), and two triplet states, $^3\Gamma_g$(CTC) and $^3\Gamma_u$(CTC), arise from the charge transfer configuration a_1a_2 (or b_1b_2). Only the $^1\Gamma_g$(CTC) and $^3\Gamma_u$(CTC) excited states may couple with the low-lying states. The orbital parts of the wavefunctions are

$$\Psi_{S(T)} = (1/2)[a_1(1)a_2(2) \pm a_1(2)a_2(1) \pm b_1(1)b_2(2) + b_1(2)b_2(1)] \qquad (9.4.1)$$

where the positive sign holds for $^1\Gamma_g$(CTC) and the negative for $^3\Gamma_u$(CTC). The coupling matrix elements may be calculated easily as

$$<^1\Gamma_g(\mathrm{CTC})\ |\mathbf{H}|\ ^1\Gamma_g(\mathrm{GC})> = (\sqrt{2}/2)[2\beta_{12} + 2l + (l' + l'')] \tag{9.4.2}$$

$$<^3\Gamma_u(\mathrm{CTC})\ |\mathbf{H}|\ ^3\Gamma_u(\mathrm{GC})> = (\sqrt{2}/2)[2\beta_{12} + 2l - (l' + l'')] \tag{9.4.3}$$

with

$$\beta_{12} = <a_2(1)\ |\mathbf{h}(1)|\ b_1(1)>$$

$$l = <a_1(1)a_2(2)\ |r_{12}^{-1}|\ a_1(1)b_1(2)>$$

$$l' = <a_1(1)a_1(2)\ |r_{12}^{-1}|\ a_2(1)b_1(2)>$$

$$l'' = <a_1(1)a_2(2)\ |r_{12}^{-1}|\ a_2(1)b_1(2)> \tag{9.4.4}$$

The energy gaps $\Delta(S{=}0)$ between the two $^1\Gamma_g$ states, and $\Delta(S{=}1)$ between the two $^3\Gamma_u$ states are obtained as

$$\Delta(S{=}0) = U + K^0$$

$$\Delta(S{=}1) = U - K^0 \tag{9.4.5}$$

with

$$U = \alpha_2 - \alpha_1 + <a_1(1)a_2(2)\ |r_{12}^{-1}|\ a_1(1)a_2(2)>$$

$$- <a_1(1)b_1(2)\ |r_{12}^{-1}|\ a_1(1)b_1(2)> + <a_1(1)b_1(2)\ |r_{12}^{-1}|\ a_2(2)b_2(1)> \tag{9.4.6}$$

$$K^0 = <a_1(1)a_2(2)\ |r_{12}^{-1}|\ a_1(2)a_2(1)> + <a_1(1)b_2(2)\ |r_{12}^{-1}|\ a_2(2)b_1(1)>$$

$$- <a_1(1)b_1(2)|r_{12}^{-1}|a_1(2)b_1(1)> \tag{9.4.7}$$

U is the mean cost in energy of transferring an electron from b_1 to a_2, and K^0, which is dominated by a one-center exchange integral, accounts for Hund's rule. If the ionic integrals l, l', and l'' are neglected, then the low-lying singlet is stabilized by $-2\beta_{12}^2/(U + K^0)$ and the low-lying triplet by $-2\beta_{12}^2/(U - K^0)$. Since both U and K^0 are positive, the triplet is lower than the singlet, the S–T energy gap J being

$$J = \frac{4\beta_{12}^2 K^0}{(U^2 - K^{02})} \tag{9.4.8}$$

This effect opposes the stabilization of the low-lying singlet, due to the interaction between the ground configuration and the a_1a_1 (or b_1b_1) charge-transfer configuration (see Section 8.4). This stabilization of the singlet is roughly $J_{AF} = -4\beta_{11}^2/U$, with

$$\beta_{11} = <a_1|\mathbf{h}(1)|b_1(1)> \tag{9.4.9}$$

K_0 in (9.4.8) is much smaller than U; it turns out that J due to this novel mechanism is smaller than the normal antiferromagnetic contribution J_{AF} by approximately a

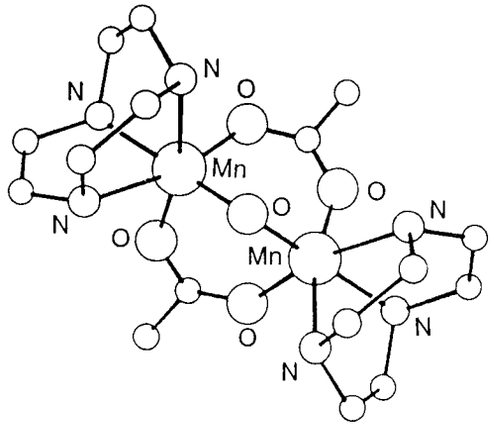


Figure 9.4. Molecular structure of $[LMnO(CH_3O_2)MnL]^{2+}$ with L = 1,4,7-triazacyclononane (from ref. [9.23]).

factor K^0/U. J in (9.4.8) becomes a significant contribution with respect to J_{AF} only when $|\beta_{12}|$ is much larger than $|\beta_{11}|$.

It is rather difficult to see whether this mechanism remains valid when the ionic integrals are no longer neglected. However, we can note that l is most probably larger than l' and l'' in absolute value. Indeed, l corresponds to the interaction between the electron density a_1^2 and the overlap density a_2b_1, whereas l' and l'' correspond to the interactions between the one-center overlap density a_1a_2 and the two-center overlap densities a_1b_1 and a_2b_1, respectively. If l' and l'' in (9.4.3) and (9.4.4) are neglected, but not l, then in (9.4.8) β_{12} must be replaced by $(\beta_{12} + l)$, and the stabilization of the triplet is preserved.

This mechanism, which affords a ferromagnetic contribution, has been invoked to justify the ferromagnetic interaction observed in $[LMnO(CH_3CO_2)_2MnL](ClO_4)_2 \cdot 2H_2O$ [9.27]. L stands for 1,4,7-triazacyclononane. Within the dinuclear cation, the two Mn(III) ions are bridged by two acetato groups and an oxo group as shown in Figure 9.4 [9.32]. This compound has been reported to exhibit an $S = 4$ ground state

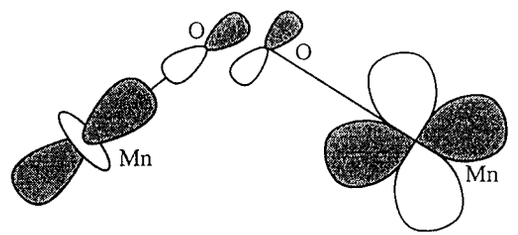


Figure 9.5. Singly occupied and empty orbitals in $[LMnO(CH_3O_2)MnL]^{2+}$ whose interaction is invoked to explain the weak ferromagnetic interaction.

with a J interaction parameter equal to 18 cm^{-1}. It has been suggested that the interaction takes place through the oxo bridge, and that the singly occupied and empty orbitals involved in the process are of the d_{xy}- and d_{z^2}-type, respectively, as schematized in Figure 9.5. Both orbitals are delocalized toward the oxo bridge, the former in a π-fashion, and the latter in a σ-fashion. The value of the MnOMn bridging angle is 117.9°, which authorizes an efficient crossed interaction β_{xy,z^2}.

9.5 Mechanism of the Anisotropic and Antisymmetric Interactions

In the orbital models describing the interaction between two spin carriers A and B discussed in this and the preceding chapters, we ignored the spin-orbit coupling. In other words, we implicitly assumed not only that the local ground states had no first-order angular momentum, but also that the local excited states were much too high in energy to couple with these local ground states. If it was so, the local ground states would not be split in zero field, and the local $\mathbf{g}_A$ and $\mathbf{g}_B$ tensors would be isotropic with the principal value $g = g_e = 2.0023$. This is usually not the case. The mixing of local and excited states through spin-orbit coupling is at the origin of the anisotropic interaction characterized by the $\mathbf{D}_{AB}$ tensor, and the antisymmetric interaction characterized by the $\mathbf{d}$ vector when the symmetry of the pair is low enough.

The expression of the components of $\mathbf{D}_{AB}$ and $\mathbf{d}$ may, in principle, be derived by replacing the zeroth-order magnetic orbitals a_μ and b_ν used in the theory of the isotropic interaction, by new local functions taking into account the spin-orbit perturbation up to the second order, and then building new Heitler–London functions using the perturbed local functions as a basis set. Such a derivation can be performed within the framework of either orthogonalized or natural orbital approaches. The calculation is, however, rather tedious. For instance, it is no longer possible to work only in the $M_S = 0$ (or 1/2) subspace since $d \cdot S_A \wedge S_B$ mixes the pair states. So far only one specific case has been treated, with $S_A = S_B = 1/2$ local spins [9.33].

Kanamori gave general expressions for the principal values D_{uu} of $\mathbf{D}_{AB}$ and the components d_u of $\mathbf{d}$ with $u = x, y, z$ [9.34, 9.35]. These expressions are

$$D_{uu} = D_{uu}^{A} + D_{uu}^{B} \tag{9.5.1}$$

$$d_u = d_u^{A} - d_u^{B} \tag{9.5.2}$$

with

$$D_{uu}^{A} = \sum_e (\lambda^A/\Delta_e^A)^2 \langle \Psi_g^A | \mathbf{L}_u^A | \Psi_e^A \rangle \langle \Psi_e^A | \mathbf{L}_u^A | \Psi_g^A \rangle$$

$$\langle \Psi_e^A(i) \Psi_g^B(j) \, | \mathbf{H} | \, \Psi_e^A(j) \, \Psi_g^B(i) \rangle \tag{9.5.3}$$

$$d_u^{A} = 2 \sum_e (\lambda^A/\Delta_e^A) \langle \Psi_g^A | \mathbf{L}_u{}^A | \Psi_e^A \rangle \langle \Psi_e^A(i) \Psi_g^B(j) | \mathbf{H} | \Psi_g^A(j) \Psi_g^B(i) \rangle \tag{9.5.4}$$

$\Psi_g{}^A$ is the unperturbed wavefunction for the local ground state of A, and $\Psi_e{}^A$ the wavefunction for one of the excited states located at an energy $\Delta_e{}^A$ above the ground state. $\mathbf{L}_i^A$ is a component of the orbital momentum, and λ^A the local spin-orbit coupling parameter assumed to be isotropic. $\mathbf{H}$, which appears in the integrals $\langle \Psi_e^A(i) \Psi_g^B(j) \, | \mathbf{H} | \, \Psi_e^A(j) \Psi_g^B(i) \rangle$ and $\langle \Psi_e^A(i) \Psi_g^B(j) \, | \mathbf{H} | \, \Psi_g^A(j) \Psi_g^B(i) \rangle$, is the electrostatic Hamiltonian without spin-orbit coupling term. These integrals contain one- and two-electron terms, and their signs cannot be known a priori. In the peculiar case where $\Psi_g{}^A$ is orthogonal to all $\Psi_e{}^B$ functions, and $\Psi_g{}^B$ orthogonal to all $\Psi_e{}^A$ functions, the exchange integrals occurring in $D_{uu}{}^A$ are positive together with the D_{uu} principal values of $\mathbf{D}_{AB}$. When the AB pair is centrosymmetric, has a C_n ($n \geq 2$) rotation axis along AB and a mirror-plan perpendicular to this axis, or has the C_{2v} symmetry with the twofold axis along AB, then $d_u{}^A$ and $d_u{}^B$ are equal, and the antisymmetric interaction vanishes. The anisotropic interaction, however, remains present. The formulas (9.5.1)–(9.5.4) are valid for both homo- and heteropairs.

The most important point emerging from Eqs. (9.5.3) and (9.5.4) is that the anisotropic and antisymmetric interactions are not directly related to the isotropic interaction. The latter involves the two local ground states $\Psi_g{}^A$ and $\Psi_g{}^B$, whereas the former involve both ground and excited local states. More precisely, the anisotropic interaction arises from the synergistic effect of the local spin-orbit coupling perturbations and the interaction between the ground state of an ion and the excited states of another. This interaction may be quite different in both magnitude and sign to the isotopic interaction. We have seen that the isotropic interaction in an AB pair is most often antiferromagnetic due to overlap of magnetic orbitals; if the symmetry of that pair is high enough, then the ground state of an ion may be orthogonal to all the excited states of the other, so that the interaction terms involved in the anisotropic interaction would all be ferromagnetic.

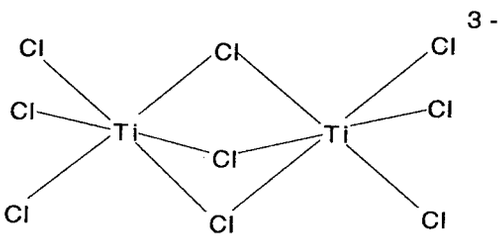
An elegant demonstration of this was given by Gatteschi et al. who studied the single-crystal EPR spectra of some of the hydroxo-bridged copper(II) dimers mentioned in Table 8.2, and determined the $\mathbf{D}_{CuCu}$ tensors [9.35–9.36]. In Section 7.1 we indicated that $\mathbf{D}_{AB}$ contains a dipolar contribution (which can be fairly well estimated from the point dipole approximation), and an anisotropic interaction contribution. The latter has been found to be preponderant. Each copper(II) ion is in square planar surroundings with a C_{2v} site symmetry. Using the reference axes in Figure 9.1, the local ground state may be labeled as $^2B_1(xy)$, and the local excited states as $^2A_1(x^2 - y^2)$, $^2A_1(z^2)$, $^2A_2(yz)$, and $^2B_2(zx)$. $^2B_1$ couples with $^2A_1(x^2-y^2)$ through $\mathbf{L}_z$, $^2A_2(yz)$ through $\mathbf{L}_x$, and $^2B_2(zx)$ through $\mathbf{L}_y$. The magnitude of the zero-field splitting within the low-lying triplet state of the dimeric molecule is governed by the interaction between the $b_1(xy)$ orbital centered on one copper(II) ion and the other orbitals centered on the other copper(II) ion. Table 9.1 tells us that all $J_{b_1\upsilon}$ ($\upsilon \neq b_1$) terms are ferromagnetic, one of these, $J_{b_1a_1}$, being dominant. The EPR data are consistent with the fact that the zero-field splitting is governed by the positive $J_{b_1a_1}$ term. Its magnitude varies according to the geometry of the dinuclear cation, but remains positive, in striking contrast with the singlet–triplet energy gap which is positive or negative, according to whether the bridging angle CuOCu is smaller or larger than 95.7° (see Section 8.6). It is interesting to note that the zero-field splitting within the triplet state

of a hydroxo-bridged copper(II) dimer is governed by the same interaction term $J_{b_1a_1}$ as the singlet–triplet splitting in the heterodinuclear compound Cu(II)VO(II) described in Section 8.6.

9.6 Interaction Involving Ions with Unquenched Orbital Momentum

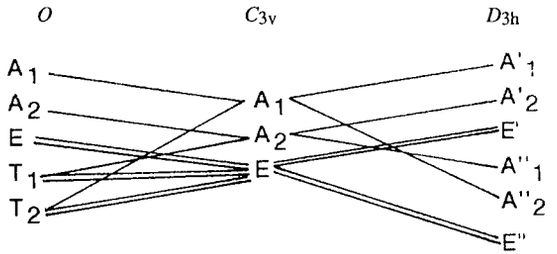
Since the beginning of Chapter 6, one of our basic hypotheses has been that the local ground states of the interacting ions within the AB pair, $^{2S_A+1}\Gamma_A$ and $^{2S_B+1}\Gamma_B$, had no first-order angular momentum. It is only within this framework that it is valid to describe the isotropic interaction using the HDVV Hamiltonian $-J\mathbf{S}_A\cdot\mathbf{S}_B$. It is probably worthwhile stressing here that orbital degeneracy does not necessarily imply first-order angular momentum. In cubic symmetry the T_1 and T_2 states have an orbital momentum, but not the E states. In this section we approach the case where, for at least one of the ions, the angular momentum is not totally quenched at the first order. We do not intend to spend much time on this rather complicated problem which is far from being fully solved yet. Quite a few approaches have already been proposed [9.37–9.47]; most often they lead to overparametrized models that are difficult to test in terms of their range of validity. However, we will point out the nature of the difficulties encountered with unquenched angular momenta, and will present some simple ideas that permit, at least in part, their solution.

To begin with, we consider one of the simplest cases where unquenched orbital momenta are involved, namely that of the dinuclear species $Ti_2Cl_9^{3-}$. This species is found in the $Cs_3Ti_2Cl_9$ phase. Its structure results from the sharing of a trigonal face between two slightly distorted $TiCl_6$ octahedra:



The overall symmetry of $Ti_2Cl_9^{3-}$ is rigorously D_{3h} the threefold axis being coincident with the trigonal axis of the crystal. The site symmetry of each Ti(III) ion is C_{3v}.

If we ignore the distortion of the $TiCl_6$ site, then the local ground state is $^2T_2$. In Section 3.2 we saw that such a state possesses an angular momentum, and that its degeneracy is partially removed by the spin-orbit coupling. To determine the pair states arising from the interaction of the two $^2T_2$ local states, it is convenient to use the correlations between irreducible representations of O and C_{3v} on the one hand and D_{3h} and C_{3v} on the other hand:



C_{3v} is called the cofactor group of O and D_{3h}. The $^2T_2$ local state splits into $^2A_1$ and ^2E under C_{3v} site symmetry. The interaction between the two $^2A_1$ local states in C_{3v} gives rise to $^1A'_1$ and $^3A''_2$ pair states in D_{3h}. The spin singlet is totally symmetric orbitally ($^1A'_1$) because it has the same symmetry as the closed-shell state of lowest energy. The interaction between the $^2A_1$ state of one ion and the ^2E state of the other gives rise to the $^1E' + {}^1E'' + {}^3E' + {}^3E''$ pair states. Finally, the interaction between the two ^2E local states gives rise to $^1A'_1 + {}^3A''_2 + {}^1A''_1 + {}^3A'_2 + {}^1E' + {}^3E''$. The correlation between local and pair states is schematized in a qualitative fashion in Figure 9.6. We end up with six spin-singlet and six spin-triplet pair states, a situation that obviously cannot be accounted for with a simple HDVV Hamiltonian. Moreover, the spin-triplet states are further split into singlets and doublets by the spin-orbit coupling according to

$$^3A'_2 \rightarrow A'_1 + E''$$

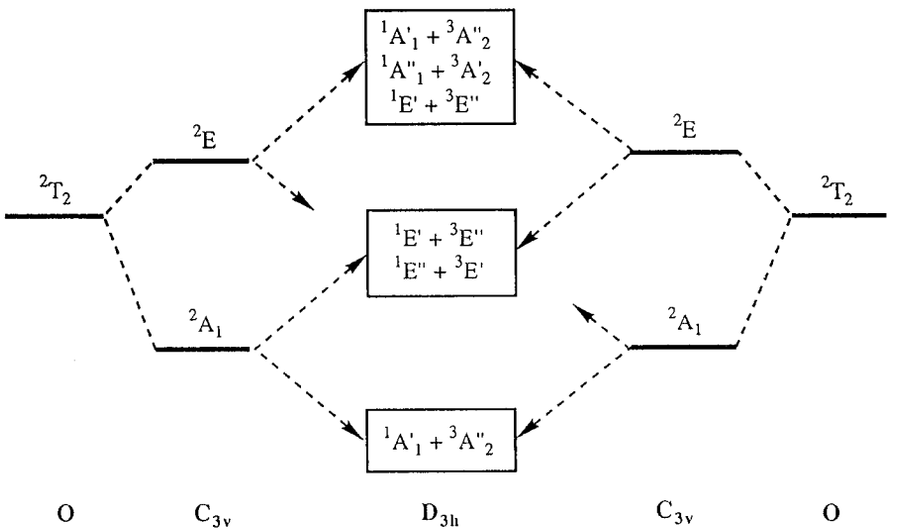


Figure 9.6. Correlations between local and pair states in the dinuclear trianion $Ti_2Cl_9^{3-}$.

$$^3A''_2 \rightarrow E' + A''_1$$

$$^3E' \rightarrow E' + A''_1 + A''_2 + E''$$

$$^3E'' \rightarrow A'_1 + A'_2 + E' + E'' \qquad (9.6.1)$$

The correlations in (9.6.1) result from the fact that the functions associated with the spin triplet transform as $A'_2 + E''$ in D_{3h} [9.47].

Two formalisms have been developed to describe the spectrum of the low-lying states and the magnetic properties of $Ti_2Cl_9^{3-}$. The simplest consists of writing the zero-field Hamiltonian as [9.46]

$$\mathbf{H} = \mathbf{H}_A + \mathbf{H}_B + \mathbf{V}_A + \mathbf{V}_B + \mathbf{H}_{SS} + \mathbf{H}_{OO} + \mathbf{H}_{SO} \qquad (9.6.2)$$

$\mathbf{H}_A$ and $\mathbf{H}_B$ are the electronic Hamiltonians for each of the metallic centers noted A and B in pure octahedral ligand fields. The eigenfunctions of $\mathbf{H}_A$ (and $\mathbf{H}_B$) are the functions associated with the $^2T_2$ local ground state. The terms $\mathbf{V}_A$ and $\mathbf{V}_B$ describe the trigonal distortion around A and B, respectively. $\mathbf{V}_A$ (and $\mathbf{V}_B$) splits the $^2T_2$ states into $^2A_1$ and ^2E separated by Δ (Δ is positive if the ground local state is the orbital singlet). $\mathbf{V}_A$ has a form which we encountered in Section 3.3, namely

$$\mathbf{V}_A = \Delta(2 - \mathbf{L}_{zA})/3 \qquad (9.6.3)$$

the z axis being the threefold rotation axis. $\mathbf{H}_{SS}$ is a spin–spin coupling operator that removes the spin degeneracy of the eigenstates of $\mathbf{H}_A + \mathbf{H}_B$. The eigenfunctions $|\Gamma,i,S,M_S\rangle$ associated with these states transform as the irreducible representations Γ, which are contained in the direct product $T_2 \times T_2$, namely $A_1 + E + T_1 + T_2$. A satisfying description of the splittings between the spin singlets and spin triplets is given by

$$\mathbf{H}_{SS} = -\mathbf{J}\mathbf{S}_A \cdot \mathbf{S}_B \qquad (9.6.4)$$

where $\mathbf{J}$ is an orbital operator such as

$$\langle\Gamma,i|\mathbf{J}|\Gamma',j\rangle = J_\Gamma \delta_{\Gamma\Gamma'}\delta_{ij} \qquad (9.6.5)$$

$\mathbf{H}_{OO}$ is an orbit–orbit coupling parameter that removes partially the orbital degeneracy of the eigenstates of $\mathbf{H}_A + \mathbf{H}_B$. It may take the form

$$\mathbf{H}_{OO} = -\mathbf{K}\mathbf{L}_A \cdot \mathbf{L}_B \qquad (9.6.6)$$

where $\mathbf{K}$ is a spin operator such as

$$\langle S,M_S|\mathbf{K}|S',M_{S'}\rangle = K_{2S+1}\,\delta_{SS'}\delta_{M_SM_{S'}} \qquad \text{with } 2S+1 = 1 \text{ or } 3 \qquad (9.6.7)$$

The δ_{xy}s appearing in Eqs. (9.6.5) and (9.6.7) are Kronecker symbols. Finally, $\mathbf{H}_{SO}$ is the nonphenomenological spin-orbit coupling operator defined as

$$\mathbf{H}_{SO} = \kappa\lambda(\mathbf{L}_A \cdot \mathbf{S}_A + \mathbf{L}_B \cdot \mathbf{S}_B) \qquad (9.6.8)$$

λ is the spin-orbit coupling parameter for the $^2T_2$ state, and κ is the orbital reduction factor (see Section 3.1). The Hamiltonian (9.6.2) is diagonalized by using of the 36

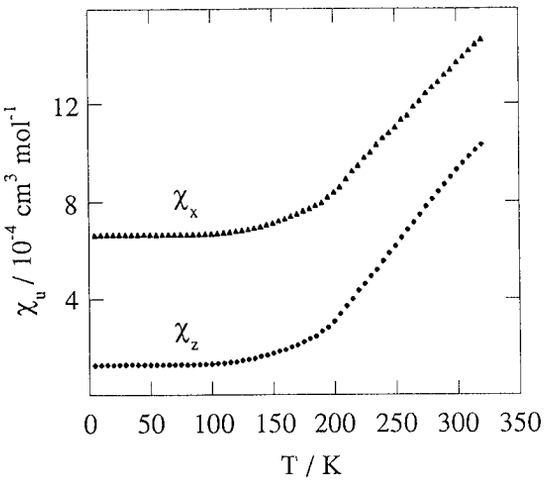


Figure 9.7. Temperature dependences of the χ_z and χ_x principal magnetic susceptibilities for $Cs_3Ti_2Cl_9$.

$|\Gamma, i, S, M_S\rangle$ functions built as products of the local functions as a basis set. To compute the principal magnetic susceptibilities, the parallel and perpendicular Zeeman perturbations are diagonalized using the eigenfunctions of $\mathbf{H}$ as a new basis set. The first- and second-order Zeeman coefficients are then calculated and introduced into the Van Vleck formula giving χ_z and χ_x (= χ_y). The calculation, therefore, requires the successive diagonalization of 36×36 matrices. The temperature dependences of the principal susceptibilities have been accurately determined [9.45]. The curves are shown in Figure 9.7. Both χ_z and χ_x decrease as the temperature is lowered. The magnetic anisotropy defined as $\chi_x - \chi_z$ remains positive and increases slightly on cooling.

Another effective Hamiltonian, based on the second quantification formalism, was proposed by Drillon and Georges [9.47]. We will not discuss this formalism in detail since we chose not to introduce second quantification in this book. This and the previous model have been tested on the same data set. In spite of the apparent simplicity of the problem, the conclusions are not strictly identical. Both approaches agree on the fact that the ground state is a diamagnetic singlet, well separated in energy from the first excited states. According to the first model, the trigonal distortion Δ is large, about 1400 cm^{-1}, and the two pair states of lowest energy are $^1A'_1$ and $^3A''_2$ arising from the $^2A_1 \times {}^2A_1$ coupling; the $^1A'_1$–$^3A''_2$ energy gap is of the order of $J_{A1} = -525\text{cm}^{-1}$. The $^3A''_2$ state is further split in zero field through the second-order spin-orbit perturbation. The other pair states arising from the $^2A_1 \times {}^2E$ and $^2E \times {}^2E$ interactions are found to be too high in energy to be thermally populated, even at 300 K. According to the second model using a second-quantification formalism, the trigonal distortion Δ is much smaller, about 800 cm^{-1}, and there are two doublet states at 494 and 542 cm^{-1} above the diamagnetic ground singlet.

It is not our intention to discuss the case of $Ti_2Cl_9^{3-}$ in a thorough manner, but only to point out that even for a compound of high symmetry, involving only two magnetic electrons, for which the thermal variations of the parallel and perpendicular magnetic susceptibilities are accurately known, the interpretation of the experimental data is ambiguous and controverted. It is far from an easy task to deduce the spectrum of the low-lying states from the magnetic data. This confirms what we stated at the beginning of this section, namely that the interaction between ions with unquenched angular momenta remains essentially an unsolved problem, in the sense where there is no widely accepted model. Each case has been treated in a specific manner, with a specific model.

Let us finish this section by looking at some of the approximations that have been used fruitfully in the case of unquenched angular momentum. The most severe consists of assuming that the distortion is large enough for the orbital degeneracy to be fully removed. In most polynuclear compounds the site symmetries of the spin carriers are rather low, which may lead to local ground states in which the orbital momentum is essentially quenched, and local excited states too high in energy to be thermally populated. Thus, the only orbital momentum effects are local anisotropies, and temperature independent paramagnetism (see Section 2.5). When the zero-field splitting within the local ground state is important, only the lowest component is thermally populated in a given temperature range. It is then possible to use the effective spin formalism described in Section 3.4. This approximation is obviously restricted to the case where the interaction parameter is very small with respect to the energy gap between the two components of lowest energy arising from the local state. Moreover, the interaction may be strongly anisotropic, and the HDVV effective Hamiltonian no longer valid. Some further information concerning this problem will be given in Section 11.3.

References

9.1. J.H. Van Vleck, "The Theory of Electric and Magnetic Susceptibilities." Oxford University Press, Oxford, 1932.

9.2. I. Morgenstern-Badarau, M. Rerat, O. Kahn, J. Jaud, J. Galy, *Inorg. Chem.* 21 (1982), 3050.

9.3. J.P. Beale, J.A. Cunningham, D. Phillips, *Inorg. Chim. Acta* 33 (1979) , 113.

9.4. P.W. Anderson, *Phys. Rev.* 115 (1959), 2.

9.5. P.W. Anderson, in "Magnetism," G.T. Rado, H. Suhl, eds., Vol. 1, p. 25. Academic Press, New York, 1963.

9.6. P.J. Hay, J.C. Thibeault, R.H. Hoffmann, *J. Am. Chem. Soc.* 97 (1975), 4884.

9.7. J.J. Girerd, M.F. Charlot, O. Kahn, *Mol. Phys.* 34 (1977), 1063.

9.8. O. Kahn, *Struct. Bonding (Berlin)* 68 (1987), 89.

9.9. J.B. Goodenough, *Phys. Rev.* 100 (1955), 564; *J. Phys. Chem. Solids* 6 (1958), 287.

9.10. J.B. Goodenough, "Magnetism and the Chemical Bond." Interscience, New York, 1963.

9.11. J. Kanamori, *J. Phys. Chem. Solids* 10 (1959), 87.

9.12. A.P. Ginsberg, *Inorg. Chim. Acta Rev.* 5 (1971), 45.

9.13. M.F. Charlot, O. Kahn, M. Drillon, *Chem. Phys.* 70 (1982), 177.

9.14. D.J. Hodgson, in "Magneto-Structural Correlation in Exchange Coupled Systems," R.D. Willett, D. Gatteschi, O. Kahn, eds., p. 497, NATO ASI Series. Reidel, Dordrecht, 1985.

9.15. J. Glerup, D.J. Hodgson, E. Pedersen, *Acta Chem. Scand.* A37 (1983), 161.

9.16. A.P. Ginsberg, M.E. Lines, *Inorg. Chem.* 11 (1972), 2289.

9.17. AP. Ginsberg, R.L. Martin, R.W. Brookes, R.C. Sherwood, *Inorg. Chem.* 11 (1972), 2884.

9.18. D.M. Duggan, D.N. Hendrickson, *Inorg. Chem.* 12 (1973), 2422; 13 (1974) , 2929.

9.19. D. Knetsch, W.L. Groeneveld, *Inorg. Nuclear Chem. Lett.* 12 (1976), 27.

9.20. E.J. Laskowski, T.R. Felthouse, D.N. Hendrickon, G. Long, *Inorg. Chem.* 15 (1976), 2908.

9.21. Y. Journaux, O. Kahn, *J. Chem. Soc. Dalton Trans.* (1979), 1575.

9.22. M.I. Arriortua, R. Cortés, L. Lezama, T. Rojo, X. Solanz, M. Font-Bardia, *Inorg. Chim. Acta* 174 (1990), 263.

9.23. R. Hotzelmann, K. Wieghardt, U. Flörke, H.J. Haupt, D.C. Weatherburn, J. Bonvoisin, G. Blondin, J.J. Glrerd, *J. Am. Chem. Soc.* 114 (1992), 1681.

9.24. Y. Pei, Y. Journaux, O. Kahn, *Inorg. Chem.* 28 (1989), 100.

9.25. Y. Journaux, O. Kahn, J. Zarembowitch, J. Galy, J. Jaud, *J. Am. Chem. Soc.* 105 (1983), 7585.

9.26. Z.J. Zhong, H. Okawa, N. Matsumoto, H. Sakiyama, S. Kida, *J. Chem. Soc. Dalton Trans.* (1991), 497.

9.27. Z.J. Zhong, N. Matsumoto, H. Okawa, H. Sakiyama, S. Kida, *Inorg. Chem.* 30 (1991), 436.

9.28. T.R. Holman, C. Juarez-Garcia, M.P. Hendrich, L. Que, Jr., E. Münck, *J. Am. Chem. Soc.* 112 (1990), 7611.

9.29. Y. Journaux, O. Kahn, I. Morgenstern-Badarau, J. Galy, J. Jaud, A. Bencini, D. Gatteschi, *J. Am. Chem. Soc.* 107 (1985), 6305.

9.30. A. Benchini, A. Caneschi, A. Dei, D. Gatteschi, C. Zanchini, O. Kahn, *Inorg. Chem.* 25 (1986), 1374.

9.31. S.M. Gorun, S.J. Lippard, *Inorg. Chem.* 30 (1991), 1625.

9.32. K. Wieghardt, U. Bossek, D. Ventur, J. Weiss, *J. Chem. Soc., Chem. Commun.* (1985), 347.

9.33. O. Kahn, in "Magneto-Structural Correlations in Exchange Coupled Systems," R.D. Willett, D. Gattesch, O. Kahn, eds., p.37, NATO ASI Series. Reidel, Dordrecht, 1985.

9.34. J..Kanamori, in "Magnetism," G.T. Rado, H. Suhi, eds., Vol. 1, p. 127. Academic Press, New York, 1963.

9.35. D. Gatteschi, A. Bencini, in "Magneto-Structural Correlations in Exchange Coupled Systems," R.D. Willett, D. Gatteschi, O. Kahn, eds., NATO ASI Series, Ser. C, Vol. 140, p. 241. Reidel, Dordrecht, (1985).

9.36. L. Banci, A. Bencini, D. Gatteschi, *J Am. Chem. Soc.* 105 (1983), 761.

9.37. F. Hartmann-Boutron, *J. Phys. Paris* 29 (1968), 212.

9.38. R.J. Elliott, M.F. Thorpe, *J. Appl. Phys.* 39 (1968), 802.

9.39. P.M. Levy, *Phys. Rev.* 117 (1969), 509.

9.40. M.E. Lines, *J. Chem. Phys.* 55 (1971), 2977.

9.41. M.E. Lines, A.P. Ginsberg, R.L. Martin, R.C. Sherwood, *J. Chem. Phys.* 57 (1972), 1.

9.42. C.G. Barraclough, A.K. Gregson, *J. Chem. Soc. Faraday Trans. II* (1972), 177.

9.43. O. Kahn, *Mol. Phys.* 29 (1975), 1039.

9.44. K.W. H. Stevens, *Phys. Rep.* 24 (1976), 1.

9.45. O. Kahn, P. Tola, H. Coudanne, *Chem. Phys.* 42 (1979), 355.

9.46. B. Briat, O. Kahn, I. Morgenstern-Badarau, *Inorg. Chem.* 20 (1981), 4193.

9.47. M. Drillon, R. Georges, *Phys. Rev. B* 24 (1981), 1278; 26 (1982), 3882.

9.48. G. Herzberg, "Molecular Spectra and Molecular Structure," Vol. 3, p. 569. van Nostrand, New York, 1966.

CHAPTER

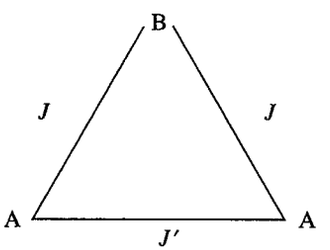
10

Trinuclear Compounds and Compounds of Higher Nuclearity

This chapter concerns discrete molecular species containing more than two magnetic centers. In such systems the microscopic nature of the interaction between adjacent centers is rigorously the same as in dinuclear compounds but the presence of more than two spin carriers may lead to new magnetic behaviors. We will thus introduce the concepts of irregular spin state structure and of spin frustration, situations that are not encountered in dinuclear species. We will also distinguish two methods of calculation of molar magnetic susceptibility, one of which may be performed from the vectorial model without writing explicitly the wavefunctions and one where this vectorial model fails.

10.1 Symmetrical ABA Compounds

A relatively large number of triangular or linear ABA trinuclear species have been described. Their basic structure is



with two equal A–B interaction pathways and an A–A pathway. The local spins are denoted S_{A_1}, S_{A_2}, ($S_{A_1} = S_{A_2} = S_A$), and S_B, and the interaction parameters are denoted J and J'. The spin Hamiltonian describing the low-lying states in zero field may be written as

$$\mathbf{H} = -J(\mathbf{S}_{A_1}\cdot\mathbf{S}_B + \mathbf{S}_{A_2}\cdot\mathbf{S}_B) - J'\mathbf{S}_{A_1}\cdot\mathbf{S}_{A_2} \tag{10.1.1}$$

where, at first, we neglect local anisotropies and anisotropic interactions. Equation (10.1.1) may be rewritten as

$$\mathbf{H} = -J(\mathbf{S}_{A_1}\cdot\mathbf{S}_B + \mathbf{S}_{A_2}\cdot\mathbf{S}_B + \mathbf{S}_{A_1}\cdot\mathbf{S}_{A_2}) - (J' - J)\mathbf{S}_{A_1}\cdot\mathbf{S}_{A_2} \tag{10.1.2}$$

or

$$\mathbf{H} = -\frac{J}{2}(\mathbf{S}^2 - \mathbf{S}_{A_1}^2 - \mathbf{S}_{A_2}^2 - \mathbf{S}_B^2) - \frac{J'-J}{2}(\mathbf{S}'^2 - \mathbf{S}_{A_1}^2 - \mathbf{S}_{A_2}^2) \tag{10.1.3}$$

with

$$\mathbf{S}' = \mathbf{S}_{A_1} + \mathbf{S}_{A_2} \tag{10.1.4}$$

$$\mathbf{S} = \mathbf{S}' + \mathbf{S}_B \tag{10.1.5}$$

The relative energies in zero field are expressed as

$$E(S,S') = -\frac{J}{2}S(S+1) - \frac{J'-J}{2}S'(S'+1) \tag{10.1.6}$$

S' varies by an integer value from 0 to $2S_A$, and for each S' value, S varies by an integer value from $|S'-S_B|$ to $S'+S_B$. To determine the magnetic susceptibility, we must add the Zeeman perturbation $\mathbf{H}_{ZE}$ to the zero-field Hamiltonian (10.1.1). $\mathbf{H}_{ZE}$ is

$$\mathbf{H}_{ZE} = \beta[g_A(\mathbf{S}_{A_1} + \mathbf{S}_{A_2}) + g_B\mathbf{S}_B]\cdot H \tag{10.1.7}$$

The local **g**-tensors are assumed to be isotropic with the g_A and g_B principal values. A $g_{S,S'}$ factor corresponds to each of the spin states with the energy $E(S,S')$, which is related to g_A and g_B through [10.1]

$$g_{S,S'} = \frac{g_A[S(S+1) + S'(S'+1) - S_B(S_B+1)] + g_B[S(S+1) - S'(S'+1) + S_B(S_B+1)]}{2S(S+1)} \tag{10.1.8}$$

deduced from (6.3.13) and (6.3.14). The expression for the molar magnetic susceptibility χ may easily be obtained by applying the simplified Van Vleck formula (1.4.11). Indeed, the $g_A\beta H$ and $g_B\beta H$ energies are certainly negligible with respect to the energy gaps between the low-lying states, such that all the second-order Zeeman coefficients vanish. The general expression for χ is

$$\chi = \frac{N\beta^2}{3kt}\frac{\sum_{S'=0}^{2S_A}\sum_{S=|S'-S_B|}^{S'+S_B} g_{S,S'}^2\, S(S+1)(2S+1)\exp[-E(S,S')/kT]}{\sum_{S'=0}^{2S_A}\sum_{S=|S'-S_B|}^{S'+S_B}(2S+1)\exp[-E(S,S')/kT]} \tag{10.1.9}$$

We will consider successively five cases, namely (1) Cu(II) trinuclear species, (2) linear Cu(II)Mn(II)Cu(II) or Cu(II)Fe(III)Cu(II) species in which the central ion is high-spin, (3) a linear Cu(II)Fe(III)Cu(II) compound in which iron(III) is low spin, (4) linear Mn(II)Cu(II)Mn(II) species, and (5) a linear Mn(II) species.

10.1.1 Copper(II) Trinuclear Species

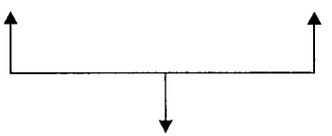
We have $S_A = S_B = 1/2$, and the relative energies of the states deduced from (10.6) are

$$E(1/2,1) = 0$$

$$E(3/2,1) = -3J/2$$

$$E(1/2,0) = -J + J' \tag{10.1.10}$$

The level $E(1/2,1)$ was taken as the energy origin. When the structure is that of an equilateral triangle with $J = J'$, the two doublet states are degenerate. The molecular symmetry is then at least C_3 and these doublet states correspond to a spectroscopic term ^{2}E. Since there is no fourfold degenerate irreducible representation in the double group C'_3, this ^{2}E term should be split into two Kramers doublets in zero field (see Section 2.4). When the system is linear, with a negligible interaction between the terminal copper(II) ions, the $E(1/2,1)$ state is the lower in energy if J is negative, which corresponds schematically to the spin structure



For J positive, the ground state is $E(3/2,1)$; the three local spins are aligned in the same direction:

In all cases, the magnetic susceptibility is given by

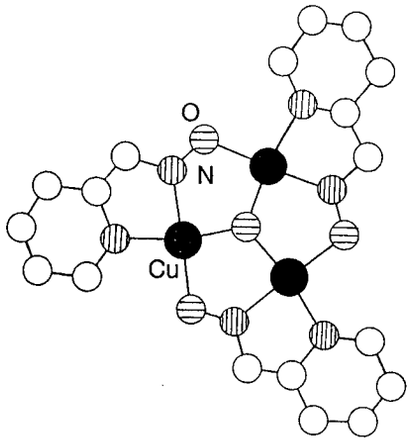


Figure 10.1. Molecular structure of the triangular copper(II) cation $[Cu_3(pao)_3OH]^{2+}$ with Hpao = pyridine-2-aldehyde-oxime. The structure is projected down the threefold axis. The μ_3-OH group is pulled out from the Cu_3 plane toward the sulfato counteranion (from ref. [10.2]).

$$\chi = \frac{N\beta^2}{4kT} \frac{g_{1/2,1}^2 + g_{1/2,0}^2 \exp[(J-J')/kT] + 10g_{3/2,1}^2 \exp[3J/2kT]}{1 + \exp[(J-J')/kT] + 2\exp[3J/2kT]} \qquad (10.1.11)$$

with, from (10.1.8)

$$g_{1/2,1} = (4g_A - g_B)/3$$

$$g_{3/2,1} = (2g_A + g_B)/3$$

$$g_{1/2,0} = g_B \qquad (10.1.12)$$

(10.1.12) is no longer valid for the equilateral case.

Two examples of copper(II) trinuclear units are represented in Figures 10.1 and 10.2. Figure 10.1 concerns the triangular cation $[Cu_3(pao)_3OH]^{2+}$ with Hpao = pyridine-

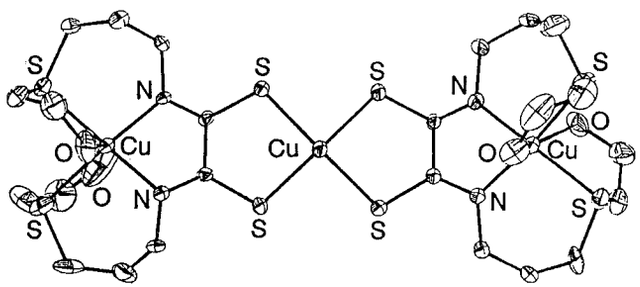


Figure 10.2. Molecular structure of the linear trinuclear cation $\{Cu_3[C_2S_2(NCH_2CH_2SCH_2CH_2OH)_2]_2\}^{2+}$ (from ref. [10.4]).

2-aldehyde-oxime. This cation has a crystallographic threefold rotation axis [10.2]. χT for $[Cu_3(pao)_3(OH)](SO_4)$ is constant and equal to 0.39 cm^3 K mol^{-1} in the whole temperature range below 400 K. This indicates that only the twofold orbitally degenerate term ^2E is thermally populated. The excited spin triplet state is located at more than 1000 cm^{-1} above. Using magnetization versus field measurements, the zero-field splitting within this ^2E term was estimated as 12 cm^{-1} [10.3]. Figure 10.2 shows the linear trinuclear cation $\{Cu_3[C_2S_2(NCH_2CH_2CH_2SCH_2CH_2OH)_2]_2\}^{2+}$ where, in spite of the large separation between the nearest neighbor copper(II) ions, the interaction is strongly antiferromagnetic [10.4]. This is due to the nature of the bridging ligands (see Section 8.6). When the anion is perchlorate, χT is equal to 0.52 cm3 K mol^{-1} at room temperature. This value slightly decreases upon cooling, and remains constant below 200 K, at 0.40 cm^3K mol^{-1}, a value expected for an isolated doublet state. J is found equal to –483 cm^{-1}.

10.1.2 Linear Cu(II)Mn(II)Cu(II) Species

The local spins are $S_A = 1/2$ and $S_B = 5/2$ and, neglecting the interaction between the terminal copper(II) ions, the relative energies of the states are

$$E(3/2,1) = 0$$

$$E(5/2,1) = -5J/2$$

$$E(7/2,1) = -6J$$

$$E(5/2,0) = -7J/2 \qquad (10.1.13)$$

The distribution in energy of these states is schematized in Figure 10.3. Each state is represented by an arrow, the length of which is equal to the spin associated with this state. Whatever the nature ferro- or antiferromagnetic of the Cu(II)–Mn(II) interaction, the spin varies monotonically versus the energy. The spin state structure is regular (see Section 6.2). For $J < 0$, which is usually the case, the ground state has the smallest spin and the most excited state has the highest spin. The $g_{S,S'}$ factors may be calculated from (10.1.8) as

$$g_{3/2,1} = (7g_{Mn} - 2g_{Cu})/5$$

$$g_{5/2,1} = (31g_{Mn} + 4g_{Cu})/35$$

$$g_{7/2,1} = (5g_{Mn} + 2g_{Cu})/7$$

$$g_{5/2,0} = g_{Mn} \qquad (10.1.14)$$

and the magnetic susceptibility may be obtained from Eq. 10.1.9. The high-temperature limit $(\chi T)_{HT}$ of χT is the sum of that expected for a Mn(II) and two Cu(II) ions, i.e.,

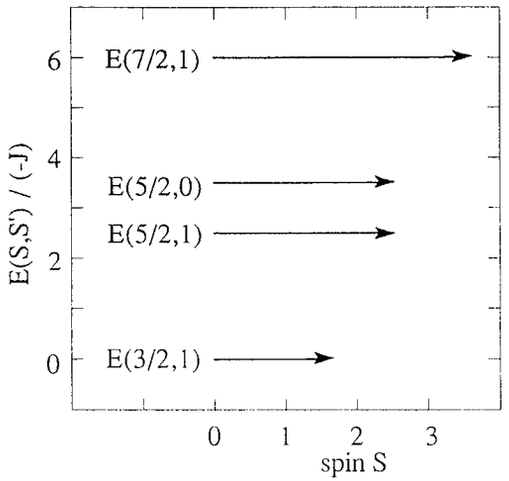


Figure 10.3. Energy spectrum of the low-lying states for a linear Cu(II)Mn(II)Cu(II) trinuclear species. Each state is represented by an arrow the length of which is equal to the spin S associated with this state. The Cu(II)–Mn(II) interaction parameter is noted J, and the interaction between terminal Cu(II) ions is assumed to be negligible.

$$(\chi T)_{HT} = \frac{N\beta^2}{12k}(35g_{Mn}^2 + 6g_{Cu}^2) \qquad (10.1.15)$$

When the temperature is lowered, the first state to be thermally depopulated is $E(7/2,1)$, which has the highest spin, and then the two states $E(5/2,0)$ and $E(5/2,1)$. The low-temperature limit $(\chi T)_{LT}$ is finally reached when only the ground state with the smallest spin is populated. The low-temperature limit $(\chi T)_{LT}$ is then expressed according to

$$(\chi T)_{LT} = \frac{5Ng_{3/2,1}^2\beta^2}{4k} \qquad (10.1.16)$$

It follows that χT continuously decreases on cooling, which is a direct consequence of the regular spin state structure.

Actually, the situation may be a little more complicated due to the zero-field splittings within the pair states with $S > 1/2$. If both the local anisotropies and the anisotropic interactions are much smaller than the isotropic interaction, then only the zero-field splitting within the ground state has an influence on the magnetic properties. Instead of exhibiting a plateau in the temperature range where only the ground state is thermally populated, the χT versus T plot may tend to a value smaller than $(\chi T)_{LT}$ given in (10.1.16) on cooling to the lowest temperatures.

To our knowledge there is no example of a symmetrical Cu(II)Mn(II)Cu(II) compound but some Cu(II)Fe(III)Cu(II) species have been described, in which Fe(III) is in a high-spin local state with $S_{Fe} = 5/2$. One of these is $\{[Cu(Mesalen)]_2Fe(acac)\}(NO_3)_2$ with Mesalen = *N,N'*-bis(methylsalicylaldiminato)ethylenediamine and acac = acetylacetonato [10.5]. The proposed structure of the trinuclear cation is shown in Figure 10.4. The Cu—Fe—Cu linkage is certainly not linear, however the interaction between the terminal Cu(ll) ions may be assumed to be negligible. The molar magnetic susceptibility of this compound was measured down to 4.2 K; χT decreases as T is lowered, which indicates that the Cu(II)–Fe(III) interaction is antiferromagnetic. However, after having reached the beginning of a plateau around 40 K, χT decreases on further cooling. This is due to a zero-field splitting within the quartet ground state, arising from both local anisotropy around iron(III) and Cu(II)–Fe(III) anisotropic interaction. The appropriate spin Hamiltonian in zero field is then

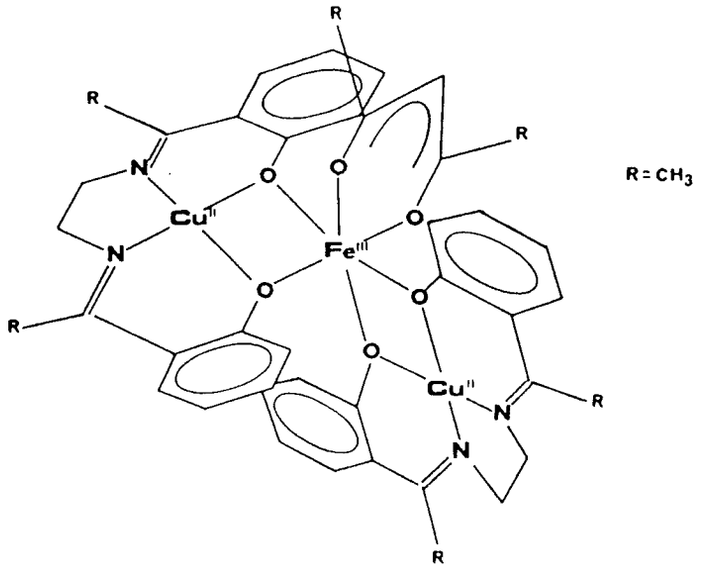


Figure 10.4. Schematic structure of the trinuclear cation $\{[Cu(Mesalen)]_2Fe(acac)\}^{2+}$ with Mesalen = *N,N'*-bis(methylsalicylaldiminato)ethylenediamine and acac = acetylacetonato (from ref. [10.5]).

$$\mathbf{H} = -J(\mathbf{S}_{Cu1}\cdot\mathbf{S}_{Fe} + \mathbf{S}_{Cu2}\cdot\mathbf{S}_{Fe}) + \mathbf{S}_{Fe}\cdot\mathbf{D}_{Fe}\cdot\mathbf{S}_{Fe} + \mathbf{S}_{Cu1}\cdot\mathbf{D}_{Cu1Fe}\cdot\mathbf{S}_{Fe} + \mathbf{S}_{Cu2}\cdot\mathbf{D}_{Cu2Fe}\cdot\mathbf{S}_{Fe} \qquad (10.1.17)$$

Assuming, for the sake of simplicity, that the zero-field splitting is axial and denoting D the energy gap between the $\pm 3/2$ and $\pm 1/2$ Kramers doublets arising from the $E(3/2,1)$ ground state, the parallel and perpendicular magnetic susceptibilities are expressed according to

$$\chi_z = (N\beta^2/4kT)\{g_{3/2,1}^2[\exp(D/kT) + 9\exp(-D/kT)] + 35g_{5/2,1}^2\exp(5J/2kT) + 35g_{5/2,0}^2\exp(7J/2kT)$$

$$+ 84g_{7/2,1}^2 \exp(6J/kT)\}/[\exp(D/kT) + \exp(-D/kT) + 3\exp(5J/2kT) + 3\exp(7J/2kT) + 4\exp(6J/kT)] \quad (10.1.18)$$

$$\chi_x = (N\beta^2/4)(4g_{3/2,1}^2/kT)\exp(D/kT) + (3g_{3/2,1}^2/D)[\exp(D/kT) - \exp(-D/kT)] + (35g_{5/2,1}^2/kT)\exp(5J/2kT) + (35g_{5/2,0}^2/kT)\exp(7J/2kT) + (84g_{7/2,1}^2/kT)\exp(6J/kT)\}/[\exp(D/kT) + \exp(-D/kT) + 3\exp(5J/2kT) + 3\exp(7J/2kT) + 4\exp(6J/kT)] \quad (10.1.19)$$

with $D = 3D_{zz}/2$, D_{zz} being one of the principal values of the zero-field splitting tensor $\mathbf{D}_{3/2,1}$ for the ground state. Gatteschi and Bencini have derived general expressions relating the tensors of the spin Hamiltonian to those of the effective spin Hamiltonians associated with each of the low-lying states [10.1]. In this case the relationship between $\mathbf{D}_{3/2,1}$ and the tensors appearing in (10.1.17) is

$$\mathbf{D}_{3/2,1} = -7(\mathbf{D}_{Cu1Fe} + \mathbf{D}_{Cu2Fe})/30 + 28\mathbf{D}_{Fe}/15 \quad (10.1.20)$$

As we have already mentioned, $\mathbf{D}_{Fe}$ is probably the main contribution to $\mathbf{D}_{3/2,1}$. The reported values of J and $|D|$, deduced from the magnetic data, are –63 and 7.3 cm^{-1}, respectively. The sign of D could not be determined unambiguously from powder magnetic susceptibility data. However, at 4.2 K the EPR spectrum displays features characteristic of a $|\pm 1/2>$ Kramers doublet, which arises from the splitting in zero-field of the spin quartet state. The anisotropy of this spectrum indicates that D is positive, the $|\pm 1/2>$ level being lower than the $|\pm 3/2>$.

10.1.3 Linear Cu(II)Fe(III)Cu(II) Compound with a Low-Spin Central Ion

If the symmetry of the central Fe(III) in a Cu(II)Fe(III)Cu(II) triad with $S_{Fe} = 1/2$ is low enough, the orbital degeneracy of this ion is largely removed, and the problem becomes similar to that of a copper(II) trinuclear species. Indeed, all three interacting ions have 1/2 local spins. We mention this case, however, because an interesting compound of this type has been reported [10.6, 10.7]. Its structure is shown in Figure 10.5. The iron(III) ion at the center of a tetraphenylphorphyrinate cycle interacts through imidazolate bridging ligands with the two copper(ll) ions situated on both sides of the cycle. In the quasi D_{4h} site symmetry the t_{2g} orbitals of Fe(III) are split into b_{2g} and e_g, with b_{2g} lower in energy; e_g is further split by spin-orbit coupling. It turns out that the unpaired electron of Fe(III) occupies one of the e_g orbitals. This orbital has a π symmetry with regard to the pseudo fourfold axis perpendicular to the plane of the cycle. The unpaired electron of each Cu(II) ion in square planar surroundings occupies an orbital localized in the copper basal plane which points along the Cu—N bond directions. This orbital has essentially σ symmetry with regard to the

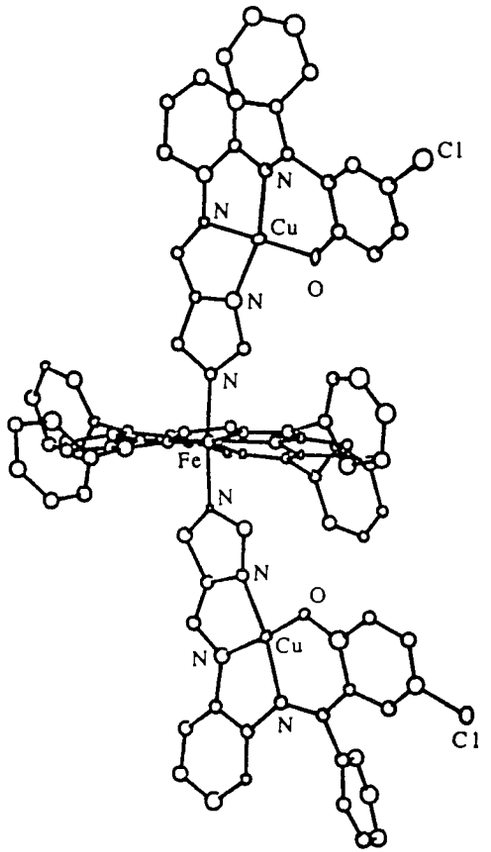


Figure 10.5. Molecular structure of the cation $[Fe(TTP)(CulM)_2]^+$, with TPP = tetraphenylporphyrinate and CulM = a copper(II) complex of the Schiff base formed by sequential condensation of 5-chloro-2-hydroxybenzophenone, 1,2-diaminobenzene, and imidazole-4-carbaldehyde (from ref. [10.7]).

fourfold axis. We are therefore faced with a novel case of orthogonality, and Fe(III)–Cu(II) ferromagnetic interactions may be expected, with an $S = 3/2$ ground state. Magnetic susceptibility together with Mössbauer and EPR data confirm this. The χT versus T plot increases as the temperature is lowered down to ca. 10 K. The Fe(III)–Cu(II) interaction parameter is reported as 44 cm^{-1}.

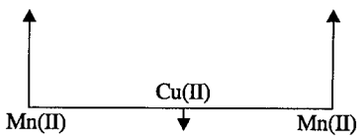
The broader significance of this work lies in its relation with results concerning the cyanide form of the enzyme cytochrome *c* oxidase. In its oxidized form, this enzyme contains an isolated Fe(III), an isolated Cu(II) ion and a Fe(III)Cu(II) pair. In presence of cyanide the Fe(III) ion of the pair is low spin, and magnetic circular dichroism and Mössbauer data have been interpreted in terms of a ferromagnetic interaction [10.8, 10.9]. A σ/π orthogonality of magnetic orbitals with respect to the Fe—CN—Cu linkage could explain the nature of the interaction [10.10].

10.1.4 Linear Mn(II)Cu(II)Mn(II) Species

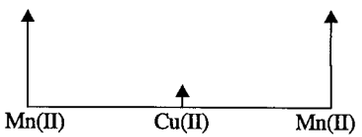
This type of trinuclear species with $S_{Mn} = 5/2$ and $S_{Cu} = 1/2$ allows us to introduce the concept of irregular spin state structure. The relative energies of the low-lying states are easily calculated from (10.1.6) as

$$\begin{aligned} E(9/2,5) &= 0 \\ E(11/2,5) &= -11J/2 \\ E(7/2,4) &= -J/2 \\ E(9/2,4) &= -5J \\ E(5/2,3) &= -J \\ E(7/2,3) &= -9J/2 \\ E(3/2,2) &= -3J/2 \\ E(5/2,2) &= -4J \\ E(1/2,1) &= -2J \\ E(3/2,1) &= -7J/2 \\ E(1/2,0) &= -3J \end{aligned} \qquad (10.1.21)$$

The spin state structure, i.e., the energy versus spin diagram, is shown in Figure 10.6. This spin state structure is no longer regular; the spin does not vary monotonicalIy versus the energy. On the contrary, for $J < 0$, the ground state has the spin 9/2 which may be schematically represented as



The two terminal local spins are aligned in the same same direction owing to the Mn(II)–Cu(II) antiferromagnetic interactions. On going up in energy, the spin of the first excited states diminishes, reaches 1/2 twice then increases. The most excited state has the highest spin, namely $S = 11/2$, and corresponds to a situation where the three local spins are aligned in the same direction as



This irregularity of the spin state structure has quite drastic consequences on the magnetic behavior. The high-temperature limit $(\chi T)_{HT}$ of χT is equal to the sum of the contributions of the isolated ions, hence

$$(\chi T)_{HT} = \frac{N\beta^2}{12k}(70g_{Mn}^2 + 3g_{Cu}^2) \qquad (10.1.22)$$

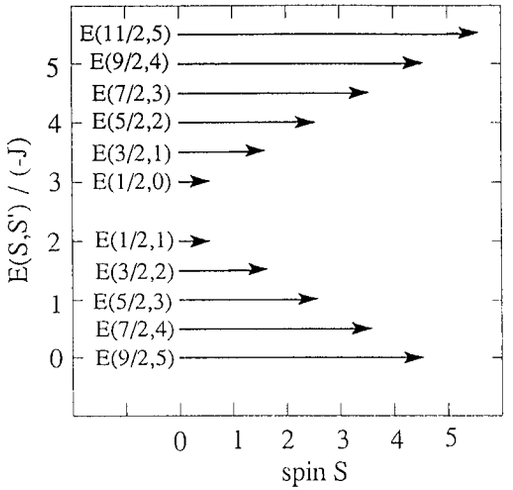


Figure 10.6. Energy spectrum of the low-lying states in a linear Mn(II)Cu(II)Mn(II) trinuclear species. Each state is represented by an arrow whose length is equal to the spin S associated with this state. The Cu(II)–Mn(II) interaction parameter is noted J, and the interaction between terminal Mn(II) ions is assumed to be negligible.

This high-temperature limit is close to 9.1 cm^3 K mol^{-1} (for g_{Mn} and g_{Cu} close to 2). When the temperature is lowered from high temperatures, the first state to be thermally depopulated is $E(11/2,5)$ which has the highest spin, and χT decreases. In the low-temperature range, when only a few excited states are significantly populated, further cooling depopulates states with a spin lower than that of the ground state, and χT increases. At very low temperatures, when only the ground state is populated, χT reaches a plateau, provided that this state is not split in zero-field, with a $(\chi T)_{LT}$ value given by

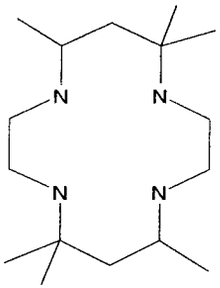
$$(\chi T)_{LT} = \frac{33Ng_{9/2,5}^2\beta^2}{4k} \qquad (10.1.23)$$

which is close to 12.4 cm^3 K mol^{-1}. The $g_{9/2,5}$ in the ground state is related to g_{Mn} and g_{Cu} through

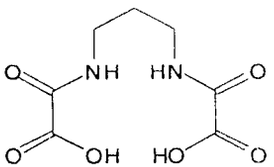
$$g_{9/2,5} = (12g_{Mn} - g_{Cu})/11 \qquad (10.1.24)$$

These high- and low-temperature behaviors result in a minimum for the χT versus T plot. The more pronounced the antiferromagnetic interaction is, the higher the temperature of this minimum.

So far, only one compound containing the Mn(II)Cu(II)Mn(II) triad has been synthesized and investigated [10.11, 10.12]. Its formula is $[Mn(Me_6\text{-}[14]ane\text{-}N_4)]_2$ Cu(pba)](CF3SO3)2·2H2O. Me_6-[14]ane-N_4 is tetradentate ligand ±5,7,7,12,14,14-hexamethyl-1,4,8,11-tetraazacyclotetradecane:



and pba^{4-} is the fourfold deprotonated form of 1,3-propylenebis(oxamate):



The skeleton of the trinuclear cation is schematized in Figure 10.7 where the terminal macrocyclic ligands have been omitted, except the coordinating nitrogen atoms. Due to the nature of the bridges, the Mn(II)–Cu(II) interaction is expected to be strongly antiferromagnetic (see Section 8.5), and the $E(9/2,5)$ ground state is expected to be well separated in energy from the first excited states. The χT versus T plot for this compound is shown in Figure 10.8. It exhibits all the features described above; χT first decreases on cooling with a very rounded minimum around 170 K, then increases in a rather abrupt fashion up to a plateau with $\chi T = 12.1$ cm^3 K mol^{-1}. The value of $g_{9/2,5}$ deduced from this $(\chi T)_{LT}$ limit is 1.98. The zero-field splitting within the ground state is certainly very weak in this compound since χT does not show any marked deviation from the Curie law in the temperature range $T < 10$ K where only the ground state is populated. The magnetic data in Figure 10.8 has been interpreted with $J = -36.6$ cm^{-1}.

A minimum in the χT versus T plot is anticipated for all ABA linear compounds with local spins $S_B = 1/2$ and $S_A > 1/2$. The temperature T_{min} of this minimum gives

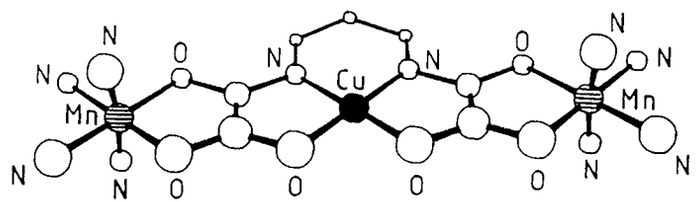


Figure 10.7. Skeleton of the trinuclear cation $[Mn(Me_6\text{-}[14]ane\text{-}N_4)]_2Cu(pba)]^{2+}$ with Me_6-[14]ane-N_4 = ±5,7,7,12,14,14-hexamethyl-1,4,8,11-tetraazacyclotetradecane, and pba = 1,3-propylenebis(oxamato) (from ref. [10.12]).

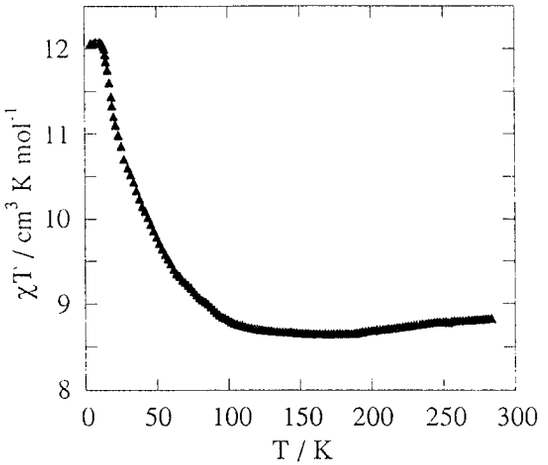


Figure 10.8. χT versus T plot for $[Mn(Me_6\text{-}[14]ane\text{-}N_4)]_2Cu(pba)](CF_3SO_3)_2{\cdot}2H_2O$ (from ref. [10.12]).

us a first estimation of the A–B interaction parameter noted J. The values of $kT_{min}/|J|$ *for* $S_A = 1$, 3/2, 2, and 5/2 are given in Table 10.1. These values have been calculated in the approximation where the g_A and g_B local Zeeman factors are equal [10.13].

10.1.5 A Linear Mn(II) Trinuclear Species

Our last example of trinuclear species concerns the compound of formula $Mn_3(CH_3CO_2)_6bipy_2$ shown in Figure 10.9. A central Mn(II) ion is bridged to each of the two terminal Mn(II) ions by three acetato groups, two of them acting as bidentate bridges and the third one bridging by just one oxygen atom. The energies $E(S,S')$ of

Table 10.1. Relationships between the Temperature T_{min} of the Minimum in χT and the Isotropic Interaction Parameter J for an ABA Linear Trinuclear Species with $S_B = 1/2$ and Various Values of S_A

S_A	$kT_{min}/\|J\|$
1	0.6435
3/2	1.420
2	2.131
5/2	3.374

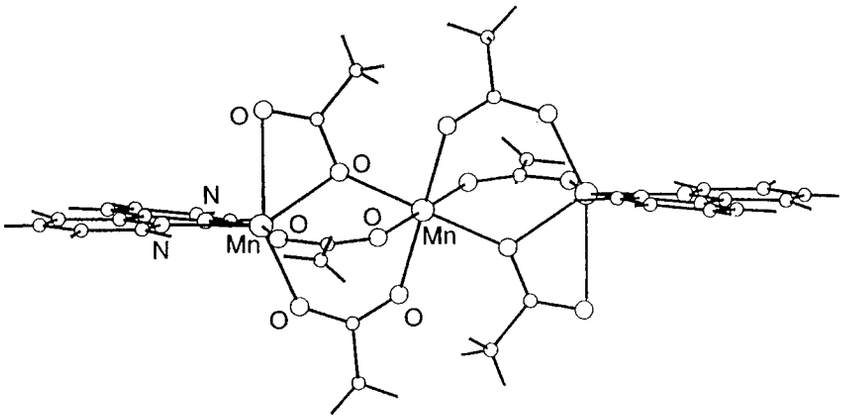


Figure 10.9. Molecular structure of the trinuclear compound $Mn_3(CH_3COO_2)_6(bipy)_2$ with bipy = 2,2′-bipyridine (from ref. [10.14]).

the 27 low-lying states calculated from Eq. (10.1.6) are tabulated in Table 10.2 in the case where the interaction between the terminal ions is neglected.

What is remarkable in the investigation of this compound is that the interaction parameter between adjacent magnetic centers has been determined independently through both low-field magnetic susceptibility and high-field magnetization measurements [10.14]. As the temperature is lowered from 285 to 4.2 K, the χT versus T plot decreases from 11.9 to 4.32 cm^3 K mol^{-1}, which indicates an antiferromagnetic interaction between adjacent Mn(II) ions. The low-temperature limit of χT corresponds to an $S = 5/2$ molecular state. The magnetic susceptibility data have been interpreted using Eq. (10.1.9). The local g factors were assumed to be equal. The magnetization M versus field H curve at 4.2 K is shown in Figure 10.10. Up to ca. 10^5 G (10 T), M follows Eq. (2.2.7) with $S = 5/2$. When the field is further increased, M continuously increases instead of showing a saturation equal to 5 $N\beta$, and reaches 7.2 $N\beta$ at 2×10^5 G (20 T), with a change of sign for the second derivative d^2M/dH^2 around 10 T. This behavior arises from the crossover between the Zeeman component $M_S = -5/2$ associated with the ground state $S = 5/2$ and the Zeeman component $M_S = -7/2$ associated with the second excited state $S = 7/2$. This crossover occurs for a field value equal to $-7J/2g\beta$. The magnetization data were interpreted by taking into account all Zeeman components of the $S = 5/2$ ground state and S = 3/2 first excited state as well as the Zeeman component $M_S = -7/2$ of the $S = 7/2$ state, located in zero field at $-3J/2$ above the ground state, and the Zeeman component $M_S = -9/2$ of the $S = 9/2$ state, located in zero field at $-8J$ above the ground state. The other levels are not significantly populated at 4.2 K and in the 0–20 T field range. The magnetization was expressed from Eq. (1.1.3). Least-squares fittings of the susceptibility and the magnetization lead to the same J value, namely $J = -4.4$ cm^{-1}. The field dependence of the 12 levels involved in the interpretation of the magnetization data is shown in Figure 10.11.

Table 10.2. Energies of the Low-Lying States in a Linear Mn(II) Trinuclear Species[a]

S	I	S'	$E(S,S')/(-J)$
15/2		5	27.5
13/2		5	20.0
		4	25.0
11/2		5	13.5
		4	18.5
		3	22.5
9/2		5	8.0
		4	13.0
		3	17.0
		2	20.0
7/2		5	3.5
		4	8.5
		3	12.5
		2	15.5
		1	17.5
5/2		5	0
		4	5.0
		3	9.0
		2	12.0
		1	14.0
		0	15.0
3/2		4	2.5
		3	6.5
		2	9.5
		1	11.5
1/2		3	5.0
		2	8.0

[a]The interaction between the terminal Mn(II) ions has been neglected.

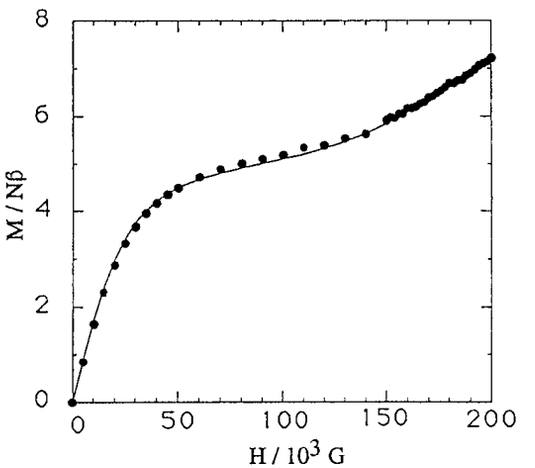


Figure 10.10. Magnetization versus magnetic field curve at 4.2 K for $Mn_3(CH_3COO_2)_6(bipy)_2$ (from ref. [10.14]).

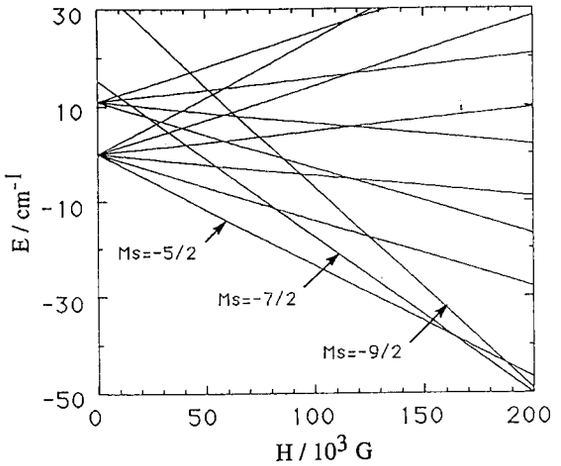
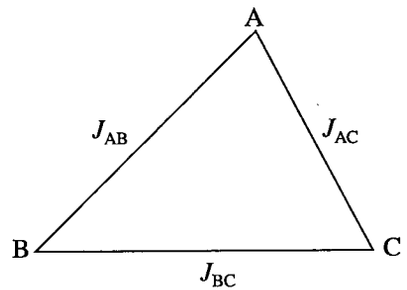


Figure 10.11. Energies of the lowest spin states as a function of the applied magnetic field for $Mn_3(CH_3COO_2)_6(bipy)_2$ (from ref. [10.14]).

10.2 Nonsymmetrical Trinuclear Compounds

In all the examples studied in this chapter so far, it has been possible to derive the theoretical expression of the molar magnetic susceptibility by combining the local spin operators in an appropriate fashion. In actual fact, this vectorial approach is restricted to rather simple cases. For most magnetic clusters, in particular those of low symmetry, it cannot be used. It is thus necessary to determine the energies of the low-lying states by diagonalizing explicitly the spin Hamiltonian using the spin functions as a basis set. Let us first treat of a rather simple example, that of a copper(II) trinuclear species ABC with three different interaction parameters J_{AB}, J_{BC}, and J_{AC}:



The spin Hamiltonian in zero field describing the isotropic interaction is

$$\mathbf{H} = -J_{AB}\mathbf{S}_A\cdot\mathbf{S}_B - J_{BC}\mathbf{S}_B\cdot\mathbf{S}_C - J_{AC}\mathbf{S}_A\cdot\mathbf{S}_C \qquad (10.2.1)$$

The interaction between three local doublets leads to a molecular quartet and two molecular doublets. We note the microstates forming the basis set $|M_{S_A}, M_{S_B}, M_{S_C}>$.

The component $M_S = 3/2$ of the quartet state is |1/2,1/2,1/2> and the energy of this state is given by

$$E(3/2) = <1/2,1/2,1/2\,|\,\mathbf{H}\,|\,1/2,1/2,1/2>$$

$$= -(J_{AB} + J_{BC} + J_{AC})/4 \qquad (10.2.2)$$

The matrix associated with **H** in the subspace $M_S = 1/2$ is

|1/2, 1/2, −1/2>	|1/2, −1/2, 1/2>	|−1/2, 1/2, 1/2>
$(-J_{AB} + J_{BC} + J_{AC})/4$	$-J_{BC}/2$	$-J_{AC}/2$
	$(J_{AB} + J_{BC} - J_{AC})/4$	$-J_{AB}/2$
		$(J_{AB} - J_{BC} + J_{AC})$

(10.2.3)

One of the roots of the $|\mathbf{H}-E| = 0$ secular determinant is obviously $-(J_{AB} + J_{BC} + J_{AC})/4$. This corresponds to the component $M_S = 1/2$ of the quartet state. The other two roots which give the energies of the doublet states are

$$E(1/2,\pm) = (J_{AB} + J_{BC} + J_{AC})/4$$

$$\pm\sqrt{[(J_{AB} - J_{BC})^2 + (J_{BC} - J_{AC})^2 + (J_{AC} - J_{AB})^2]/8} \qquad (10.2.4)$$

We can easily check that for $J_{AB} = J_{BC} = J$ and $J_{AC} = J'$, we can retrieve the energies given in (10.1.10).

It is important to point out here that the relative energies of the three states depend on only two energy gaps, Δ_1 and Δ_2 defined by

$$\Delta_1 = E(1/2,-) - E(1/2,+)$$

$$\Delta_2 = E(3/2) - E(1/2,+) \qquad (10.2.5)$$

such that the three interaction parameters cannot be determined unequivocally from the magnetic data. A simplifying hypothesis must be made, for instance assuming that one of the parameters is negligible with respect to the other two due to the structure of the trinuclear unit. The magnetic susceptibility may be expressed versus Δ_1 and Δ_2 as

$$\chi T = \frac{N\beta^2}{4k}\,\frac{g_{1/2,+}^2 + g_{1/2,-}^2\exp(-\Delta_1/kT) + 10g_{3/2}^2\exp(-\Delta_2/kT)}{1 + \exp(-\Delta_1/kT) + 2\exp(-\Delta_2/kT)} \qquad (10.2.6)$$

$g_{1/2,+}$, $g_{1/2,-}$, and $g_{3/2}$ are the molecular g-factors. In the absence of symmetry, they cannot be easily related to the local g-factors. Actually, the relationships between molecular and local g-factors depend on the interaction parameters such that, in principle, it is possible to determine these parameters from the EPR spectrum of the trinuclear species without measuring the temperature dependence of the signal intensities. This would require, of course, an accurate knowledge of the local g-factors.

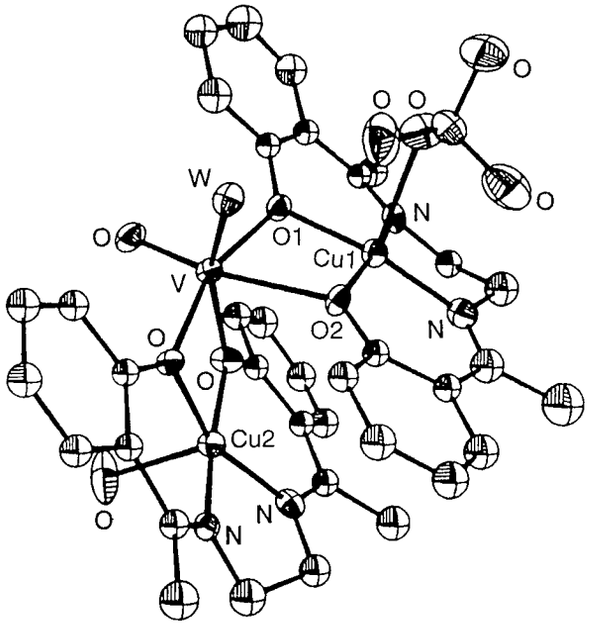
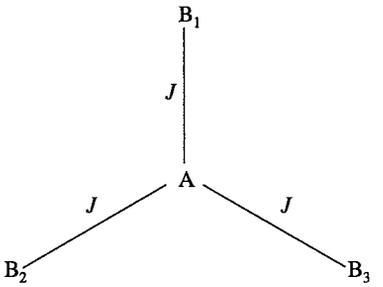


Figure 10.12. Molecular structure of the cation $\{[Cu(Mesalen)]_2VO\}^{2+}$ (from ref. [10.15]).

An interesting example of a trinuclear compound that involves three local doublets has been reported by Gatteschi et al. [10.15]. This compound is $\{[Cu(Mesalen)]_2(H_2O)VO\}(ClO_4)_2$. The structure of the cation is shown in Figure 10.12. The vanadyl ion is linked to the two copper(II) ions through the phenolic oxygen atoms of the Mesalen Schiff base, and there are two different CuO_2V bridging networks. In Cu1—O1—O2—V, O1 and O2 are in the basal plane of copper in square pyramidal surroundings; 01 is also in the basal plane of the vanadyl ion, but O2 occupies the apical position, opposite to that of the vanadyl oxygen atom. The Cu1—VO interaction parameter is noted J_{Cu1V}. The other bridging network is similar to that encountered in $CuVO(fsa)_2en{\cdot}CH_3OH$ (see Section 8.6); the two basal planes around copper and vanadium, respectively, share an edge. This bridging network has all the symmetry requirements for strict orthogonality of the magnetic orbitals, hence a ferromagnetic interaction. Therefore, the J_{Cu2V} parameter is expected to be positive and of the same order of magnitude as in $CuVO(fsa)_2en{\cdot}CH_3OH$. The magnetic susceptibility data have been analyzed with Eq. (10.2.6). Δ_1 and Δ_2 are defined as in (10.2.5); J_{AB} stands for J_{Cu1V}, J_{BC} for J_{Cu2V}, and $J_{AC} = J_{Cu1Cu2}$ is assumed to be negligible. The best fitting is obtained for $\Delta_1 \approx \Delta_2 \approx -90$ cm^{-1}. This means that one of the doublet states and the quartet state are almost degenerate, and that the second doublet state is located at roughly 90 cm^{-1} above. This is consistent with J_{Cu1V} close to zero and J_{Cu2V} close to +90 cm^{-1}.

10.3 Tetranuclear Species with a Central Magnetic Ion

We will present three examples of tetranuclear compounds in which a central ion A interacts with three peripheral ions B_i (i = 1 to 3). In two of these, the three interaction pathways A–B_i are equivalent and characterized by the J parameter:



The energies of the spin states together with the expression of the magnetic susceptibility can then be derived from the vectorial model. The third example will deal with an hypothetical system of the same type, but one of A–B interaction pathway is different from the other two. Even if the interacting centers are local doublets, the vectorial model can no longer be used.

In the case where the three interaction pathways are equivalent, the spin Hamiltonian describing the isotropic interaction is

$$\mathbf{H} = -J\mathbf{S}_A{\cdot}(\mathbf{S}_{B_1} + \mathbf{S}_{B_2} + \mathbf{S}_{B_3}) \tag{10.3.1}$$

which can be rewritten as

$$\mathbf{H} = -\frac{J}{2}(\mathbf{S}^2 - \mathbf{S}'^2 - \mathbf{S}_A{}^2) \tag{10.3.2}$$

with

$$\mathbf{S}' = \mathbf{S}_{B_1} + \mathbf{S}_{B_2} + \mathbf{S}_{B_3} \tag{10.3.3}$$

$$\mathbf{S} = \mathbf{S}' + \mathbf{S}_A \tag{10.3.4}$$

The relative energies are then given by

$$E(S,S') = -\frac{J}{2}[S(S+1) - S'(S'+1)] \tag{10.3.5}$$

where the constant energy term $-JS_A(S_A+1)/2$ has been ignored. In addition to the relative energies $E(S,S')$, we need the $g_{S,S'}$ factors which can be obtained from Eq. (6.3.13) where S_B stands for S'. The magnetic susceptibility is then obtained from the simplified Van Vleck formula (1.4.11).

Let us apply these considerations above to the compound $\{Mn[Cu(oxpn)]_3\}(ClO_4)_2$ that was studied by Lloret et al. [10.16] and that contains

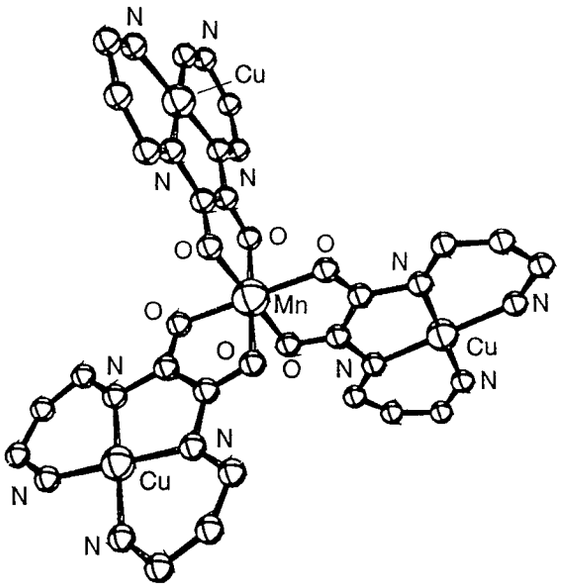


Figure 10.13. Schematized structure of the tetranuclear cation $\{Mn[Cu(oxpn)]_3\}^{2+}$ with oxpn = oxamido-*N*,*N*′-3-aminopropane (from ref. [10.16]).

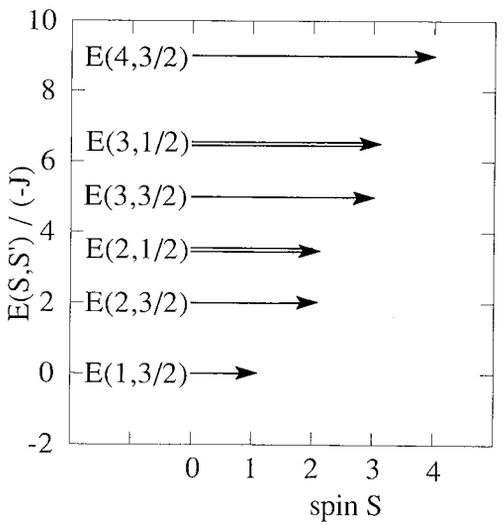


Figure 10.14. Energy spectrum of the low-lying states in a $Mn(II)Cu(II)_3$ tetranuclear species. Each state is represented by an arrow whose length is equal to the spin S associated with this state. The Cu(II)–Mn(II) interaction parameter is noted *J*, and the interaction between terminal Cu(II) ions is assumed to be negligible.

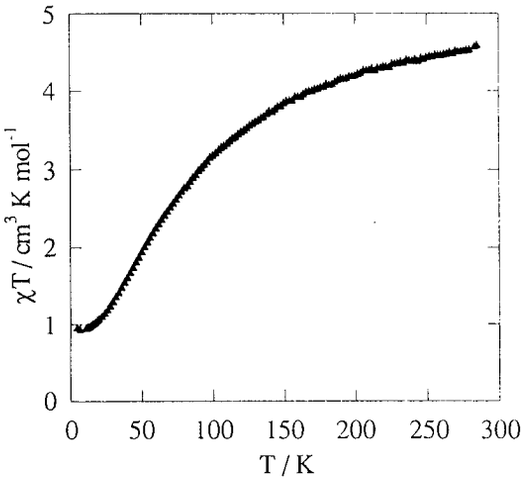


Figure 10.15. χT versus T plot for $\{Mn[Cu(oxpn)]_3\}(ClO_4)_2$ (from ref. [10.16]).

the tetranuclear cation represented schematically in Figure 10.13. H_2oxpn is oxamide-*N*,*N*′-3-aminopropane. The coupling of the three copper(II) ions gives rise to a quartet and two doublets (see Section 10.1) such that S' takes the values 3/2 and twice 1/2. For $S' = 3/2$, S takes the values 1, 2, 3, and 4, and for $S' = 1/2$, S takes the values 2 and 3. The relative energies of the molecular states and the $g_{S,S'}$ factors are calculated as

$$
\begin{array}{ll}
E(1, 3/2) = 0 & g_{1,3/2} = (7g_{Mn} - 3g_{Cu})/4 \\
E(2, 3/2) = -2J & g_{2,3/2} = (11g_{Mn} + g_{Cu})/12 \\
E(3, 3/2) = -5J & g_{3,3/2} = (17g_{Mn} + 7g_{Cu})/24 \\
E(4, 3/2) = -9J & g_{4,3/2} = (25g_{Mn} + 15g_{Cu})/40 \\
E(2, 1/2) = -7J/2 \text{ twice} & g_{2,1/2} = (14g_{Mn} - 2g_{Cu})/12 \\
E(3, 112) = -13J/2 \text{ twice} & g_{3,1/2} = (20g_{Mn} + 4g_{Cu})/24
\end{array}
\qquad (10.3.6)
$$

Figure 10.14 shows the energy versus spin diagram. The spin state structure is regular. For $J < 0$, the ground state is a triplet; on going up in energy, we encounter first three quintets, then three septets and finally a nonet. The χT versus T plot for $\{Mn[Cu(oxpn)]_3\}(ClO_4)_2$ is displayed in Figure 10.15. As the sample is cooled, χT decreases continuously, which indicates that the Mn(II)–Cu(II) isotropic interaction is antiferromagnetic, and reaches a plateau below 7 K with $\chi T = 0.98$ cm^3 K mol^{-1} in the temperature range where only the $E(1,3/2)$ state is thermally populated. These data have been fitted with $J = -26.6$ cm^{-1}, $g_{Mn} = 1.98$ and $g_{Cu} = 2.00$.

The second example deals with the compound of formula $\{Cr[(ox)Ni(Me_6\text{-}[14]ane\text{-}N_4)]_3\}(ClO_4)_3$ containing the Cr(III)Ni((II)$_3$ cation, the skeleton of which is schematized in Figure 9.2; ox stands for oxalato and Me_6-[14]ane-N_4 is the tetradentate macrocycle ± 5,7,7,12,14,14-hexamethyl-1,4,8,11-tetraazacyclotetradecane, which was mentioned in Section 9.3. The local spins are $S_{Cr} = 3/2$ for the central chro-

mium(III) ion and $S_{Ni} = 1$ for the peripheral nickel(II) ions in distorted octahedral environments. The method for determining the relative energies $E(S,S')$ of the 22 low-lying states and the $g_{S,S'}$ factors is the same as that used above for the Mn(II)Cu(II)$_3$ cluster. This leads to

$$
\begin{array}{ll}
E(9/2,3) = 0 & g_{9/2,3} = (3g_{Cr} + 6g_{Ni})/9 \\
E(7/2,2) = 3J/2 \text{ twice} & g_{7/2,2} = (3g_{Cr} + 4g_{Ni})/7 \\
E(5/2,1) = 3J \text{ three times} & g_{5/2,1} = (3g_{Cr} + 2g_{Ni})/5 \\
E(3/2,0) = 9J/2 & g_{3/2,0} = g_{Cr} \\
E(7/2,3) = 9J/2 & g_{7/2,3} = (5g_{Cr} + 16g_{Ni})/21 \\
E(5/2,2) = 5J \text{ twice} & g_{5/2,2} = (13g_{Cr} + 22g_{Ni})/35 \\
E(3/2,1) = 11J/2 \text{ three times} & g_{3/2,1} = (11g_{Cr} + 4g_{Ni})/15 \\
E(1/2,1) = 7J \text{ three times} & g_{1/2,1} = (5g_{Cr} - 2g_{Ni})/3 \\
E(3/2,2) = 15J/2 \text{ twice} & g_{3/2,2} = (g_{Cr} + 4g_{Ni})/5 \\
E(5/2,3) = 8J & g_{5/2,3} = (g_{Cr} + 34g_{Ni})/35 \\
E(1/2,2) = 9J \text{ twice} & g_{1/2,2} = -g_{Cr} + 2g_{Ni} \\
E(3/2,3) = 21J/2 & g_{3/2,3} = (-3g_{Cr} + 8g_{Ni})/5
\end{array}
\qquad (10.3.7)
$$

where, in contrast with the preceding example, the state of highest spin $E(9/2,3)$ was taken as the energy origin. The other states are ranked in the order of the increasing energies. This choice directly proceeds from the magnetic properties of this Cr(III)Ni(II)$_3$ species. The χT versus T plot is shown in Figure 10.16. At room temperature χT is equal to 5.5 cm^3 K mol^{-1}, which corresponds to what is expected for one Cr(III) ion and three Ni(II) ions, which are magnetically isolated. When the compound is cooled, χT increases in a continuous fashion to ca. 6 K, and then reaches a plateau with $\chi T = 11.3(1)$ cm^3 K mol^{-1}.

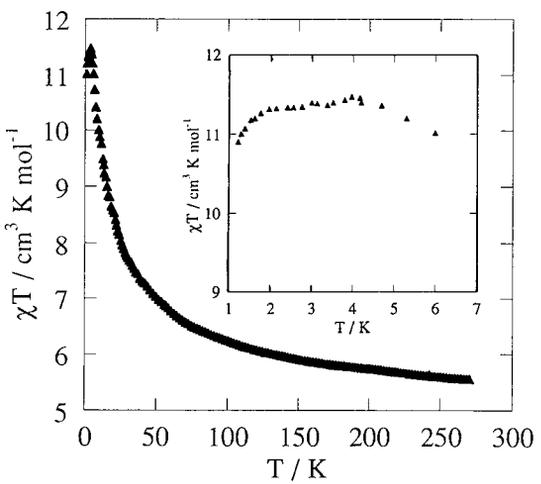


Figure 10.16. χT versus T plot for {Cr[(ox)Ni(Me$_6$-[14]ane-N$_4$)]$_3$}(ClO$_4$)$_3$. The insert shows the detail of the data in the 1–7 K temperature range (from ref. [10.17]).

This behavior clearly indicates that the Cr(III)–Ni(II) interaction is ferromagnetic ($J > 0$) such that in the ground state the four local spins are aligned parallel, the resulting spin being $S = 9/2$. The value of the plateau in the low temperature range is close to what is predicted for an isolated $S = 9/2$ state. The magnetic data have been interpreted with $J = 5.3$ cm^{-1}, $g_{Cr} = 2.05$ and $g_{Ni} = 2.17$. Magnetization versus magnetic field data carried out in the temperature range where only the ground state is thermally populated has confirmed that the spin of this state is $S = 9/2$ [10.17].

Again the nature of the Cr(III)–Ni(II) isotropic interaction may be understood from symmetry considerations. In a simple description we could say that Cr(III) in octahedral surroundings has three unpaired electrons occupying the three low-lying t_{2g} orbitals and that Ni(II), also in octahedral surroundings, has two unpaired electrons occupying the two high-lying e_g orbitals. The ferromagnetic nature of the interaction would then arise from the orthogonality of the t_{2g} and e_g orbitals. Actually, the situation is more complex because we have to consider the symmetry of a Cr(ox)Ni bridging network which is, at the most, C_{2v}. Under C_{2v} symmetry the unpaired electrons of Cr(III) occupy orbitals transforming as $a_1 + a_2 + b_2$ while the unpaired electrons of Ni(II) occupy orbitals transforming as $a_1 + b_1$. These orbitals are represented sche-

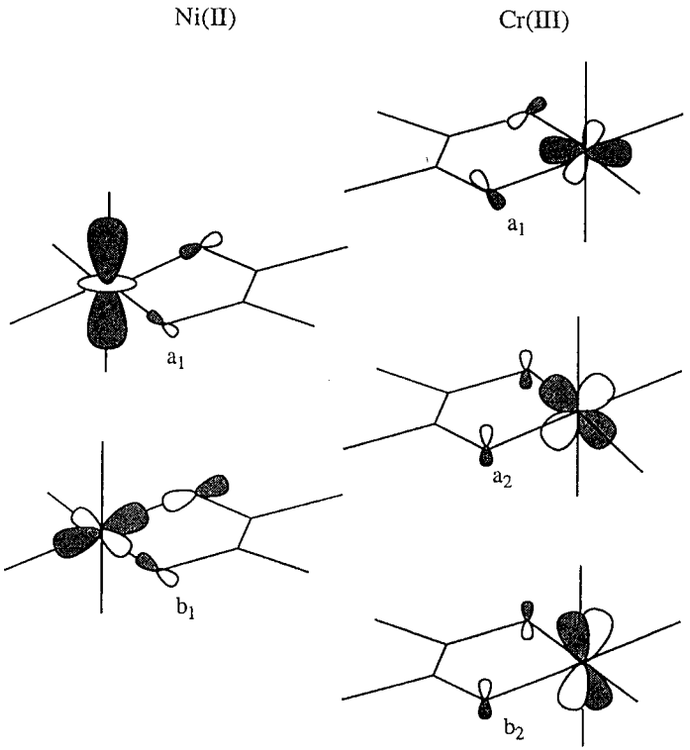


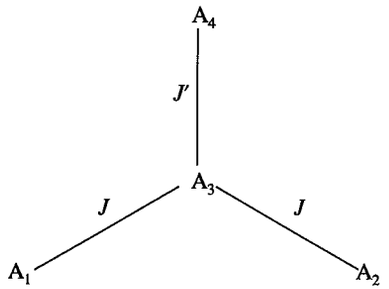
Figure 10.17. Schematic representation of the magnetic orbitals centered on the Cr(III) and Ni(II) ions in the tetranuclear cation {Cr[(ox)Ni(Me$_6$-[14]ane-N$_4$)]$_3$}$^{+3}$.

matically in Figure 10.17. The isotropic interaction parameter J may be expressed as the sum of $J_{\mu\nu}$ contributions involving pairs of interacting orbitals as (see Section 9.2)

$$J = (J_{a_1a_1} + J_{a_1b_1} + J_{a_2a_1} + J_{a_2b_1} + J_{b_2a_1} + J_{b_2b_1})/6 \tag{10.3.8}$$

where all $J_{\mu\nu}$s are positive since they involve orthogonal orbitals, except $J_{a_1a_1}$. This contribution, however, may be expected to be rather small. Indeed, the a_1 orbital of z^2-type centered on nickel is weakly delocalized in the plane of the oxalato bridge.

Our last example of tetranuclear species concerns the system



with $S_{A_i} = 1/2$ for $i = 1$ to 4, and $J \neq J'$. The spin Hamiltonian describing the isotropic interaction in zero field is then

$$\mathbf{H} = -J(\mathbf{S}_{A_1}\cdot\mathbf{S}_{A_3} + \mathbf{S}_{A_2}\cdot\mathbf{S}_{A_3}) - J'\mathbf{S}_{A_3}\cdot\mathbf{S}_{A_4} \tag{10.3.9}$$

It is easy to check that the vectorial model cannot be used. The derivation of the energies of the states that arise from the interaction between the four local doublets is rather tedious. However, we will perform it here in order to show how this kind of problem can be solved. The system may be considered as being formed of the two pairs A_1A_2 and A_3A_4. The pair states are a triplet and a singlet with the spin functions

$$S_{AA} = 1 \begin{cases} |{+}{+}> \\ (\sqrt{2}/2)(|{+}{-}> + |{-}{+}>) \\ |{-}{-}> \end{cases}$$

$$S_{AA} = 0 \quad (\sqrt{2}/2)(|{+}{-}> - |{-}{+}>) \tag{10.3.10}$$

In (10.3.10) the kets stand for $|M_{SA_1}, M_{SA_2}>$ or $|M_{SA_3}, M_{SA_4}>$, and only the + or – sign of M_{SA_i} is indicated explicitly. The interaction between two pair triplets gives rise to states with S = 2, 1 and 0. The interaction between a pair triplet and a pair singlet gives rise to an $S = 1$ state. Finally, the interaction between two pair singlets gives rise to an $S = 0$ state. It follows that we have one $S = 2$, three $S = 1$, and two $S = 0$ states. The spin functions associated with these molecular states are obtained by combining the functions (10.3.10) and using the Clebsch–Gordon coefficients (see Appendix 6). In fact, we need only one function per state; we take that of highest M_S:

$$S = 2 \quad \Psi_1 = |{+}\,{+}\,{+}\,{+}>$$

$$S = 1 \begin{cases} \Psi_2 = (1/2)(|{+}{+}{+}{-}> + |{+}{+}{-}{+}> - |{+}{-}{+}{+}> - |{-}{+}{+}{+}>) \\ \Psi_3 = (\sqrt{2}/2)(|{+}{-}{+}{+}> - |{+}{+}{-}{+}>) \\ \Psi_4 = (\sqrt{2}/2)(|{+}{-}{+}{+}> - |{-}{+}{+}{+}>) \end{cases} \tag{10.3.11}$$

$$S = 0 \begin{cases} \Psi_5 = (\sqrt{3}/6)(2|{+}{+}{-}{-}> + 2|{-}{-}{+}{+}> \\ \qquad - |{+}{-}{+}{-}> - |{+}{-}{-}{+}> - |{-}{+}{+}{-}> - |{-}{+}{-}{+}> - |{-}{-}{+}{+}>) \\ \Psi_6 = (1/2(|{+}{-}{+}{-}> - |{+}{-}{-}{+}> - |{-}{+}{+}{-}> + |{-}{+}{-}{+}>) \end{cases}$$

In (10.3.11) the kets stand for $|M_{SA_1}, M_{SA_2}, M_{SA_3}, M_{SA_4}>$ and only the + or – sign of M_{SA_i} is indicated explicitly. The energy E_1 of the $S = 2$ state is given by

$$E_1 = <\Psi_1|\mathbf{H}|\Psi_1> = -(2J + J')/4 \tag{10.3.12}$$

The energies of the $S = 1$ states are obtained by diagonalizing the spin Hamiltonian (10.3.9) and using the Ψ_i ($i = 2$ to 4) functions as a basis set. The matrix associated with **H** is

Ψ_2	Ψ_3	Ψ_4
$(2J–J')/4$	$-\sqrt{2}J/2$	0
	$3J'/4$	0
		$-J'/4$

(10.3.13)

and the eigenvalues are found as

$$E_2 = (J + J' + \sqrt{9J^2 - 4JJ' + 4J'^2})/4$$

$$E_3 = (J + J' - \sqrt{9J^2 - 4JJ' + 4J'^2})/4$$

$$E_4 = -J'/4 \tag{10.3.14}$$

The energies of the $S = 0$ states are obtained in a similar way, by diagonalizing the spin Hamiltonian (10.3.9) and using the Ψ_i ($i = 5, 6$) functions as a basis set. The matrix associated with **H** is

Ψ_5	Ψ_6
$(4J–J')/4$	0
	$3J'/4$

(10.3.15)

The matrix is diagonal so that the energies of the singlet states are

$$E_5 = (4J–J')/4$$

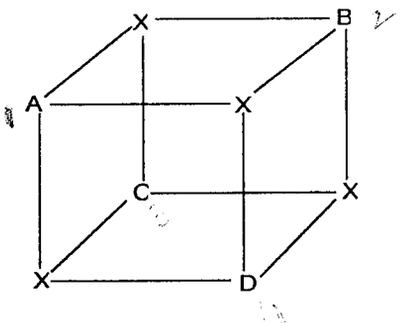
$$E_6 = 3J'/4$$

From these energies it is straightforward to derive an expression for the molar magnetic susceptibility, assuming that all the g factors are isotropic and equal. This expression is

$$\chi T = (2Ng^2\beta^2/k)[5\exp(-E_1/kT) + \exp(-E_2/kT) + \exp(-E_3/kT) + \exp(-E_4/kT)/[5\exp(-E_1/kT) + 3\exp(-E_2/kT) + 3\exp(-E_3/kT) + 3\exp(-E_4/kT) + \exp(-E_5/kT) + \exp(-E_6/kT)] \quad (10.3.16)$$

10.4 Cubane-Like Tetranuclear Species

An important class of transition metal clusters is characterized by a cubane-like structure. The ideal cubane structure is



Four metal ions noted A, B, C, and D occupy the vertices of a regular tetrahedron. Each ion is bridged to the other three through the atoms X, which occupy the vertices of a second regular tetrahedron. The simplest, but rather exceptional case, is that where all transition metal ions are of the same nature and the interaction pathways are equal. The symmetry is then T_d. If so, the spin Hamiltonian in zero field is

$$\mathbf{H} = -J(\mathbf{S_A}\cdot\mathbf{S_B} + \mathbf{S_A}\cdot\mathbf{S_C} + \mathbf{S_A}\cdot\mathbf{S_D} + \mathbf{S_B}\cdot\mathbf{S_C} + \mathbf{S_B}\cdot\mathbf{S_D} + \mathbf{S_C}\cdot\mathbf{S_D}) \quad (10.4.1)$$

which can be rewritten as

$$\mathbf{H} = -\frac{J}{2}(\mathbf{S}^2 - \mathbf{S_A}^2 - \mathbf{S_B}^2 - \mathbf{S_C}^2 - \mathbf{S_D}^2) \quad (10.4.2)$$

with

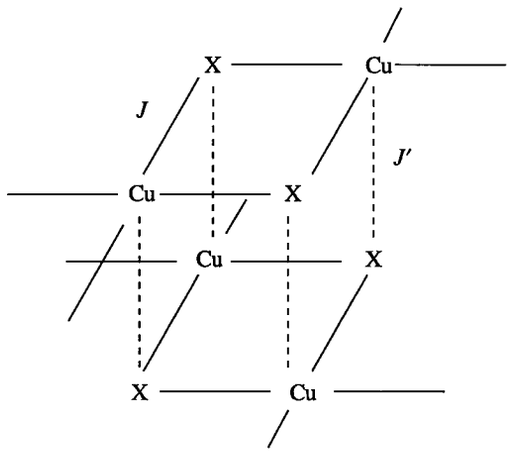
$$S = \mathbf{S_A} + \mathbf{S_B} + \mathbf{S_C} + \mathbf{S_D} \quad (10.4.3)$$

The relative energies which depend only on S are simply given by

$$E(S) = -\frac{J}{2}S(S+1)/2 \quad (10.4.4)$$

so that all states with the same S are degenerate.

A much more realistic situation in copper(II) chemistry is that where the tetranuclear structure results from the association of two planar bibridged copper(II) dimers through weak Cu-bridge apical interactions as



The interactions between A and B on the one hand and between C and D on the other hand (noted J) are certainly much larger than those between two ions belonging to different dimeric units (noted J'). The zero-field spin Hamiltonian is then

$$\mathbf{H} = -J(\mathbf{S_A}\cdot\mathbf{S_B} + \mathbf{S_C}\cdot\mathbf{S_D}) - J'(\mathbf{S_A}\cdot\mathbf{S_C} + \mathbf{S_A}\cdot\mathbf{S_D} + \mathbf{S_B}\cdot\mathbf{S_C} + \mathbf{S_B}\cdot\mathbf{S_D}) \quad (10.4.5)$$

or

$$\mathbf{H} = -[(J-J')/2](\mathbf{S}_{AB}^2 + \mathbf{S}_{CD}^2 - \mathbf{S}_A^2 - \mathbf{S}_B^2 - \mathbf{S}_C^2 - \mathbf{S}_D^2) - (J'/2)(\mathbf{S}^2 - \mathbf{S}_A^2 - \mathbf{S}_B^2 - \mathbf{S_C}^2 - \mathbf{S}_D^2) \quad (10.4.6)$$

with

$$\mathbf{S_{AB}} = \mathbf{S_A} + \mathbf{S_B}$$
$$\mathbf{S_{CD}} = \mathbf{S_C} + \mathbf{S_D}$$
$$\mathbf{S} = \mathbf{S_{AB}} + \mathbf{S_{CD}} \quad (10.4.7)$$

The energies $E(S,S_{AB},S_{CD})$ are expressed as

$$E(S, S_{AB}, S_{CD}) = -\frac{J-J'}{2}[S_{AB}(S_{AB}+1) + S_{CD}(S_{CD}+1)] - \frac{J'}{2}S(S+1) \quad (10.4.8)$$

with $E(0,0,0)$ as the energy origin. It follows that

$$E(0,0,0) = 0$$

$$E(1,0,1) = -J$$

$$E(1,1,0) = -J$$

$$E(0,1,1) = -2J+2J'$$

$$E(1,1,1) = -2J+J'$$

$$E(2,1,1) = -2J - J' \qquad (10.4.9)$$

For $J = J'$, we find the degeneracy of the levels of the same S. For $J' = 0$, we have two independent copper(II) dimers. In the general case the magnetic susceptibility is derived as

$$\chi = (2Ng^2\beta^2/kT)\{2\exp(J/kT) + \exp[(2J-J')/kT] + 5\exp[(2J+J')/kT]\}/\{1 + 6\exp(J/kT) + \exp[(2J-2J')/kT] + 3\exp[(2J-J')/kT] + 5\,exp[(2J+J')/kT]\} \qquad (10.4.10)$$

The magnetic properties of a number of cubane-like copper(II) clusters have been investigated [10.18–10.20]. One of them is $[Cu(bipy)(OH)]_4(PF_6)_4$. Its structure consists of two almost planar di-μ-hydroxo copper(II) dimers held together by long out-of-plane Cu—OH bonds. The skeleton of this cubane-like cluster is represented in Figure 10.18. Interestingly, both J and J' for this compound have been found to be positive so that the ground state is a spin quintet.

Quite an interesting class of cubane-type clusters is the $[Fe_4S_4]$ proteins and their synthetic models [10.21–10.23]. These clusters consist of ferrous and ferric ions

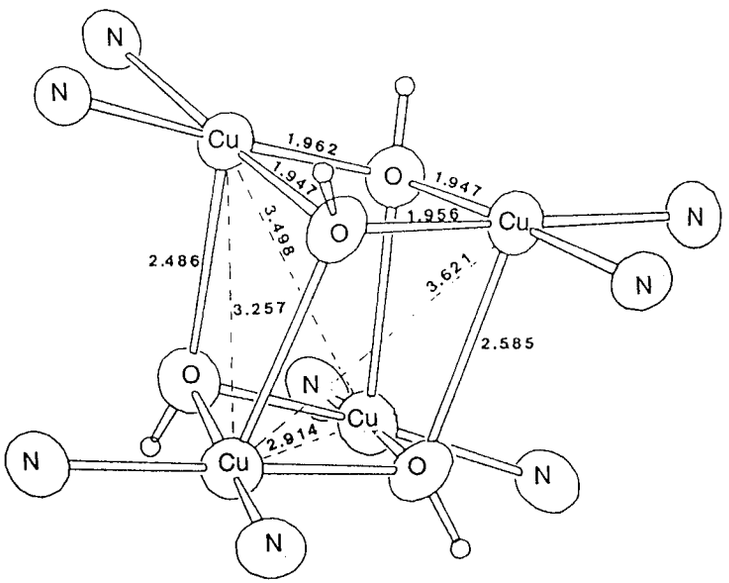


Figure 10.18. Molecular structure of the tetranuclear copper(II) compound $[Cu(bipy)(OH)]_4^{4+}$ (from ref. [10.20]).

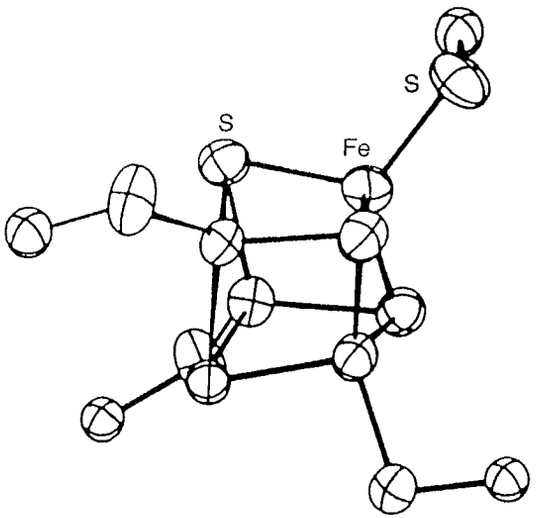


Figure 10.19. Molecular structure of the $[Fe_4S_4(SPh)_4]^{2-}$ cluster containing formally two Fe(III) and two Fe(II) ions. The structure of the $[Fe_4S_4(SPh)_4]^{3-}$ cluster is very similar (from ref. [10.24]).

coordinated in a tetrahedral fashion with sulfide bridges and cystein thiolate ligands binding the structure cores to the proteins. In the synthetic clusters terminal thiolates SR^- replace the cysteine thiolates. The protein clusters can occur in three different oxidation states, $[Fe_4S_4]^{1+,2+,3+}$, with formally three Fe(II) and one Fe(III), two Fe(II) and two Fe(III), and one Fe(II) and three Fe(III), respectively. The structure of one of these clusters, $[Fe_4S_4(SPh)_4]^{2-}$, is shown in Figure 10.19 [10.24]. Nature most often uses only the first two oxidation states. $[Fe^4S^4]^{3+}$ is generally confined to proteins referred to as "high potential iron proteins" (HPIP). For the synthetic clusters, it has not yet been possible to isolate the three compounds $[Fe_4S_4(SR)_4]^{1-,2-,3-}$, whatever the type of thiolate SR^- used. With most thiolates, tris- and dianions can be isolated whereas the monoanion, which can be generated electrochemically, is too unstable to be isolated. With very bulky thiolates like 2,4,6-iPr$_3$-SPh$^-$, the di- and monoanions can be isolated [10.25].

The interpretation of the magnetic properties of these clusters raises a problem that is far from being completely solved. The crystallographic data concerning both the proteins and the synthetic models suggest that the Fe(II) and Fe(III) ions cannot be distinguished. Even when two crystallographic sites are found, their proportion may not correspond to the proportions of Fe(II) and Fe(III). The obvious interpretation is that the valences are not fully localized. Each iron site has an oxidation state somewhere between 2+ and 3+. If this is so, the orbital or phenomenological models we have already used fail. Thus, the HDVV scheme of formal spin–spin coupling supposes that the active electrons are localized in their magnetic orbitals; they may exchange, but none of them is allowed to move from one site to another. At this stage,

it clearly appears that a new model is necessary, which takes explicitly into account the electron delocalization. Such a model does exist, and Chapter 13 is devoted to this subject, but it has not yet been fully extended to $[Fe_4S_4]$ clusters. The first attempts to use this new model involving the electron delocalization were performed by Noodelman [10.26]. We will discuss this work in Section 13.6. Otherwise, all authors used the HDVV scheme even though some of them pointed out that this scheme could be of limited validity due to the electron delocalization.

The most thoroughly investigated oxidation state is $[Fe_4S_4]^+$ with formally three Fe(II) noted A, B, and C, and one Fe(III) noted D. The zero-field Hamiltonian in the HDVV scheme is

$$\mathbf{H} = -J(\mathbf{S_A}\cdot\mathbf{S_B} + \mathbf{S_B}\cdot\mathbf{S_C} + \mathbf{S_A}\cdot\mathbf{S_C}) - J'\mathbf{S_D}\cdot(\mathbf{S_A} + \mathbf{S_B} + \mathbf{S_C}) \qquad (10.4.11)$$

where J is the Fe(II)–Fe(II) interaction parameter, and J' is the Fe(II)–Fe(III) one. **H** in (10.4.11) can be transformed as

$$\mathbf{H} = -\frac{J}{2}(\mathbf{S}^2_{ABC} - \mathbf{S}^2_A - \mathbf{S}^2_B - \mathbf{S}^2_C) - \frac{J'}{2}(\mathbf{S}^2 - \mathbf{S}^2_{ABC} - \mathbf{S}^2_D) \qquad (10.4.12)$$

with

$$\mathbf{S}_{ABC} = \mathbf{S}_A + \mathbf{S}_B + \mathbf{S}_C$$

$$\mathbf{S} = \mathbf{S}_{ABC} + \mathbf{S}_D \qquad (10.4.13)$$

If the eigenvalues depend only on the two quantum numbers S_{ABC} and S, the labeling of the states requires an additional quantum number, S_{AB} ($\mathbf{S}_{AB} = \mathbf{S}_A + \mathbf{S}_B$). The relative energies are then

$$E(S, S_{ABC}, S_{AB}) = -\frac{J-J'}{2}S_{ABC}(S_{ABC}+1) - \frac{J'}{2}S(S+1) \qquad (10.4.14)$$

All together there are 90 states; those having the same S and S_{ABC} but different S_{AB} values are degenerate. Using this model several fits of the experimental data were proposed, a typical fit for $(Et_4N)_3[Fe_4S_4(SPh)_4]$ being $J = -110$ cm^{-1} and $J' = -150$ cm^{-1}. The two types of interaction are antiferromagnetic and the ground state is $S = 1/2$. The nature of the ground state is more interesting than the calculated values of J and J' because it depends in a rather complicated fashion on the ratio J'/J. This ground state may be either $S = 1/2$ or $S = 3/2$ [10.27, 10.28]. It appears that only minor structural changes are sufficient to effect a change in the nature of the ground state. So, for $(Me_4N)_3[Fe_4S_4(SPh)_4]\cdot 2CH_3CN$, the ground state is apparently $S = 3/2$. This duality 3/2–1/2 occurs for both natural and model clusters. A 3/2–1/2 spin-admixed case has also been reported [10.28]. Furthermore, if the bridging sulfur atoms are replaced by Se or Te, an $S = 3/2$ ground state is observed [10.28, 10.29]. This ground state is split in zero field into two Kramers doublets separated by a few wavenumbers [10.30].

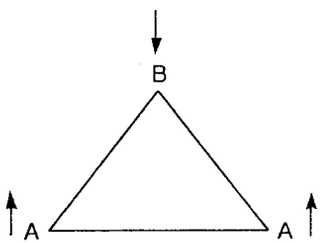
The nature of the ground state is more easily understood for the $[Fe_4S_4(SR)_4]^{2-}$ species containing formally two Fe(II) and two Fe(III). The magnetic susceptibility corrected for uncoupled impurities decreases as the temperature is decreased, and tends

to zero as T approaches absolute zero, which indicates that the ground state is a spin singlet [10.27]. A fitting of the experimental data has been proposed which takes into account three interaction parameters [Fe(II)–Fe(II), Fe(III)–Fe(III), and Fe(II)–Fe(III)]. The vectorial model cannot be used. The energies are determined by diagonalization of the spin Hamiltonian using the 900 microstates as a basis set. In fact, the matrix can be block-diagonalized thereby reducing the problem to that of separately diagonalizing a number of matrices of smaller size. Interaction parameters values for $(Et_4N)_2[Fe_4S_4(SPh)_4]$ have been proposed which are as large as $J_{Fe(II)Fe(II)} = -450$ cm^{-1}, $J_{Fe(III)Fe(III)} = -550$ cm^{-1}, and $J_{Fe(II)Fe(III)} = (J_{Fe(II)Fe(II)} + J_{Fe(III)Fe(III)})/2 = -500$ cm^{-1}. These values are not very convincing. Indeed, the interactions are so strongly antiferromagnetic that only a few excited states are significantly populated at room temperature or below, such that it appears difficult to evaluate several interaction parameters.

As far as the most oxidized species is concerned, we have already mentioned that the $[Fe_4(2,4,6,\text{-}i\text{Pr}_3\text{-SPh})_4]^-$ monoanion was isolated (with the NBu_4^+ counterion). EPR and Mössbauer spectra demonstrated that the ground state is a spin doublet that is well separated in energy from the first excited states [10.31]. In 1990 the magnetic susceptibility of (NBu_4) $[Fe_4(2,4,6,\text{-}i\text{Pr}_3\text{-SPh})_4]$ was measured and interpreted with a model that takes explicitly the electron delocalization into account for the first time [10.26]. The data and the model will be discussed in Chapter 13.

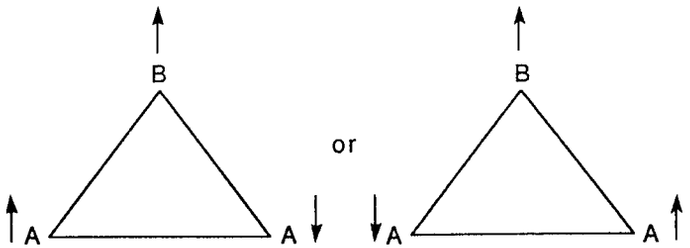
10.5 Competing Interactions and Spin Frustration

We turn back to the isosceles triangle ABA with J and J' interaction parameters presented in Section 10.1 to introduce the concept of competing interactions in the context of molecular magnetism. Let us first consider the case where the three local spins S_{A_1}, S_{A_2}, and S_B are equal to 1/2. If both J and J' are negative, the exact nature of the ground state depends on the ratio $\rho = J'/J$. For $\rho \leq 1$, this ground state is $E(1/2,1)$, and may be represented schematically as



$E(1/2,1)$

Although the A_1–A_2 interaction is antiferromagnetic, the local spins S_{A_1} and S_{A_2} are aligned parallel. This is due to the large B–A_1 and B–A_2 antiferromagnetic interactions that polarize S_{A_1} and S_{A_2} ferromagnetically. For $p \geq 1$, the ground state becomes E(1/2,0), and may be represented as



$E(1/2,0)$

Along one of the two B–A edges the local spins are aligned parallel, due to the large A_1–A_2 antiferromagnetic interaction. In such a triangular topology it is not possible to draw antiparallel arrows along each of the three edges of the triangle.

When the local spins at the corners of the isosceles triangle are larger than 1/2, the situation is more complicated. We examine here the case where $S_{A_1} = S_{A_2} = S_B = 1$. The energies of the low-lying states calculated from (10.1.6) are

$$E(3,2) = -3J - 3J'$$

$$E(2,2) = -3J'$$

$$E(1,2) = 2J - 3J'$$

$$E(2,1) = -2J - J'$$

$$E(1,1) = -J'$$

$$E(0,1) = J - J'$$

$$E(1,0) = -J \qquad (10.5.1)$$

The variation of these energies versus the ratio $\rho = J'/J$ for both J and J' negative is shown in Figure 10.20, where the energy of the ground state is taken as the energy origin. For $\rho \leq 1/2$, the ground state is $E(1,2)$, and for $\rho \geq 2$, the ground state is $E(1,0)$. $E(1,2)$ and $E(1,0)$ for a triangle of local triplets may be represented exactly as $E(1/2,1)$ and $E(1/2,0)$, respectively, for a triangle of local doublets. On the other hand, for $1/2 \leq \rho \leq 2$, the ground state is $E(0,1)$. This state cannot be represented by combining local spins assimilated to classical vectors. As a whole the situation may be described as follows: the antiferromagnetic interaction along B–A_1 and B–A_2 polarizes the spins around A_1 and A_2 in a ferromagnetic fashion. Any antiferromagnetic interaction along A_1–A_2 opposes this effect. When $|J'|$ is small enough ($\rho \leq 1/2$), the ferromagnetic polarization takes over. When $|J'|$ is large enough ($\rho \geq 2$), the A_1–A_2 antiferrromagnetic interaction takes over. When the ferromagnetic polarization and the A_1–A_2 antiferromagnetic interaction are of the same order of magnitude ($1/2 \leq \rho \leq 2$), the system looks for a compromise. The spin vectors around A_1 and A_2 are aligned neither parallel nor antiparallel; the spin for the A_1–A_2 edge takes the intermediate value $S' = 1$. Particularly interesting are the two situations $\rho = 1/2$ and $\rho = 2$. For both values the

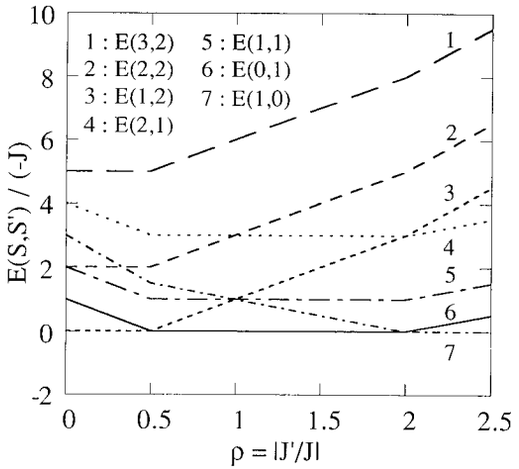
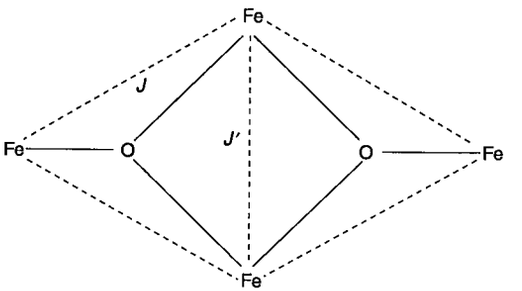


Figure 10.20. Variation of the spin state energies for an isosceles triangle ABA with local spins $S_{A_1} = S_{A_2} = S_B = 1$ versus the ratio J'/J. J is the interaction parameter along the A_1–B and A_2–B edges, and J' is the interaction parameter along the A_1–A_2 edge. Both J and J' are assumed to be negative.

ground state is accidentally degenerate and the spins are unable to decide which state to be. The system is said to be frustrated.

Competing interactions are evidently not limited to triangular topologies. They occur in all the cases where there is competition between ferromagnetic polarization and antiferromagnetic interaction. Several examples have been discussed by Hendrickson et al. [10.32–10.34]. One of these concerns the tetranuclear species $[Fe_4O_2(CH_3CO_2)_7(bpy)_2]^+$ shown in Figure 10.21. This species contains a core consisting of a tetranuclear bis(μ_3-O) high-spin iron(III) cluster disposed in a butterfly arrangement:



The magnetic susceptibility data reveal an $S = 0$ ground state. The fitting of these data leads to $J = -90$ cm^{-1}, J referring to the interaction between wing and body Fe(III) ions. On the other hand, the interaction parameter between the body Fe(III) ions J'

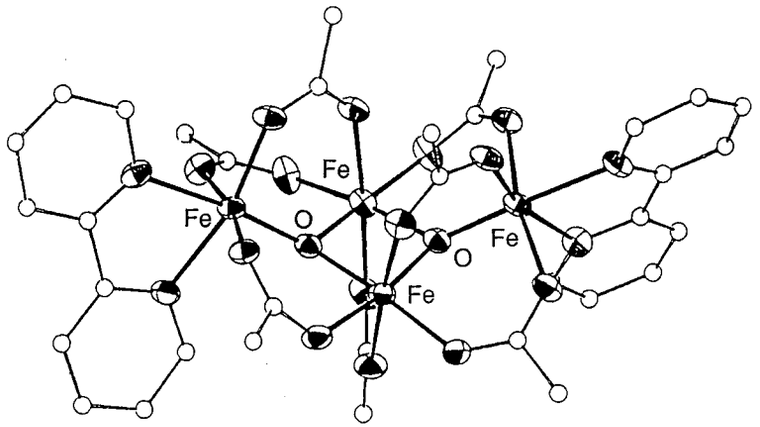
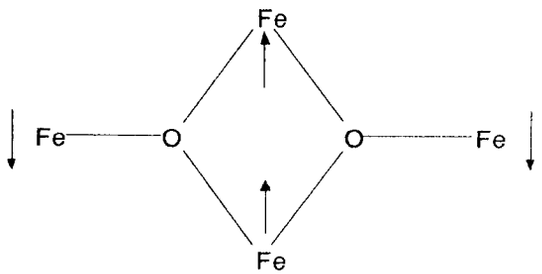


Figure 10.21. Molecular structure of the cation $[Fe_4O_2(CH_3CO_2)(bpy)_2]^+$ with bpy = 2,2′-bipyridine. For clarity, only the Fe and μ_3-O atoms are labeled (from ref. [10.31]).

remains indeterminate; it can only be said that J' is more positive than $-30\ cm^{-1}$. The large $|J|$ value gives rise to a spin structure within the Fe_4O_2 core as



in which the local spins of the body Fe(III) ions are polarized ferromagnetically, although the intrinsic interaction between these ions is antiferromagnetic. In a certain sense the uncertainty in J' is the mathematical response to this competition between antagonist factors.

To conclude this section it is probably worth underlining that the situation of competing interactions is not a new phenomenon in molecular chemistry. The nature and the relative energies of the low-lying states are normally deduced from the HDVV Hamiltonian. We should remember that the topology sometimes creates a ferromagnetic polarization between two antiferromagnetically coupled spin carriers. The competition between these two opposite forces may lead to ground states that cannot be described in the simple fashion of combining the local spins assimilated to classical vectors. In contrast, spin frustration in molecular species might lead to quite new phenomena. Since two spin states are accidentally degenerate, a small external perturbation might modify the nature of the ground state. This situation might have

similarity with the high spin–low spin transition; the system hesitates as for the nature of its ground state.

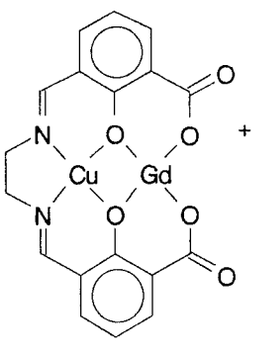
The concept of spin frustration, introduced by solid state physicists at the end of the 1970s [10.35, 10.36], has sometimes been understood in a too broad sense by molecular chemists. Spin frustration refers to a system in which competing interactions lead to a degenerate ground state. The mere presence of competing interactions does not mean that the system is frustrated. Only if the interactions have exactly the right relative size does the frustrated state arise.

10.6 High-Spin Molecules

We have emphasized that the isotropic interaction between two magnetic centers was much more frequently antiferromagnetic than ferromagnetic. The antiferromagnetic coupling, to some extent, represents a border-line case of a very weak bond. It corresponds to the trend for magnetic electrons to pair in molecular orbitals of low energy (see Section 9.2). After all, such electron pairing is the cornerstone of the chemical bond, and most molecules have closed-shell ground states. This is why the design of high-spin molecules for some years has been a challenge. In this section we will mention some of the latest achievements in this area.

The first idea that comes to mind to obtain a high-spin molecule is to use a high-spin carrier. The best candidate for this is the Gd(III) ion with an $4f^7$ configuration; its local spin is 7/2. Since quite a few Gd(III) mononuclear species are known, only the systems with a ground state spin larger than 7/2 are really interesting in terms of high-spin molecules. The obvious way to obtain such systems is to impose ferromagnetic interactions between nearest neighbor magnetic centers. The tetranuclear species in Figure 9.2, in which a Cr(III) ion is surrounded by three Ni(II) ions, is an example of this type of system. The three Cr(III)–Ni(II) interactions are ferromagnetic due to the orthogonality of the magnetic orbitals, and the ground state possesses the spin 9/2.

Another case of ferromagnetic interaction, which is not directly related to the symmetry of the magnetic orbitals, is offered by the Gd(III)–Cu(II) pair. So, the cation $[GdCu(fsa)_2en(H_2O)_4]^+$ with the bridging network



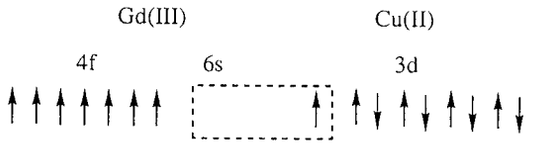


Figure 10.22. Spin delocalization mechanism proposed to account for the Gd(III)-Cu(II) ferromagnetic interaction.

shows an $S = 4$ round state stabilized by ca. 6 cm^{-1} with respect to the $S = 3$ excited state [10.37]. It has been suggested that the ferromagnetic interaction was due to a partial delocalization of the copper(II) unpaired electron toward the empty 6*s* orbital of Gd(III), which would force the seven f electrons of the rare earth to align parallel to this copper(II) electron on the basis of Hund's rule, as schematized in Figure 10.22 [10.368]. Several $Gd(III)Cu(II)_2$ trinuclear species have been investigated [10.38–

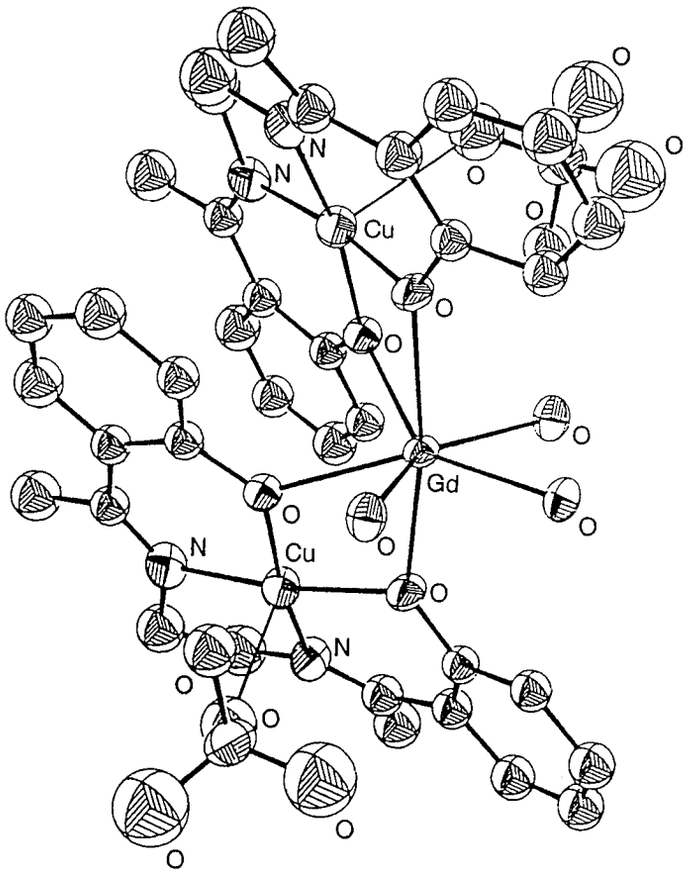


Figure 10.23. Molecular structure of the trinuclear cation $\{[Cu(Mesalen)]_2Gd(H_2O)_3\}^{3+}$ with an $S = 9/2$ ground state spin (from ref. [10.38]).

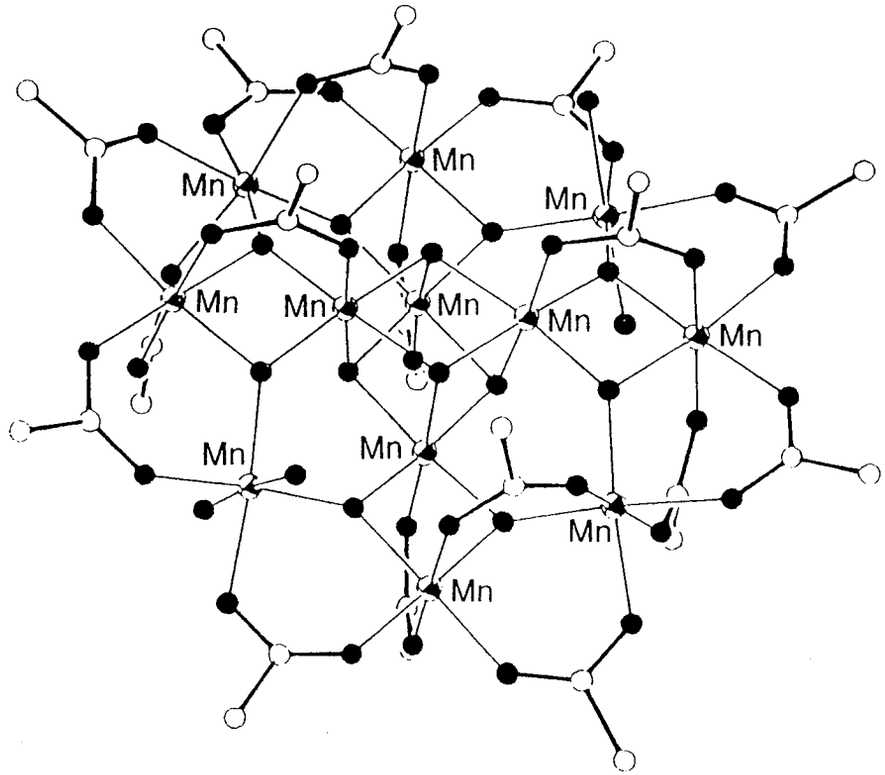


Figure 10.24. Molecular structure of the high-spin molecule $Mn_{12}O_{12}(PhCO_2)_{16}(H_2O)_4$. For clarity, the phenyl rings are omitted. Only the manganese atoms are labeled; the oxygen atoms are represented in black (from ref. [10.40]).

10.40]. The structure of one of these, $\{[Cu(Mesalen)]_2Gd(H_2O)_3\}^{3+}$, is shown in Figure 10.23. Gd(III) has a distorted pentagonal bipyramidal environment and Cu(II) a square pyramidal environment. All these $Gd(III)Cu(II)_2$ species show an $S = 9/2$ ground state with Gd(III)–Cu(II) interaction parameters of a few wavenumbers.

High-spin molecules may also have been obtained without imposing ferromagnetic interactions between nearest neighbors, by playing with spin topology. The Mn(II)Cu(II)Mn(II) species in Figure 10.7, with again an $S = 9/2$ ground state spin, illustrates the possibilities in this field. A beautiful example of a molecule with an $S = 12$ ground state spin in which there is alternation of high and small local spins will be discussed in Section 11.6 (see also Figure 11.20). This compound might well exhibit the largest ground state spin value for any molecule.

Another intriguing high-spin molecule is $Mn_{12}O_{12}(PhCO_2)_{16}(H_2O)_4$ whose structure is shown in Figure 10.24. This compound consists of a central $Mn(IV)_4O_4$ cubane-like core held within a nonplanar ring of eight Mn(III) ions by eight μ_3-oxide atoms. Magnetic susceptibility data within a field of 1 T, and magnetization measurements performed between 1.63 K and 20 K in various fields up to 4.8 T have been interpreted as resulting from an $S = 14$ ground state spin [10.41]. Very low-field magnetic susceptibility data obtained in other laboratories do not confirm this interpretation, but rather suggest that the ground state spin is between 8 and 10. As the applied magnetic field increases, crossovers occur between the ground state and very nearby excited states with a larger spin, which makes a precise determination of the ground state spin uncertain. Interestingly, the compound $[Mn_{12}O_{12}(CH_3CO_2)_{16}(H_2O)_4]\cdot 2CH_3CO_2H\cdot 4H_2O$ has a structure rather similar to that of $Mn_{12}O_{12}(PhCO_2)_{16}(H_2O)_4$. Zero-field susceptibility and high-field magnetization data have been convincingly interpreted as resulting from an $S = 10$ ground state, which is compatible with the eight Mn(III) spins up and the four Mn(IV) spins down [10.42].

Other important classes of high-spin molecules will be presented in Chapters 12 and 13.

References

10.1. D. Gatteschi, A. Bencini, in "Magneto-Structural correlations in Exchange Coupled Systems," R.D. Willett, D. Gatteschi, O. Kahn, eds., p. 241, NATO ASI Series. Reidel, Dordrecht, 1985.

10.2. R. Beckett, R. Colton, F.F. Hoskins, R.L. Martin, D.G. Vince, *Aust. J. Chem.* 22 (1969), 2527.

10.3. B.S. Tsukerblat, B. Ya. Kuyavskaya, M.I. Belinskii, A.V. Ablov, *Theoret. Chim. Acta(Berl.)* 38 (1975), 131.

10.4. R. Veit, J.J. Girerd, O. Kahn, F. Roberts, Y. Jeannin, *Inorg. Chem.* 25 (1986), 4175.

10.5. I. Morgenstern-Badarau, H.H. Wickman, *Inorg. Chem.* 24 (1985), 1889; I. Morgenstern-Badarau, D. Laroque, E. Bill, H. Winkler, A.X. Trauwein, F. Robert, Y. Jeannin, *Inorg. Chem.* 30 (1991), 3180.

10.6. C.A. Koch, C.A. Reed, G. Brewer, N.P. Rath, W.R. Scheidt, G.P. Gupta, G. Lang, *J. Am. Chem. Soc.* 111 (1989), 7645.

10.7. G.P. Gupta, G. Lang, C.A. Koch, B. Wang, W.R. Scheidt, C.A. Reed, *Inorg. Chem.* 29 (1990), 4234.

10.8. A.J. Thomson, M.K. Johnson, C. Greenwood, P.E. Gooding, *Biochem. J.* 193 (1980), 687.

10.9. T.A. Kent, E. Münck, W.R. Dunham, W.F. Filter, K.L. Findling, T. Yoshida, J.A. Fee, *J. Biol. Chem.* 257 (1982), 12489.

10.10. O. Kahn, *Struct. Bond.(Berlin)* 68 (1987), 89.

10.11. Y. Pei, Y. Journaux, O. Kahn, A. Dei, D. Gatteschi, *J. Chem. Soc., Chem. Commun.* (1986), 1300.

10.12. Y. Pei, Y. Journaux, O. Kahn, *Inorg. Chem.* 27 (1988), 399.

10.13. J. Ribas, C. Diaz, R. Costa, Y. Journaux, C. Mathonière, O. Kahn, *Inorg. Chem.* 29 (1990), 2042.

10.14. S. Menage, S.E. Vitols, P. Bergerat, E. Codjovi, O. Kahn, J.J. Girerd, M. Guillot, X. Solans, T. Calvet, *Inorg. Chem.* 30 (1991), 2666.

10.15. A. Bencini, C. Benelli, A. Dei, D. Gatteschi, *Inorg. Chem.* 24 (1985), 695.

10.16. F. Lloret, J. Journaux, M. Julve, *Inorg. Chem.* 29 (1990), 3967.

10.17. Y. Pei, Y. Journaux, O. Kahn, *Inorg. Chem.* 28 (1989), 100.

10.18. J.W. Hall, W.E. Estes, E.D. Estes, R.P. Scaringe, W.E. Hatfield, *Inorg. Chem.* 16 (1977), 1572.

10.19. L. Merz, W. Haase, *J. Chem. Soc., Dalton Trans.* (1978), 1594; (1980), 875.

10.20. J. Sletten, A. Sorensen, M. Julve, Y. Journaux, *Inorg. Chem.* 29 (1990), 5054.

10.21. T.G. Spiro ed., "Iron-Sulfur Proteins," Wiley-Interscience, New York, 1982.

10.22. R.H. Holm, S. Ciurli, J.A. Weigel, *Prog. Inorg. Chem.* 38 (1990), 1.

10.23. P.A. Lindahl, J.A. Kovacs, *Clusters* 1 (1990), 29.

10.24. L. Que, Jr., M.A. Bobrik, J.A. Ibers, R.H. Holm, *J. Am. Chem. Soc.* 96 (1974), 4168.

10.25. T. O'Sullivan, M.M. Millar, *J. Am. Chem. Soc.* 107 (1985), 4096.

10.26. J. Jordanov, E.K.H. Roth, P.H. Fries, L. Noodleman, *Inorg. Chem.* 29 (1990), 4288.

10.27. G.C. Papaefthymiou, E.J. Laskowsky, S. Frota-Pessoa, R.B. Frankel, R.H. Holm, *Inorg. Chem.* 21 (1982), 1723.

10.28. M.J. Carney, G.C. Papaefthymiou, M.A. Whitener, K. Spartalian, R.B.Frankel, R.H. Holm, *Inorg. Chem.* 27 (1988), 346.

10.29. P. Barbaro, A. Bencini, I. Bertini, F. Briganti, S. Midollini, *J. Am. Chem. Soc.* 112 (1990), 7238.

10.30 M.J. Carney, G.C. Papaefthymiou, R.B. Frankel, R.H. Holm, *Inorg. Chem.* 28 (1989), 1497.

10.31. V. Papaefthymiou, M.M. Millar, E. Münck, *Inorg. Chem.* 25 (1986), 3010.

10.32. J.K. McCusker, J.B. Vincent, E.A. Schmitt, M.L. Mino, K. Shin, D.K. Coggin, P.M. Hagen, J.C. Huffman, G. Christou, D.N. Hendrickson, *J. Am. Chem. Soc.* 113 (1991), 3012.

10.33. J.K. McCusker, C.A. Chrismas, P.M. Hagen, R.K. Chadha, D.F. Harvey, D.N. Hendrickson, *J. Am. Chem. Soc.* 113 (1991), 6114.

10.34. J.K. McCuster, E.A. Schmitt, D.N. Hendrickson, in "Magnetic Molecular Materials," D. Gatteschi, O. Kahn, J.S. Miller, F. Palacio, eds., p. 297, NATO ASI Series, Kluwer, Dordrecht, 1991.

10.35. G. Toulouse, *Commun. Phys.* 2 (1977), 115.

10.36. J. Villain, *J. Phys. C* 10 (1977), 1717.

10.37. N. Matsumoto, M. Sakamoto, H. Tamaki, H. Okawa, S. Kida, *Chem. Lett.* (1989), 853.

10.38. C. Benelli, A. Caneschi, D. Gatteschi, Olivier Guillou, L. Pardi, *Inorg. Chem.* 29 (1990), 1750.

10.39. A. Bencini, C. Benelli, A. Caneschi, R.L. Carlin, A. Dei, D. Gatteschi, *J. Am. Chem. Soc.* 107 (1985), 8128.

10.40. A. Bencini, C. Benelli, A. Caneschi, A. Dei, D. Gatteschi, *Inorg. Chem.* 25 (1986), 572.

10.41. P.D.W. Boyd, Q. Li, J.B. Vincent, K. Folting, H.R. Chang, W.E. Streib, J.C. Huffman, G. Christou, D.N. Hendrickson, *J. Am. Chem. Soc.* 110 (1988), 8537.

10.42 A. Caneschi, D. Gatteschi, R. Sessoli, *J. Am. Chem. Soc.* 113 (1991), 5873.

CHAPTER

11

Magnetic Chain Compounds

One-dimensional magnetic compounds occupy an intermediate situation between magnetic clusters of high nuclearity and three-dimensional extended lattices. Furthermore, one-dimensional physics, as a whole, has been a very active area of research for two or three decades. These two reasons, among others, probably account for the flood of studies devoted to magnetic chain compounds. Several review papers or book chapters on this subject have been published [11.1–11.7].

In this chapter, we do not intend to review, or even to survey the field. Rather we will focus on some specific aspects that, we feel, are directly related to molecular chemistry. We will treat successively equally spaced chains, particularly the copper(II) chains, then alternating chains, and finally bimetallic chains. In each case we will select a few theoretical considerations of interest to the chemists, and present some examples. Quite a few studies performed by solid state physicists deal with compounds in which the interactions between spin carriers are extremely weak. Such a situation has some advantages; in particular the heat capacity features of magnetic origin are not masked by the lattice heat capacity, which roughly varies as T^3. However, chemists are certainly more interested by species where the magnetic interactions are rather large, and we will select our examples from the compounds of that kind.

11.1 Chains of Equally Spaced Copper(II) Ions

The simplest case of a magnetic chain is provided by an array of equally spaced copper(II) ions with $S_{Cu} = 1/2$ local spins, as schematized by

$$\text{——}\ A_i \overset{J}{\text{———}} A_{i+1} \overset{J}{\text{———}} A_{i+2}\ \text{——}$$

The spin Hamiltonian in zero-field adapted to describe the isotropic interaction between nearest neighbor ions is

$$\mathbf{H} = -J \sum_{i=1}^{n-1} \mathbf{S}_{A_i} \cdot \mathbf{S}_{A_{i+1}} \qquad (11.1.1)$$

where the summation runs over the n sites of the chain. When n tends to infinite, there is no analytical method that can be used to determine the energies of the low-lying states and the magnetic susceptibility. Nonetheless, the problem may be solved numerically by considering ring chains of increasing size and extrapolating for the case where n becomes infinite. The ring structure allows the introduction of the limit conditions:

$$\mathbf{S}_{A_{i+n}} = \mathbf{S}_{A_i} \qquad \text{for any } n \tag{11.1.2}$$

This method was first applied by Bonner and Fisher in 1964, who calculated explicitly the magnetic susceptibility of ring chains up to $n = 11$, and proposed an extrapolation for the infinite ring [11.8]. More precisely, they followed an approach proposed by Orbach [11.9], and worked in the various subspaces M_S varying by an integer value from $n/2$ to 0 or 1/2, according to whether n is even or odd. The diagonalization of **H** in (11.1.1) in the subspace $M_S = n/2$ gives the energy of the unique state of highest spin $S = n/2$. The diagonalization of **H** in the subspace $M_S = (n/2)–1$ also gives the energy of this S = n/2 state as well as the energies of the $S = (n/2)–1$ states, etc. The diagonalization of **H** in the $M_S = 0$ (or 1/2) subspace gives obviously all energies. As far as the extrapolation is concerned, a difficulty appears in the case where J is negative, related to the parity of n. If the interaction between nearest neighbors is antiferromagnetic, then the ground state for n even is diamagnetic and the magnetic susceptibility χ (per copper site) tends to zero as T approaches zero. On the other hand, the ground state for n odd is a doublet $S = 1/2$, and χ diverges as T approaches zero. The problem is then of determining the low-temperature limit $(\chi)_{LT}$ of χ when n becomes infinite. Bonner and Fisher demonstrated that this limit is actually finite and given by

$$(\chi)_{LT} = 0.14692 N g^2 \beta^2 / |J| \tag{11.1.3}$$

This result may be understood as follows: when n is infinite and J is negative, the energy levels form a continuum from one of the $S = 0$ levels up to the unique $S = n/2$ level. At absolute zero only the bottom of this continuum is thermally populated but, since there is no gap between the $S = 0$ level and the levels immediately above it, χ does not tend to zero.

Bonner and Fisher's results for $J < 0$ are represented in Figure 11.1 in the form of a $|J|\chi/Ng^2\beta^2$ versus $kT/|J|$ plot. χ passes through a rounded maximum at a temperature T_{max} defined by

$$kT_{max}/|J| = 0.641 \tag{11.1.4}$$

These results have been fitted by the following numerical expression [11.10]:

$$\chi = \frac{Ng^2\beta^2}{kT} \frac{0.25 + 0.074975x + 0.075235x^2}{1.0 + 0.9931x + 0.172135x^2 + 0.757825x^3} \tag{11.1.5}$$

with

$$x = |J|/kT \tag{11.1.6}$$

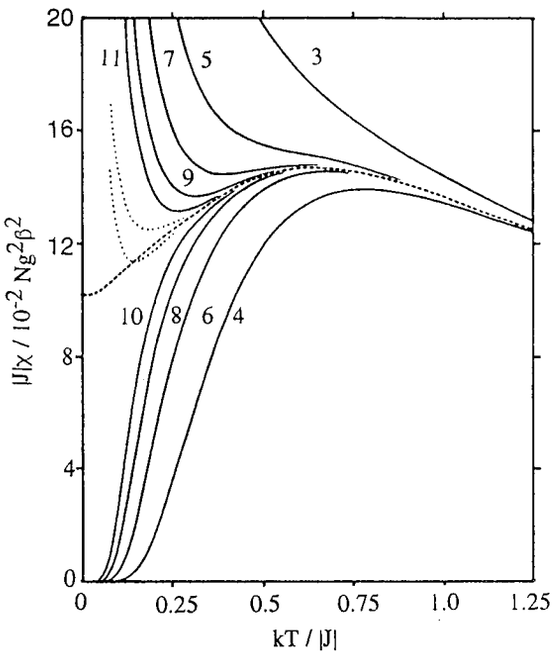


Figure 11.1. $|J|\chi/Ng^2\beta^2$ versus $kT/|J|$ plot for ring chains of n antiferromagnetically coupled copper(II) ions. The dotted curves are the averages of $n = 9$ and 10, and $n = 10$ and 11, respectively. The dashed line is the estimated limit for $n \to \infty$ (from ref. [11.8]).

Equation (11.1.5) is clearly much more convenient to work with than a listing of the numerical values when we need to interpret experimental data quantitatively. For $J > 0$, both χ and χT increase continuously as the temperature decreases, and diverge as T approaches zero. To our knowledge, no analytical expression has yet been proposed for the ferromagnetic situation. However, a high-temperature series expansion, valid for both positive and negative J values, has been proposed [11.11]. This expansion is

$$\chi = \frac{Ng^2\beta^2}{4kT}\left[\frac{N}{D}\right]^{2/3} \tag{11.1.7}$$

with

$$N = 1.0 + 5.7979916y + 16.902653y^2 + 29.376885y^3 + 29.832959y^4 + 14.036918y^5$$

$$D = 1.0 + 2.7979916y + 7.0086780y^2 + 8.6538644y^3 + 4.5743114y^4 \tag{11.1.8}$$

and

$$y = J/2kT \tag{11.1.9}$$

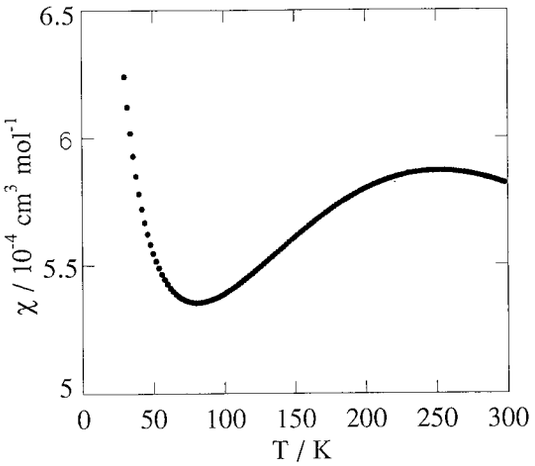
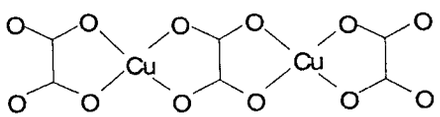


Figure 11.2. χ (per copper site) versus T plot for $Cu(C_2O_4)\cdot 1/3H_2O$. The magnetic data have been fitted using a model that takes into account a proportion $\rho = 0.0165$ of noncoupled copper(II) ions in addition to the proportion $1-\rho = 0.9835$ of infinite chain compound. J was found as -291 cm^{-1}, and g as 2.08.

What is quite important to realize is that whatever the nature ferro- or antiferromagnetic of the interaction, a one-dimensional magnetic compound does not order magnetically unless its temperature is at absolute zero. This result was first stated by Ising in 1925 [11.12]. In a real solid, however, the chains are never perfectly isolated from each other; interchain interactions superimpose onto dominant intrachain interactions, which leads to a three-dimensional magnetic ordering at a finite temperature. This important question will be discussed in more detail in the following chapter.

Let us give two examples of molecular chain compounds with copper(II) ions, one with $J < 0$ and the other with $J > 0$. The former example concerns a compound that has been known for quite a long time, but whose structure has only recently been solved. This compound is copper(II) oxalate $Cu(C_2O_4)\cdot 1/3H_2O$. Its structure was determined by extended X-ray absorption fine structure (EXAFS) spectroscopy, and is that of a planar ribbon, as schematized [11.13]:



The copper(II) ions are bridged by the oxalato ligand in the same way as in the oxalato dinuclear complexes presented in Section 8.6. The Cu—Cu intrachain separation is estimated at 5.14 Å. The magnetic behavior shown in Figure 11.2 is typical of a copper(II) chain with a strong antiferromagnetic interaction. The temperature dependence of the magnetic susceptibility exhibits a maximum around 270 K. The increase in magnetic susceptibility as T is lowered below ca. 70 K is assigned to a noncoupled

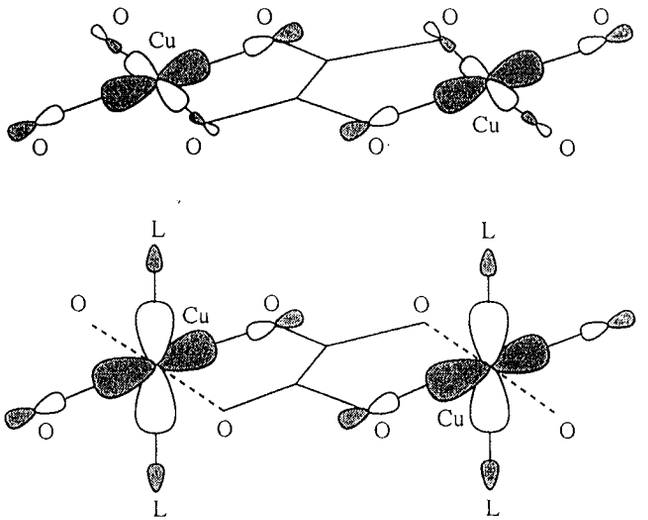


Figure 11.3. Schematic representation of the interaction between two magnetic orbitals centered on nearest neighbor copper(II) ions in $Cu(C_2O_4)\cdot 1/3H_2O$ (top) and $Cu(C_2O_4)L_2$ (bottom); L stands for a nitrogen containing ligand (from ref. [11.13]).

copper(II) impurity and ends of chains, such that χ does not tend to the limit expected for perfect one-dimensional copper(II) species [11.12]. J can be deduced from these magnetic data and is found to be equal to -291 cm^{-1}. A strong interaction exists in spite of the rather large Cu—Cu distance, and is attributed to the fact that the magnetic orbitals centered on nearest neighbor copper(II) ions are located in the plane of the oxalato bridge as shown in Figure 11.3. Thus, they are favorably oriented to overlap on both sides of the oxalato group as we pointed out in Section 8.6.

Copper(II) oxalate can react with most nitrogen-containing ligands L, to afford new chain compounds of formula $Cu(C_2O_4)L_2$ in which the ligands L are bonded on both sides of the oxalato ribbon plane. The environment of copper then adopts an elongated rhombic geometry with two short Cu—N, two short Cu—O, and two long Cu—O distances. Each magnetic orbital that points toward the four nearest neighbors of the copper atom is now located in a plane perpendicular to the plane of the oxalato bridge as shown in Figure 11.3. Any two magnetic orbitals of this kind, which are centered on adjacent copper atoms, will now be unfavorably oriented to overlap, and the interaction along the chain is expected to decrease dramatically. Such a behavior is encountered, for instance, in $Cu(C_2O_4)(NH_3)_2\cdot H_2O$ where two ammonia molecules are bonded to the copper atom with a trans configuration. The Cu—N bond length and short and long Cu—O bond lengths are 1.96, 2.16, and 2.33 Å, respectively [11.14]. In this compound the maximum in χ is observed around 15 K, and J is found to be equal to -15.4 cm^{-1}. The drastic reduction in the magnitude of the interaction due to such a subtle chemical change suggests that the mechanism of this interaction is rather well understood, and underlines the possibilities of molecular engineering in the field

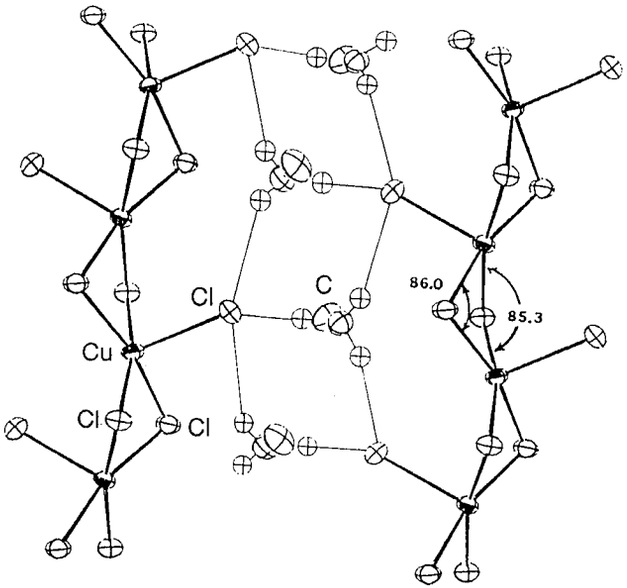


Figure 11.4. Structure of the chain compound cyclohexylammonium copper(II) trichloride $(C_6H_{11})CuCl_3$. The chains along the c axis are hydrogen bonded together through the hexylammonium cations (from ref. [11.15]).

of the magnetism. Other aspects of the molecular engineering of magnetic compounds will be exemplified later on.

The second example of a copper(II) chain is provided by cyclohexylammonium copper(II) trichloride, $(C_6H_{11}NH_3)CuCl_3$, often abbreviated as CHAC [11.15–11.17]. The structure of this compound is represented in Figure 11.4. It consists of symmetrically bibridged linear chains that lie parallel to the c axis of an orthorhombic lattice. Copper(II) is in square pyramidal surroundings with a nonbridging chlorine atom at the apex. The cyclohexylammonium cations hydrogen bond the chains together into sheets in the bc plane through the apical chlorine atoms. They also efficiently separate these sheets from each other along the a direction. The CuClCu bridging angles are equal to 85.3° and 86°, which gives rise to accidental orthogonality of the magnetic orbitals centered on adjacent copper(II) ions (see Section 8.6). It follows that a ferromagnetic interaction along the chain may be anticipated, as well as a weak interchain coupling in the bc sheets. The interaction along the a direction may be expected to be very small, and rather dipolar in nature. The χT versus T plot for a powder sample of CHAC increases rapidly as T decreases. At 2.2 K, χT is approximately 30 times as large as at room temperature. An intrachain interaction parameter of 100 cm^{-1} has been extracted from these data. At 2.18 K, χT shows a sharp maximum corresponding to the onset of long range magnetic order.

Magnetization studies have revealed a metamagnetic behavior, i.e., a change in the nature of the ground state when applying a magnetic field stronger than a critical value H_c close to 100 G. For $H < H_c$, below 2.18 K the ground state is antiferromagnetic,

with a small canting of the spins, leading to a weak residual moment along the a direction. From the magnitude of this residual moment, the angle between the spins is estimated as 17°. For $H > H_c$, the ground state is ferromagnetic-like, with a parallel alignment of all the spins. The term ferromagnetic-like means that the compound is not a genuine ferromagnet since it does not exhibit a spontaneous magnetization in zero-field The small value of H_c is due to the fact that the antiferromagnetic interchain coupling is very small. This coupling is actually found along a with the ratio $|J_a/J_c|$ between the interaction parameters along the a and c (chain) axes inferior to 10^{-4}. As for the interaction along c, it is ferromagnetic with J_b/J_c of the order of 10^{-3}.

In summary, the compound CHAC may be described as follows. Along c, we have ferromagnetic chains ($J_c = 100$ cm^{-1}). These chains interact ferromagnetically along b ($J_b \approx 10^{-1}$ cm^{-1}) to give ferromagnetic planes bc. Finally, these planes are very weakly antiferromagnetically coupled along a ($J_a < -10^{-2}$), and the three-dimensional ground state in zero-field is antiferromagnetic (ignoring the small canting). An applied field of 100 G is sufficient to overcome J_a.

11.2 Chains of Equally Spaced Magnetic Centers with Local Spins Larger Than 1/2

The method that was introduced by Bonner and Fisher may also be used to determine the energy level spectrum and the magnetic susceptibility of a chain compound with local spins larger than 1/2, provided that the Hamiltonian (11.1.1) remains valid [11.18–11.20]. However, as the value of the local spin becomes larger, the number n of sites that it is possible to take into account in the ring chain calculation becomes smaller. For instance, with $S_A = 5/2$, the calculation so far has been limited to five sites such that the accuracy of the extrapolation for infinite n is questionable. An analytical expression fitting the numerical results of ring chain calculations [11.18] has been proposed for $S_A = 1$, but not for larger spins. This expression which is valid for $J < 0$, is [11.21]

$$\chi = \frac{Ng^2\beta^2}{kT} \frac{2.0 + 0.0194x + 0.777x^2}{3.0 + 4.346x + 3.232x^2 + 5.834x^3} \tag{11.2.1}$$

with

$$x = |J|/kT \tag{11.2.2}$$

The local anisotropy was neglected in (11.1.1), which is of course rigorously correct for $S_A = 1/2$ ions, but may no longer be valid for other magnetic ions, particularly when the zero-field splitting is of the same order of magnitude as the isotropic interaction, or, for $|D| << |J|$, when analyzing low-temperature magnetic data. The zero-field Hamiltonian that needs to be considered for a ring chain is then

$$\mathbf{H} = -J\sum_{i=1}^{n-1} \mathbf{S}_{A_i} \cdot \mathbf{S}_{A_{i+1}} + D\sum_{i=1}^{n} [\mathbf{S}^2_{A_{i(z)}} - S_A(S_A+1)/3] \tag{11.2.3}$$

where the local anisotropy is assumed to be axial. Some numerical calculations of parallel and perpendicular susceptibilities using the ring chain technique and Hamiltonian (11.2.3) have been reported, notably by Weng [11.18], Blöte [11.19], and de Neef [11.20]. These calculations are rather lengthy, even for $S_A = 1$. They require tabulation of the principal susceptibilities χ_z and χ_x versus kT/J for a large range of D/J values, and the results are difficult to use when real cases have to be analyzed quantitatively.

When the value of the local spin S_A becomes large, this spin may be treated as a classical vector. An analytical expression for the magnetic susceptibility of an infinite chain of classical spins has been derived by Fisher [11.22]. This expression, in which the classical spin has been scaled to a real quantum spin S, is

$$\chi = \frac{Ng^2\beta^2 S(S+1)}{3kT} \frac{1+u}{1-u} \tag{11.2.4}$$

with

$$u = \coth\left[\frac{JS(S+1)}{kT}\right] - \left[\frac{kT}{JS(S+1)}\right] \tag{11.2.5}$$

Of course, the larger S_A is, the better this classical approximation. In practice Eq. (11.2.4) seems fairly good for $S_A = 5/2$ and above; it has often been used for Mn(II) chains (see below). Smith and Friedberg have extended this classical spin approach by introducing an axial anisotropy $D\Sigma_i \mathbf{S}^2_{A_{i(z)}}$ in the interaction Hamiltonian, the axis of the local distortion being taken along the chain axis. Thus, they derived the following expressions [11.23]:

$$\Delta\chi = \chi_z - \chi_x$$
$$= \frac{2Ng^2\beta^2}{15kT} S(S+1) \frac{DS(S+1)}{kT} F \tag{11.2.6}$$

with

$$F = \frac{(1+u)(1+v)}{(1-u)(1-v)} + \frac{2u}{1-u} \tag{11.2.7}$$

u being defined as in (11.2.5), and v by

$$v = 1 - \frac{3ukT}{JS(S+1)} \tag{11.2.8}$$

The only distinction between chains of equally spaced local spins we have made so far is between $S_A = 1/2$ and $S_A > 1/2$. In the former case there is no local anisotropy, and the possible (and very weak) magnetic anisotropy is due exclusively to the

anisotropy of the **g**-tensor. In the latter case local anisotropy may be operative and induce a much more pronounced magnetic anisotropy. In 1983 Haldane suggested that another distinction should be made between antiferromagnetically coupled chains, namely between those with half-integer local spins and those with integer local spins [11.24]. According to Haldane, the energy level spectrum for a chain with half-integer spins is a continuum with a singlet ground state infinitely close to the first excited states (in the limit $n \to \infty$). This explains why the low-temperature limit of χ is finite in the absence of local anisotropy, and not zero. In contrast, the spectrum for the chains with integer spins would present a gap between a singlet ground state and the first excited states such that, even in the absence of local anisotropy, the low-temperature limit of χ would be zero. We will see below that recent experimental data on nickel(II) chains seem to confirm this apparently very surprising conjecture of Haldane.

Let us present two examples of chain compounds with local spins larger than 1/2. The first concerns the compound $[Ni(en)_2(NO_2)](ClO_4)$ with en = ethylenediamine. This compound, abbreviated as NENP by the physicists, has recently been investigated in a very thorough manner, with respect to the existence of the Haldane gap. The structure consists of infinite chains $Ni(en)_2(NO_2)$ running along the b axis of an orthorhombic lattice. The nickel(II) ions are bridged by NO_2 groups in an antisymmetric manner; NO_2 is bound to one nickel(II) ion by its nitrogen atom, and to the other nickel(II) ion by one of its oxygen atoms. The Ni—Ni separation along the chain is 5.150 Å. The chains are separated from each other by noncoordinated perchlorate anions [11.21]. The structure is shown in Figure 11.5. The three principal susceptibilities exhibit a rounded maximum around 60 K, and tend to values that are very close to zero as the temperature approaches absolute zero, which is consistent with the existence of a gap between the diamagnetic ground state and the first excited states. The intrachain interaction parameter is found to be $J = -33.5$ cm^{-1}, and the local anisotropy to be rather weak with regard to J ($|D/J| = 0.2$). The fitting of the susceptibility data below 5 K allows an estimate of the gap, namely 9 cm^{-1}. This value has been confirmed by magnetization measurements. At a critical field H_c, there is a crossover between the most negative Zeeman component of the first excited state and the singlet ground state. This critical field, which breaks the gap, is estimated to be 9 T. The corresponding gap is $E = g\beta H_c$, i.e., g cm^{-1} for $g = 2.2$. Inelastic neutron scattering (INS) measurements have confirmed the existence of such a gap. INS actually shows two gaps attributed to the splitting of the Haldane gap due to the local anisotropy. The dispersion law of the magnetic excitations in the INS spectra also gave an estimate of the ratio between inter- and intrachain interaction parameters which is about 10^{-4} [11.24–11.27]. $[Ni(en)(NO_2)](CPO_4)$ is an excellent one-dimensional system. A last point concerning $[Ni(en)(NO_2)](CPO_4)$ deserves a comment, namely the magnitude of the interaction in spite of the rather large Ni—Ni separation. This large magnitude is attributed to the fact that the z^2-type magnetic orbitals are strongly delocalized toward the NO_2 bridges as shown in Figure 11.6, z referring here to the chain axis. It results that two magnetic orbitals centered on nearest neighbor nickel(II) ions may efficiently overlap [11.21].

The second example deals with one of the most extensively investigated magnetic chain compounds, namely $[N(CH_3)_4]MnCl_3$, often abbreviated as TMMC [11.28–

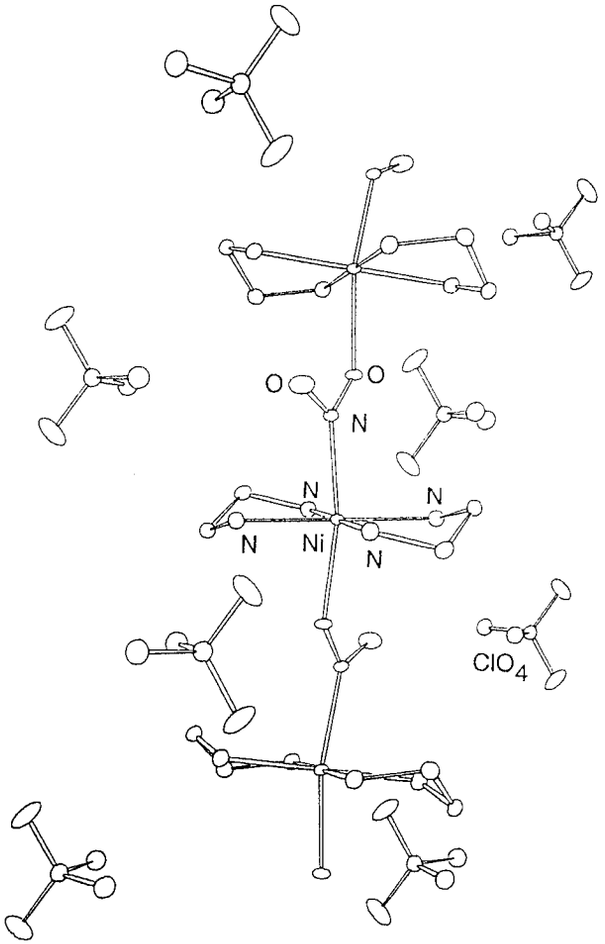


Figure 11.5. Structure of the chain compound $[Ni(en)_2(NO_2)](ClO_4)$ with en = ethylenediamine. This compound is generally abbreviated as NENP by the physicists (from ref. [11.21]).

11.31]. The structure consists of $MnCl_3$ chains resulting from the sharing of trigonal faces between $MnCl_6$ octahedra, and tetramethylammonium cations. The intrachain Mn—Mn separation is 3.10 Å, while the shortest interchain Mn—Mn separation is 9.15 Å. TMMC may be considered as an almost ideal one-dimensional antiferromagnet in the sense that the ratio J_{intra}/J_{inter} between intra- and interchain interaction parameters is of the order of 10^{-4}. The χ versus T plot shows the expected maximum around 55 K. These data were fitted with the classical-spin expression (11.2.4), and J_{intra} was found to be equal to $-9.3\ cm^{-1}$. Single crystal magnetic measurements have revealed that on cooling down below ca. 50 K, the susceptibility along the chain axis

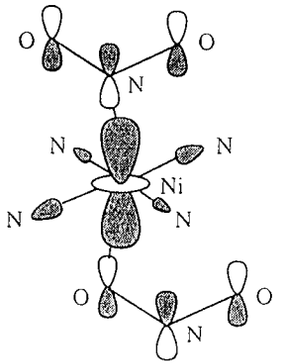


Figure 11.6. Schematic representation of the z^2-type magnetic orbital for $[Ni(en)_2(NO_2)](ClO_4)$.

becomes smaller than the susceptibility perpendicular to the chain axis. This anisotropy has been attributed to dipolar effects that favor alignment of the local spins perpendicular to the chain axis. Three-dimensional antiferromagnetic ordering occurs at $T_N = 0.85$ K. T_N was determined by several techniques including proton magnetic resonance, ^{35}Cl nuclear quadrupole resonance, neutron diffraction, and heat capacity measurements. Several papers focused on solitons in TMMC [11.32–11.33]. This important aspect of one-dimensional physics is, however, beyond the scope of this book.

11.3 Ising and XY Models

Up to this point in this chapter, we have implicitly assumed that the local ground state of the magnetic centers was an orbital singlet without first-order angular momentum. This condition is satisfied for copper(II) ions in tetragonal or trigonal environments, as well as for nickel(II) and manganese(II) ions in distorted octahedral environments. In such cases the HDVV Hamiltonian properly describes the isotropic interaction. In both chain compounds and discrete units this isotropic interaction is, of course, purely orbital in nature, and only an orbital approach may lead to the energies of the states. However, it has been shown that these energies follow the same interval rule as that deduced from the HDVV Hamiltonian, at least to a first-order approximation [11.34–11.35]. However, when the local ground state of the interacting centers has first-order angular momentum, the HDVV model is no longer a good phenomenological model. A more general expression of the spin Hamiltonian that describes the interaction in a chain compound may be written as

$$\mathbf{H} = -J \sum_{i=1}^{n-1} [\alpha \mathbf{S}_{A_{i(x)}} \cdot \mathbf{S}_{A_{i+1(x)}} + \beta \mathbf{S}_{A_{i(y)}} \cdot \mathbf{S}_{A_{i+1(y)}} + \gamma \mathbf{S}_{A_{i(z)}} \cdot \mathbf{S}_{A_{i+1(z)}}] \qquad (11.3.1)$$

Three limiting situations have been investigated in more depth [11.5], namely

$\alpha = \beta = \gamma = 1$ corresponding to the HDVV model
$\alpha = \beta = 0$ and $\gamma = 1$ corresponding to the Ising model
$\alpha = \beta = 1$ and $\gamma = 0$ corresponding to the XY model

Let us first consider the Ising model. For $S_A = 1/2$ local Ising spins, analytical expressions of the parallel and perpendicular magnetic susceptibilities were derived by Fisher as [11.36]

$$\chi_z = \frac{Ng_z^2\beta^2}{4kT}\exp(J/2kT) \tag{11.3.2}$$

$$\chi_x = \frac{Ng_x^2\beta^2}{2J}[\tanh(J/2kT) + (J/4kT)\mathrm{sech}^2(J/4kT)] \tag{11.3.3}$$

The χ_z versus T plot exhibits a maximum for $J < 0$, and the χ_x versus T plot exhibits a maximum for both $J < 0$ and $J > 0$. χ_x is an even function of J, and does not depend on its sign.

The best documented examples of chain compounds with an Ising behavior are certain cobalt(II) derivatives. In octahedral environment cobalt(II) has a $^4T_1$ ground state. If the symmetry is lowered, then the threefold orbital degeneracy is partially or totally removed, and the ground state may become a $^4\Gamma$ orbital singlet. This state is split in zero field into two Kramers doublets (see Section 3.4). The resonance within the Kramers doublet of lower energy usually gives $g_z > g_x$. The Ising model with an effective local spin $S_A = 1/2$ then applies in the temperature range where only this ground Kramers doublet is thermally populated. $[(CH_3)_3NH]CoCl_3 \cdot 2H_2O$ [11.37] and $Co(py)_2Cl_2$ with py = pyridine [11.38] are two examples of cobalt(II) chains of this kind.

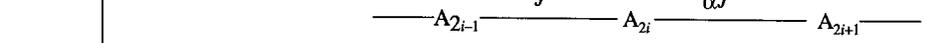
In principle the XY model applies for cobalt(II) chains with 1/2 effective local spins when the local ground state is such that $g_x >> g_z$. To our knowledge, there is only one convincing report of compound with an XY behavior. This compound is $(N_2H_5)_2Co(SO_4)_2$ where adjacent cobalt atoms are bridged by two sulfato groups [11.39]. Resonance within the ground Kramers doublet yields $g_x = 4.9$ and $g_z = 2.2$. The interaction parameter, deduced from the temperature dependence of the magnetic specific heat, was found to be $J = -9.8\ cm^{-1}$.

To conclude this section, we will stress that quantitative analysis of magnetic properties for systems involving ions with a significant orbital contribution remains a very difficult task. As a matter of fact, in most cases, neither a pure Ising nor a pure XY model is appropriate. Moreover, even in the ideal and rather exceptional case of a pure Ising (or XY) chain, this situation is restricted to the temperature range where only the ground Kramers doublet is thermally populated. The energy gap between the two Kramers doublets arising from the $^4\Gamma$ ground state is, at the most, of the order of a few tens of cm^{-1}. It follows that the Ising (or XY) model is no longer valid above a few tens of K. This is not dramatic if $|J|$ is very small. Most of the information may

then be deduced from low-temperature magnetic data. On the other hand, when J is the same order of magnitude as the energy gap between the two low-lying Kramers doublets, the low-temperature magnetic data are much less informative. In other respects, it is probably worthwhile to point out here that the Ising model has often been used when it was clearly not valid, the only justification for its use being that the calculation of the interaction parameter would otherwise be intractable. It is also useful to remind the reader that the site symmetries of the molecular species are generally rather low such that the angular momenta are largely quenched. In such cases the isotropic HDVV Hamiltonian is appropriate.

11.4 Alternating Chains and Spin-Peierls Transition

To now, we have only discussed the case of chains with equally spaced magnetic centers, called uniform chains. A somewhat more complicated situation is offered by alternating chain compounds with two interaction parameters J and αJ:



The spin Hamiltonian in zero field appropriate for such systems is

$$\mathbf{H} = -J\sum_{i=1}^{n/2}[\mathbf{S}_{A_{2i}}\cdot\mathbf{S}_{A_{2i-1}} + \alpha\mathbf{S}_{A_{2i}}\cdot\mathbf{S}_{A_{2i+1}}] \tag{11.4.1}$$

α is the alternation parameter ≤ 1; $\alpha = 1$ corresponds to a uniform chain whereas $\alpha = 0$ corresponds to isolated pairs. Most studies have been limited to positive values of α; the two interaction parameters then have the same sign. In principle, α may also be negative such that the interactions are alternatively ferro- and antiferromagnetic. We will limit this section to the 1/2 local spins which is the only case that has been studied in detail. The energy levels and the reduced magnetic susceptibility $|J|\chi/Ng^2\beta^2$ of alternating chains have been determined for various α values, using the ring chain technique [11.40]. When J is negative and for any α value in the $0 \leq \alpha < 1$ range, there is a gap in the energy level spectrum between a singlet ground state and the first excited state. Consequently, χ tends to zero when the temperature approaches absolute zero, which is in contrast with what happens for a uniform chain.

Hatfield has fitted the numerical values of magnetic susceptibility for an alternating ring chain of $n = 10$ local spins 1/2 using an analytical expression of the form [11.40, 11.41]

$$\chi = \frac{Ng^2\beta^2}{kT}\frac{A + Bx + Cx^2}{1 + Dx + Ex^2 + Fx^3} \tag{11.4.2}$$

with

$$x = |J|/kT \tag{11.4.3}$$

The A–F coefficients are functions of α. Two sets of coefficients have been proposed according to the value of α. For $0 \leq \alpha \leq 0.4$, these coefficients are

$$A = 0.25$$

$$B = -0.062935 + 0.11376\alpha$$

$$C = 0.0047778 - 0.033268\alpha + 0.12742\alpha^2 - 0.32918\alpha^3 + 0.25203\alpha^4$$

$$D = 0.053860 + 0.70960\alpha \qquad (11.4.4)$$

$$E = -0.00071302 - 0.10587\alpha + 0.54883\alpha^2 - 0.20603\alpha^3$$

$$F = 0.047193 - 0.0083778\alpha + 0.87256\alpha^2 - 2.7098\alpha^3 + 1.9798\alpha^4$$

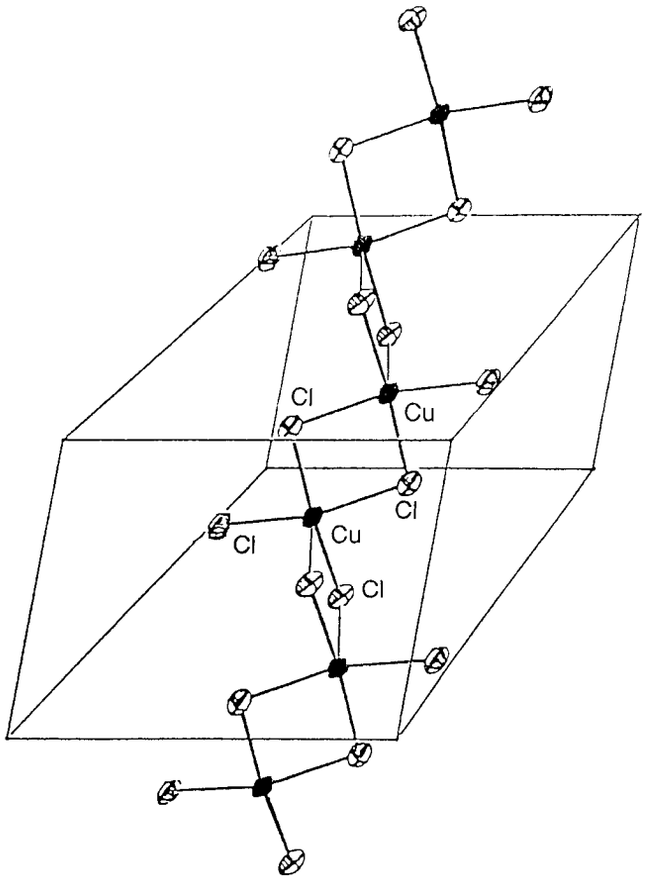


Figure 11.7. Structure of the alternating chains $(CuCl_3)_n$ in (ipa)$CuCl_3$ with ipa = isopropylammonium (from ref. [11.44]).

and for $0.4 < \alpha \leq 1$, they are

$$A = 0.25$$

$$B = -0.068475 + 0.13194\alpha$$

$$C = 0.0042563 - 0.031670\alpha + 0.12278\alpha^2 - 0.29943\alpha^3 + 0.21814\alpha^4$$

$$D = 0.035255 + 0.65210\alpha \qquad (11.4.5)$$

$$E = -0.00089418 - 0.10209\alpha + 0.87155\alpha^2 - 0.18472\alpha^3$$

$$F = 0.045230 - 0.0081910\alpha + 0.83234\alpha^2 - 2.6181\alpha^3 + 1.92813\alpha^4$$

With the two sets of A–F coefficients Eq. (11.4.2) is valid for $kT/|J| \geq 0.25$ and $J \leq 0$. It reproduces the numerical values of χ for $n = 10$ very well, and is expected to be a good approximation for infinite systems. No analytical expression has been proposed for $J > 0$.

In 1987 Landee reviewed the known examples of alternating chains with 1/2 local spins, and distinguished between alternating spacings and alternating ligands [11.43]. The compound (ipa)$CuCl_3$ with ipa = isopropylammonium is an example of the former class [11.44, 11.45]. The structure consists of $CuCl_3$ alternating chains as shown in Figure 11.7. Each copper atom is fivefold coordinate with four short Cu—Cl bonds (2.30 and 2.31 Å) in the basal plane, and a long Cu—Cl apical bond (2.70 Å). The fitting of the magnetic data with the alternating chain model is shown in Figure 11.8,

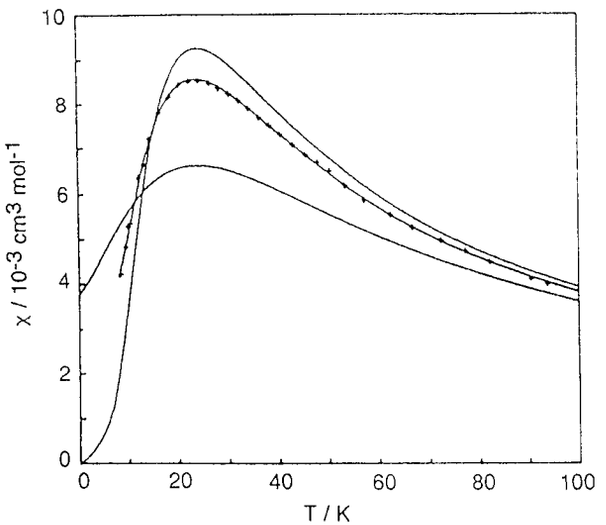


Figure 11.8. χ versus T plot for (ipa)$CuCl_3$. The solid line through the data corresponds to the best fitting with the alternating chain law. The upper and lower solid lines correspond to the predictions of magnetic susceptibility for the dimer ($\alpha = 0$) and the equally spaced chain ($\alpha = 1$) models, using the best-fit parameters for the alternating chain model (from ref. [11.45]).

along with the calculated curves for a uniform chain ($\alpha = 1$) and noninteracting pairs ($\alpha = 0$). The parameters have been found to be $J = -27.4$ cm^{-1} and $\alpha = 0.35$. There is no doubt that the more pronounced interaction occurs in the Cu_2Cl_2 bridging network involving only short Cu—Cl bonds. The two magnetic orbitals are then located in the plane of this bridging network, and may interact efficiently. Chains with alternating spacings may also be obtained with organic radicals like $TCNQ^-$ (TCNQ = tetracyanoquinodimethane) [11.46].

The alternating chain model has also been used to interpret magnetic data obtained with compounds whose room temperature structure does not show any alternation. These compounds are then assumed to undergo a phase transition with an alternation of the interaction pathways below a critical temperature. In the absence of experimental evidence, in particular of low-temperature X-ray data proving unambiguously the alternation, the relevance of these studies is questionable.

Alternation in chains with alternating bridges L and L′ of the type ...L′ALAL′AL... is certainly easier to control than in chains with alternating spacings, and numerous examples have been reported. One of these is Cu(3,6-dto)Cl_2 with 3,6-dto = 3,6-dithiaoctane [11.47]. The copper(II) ions are bibridged by alternating chlorine atoms and sulfur atoms of 3,6-dto as shown in Figure 11.9, leading to Cu—Cu separations

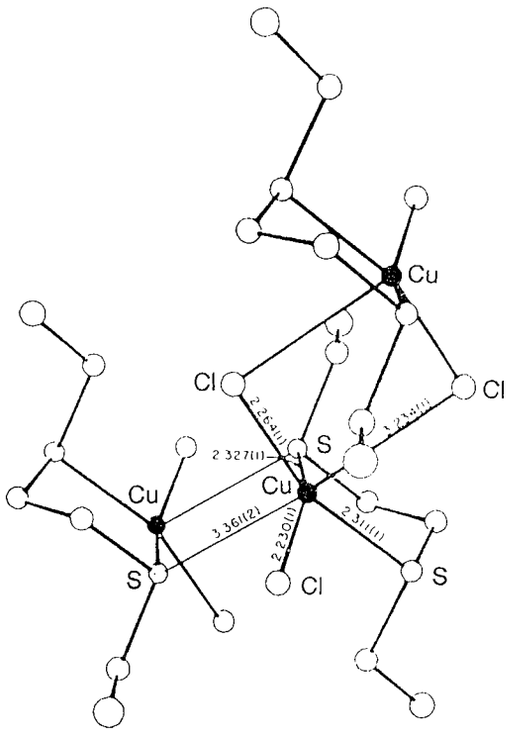


Figure 11.9. Structure of the alternating chain compound Cu(3,6-dto)Cl_2 with 3,6-dto = 3,6-dithiaoctane (from ref. [11.47]).

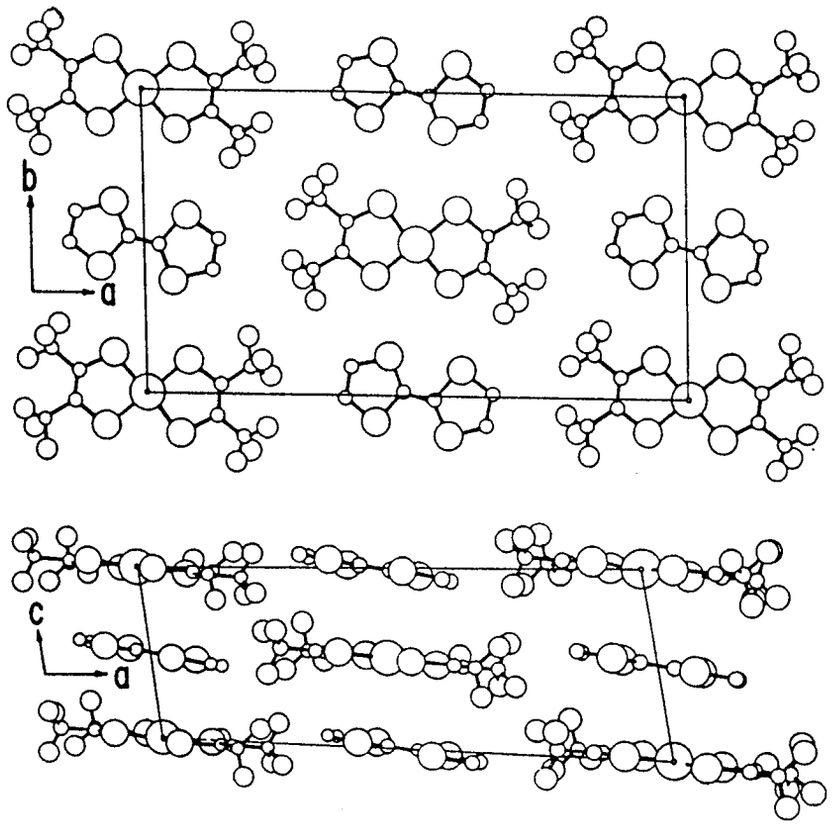


Figure 11.10. Structure of the donor–acceptor compound $(TTF)^+[CuS_4C_4(CF_3)_4]^-$ with TTF^+ = tetrathiafulvalinium and $[CuS_4C_4(CF_3)_4]^-$ = bis-cis-(1,2-perfluoromethylethylene-1,2-dithiolato)-copper(III). The $CuS_4C_4(CF_3)_4$ anions are at the corners and face centers. The TTF cations are at the midpoint of each edge (from ref. [11.49]).

of 4.406 and 4.679 Å, respectively. The Cu_2Cl_2 and Cu_2S_2 planes are mutually perpendicular. The magnetic data have been satisfactorily interpreted with $J = -5.4$ cm^{-1} and $\alpha = 0.69$. It is not obvious how we should associate the interaction parameters J and αJ with the bridging networks. The more pronounced interaction probably might correspond to the Cu_2S_2 network. Indeed, sulfur-containing bridges are known to favor rather strong antiferromagnetic interactions (see Section 8.6).

The alternating situation has some relevance with the spin-Peierls transition phenomenon [11.48]. In 1955, using a purely theoretical approach, Peierls demonstrated that an uniform magnetic chain was energetically unstable with respect to dimerization at low temperatures, so that such a chain was expected to dimerize at a critical temperature T_{sp}. The transition is predicted to be of second-order. Below T_{sp} the system has a singlet ground state with a gap between this state and the first excited states.

While an isolated uniform chain should order magnetically at $T = 0$ K, there is no long-range order for a spin-Peierls chain, even at 0 K. The driving force is the coupling between one-dimensional electronic structure and three-dimensional lattice vibrations. The spin-Peierls transition has the same origin as the Peierls distortion for one-dimensional conducting compounds, with a conducting–semiconducting transition. Experimentally, the spin-Peierls transition is evidenced by a knee in the susceptibility versus temperature curve at T_{sp}, with a rather abrupt fall of χ below T_{sp}.

Apparently the first, and best documented example of spin-Peierls transition is the donor-acceptor compound $(TTF)^+[CuS_4C_4(CF_3)_4]^-$ where TTF^+ is the tetrathiafulvalinium ion and $[CuS_4C_4(CF_3)_4]^-$ is bis-cis-(1,2-perfluoromethylethylene-1,2-dithiolato)copper(III) [11.49–11.50]. The molecular arrangement at 200 K is represented in Figure 11.10. It consists of alternate stacks of the two kinds of ions along the c axis. The spin carriers are the TTF^+ cations; they may interact along the chain axis through the diamagnetic $[CuS_4C_4(CF_3)_4]^-$ anions. The χ versus T plot is shown in Figure 11.11. It exhibits a break at 12 K and a sharp decrease of χ below this critical temperature. In the 250–12 K temperature range the magnetic data are accurately fitted by the one-dimensional Heisenberg model, and the intrachain interaction parameter J is found to be equal to -107 cm^{-1}. The break at 12 K is associated with the spin dimerization. The molecular displacements that accompany the transition, studied by X-ray diffraction at 4.2 K, are very weak in magnitude. Effectively, they correspond to a dimerization of the magnetic units TTF^+. Surprisingly, the dimerization is not found to be along the chain axis but in a direction perpendicular to this axis [11.51].

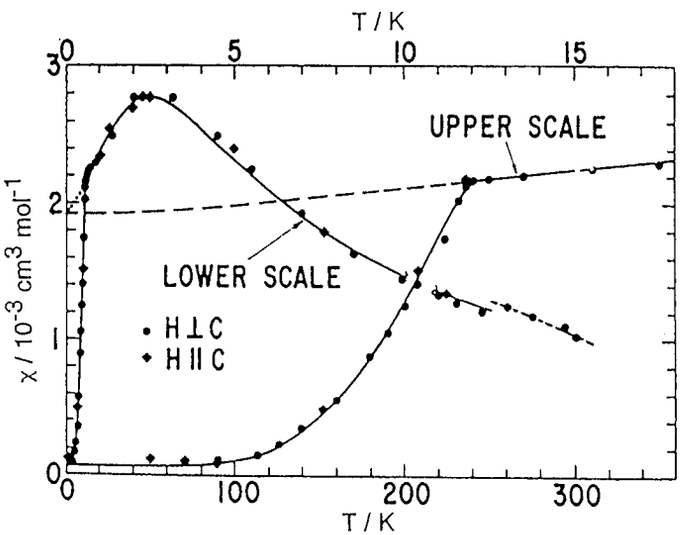


Figure 11.11. χ versus T plot for $(TTF^+[CuS_4C_4(CF_3)_4]^-$. The solid lines are calculated from an equally spaced chain model above 12 K, and a temperature dependent alternating chain model below 12 K (from ref. [11.49]).

The isostructural compound $(TTF)^+[AuS_4C_4(CF_3)_4]^-$ exhibits a spin-Peierls transition at 2.1 K.

After the discovery of spin-Peierls transition in $(TTF)^+[CuS_4C_4(CF_3)_4]^-$ and in its gold analogue, several groups have attempted to design new one-dimensional compounds exhibiting this phenomenon. To a large extent, these efforts have been unsuccessful. Apparently it is difficult to fulfill the conditions required for the observation of a spin dimerization. The first condition is that the compound must be a very good one-dimensional antiferromagnet, with a ratio between inter and intrachain interaction parameters of the order of 10^{-4}, or even smaller. Another condition is that the structure of the chain must be flexible, in the sense that the dimerization should not require too large an energy. In $(TTF)^+[MS_4C_4(CF_3)_4]^-$, M = Cu or Au, such a flexibility arises from the stacking of the planar molecular units along the chain. In this respect, it is worth noting that other cases of spin dimerization have been observed in $TCNQ^-$-containing organic compounds in which the $TCNQ^-$ units are equally spaced above a critical temperature and dimerized below it [11.52]. Apparently, in one-dimensional copper(II) polymers, the skeleton of the chains is too rigid.

11.5 Regular Ferrimagnetic Chains: Theory

The newest aspect in the field of magnetic chain compounds concerns the design and investigation of systems in which two kinds of magnetic centers A and B regularly alternate. These compounds are called either ordered bimetallic chains, or ferrimagnetic chains; they may be schematized as

$$\text{—}A_i\overset{J}{\text{———}}B_{i+1}\overset{J}{\text{———}}A_{i+2}\overset{J}{\text{———}}B_{i+3}\text{—}$$

The name bimetallic chain implicitly supposes that the magnetic centers are metal ions, which is not always the case; the name ferrimagnetic chain supposes that the interaction parameter J is negative. The first compound of this type was reported in 1981 [11.53]. Several families of ferrimagnetic chains have since been described and the main theoretical concepts adapted to this new class of materials have been discussed. The design of ferrimagnetic chains probably represents the most significant contribution of synthetic molecular chemistry to the area of magnetic materials. For this reason we devote a large amount of space to those compounds. We will first focus on the theoretical aspects, at both the qualitative and the quantitative levels, and then we will present some experimental results.

11.5.1 Qualitative Approach

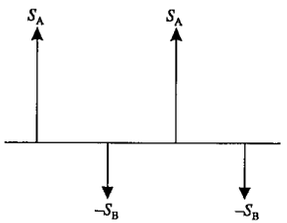
To introduce the basic concepts [11.54], we consider an ordered and regular ring chain $(AB)_n$ where n may become infinite. A and B carry the local spins S_A and S_B, respectively, with $S_A \neq S_B$. The spin Hamiltonian in zero-field appropriate for the description of the isotropic interaction between nearest neighbors is

$$\mathbf{H} = -J \sum_{i=1}^{2n} \mathbf{S}_i \cdot \mathbf{S}_{i+1} \tag{11.5.1}$$

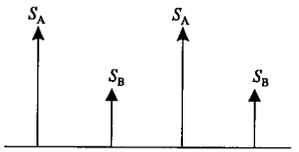
with

$$\begin{aligned} S_{2i-1} &= S_A \\ S_{2i} &= S_B \\ S_{2n+i} &= S_i \end{aligned} \tag{11.5.2}$$

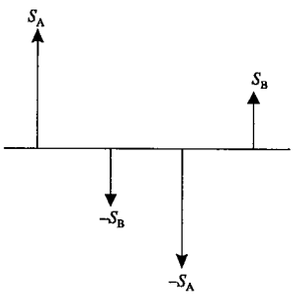
First, we suppose that J is negative. Assuming that the local spins are preferentially oriented perpendicular to the chain axis, the state of lowest energy E_g may then be schematized as



Its spin is $S_g = n(|S_A - S_B|)$. The state of highest energy E_e is as



and its spin is $S_e = n(S_A+S_B)$. Between the two limits E_g and E_e, states with spins less than S_g do exist. Notably there are some states with $S = 0$ if the total number of unpaired electrons $2n(S_A+S_B)$ is even; one of them may be schematized as



or with $S = 1/2$ if $2n(S_A+S_B)$ is odd. In other words, for any $n \neq 1$, the spin state structure is irregular (see Section 10.1). This irregularity increases with n and $|S_A - S_B|$ since S_g is proportional to these two quantities whereas the lowest spin is always 0 or 1/2.

Let us look for the consequences of this irregularity in the temperature dependence of χT, χ being the molar magnetic susceptibility per AB unit. The Zeeman perturbation which we need to add to (11.5.1) is

$$\mathbf{H}_{ZE} = \beta \sum_{i=1}^{n} (\mathbf{S}_{2i-1} \cdot \mathbf{g}_{A_{2i-1}} + \mathbf{S}_{2i} \cdot \mathbf{g}_{B_{2i}}) \cdot H \tag{11.5.3}$$

We assume that the local tensors $\mathbf{g}_A$ and $\mathbf{g}_B$ are isotropic with g_A and g_B principal values, respectively. The high-temperature limit $(\chi T)_{HT}$ of χT corresponds to the sum of the local contributions, i.e.

$$(\chi T)_{HT} = \frac{N\beta^2}{3k} [g_A{}^2 S_A(S_A + 1) + g_B{}^2 S_B(S_B + 1)] \tag{11.5.4}$$

The low-temperature limit $(\chi T)_{LT}$ of χT is reached when only the ground state is thermally populated. This limit is

$$(\chi T)_{LT} = \frac{Ng^2\beta^2}{3k} [n(S_A - S_B)^2 + |S_A - S_B|] \tag{11.5.5}$$

where the Zeeman factor g for the ground state is a linear combination of g_A and g_B. For a particular ring chain size n_0 $(\chi T)_{HT}$ and $(\chi T)_{LT}$ are equal. In the approximation where g_A and g_B are equal, n_0 is given by

$$n_0 = \frac{S_A{}^2 + S_B{}^2 + 2S_<}{(S_A - S_B)^2} \tag{11.5.6}$$

where $S_<$ is the smaller of S_A and S_B. For $n > n_0$, $(\chi T)_{LT}$ is larger than $(\chi T)_{HT}$, and, as n tends to infinite, $(\chi T)_{LT}$ diverges. On cooling from high temperatures, we first depopulate the state of highest spin S_e, and χT decreases. We are thus led to the following result for an AB ferrimagnetic chain, χT first decreases on cooling, reaches a minimum for a finite temperature, then diverges as T approaches zero. This behavior is valid for any couple of S_A and $S_B \neq S_A$ local spins, provided that there is no accidental compensation of the local magnetic momenta. Another way of expressing the same result is to say that at high temperatures χT tends to the paramagnetic limit. The minimum of χT corresponds to a short range order state where the spins S_A and S_B of adjacent magnetic centers are antiparallel, but where there is no correlation between neighboring AB units. As T decreases, the correlation length within the chain increases. The divergence of χT at low temperatures may be associated with the onset of magnetic ordering at 0 K. Below the temperature of the minimum of χT, the magnetic behavior is qualitatively equivalent to what happens for a chain of n spins $|S_A - S_B|$, which are ferromagnetically coupled.

If the intrachain interaction is ferromagnetic, i.e., $J > 0$, the order of the spin levels is reversed and a priori no extremum of the χT versus T plot can be predicted.

11.5.2 Quantitative Approach

Several quantitative approaches to the magnetic susceptibility of ferrimagnetic chains have been proposed to date. The first consists of carrying out the calculation for $(AB)_n$ ring chains of increasing size with quantum numbers S_A and S_B, and extrapolating to $n \rightarrow \infty$. In the hypothesis where the local Zeeman factors g_A and g_B are taken to be equal, the calculation is straightforward, since all the spin states characterized by S have the same g-factor and the labels S are good quantum numbers, even in the presence of the Zeeman perturbation. The method is then an extension of the method first used by Bonner and Fisher for uniform chains of local spins 1/2 (see Section 11.1) [11.54–11.56]. On the other hand, for $g_A \neq g_B$, each spin state has a given g_S-factor and furthermore the Zeeman perturbation couples the components of the same M_S belonging to different spin states, which introduces second-order Zeeman coefficients into the calculation of the susceptibility. Even if the full D_n symmetry of the problem is taken into account, this method is severely limited by both the storage capacity of the computer and the computing time. The first calculation of this kind was for $S_A = 1$ and $S_B = 1/2$. Drillon et al. performed the calculation up to $n = 5$, and extrapolated to $n \rightarrow \infty$ [11.56]. The results for $g_A = g_B$ are shown in Figure 11.12. The minimum of χT is shifted toward higher temperatures as n increases; the minimum occurs for $kT/|J| = 0.570$ when n is extrapolated to infinity. The divergence of χT as T tends to zero then follows a law in $T^{-0.80}$, which coincides with the law found for a ferromagnetic chain of 1/2 local spins. These results have been fitted by the analytical expression

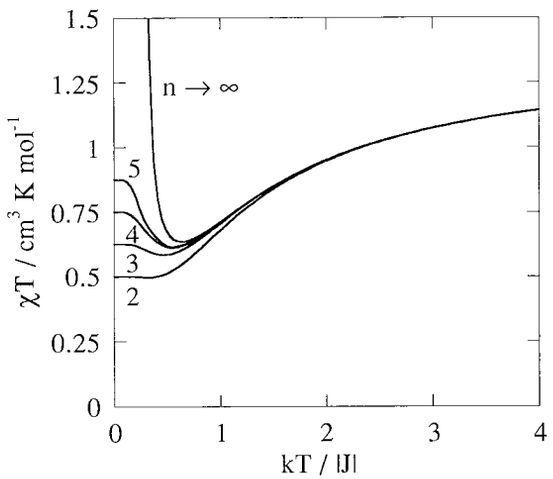


Figure 11.12. χT versus $kT/|J|$ curves for antiferromagnetically coupled ring chains $(AB)_n$ with local spins $S_A = 1$ and $S_B = 1/2$. The local Zeeman factors g_A and g_B are both taken equal to 2.00 (from ref. [11.46]).

$$\chi T = \frac{N g_A^2 \beta^2}{k} \frac{ax^3 + bx^2 + cx + d}{ex^2 + fx + g} \tag{11.5.7}$$

with

$$a = -0.0341468$$
$$b = 2.81693$$
$$c = -7.23100$$
$$d = 11$$
$$e = 1.29663$$
$$f = 0.697190$$
$$g = 12 \tag{11.5.8}$$

and

$$x = |J|/kT \tag{11.5.9}$$

Equation (11.5.7) is valid only for $J < 0$. Drillon et al. also investigated the effect $g_B/g_A \neq 1$ and found that for $g_B/g_A = 2.66$, there was an exact compensation of the local magnetic moments, such that the ground state became nonmagnetic. The $(AB)_n$ ring chain technique has been extended to several other cases, notably for $S_A = 1/2$ and S_B taking all other values up to 5/2 [11.55]. The calculation was only performed up to $n = 3$ except for $S_B = 1$.

The simplest method consists of considering that both S_A and S_B are classical spins with g_A^e and g_B^e effective g-factors and a J^e effective interaction parameter. These effective parameters are related to the actual parameters through

$$g_A^e = g_A[S_A(S_A + 1)]^{1/2} \tag{11.5.10}$$

and a similar expression for g_B^e, and

$$J^e = J[S_A(S_A + 1)S_B(S_B + 1)]^{1/2} \tag{11.5.11}$$

An analytical expression for the magnetic susceptibility was derived by Drillon et al. [11.57]. This expression is

$$\chi T = \frac{N\beta^2}{3k}\left[g^2 \frac{1+u}{1-u} + \delta^2 \frac{1-u}{1+u}\right] \tag{11.5.12}$$

with

$$g = (g_A^e + g_B^e)/2$$
$$\delta = (g_A^e - g_B^e)/2$$
$$u = \coth(J^e/kT) - (kT/J^e) \tag{11.5.13}$$

For $\delta = 0$, (11.5.12) reduces to expression (11.2.4), which is valid for a regular chain of classical spins. For $\delta \neq 0$ and $J < 0$, a minimum in the χT versus T plot appears, and as T approaches zero, χT diverges in a ferromagnetic-like fashion, according to a law in $\delta^2 T^{-1}$. The validity of this purely classical approach is better when S_A and S_B are larger. For $S_A = 5/2$, (11.5.12) is probably valid if $S_B = 2$, 3/2, or even 1 but may become a very poor approximation when $S_B = 1/2$.

Another approach, of particular interest when S_A–S_B is large, consists of treating S_A as a classical spin and S_B as a quantum spin. This approach was first introduced by Seiden in the case where $S_A = 5/2$ and $S_B = 1/2$ with $g_A = g_B$ [11.58]. It was then generalized by Georges et al. to any S_A, S_B couple with $g_A \neq g_B$, and also to the case of alternating ferrimagnetic chains with two interaction parameters $J_{AB}(1 + \alpha)$ and $J_{AB}(1-\alpha)$ [11.59]. Georges et al. derived analytical expressions for magnetic susceptibility, which will be given in Section 11.6.

The following empirical expression has been proposed to fit the results of Seiden ($S_A = 5/2$, $S_B = 1/2$, $g_A = g_B = g$), when J is negative [11.60]:

$$\chi T = \frac{g^2}{4} \frac{4.75 - 1.62370x + 2.05042x^2 - 4.52588x^3 - 8.64256x^4}{1 + 0.77968x - 1.56527x^2 - 1.57333x^3 - 0.11666x^{4.5}} \qquad (11.5.14)$$

with

$$x = |J|/kT \qquad (11.5.15)$$

χT in (11.15.14) is expressed in $cm^3\ K\ mol^{-1}$. Equation (11.5.14) leads to a curve with a rounded maximum for $kT/|J| = 2.98$. On cooling below the temperature of the minimum, χT diverges in an abrupt fashion, following a law in $T^{-0.69}$.

In all the approaches presented in this section, we have neglected local anisotropies, which could lead to a rather poor description of χT in the low temperature range if $|D/J|$ is not very small. In the specific case $S_A = 1$, $S_B = 1/2$, it has been shown that increasing $|D/J|$ shifts the minimum of χT toward lower temperatures. The anisotropic interaction has the same effect [11.54]. The local anisotropy has been taken into account in a treatment where the interaction was assumed to be of the Ising-type. Such a treatment may be valid for ferrimagnetic chains involving orbitally degenerate ions [11.61, 11.62]. Finally, all the calculations deal with perfectly isolated chains. In fact, much above the critical temperature where three-dimensional order appears, interchain interactions may substantially perturb the magnetic behavior. If the interchain interactions are predominantly antiferromagnetic, as is often the case, the divergence of χT is stopped by a sharp maximum of χT occurring just above the critical temperature, and of χ occurring at the critical temperature. If the interchain interactions are predominantly ferromagnetic, a zero-field magnetization may be obtained below the critical temperature. This aspect will be investigated in a more thorough manner in Section 12.6.

11.6 Regular Ferrimagnetic Chains: Some Examples

The first structurally characterized ferrimagnetic chain is $MnCu(dto)_2(H_2O)_3 \cdot 4.5H_2O$ with dto = dithiooxalato. This compound was investigated by Gleizes, Verdaguer et al. [11.60, 11.63]. The structure of two adjacent chains together with the detail of the structure of one of the chains is shown in Figure 11.13. Copper(II) is tetracoordinated with a planar environment and manganese(II) is hepta-coordinated in a rather unusual fashion. The metal ions are bridged by the dithiooxalato ligand, the oxygen atoms being bound to the manganese and the sulfur atoms to the copper. The magnetic data closely follow the theoretical predictions. On cooling, χT decreases, reaches a weakly pronounced minimum around 130 K, and then increases as T is lowered further to 7.5 K. Finally, below 7.5 K, χT decreases rapidly. In the range 7.5–300 K, these data are well fitted by the classical-quantum model, with $J = -30.3\ cm^{-1}$. The maximum of χT at 7.5 K is clearly due to antiferromagnetic interchain interactions. Magnetization studies at 1.3 and 4.2 K show a saturation corresponding

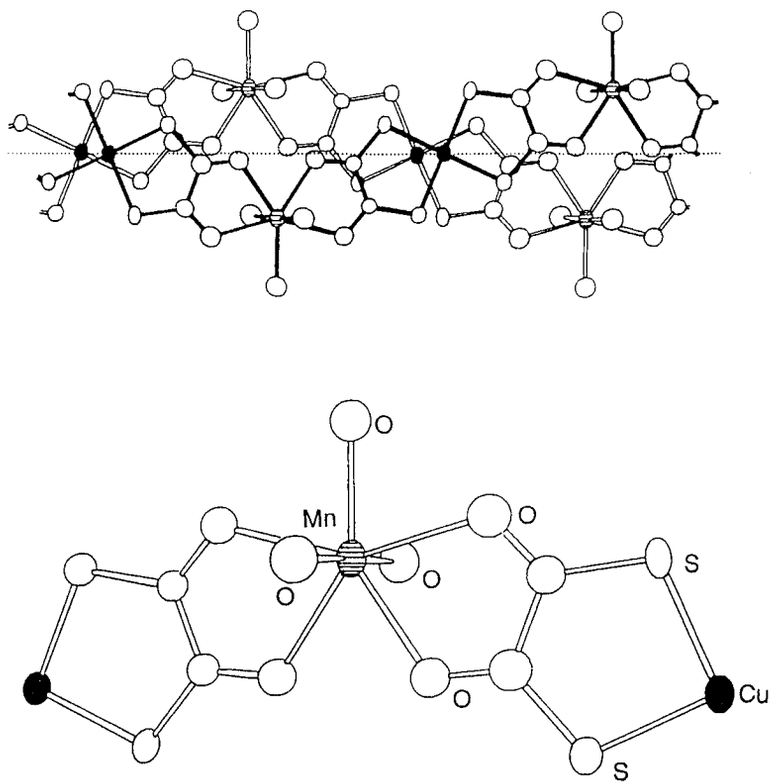


Figure 11.13. Structure of the ferrimagnetic chain compound $MnCu(dto)_2(H_2O)_3 \cdot 4.5H_2O$ with dto = dithiooxalato. The figure shows two adjacent chains as well and the detail of one of the chains (from ref. [11.63]).

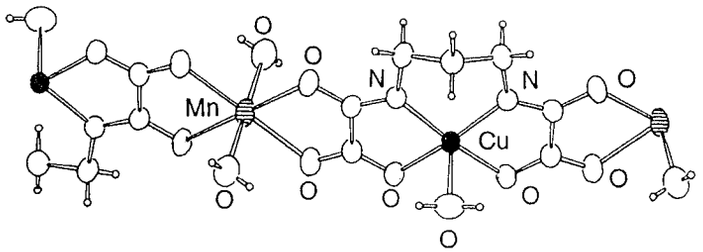


Figure 11.14. Structure of the ferrimagnetic chain compound $MnCu(pba)(H_2O)_3 \cdot 2H_2O$ with pba = 1,3-propylene-bis(oxamato) (from ref. [11.64]).

to a spin $S_A - S_B = 2$ per MnCu unit. The relatively large value of J in $MnCu(dto)_2(H_2O)_3 \cdot 4.5H_2O$ confirms the efficiency of the dithiooxalato bridge in propagating strong antiferromagnetic interactions between metal centers separated by 5.75 Å.

Another example of a Mn(II)Cu(II) bimetallic chain is provided by $MnCu(pba)(H_2O)_3 \cdot 2H_2O$, with pba = 1,3-propylenebis(oxamato) [11.64, 11.65]. The structure of the chain is shown in Figure 11.14. Manganese(II) is in distorted octahedral surroundings and copper(II) is in square pyramidal surroundings. The metal ions are aligned along the *b*-axis of the orthorhombic structure, while in the previous example they formed zig-zag chains. Within the chains, two nearest neighbor ions are bridged by an oxamato group with an Mn—Cu separation of 5.41 Å. As expected, the χT versus T plot shown in Figure 11.15 exhibits a rounded minimum about 115 K, and a very rapid increase as the temperature is lowered further. A maximum of χT is observed,

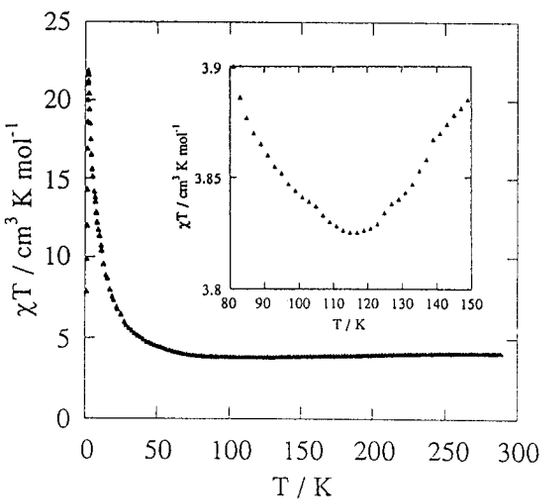


Figure 11.15. χT versus T plot for $MnCu(pba)(H_2O)_3 \cdot 3H_2O$. The inset shows the rounded minimum of the curve at 115 K, which is the fingerprint of the ferrimagnetic chain compounds (from ref. [11.64]).

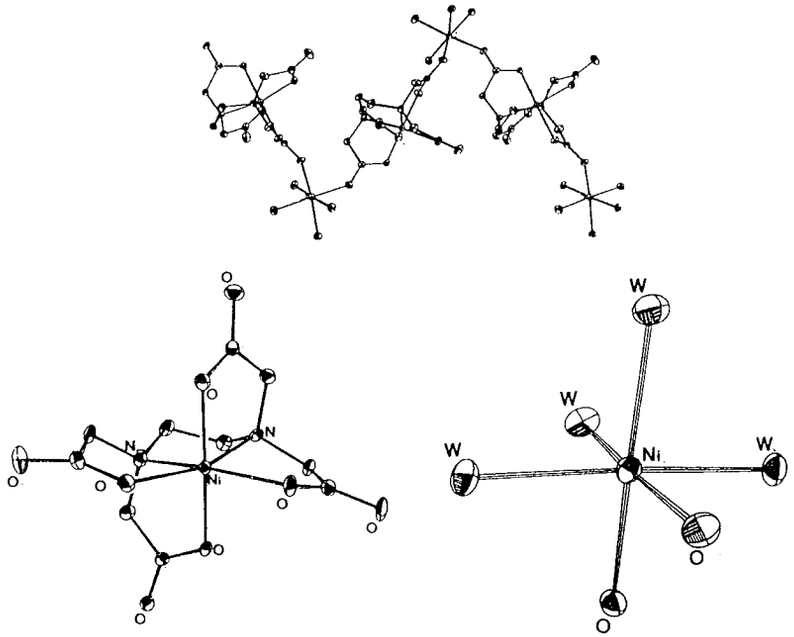


Figure 11.16. Structure of $Ni_2(edta)(H_2O)_4 \cdot 2H_2O$ with edta = ethylenediaminetetracetato. The figure shows the chain itself and the two Ni(II) coordination spheres (from ref. [11.67]).

but at a much lower temperature than in the previous example, namely 2.3 K instead of 7.5 K. The one-dimensional character is more pronounced. The fitting of the magnetic data above 4.2 K leads to $J = -23.4$ cm^{-1}. The maximum of χT at 2.3 K is related to the onset of long-range antiferromagnetic ordering. This point will be treated further in Section 12.6.

A completely different family of ferrimagnetic chains was investigated by Coronado et al. [11.66, 11.67]. The general formula is $AB(edta)(H_2O)_4 \cdot 2H_2O$ where edta is ethylenediaminetetracetato and A and B are two divalent ions. All of these compounds are reported to be isomorphous with $A_2(edta)(H_2O)_4 \cdot 2H_2O$, A = Ni or Zn [11.66, 11.68], the crystal structures of which have been solved. The structure of the nickel(II) derivative is shown in Figure 11.16. There are two distinct coordination sites. In one coordination site, a metal ion, say A, is in distorted octahedral environment involving the oxygen atoms of the four carbonyl groups and the nitrogen atoms of edta. In the other coordination site, B is in a less distorted octahedral environment formed by four water molecules and two oxygen atoms of adjacent A(edta) units. Thus, two of the four carboxylic groups of edta are bidentate bridges, and zig-zag chains $[A(edta)–B(H_2O)_4]_n$ are formed. The two coordination sites have some selectivity. For instance, Cu(II) and Ni(II) prefer the chelated sites A, and when these sites are fully occupied, Co(II) or Mn(II) goes into the hydrated sites B. One of the most accurately investigated compound of this edta family is the Ni(II)Mn(II) species [11.67]. The χT versus T plot

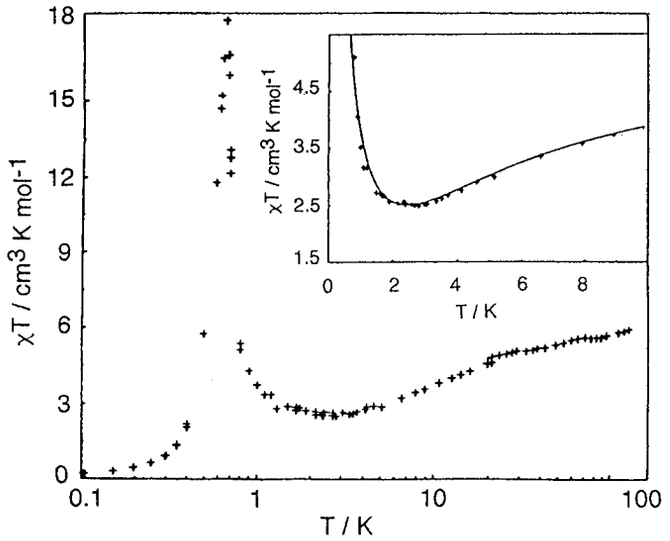
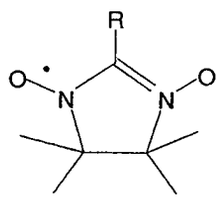


Figure 11.17. χT versus T plot for NiMn(edta)$(H_2O)_4 \cdot 2H_2O$. Note that the temperature scale is logarithmic. The full line in the inset corresponds to the fit with the ferrimagnetic chain model (from ref. [11.67]).

is shown in Figure 11.17. It exhibits the expected minimum, but at very low temperature, namely 2.5 K, then a rapid increase when the temperature is lowered further, and a sharp maximum at 0.665 K. The low temperature of the minimum of χT indicates that the intrachain interaction is much weaker than in the previous two examples, which is not surprising considering the structure of these edta compounds. The maximum of χT is related to the onset of a three-dimensional antiferromagnetic ordering, which is confirmed by a λ-peak at 0.660 K in the heat capacity curve. The magnetic data have been fitted with both the classical spin approach [Eq. (11.5.12)] and the ring chain approach. The results of the latter are $J = -1.04$ cm^{-1} $g_{Ni} = 2.39$, and $g_{Mn} = 1.95$.

Our last example of ferrimagnetic chain concerns a compound in which the spin carriers are alternately Mn(II) ions with local spins 5/2 and nitronyl nitroxide radicals with local spins 1/2 [11.69, 11.70]. The structure of these organic radicals is



where R may be an alkyl or an aromatic group. These radicals can bind to two different metal atoms with their two equivalent oxygen atoms, provided that the metal atoms are strong Lewis acids. The crystal structure of Mn(hfa)$_2$NIT-isopropyl is shown in

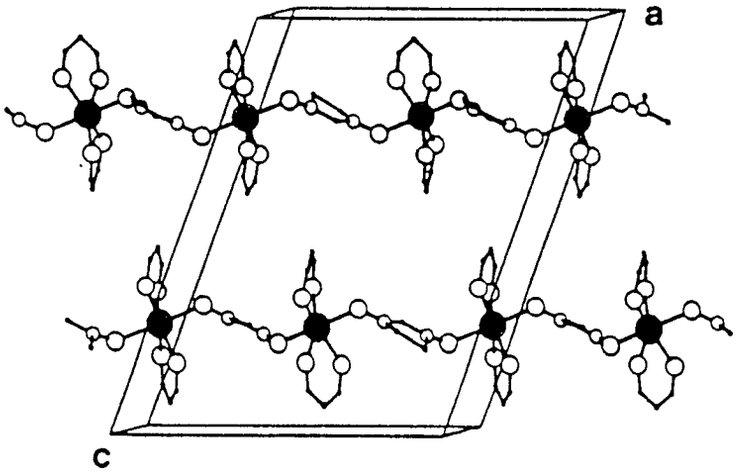


Figure 11.18. Structure of the chain compound Mn(hfa)$_2$NIT-isopropyl with hfa = hexafluoroacetylacetonato and NIT-isopropyl = 2-isopropyl-4,4,5,5-tetramethyl-4,5-dihydro-1 *H*-imidazolyl-1-oxy-3-oxide (from ref. [11.69]).

Figure 11.18; hfa stands for hexafluoroacetylacetonato and NIT-isopropyl for the radical whose exact name is 2-isopropyl-4,4,5,5-tetramethyl-4,5-dihydro-1 *H*-imidazolyl-1-oxy-3-oxide. This structure consists of chains of Mn(hfa)$_2$ units bridged by NIT-isopropyl radicals. The Mn(II) ions are hexacoordinated by four oxygen atoms from two hfa groups and two oxygen atoms from two radicals in trans positions. The χT versus T plot increases continuously on cooling from room temperature to 7.6 K. At this temperature, χT reaches a sharp maximum, due to three-dimensional effects, with a value of the order of 350 cm^3 K mol^{-1}. We will discuss later the three-dimensional ordering in compounds of this kind. Here, we only discuss the one-dimensional behavior. Mn(hfa)$_2$NIT-isopropyl is a very good example of ferrimagnetic chain compound with a large interaction parameter. The absence of a minimum in the χT variation is due to the fact that $|J|$ is larger than $kT_{max}/2.98$ where T_{max} is the highest temperature investigated, namely room temperature. The fitting of the magnetic data with Eq. (11.5.14) leads to a J value of the order of –250 cm^{-1}.

In addition to Mn(hfa)$_2$NIT-R ferrimagnetic chains, a ring chain compound of formula [Mn(hfa)$_2$NIT-Ph]$_6$ with Ph = phenyl was described, which contains six Mn(II)-organic radical pairs [11.71]. Its superb structure is displayed in Figure 11.19. The six metal atoms and the O—N—C—N—O atoms of the six radicals define a 36-membered macrocycle, the diameter of which can be estimated at 14.7 Å from the separation between two opposite manganese atoms. The χT versus T plot for this compound is shown in Figure 11.20. Below ca. 50 K, χT is almost constant and equal to 12.5 cm^3 K mol^{-1} per Mn-radical pair, which corresponds to a spin S = 12 for the ground state of the macrocycle. Magnetization measurements confirm the ferrimagnetic nature of the ground state with an alternation of high spins up and low spins down.

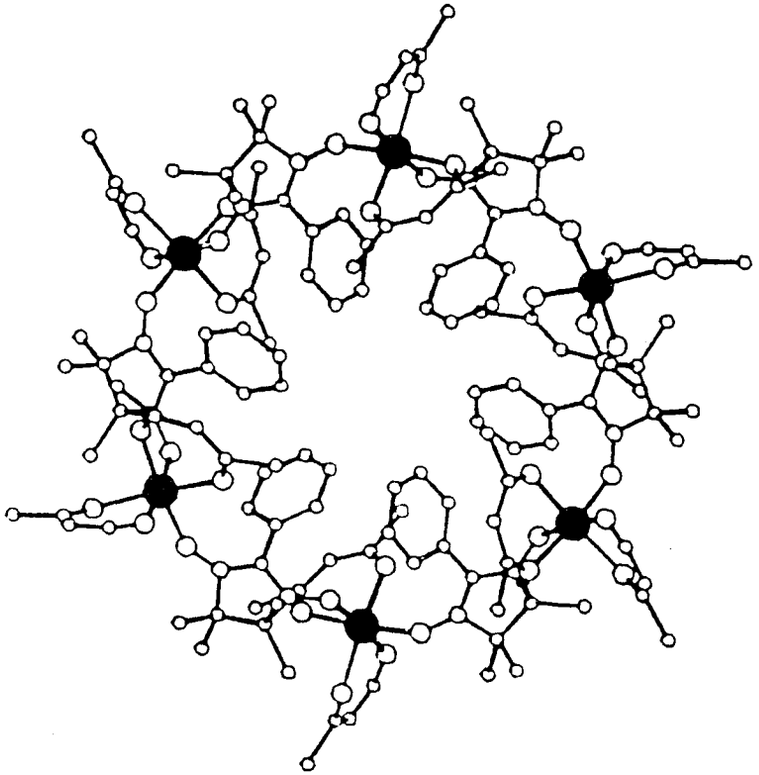


Figure 11.19. Molecular structure of $[Mn(hfa)_2NIT\text{-}Ph]_6$ with NIT-Ph = 2-phenyl-4,4,5,5-tetramethyl-4,5-dihydro-1*H*-imidazolyl-1-oxy-3-oxide (from ref. [11.71]).

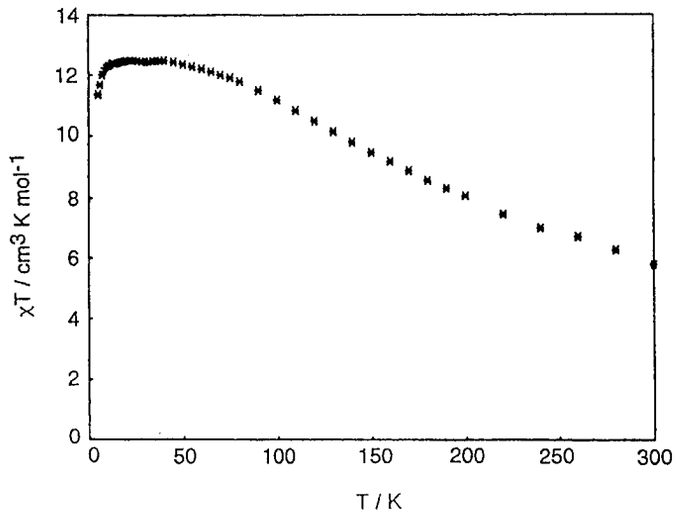


Figure 11.20. χT for Mn-radical unit versus T plot for $[Mn(hfa)_2NIT\text{-}Ph]_6$ (from ref. [11.71]).

11.7 Alternating Ferrimagnetic Chains

Since the possibilities of synthesis in molecular chemistry are almost limitless, it was foreseeable that shortly after the regular ferrimagnetic chains, the first alternating ferrimagnetic chains would be reported. This happened in 1988 along with a theoretical model which is appropriate for describing the magnetic properties of these rather exotic systems [11.72]. Let us start by examining this model. We will not detail the various steps of the calculation, but just give the final result [11.73]. For this, we consider a chain compound like

$$\text{—}A_i \overset{J_{AB}(1+\alpha)}{\text{———}} B_i \overset{J_{AB}(1-\alpha)}{\text{———}} A_{i+1} \overset{J_{AB}(1+\alpha)}{\text{———}} B_{i+1}\text{—}$$

where there is alternation not only of spin carriers but also of interaction pathways. The local spins are S_A and S_B, the local Zeeman factors are g_A and g_B and the interaction parameters between nearest neighbors $J_{AB}(1+\alpha)$ and $J_{AB}(1-\alpha)$. The spin Hamiltonian in zero-field is

$$\mathbf{H} = -J\sum_i \mathbf{S}_{B_i}\cdot[(1+\alpha)\mathbf{S}_{A_i} + (1-\alpha)\mathbf{S}_{A_{i+1}}] \tag{11.7.1}$$

S_A is assumed be large enough to be treated of as a classical spin. S_B is treated as a quantum spin. We write

$$J = J_{AB}[S_A(S_A+1)]^{1/2}$$

$$G = g_A[S_A(S_A+1)]^{1/2}$$

$$g = g_B$$

$$s = S_B$$

$$x = J/kT \tag{11.7.2}$$

The molar magnetic susceptibility is then given by

$$\chi = \frac{N\beta^2}{3kT}\frac{g^2[s(s+1)(1-P)+2QR] + 2gG(Q+R) + G^2(1+P)}{1-P} \tag{11.7.3}$$

with

$$P = A_1/A_0$$

$$Q = x[(1+\alpha)B_0 + (1-\alpha)B_1]/A_0$$

$$R = x[(1-\alpha)B_0 + (1+\alpha)B_1]/A_0 \tag{11.7.4}$$

and

$$A_0 = (2\pi/\Lambda^2) \sum_{\sigma=-s}^{s} \sum_{\varepsilon=\pm} [\varepsilon \exp(\sigma\lambda_\varepsilon)/\sigma^2]\sigma\lambda_\varepsilon - 1)$$

$$A_1 = (\pi/\Lambda^4) \sum_{\sigma=-s}^{s} \sum_{\varepsilon=\pm} [\varepsilon \exp(\sigma\lambda_\varepsilon)/\sigma^4][\sigma^3\lambda_\varepsilon^3 - 3\sigma^2\lambda_\varepsilon^2 + (6 - \sigma^2\lambda^2)\sigma\lambda_\varepsilon + \sigma^2\lambda^2 - 6]$$

$$B_0 = (2\pi/\Lambda^2) \sum_{\sigma=-s}^{s} \sum_{\varepsilon=\pm} \varepsilon \exp(\sigma\lambda_\varepsilon)$$

$$B_1 = (\pi/\Lambda^4) \sum_{\sigma=-s}^{s} \sum_{\varepsilon=\pm} [\varepsilon \exp(\sigma\lambda_\varepsilon)/\sigma^2][\sigma^2\lambda_\varepsilon^2 - 2\sigma\lambda_\varepsilon + 2 - \sigma^2\lambda^2] \qquad (11.7.5)$$

λ_+, λ_-, λ_2, and Λ^2 are defined as

$$\lambda_+ = -2x$$

$$\lambda_- = \alpha\lambda_+$$

$$\lambda^2 = 2x^2(1 + \alpha^2)$$

$$\Lambda^2 = x^2(1 - \alpha^2) \qquad (11.7.6)$$

Equations (11.7.3)–(11.7.6) are also valid for the regular ferrimagnetic chains which were studied in the previous two sections, in which case we have $\alpha = 0$, hence $P = Q$ and $\lambda_- = 0$.

The case $S_A = 5/2$, $S_B = 1/2$, for instance Mn(II)Cu(II), is obviously the most favorable to apply Eqs. (11.7.3)–(11.7.6). Let us examine the role of the alternation parameter α in that system [11.7.4]. We assume that the most important exchange parameter (in absolute value) $J_1 = J_{AB}(1+\alpha)$ is constant and that the other one $J_2 = J_{AB}(1-\alpha)$ may vary. We define ρ as

$$\rho = J_2/J_1 \qquad (11.7.7)$$

such that $\rho = 1$ corresponds to the regular ferrimagnetic chain and $\rho = 0$ to the AB isolated pair with J_1 interaction parameter. Figure 11.21 shows the χT versus $kT/|J|$ curves for ρ =1, 2/3, 1/3, and 0. For the sake of simplicity g_A and g_B are taken both equal to 2. For $\rho = 1$, the result is in principle similar to the result arising from Eq. (11.5.14). However, the position of the minimum of χT is slightly different; this minimum is obtained for $kT/|J_1| = 3.12$. *As* ρ decreases, the minimum of χT is shifted toward lower temperatures. For instance, it occurs at $kT/|J_1| = 2.52$ for $\rho = 2/3$ and $kT/|J_1| = 1.70$ for $\rho = 1/3$. When ρ becomes too small, the classical-quantum approach fails. Indeed, it leads to $\chi T = 3.27$ cm^3 K mol^{-1} at 0 K for $\rho = 0$, whereas for an antiferromagnetically coupled $S_A = 5/2$, $S_B = 1/2$ pair, the low-temperature limit of χT when only the ground quintet state is thermally populated is $\chi T = 2Ng^2\beta^2/k$ (= 3.00

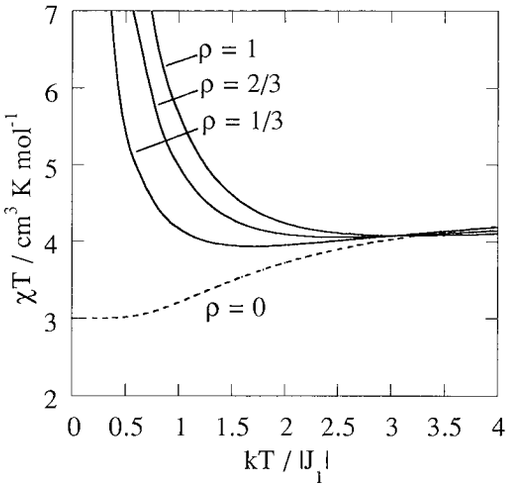


Figure 11.21. χT versus $kT/|J_1|$ curves for alternating ferrimagnetic chains with local spins $S_A = 5/2$ and $S_B = 1/2$ and $\rho = 1$, 2/3, 1/3, and 0. The parameter ρ is defined as J_2/J_1 such that $\rho = 1$ corresponds to the regular ferrimagnetic chain, and $\rho = 0$ to the isolated pair. The curves for $\rho = 1$, 2/3, and 1/3 are calculated from the classical-quantum model, whereas the curve for $\rho = 0$ in dotted line is calculated from the quantum pair law (from ref. [11.74]).

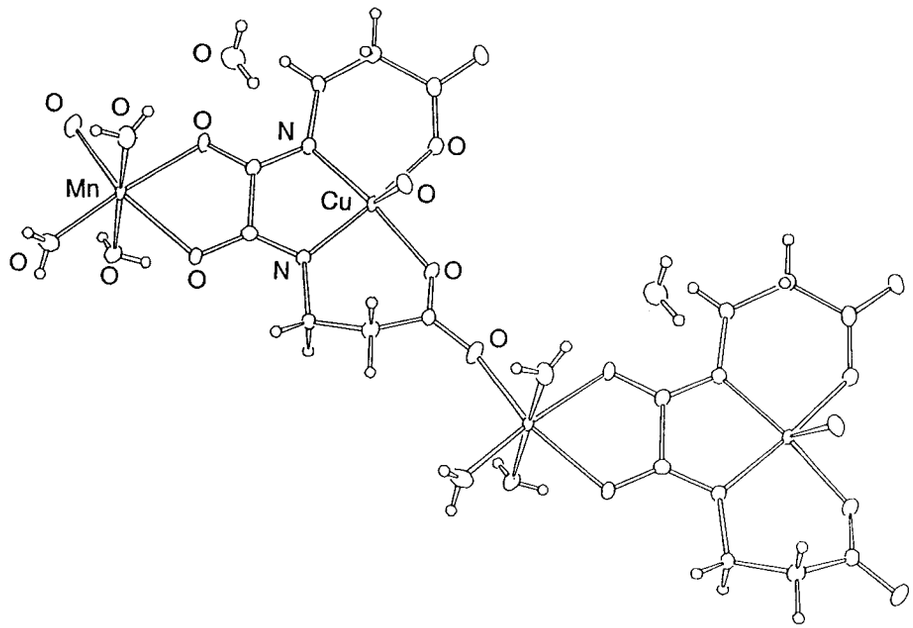


Figure 11.22. Structure of the alternating ferrimagnetic chain compound MnCu(obp)$(H_2O)_3 \cdot 2H_2O$ with obp = oxamido-bis(*N*,*N*′-propionato) (from ref. [11.72]).

cm^3 K mol^{-1} for $g = 2$). The curve (dotted line) for $\rho = 0$ in Figure 11.21 is deduced from the expression

$$\chi T = \frac{2Ng^2\beta^2}{k} \frac{5 + 14\exp(3J_1/kT)}{5 + 7\exp(3J_1/kT)} \tag{11.7.8}$$

which is appropriate for such a pair of quantum spins.

The first reported example of alternating ferrimagnetic chain is $MnCu(obp)(H_2O)_3 \cdot H_2O$ with obp = oxamidobis(propionato) [11.72]. The chain structure shown in Figure 11.22 consists of octahedral Mn(II) and planar Cu(II) ions alternately bridged by oxamido and carboxylato groups. The Mn—Cu separations are equal to 5.452 and 6.066 Å, respectively. The variation of χT versus T presents a rounded minimum about 40 K, characteristic of the one-dimensional ferrimagnetic behavior, and a sharp maximum at 2.9 K, related to the onset of three-dimensional antiferromagnetic ordering at 2.3 K. The magnetic data above 10 K closely follow Eqs. (11.7.3)–(11.7.6) with $J_{AB}(1+\alpha) = -32(1)$ cm^{-1} and $J_{AB}(1-\alpha) = -7(1)$ cm^{-1}. There is no doubt that the interaction parameter $J_{AB}(1+\alpha)$ has to be associated with the oxamido bridge and that $J_{AB}(1-\alpha)$ with the carboxylato bridge.

References

11.1. L.J. de Jongh, A.R. Miedema, *Adv. Phys.* 23 (1974), 1.

11.2. L.J. de Jongh, in "Recent Developments in Condensed Matter Physics," J.T. Devrease, ed., Vol. 1, p. 343. Plenum, New York, 1981.

11.3. J.S. Miller, ed., "Extended Linear Chain Compounds," Vol. 3. Plenum, New York, 1983.

11.4. J.S. de Jongh, in "Magneto-Structural Correlations in Exchange Coupled Systems," R.D. Willett, D. Gatteschi, O. Kahn, eds., p. 1, NATO ASI Series. Reidel, Dordrecht, 1985.

11.5. J.C. Bonner, in "Magneto-Structural Correlations in Exchange Coupled Systems," R.D. Willett, D. Gatteschi, O. Kahn eds., p. 15, NATO ASI Series. Reidel, Dordrecht, 1985.

11.6. Several papers concerning magnetic chain compounds appeared in L.V. Interrante, ed., "Extended Interactions between Metal Ions in Transition Metal Complexes," ACS Symposium Series, American Chemical Society, Washington, D.C., 1974.

11.7. R.L. Carlin, "Magnetochemistry," chap. 7, Springer-Verlag, Berlin, 1986.

11.8. J.C. Bonner, M.E. Fisher, *Phys. Rev. A* 135 (1964), 640.

11.9. R.L. Orbach, *Phys. Rev.* 112 (1958), 309.

11.10. W.E. Estes, D.P. Gavel, W.E. Hatfield, D. Hodgson, *Inorg. Chem.* 17 (1978), 1415.

11.11. G.A. Baker, Jr, G.S. Rushbrooke, H.E. Gilbert, *Phys. Rev.* 135 (1964), A1272.

11.12. E. Ising, *Z. Phys.* 31 (1925), 253.

11.13. A. Michalowicz, J.J. Girerd, J. Goulon, *Inorg. Chem.* 18 (1979), 3004. J.J. Girerd, O. Kahn, M. Verdaguer, *Inorg. Chem.* 19 (1980), 274.

11.14. J. Garas, M. Langfelderova, G. Lundgren, J. Gazo, *Collect. Czech. Chem. Commun.* 37 (1972), 3181.

11.15. R.D. Willett, C.P. Landee, R.M. Gaura, D.D. Swank, H.A. Groenendijk, A.J. van Duynevelt, *J. Magn. Magn. Mat.* 15–18 (1980), 1055.

11.16. H.A. Groenendijk, A.J. van Duynevelt, H.W.J. Blöte, R.M. Gaura, R.D. Willett, *Physica* 106B (1981), 47.

11.17. R.D. Willett, R.M. Gaura, C.P. Landee, in ref. 11.3, p. 143.

11.18. C.Y. Weng, Ph.D. thesis, Carnegie-Mellon University, 1968.

11.19. H.W.J. Blöte, *Physica* 79B (1975), 427.

11.20. T. de Neef, *Phys. Rev. B* 13 (1976), 4141.

11.21. A. Meyer, A. Gleizes, J.J. Girerd, M. Verdaguer, O. Kahn, *Inorg. Chem.* 21 (1982), 1729.

11.22. M.E. Fisher, *Am. J. Phys.* 32 (1964), 343.

11.23. T. Smith, S.A. Friedberg, *Phys. Rev.* 176 (1968), 660.

11.24. F.D.M. Haldane, *Phys. Lett.* 93A (1983), 464; *Phys. Rev. Lett.* 50 (1983), 1153.

11.25. J.P. Renard, S. Clement, M. Verdaguer, *Proc. Indian Acad. Sci. (Chem. Sci.)* 98 (1987), 131.

11.26. J.P. Renard, M. Verdaguer, L.P. Regnault, W.A.C. Erkelens, J. Rossat-Mignod, J. Ribas, W.G. Stirling, C. Vettier, *J. Appl. Phys.* 63 (1988), 3538.

11.27. J.P. Renard, L.P. Regnault, M. Verdaguer, *J. Phys. (Les Ulis)* 49 (1988), C8-1425.

11.28. R. Dingle, M.E. Lines, S.L. Holt, *Phys. Rev.* 187 (1969), 643.

11.29. B. Morosin, E.J. Graeber, *Acta Crystallogr.* 23 (1967), 766.

11.30. C. Dupas, J.P. Renard, *Phys. Lett. A* 43 (1973), 119.

11.31. C. Dupas, J.P. Renard, *Phys. Rev. B* 18 (1978), 401.

11.32. L.P. Regnault, J.P. Boucher, J. Rossat-Mignod, J.P. Renard, J. Bouillot, W.G. Stirling, *J. Phys. C* 15 (1982), 1261.

11.33. J.P. Boucher, L.P. Regnault, J. Rossat-Mignod, J.P. Renard, J. Bouillot, W.G. Stirling, *J. Appl. Phys.* 52 (1981), 1956.

11.34. J.J. Girerd, M.F. Charlot, O. Kahn, *Mol. Phys.* 34 (1977), 1063.

11.35. M.F. Charlot, J.J. Girerd, O. Kahn, *Phys. Stat. Sol. B* 86 (1978), 497.

11.36. M.E. Fisher, *J. Math. Phys.* 4 (1963), 124.

11.37. D.B. Losee, J.N. McElearney, G.E. Shankle, R.L. Carlin, P.J. Creswell, W.T. Robinson, *Phys. Rev. B* 8 (1973), 2185.

11.38. S. Foner, R.B. Frankel, W.M. Reiff, H. Wong, G.J. Long, *J. Chem. Phys.* 68 (1978), 4781.

11.39. F.W. Klaijsen, H. denAdel, Z. Zokoupil, W.J. Huiskamp, *Physica B* 79 (1975), 113.

11.40. W. Duffy, K.P. Barr, *Phys. Rev.* 165 (1968), 647.

11.41. J.W. Hall, W.E. Marsh, R.R. Weller, W.E. Hatfield, *Inorg. Chem.* 20 (1981), 1033.

11.42. W.E. Hatfield, in "Magneto-Structural Correlations in Exchange Coupled Systems," R.D. Willett, D. Gatteschi, O. Kahn, eds., p. 555, NATO ASI. Reidel, Dordrecht, 1985.

11.43. C.P. Landee, in "Organic and Inorganic Low-Dimensional Crystalline Materials," P. Delhaes, M. Drillon, eds., p. 75, NATO ASI. Plenum, New York, 1987.

11.44. S.A. Roberts, D.R. Bloomquist, R.D. Willett, H.W. Dodgen, *J. Am. Chem. Soc.* 103 (1981), 2603.

11.45. L.W. Ter Haar, W.E. Hatfield, *Inorg. Chem.* 24 (1985), 1022.

11.46. Z.G. Soos, D.J. Klein, in "Molecular Association," R. Foster, ed., Vol. 1, p. 1. Academic Press, New York, 1975.

11.47. M.M. Olmstead, W.K. Musker, L.W. Ter Haar, W.E. Hatfield, *J. Am. Chem. Soc.* 104 (1982), 6627.

11.48. R.E. Peierls, "Quantum Theory of Solids." Oxford University Press, London, 1955.

11.49. J.W. Bray, H.R. Hart, Jr., L.V. Interrante, I.S. Jacobs, J.S. Kasper, G.D. Watkins, S.H. Wee, J.C. Bonner, *Phys. Rev. Lett.* 35 (1975), 744.

11.50. I.S. Jacobs, J.W. Bray, H.R. Hart, Jr., L.V. Interrante, J.S. Kasper, G.D. Watkins, D.E. Prober, J.C. Bonner, *Phys. Rev. B* 14 (1976), 3036.

11.51. J.S. Kasper, D.E. Moncton, *Phys. Rev. B* 20 (1979), 2341.

11.52. B. van Bodegom, B.C. Larson, H.A. Mook, *Phys. Rev. B* 24 (1981), 1520.

11.53. A. Gleizes, M. Verdaguer, *J. Am. Chem. Soc.* 103 (1981), 7373.

11.54. M. Verdaguer, M. Julve, A. Michalowicz, O. Kahn, *Inorg. Chem.* 22 (1983), 2624.

11.55. M. Drillon, J.C. Gianduzzo, R. Georges, *Phys. Lett. A* 96 (1983), 413.

11.56. M. Drillon, E. Coronado, R. Georges, J.C. Gianguzzo, J. Curely, *Phys. Rev. B* 40 (1989), 10992.

11.57. M. Drillon, E. Coronado, D. Beltran, R. Georges, *Chem. Phys.* 79 (1983), 449

11.58. J. Seiden, *J. Phys. (Paris) Lett.* 44 (1983), L947.

11.59. R. Georges, J. Curely, J.C. Gianduzzo, Q. Xu, O. Kahn, Y. Pei, *Physica B* 153 (1988), 77.

11.60. M. Verdaguer, A. Gleizes, J.P. Renard, J. Seiden, *Phys. Rev. B* 29 (1984), 5144.

11.61. R. Georges, J. Curely, M. Drillon, *J. Appl. Phys.* 58 (1985), 914.

11.62. J. Curely, R. Georges, M. Drillon, *Phys. Rev. B* 33 (1986), 6243.

11.63. A. Gleizes, M. Verdaguer, *J. Am. Chem. Soc.* 106 (1984), 3727.

11.64. Y. Pei, M. Verdaguer, O. Kahn, J. Sletten, J.P. Renard, *Inorg. Chem.* 26 (1987), 138.

11.65. O. Kahn, Y. Pei, M. Verdaguer, J.P. Renard, J. Sletten, *J. Am. Chem. Soc.* 110 (1988), 782.

11.66. E. Coronado, M. Drillon, A. Fuertes, D. Beltran, A. Mosset, J. Galy, *J. Am. Chem. Soc.* 108 (1986), 900.

11.67. E. Coronado, M. Drillon, P.R. Nugteren, L.J. de Jongh, D. Beltran, R. Georges, *J. Am. Chem. Soc.* 111 (1989), 3874.

11.68. A.I. Pozhidaev, T.N. Polynova, M.A. Porai-Koshits, N.N. Neronova, *Zh. Strukt. Khim.* 14 (1973), 570.

11.69. A. Caneschi, D. Gatteschi, P. Rey, R. Sessoli, *Inorg. Chem.* 27 (1988), 1756.

11.70. A. Caneschi, D. Gatteschi, J.P. Renard, P. Rey, R. Sessoli, *Inorg. Chem.* 28 (1989), 1976.

11.71. A. Caneschi, D. Gatteschi, J. Laugier, P. Rey, R. Sessoli, C. Zanchini, *J. Am. Chem. Soc.* 110 (1988), 2795.

11.72. Y. Pei, O. Kahn, J. Sletten, J.P. Renard, R. Georges, J.C. Gianduzzo, J. Curely, Q. Xu, *Inorg. Chem.* 27 (1988), 47.

11.73. K. Nakatani, J.Y. Carriat, Y. Journaux, O. Kahn, F. Llioret, J.P. Renard, Y. Pei, J. Sletten, M. Verdaguer, *J. Am. Chem. Soc.* 111 (1989), 5739.

11.74. O. Kahn, Y. Pei, K. Nakatani, J. Sletten, in "Organometallic Polymers with Special Properties," R.M. Laine, ed., p. 131, NATO ASI. Kluwer, Dordrecht, 1992.

CHAPTER

12

Magnetic Long-Range Ordering in Molecular Compounds: Design of Molecular-Based Magnets

12.1 Three-Dimensional Magnetic Ordering

The essential purpose of this book is to describe the magnetic properties of molecular entities. In most compounds those entities are well isolated from each other in a crystal lattice. However, at low temperatures, intermolecular interactions cannot be ignored. For any assembly of molecular species with a magnetic ground state, a three-dimensional magnetic ordering is expected to occur at a finite temperature T_c. When the intermolecular contacts are only of the van der Waals type, T_c may be very low, sometimes in the 10^{-2} K range. However, when hydrogen bonds are present to provide some interaction pathways, T_c may reach a few Kelvin. At temperatures above T_c the orientation of each molecular spin is independent of that of its neighbors. The length along which two adjacent molecular spins are correlated, or the correlation length, is zero. As T is lowered and approaches the critical temperature, the correlation length increases. Just above T_c the correlation length becomes very large, but is still finite. At T_c it becomes infinite. If the interaction between the nearest-neighbor molecules is ferromagnetic, then the molecular spins are oriented in a parallel fashion. Conversely, if the interaction is antiferromagnetic, the spins are oriented in a way that two adjacent molecular spins are antiparallel. In a few cases, when the symmetry of the crystal structure is low enough, the molecular spins may not be rigorously antiparallel but canted, which gives rise to weak ferromagnetism. This canting may also occur with predominantly parallel spins, which leads to the canted ferromagnetism.

The transition from a paramagnetic state to an ordered magnetic state is generally a second-order phase transition with a typical λ-like discontinuity in the heat-capacity curve, and characteristic susceptibility and magnetization behaviors. These aspects are treated in detail in numerous books dealing with magnetic materials [12.1–12.4]. As

far as we are concerned, we will focus on the behavior of assemblies of molecules exhibiting three-dimensional ferromagnetic behavior. Propositions or speculations were often put forward to favor ferromagnetic interactions between magnetic molecules, and to design molecular-based ferromagnets, but it was only in 1986 that compounds of this kind were really first characterized.

12.2 Molecular-Based Magnets

In this section we will provide some information on the magnetic behavior of molecular-based magnets. We consider an assembly of magnetic molecules interacting ferromagnetically, and we assume that these molecules are all of the same type. At temperatures much higher than T_c intermolecular interactions do not influence the magnetic properties. If each molecule contains a unique magnetic center, then the magnetic susceptibility χ varies according to the Curie law, provided that there is no first-order orbital momentum. If each molecule contains several magnetic centers, the intramolecular interactions generally dominate. These may be either ferro- for antiferromagnetic, but, in this latter case, they must lead to a nondiamagnetic ground state. Such a situation is achieved, for instance, for an odd number of equivalent local spins, or for a pair of nonequivalent local spins. If we ignore intermolecular interactions for a moment, then the molar magnetic susceptibility χ should follow the Curie law in the temperature range where only the molecular ground state is thermally populated, with the Curie constant depending on the ground state spin. Actually, we assume that significant intermolecular ferromagnetic effects are operative. If this is so, as T decreases and approaches T_c, χ increases faster than anticipated for the Curie law, which is easily detected when plotting χT versus T. χT increases more and more rapidly on cooling. This increases in χT is related to an increase in the correlation length. At temperatures not too close to T_c the intermolecular interactions can be fairly well accounted for by the mean-field approximation (see Sections 2.5 and 6.5). At T_c, χ and χT should in principle diverge, and a spontaneous magnetization should appear. On cooling further, the magnetization increases to a saturation value $M_S = N\beta gS$, where S is the spin associated with the molecular ground state and g is the Zeeman factor.

In practice, when the magnetic field is rigorously zero, no magnetization is usually detected. This is due to two factors, domain formation and demagnetizing field [12.6, 12.7]. A sample of a ferromagnetic material is generally broken into domains. Each domain has a net magnetization in zero field, directed in a given direction. The magnetic moments of the domains are randomly oriented within the sample such that the resulting magnetization is zero. An external magnetic field provokes a displacement of the domain walls and the formation of a new domain structure. The magnetic moments of the domains are no longer randomly oriented but tend to align along the field. A magnetization M is then observed. If the sample was a monodomain, the $M = f(H)$ variation at $T \leq T_c$ would be as shown in Figure 12.1 with an infinite zero-field susceptibility $(dM/dH)_{H=0}$. Since domain formation is taking place, the slope of the M versus H plot for $H = 0$ is very large but not infinite. It must be stressed here that the

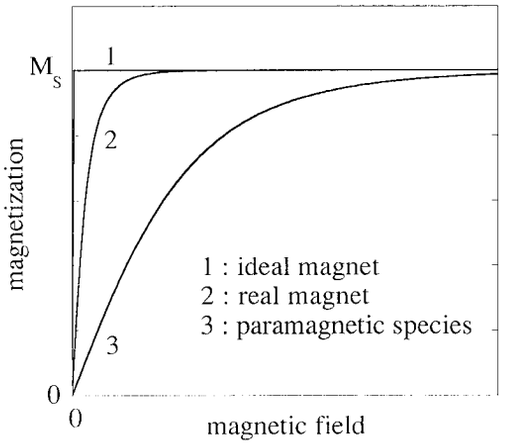


Figure 12.1. Field dependence of the first magnetization for a magnet below T_c.

increase of M versus H is much faster for a ferromagnetic material than for a paramagnetic system. However, if the ferromagnetic and paramagnetic samples contain the same spin carriers, then the saturation magnetizations are identical; this saturation is reached for a much lower field in ferromagnetic samples than in paramagnetic samples (see Figure 12.1). The other factor that limits the zero-field susceptibility, the demagnetizing field, is due to uncompensated moments at the surface of the sample, in the opposite direction to that of the magnetization of this sample. This demagnetizing field can be estimated in principle if we know the geometry of the sample.

Up to this point, in this section, we have ignored the anisotropic effects due to spin-orbit coupling (see Chapter 7). These effects are anticipated to be negligibly small for purely organic materials, the magnetic properties of which are associated with s or p electrons. EPR spectroscopy of organic radicals clearly shows that the orbital contribution is extremely weak. The resonance is always observed for g-values very close to $g_e = 2.0023$. On the other hand, magnetic anisotropy may be important for metal ion-containing compounds. It leads to a preferred spin orientation, called an easy magnetization axis. For $T \leq T_c$, the zero-field susceptibility $(dM/dH)_{H=0}$ is larger and the saturation magnetization M_S is reached at a weaker field when this field is applied along the easy magnetization axis.

There is another aspect that is quite typical of ferromagnetic materials, including the molecular-based ferromagnets, namely, remnance and hysteresis. When the magnetic field is switched on below T_c, then switched off, the field-induced magnetization does not totally disappear, which is in contrast with what happens for paramagnetic systems. In some cases, this remnant magnetization may be equal to the field-induced magnetization. All the information is retained. To suppress the remnant magnetization, it is necessary to apply a coercive field in the opposite direction. Remnant magnetization and coercive field define the hysteresis loop and confer a memory effect to the

material. The mechanism of the hysteresis phenomenon for a ferromagnet is beyond the scope of this book. We will just say that it takes much more time for the magnetic moments of the ferromagnetic domains to reorient randomly when the field is switched off than for those of paramagnetic molecules.

12.3 Orbital Degeneracy and Ferromagnetic Interaction

In Section 8.4 we saw that the antiferromagnetic interaction in an AB pair may be interpreted as resulting from a coupling between the singlet state arising from the ground configuration and one of the singlet states arising from the charge-transfer configuration. This is the orthogonalized magnetic orbital view. In such an approach it is supposed that the two magnetic orbitals a and b, centered on A and B, respectively, are well separated in energy from both the doubly occupied orbitals of lower energy and the vacant orbitals of higher energy. Quite a different situation might be anticipated if the two interacting local states are orbitally degenerate or quasi-degenerate. Let us assume that there are two degenerate orbitals a_1 and a_2 around A, and b_1 and b_2 around B, with one unpaired electron around each center (see Figure 12.2). In the ground configuration the two electrons may have parallel or antiparallel spins. The state of lowest energy arising from the charge transfer configuration A^-B^+ (or A^+B^-) is certainly a triplet state; the electrons occupy different orbitals in order to respect Hund's rule. The coupling between the triplet states arising from ground and excited configurations stabilizes the low-lying triplet state and may favor a ferromagnetic interaction, as schematized in Figure 12.2.

These arguments have been tested theoretically on a simple system, the benzene dimer dianion with parallel rings and D_{6h} symmetry [12.8–12.9]. Each $(C_6H_6)^-$ unit has an unpaired electron occupying the π^* twofold degenerate orbitals. The energies of the 10 singlet and 6 triplet states arising from the various occupations of the π^* orbitals have been calculated using a full configuration interaction. The ground state is found to be a triplet for any separation between the rings. A separation of 3.2 Å

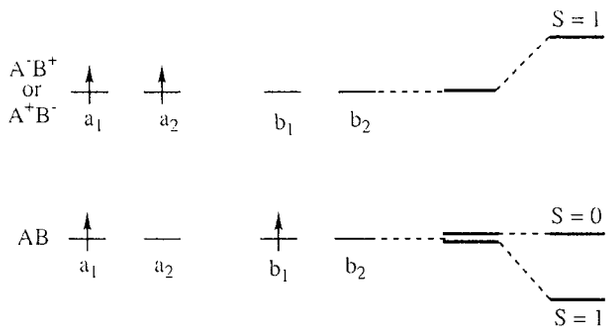


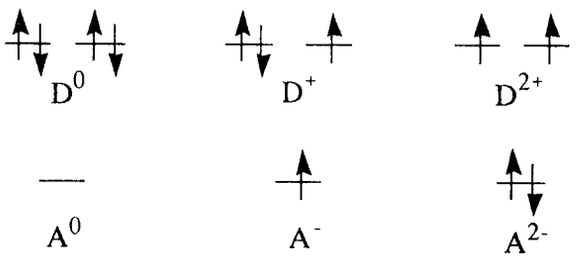
Figure 12.2. A mechanism suggested to stabilize a triplet ground state in a pair of orbitally degenerate units.

corresponds to the equilibrium geometry, and here the energy gap between the ground triplet and the first excited singlet is about 1100 cm^{-1}. This excited singlet is thus completely depopulated, even at ambient temperature. Actually, the first excited state is not the singlet of lowest energy but another triplet.

If, instead of a pair of benzene anions, we consider an infinite stack, the same arguments suggest that the system might behave as a ferromagnetic chain. In principle the requirements for applying these ideas are rather simple; we need a stack of radicals whose symmetry is high enough to give an orbital degeneracy. This happens if the radical has a C_n symmetry axis with $n \geq 3$. A major difficulty, however, is the Jahn–Teller distortion, which tends to lower the molecular symmetry, and stabilize a singlet state. This could explain, among other reasons (see below), why all efforts to synthesize such stacks of orbitally degenerate radicals have so far been unsuccessful.

Much the same idea was put forward by McConnell [12.10] as early as 1967, which was then reformulated in a slightly different context by Breslow [12.11] in 1982. In the perspective of designing organic ferromagnets, McConnell and Breslow considered alternating stacks of donor cations D^+ and acceptor anions A^- where either A or D, but not both, has a symmetry high enough to allow an orbital degeneracy. The two situations to which McConnell explicitly referred are (1) D^0 is a triplet with two unpaired electrons occupying two degenerate orbitals, and A^0 is a closed-shell singlet; (2) D^0 is a closed-shell singlet, and A^0 is a triplet with two unpaired electrons occupying two degenerate orbitals. In both cases D^+ and A^- are doublets and, if a D^+A^- pair is singled out from the stack, the interaction within D^+A^- gives rise to a low-lying singlet and a low-lying triplet. According to McConnell, coupling between the ground configuration D^+A^- and the backward charge-transfer configuration D^0A^0 stabilizes the low-lying triplet, and therefore favors ferromagnetic coupling, as schematized in Figure 12.3.

Breslow assumed that D^0 and A^0 were such efficient donor and acceptor, respectively, that $D^{2+}A^{2-}$ is lower in energy than D^0A^0. In other words, the forward charge transfer is favored with respect to the backward charge transfer. He then considered the two following situations: (1) D^0 is highly symmetrical, and has two orbitally degenerate highest occupied molecular orbitals (HOMO), and A, which has a lower symmetry, has a nondegenerate lowest unoccupied molecular orbital (LUMO). The ground states of D^+ and D^{2+} are then an orbitally degenerate spin doublet and an orbitally nondegenerate spin triplet, respectively. Those of A^- and A^{2-} are an orbitally nondegenerate spin doublet and a closed-shell spin singlet, respectively, as



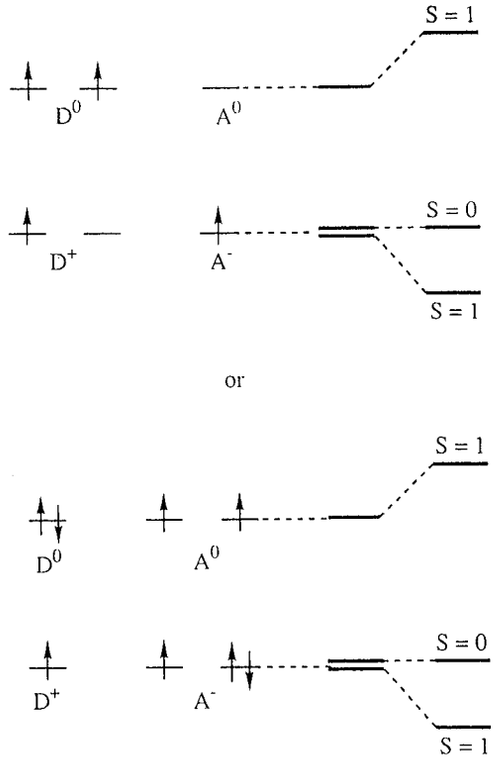
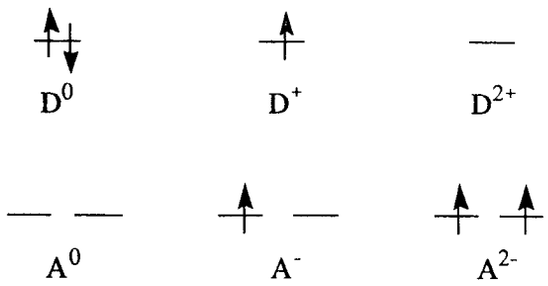


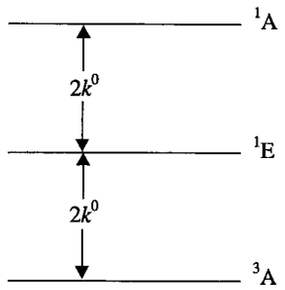
Figure 12.3. The McConnell mechanism supposed to favor a ferromagnetic interaction in a D^+A^- pair singled out from a D^+A^- alternated stack. It must be stressed here that the validity of the mechanism is questionable. Indeed, even if they are higher in energy, the singlet states arising from the backward charge-transfer configuration may couple efficiently with the low-lying singlet state, and stabilize this state.

(2) A^0 is highly symmetric, and has two degenerate LUMOs, and D^0, which has a lower symmetry, has a nondegenerate HOMO. The situation is then as



Due to Hund's rule, in both cases the lowest state arising from $D^{2+}A^{2-}$ is a triplet and coupling between D^+A^- and $D^{2+}A^{2-}$ stabilizes the triplet of lower energy. At the scale of an infinite stack this would lead to one-dimensional ferromagnetic behavior.

To test this strategy, Breslow et al. synthesized a large number of new aromatic molecules of high symmetry. However, they never observed any ferromagnetic interaction [12.12–12.15]. Actually, all synthetic efforts to test this McConnell–Breslow mechanism up to now have failed, a situation that pushed the theoreticians to investigate the status of this mechanism in a more thorough manner. Indeed, the arguments above are far oversimple. To show this, let us examine the situations proposed by Breslow (the discussion would be quite similar if we examined the McConnell schemes). It is true that the lowest state arising from $D^{2+}A^{2-}$ is a spin triplet, but, in addition to this triplet state, there are three singlet states, two of which are possibly degenerate. This situation is reminiscent of what happens in the dioxygen molecule. The states arising from the $(\pi_g)^2$ ground configuration for O_2 are $^3\Sigma_g^-$, which is the lowest, and $^1\Delta_g$ and $^1\Sigma_g^+$. The energy separations between triplet and singlet states arising from the $D^{2+}A^{2-}$ configuration when the symmetry group contains a C_3 or C_6 rotation axis are as



where k^0 is the one-site exchange integral $< a_1(i)a_2(j)|1/r_{ij}|a_1(j)a_2(i) >$, and a_1 and a_2 are the two degenerate orbitals for D^0 or A^0. The coupling between the singlet states arising from $D^{2+}A^{2-}$, and D^+A^- stabilizes the low-lying singlet state, in the same way as the coupling between the two triplet states stabilizes the low-lying triplet state. It has been demonstrated that if k^0 is smaller than the energy gap U between $D^{2+}A^{2-}$ and D^+A^-, then the low-lying singlet state is more stabilized than the low-lying triplet, and the overall interaction is antiferromagnetic. The condition $k^0 < U$ seems to occur in most real situation. If this is so, the McConnell–Breslow mechanism does not operate, as emphasized in Figure 12.4. If the symmetry of the D^+A^- pair is described by groups containing fourfold rotation or rotation-reflexion operations, then the situation is a little more complicated. The states arising from $D^{2+}A^{2-}$ are all orbitally nondegenerate, and may be labeled as $^3A + {}^1B + {}^1B + {}^1A$. In this situation it has also been shown that the validity of the McConnell–Breslow mechanism is rather unlikely [12.16].

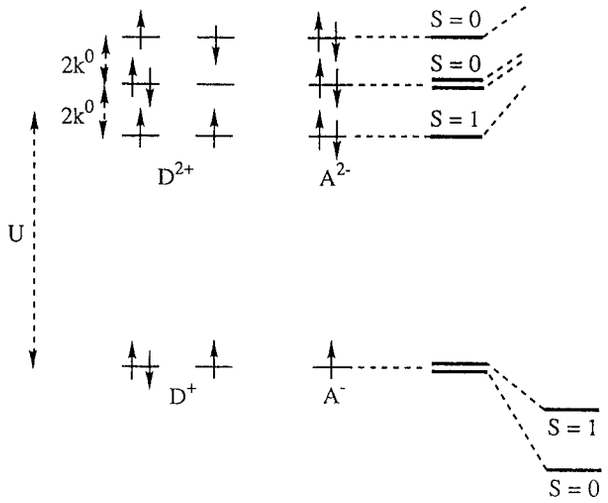
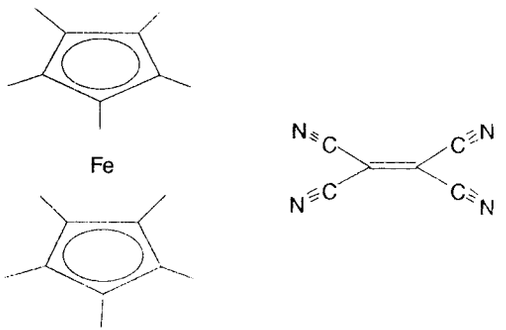


Figure 12.4. Criticism of the McConnell–Breslow mechanism in the case where the donor unit has two degenerate highest occupied molecular orbitals, and that the forward charge-transfer configuration $D^{2+}A^{2-}$ is lower in energy than the backward one. In the original formulation of this mechanism the four singlet states arising from $D^{2+}A^{2-}$ are neglected, so that the low-lying triplet is lower than the low-lying singlet.

12.4 Ferromagnetic Transitions in Decamethylferrocenium Tetracyanoethenide and Other Decamethylmetallocenium Charge-Transfer Salts

In 1987 Miller et al. characterized a ferromagnetic transition in the organometallic donor–acceptor salt decamethylferrocenium tetracyanoethenide [12.17–12.19]. This compound is obtained by a simple reaction in acetonitrile between decamethylferrocene $Fe(Me_5C_5)_2$ and tetracyanoethylene (TCNE), which play the role of electron donor and acceptor, respectively:



The crystal structure of an acetonitrile solvate $[Fe(Me_5C_5)_2][TCNE]\cdot MeCN$ consists of chains of alternating $[Fe(Me_5C_5)_2]^+$ and $[TCNE]^-$ units with a Fe—Fe separation along the chain of 10.42 Å. $[TCNE]^-$ lies between the essentially parallel Me_5C_5 rings, while the adjacent TCNE and Me_5C_5 planes are not rigorously parallel, making a dihedral angle of 2.8°. The shortest interchain separation is equal to 8.23 Å and is associated with pairs of chains in which the cation of one of the chains has the anion of the other chain as nearest neighbor, as schematized in Figure 12.5. It should be noted that this MeCN solvate is not the compound whose physical properties were investigated. This solvate easily loses its MeCN molecule with a transformation of the lattice from monoclinic to orthorhombic. The structure of the orthorhombic phase is strongly disordered and it has not been possible to refine it properly. All magnetic studies, however, were performed with this desolvated orthorhombic phase.

Let us now present the main physical properties of $[Fe(Me_5C_5)_2][TCNE]$. The χT versus T plot increases as T is lowered, and follows the expected behavior of a Heisenberg ferromagnetic chain compound. The intrachain interaction parameter is found to be $J = 26$ cm^{-1}. The temperature dependence of the magnetization of a polycrystalline sample within the magnetic field of the earth shows a break at $T_c = 4.8$ K, then saturates at a value close to 2×10^3 cm^3 G mol^{-1} (see Figure 12.6) when cooling below T_c. The magnetization of a single crystal at 4.2 K saturates at a value of 1.6×10^4 cm^3 G mol^{-1} for a magnetic field parallel to the chain axis of a few tens of gauss. This value M_S of the saturation magnetization is interpreted as resulting from the

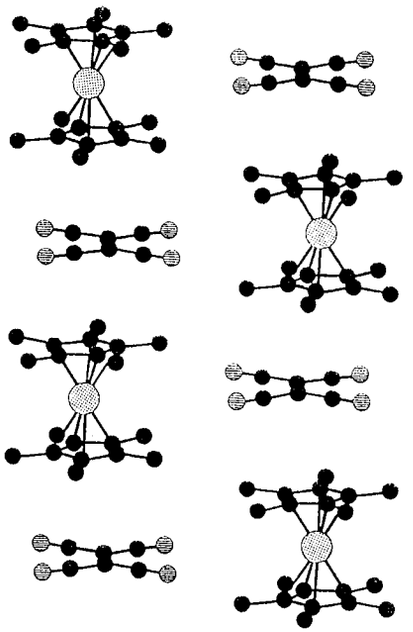


Figure 12.5. Crystal structure of $[Fe(Me_5C_5)_2][TCNE]$. The figure shows only two stacks with the shortest interstack separation (from ref. [12.17]).

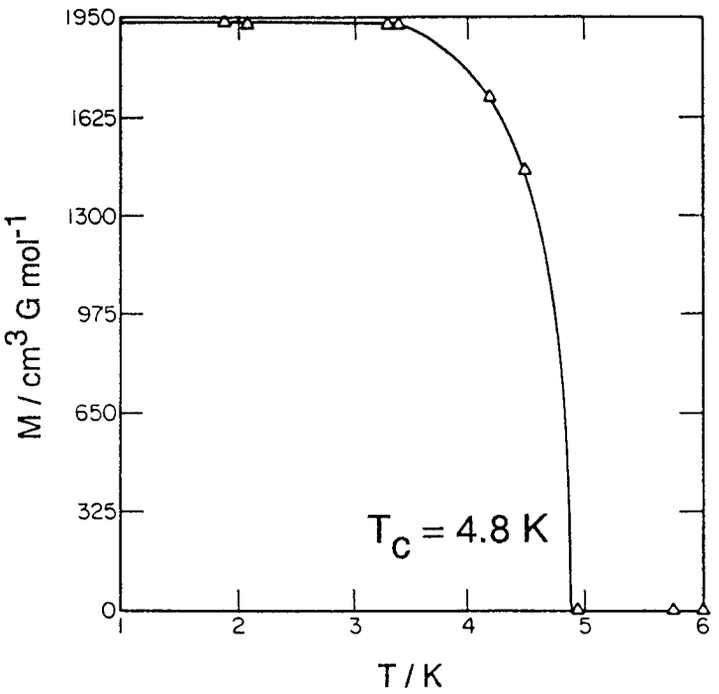


Figure 12.6. Temperature dependence of the magnetization for $[Fe(Me_5C_5)_2][TCNEl$ within the magnetic field of the earth (from ref. [12.17]).

ferromagnetic alignment of the spins $S = 1/2$ of the donor D^+ and acceptor A^- units with parallel g-factors $g_D = 4$ and $g_A = 2$:

$$M_S = N(g_D + g_A)\beta/2 \qquad (12.4.1)$$

The magnetization versus field loop at 2 K shows a pronounced hysteresis effect with a coercive field of the order of 10^3 G (see Figure 12.7).

All these experimental data are consistent with the following description of $[Fe(Me_5C_5)_2]$ [TCNE]. The 1/2 spins associated with both donor and acceptor units are strongly coupled along the chains in a ferromagnetic fashion ($J = 26$ cm^{-1}); the ferromagnetic chains are weakly coupled on the scale of the crystal lattice, again in a ferromagnetic fashion. It results that the compound exhibits bulk ferromagnetic properties with a spontaneous magnetization below $T_c = 4.8$ K.

The discovery of a ferromagnetic transition in $[Fe(Me_5C_5)_2$[TCNE] provoked a great impulse among the chemists working in the field of molecular magnetism, and soon after this discovery at least three novel decamethylmetallocenium charge-transfer salts ordering ferromagnetically were characterized. These are two decamethylmanganocenium derivatives, namely $[Mn(Me_5C_5)_2]$ [TCNQ] (TCNQ = tetracyanoquinodimethane) [12.20], and $[Mn(Me_5C_5)_2]$ [TCNE] [12.21] with T_c = 6.2 and 8.8 K,

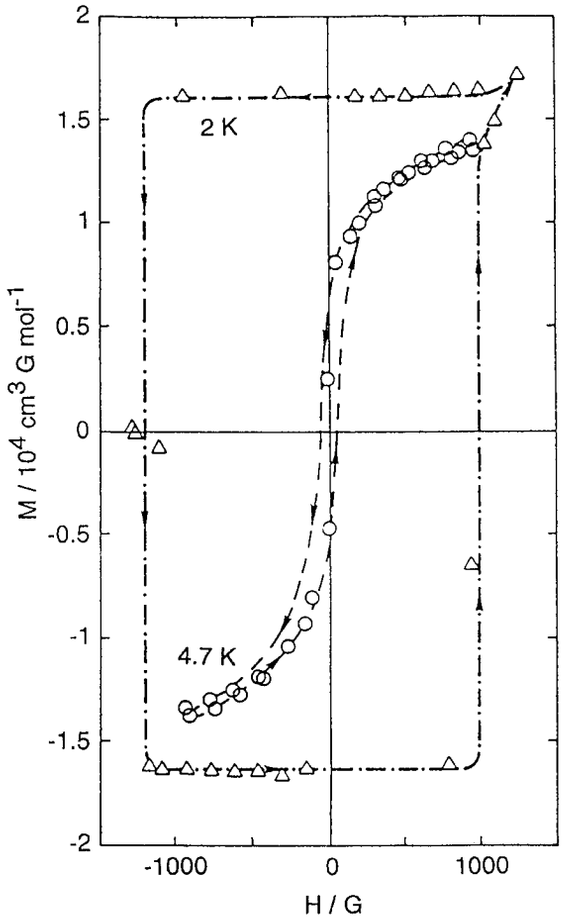


Figure 12.7. Field-dependent magnetization at 2 K and 4.7 K for a single-crystal of $[Fe(Me_5C_5)_2][TCNE]$ oriented with the stacking axis parallel to the applied magnetic field. The hysteresis loop shows a coercive field of 1000 G at 2 K, and 30 G at 4.7 K (from ref. [12.19]).

respectively, and the chromocenium derivative $[Cr(Me_5C_5)_2]$ [TCNE] with $T_c = 3.1$ K [12.22]. In addition, other decamethylmetallocenium derivatives were found to exhibit metamagnetic behavior, i.e., stabilization of a ferromagnetic-like state when applying a magnetic field. This is the case for $[Fe(Me_5C_5)_2][TCNQ]$[12.23] and $[Mn(Me_5C_5)_2][M(tfd)_2]$, where M is Ni, Pd, or Pt, and tfd is bis(trifluoromethyl)ethylenedithiolato [12.24].

The problem now is how to understand the mechanism of intra- and intermolecular ferromagnetic interactions. Miller et al. attributed the ferromagnetic behavior of $[Fe(Me_5C_5)_2]$ [TCNE] to the McConnell–Breslow mechanism. The $3d$ metal orbitals of iron in the crystal field due to the Me_5C_5 rings are split into $a_1(d_{z^2})$, $e_2(d_{x^2-y^2}$ and $d_{xy})$, and $e_1(d_{yz}$ and $d_{zx})$ with the relative energies as

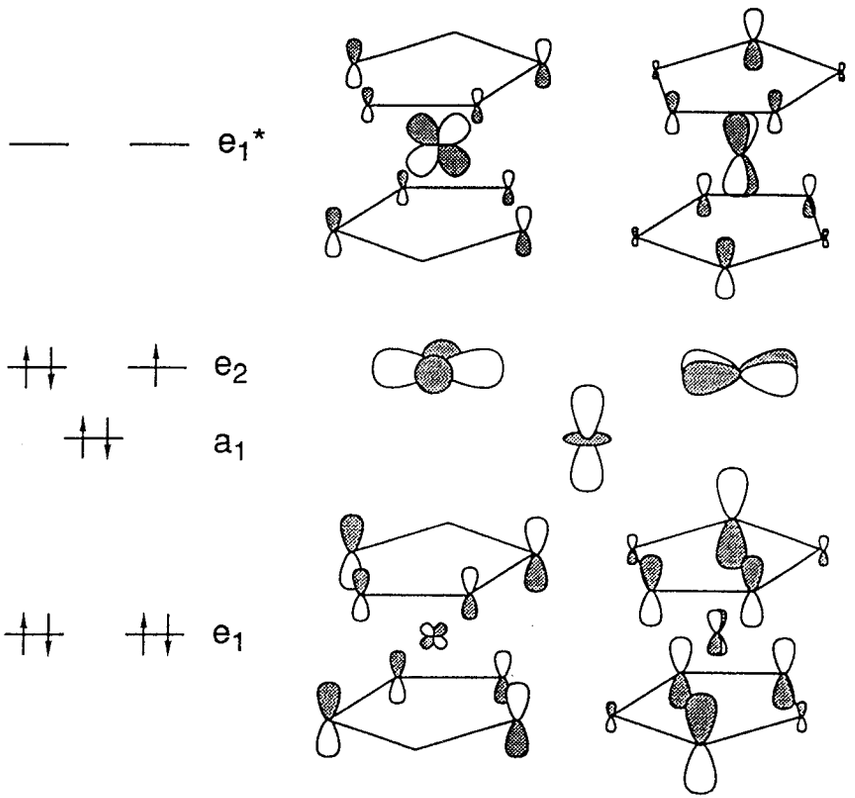
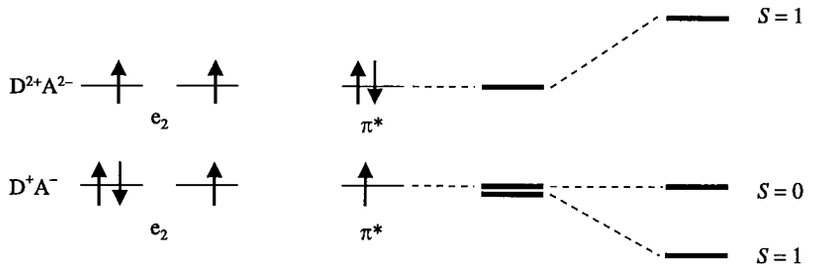


Figure 12.8. Energy scheme and spatial distribution for the molecular orbitals of g symmetry for the decamethylferrocenium cation (from ref. [12.28])

$\varepsilon(a_1) \leq \varepsilon(e_2) << \varepsilon(e_1)$; a_1 is very close to, or possibly accidentally degenerate with e_2. The z axis is along the fivefold symmetry axis, and the x and y axes are perpendicular to the ring planes (see also Figure 12.8). For the sake of simplicity the subscripts g of the irreducible representations of D_{5d} have been omitted. For neutral $Fe(Me_5C_5)_2$ the ground configuration is $(a_1)^2(e_2)^4$, and for the monocation $[Fe(Me_5C_5)_2]^+$ the ground configuration is $(a_1)^2(e_2)^3$ with a local spin doublet state. For neutral TCNE the first antibonding orbital π^* is empty, and for the monoanion $[TCNE]^-$ it is singly occupied such that the local ground state is also a spin doublet. The interaction between the unpaired electrons of D^+ and A^- leads to singlet and triplet states. Furthermore, it was assumed that $D^{2+}A^{2-}$ was low enough in energy to interact with D^+A^-. The configuration of lower energy is $(a_1)^2(e_2)^2$ for D^{2+}, and $(\pi^*)^2$ for A^{2-}, which leads to a triplet state for a $D^{2+}A^{2-}$ pair singled out from the stack. The interaction between the two triplets, arising from D^+A^- and $D^{2+}A^{2-}$, stabilizes the former, which corresponds to a ferromagnetic interaction:



This mechanism was assumed to work both for intra- and interchain interactions.

At first sight, this interpretation seems to be satisfying. However, some objections may be raised, which actually makes its validity quite doubtful. First, we pointed out in Section 12.3 that this McConnell–Breslow mechanism may not be of the general validity that was assumed during two decades. Furthermore, nothing proves that $D^{2+}A^{2-}$ is lower in energy than D^0A^0. Finally, and this may well be the most serious objection, the stabilization of the low-lying triplet state due to the $D^+A^-/D^{2+}A^{2-}$ coupling is roughly proportional to the square of a transfer integral of the form $<e_2(i)|\mathbf{h}(i)|\pi^*(i)>$ (the e_2 orbitals may have both x^2-y^2 and xy symmetry). If we neglect the small dihedral angle between Me_5C_5 and TCNE planes, then the e_2 orbitals of $Fe(Me_5C_5)_2$ and π^* orbital of TCNE are strictly orthogonal, and the two integrals $<e_2(i)|\mathbf{h}(i)|\pi^*(i)>$ are zero [12.25]. Even by taking into account this angle, the values of these integrals remain extremely small, and does not explain the stabilization of the low-lying triplet of ca. 26 cm^{-1}. An alternative and more simple interpretation would consist of saying that in $[Fe(Me_5C_5)_2][TCNE]$ the magnetic orbitals e_2 for D^+ and π^* for A^- are orthogonal (if we neglect the dihedral angle between Me_5C_5 and TCNE planes) or quasi-orthogonal, which favors a ferromagnetic interaction (see Section 8.6). But, here again, a severe objection may be raised. The stabilization of the triplet state with respect to the singlet state in the case of strict orthogonality of the magnetic orbitals is governed by the two-electron exchange integrals $<e_2(i)\pi^*(j)|1/r_{ij}|e_2(j)\pi^*(i)>$. Such integrals are nonvanishing only when there are some zones of space where the overlap densities $e_2(i)\pi^*(i)$ have significant values (see, for instance, Figure 8.24). Apparently this is not the case. The two e_2 orbitals are very weakly delocalized toward the Me_5C_5 rings, and therefore also toward the TCNE units. In turn, the π^* orbital vanishes in the vicinity of the iron atom.

At this stage, it is quite informative to examine what the McConnell–Breslow mechanism predicts for the decamethylchromocenium charge-transfer salts. The a_1 and e_2 orbitals are quasi-degenerate such that the ground configuration for $D^+ = [Cr(C_5Me_5)_2]^+$ is $(a_1e_2)^3$ with a 3/2 spin. The interaction between D^+ and A^- leads to $S = 1$ and 2 states. Let us now consider $D^{2+}A^{2-}$. The configuration of lower energy is $(a_1e_2)^2$ for D^{2+}, and $(\pi^*)^2$ for A^{2-}, which leads to an $S = 1$ state for a $D^{2+}A^{2-}$ pair. The coupling between the two $S = 1$ states, arising from D^+A^- and $A^{2+}A^{2-}$, stabilizes the former. This corresponds to an antiferromagnetic interaction between D^+ and A^-, and since there is no compensation of the magnetic moments, to ferrimagnetic D^+A^- stacks. We have seen that $[Cr(Me_5C_5)_2]$ [TCNE] presents both intra- and interstack ferromag-

netic interactions, and orders ferromagnetically at $T_c = 3.1$ K. One objection might be that in that case the charge-transfer configuration of interest is D^0A^0 instead of $D^{2+}A^{2-}$. Unfortunately, the configurations of lowest energy for D^0 and A^0 are $(a_1e_2)^4$ and $(\pi^*)^0$, respectively, and the lowest state arising from D^0A^0 is again a triplet. Therefore, the $S = 1$ low-lying state may also be stabilized by the D^+A^-/D^0A^0 coupling.

Two alternative interpretations of the ferromagnetism in $[Fe(Me_5C_5)_2]$ [TCNE] have been put forward. Before examining these interpretations, it is worth taking a more careful look at the orbital energy scheme of the ferrocenium cation depicted in Figure 12.8. Only orbitals of g symmetry are represented; these of u symmetry cannot combine with the d metal orbitals. The a_1 and e_2 orbitals have only negligible contributions from the C_5Me_5 rings. The e_1 orbitals just above are mainly metallic orbitals, but have rather large antibonding contributions from the rings, and hence the e_1* notation in Figure 12.8. The orbitals just below a_1 have also the e_1 symmetry with a dominant ring character and rather large bonding contributions from the metal.

One of the possible mechanisms for the ferromagnetic interaction involves the coupling between D^+A^- and an excited configuration of D^0A^0 corresponding to the electron transfer from π^* of A^- toward e_1* of D^+. In Section 9.4 we saw that such a transfer from a singly occupied orbital of one unit toward an empty orbital of the other unit favors the ferromagnetic interaction. The matrix elements $\langle e_1^*(i)|\mathbf{h}(i)|\pi^*(i)\rangle$ may be rather large due to delocalization of the e_1* orbitals toward the rings, and the subsequent ring–TCNE overlaps [12.26].

Another attractive mechanism is based on spin polarization effects (see Section 12.6). Since the e_2 orbitals are almost pure metal orbitals, the unpaired electron of $[Fe(Me_5C_5)_2]^+$ in a molecular orbital description (at the SCF level) is essentially localized on the iron atom. In fact, if a CI treatment is performed, significant negative spin densities are found on the rings, in addition to the dominant positive spin density on the metal, a situation which is substantiated by NMR experiments. The excited configuration of $[Fe(C_5Me_5)_2]^+$ that plays the key role in the appearance of negative spin density on the rings is that where an electron is promoted from e_1 to e_1* orbitals. The ferromagnetic interaction along the stacks of $[Fe(C_5Me_5)_2]$[TCNE] is then attributed to the coupling between the negative spin densities on the pentamethylcyclopentadienyl rings and the positive spin densities on the nearest neighbor $TCNE^-$ units [12.27, 12.28].

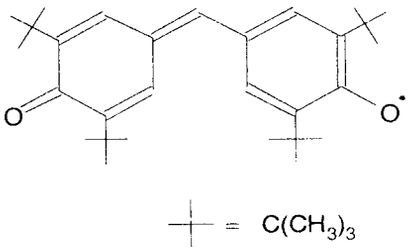
Both mechanisms we have just described are plausible. Furthermore, and this is an important point, they lead to predictions which are in total agreement with the experimental findings for the other decamethylmetallocenium charge-transfer salts [12.29].

12.5 Intermolecular Ferromagnetic Interactions and Ferromagnetic Transition in Organic Radicals

The metallocenium charge-transfer salts we described in the previous section cannot be considered as purely organic ferromagnets. Whatever the exact mechanism of the

interaction between the D^+ and A^- units, the metal ions with their d orbitals play an essential role. In contrast, this section will be devoted to π radical compounds with unpaired electrons arising from atomic $2p$ orbitals in which intermolecular ferromagnetic interactions are observed.

One of the most interesting although not yet fully elucidated cases of intermolecular ferromagnetic interaction is provided by the galvinoxyl radical. The molecular structure is as



with an unpaired electron shared between the two symmetrically related halves of the molecule. At room temperature, in the solid state, galvinoxyl forms one-dimensional stacking columns along the c direction of the monoclinic lattice as shown in Figure 12.9 [12.30]. The molecular units are nearly planar. Magnetic measurements reveal rather large intermolecular ferromagnetic interactions in the 300–85 K temperature range [12.31, 12.32]. The magnetic susceptibility increases much faster than expected for isolated radicals as the temperature is lowered. The data have been fitted by the

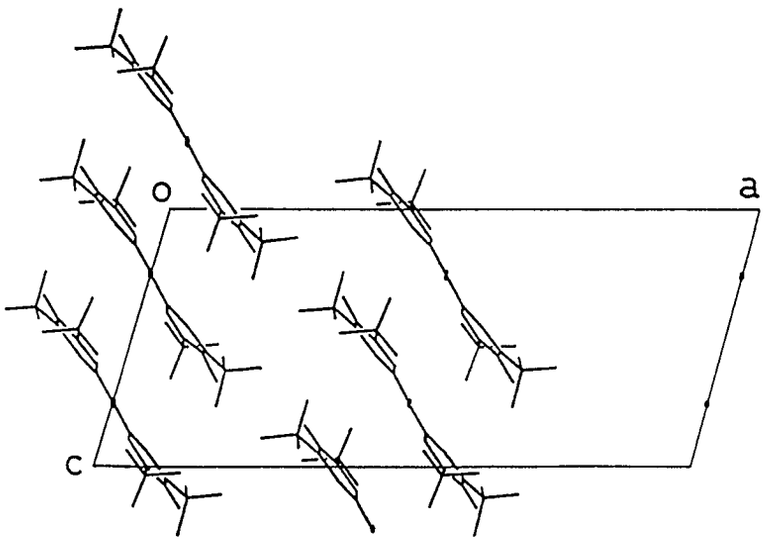


Figure 12.9. Crystal structure of galvinoxyl at room temperature (from ref. [12.30]).

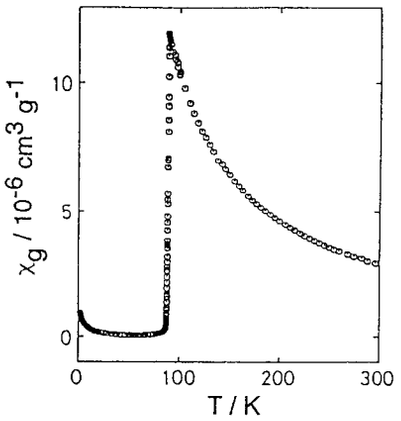
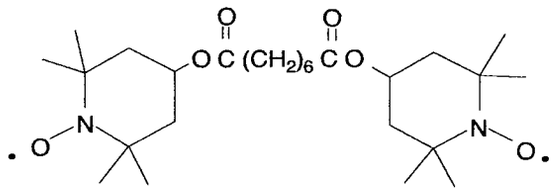


Figure 12.10. Temperature dependence of the massic magnetic susceptibility χ_g for galvinoxyl (from ref. [12.31]).

Curie–Weiss law (see Section 2.6) using a positive Weiss constant of 11 K, which corresponds to an interaction parameter J of the order of 10 cm^{-1}, assuming that galvinoxyl behaves magnetically as a chain compound. At 85 K quite an abrupt phase transition occurs, and most of the paramagnetism vanishes. These unusual susceptibility data are shown in Figure 12.10. Heat capacity measurements indicate that this phase transition is of the first order with a large entropy change $\Delta S = 18.87$ J K^{-1} mol^{-1} [12.33]. The crystal structure below 85 K has not yet been determined. We can anticipate, however, that the phase transition is accompanied by a dimerization of the galvinoxyl units, affording antiferromagnetically coupled pairs. This interpretation is substantiated by the fact that at 77 K the EPR spectrum shows a fine structure characteristic of a triplet state [12.34, 12.35]. The intermolecular interaction between adjacent units is therefore ferromagnetic above 85 K and antiferromagnetic below 85 K, which suggests a very subtle mechanism. Perhaps, a small change in the separation between the molecular units along the columns may be sufficient to modify the sign of the interaction parameter.

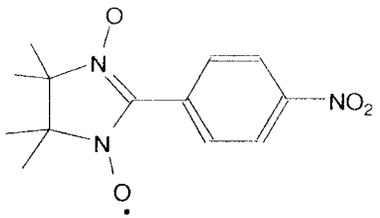
The phase transition at 85 K can be suppressed by mixing some hydrogalvinoxyl into galvinoxyl. Hydrogalvinoxyl is the closed-shell precursor of galvinoxyl [12.36, 12.37]. The crystal structures of both compounds are isomorphous. For a 6/1 mixed crystal of galvinoxyl and hydrogalvinoxyl the magnetic susceptibility follows the Curie–Weiss law with a positive Weiss constant down to 2 K; the intermolecular ferromagnetic interaction is retained. However, there is no three-dimensional ferromagnetic ordering, perhaps because the interchain interactions are negligibly weak.

Some other cases of intermolecular ferromagnetic interaction between organic radicals have been reported. The interaction parameters, however, are very small, at the most of the order of 1 cm^{-1}. Among these cases, we can mention the tanol suberate whose structure is as



The tanol radicals in the crystallographic *ac* plane are ferromagnetically coupled while antiferromagnetic interaction operates along the *b* axis. Below 0.39 K tanol suberate orders antiferromagnetically in zero field. When a field of 60 G is applied, the ground state becomes ferromagnetic-like. Tanol suberate may be described as an organic metamagnet [12.38, 12.39].

In 1991 Kinoshita et al. reported the first purely organic molecular compound (containing only C, H, N, and O elements) exhibiting a ferromagnetic transition. This compound is the *p*-nitrophenyl nitronyl nitroxide radical:



Four crystallographic phases of this compound have been characterized [12.40–12.44]. Their crystal structures consist of two-dimensional arrangements of molecular units with rather short contacts between the nitro group of a unit and the nitroxide group of adjacent units. All phases display a Curie–Weiss behavior down to 2 K with positive Weiss constants of the order of one or two K. The zero-field magnetic susceptibility below 3 K for the γ phase, shown in Figure 12.11, presents a break at $T_c = 0.60$ K, which indicates a ferromagnetic transition. The heat capacity curve also shown in Figure 12.11 displays a λ-shaped anomaly at T_c, characteristic of a three-dimensional phase transition [12.45]. In 1993 another purely organic ferromagnet was reported, with $T_c = 1.48$ K [10.111].

12.6 Spin Polarization and Intermolecular Ferromagnetic Interactions

For all the examples reported in the preceding section the mechanism of the intermolecular ferromagnetic interactions is far from being perfectly understood even if some suggestions have been put forward [12.46].

At present, the most attractive strategy for obtaining such ferromagnetic interactions was proposed by McConnell as early as 1963 [12.47]. This strategy is based on the concept of spin polarization. We already mentioned this concept in Chapter 8 and Section 12.4, but it seems important to give more information.

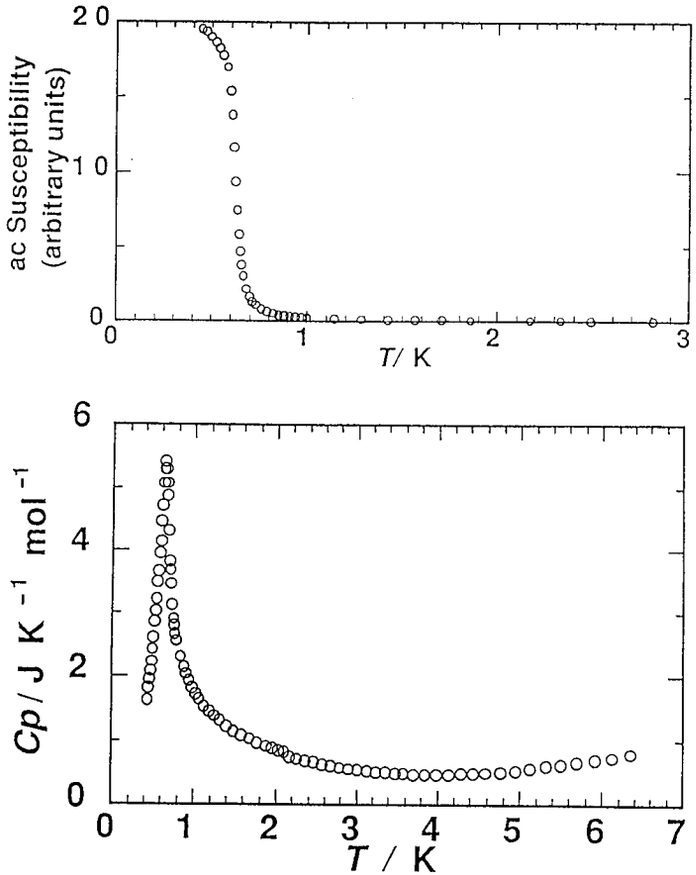


Figure 12.11. Temperature dependence of the zero-field magnetic susceptibility (top) and of the heat capacity (bottom) for the γ phase of *p*-nitrophenyl nitronyl nitroxide (from ref. [12.45]).

Let us consider a radical with one unpaired electron. In molecular orbital theory, at the SCF level, this electron is described by the highest occupied molecular orbital, also called the singly occupied molecular orbital (SOMO). This orbital may be delocalized on the whole molecule or on a part of the molecule. If so, the spin density is also delocalized, but is positive (or zero) in any point of space. At this stage, it is probably worth specifying that the spin density is expressed in $N\beta$ Å^{-3}. The integration of the spin density over the whole space gives the saturation magnetization $N\beta gS$, which is equal to 1 $N\beta$ for an organic monoradical. The molecular space is usually divided into atomic regions. The integration of the spin density over those regions gives the atomic spin populations. The sum of the atomic spin populations within a molecule is again equal to $N\beta gS$. This SCF view in which the spin density is positive or zero everywhere is far too oversimplified. It is experimentally well known, from EPR spectroscopy or polarized neutron diffraction (PND) studies for instance, that the spin population on

certain atoms may be negative, i.e., in opposition to the total spin of the unpaired electron. From a theoretical point of view, there are two ways of calculating negative spin densities, namely the spin-unrestricted Hartree–Fock (UHF) approach, and the spin-restricted Hartree–Fock approach followed by configuration interaction (RHF + CI). It is not the goal of this book to discuss thoroughly these approaches. However, it might be useful to give some additional information.

The UHF approach consists of assuming that for the *i*th doubly occupied orbital level of an open-shell molecule the φ_i^{α} orbital occupied by an electron with α spin has neither the same spatial distribution nor the same energy as the φ_i^{β} orbital occupied by an electron with β spin. The aim of such an assumption is to account for the fact that the α unpaired electron in the SOMO does not interact with the α electron of φ_i^{α} in exactly the same way as with the β electron of φ_i^{β}. Thus, a monodeterminental wavefunction built from the occupied orbitals may give positive as well as negative spin populations. The calculation does not require any coupling with excited states, which is certainly an advantage compared to the RHF + CI approach. This UHF description, however, has a serious defect; the wavefunctions are no longer exact eigenfunctions of the spin operator $\mathbf{S}^2$. In other words, for a monoradical the description of the spin doublet ground state is contaminated by some contributions of higher spin multiplicity, spin quartet, sextet. . . . Some techniques exist that annihilate this contamination, but they are rather difficult to implement, and do not provide an unique solution. A consequence of this contamination is that the spin-unrestricted approach systematically overestimates the spin polarization. The negative spin populations which are calculated can be as much as twice as large as the values deduced from EPR or PND experiments.

In the RHF method each doubly occupied orbital has a unique energy. As we mentioned, the unpaired electron is entirely described by the highest occupied molecular orbital, and the spin density is positive or zero everywhere. A more realistic view of the spin distribution is obtained through the interaction of the SCF ground state with excited states of the same symmetry. This is called the configuration interaction. The calculation may be tedious, but in general leads to atomic spin populations in fairly good agreement with the values deduced from the experiment.

A classical example of spin polarization is that of the allyl radical CH_2CHCH_2. The three π molecular orbitals φ_i $i = 1$ to 3, built from the three $2p_z$ atomic orbitals noted ϕ_A, ϕ_B, and ϕ_C (A and C note the terminal carbon atoms, and B the central one) are in the Hückel approximation [12.48]:

$$\varphi_1 = (1/2)(\phi_A + \sqrt{2}\phi_B + \phi_C)$$

$$\varphi_2 = (\sqrt{2}/2)(\phi_A - \phi_C)$$

$$\varphi_3 = (1/2)(\phi_A - \sqrt{2}\phi_B + \phi_C) \qquad (12.6.1)$$

These molecular orbitals are shown in Figure 12.12; φ_1 and φ_3 are symmetrical, and φ_2 is antisymmetrical with regard to the mirror-plane perpendicular to the molecular plane; in other respects, φ_1 is bonding, φ_2 is nonbonding, and φ_3 is antibonding. At the SCF level, the ground configuration is $\varphi_1^2\,\varphi_2^1$. The unpaired electron is described

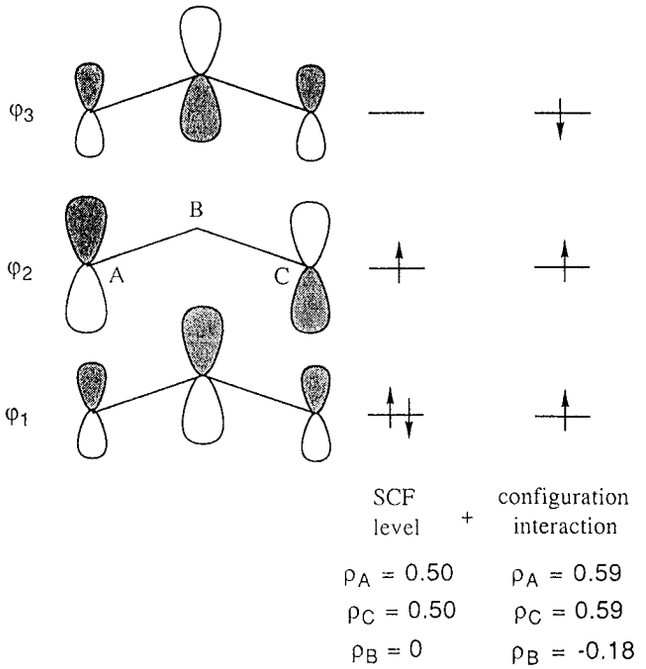


Figure 12.12. Schematic representation of the π molecular orbitals, and atomic spin populations in the allyl radical CH_2CHCH_2.

by φ_2, and the spin density on the central carbon atom is zero. This description neglects the fact that the interaction between the α electron in φ_2 and the α electron in φ_1 is slightly weaker than the interaction between the α electron in φ_2 and the β electron in φ_1. A more accurate description of the spin distribution in the ground state is obtained by coupling the state arising from the ground configuration and the state of the same symmetry arising from the monoexcited configuration $\varphi_1^1\varphi_2^1\varphi_3^1$. If we do this, a negative spin density appears on the central carbon atom, and in turn the positive spin densities on the terminal carbon atoms are enhanced. Experiments and accurate calculations give the atomic spin populations $\rho_A = \rho_C = 0.59$ and $\rho_B = -0.18$ [12.49].

It is worth making a short digression here. The atomic spin populations may be also deduced from the HDVV formalism, which is strictly equivalent to the Heitler–London approach. The spin Hamiltonian is then

$$\mathbf{H} = -J(\mathbf{S}_A \cdot \mathbf{S}_B + \mathbf{S}_B \cdot \mathbf{S}_C) \tag{12.6.2}$$

where $\mathbf{S}_A$, $\mathbf{S}_B$, and $\mathbf{S}_C$ are the local spin operators for the carbon atoms, each carrying an unpaired electron. The A–B and B–C bonds correspond to a negative J value; the π electrons tend to align antiparallel. The interaction between the three π electrons gives rise to two doublets and a quartet (see Section 10.1). The lowest doublet arises from the antiferromagnetic coupling between the triplet pair state for

AC and the local doublet for B. The component $M_S = 1/2$ of the ground state wavefunction is obtained by using the Clebsh–Gordon coefficients (see Annex 6) as

$$\Psi = (\sqrt{2/3})\,|\phi_A\bar{\phi}_B\phi_C| - (\sqrt{1/6})(|\bar{\phi}_A\phi_B\phi_C| + |\phi_A\phi_B\bar{\phi}_C|) \tag{12.6.3}$$

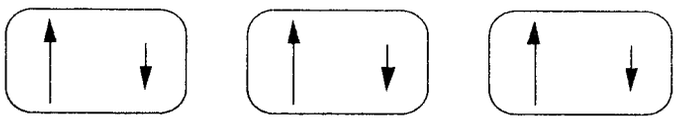
so that the atomic spin populations are found as

$$\rho_A = \rho_C = 2/3$$

$$\rho_B = -1/3 \tag{12.6.4}$$

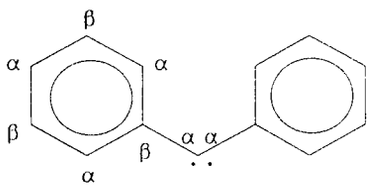
In contrast with the molecular orbital approach at the SCF level, the Heitler–London approach overestimates the negative spin density on the central carbon atom.

After this digression let us return to the McConnell strategy. This requires molecular units showing noncompensating regions of positive and negative spin densities, a situation that may arise from the spin polarization effect. The second step consists of assembling these units within the crystal lattice in such a way that the positive spin density of a unit interacts in a up-down fashion with the negative spin density of the adjacent unit. Thus, all the molecular spins are aligned along the same direction as



which corresponds to intermolecular ferromagnetic interactions.

The efficiency of this mechanism was nicely demonstrated by Iwamura et al. as follows [12.50]: The diphenylcarbene has a triplet ground state (see Section 12.7). Furthermore, spin polarization results in an alternation of the spin densities (see Section 12.7) as



Iwamura incorporated two diphenylcarbene units in the [2,2]paracyclophane skeleton. Pseudo*ortho*, pseudo*meta*, and pseudo*para* dicarbenes were obtained by photolysis of the corresponding diazo compounds (see Figure 12.13). By analysis of the temperature dependence of the EPR signal intensities it was found that the pseudo*ortho* and pseudo*para* derivatives have quintet ground states while the pseudo*meta* derivative has a singlet ground state. For both the pseudo*ortho* and pseudo*para* compounds the

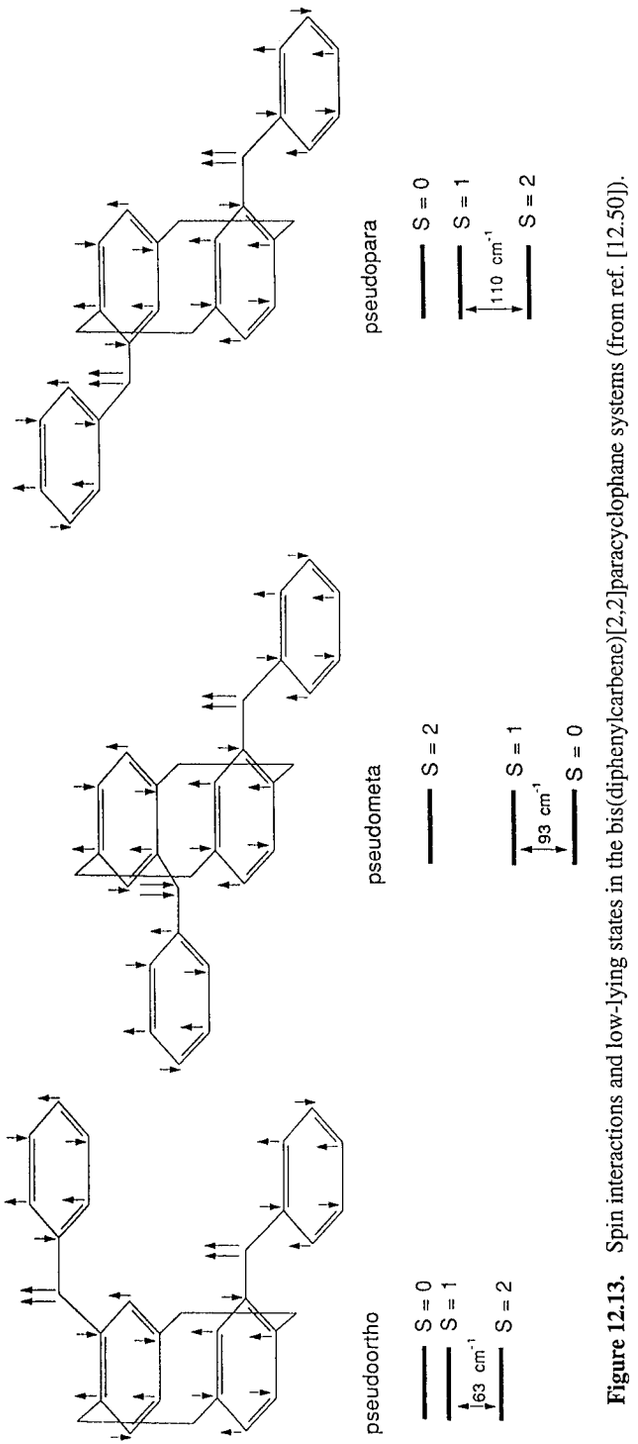
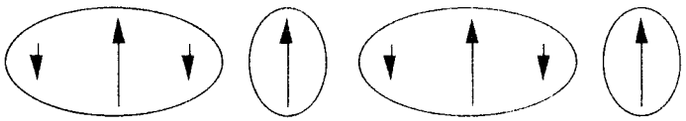


Figure 12.13. Spin interactions and low-lying states in the bis(diphenylcarbene)[2,2]paracyclophane systems (from ref. [12.50]).

first excited state is the triplet state; in the former this triplet is located at 63 cm^{-1} above the quintet, which corresponds to an interaction parameter between the two carbenes $J = 31.5$ cm^{-1}; in the latter the triplet is located at about 100 cm^{-1} above the quintet, and J is equal to 50 cm^{-1}. For the pseudo*meta* derivative the triplet is located at 93 cm^{-1} above the singlet, and J is equal to –93 cm^{-1}. The McConnell mechanism based on spin polarization is perfectly in line with these findings, as emphasized in Figure 12.13. For the pseudo*ortho* and pseudo*para* derivatives the spin up–spin down interactions between nearest-neighbor carbon atoms of different carbene units favor overall parallel alignment of molecular spins.On the contrary, for the pseudo*meta* derivative the intermolecular spin up–spin down interactions favor antiparallel alignment of molecular spins.

In spite of its elegance this interpretation has a weak point that would deserve to be examined in a more thorough manner; the intermolecular interactions between second nearest neighbors are totally neglected, while they act in a way that opposes the interactions between first nearest neighbors.

Spin polarization may also lead to a large positive spin density at the center of the molecular unit and some negative spin density at the periphery. Such a unit may couple ferromagnetically with a radical, for instance in a donor–acceptor stack, as

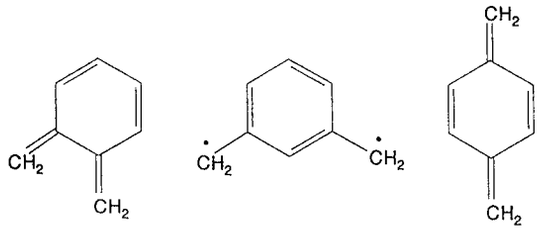


In Section 12.4 we saw that such a mechanism has been proposed to account for the ferromagnetic behavior of $[Fe(C_5Me_5)_2][TCNE]$ and other decamethylmetallocenium charge-transfer salts.

12.7 Topological Degeneracy and High-Spin Polycarbenes

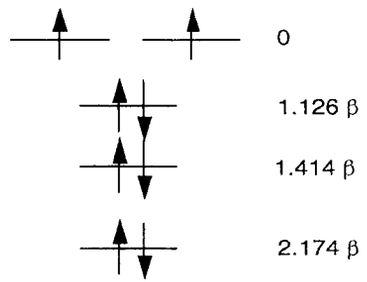
A very important approach to purely organic ferromagnets has been developed since the beginning of the sixties, essentially by Japanese groups. The pioneering papers along this line were written by Higuchi [12.51]. The field was recently reviewed by Iwamura [12.52]. The approach is based on the concept of topological degeneracy in alternant hydrocarbons, which was first introduced by Longuet–Higgins [12.53]. The possibility of designing ferromagnetic polymers using this concept was first recognized by Mataga [12.54].

Simple molecular orbital theory at the Hückel level predicts that an alternant hydrocarbon with n carbon atoms has at least $n - 2t$ singly occupied nonbonding π molecular orbitals, t being the maximum number of double bonds occurring in any resonance structure. Let us apply this idea to the three isomers of benzoquinodimethane:



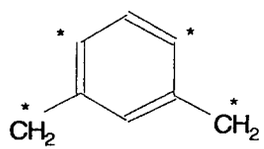
$S = 0$ $S = 1$ $S = 0$

The number of carbon atoms is eight, and the maximum number of double bonds is four for the *ortho*-, three for the *meta*-, and four for the *para*-isomers. It follows that the *ortho*- and *para*-isomers should have a closed-shell ground state with all of the π electrons occupying bonding molecular orbitals, and that the *meta*-isomer should have a triplet ground state with two unpaired electrons occupying nonbonding molecular orbitals. A Hückel calculation leads to the following molecular orbital scheme:

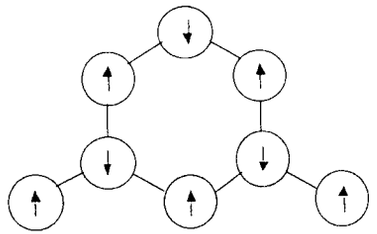


where β is the transfer integral of the Hückel method. A more sophisticated molecular orbital calculation with an extensive CI indicates that the triplet state is stabilized by almost 3000 cm^{-1} with respect to the first excited singlet state [12.55].

The nature of the ground state in alternant hydrocarbons can also be predicted within the framework of valence-bond approaches. Ovchinnikov [12.56], and then Klein [12.57, 12.58] proposed the following formulation: the carbon atoms in alternant hydrocarbons are starred in such a way that there are as many stars created as possible and no two stars come next to each other. The numbers of starred and unstarred atoms are then n^* and n, respectively. If we do so, the spin associated with the ground state is $S = (n^* - n)/2$. We can easily check that for *ortho*- and *para*-benzoquinodimethane n^* and n are equal to 4, and S is equal to zero. On the other hand, for the *meta*-derivative, n^* and n are equal to 5 and 3, respectively, and S is equal to 1 as schematized:

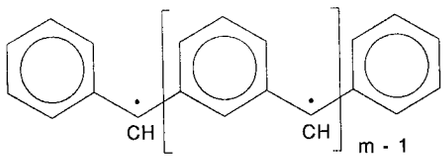


This formulation may also be viewed in another way. A starred atom corresponds to an up-spin and an unstarred atom to a down-spin,



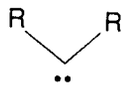
which again leads to an $S = 1$ molecular spin for the *meta*-benzoquinodimethane. The alternation of up- and down-spins results from the spin polarization effect. ^1H ENDOR data nicely confirms this spin alternation [12.59, 12.60].

Both molecular orbital and valence-bond theories predict that the polyradicals of the type

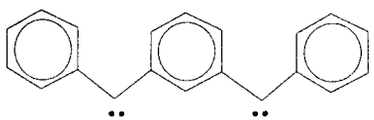


consisting of methylphenyl units connected in *meta*-positions will have an $S = m/2$ ground state. In the diradical with $m = 2$ the stabilization of the spin triplet ground state with respect to the spin singlet excited state is estimated to be as large as $J = 3300$ cm^{-1} [12.61].

In the metaphenyl polyradicals drawn above the unpaired electrons occupy π orbitals. If the —CH— groups between two phenyl rings are replaced by —C— carbene functions, there are m new unpaired electrons occupying σ-type localized orbitals in addition to the m unpaired electrons occupying π orbitals. On each carbenic site the σ and π orbitals are orthogonal, and the local spin state is a triplet. For instance, in a monocarbene like

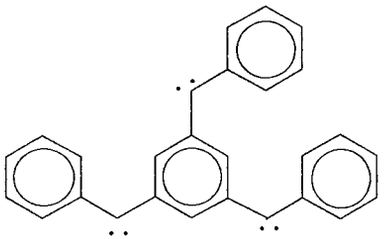


there are two unpaired electrons occupying the nonbonded orbitals of σ and π symmetry, respectively, and the ground state is actually a triplet. Such a carbene is relatively stable if R is a phenyl group. We now consider the benzene-1,3-bis-phenyl-methylene:



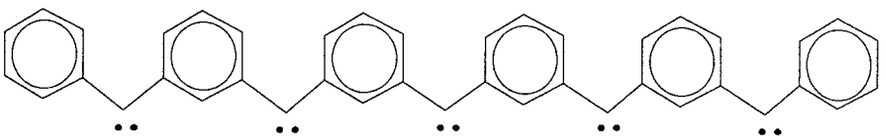
with a *meta* substitution on the central ring. One unpaired electron of each divalent carbon atom remains in a σ orbital essentially localized on the carbon atom, while the other participates in conjugation with the π system of the aromatic rings. There are two π orbitals of this kind, which may be nearly degenerate with the two singly occupied σ orbitals. It follows that the four electrons occupy four quasi-degenerate orbitals, and that the ground state is expected to be a quintet. Independently, Itoh [12.62] and Wassermann [12.63] demonstrated that the EPR spectrum of this dicarbene was characteristic of an $S = 2$ ground state.

Along the same line, the benzene-1,3,5-tris-phenylmethylene

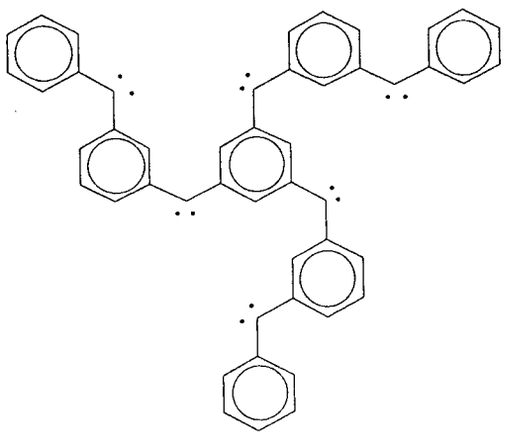


was found to exhibit an $S = 3$ ground state [12.64, 12.65].

In a very impressive synthetic effort Iwamura et al. were able to develop this strategy up to the linear tetracarbene [12.66–12.69], to the linear pentacarbene [12.70]:



then to the branch-chain hexacarbene [12.71]:



They proved unambiguously that the ground state is a spin nonet, $S = 4$, for the tetracarbene, a spin undecet, $S = 5$, for the pentacarbene, and a spin tridecet, $S = 6$, for the hexacarbene. Let us detail a little the physical data concerning the tetracarbene. It should be pointed out that this tetracarbene is very unstable and needs to be stabilized in a matrix of benzophenone, and even then, it decomposes above 100 K. The EPR spectrum of the doped matrix shows an eight-line pattern corresponding to the $\Delta M_S = \pm 1$ allowed transitions $A_\pm$ ($M_S = \pm 4 \rightarrow M_S = \pm 3$), $B_\pm$ ($M_S = \pm 3 \rightarrow M_S = \pm 2$), $C_\pm$ ($M_S = \pm 2 \rightarrow M_S = \pm 1$), and $D_\pm$ ($M_S = \pm 1 \rightarrow M_S = 0$). The relative separations are $(A_- - A_+)/(B_- - B_+)/(C_- - C_+)/(D_- - D_+)$ close to 7/5/3/1, and the relative integrated intensities are $A_\pm/B_\pm/C_\pm/D_\pm = 4/7/9/10$ as expected for an $S = 4$ state with a nearly axial zero-field splitting. The product of intensity and temperature is constant up to 100 K, which indicates that only the nonet ground state is thermally populated. The EPR spectrum does not actually show any signal associated with thermally populated triplet, quintet or septet states in this temperature range. The magnetization curve for this tetracarbene closely follows Eq. (2.2.7) with $S = 4$. At 2.1 K a saturation magnetization of 8 $N\beta$ is reached for magnetic fields higher than 2.5×10^4 G. The magnetic susceptibility of the tetracarbene doped in crystalline benzophenone follows the Curie law expected for an $S = 4$ ground state well separated in energy from the first excited states.

Until now, all efforts to assemble these high-spin polycarbenes within a matrix in a ferromagnetic fashion have failed. Instead, weak intermolecular antiferromagnetic interactions were detected. Despite this, there is no doubt that the stabilization of a high-spin state by topological degeneracy is a breakthrough in molecular magnetism. It is quite remarkable that only the high-spin state is thermally populated; the interaction parameter for the benzene-1,3-bis-phenymethylene is of the order of 10^3 cm^{-1}. The two directions in which progress is expected are the chemical stabilization of compounds and their polymerization.

12.8 Magnetic Ordering of Ferrimagnetic Chains

Another approach in the design of molecular-based compounds exhibiting a magnetic transition consists of assembling ferrimagnetic chains within a crystal lattice in a ferromagnetic fashion. Such an approach was first applied successfully in the case of the compound $MnCu(pbaOH)(H_2O)_3$ with pba = 2-hydroxy-1,3-propylenebis(oxamato) [12.72]. The structure of this compound consists of Mn(II)Cu(II) chains which are almost identical to those encountered in $MnCu(pba)(H_2O)_3 \cdot 2H_2O$ (see Section 11.6 and Figure 11.14). These two compounds are abbreviated hereafter as MnCu(pbaOH) and MnCu(pba), respectively. The crystal lattices, however, are different (see Figure 12.14). For both compounds the structure is orthorhombic, and the chains run along the b axis. Along the *c* axis the chains are related by a unit-cell translation with Mn—Mn and Cu—Cu shortest interchain separations. The structural difference between the two compounds concerns the relative positions of the chains along the *a* axis. The shortest metal–metal separations between neighboring chains in the *a* direction are Cu—Cu and Mn—Mn in MnCu(pba), and Mn—Cu in MnCu(pbaOH).

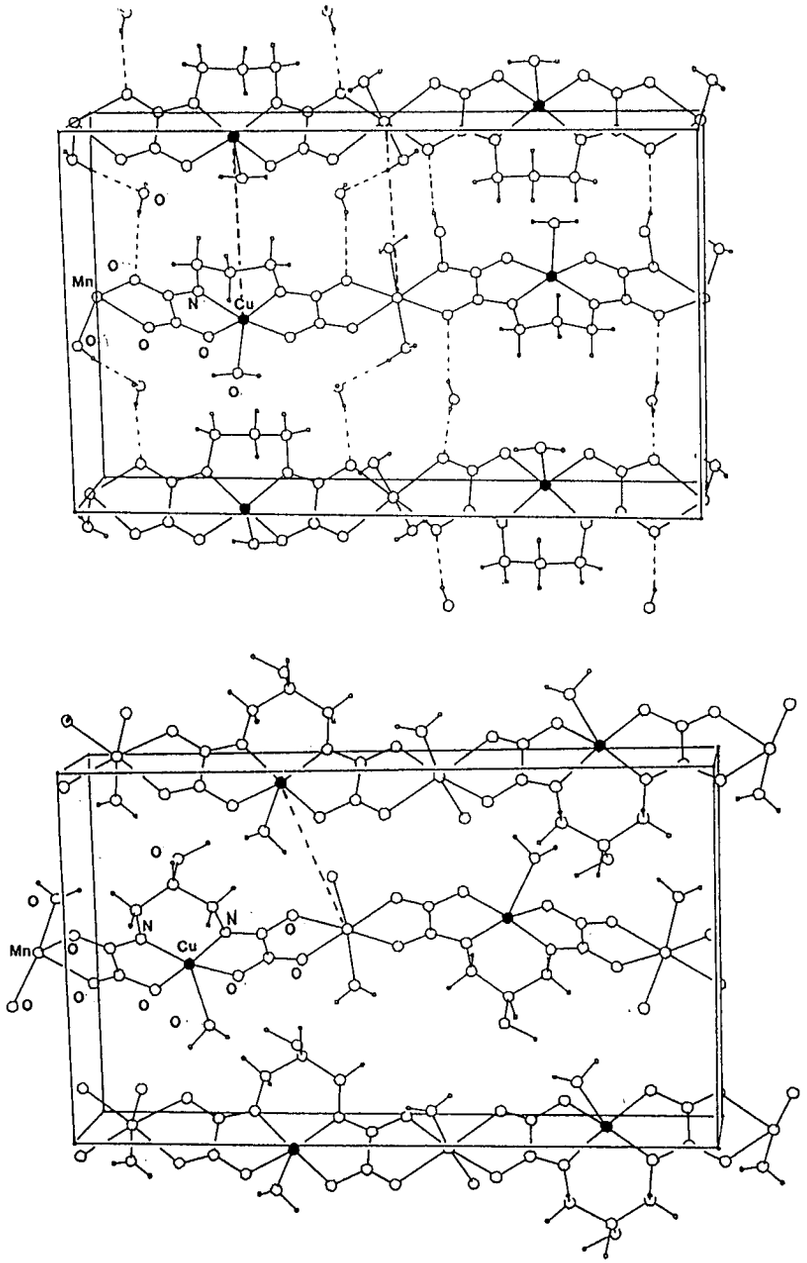
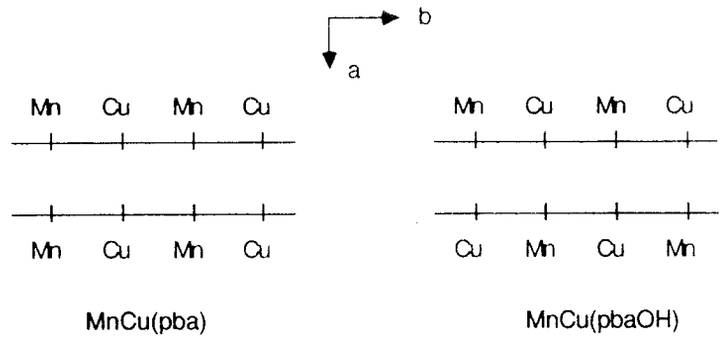


Figure 12.14. Relative positions of the chains along the *a* axis for $MnCu(pba)(H_2O)_3 \cdot 2H_2O$ (top) and $MnCu(pbaOH)(H_2O)_3$ (bottom). The shortest interchain metal–metal separations along the *a* direction are shown as dotted lines (from ref. [12.72]).

Such a difference may be described as follows: in the pbaOH compound, every other chain is displaced by almost half of a repeat unit along the b-axis compared to the MnCu(pba) compound, as schematized by



The magnetic properties of MnCu(pbaOH) down to 30 K are identical to those of MnCu(pba) within experimental uncertainties. The χT versus T plot exhibits a minimum around 115 K, characteristic of one-dimensional ferrimagnetic behavior. However, as T is lowered below 30 K, the increase of χT is much more abrupt for the MnCu(pbaOH) compound than for the MnCu(pba) compound. Below 5 K the two compounds behave quite differently. We have already seen that MnCu(pba) exhibits a maximum of χT at 2.3 K and a maximum of χ at 2.2 K, due to the onset of a three-dimensional antiferromagnetic ordering. In contrast, for MnCu(pbaOH) χT reaches values as high as 100 cm^3 K mol^{-1} and becomes strongly field-dependent. The variation of the molar magnetization M versus T within a field of 3×10^{-2} G for a polycrystalline sample confirms that a ferromagnetic transition occurs (see Figure 12.15). The field-cooled magnetization (FCM) obtained on cooling within the field shows the typical features of a ferromagnetic transition, i.e., a rapid increase in M when T decreases below 5 K, then a break in the curve at $T_c = 4.6$ K, and finally the beginning of the saturation. If the field is switched off below T_c, then a remnant magnetization is observed. The zero-field-cooled magnetization (ZFCM) is obtained by cooling below T_c in zero field and then applying the field and heating. At any temperature below T_c the ZFCM is smaller than the FCM, due to the fact that in this temperature range the applied field is too weak to move the domain walls. The ZFCM displays a maximum at T_c, as expected for a polycrystalline ferromagnet [12.73]. The field dependence of the magnetization displays very large zero-field susceptibility $(dM/dH)_{H=0}$, then rapid saturation with a saturation magnetization value of 4 $N\beta$. This corresponds to what is expected if all the $S_{Mn} = 5/2$ spins are aligned in one direction and all the $S_{Cu} = 1/2$ spins are aligned in the opposite direction. MnCu(pbaOH) also exhibits an hysteresis loop $M = f(H)$ characteristic of a soft ferromagnet.

Magnetic anisotropy measurements and single-crystal EPR data have shown that the preferred spin orientation is along the *c*-axis for both MnCu(pba) and MnCu(pbaOH) [12.72, 12.74]. The dramatic difference in the magnetic behaviors has been related to the difference in the crystal packings. Assuming that the interactions

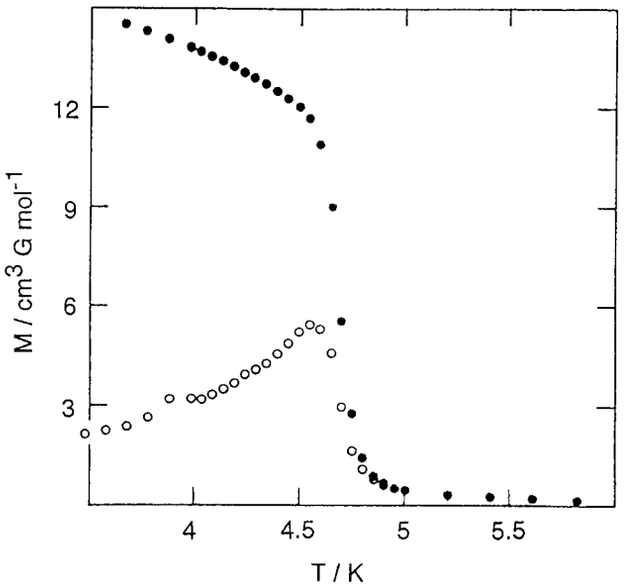
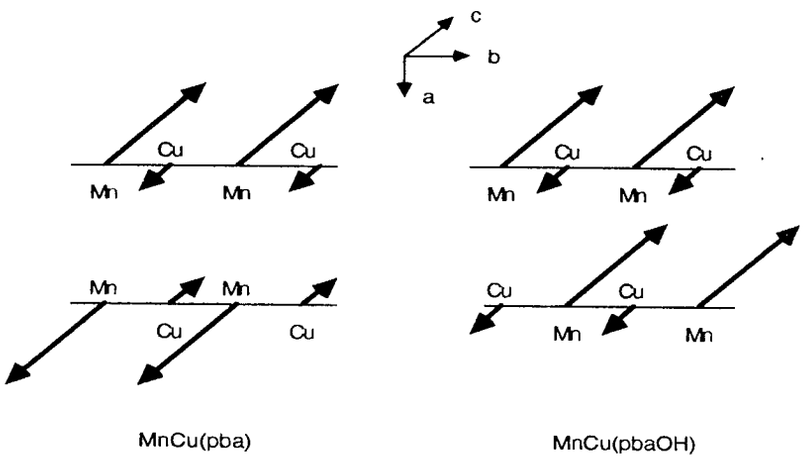


Figure 12.15. Magnetization versus temperature curves for $MnCu(pbaOH)(H_2O)_3$ within a field of 3×10^{-2} G; (o) FCM, (o) ZFCM (from ref. [12.72]).

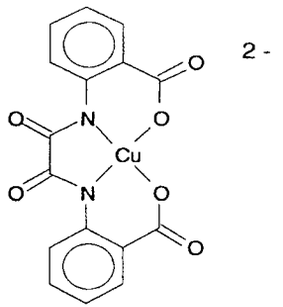
between nearest-neighbor magnetic ions are antiferromagnetic, both along the chain axis *b* and along the *a* axis, the relative positions of the chains along *a* leads to a cancellation of the resulting spin for MnCu(pba). In contrast, the displacement of every other chain in the *ab* plane in MnCu(pbaOH) by roughly half of a repeat unit leads to a parallel alignment of the $S_{Mn} = 5/2$ local spins:



It has been suggested that the interchain interaction along *c* is dominated by the dipolar contribution, and favors the parallel alignment of the spins.

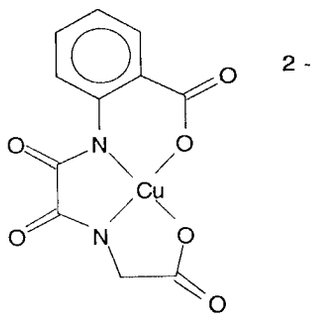
The water molecule occupying the apical site in the copper(II) coordination sphere of MnCu(pbaOH) may be selectively removed through a mild thermal treatment. The transition temperature is then shifted toward $T_c = 30$ K. The chains are probably closer to each other after partial dehydration, and the interchain interactions are enhanced [12.75].

Another Mn(II)Cu(II) derivative exhibiting a ferromagnetic transition has been reported, namely $MnCu(obbz)(H_2O)$ where obbz stands for oxamido-*N*,*N*′-bis(2-benzoato). This compound is obtained through the reaction of the Mn(II) ion on the copper(II) precursor $[Cu(obbz)]^{2-}$:



This affords an alternating bimetallic chain compound rather similar to $MnCu(obp)(H_2O)_3 \cdot H_2O$ (see Figure 11.22). The crystal packing is not yet known. The critical temperature is found as $T_c = 14$ K [12.76].

The final result we discuss concerning Mn(II)Cu(II) compounds arises from the reaction of the precursor $[Cu(obze)]^{2-}$ with obze = oxamido-*N*-benzoato-*N*′-ethanoato



on Mn(II). This affords a compound whose structure consists of $MnCu(obze)(H_2O)_4$ discrete units and noncoordinated water molecules as shown in Figure 12.16 [12.77]. The magnetic behavior of this compound is that of antiferromagnetically coupled Mn(II)Cu(II) pairs; χT decreases continuously on cooling. When this compound is

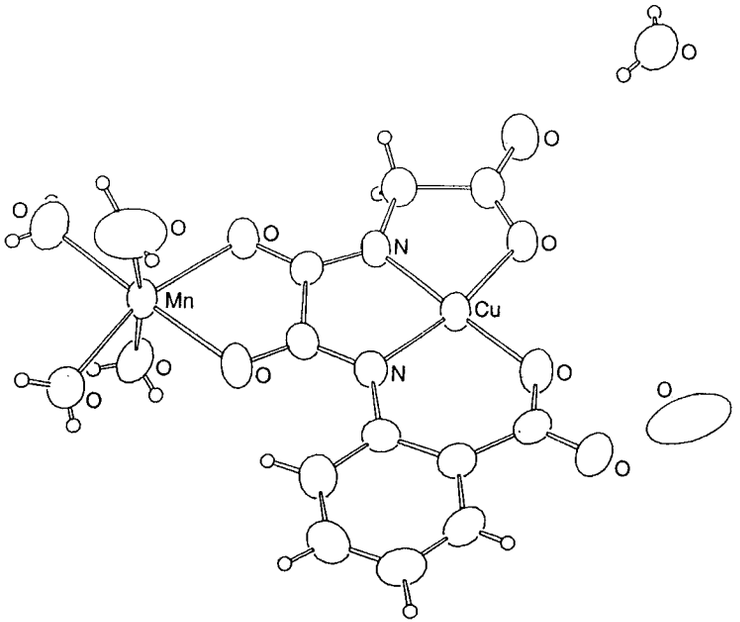
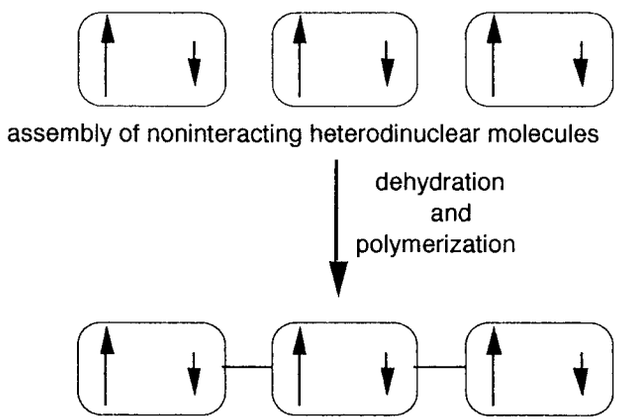


Figure 12.16. Molecular structure of MnCu(obze)$(H_2O)_4$,$2H_2O$ (from ref. [12.77]).

kept under vacuum at room temperature, the noncoordinated as well as two of the coordinated water molecules are removed, and a polymerization reaction in the solid state takes place. This provides some exchange pathway between the Cu(II) ion of one molecule with a negative spin density and the Mn(II) ion of another molecule with a positive spin density. The magnetic behavior of the resulting material is completely modified; a ferromagnetic transition is observed at $T_c = 4.6$ K. The process may be schematized as



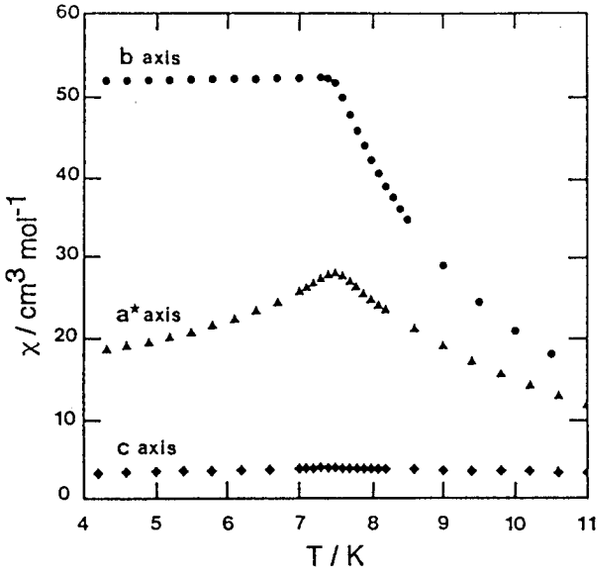


Figure 12.17. Molar magnetic susceptibility of oriented crystals of $Mn(hfa)_2$NIT-isopropyl within a magnetic field of 0.5 G (from ref. [12.78]).

where for the sake of simplicity we have represented a one-dimensional array of local spins while the actual structure of the polymer must be two- or three-dimensional. This process is quite reminiscent of the McConnell mechanism we discussed in Section 12.6.

Rather similar results were reported by Caneschi et al., concerning the metal-nitronyl nitroxide ferrimagnetic chain compounds discussed in Section 11.6. Thus, the compound $Mn(hfa)_2$NIT-R with R = isopropyl exhibits a three-dimensional ferromagnetic transition at $T_c = 7.61$ K [12.78, 12.79]. This compound crystallizes in the monoclinic system, the ferrimagnetic chains running along the c axis (see Figure 11.18). It should be noted that in one of the directions perpendicular to the chain axis, the shortest interchain separation involves the Mn(II) ion of one chain and the organic radical of the adjacent chain, which is reminiscent of the situation encountered in MnCu(pbaOH). Single crystal magnetic susceptibilities are shown in Figure 12.17. The preferred spin orientation is along b. The temperature independence of the magnetic susceptibility below T_c in this direction is due to demagnetizing effects, and is typical of a bulk ferromagnet. From single crystal EPR data, it is postulated that the interchain interactions are dominated by dipolar effects. Along this line, it can be noted that the analogous compound in which Mn(II) is replaced by Ni(II) orders at a slightly lower temperature, namely 5 K; the smaller local spin of Ni(II) as compared to Mn(II) makes the dipolar interactions between ferrimagnetic chains less effective [12.79]. The intrachain interactions are larger and the interchain interactions are smaller in these metal-nitronyl nitroxide chains compared to the Mn(II)Cu(II) chains, which altogether results in similar critical temperatures.

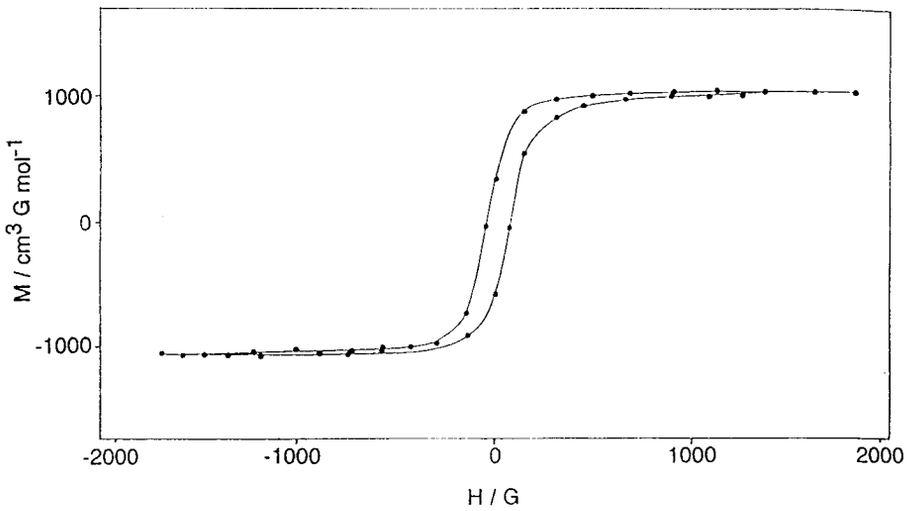
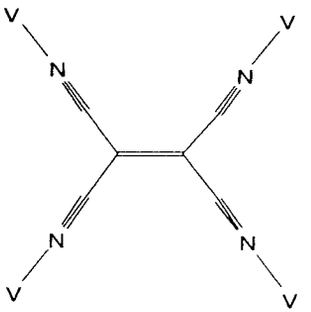


Figure 12.18. Field dependence of the magnetization for $V(TCNE)_2 \cdot 1/2CH_2Cl_2$ at room temperature (from ref. [12.81]).

Three-dimensional ferromagnetic transitions above 20 K were also observed for compounds of formula $[Mn(F_5benz)_2]_2$NIT-R where F_5benz is pentafluorobenzoato and R an alkyl group. The structure of these compounds is not known. The χT versus T plot for the R = Et derivative continuously increases as T is lowered down to 20.5 K and reaches a very high value, close to 800 cm^3 K mol^{-1}, at this temperature. The field-cooled magnetization curve shows the typical break at T_c = 20.5 K, and below T_c the $M = f(H)$ curves show hysteresis loops characteristic of a soft ferromagnet [12.80].

It is not clear whether the very spectacular result reported by Miller et al. in mid-1991 should be included in this section dealing with ferrimagnetic systems. This is due to the fact that at present little is known about this novel molecular-based material, which behaves as a magnet at room temperature. This compound is obtained as an insoluble, amorphous, black solid from the reaction of bisbenzene vanadium $V(C_6H_6)_2$ with an excess of TCNE in dichloromethane [12.81]. The empirical composition is $V(TCNE)_2 \cdot 1/2CH_2Cl_2$. The first step in the reaction is electron transfer from $V(C_6H_6)_2$ to TCNE followed by the unexpected loss of the benzene ligands. The compound is extremely unstable, decomposing rapidly in air and becoming nonmagnetic. The remarkable feature of $V(TCNE)_2 \cdot 1/2CH_2Cl_2$ is the field-dependent magnetization and the hysteresis loop at room temperature with a coercive field of 60 G shown in Figure 12.18. The critical temperature exceeds 350 K, which is the temperature of thermal decomposition of the material.

Although the structure of $V(TCNE)_2 \cdot 1/2CH_2Cl_2$ is as yet unknown, it is interesting to formulate some hypotheses. The compound might have a three-dimensional polymeric structure with $TCNE^-$ bridging V(II) ions, TCNE being bonded to the metal through its nitrogen atoms as



The local spin of V(II) is 3/2 and that of $TCNE^-$ is 1/2. Due to the direct bonding of TCNE to the metal, a strong interaction between the two spin carriers may be anticipated. If this interaction is antiferromagnetic, a ferrimagnetic behavior is expected because of the noncompensation of magnetic moments, with an $S = 1/2$ resulting spin per molecular unit. The value of the saturation magnetization, 6000 cm^3 G mol^{-1} (or 1.1 $N\beta$), is in line with this interpretation. In the absence of additional information, it would be dangerous to go further in these speculations concerning the structure and the type of magnetic behavior for this room-temperature molecular-based magnet. Interestingly, the N-coordination of $TCNE^-$ has been found in [Mn(TPP)](TCNE) with TTP = meso-tetraphenylporphyrinato. This compound has a chain structure with $TCNE^-$ bridging two Mn(III) ions through two nitrogen atoms in trans position. [Mn(TPP)](TCNE) exhibits a spontaneous magnetization below T_c = 18 K [12.82].

12.9 Spin Canting and Magnetic Ordering

All situations investigated since the beginning of this chapter correspond to parallel or antiparallel spin alignments. In the latter, a molecular-based magnet may be obtained if there is no compensation of the magnetic moments. Another situation may occur, in which the local spins in the ordered magnetic state are neither perfectly parallel, nor antiparallel, but are canted. We already studied the microscopic origin of this phenomenon in Section 7.2. Here we will speak briefly about some molecular compounds in which this kind of canting leads to three-dimensional magnetic ordering. In fact, in Section 7.2, we assumed implicitly that the antisymmetric interaction was a weak perturbation superimposing onto dominant antiferromagnetic interactions as

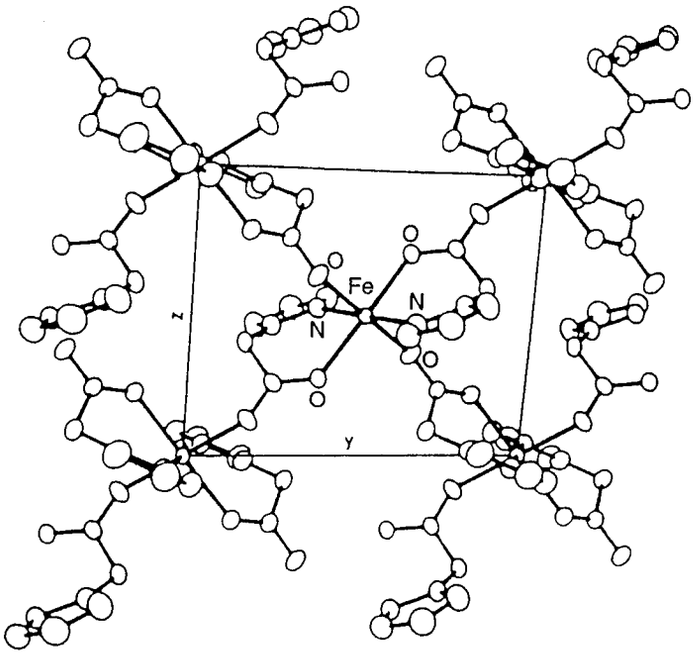


Figure 12.19. Structure of a sheet of [Fe(4-imidazoleacetate)$_2$]·2CH$_3$OH. For clarity the methanol molecules have been omitted (from ref. [12.82]).

This situation is the most frequent, and provokes weak ferromagnetism. Below a critical temperature the material possesses a net magnetic moment. A very weak magnetic field is sufficient to induce a magnetization M_w. When the field is increased further, the magnetization progressively increases provided that the antiferromagnetic interaction characterized by the J parameter is not too large, and tends to the M_S saturation value corresponding to the parallel alignment of all of the spins. The canting angle α is related to M_w and M_S through

$$\sin\alpha = M_w/M_S \qquad (12.9.1)$$

In 1991 a well-documented example of long-range ordering with a net magnetic moment resulting from a weak ferromagnetism was described. It concerns the compound [Fe(4-imidazoleacetate)$_2$]·2CH$_3$OH whose structure is represented in Figure 12.19 [12.83]. The Fe(II) ion surrounded by two nitrogen atoms of imidazole groups in trans positions and four oxygen atoms from acetato groups is linked to four neighboring molecules through acetato bridges with a Fe—Fe separation of 6.265 Å. This leads to intertwined linear chains and results in a sheet structure. Both the χ and χT versus T plots display rapid increases, then breaks around T_c = 15 K. The field dependence of the magnetization reveals a magnetization of ca. 1 $N\beta$ at 3 T, whereas the saturation magnetization expected for isolated high-spin Fe(II) ions is above 4 $N\beta$. Furthermore beautiful hysteresis is observed at 4.2 K with a coercive field as large

as 6×10^3 G, the remnant magnetization being equal to 0.22 $N\beta$. These data may be interpreted as follows: within the sheets the Fe(II) ions are coupled antiferromagnetically, but antisymmetric exchange is favored by the pronounced spin-orbit coupling for the $^5T_2$ ground state of Fe(II) in octahedral environment. This creates a canting between the $S_{Fe} = 2$ local spins which would otherwise align antiparallel below 15 K. Finally, it is assumed that the interactions between the sheets connected to each other by hydrogen bonds promote the observed three-dimensional weak ferromagnetism .

Another situation may occur, in which the spin canting superimposes onto the dominant ferromagnetic interactions as

and results in a canted ferromagnetism. The β polymorph of Mn(II) phthalocyanine MnPc offers one of the most elegant cases of canted ferromagnetism. The structure is schematized in Figure 12.20 and consists of two types of stacking chains of MnPc units, the planes of the units belonging to the two adjacent chains making an angle of ca. 90°. The shortest intermolecular separation in MnPc is between the metal ion of a

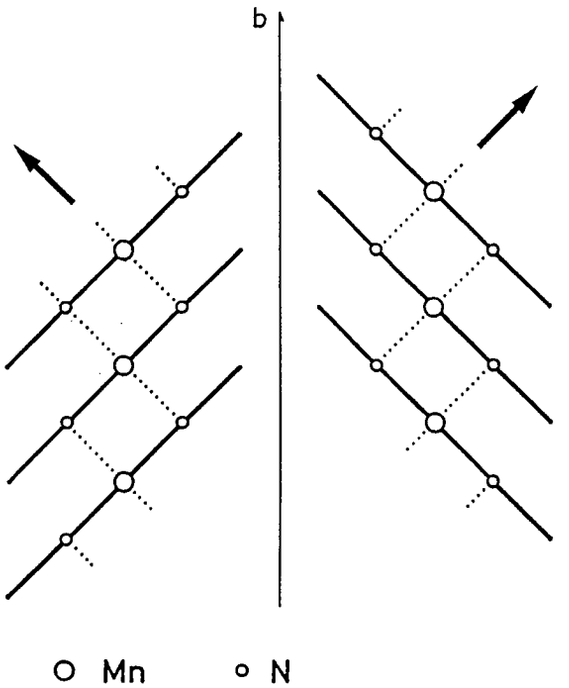


Figure 12.20. Stacking of the MnPc planar units in the β polymorph of Mn(II) phthalocyanine. The black arrows indicate the spin orientation for both types of MnPc units (from ref. [12.90]).

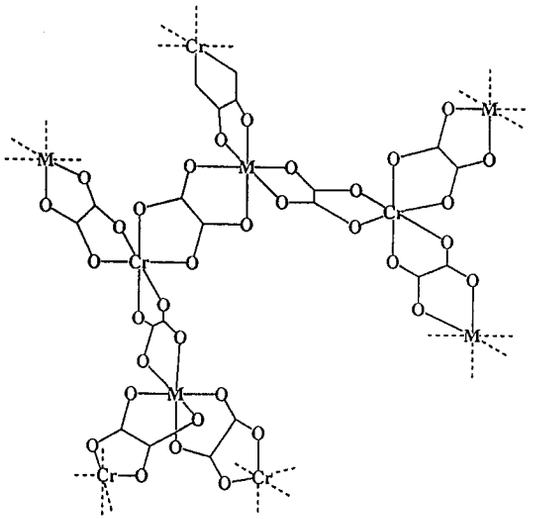


Figure 12.21. Postulated three-dimensional structure for $[MCr(ox)_3]^-$ in $(NBu_4)MCr(ox)_3]$ with ox = oxalato (from ref. [12.91]).

unit and one of the nitrogen atoms of the adjacent unit in the same chain (see Figure 12.21) [12.84, 12.85]. The MnPc unit has an intermediate-spin ground state with $S_{Mn} = 3/2$. The magnetic moment on an MnPc molecule is aligned perpendicular to the molecular plane, reflecting the strong magnetic anisotropy due to the orbital contribution. Principal susceptibility, magnetization, and PND measurements indicate that the magnetic moments are coupled ferromagnetically along each chain, and that the ferromagnetically coupled moments on one chain are almost perpendicular to those of its neighbor resulting in a canted ferromagnetism with the critical temperature T_c of 8.6 K [12.86–12.90]. T_c increases when applying a hydrostatic pressure [12.91].

Although this is not yet absolutely sure, the findings reported by Matsumoto, Okawa et al. in 1990 might represent a new example of canted ferromagnetism. The trisanion $[Cr(ox)_3]^{3-}$ reacts with M(II) ions in water in the presence of tetra(n-butyl)ammonium bromide to afford compounds of formula $(NBu_4)[MCr(ox)_3]$; ox stands for oxalato, and M may be Mn, Fe, Co, Ni, or Cu [12.92, 12.93]. Those compounds exhibit the characteristic features of a ferromagnetic transition in the 6–14 K temperature range. The highest critical temperature, T_c = 14 K, is obtained for $(NBu_4)[CuCr(ox)_3]$. Below T_c the magnetization versus magnetic field curves display hysteresis loops, with a coercive field of the order of 100 G at 5 K. The structure of these compounds is still unknown; a three-dimensional network of oxalato-bridged Cr(III) and Cu(II) ions has been postulated (see Figure 12.21) with the NBu_4^+ cations located within the cavities. The idea that we are dealing with canted ferromagnetism is suggested by the fact that the magnetization in high field reach values that are weaker than expected for all spins aligned parallel, and stronger than expected for S_{Cr} and S_M local spins aligned antiparallel. For instance, in the $(NBu_4)[CuCr(ox)_3]$

compound, the saturation magnetization M_S seems to be about 3 $N\beta$. If the Cr(III)-Cu(II) interaction were ferromagnetic, then M_S would be $Ng\beta(S_{Cr} + S_{Cu}) = 4\,N\beta$, whereas if it were ferrimagnetic, then M_S would be $Ng\beta(S_{Cr} - S_{Cu}) = 2\,N\beta$. This result is surprising. Indeed, due to the orthogonality of the magnetic orbitals in $(NBu_4)[CuCr(ox)_3]$ (see Section 9.3), the Cr(III)–Cu(II) interaction is anticipated to be ferromagnetic, which should lead to a genuine ferromagnetic behavior.

To finish this section, we would like to mention quite an interesting result announced by Clement et al., which is situated at the frontier between solid state chemistry and molecular chemistry. The MPS_3 phases, where M is a divalent metal ion, form a family of lamellar materials [12.94], best described as two-dimensional arrays of M(II) ions assembled by $P_2S_6^{4-}$ bridging ligands. These display a unique reactivity in solution, as they take up cations from the solution and release M(II) cations from the layers into the solution [12.95]. The pure phases $MnPS_3$ and $FePS_3$ order antiferromagnetically, with ordering temperatures of 78 and 120 K, respectively. Intercalating these phases with various guest molecular cations results in a dramatic modification of the magnetic properties. Whereas the behavior above the ordering temperature is still dominated by antiferromagnetic interactions, the intercalated material shows a magnetic transition at a critical temperature T_c below the ordering temperature of the pure phase, with spontaneous magnetization and hysteresis loop below T_c [12.96, 12.97]. The highest T_c reported so far for this class of compounds was obtained by intercalating a pyridine-pyridinium (py+pyH) mixture into $FePS_3$. The resulting material of empirical formula $Fe_{0.88}PS_3(py+pyH)_{0.36}(H_2O)_{0.5}$ shows a magnetic transition at T_c = 90 K. The saturation magnetization is only 3% or so of the value expected if all spins were ordered parallel. Nevertheless, the material is attracted by a permanent magnet. The reason why intercalation of diamagnetic species leads to such a spectacular modification of the magnetic properties is not yet perfectly understood. The phenomenon may arise from a canting of the antiferromagnetically coupled spins induced by the intercalation. The spontaneous magnetization may also be due to uncompensated magnetic moments around the metal vacancies, which would introduce some spin frustration into the M(II) lattice.

12.10 Some Additional Results

In this last section we will present a few additional results that, in our opinion, provide new information in the area of molecular magnetism, or at its frontier, but that could not be easily included in the other sections of this chapter.

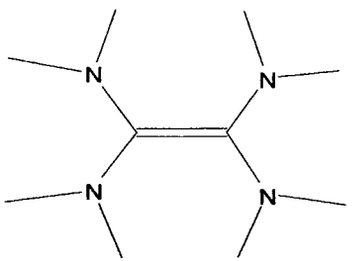
First, we should point out that the first genuine molecular compound displaying a ferromagnetic transition was described by Wickman et al. as early as 1967 [12.98, 12.99]. This compound, which we already mentioned in Section 5.3, is chloro-bis(diethyldithiocarbamato)iron(III) with an intermediate spin S = 3/2. It orders ferromagnetically at 2.46 K.

In Sections 8.6 and 9.3 we pointed out that strict orthogonality of magnetic orbitals was an efficient means of achieving a ferromagnetic interaction. This mechanism nicely applies for the molecular composites $[R(NH_3)_2]CuC1_4$, where R is an organic

chain $(—CH_2—)_n$, the organic diamonium groups being interleaved between layers of inorganic ions. The basic structure of the inorganic part $CuCl_4^{2-}$ closely resembles the perovskite structure, and consists of layers of $CuC1_6$ octahedra linked into a square array by sharing equilateral vertices. A cooperative Jahn–Teller distortion occurs due to the presence of Cu(II) ions, with a displacement of the chloride ions within the basal plane such that the long axes of two nearest neighbor metal ions are mutually orthogonal, as schematized in Figure 12.22. The unpaired electron of each Cu(II) occupies a *d*-type orbital pointing toward the four nearest chlorine atoms, two of them being on either side of the basal plane, the other two being in the basal plane. Because of the Jahn–Teller distortion, two of these orbitals centered on adjacent metal ions are strictly orthogonal, and all interactions are expected to be ferromagnetic. Actually, a long-range ordered ferromagnetic state is observed around 8 K [100, 101]. We will not discuss the mechanism of the long-range order in an essentially two-dimensional magnetic lattice. Let us merely say that both dipolar effects and magnetic anisotropy play a role.

When Cu(II) is replaced by Cr(II), which is also a Jahn–Teller ion, long-range ferromagnetic ordering takes place for temperatures around 40 K. Again, orthogonality of magnetic orbitals was assumed to be responsible for the intraplane ferromagnetic interactions [12.102, 12.103].

Among the tremendous amount of studies dealing with fullerene C_{60} (see Figure 12.23) that have appeared since the first report of a convenient way to synthesize this new form of carbon [12.104–12.106], there are some concerning the magnetic properties. In 1991 Wudl et al. described a new compound resulting from the reaction of C_{60} with the strong organic reducing agent TDAE = tetrakis(dimethylamino)ethylene:



A black powder of empirical formula $C_{60}(TDAE)_x$ was isolated. This compound apparently exhibits a ferromagnetic transition at T_c = 16.1 K. Quite surprisingly, the field dependence of the magnetization below T_c does not show any sign of hysteresis [12.107]. In the absence of additional information—neither the exact formula nor the value of the saturation magnetization are known at present—it would be hazardous to speculate on these quite preliminary data. What can be said, however, is that C_{60} has low-lying lowest unoccupied molecular orbitals (LUMO), which causes it to be a soft electrophile. Therefore, some partial electron transfer from TDAE toward C_{60} providing spin densities on both units is not inconceivable.

We deliberately decided not to speak about the various reports of organic-derived polymers that are claimed to exhibit some ferromagnetic behavior. It is now clear that

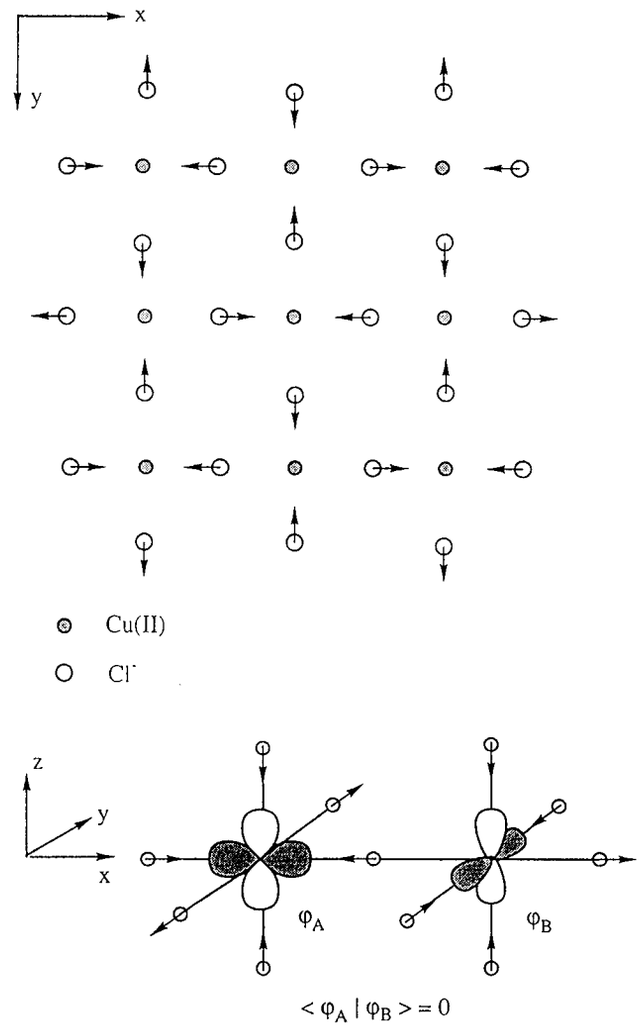


Figure 12.22. Top: Basal plan (*xy*) in $\{R(NH_3)_2]CuCl_4$ molecular composites. The Cu(II) ion is sixfold coordinated with two short and two long Cu—Cl bond lengths within the basal plane, and two other short bond lengths on either side of the basal plane (along the *z* axis). Bottom: Orthogonality of the magnetic orbitals centered on nearest-neighbor Cu(II) ions within the basal plane of $[R(NH_3)_2]CuCl_4$. The arrows symbolize the cooperative Jahn–Teller distortion.

in most cases the behavior is not intrinsic, but arises from ferromagnetic impurities, essentially iron or iron oxides such as Fe_2O_3. This field has been convincingly reviewed in 1992 by Miller [12.108]. There is no doubt that the design of polymers incorporating magnetic particles may play quite an interesting role in material science, and complement conventional magnetic alloys. This is particularly true if those

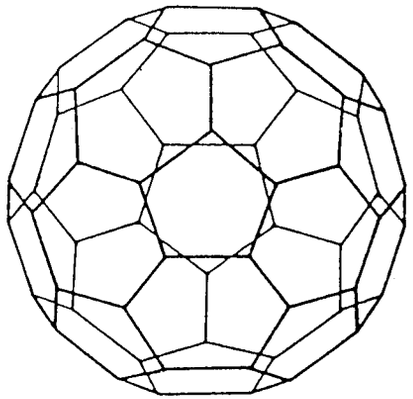


Figure 12.23. Structure of fullerene C_{60}.

polymers are processable and transparent. The study of this type of materials, however, does not fit the spirit of this book. In other respects, the design of organic-based polymers with intrinsic ferromagnetic behavior is certainly not hopeless, even if the goal has not yet been reached. We already discussed at length of Mataga's ideas in Section 12.7, based on the polymerization of high-spin polycarbenes. Other strategies have been put forward. Among them, we may mention the very original polaronic approach proposed by Fukutome [12.109] and tested by Dougherty [12.110]. This approach consists of doping a conjugated polymer to produce polarons, and enforcing the ferromagnetic coupling by inserting ferromagnetically coupling bridges between the polarons, for instance metaphenylene units.

References

12.1. R.L. Carlin, "Magnetochemistry." Springer, Berlin, 1986.

12.2. A.H. Morrish, "The Physical Principles of Magnetism." Krieger, New York, 1980.

12.3. A. Herpin, "Théorie du Magnétisme." Press. Univ. France, Paris, 1968.

12.4. D.H. Martin, "Magnetism in Solids." MIT Press, Cambridge, 1967.

12.5. D. Gatteschi, O. Kahn, J.S. Miller, F. Palacio, eds., "Magnetic Molecular Materials." NATO ASI, Kluwer, Dordrecht, 1991.

12.6. F. Palacio, F. J. Lazaro, A.J. van Duyneveldt, *Mol. Cryst. Liq. Cryst.* 176 (1989), 289.

12.7. F. Palacio, in reference 12.5, p. 1.

12.8. J.B. Torrance, P.S. Bagus, I. Johannsen, A.I. Nazzal, S.S.P. Parkin, P. Batail, *J. Appl. Phys.* 63 (1988), 2962.

12.9. P.S. Bagus, J.B. Torrance, *Phys. Rev. B* 39 (1989), 7301.

12.10. H.M. McConnel, *Proc. R A. Welch Found. Conf.* 11 (1967), 144.

12.11. R. Breslow, B. Jaun, R.Q. Kluttz, C.Z. Xia, *Tetrahedron* 38 (1982), 863.

12.12. R. Breslow, *Pure Appl. Chem.* 54 (1982), 927.

12.13. R. Breslow, P. Maslak, J. Thomaides, *J. Am. Chem. Soc.* 106 (1984), 6453.

12.14. R. Breslow, *Mol. Cryst. Liq. Cryst.* 125 (1985), 261.

12.15. T.J. LePage, R. Breslow, *J. Am. Chem. Soc.* 109 (1987), 6412.

12.16. C. Kollmar, O. Kahn, *J. Am. Chem. Soc.* 111 (1991), 7987.

12.17. J.S. Miller, J.C. Calabrese, H. Rommelmann, S.R. Chittipeddi, J.H. Zhang, W.M. Reiff, A.J. Epstein, *J. Am. Chem. Soc.* 109 (1987), 769.

12.18. J.S. Miller, A.J. Epstein, *J. Am. Chem. Soc.* 109 (1987), 3850.

12.19. J.S. Miller, A.J. Epstein, W.M. Reiff, *Science* 240 (1988), 40; *Chem. Rev.* 88 (1988), 201; *Acc. Chem. Res.* 21 (1988), 114.

12.20. W.E. Broderick, J.A. Thomson, E.P. Day, B.M. Hoffman, *Science* 249 (1990), 401.

12.21. G.T. Yee, J.M. Manriquez, D.A. Dixon, R. S. McLean, D.M. Groski, R.B. Flippen, K.S. Narayan, A.J. Epstein, J.S. Miller, *Adv. Mat.* 3 (1991), 309.

12.22. W.E. Broderick, B.M. Hoffman, *J. Am. Chem. Soc.* 113 (1991), 6334. D.M. Eichhorn, D.C. Skee, W.E. Broderick, B.M. Hoffman, *Inorg. Chem.* 32 (1993), 491.

12.23. G.A. Candela, L.J. Swartzendruber, J.S. Miller, M.J. Rice, *J. Am. Chem. Soc.* 101 (1979), 2755.

12.24. W.E. Broderick, J.A. Thompson, B.M. Hoffman, *Inorg. Chem.* 30 (1991), 2958.

12.25. Z.G. Soos, P.C.M. McWilliams, *Mol. Cryst. Liq. Cryst.* 176 (1989), 369.

12.26. A.L. Tchougreeff, *J. Chem. Phys.* 96 (1992), 6026.

12.27. A.L. Buchachenko, *Mol. Cryst. Liq. Cryst.* 176 (1989), 307.

12.28. C. Kollmar, M. Couty, O. Kahn, *J. Am. Chem. Soc.* 111 (1991), 7994.

12.29. C. Kollmar, O. Kahn, *J. Chem. Phys.* 96 (1992), 2988.

12.30. D.E. Williams, *Mol. Phys.* 16 (1969), 145.

12.31. K. Mukai, H. Nishiguchi, Y. Deguchi, *J. Phys. Soc. Jpn.* 23 (1967), 125.

12.32. K. Mukai, *Bull. Chem. Soc. Jpn.* 42 (1969), 40.

12.33. A. Kosaki, H. Suga, S. Seki, K. Mukai, Y. Deguchi, *Bull. Chem. Soc. Jpn.* 42 (1969), 1525.

12.34. K. Mukai, A. Sogabe, *J. Chem. Phys.* 72 (1980), 598.

12.35. K. Mukai, K. Ueda, K. Ishizu, Y. Deguchi, *J. Chem. Phys.* 77 (1982), 1606.

12.36. K. Awaga, T. Sugano, M. Kinoshita, *Solid State Commun.* 57 (1986), 453.

12.37. K. Awaga, T. Sugano, M. Kinoshita, *J. Chem. Phys.* 85 (1986), 2211.

12.38. G. Chouteau, C. Veyret-Jeandey, *J. Phys. (Paris)* 42 (1981), 1441.

12.39. A. Benoit, J. Flouquet, B. Gillon, J. Schweitzer, *J. Mag. Mag. Mater.* 31–34 (1983), 1155.

12.40. K. Awaga, Y. Maruyama, *J. Phys. Chem.* 91 (1989), 2743.

12.41. K. Awaga, Y. Maruyama, *Chem Phys. Lett.* 158 (1989), 556.

12.42. K. Awaga, T. Inabe, U. Nagashima, Y. Maruyama, *J. Chem. Soc., Chem. Commun.* (1989), 1617.

12.43. P.M. Allemand, C. Fite, P. Canfield, G. Srdanov, N. Keder, F. Wudl, *Proceedings of the International Conference on Science and Technology of Synthetic Metals* (ICSM 90), Tübingen, p. 452 (1990).

12.44. P. Turek, K. Nozawa, D. Shiomi, K. Awaga, T. Inabe, Y. Maruyama, M. Kinoshita, *Chem. Phys. Lett.* 180 (1991), 327.

12.45. M. Tamura, Y. Nakazawa, D. Shiomi, K. Nozawa, Y. Hosokoshi, M. Ishikawa, M. Takahashi, M. Kinoshita, *Chem. Phys. Lett.* 186 (1991), 401.

12.46. A.L. Buchachenko, *Uspeckhi Khim.* 59 (1990), 529; *Engl. Transl. Russian Chem. Rev.* 59 (1990), 307.

12.47. H.M. McConnell, *J. Chem. Phys.* 39 (1963), 1910.

12.48. See, for instance, L. Salem, "The Molecular Orbital Theory of Conjugated Systems." Benjamin, New York, 1966.

12.49. J. E. Wertz, J.R. Bolton, "Electron Spin Resonance; Elementary Theory and Practical Applications," p. 118. McGraw-Hill, New York, 1972.

12.50. A. Izuoka, S. Murata, T. Sagawara, H. Iwamura, *J. Am. Chem. Soc.* 109 (1987), 2631.

12.51. J. Higuchi, *J. Chem. Phys.* 38 (1963), 1237; 39 (1963), 1847.

12.52. H. Iwamura, *Adv. Phys. Org. Chem.* 26 (1990), 179.

12.53. H.C. Longuet-Higgins, *J. Chem. Phys.* 18 (1950), 265.

12.54. N. Mataga, *Theor. Chim. Acta* 10 (1968), 372.

12.55. J. Baudet, *J. Chim. Phys. Chim. Biol.* 68 (1971), 191.

12.56 A.A. Ovchinnikov, *Theor. Chim. Acta* 47 (1978), 297.

12.57. D.J. Klein, *J. Chem. Phys.* 77 (1982), 3098.

12.58. D.J. Klein, C.J. Nelin, S. Alexander, F.A. Matsen, *J. Chem. Phys.* 77 (1982), 3101.

12.59. T. Takui, S. Kita, S. Ichikawa, Y. Teki, T. Kinoshita, K. Itoh, *Mol. Cryst. Liq. Cryst.* 176 (1989), 67.

12.60. M. Okamoto, Y. Teki, T. Takui, T. Kinoshita, K. Itoh, *Chem. Phys. Lett.* 173 (1990), 265.

12.61. N. Tyutyulkov, S. Karabunarliev, *Intern. J. Quantum Chem.* 29 (1986), 1325.

12.62. K. Itoh, *Chem. Phys. Lett.* 1 (1967), 235.

12.63. E. Wasserman, R.W. Murray, W.A. Jager, A.M. Trozzolo, G. Smolinski, *J. Am. Chem. Soc.* 89 (1967), 5076.

12.64. T. Takui, K. Itoh, *Chem. Phys. Lett.* 19 (1973), 120.

12.65. K. Itoh, *Pure Appl. Chem.* 50 (1978), 1251.

12.66. Y. Teki, T. Takui, K. Itoh, H. Iwamura, K. Kobayashi, *J. Am. Chem. Soc.* 105 (1983), 3722.

12.67. T. Sugawara, S. Bandow, K. Kimura, H. Iwamura, K. Itoh, *J. Am. Chem. Soc.* 106 (1984), 6449; 108 (1986), 368.

12.68. Y. Teki, T. Takui, K. Itoh, H. Iwamura, K. Kobayashi, *J. Am. Chem. Soc.* 108 (1986), 2147.

12.69. H. Iwamura, *Pure Appl. Chem.* 58 (1986), 187.

12.70. I. Fujita, Y. Teki, T. Takui, T. Kinoshita, K. Itoh, F. Miko, Y. Sewaki, H. Iwamura, A. Izuoka, T. Sagawara, *J. Am. Chem. Soc.* 112 (1990), 4074.

12.71. N. Nakamura, K. Inoué, H. Iwamura, T. Fujioka, Y. Sawaki, *J. Am. Chem. Soc.* 114 (1992), 1484.

12.72. O. Kahn, Y. Pei, M. Verdaguer, J.P. Renard, J. Sletten, *J. Am. Chem. Soc.* 110 (1988), 782.

12.73. M. Hitzfeld, P. Ziemann, W. Buckel, H. Claus, *Phys. Rev. B* 29 (1984), 5023.

12.74. D. Gatteschi, O. Guillou, C. Zanchini, R. Sessoli, O. Kahn, M. Verdaguer, Y. Pei, *Inorg. Chem.* 28 (1989), 287.

12.75. K. Nakatani, P. Bergerat, E. Codjovi, C. Mathonière, Y. Pei, O. Kahn, *Inorg. Chem.* 30 (1991), 3977.

12.76. K. Nakatani, J.Y. Carriat, Y. Journaux, O. Kahn, F. Lloret, J.P. Renard, Y. Pei, J. Sletten, M. Verdaguer, *J. Am. Chem. Soc.* 111 (1989), 5739.

12.77. Y. Pei, O. Kahn, K. Nakatani, E. Codjovi, C. Mathonière, J. Sletten, *J. Am. Chem. Soc.* 113 (1991), 6558.

12.78. A. Caneschi, D. Gatteschi, J.P. Renard, P. Rey, R. Sessoli, *Inorg. Chem.* 28 (1989), 1976; 28 (1989), 3314.

12.79. A. Caneschi, D. Gatteschi, R. Sessoli, P. Rey, *Acc. Chem. Res.* 22 (1989), 392.

12.80. A. Caneschi, D. Gatteschi, J.P. Renard, J.P. Rey, R. Sessoli, *J. Am. Chem. Soc.* 111 (1989), 785.

12.81. J.M. Manriquez, G. T. Yee, R.S. McLean, A.J. Epstein, J.S. Miller, *Science* 252 (1991), 1415.

12.82. M.A. Marinez-Lorente, J.P. Tuchagues, V. Pétrouléas, J.M Savariault, R. Poinsot, M. Drillon, *Inorg. Chem.* 30 (1991), 3587.

12.83. J.S. Miller, J.C. Calabrese, R. S. McLean, A.J. Epstein, *Adv. Mater.* 4 (1992), 498.

12.84. R. Mason, G.A. Williams, P.E. Fielding, *J. Chem. Soc. Dalton* (1979), 676.

12.85. B.N. Figgins, R. Mason, G.A. Williams, *Acta Crystallogr. B* 36 (1980), 2963.

12.86. C.G. Barraclough, R.L. Martin, S. Mitra, *J. Chem. Phys.* 53 (1970), 1638.

12.87. C.G. Barraclough, A.K. Gregson, S. Mitra, *J. Chem. Phys.* 60 (1974), 962.

12.88. H. Miyoshi, *J. Phys. Soc. Jpn.* 37 (1974), 50.

12.89. B.N. Figgis, G.A. Williams, J.B. Forsyth, R. Mason, *J. Chem. Soc. Dalton* (1981), 1837.

12.90. S. Mitra, A.K. Gregson, W.E. Hatfield, R.R. Weller, *Inorg. Chem.* 22 (1983), 1729.

12.91. K. Awaga, Y. Maruyama, *Phys. Rev. B* 44 (1991), 2589.

12.92. Z.J. Zong, N. Matsumoto, H. Okawa, S. Kida, *Chem. Lett.* (1990), 87.

12.93. H. Tamaki, Z.J. Zhong, N. Matsumoto, S. Kida, M. Koikawa, N. Achiva, Y. Hashimoto, H. Okawa, *J. Am. Chem. Soc.* 114 (1992), 6974.

12.94. R. Brec, *Solid State Ionics* 22 (1986), 3.

12.95. R. Clement, O. Garnier, J. Jegoudez, *Inorg. Chem.* 25 (1986), 1404.

12.96. R. Clement, J.P. Audière, J.P. Renard, *Rev. Chim. Minér.* 19 (1982), 560.

12.97. R. Clément, L. Lomas, J.P. Audière, *Chem. Mater.* 2 (1990), 641.

12.98. H.H. Wickman, A.M. Trozzolo, H.J. Williams, G.W. Hull, F.R. Merritt, *Phys. Rev.* 155 (1967), 563.

12.99. G.C. deFotis, F. Palacio, C.J. O'Connor, S.N. Bhatia, R.L. Carlin, *J. Am. Chem. Soc.* 99 (1977), 8314.

12.100. L.J. de Jongh, A.R. Miedema, *Adv. Phys.* 24 (1974), 1.

12.101. H. von Känel, *Physica B* 96 (1979), 167.

12.102. P. Day, *Acc. Chem. Res.* 12 (1979), 236.

12.103. P. Day, *Phil. Trans. R. Soc. London A* 314 (1985), 145.

12.104. H.W. Kroto, J.R. Heath, S.C. O'Brien, R.F. Curl, R.E. Smalley, *Nature (London)* 318 (1985), 162.

12.105. W. Krätschmer, L.D. Lamb, K. Fostiropoulos, D.R. Huffman, *Nature (London)* 347 (1990), 354.

12.106. H.W. Kroto, A.W. Allaf, S.P. Balm, *Chem. Rev.* 91 (1991), 1213.

12.107. P.M. Allemand, K.C. Khemani, A. Koch, F. Wudl, K. Holczer, S. Donovan, G. Grüner, J.D. Thompson, *Science* 253 (1991), 301.

12.108. J.S. Miller, *Adv. Mat.* 4 (1992), 298.

12.109. H. Fukutome, A. Takahashi, M. Ozahi, *Chem. Phys. Lett.* 133 (1987), 34.

12.110. D.A. Kaisaki, W. Chang, D.A. Dougherty, *J. Am. Chem. Soc.* 113 (1991), 2764.

12.111. R. Chiarelli, M.A. Novak, A. Rassat, J.L. Tholence, *Nature* 363 (1993), 147.

CHAPTER

13

Spin-Dependent Delocalization in Mixed-Valence Compounds

In Chapters 6–11 we considered only polymetallic systems in which the active electrons were perfectly localized. For instance, when we studied the Cu(II)Ni(II) compound shown in Figure 6.14, it was clear that the interaction occurred between a local doublet and a local triplet, the former being associated with the copper(II) ion and the latter with the nickel(II) ion. In this chapter, on the contrary, we will discuss the magnetic properties of polynuclear species in which the unpaired electrons are not all fully localized. Such a situation happens in the mixed-valence compounds, containing several atoms of the same nature but with formally different oxidation states, like Fe(II)Fe(III) or Ni(I)Ni(II).

13.1 Mixed-Valence Dinuclear Compounds with a Unique Unpaired Electron

A great many papers and books have been devoted to mixed-valence compounds [13.1–13.6], and we do not intend to develop the theories appropriate to these species here. We recall, however, some basic concepts valid for a mixed-valence dinuclear compound $A^{\bullet}$–B $\leftrightarrow$ A–$B^{\bullet}$ in which an electron (noted $^{\bullet}$) may move from one site to the other. When this electron is located on A, it occupies an antibonding orbital centered on A, such that the A coordination sphere is expanded with respect to the B coordination sphere. When this electron is located on B, the B coordination sphere is expanded. More generally, the metal–ligand bond length is longer in the q oxidation state than in the q+1. This role of the nuclear relaxation leads to the introduction of a vibrational coordinate Q, which corresponds to the antisymmetrical breathing of the coordination spheres, as shown in Figure 13.1. When $Q = 0$, the two coordination spheres are identical; the dinuclear compound is symmetrical. $Q < 0$ corresponds to an increase of the metal–ligand bond lengths around A, and a decrease around B. $Q > 0$ corresponds to the reverse situation.

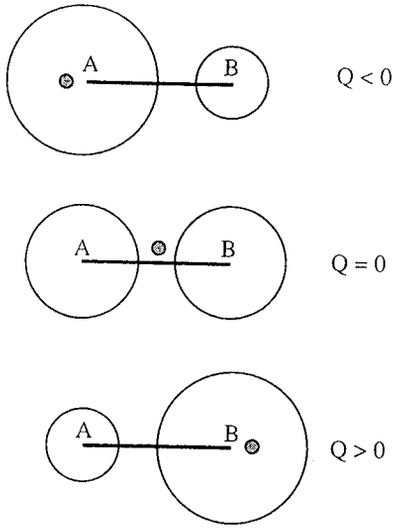


Figure 13.1. Antisymmetrical breathing vibration coordinate Q in a mixed-valence dinuclear compound $A^{\bullet}$–B ↔ A–$B^{\bullet}$ with chemically equivalent A and B sites.

Let us assume first that there is no orbital interaction between the A and B sites. In other words Ψ_A and Ψ_B are the electronic wavefunctions associated with $A^{\bullet}$–B and A–$B^{\bullet}$, respectively, with

$$\langle\Psi_A|\mathbf{H}|\Psi_A\rangle = \langle\Psi_B|\mathbf{H}|\Psi_B\rangle = E^0 \tag{13.1.1}$$

$$\langle\Psi_A|\mathbf{H}|\Psi_B\rangle = 0 \tag{13.1.2}$$

H is the electronic Hamiltonian. Thus, the two potential energy curves in the harmonic approximation are expressed as

$$E_A = E^0 + lQ + kQ^2/2$$

$$E_B = E^0 - lQ + kQ^2 \tag{13.1.3}$$

where k is the force constant associated with the Q vibration, and l is a parameter specifying the magnitude of the distortion of the dinuclear species in its equilibrium geometry. The energy minima are obtained for $Q = \pm\, l/k$. These curves are shown in Figure 13.2. In such a system in which there is no orbital interaction between the sites, the electron transfer requires an energy, often called the optical energy, given by

$$E_{op} = E_B(Q = -l/k) - E_A(Q = -l/k)$$

$$= 2l^2/k \tag{13.1.4}$$

The situation where the orbital interaction between the sites is no longer zero is much more interesting. We then have

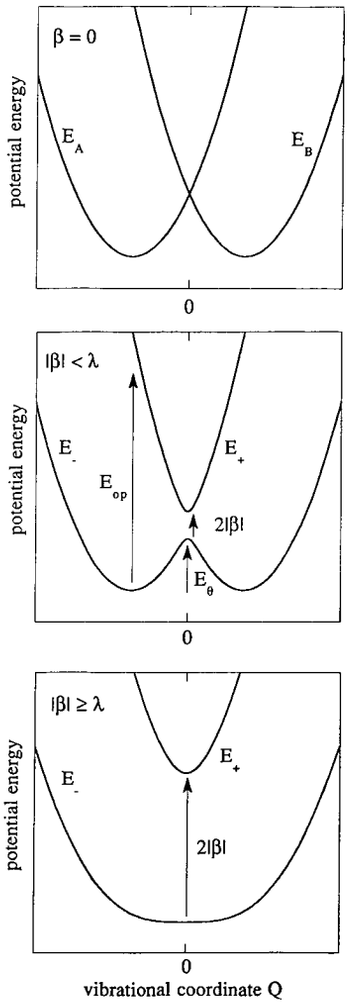


Figure 13.2. Potential energy curves for a mixed-valence dinuclear compound with diamagnetic cores.

$$\langle\Psi_A|\mathbf{H}|\Psi_B\rangle = \beta \tag{13.1.5}$$

where β is the transfer integral between the two site orbitals occupied by the additional electron. The energy matrix using Ψ_A and Ψ_B as a basis set is:

Ψ_A	Ψ_B
$E^0 + lQ + kQ^2/2$	β
	$E^0 - lQ + kQ^2/2$

(13.1.6)

The roots are

$$E_- = E^0 + kQ^2/2 - \sqrt{l^2Q^2+\beta^2}$$

$$E_+ = E^0 + kQ^2/2 + \sqrt{l^2Q^2+\beta^2} \qquad (13.1.7)$$

The crossing of the two curves E_A and E_B in $Q = 0$ is now avoided due to the transfer integral β, and a gap $2|\beta|$ appears. Both E_- and E_+ are symmetrical in Q; E_- corresponds to the ground state and E_+ to the excited state (see Figure 13.2).

For $|\beta| < l^2/k$, E_- presents an energy barrier with a maximum in $Q = 0$ between two energy minima. There is now a new mechanism for the $A^\bullet$–B $\leftrightarrow$ A–$B^\bullet$ process, consisting of the jump above the energy barrier, which requires an energy E_θ, often called thermal energy, given by

$$E_\theta = E_+(Q = 0) - E_{-,\min} \qquad (13.1.8)$$

where $E_{-,\min}$ is the minimum of E_-. E_θ is easily found as

$$E_\theta = \frac{l^2}{2k} + \frac{\beta^2 k}{2l^2} - |\beta| \qquad (13.1.9)$$

The optical process induces an absorption band called the intervalence band, which is characteristic of the mixed-valence compounds. Its energy is E_{op} given in (13.1.4), and its intensity is proportional to β^2.

For $|\beta| \geq l^2/k$, the energy barrier disappears, and E_- presents a unique energy minimum at $Q = 0$. The transition from E_- to E_+ requires an energy $2|\beta|$ (see Figure 13.2). The intensity of the intervalence band is then maximum, and no longer depends on β.

These results led to the classification of mixed-valence compounds into three classes according to the relative magnitudes of the transfer integral $|\beta|$, which favors electron delocalization, and the vibronic coupling parameter λ defined as

$$\lambda = l^2/k \qquad (13.1.10)$$

which favors electron localization with a dissymmetrical distortion of the A and B coordination spheres. This classification was proposed by Robin and Day in 1967 [13.11]. For $\lambda >> |\beta|$ the valences are trapped, and the additional electron is perfectly localized. This corresponds to class I. For $\lambda \leq |\beta|$ the additional electron is fully delocalized over the two sites. Moreover, these sites are strictly equivalent from a structural point of view. It is not possible to distinguish the valences of the metal atoms. The compounds of this kind are said to belong to class III. The intermediate situations correspond to class II. There is an energy barrier between the two energy minima corresponding to distorted geometries. The valences are essentially trapped, but there is a nonzero probability of electron hopping from one site to the other.

The mixed-valence dinuclear species with a unique unpaired electron have been extensively investigated, in particular through optical and EPR techniques. This is for instance the case for the Creutz–Taube complex $[(NH_3)_5Ru(pyrazine)Ru(NH_3)_5]^{5+}$ that contains a low-spin Ru(III) ion with one unpaired electron and a low-spin Ru(II) ion with a diamagnetic local state [13.7]. The magnetic properties of this compound do not provide information on either the degree of electron delocalization or the hopping frequency of the unpaired electron between the two ruthenium atoms. The only thermally populated state is a spin doublet, whatever the values of $|\beta|$ and λ may be, and the magnetic susceptibility follows the Curie law $\chi T = Ng^2\beta^2/4k$, g being the Zeeman factor for this ground state.

In some mixed-valence dinuclear compounds, however, the metal atoms have unpaired electrons in their two oxidation states, such that electron delocalization is superimposed onto the magnetic interaction. This situation may induce the new phenomena to which this chapter is devoted.

13.2 Spin-Dependent Delocalization in Mixed-Valence Dinuclear Compounds with Two Magnetic Sites

The electronic structure of mixed-valence compounds with magnetic cores was first investigated by Zener [13.8], and then by Anderson and Hasegawa [13.9] at the beginning of the 1950s, who introduced the concept and the term of double exchange. This term may appear rather obscure to many chemists, and we propose to coin the term spin-dependent delocalization (SDD). The first works concerning spin-dependent delocalization were carried out in the context of solid state physics. The concept was introduced in molecular chemistry from 1983, notably by Girerd et al. and Münck et al. [13.10–13.14], and by the school of Kishinev [13.15, 13.16]. Other groups independently developed similar ideas [13.17, 13.18].

We present here the spin-dependent delocalization in the simple case of an AB dinuclear system in which A and B are two symmetry-related sites. We also assume that there are three active electrons and four magnetic orbitals, a_1 and a_2 centered on A, and b_1 and b_2 centered on B as shown in Figure 13.3. Furthermore, the four orbitals are assumed to be mutually orthogonal. Let us notice here that these orthogonalized magnetic orbitals can be deduced from the three singly occupied and the lowest unoccupied molecular orbitals for the $S = 3/2$ pair state, using the localization criterion (8.3.1). This possibility arises from the fact that the two sites are symmetry-related. One of the electrons, hereafter referred as the transferable electron, may move freely from a_2 to b_2 and vice versa.

According to whether the transferable electron occupies the a_2 or b_2 orbital, the local spins are $S_A = 1$, $S_B = 1/2$ or $S_A = 1/2$, $S_B = 1$. The situation $S_A = 1$, $S_B = 1/2$ gives rise to a doublet state and a quartet state separated by $3J/2$ [see (6.1.10)]. The wavefunctions noted $|A,S,M_S>$ may be obtained by the use of the Clebsh–Gordon coefficients (see Appendix 6) as

$$|A,3/2,3/2> = |a_1a_2b_1|$$

$$|A,1/2,1/2> = (\sqrt{6}/6)(2|a_1a_2\bar{b}_1| - |\bar{a}_1a_2b_1| - |a_1\bar{a}_2b_1|) \qquad (13.2.1)$$

The situation $S_A = 1/2$, $S_B = 1$ gives rise to another doublet state and another quartet state, again with a $3J/2$ energy separation. The $|B,S,M_S>$ wavefunctions are as

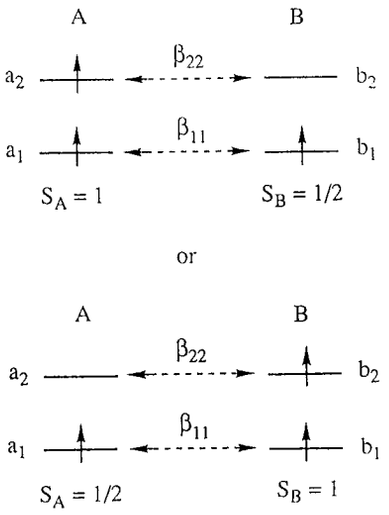


Figure 13.3. Four-orbital three-electron scheme for a symmetrical mixed-valence dinuclear compound with local doublet and triplet states.

$$|\mathrm{B},3/2,3/2\rangle = -|a_1 b_1 b_2|$$

$$|\mathrm{B},1/2,1/2\rangle = (\sqrt{6}/6)(2|\bar{a}_1 b_1 b_2| - |a_1 \bar{b}_1 b_2| - |a_1 b_1 \bar{b}_2|) \qquad (13.2.2)$$

We see, therefore, that the delocalization doubles the number of low-lying states. The electrostatic Hamiltonian of the problem may be written as

$$\mathbf{H} = \mathbf{h}(1) + \mathbf{h}(2) + \mathbf{h}(3) + 1/r_{12} + 1/r_{23} + 1/r_{31} \qquad (13.2.3)$$

with the same notations as in (9.2.1). We define the transfer integral β_{22} as

$$\beta_{22} = \langle a_2(i)\,|\,\mathbf{h}(i)\,|\,b_2(i)\rangle \qquad (13.2.4)$$

The functions $|\mathrm{A},S,M_S\rangle$ and $|\mathrm{B},S,M_S\rangle$ are not eigenfunctions of **H**. They couple due to the off-diagonal matrix elements

$$\langle \mathrm{A},3/2,M_S\,|\mathbf{H}|\,\mathrm{B},3/2,M_S\rangle = \beta_{22}$$

$$\langle \mathrm{A},1/2,M_S\,|\mathbf{H}|\,\mathrm{B},1/2,M_S\rangle = \beta_{22}/2 \qquad (13.2.5)$$

The phases of the wavefunctions in (13.2.2) have been chosen in a way that gives positive signs for these off-diagonal terms. The matrix elements (13.2.5) do not contain two-electron terms because of the orthogonality conditions between the magnetic orbitals. The matrix energy is then

$\lvert\mathrm{A},3/2,3/2\rangle$	$\lvert\mathrm{B},3/2,3/2\rangle$	$\lvert\mathrm{A},1/2,1/2\rangle$	$\lvert\mathrm{B},1/2,1/2\rangle$	
$-3J/2$	β_{22}	0	0	
	$-3J/2$	0	0	
		0	$\beta_{22}/2$	
			0	(13.2.6)

where the energy of the doublet pair states in the absence of delocalization is taken as the energy origin. The correct wavefunctions, which take into account the electron delocalization, are of the form

$$|\mathrm{AB},S,M_S,\pm\rangle = (\sqrt{2}/2)(|\mathrm{A},S,M_S\rangle \pm |\mathrm{B},S,M_S\rangle) \qquad (13.2.7)$$

and the energies are

$$E(3/2,\pm) = -3J/2 \pm \beta_{22}$$

$$E(1/2,\pm) = \pm\,\beta_{22}/2 \qquad (13.2.8)$$

Since the basis of the magnetic orbitals is assumed to be orthogonal, the J interaction parameter may be expressed as

$$J = k_{11} + k_{12} - \frac{2(\beta_{11} + l_{11})^2}{j_{11}^{\,0} - j_{11}} \qquad (13.2.9)$$

with the same notations as in (9.2.4). We may also suppose that J is negative, which implies that the second term on the right-hand side of (13.2.9) dominates. The key point of Eq. (13.2.8) is that the stabilization of the quartet state due to electron

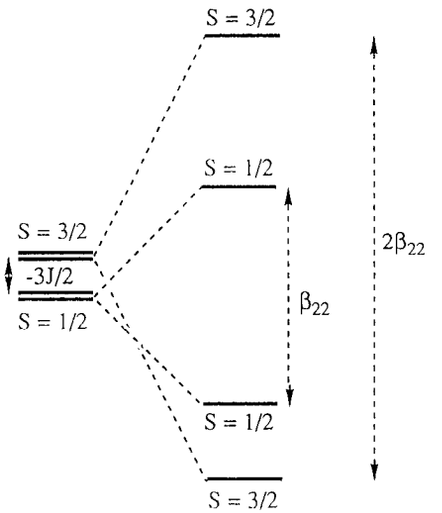


Figure 13.4. Spectrum of the low-lying states for a symmetrical mixed-valence dinuclear compound with local doublet and triplet states.

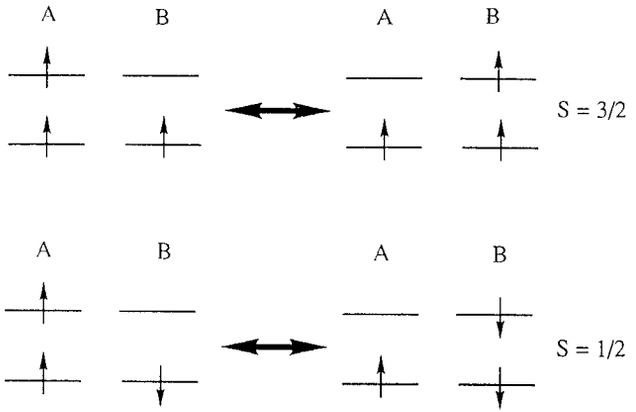


Figure 13.5. Spin delocalization in the $S = 3/2$ and $S = 1/2$ pair states for a fully delocalized symmetrical mixed-valence dinuclear compound with local doublet and triplet states.

delocalization is of the first order in β_{22}, whereas the possible stabilization of the doublet state due to antiferromagnetic interaction is of the second order in β_{11} as schematized in Figure 13.4. It follows that the electron delocalization strongly favors the state of highest spin as the ground state. This phenomenon arises from the fact that the transferable electron can move from one site to the other without spin flip in the quartet state, but not in the doublet state, and the spin flip costs energy (see Figure 13.5).

The result demonstrated above can easily be generalized to other fully delocalized dinuclear systems with a unique transferable electron. The off-diagonal term between two localized states of spin S is

$$<A,S,M_S|\mathbf{H}|B,S,M_S> = \frac{\beta(2S+1)}{2(2S_0+1)} \tag{13.2.10}$$

where S_0 is the core spin of site A (or B), i.e., the local spin in the absence of the transferable electron. S_0 is equal to 1/2 in the example treated above. For a given system the splitting within the pair of localized states of spin S becomes larger as S increases, which favors the stabilization of the highest-spin state. It is worth pointing out that the splitting of the spin states due to spin-dependent delocalization does not respect the Lande interval rule (6.1.10), and hence does not obey the HDVV Hamiltonian. The energies of these states vary as $2S + 1$ instead of $S(S + 1)$.

13.3 A Spin Hamiltonian Taking into Account the Spin-Dependent Delocalization

The combined effect of magnetic interaction and spin-dependent delocalization in a symmetrical dinuclear system AB may be described by an Hamiltonian proposed by Girerd and Münck, which is of the form

$$\mathbf{H} = -J({}^{A}\mathbf{S}_A \cdot {}^{A}\mathbf{S}_B\, \mathbf{O}_A + {}^{B}\mathbf{S}_A \cdot {}^{B}\mathbf{S}_B\, \mathbf{O}_B) + B\mathbf{T}_{AB} \tag{13.3.1}$$

${}^{A}\mathbf{S}_A$ and ${}^{A}\mathbf{S}_B$ are the local spin operators when the transferable electron is on the A site, and ${}^{B}\mathbf{S}_A$ and ${}^{B}\mathbf{S}_B$ are the local spin operators when the transferable electron is on the B site. $\mathbf{O}_A$ and $\mathbf{O}_B$ are occupation operators with the effects

$$\mathbf{O}_A\,|\,A,S,M_S> = |\,A,S,M_S>$$

$$\mathbf{O}_A\,|\,B,S,M_S> = 0 \tag{13.3.2}$$

and similar expressions when permuting A and B. The $B\mathbf{T}_{AB}$ term gives the splitting of the localized states $|A,S,M_S>$ and $|B,S,M_S>$ due to spin-dependent delocalization. $\mathbf{T}_{AB}$ acts as

$$\mathbf{T}_{AB}\,|\,A,S,M_S> = (S + 1/2)|B,S,M_S>$$

$$\mathbf{T}_{AB}|B,S,M_S> = (S + 1/2)|A,S,M_S> \tag{13.3.3}$$

with

$$B = \frac{\beta}{(2S_0 + 1)} \tag{13.3.4}$$

β being here the transfer integral between the magnetic orbitals occupied by the transferable electron. The two adjustable parameters of the model are the interaction parameter J and the electron delocalization parameter B. By choosing the energy origin properly, the solutions for the symmetrical case are found as

$$E(S,\pm) = -\frac{J}{2}S(S+1) \pm B(S+1/2) \tag{13.3.5}$$

Let us apply Eqs. (13.3.4) and (13.3.5) to the Fe(II)Fe(III) case with high-spin local states. The energies are

$$E(1/2,\pm) = \pm\,\beta/5$$

$$E(3/2,\pm) = -3J/2 \pm 2\beta/5$$

$$E(5/2,\pm) = -4J \pm 3\beta/5$$

$$E(7/2,\pm) = -15J/2 \pm 4\beta/5$$

$$E(9/2,\pm) = -12J \pm \beta \tag{13.3.6}$$

Figure 13.6 represents the $E(S,\pm)/|J|$ versus $|\beta/J|$ variations for both J negative and positive. For $J < 0$, the ground state is $S = 1/2$ when $|\beta/J|$ is weak enough. When $|\beta/J|$ increases, the ground state becomes successively $S = 3/2$, 5/2, 7/2, and finally 9/2. The delocalization then dominates the antiferromagnetic interaction and the ground state has the highest spin. The $J > 0$ situation is simpler. Both J and β favor the stabilization of an $S = 9/2$ state.

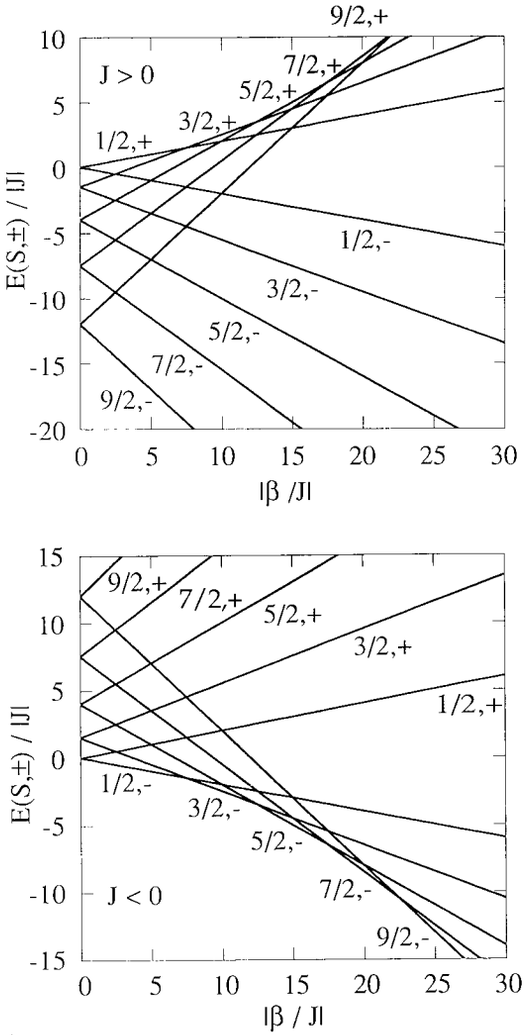


Figure 13.6. $E(S,\pm)/|J|$ *versus* $|\beta/J|$ curves for an Fe(II)Fe(III) compound, with $J<0$ and $J>0$.

The spin Hamiltonian (13.3.1) has been extended to the case where the A and B sites are no longer strictly identical. It may then be written as

$$\mathbf{H} = (E_A - {}^{A}J\,{}^{A}\mathbf{S}_A\cdot{}^{A}\mathbf{S}_B)O_A + (E_B - {}^{B}J\,{}^{B}\mathbf{S}_A\cdot{}^{B}\mathbf{S}_B)O_B + B\mathbf{T}_{AB} \qquad (13.3.7)$$

E_A is the zeroth-order energy of AB, and ${}^{A}J$ the interaction parameter when the transferable electron is on A. In the same way, E_B and ${}^{B}J$ refer to the case where the transferable electron is on B. The matrix elements of $\mathbf{H}$ in (13.3.7), using the $|A,S,M_S\rangle$ and $|B,S,M_S\rangle$ functions as a basis set are

$$\langle A,S,M_S|\mathbf{H}|A,S,M_S\rangle = E_A - {}^{A}J[S(S+1) - {}^{A}S_A({}^{A}S_A+1) - {}^{A}S_B({}^{A}S_B+1)]/2$$

$$\langle B,S,M_S|\mathbf{H}|B,S,M_S\rangle = E_B - {}^{B}J[S(S+1) - {}^{B}S_A({}^{B}S_A+1) - {}^{B}S_B({}^{B}S_B+1)]/2$$

$$\langle A,S,M_S|\mathbf{H}|B,S,M_S\rangle = \langle B,S,M_S|\mathbf{H}|A,S,M_S\rangle = B(S+1/2) \qquad (13.3.8)$$

The symmetrical case is included in (13.3.7), and corresponds to $E_A = E_B$ and ${}^{A}J = {}^{B}J = J$. The other borderline situation occurs when $E_B - E_A$ is very large with respect to the off-diagonal term. The states of lowest energy correspond to the situation where the transferable electron is on A. Their energies may be obtained through a perturbational approach as

$$E(S) = E_A - \frac{{}^{A}J}{2}[S(S+1) - {}^{A}S_A({}^{A}S_A+1) - {}^{A}S_B({}^{A}S_B+1)] - \frac{B^2(S+1/2)^2}{(E_B-E_A)} \qquad (13.3.9)$$

which can be rewritten as

$$E(S) = E_A - \frac{B^2}{4(E_B-E_A)} - \frac{{}^{A}J}{2}[S(S+1) - {}^{A}S_A({}^{A}S_A+1) - {}^{A}S_B({}^{A}S_B+1)]$$

$$- \frac{B^2S(S+1)}{(E_B-E_A)} \qquad (13.3.10)$$

or, by changing the energy origin, as

$$E(S) = -\frac{J_{eff}}{2}S(S+1) \qquad (13.3.11)$$

with

$$J_{eff} = {}^{A}J + \frac{2B^2}{(E_B-E_A)} \qquad (13.3.12)$$

which corresponds to the Lande interval rule. In other words, in the limit where $E_B - E_A$ is large, the low-lying states are again described by an HDVV Hamiltonian of the form $-J_{eff}\,{}^{A}\mathbf{S}_A\cdot{}^{A}\mathbf{S}_B$, and the effective interaction parameter contains both a contribution due to magnetic interaction and another contribution due to spin-dependent delocalization. This additional contribution is always positive. We will discuss this point further in the next section.

For completeness, it should be noticed that when the A and B sites are not identical, there is no univocal way of determining the orthogonalized orbitals (see Section 9.2). We will ignore this real difficulty in what follows.

13.4 Role of the Nuclear Relaxation

Let us return to the AB mixed-valence species in which the A and B sites are chemically equivalent, and reintroduce the vibrational coordinate Q defined in Section 13.1. The potential energy matrix between two localized states of spin S is

$\lvert A,S,M_S\rangle$	$\lvert B,S,M_S\rangle$
$-\frac{JS(S+1)}{2}+IQ+\frac{kQ^2}{2}$	$\frac{\beta(2S+1)}{2(2S_0+1)}$
	$-\frac{JS(S+1)}{2}-IQ+\frac{kQ^2}{2}$

(13.4.1)

The roots are

$$E(S,\pm)=-\frac{JS(S+1)}{2}+\frac{kQ^2}{2}\pm\left[l^2Q^2+\frac{\beta^2(2S+1)^2}{4(2S_0+1)^2}\right]^{1/2} \tag{13.4.2}$$

All the $E(S,+)=f(Q)$ curves present a minimum in $Q=0$. The $E(S,-)=f(Q)$ curves present an energy barrier between two minima for $\lambda > |\beta_{eff}(S)|$, and a unique minimum in $Q=0$ for $\lambda \leq |\beta_{eff}(S)|$, with

$$\beta_{eff}(S)=\frac{\beta(2S+1)}{2(2S_0+1)} \tag{13.4.3}$$

The effective electronic interaction between two localized states of spin S is spin dependent. It is maximum, and equal to β, for the highest-spin state, and decreases as S decreases. The valences may be delocalized for the high-spin states and trapped for the low-spin states. Such a situation is represented in Figure 13.7 for an Fe(II)Fe(III) species with $S_{Fe(II)}=2$, $S_{Fe(III)}=5/2$, and $J=-100\ cm^{-1}$, $|\beta|=3000\ cm^{-1}$, $\lambda=2000\ cm^{-1}$. In this example the vibronic coupling parameter λ is larger than $|\beta_{eff}(1/2)|$,

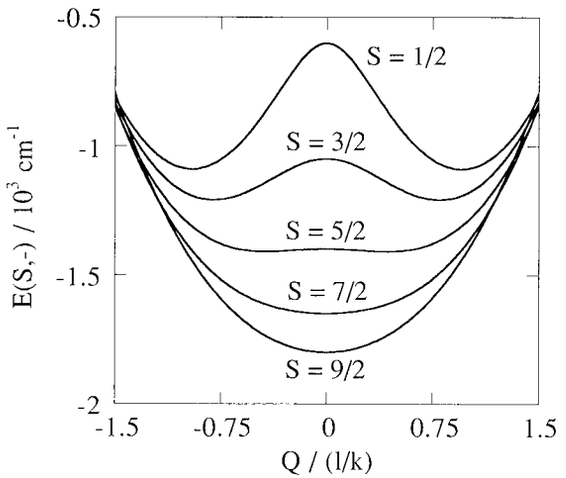


Figure 13.7. Potential energy curves for the lowest-lying spin states of a Fe(II)Fe(III) dinuclear system with $J=-100\ cm^{-1}$, $|\beta|=3000\ cm^{-1}$, and $\lambda=2000\ cm^{-1}$.

$|\beta_{eff}(3/2)|$, and $|\beta_{eff}(5/2)|$, but smaller than $|\beta_{eff}(7/2)|$ and $|\beta_{eff}(9/2)|$. In a certain sense we could say that the system belongs to class II of Robin and Day's classification in its $S=1/2$, 3/2, and 5/2 spin states, but belongs to class III in its $S=7/2$ and 9/2 spin states. The important message is that the electron transfer becomes easier as the spin increases.

Let us move to the case where the condition $\lambda > |\beta_{eff}(S)|$ holds for any S value. The vibronic coupling which favors the electron localization is large, and all energy minima $E(S,-)_{min}$ correspond to distorted geometries with inequivalent A and B sites. $E(S,-)_{min}$ may be easily calculated by looking for the Q value minimizing $E(S,-)$, and then introducing this value into (13.4.2). $E(S,-)_{min}$ is found as

$$E(S,-)_{min}=-\frac{J}{2}S(S+1)-\frac{\lambda}{2}-\frac{\beta_{eff}(S)^2}{2\lambda} \tag{13.4.4}$$

which may be reexpressed by changing the energy origin and introducing (13.4.3) into (13.4.4) as

$$E(S,-)_{min}=-\frac{J_{eff}}{2}S(S+1) \tag{13.4.5}$$

with

$$J_{eff}=J+\frac{\beta^2}{\lambda(2S_0+1)^2} \tag{13.4.6}$$

This result was demonstrated by Girerd et al. [13,14]. It indicates that when the valences are trapped, the HDVV Hamiltonian can be used again to find the relative energies of the low-lying states. The interaction parameter that occurs in the $-J_{eff}\mathbf{S}_A\cdot\mathbf{S}_B$ spin Hamiltonian then contains two contributions; one of these is the normal interaction parameter that may be negative or positive, the other is always positive, and involves both electron-transfer and vibronic-coupling parameters. The presence of this new positive term confirms that electron transfer favors ferromagnetic interaction.

13.5 Some Examples of Delocalized Mixed-Valence Pairs with Magnetic Cores

To now the discussion in this chapter is purely theoretical, and perhaps overwhelming for some readers. It may be useful to remind the reader that only the mixed-valence compounds are concerned. Moreover, when the valences are trapped, which is by far the most frequent case, the HDVV Hamiltonian introduced in Section 6.1 remains valid. However, the interaction parameter deduced from the fitting of the magnetic data contains a ferromagnetic contribution, in addition to the normal magnetic interaction contribution. This ferromagnetic contribution is related to the transfer integral between the two localized orbitals that may be occupied by the supplementary electron. It is only when the valences are delocalized that the HDVV Hamiltonian is no longer

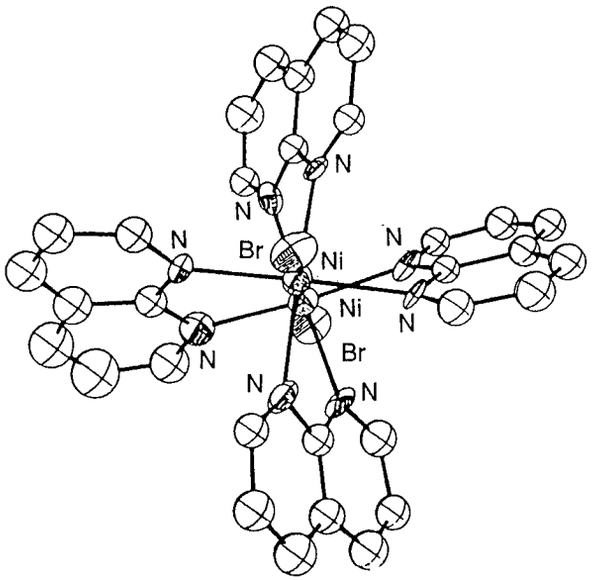
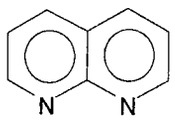


Figure 13.8. Molecular structure of the cation $[Ni_2(napy)_4Br_2]^+$ with napy = 1,8-naphthyridine (from ref. 13.20).

valid. The relative energies of the low-lying states do not respect the Lande interval rule.

Experimental data that allow substantiation of the concepts developed above are still rather rare. However, three convincing examples of the stabilization of a pair state of highest spin multiplicity through spin-dependent delocalization have been reported. The first one concerns the $[Ni_2(napy)_4Br_2][BPh_4]$ compound with napy = 1,8-naphthyridine [13.19–13.21]:



The dinuclear cation is shown in Figure 13.8. This cation formally contains a nickel(II) ion and a nickel(I) ion, both in square pyramidal surroundings, with spin triplet and spin doublet local ground states, respectively. From the X-ray crystal structure the two nickel atoms are identical, so that we may expect to be in the fully delocalized situation $S_A = 1$, $S_B = 1/2 \leftrightarrow S_A = 1/2$, $S_B = 1$ treated above. The $S = 3/2$ state has actually been found to be the ground state with a large ground quartet-excited doublet energy gap. If the valences in $[Ni_2(napy)_4Br_2]^+$ are delocalized, then the transfer integral between the two localized orbitals occupied by the traveling electron should be large (in absolute value). The symmetry of the metal sites causes this integral to involve the

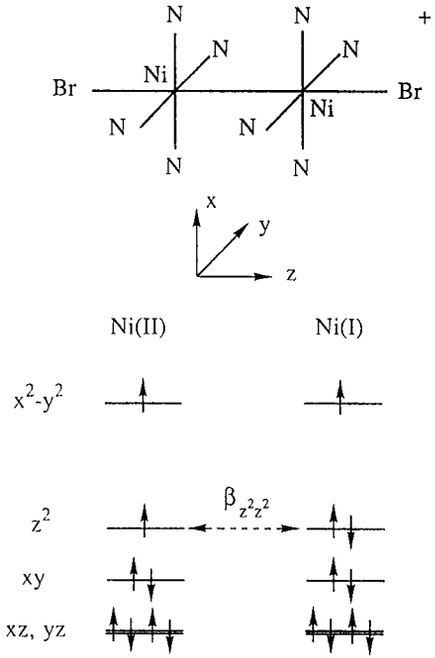


Figure 13.9. Relative energies of the metal orbitals in $[Ni_2(napy)_4Br_2]^+$.

z-type orbitals pointing toward each other, as shown in Figure 13.9. The Ni—Ni distance is only 2.41 Å, and $|\beta_{z^2z^2}|$ may indeed be large.

The second example deals with the reduced form of the Fe_3S_4 cluster contained in the ferredoxin II isolated from *Disulfovibrio gigas*. This cluster, whose structure is schematized in Figure 13.10, contains formally two iron(III) and one iron(II) ions. Mössbauer spectroscopy in zero field shows a doublet characteristic of an high-spin iron(III) with an $S_A = 5/2$ local state and another doublet corresponding to two indistinguishable iron atoms in the +2.5 average oxidation state [13.22]. These two irons form a delocalized pair $S_B = 5/2$, $S_C = 2 \leftrightarrow S_B = 2$, $S_C = 5/2$. Its ground state, also determined by Mössbauer spectroscopy, corresponds to the highest spin $S_{BC} = 9/2$. The ground state of the cluster as a whole is $S_{ABC} = 2$, which arises from the antiferromagnetic interaction between the local spin $S_A = 5/2$ and the pair spin $S_{BC} = 9/2$. Münck et al. attributed the stabilization of the $S_{BC} = 9/2$ state for the delocalized Fe(II)Fe(III) pair to spin-dependent delocalization [13.11–13.13]. Even if this interpretation seems quite plausible, why the delocalization within the trinuclear species occurred on a pair, and not on the three metal sites as a whole remained a mystery. In 1989 Borshch demonstrated that vibrations together with electron transfer within a $Fe(II)Fe(III)_2$ regular triangle could induce such a pair delocalization [13.23].

The third example also concerns a fully delocalized Fe(II)Fe(III) dinuclear compound. Its formula is $[LFe(OH)_3FeL](ClO_4)_2 \cdot 2CH_3OH \cdot 2H_2O$, and the structure of the dication is schematized in Figure 13.11. The two iron atoms are triply bridged by

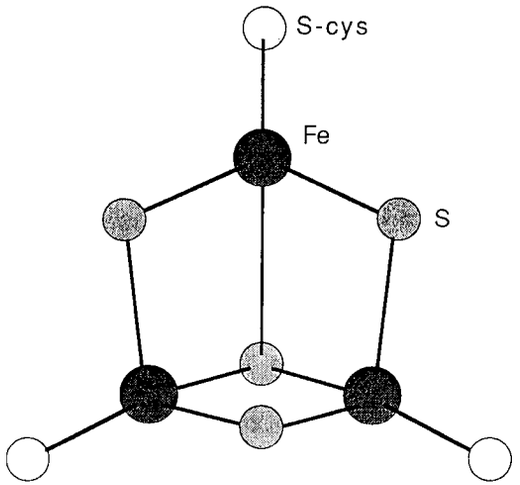


Figure 13.10. Schematic structure of the protein with the $[Fe_3S_4]$ core. S-cys stands for the sulfur atom of a cystein group.

hydroxo groups, and the peripheral ligands are *N,N',N''*-1,4,7-trimethyltriazacyclononane. From Mössbauer and EPR spectroscopies, and magnetic susceptibility measurements, the parallel spin state $S = 9/2$ was found to be the lowest in energy. Moreover, only this state is thermally populated up to room temperature. The intervalence band appears at 13,200 cm^{-1} with a strong intensity. The energy gap between the $E(9/2,+)$ and $E(9/2,-)$ states is $2|\beta|$ since the system is delocalized, which provides an accurate value for the transfer integral, $|\beta| = 6600$ cm^{-1}. The J interaction parameter, on the other hand, remains unknown [13.24, 13.25].

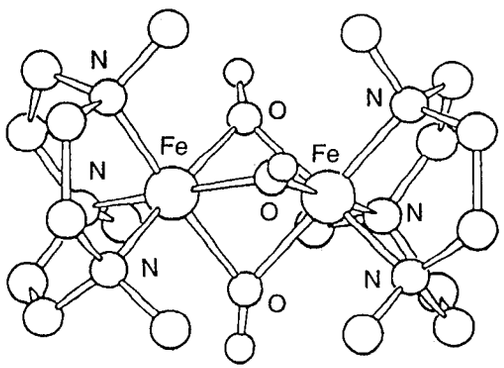


Figure 13.11. Molecular structure of $[LFe(OH)_3FeL]^{2+}$, with L = *N,N',N''*-1,4,7-trimethyltriazacyclononane. This structure is a model deduced from EXAFS data. The X-ray diffraction data were not yet published as completion of this book (from ref. [13.25]).

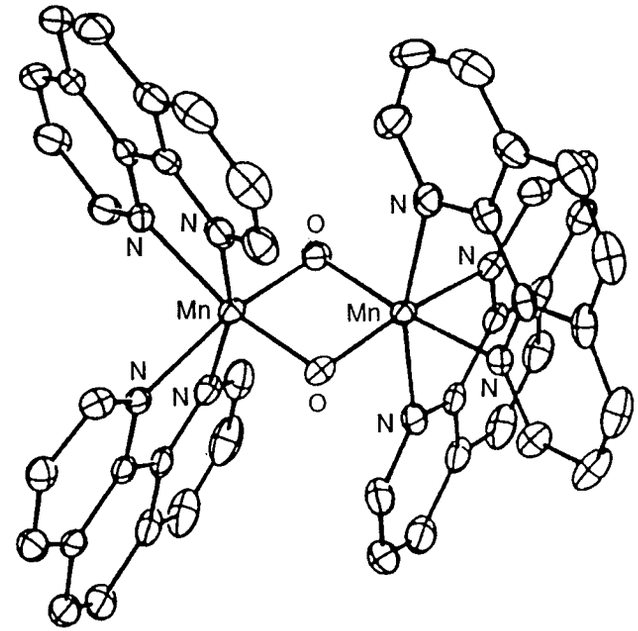


Figure 13.12. Molecular structure of the trication $[(phen)MnO_2Mn(phen)]^{3+}$ deduced from X-ray diffraction data. The two manganese sites are chemically inequivalent. They are crystallographically symmetry related through a twofold rotation axis due to a disorder effect (from ref. [13.26]).

Let us return to our first example for a moment. We mentioned that the two nickel sites were crystallographically equivalent, which suggested delocalized valences. We would like to point out here that the equivalence of the sites deduced from X-ray diffraction is not a definitive proof of electron delocalization in mixed-valence species. An example will help us to understand this problem. The crystal structure of the $[(phen)MnO_2Mn(phen)](PF_6)_3 \cdot CH_3CN$ mixed-valence compound has been solved by Bürgi et al. [13.26]. The structure of the Mn(III)Mn(IV) cation is shown in Figure 13.12. The two manganese sites are found to be strictly equivalent with a twofold rotation axis relating the two metal sites. While this could indicate that the sites are chemically equivalent, and that the compound is delocalized, it could also be due to a static disorder resulting from the crystallographic superposition of an equal number of Mn(III)Mn(IV) and Mn(IV)Mn(III) cations, or from a dynamic disorder due to rapid electron transfer between the two sites on the crystallographic time scale. A very detailed analysis of anisotropic displacement parameters allows to rule out the delocalization hypothesis. The sites within a same dinuclear unit actually are not chemically equivalent. Mn(III) has a d^4 configuration, and undergoes a Jahn–Teller distortion with a lengthening of the axial Mn—N bonds, while Mn(III) with the d^3 configuration is roughly spherical. In the disordered structure the X-ray diffraction sees an averaging of the nitrogen positions, leading to a large apparent thermal motion

of the axial nitrogen atoms along the Mn—N bond direction. The magnetic properties are perfectly in line with the fact that the valences are localized. The $S_{Mn(III)} = 2$ and $S_{Mn(IV)} = 3/2$ local spins are antiferromagnetically coupled, with a pair doublet ground state, and a J interaction parameter equal to -296 cm^{-1}. Other localized Mn(III)Mn(IV) compounds present the same averaging effect in the solid state [13.27]. In contrast, the X-ray structure of $[(bipy)MnO_2Mn(bipy)](ClO_4)_3 \cdot 3H_2O$ clearly distinguishes a Jahn–Teller distorted Mn(III) site and a much less distorted Mn(IV) site [13.28].

13.6 Mixed-Valence Compounds of Higher Nuclearity

The theories appropriate for the trinuclear or tetranuclear mixed-valence compounds with more than one unpaired electron are not yet fully established, and unambiguous experimental data are rare. Therefore, we will treat of this problem only briefly. A point deserves to be mentioned concerning the theory. In a system with three sites or more and a number of active electrons which is different from the number of sites, a new term was introduced in the description of the low-lying state, called exchange-transfer by Blondin and Girerd [13.14, 13.29]. Let us show how this term operates in the simple case of a triangle with three site orbitals a, b, and c, and only two electrons. Each electron can move from one site to another provided that the arrival site is not already doubly occupied; this is electron transfer. Two unpaired electrons occupying neighboring sites can interact; this is magnetic interaction. In addition to these two effects, there is a novel effect that operates when two electrons of opposite spins occupy two neighboring sites, say a and b. The electron on a can move toward c via the double occupation of b, as schematized in Figure 13.13. This effect depends on both the antiferromagnetic interaction between the electrons on a and b, and the transfer integral between b and c. It appears at the same perturbation order as the magnetic interaction in an orthogonalized orbital approach [13.30]. We will no longer speak about exchange-transfer. We will note, however, that a term of this kind has been taken into account by some solid states physicists working on the theory of superconductivity in copper oxides.

Let us now consider some actual compounds. In Section 10.4 we mentioned that the valences in Fe_4S_4 proteins or model derivatives were delocalized. To our knowledge, Noodelman was the first to introduce explicitly spin-dependent delocalization in the study of these systems (he coined the term resonance delocalization). He first focused on the $[Fe_4S_4]^{3+}$ core, and assumed that this species contained a localized Fe(III)Fe(III) pair with A and B sites, and a delocalized Fe(II)Fe(III) pair with C and D sites [13.31, 13.32]. Taking into account magnetic interaction and spin-dependent delocalization (or resonance delocalization) the spin Hamiltonian is

$$\mathbf{H} = -J_1\mathbf{S}_A\cdot\mathbf{S}_B - J_2\mathbf{S}_C\cdot\mathbf{S}_D - J(\mathbf{S}_A\cdot\mathbf{S}_C + \mathbf{S}_A\cdot\mathbf{S}_D + \mathbf{S}_B\cdot\mathbf{S}_C + \mathbf{S}_B\cdot\mathbf{S}_D) \pm B\mathbf{T}_{CD} \quad (13.6.1)$$

$\mathbf{T}_{CD}$ being the spin-dependent delocalization operator defined in (13.3.3), and acting within the CD pair. Equation (13.6.1) may be rewritten as

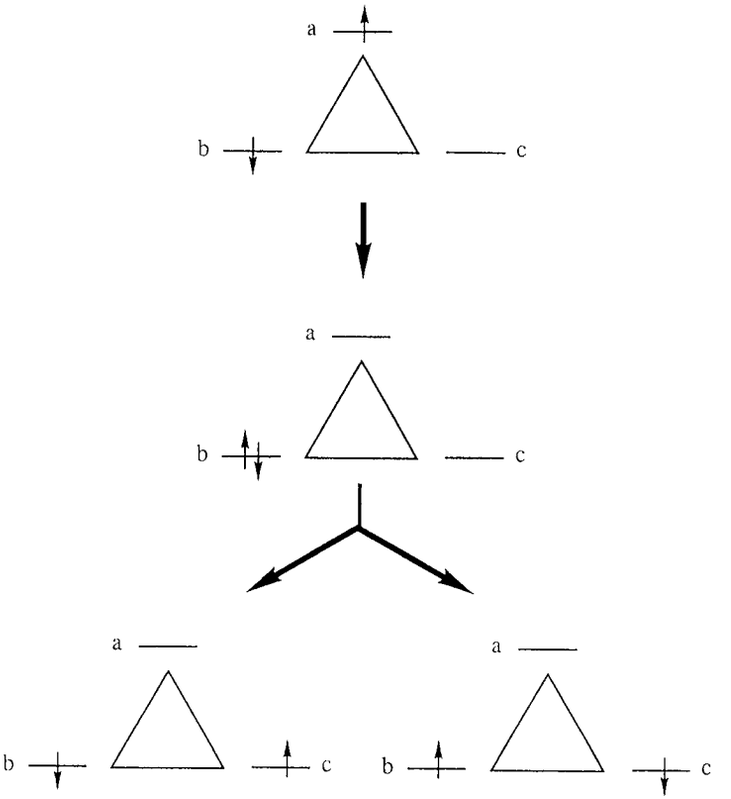


Figure 13.13. Effect of the exchange-transfer in a triangle with three site orbitals and two electrons.

$$\mathbf{H} = -J(\mathbf{S}_A\cdot\mathbf{S}_B + \mathbf{S}_A\cdot\mathbf{S}_C + \mathbf{S}_A\cdot\mathbf{S}_D + \mathbf{S}_B\cdot\mathbf{S}_C + \mathbf{S}_B\cdot\mathbf{S}_D + \mathbf{S}_C\cdot\mathbf{S}_D) - \Delta J_1\mathbf{S}_A\cdot\mathbf{S}_B - \Delta J_2\mathbf{S}_C\cdot\mathbf{S}_D \pm B\mathbf{T}_{CD} \quad (13.6.2)$$

with

$$\Delta J_1 = J_1 - J$$
$$\Delta J_2 = J_2 - J \quad (13.6.3)$$

J_1 is the interaction parameter within the localized AB pair, J_2 within the delocalized CD pair, and J between an ion from AB and an ion from CD. B is the parameter introduced in (13.3.1). S_{AB} and S_{CD} are the quantum numbers associated with the spin operators for the AB and CD pairs, respectively. The eigenvalues of (13.6.2) are

$$E(S,S_{AB},S_{CD}) = -JS(S+1)/2 \pm B(S_{CD}+1/2) - \Delta J_1 S_{AB}(S_{AB}+1)/2 - \Delta J_2 S_{CD}(S_{CD}+1)/2 \quad (13.6.4)$$

J_1 is thought to be the most negative interaction parameter. The ground state is then $E(1/2,4,9/2)$. The electron delocalization in the CD pair induces $S_{CD} = 9/2$. In spite of

the antiferromagnetic interaction in the AB pair, S_{AB} is equal to 4; this is an example of spin frustration due to the antiferromagnetic coupling characterized by the J parameter. This model was used to interpret the magnetic susceptibility data of $(Et_4N)[Fe_4S_4(2,4,6\text{-}i\text{Pr}_3\text{SPh})_4]$. Best fit parameters settled at $J = -652$ cm^{-1}, $J_1 = -797$ cm^{-1}, and $B = 592$ cm^{-1} (which corresponds to a transfer integral $|\beta| = 2960$ cm^{-1}). J_2 cannot be determined due to the delocalization in the CD pair, which favors the parallel spin alignment. With these parameters the first excited state, $E(1/2,3,7/2)$, is located at only 11 cm^{-1} above the ground state $E(1/2,4,9/2)$. Noodelman's basic hypothesis was that delocalization occurs only within a pair, and not within the tetranuclear cluster as a whole. Some recent results in Mössbauer spectroscopy do confirm this hypothesis. However, Noodelman's model does not explain why the additional electron is trapped within a pair. The understanding of the microscopic origin of this trapping would represent a major breakthrough in the theory of iron–sulfur proteins.

Subsequently, Noodelman studied the $[Fe_4S_4]^+$ and $[Fe_4Se_4]^+$ cores of proteins or synthetic analogues with formally three Fe(II) and one Fe(III) ion [13.33]. In Section 10.4 we saw that the problem is more complicated than for the $[Fe_4S_4]^{3+}$ core. As a matter of fact, if the ground state is generally $S = 1/2$, it may be also $S = 3/2$, or an $S = 1/2$–$3/2$ spin-admixed. For the reduced Se ferredoxin the coexistence of $S = 1/2$ and $S = 7/2$ has been reported. The model developed by Noodelman is much the same as for the $[Fe_4S_4]^{3+}$ core. The delocalization is again assumed to occur only in a Fe(II)Fe(III) pair, the other Fe(II)Fe(II) pair being fully localized. This model nicely accounts for the wide variety of ground state spins. However, as in the case above, the origin of the trapping in a pair is not explained yet.

References

13.1. D.B. Brown, "Mixed Valence Compounds," NATO ASI. Reidel, Dordrecht, 1980.

13.2. K. Prassides, "Mixed Valency Systems: Applications in Chemistry, Physics and Biology," NATO ASI. Kluwer, Dordrecht, 1991.

13.3. M.B. Robin, P. Day, *Adv. Inorg. Chem. Radiochem.* 10 (1967), 247.

13.4. N.S. Hush, *Prog. Inorg. Chem.* 8 (1967), 391.

13.5. S.B. Piepho, E.R. Krausz, P.N. Schatz, *J. Am. Chem. Soc.* 100 (1978), 2996.

13.6. K.Y. Wong, P.N. Schatz, Prog. *Inorg. Chem.* 28 (1981), 369.

13.7. C. Creutz, H. Taube, *J. Am. Chem. Soc.* 91 (1969), 3988; 95 (1973), 1086.

13.8. C. Zener, *Phys. Rev.* 82 (1951), 403.

13.9. P.W. Anderson, H. Hasegawa, *Phys. Rev.* 100 (1955), 675.

13.10. J.J. Girerd, *J. Chem. Phys.* 1979 (1983), 1766.

13.11. V. Papaefthymiou, J.J. Girerd, I. Moura, J.J.G. Moura, E. Münck, *J. Am. Chem. Soc.* 109 (1987), 4703.

13.12. E. Münck, V. Papaefthymiou, K.K. Surerus, J.J. Girerd, in "Metal Clusters in Proteins," L. Que, ed., ACS Symposium Series 372. ACS, Washington, D.C., 1988.

13.13. J.J. Girerd, V. Papaefthymiou, K.K. Surerus, E. Münck, *J. Pure Appl. Chem.* 61 (1989), 805.

13.14. G. Blondin, J.J. Girerd, *Chem. Rev.* 90 (1990), 1359.

13.15. M.I. Belinskii, B.S. Tsukerblat, N.V. Gérbeléu, *Sov. Phys. Solid State* 25 (1983), 497.

13.16. S.A. Borshch, I.N. Kotov, I.B. Bersuker, Sov. *J. Chem. Phys.* 3 (1985), 1009.

13.17. M. Drillon, G. Pourroy, J. Darriet, *J. Chem. Phys.* 88 (1984), 27.

13.18. L. Noodelman, E.J. Baerends, *J. Am. Chem. Soc.* 106 (1984), 2316.

13.19. D. Gatteschi, C. Mealli, L. Sacconi, *J. Am. Chem. Soc.* 95 (1973), 2736.

13.20. L. Sacconi, C. Mealli, D. Gatteschi, *Inorg. Chem.* 13 (1974), 1985.

13.21. A. Bencini, D. Gatteschi, L. Sacconi, *Inorg. Chem.* 17 (1978), 2670.

13.22. E. Münck, T.A. Kent, *Hyp. Int.* 27 (1986), 161.

13.23. S.A. Borshch, L.F. Chibotaru, *Chem. Phys.* 135 (1989), 375.

13.24. S. Drücke, P. Chaudhuri, K. Pohl, K. Wieghardt, X.Q. Ding, E. Bill, A. Sawaryn, A.X. Trautwein, H. Winkler, S.J. Gurman, *J. Chem. Soc. Chem. Commun.* (1989), 59.

13.25. X.Q. Ding, E.L. Bominaar, E. Bill, H. Winkler, A.X. Trautwein, S. Drüeke, P. Chauduri, K. Wieghardt, *J. Chem. Phys.* 92 (1990), 178.

13.26. M. Stebler, A. Ludi, H.B. Bürgi, *Inorg. Chem.* 25 (1986), 4743.

13.27. P.A. Goodson, D.J. Hodgson, K. Michelsen, *Inorg. Chim. Acta* 172 (1990), 49.

13.28. P.M. Plaksin, R.C. Stoufer, M. Mathew, G.J. Palenik, *J. Am. Chem. Soc.* 94 (1972), 2121.

13.29. G. Blondin, S. Borshch, J.J. Girerd, *Comments Inorg Chem.* 12 (1992), 315.

13.30. M.I. Belinskii, *Mol. Phys.* 60 (1987), 793.

13.31. L. Noodelman, *Inorg. Chem.* 27 (1988), 3677.

13.32. J. Jordanov, E.K.H. Roth, P.H. Fries, L. Noodelman, *Inorg. Chem.* 29 (1990), 4288.

13.33. L. Noodelman, *Inorg. Chem.* 30 (1991), 246; 30 (1991), 256.

Appendix 1
Physical Constants and Units

1.1 Some Fundamental Physical Constants in SI

Planck constant:	h	$6.6260755\ 10^{-34}$ J s
	$h/2\pi$	$1.05457267\ 10^{-34}$ J s
Elementary charge	e	$1.60217733\ 10^{-19}$ C
Electron mass	m	$9.1093897\ 10^{-31}$ kg
Proton mass	m_p	$1.6726231\ 10^{-27}$ kg
Avogadro number	N	$6.0221367\ 10^{23}$ mol^{-1}
Molar gas constant	R	8.3145121 J mol^{-1}
Boltzmann constant	k	$1.3806580\ 10^{-23}$ J K^{-1}
Bohr magneton	β	$9.27401549\ 10^{-24}$ J T^{-1}
Nuclear magneton	β_N	$5.05078647\ 10^{-27}$ T^{-1}

1.2 Values of the Boltzmann Constant and Bohr Magneton Used in this Book

As mentioned in Chapter 1, we utilize the cgsemu system throughout the book. The two essential constants in magnetism k and β are then given by

Boltzmann constant	k	$1.3806580\ 10^{-16}$ erg K^{-1}
Bohr magneton	β	$9.27401549\ 10^{-21}$ erg T^{-1}

so that the constant $N\beta^2/3k$ appearing in the Curie law (12.1.5) is equal to 0.125048612 cm^3 mol^{-1}.

If the energies are expressed in cm^{-1}, k and β become

Boltzmann constant	k	0.69503877 cm^{-1} K^{-1}
Bohr magneton	β	$4.66864374\ 10^{-5}$ cm^{-1} G^{-1}

1.3 Conversion of Energy Units

	a.u.	J	eV	kcal mol^{-1}
1 a.u.	1.	4.35974812D–18	2.72113957D+01	6.27509541D+02
1 J	2.29371049D+17	1.	6.24150636D+18	1.43932522D+20
1 eV	3.67493095D–02	1.60217733D–19	1.	2.30605423D+01
1 kcal mol^{-1}	1.59360127D–03	6.94770014D–21	4.33641146D–02	1.
1 cm^{-1}	4.55633538D–06	1.98644746D–23	1.23984245D–04	2.85914392D–03
1 MHz	1.51982989D–10	6.62607550D–28	4.13566924D–09	9.53707754D–08
1 K	3.16682974D–06	1.38065800D–23	8.61738569D–05	1.98721587D–03
1 kJ mol^{-1}	3.80879844D–04	1.66054019D–21	1.03642721D–02	2.39005736D–01

	cm^{-1}	MHz	K	kJ mol^{-1}
1 a.u.	2.19474625D+05	6.57968374D+09	3.15773213D+05	2.62549992D+03
1 J	5.03411250D+22	1.50918896D+27	7.24292330D+22	6.02213670D+20
1 eV	8.06554093D+03	2.41798834D+08	1.16044475D+04	9.64853090D+01
1 kcal mol^{-1}	3.49755041D+02	1.04853924D+07	5.03216592D+02	4.18400000D+00
1 cm^{-1}	1.	2.99792458D+04	1.43876866D+00	1.19626582D–02
1 MHz	3.33564095D–05	1.	4.79921566D–05	3.99031324D–07
1 K	6.95038770D–01	2.08367381D+04	1.	8.31451121D–03
1 kJ mol^{-1}	8.35934612D+01	2.50606892D+06	1.20271652D+02	1.

Appendix 2
Action of $\mathbf{L}_x$, $\mathbf{L}_y$, and $\mathbf{L}_z$ on the *d* Orbitals

	$\mathbf{L}_x$	$\mathbf{L}_y$	$\mathbf{L}_z$
d_{z^2}	$-i\sqrt{3}d_{yz}$	$i\sqrt{3}d_{zx}$	0
$d_{x^2-y^2}$	$-id_{yz}$	$-id_{zx}$	$2id_{xy}$
d_{xy}	id_{zx}	$-id_{yz}$	$-2id_{x^2-y^2}$
d_{yz}	$id_{x^2-y^2} + i\sqrt{3}\, d_{z^2}$	id_{xy}	$-id_{zx}$
d_{zx}	$-id_{xy}$	$-i\sqrt{3}d_{z^2} + id_{x^2-y^2}$	id_{yz}

Appendix 3
Matrix Elements of $\mathbf{L}_x$, $\mathbf{L}_y$, and $\mathbf{L}_z$ Using the *d* Orbitals as a Basis Set

	d_z^2	$d_{x^2-y^2}$	d_{xy}	d_{yz}	d_{zx}
$\mathbf{L}_x =$	0	0	0	$i\sqrt{3}$	0
	0	0	0	i	0
	0	0	0	0	$-i$
	$-i\sqrt{3}$	$-i$	0	0	0
	0	0	i	0	0
$\mathbf{L}_y =$	0	0	0	0	$-i\sqrt{3}$
	0	0	0	0	i
	0	0	0	i	0
	0	0	$-i$	0	0
	$i\sqrt{3}$	$-i$	0	0	0
$\mathbf{L}_z =$	0	0	0	0	0
	0	0	$-2i$	0	0
	0	$2i$	0	0	0
	0	0	0	0	i
	0	0	0	$-i$	0

Appendix 4
Action of the Spin Operators $\mathbf{S}_x$, $\mathbf{S}_y$, and $\mathbf{S}_z$ on the Spin Functions

4.1 S = 1/2

	$\|1/2\rangle$	$\|-1/2\rangle$
$\mathbf{S}_x =$	0	1/2
	1/2	0
$\mathbf{S}_y =$	0	$i/2$
	$-i/2$	0
$\mathbf{S}_z =$	1/2	0
	0	−1/2

4.2 S = 1

	$\|1\rangle$	$\|0\rangle$	$\|-1\rangle$
$\mathbf{S}_x =$	0	$\sqrt{2}/2$	0
	$\sqrt{2}/2$	0	$\sqrt{2}/2$
	0	$\sqrt{2}/2$	0
$\mathbf{S}_y =$	0	$-i\sqrt{2}/2$	0
	$i\sqrt{2}/2$	0	$-i\sqrt{2}/2$
	0	$i\sqrt{2}/2$	0
$\mathbf{S}_z =$	1	0	0
	0	0	0
	0	0	−1

4.3 S = 3/2

	|3/2>	|1/2>	|−1/2>	|−3/2>
S_x =	0	$\sqrt{3}/2$	0	0
	$\sqrt{3}/2$	0	1	0
	0	1	0	$\sqrt{3}/2$
	0	0	$\sqrt{3}/2$	0
S_y =	0	$-i\sqrt{3}/2$	0	0
	$i\sqrt{3}/2$	0	$-i$	0
	0	i	0	$-i\sqrt{3}/2$
	0	0	$i\sqrt{3}/2$	0
S_z =	3/2	0	0	0
	0	1/2	0	0
	0	0	−1/2	0
	0	0	0	−3/2

4.4 S = 2

	|2>	|1>	|0>	|−1>	|−2>
S_x =	0	1	0	0	0
	1	0	$\sqrt{3/2}$	0	0
	0	$\sqrt{3/2}$	0	$\sqrt{3/2}$	0
	0	0	$\sqrt{3/2}$	0	1
	0	0	0	1	0
S_y =	0	$-i$	0	0	0
	i	0	$-i\sqrt{3/2}$	0	0
	0	$i\sqrt{3/2}$	0	$-i\sqrt{3/2}$	0
	0	0	$i\sqrt{3/2}$	0	$-i$
	0	0	0	i	0
S_z =	2	0	0	0	0
	0	1	0	0	0
	0	0	0	0	0
	0	0	0	−1	0
	0	0	0	0	−2

4.5 S = 5/2

	|5/2>	|3/2>	<|1/2>	|−1/2>	|−3/2>	|−5/2>
S_x =	0	$\sqrt{5}/2$	0	0	0	0
	$\sqrt{5}/2$	0	$\sqrt{2}$	0	0	0
	0	$\sqrt{2}$	0	3/2	0	0
	0	0	3/2	0	$\sqrt{2}$	0
	0	0	0	$\sqrt{2}$	0	$\sqrt{5}/2$
	0	0	0	0	$\sqrt{5}/2$	0
S_y =	0	$-i\sqrt{5}/2$	0	0	0	0
	$i\sqrt{5}/2$	0	$-i\sqrt{2}$	0	0	0
	0	$i\sqrt{2}$	0	$-3i/2$	0	0
	0	0	$3i/2$	0	$-i\sqrt{2}$	0
	0	0	0	$i\sqrt{2}$	0	$-i\sqrt{5}/2$
	0	0	0	0	$i\sqrt{5}/2$	0
S_z =	5/2	0	0	0	0	0
	0	3/2	0	0	0	0
	0	0	1/2	0	0	0
	0	0	0	−1/2	0	0
	0	0	0	0	−3/2	0
	0	0	0	0	0	−5/2

Appendix 5
Escalator Operators $\mathbf{S}_+$ and $\mathbf{S}_-$ and $\mathbf{L}_+$ and $\mathbf{L}_-$

The spin escalator operators $\mathbf{S}_+$ and $\mathbf{S}_-$ are defined through

$$\mathbf{S}_+ = \mathbf{S}_x + i\mathbf{S}_y$$

$$\mathbf{S}_- = \mathbf{S}_x - i\mathbf{S}_y$$

hence

$$\mathbf{S}_x = (\mathbf{S}_+ + \mathbf{S}_-)/2$$

$$\mathbf{S}_y = (\mathbf{S}_+ - \mathbf{S}_-)/2i$$

$\mathbf{S}_+$ and $\mathbf{S}_-$ act on the spin functions $|S, M_S>$ as follows:

$$\mathbf{S}_+ \,|\, S,M_S> = [(S + M_S + 1)(S - M_S)]^{1/2} \,|\, S,M_S + 1> \text{ if } M_S < S$$

$$= 0 \text{ if } M_S = S$$

$$\mathbf{S}_- \,|\, S,M_S> = [(S - M_S + 1)(S + M_S)]^{1/2} \,|\, S,M_S - 1> \text{ if } M_S > -S$$

$$= 0 \text{ if } M_S = -S$$

The action of the orbital escalator operators $\mathbf{L}_+$ and $\mathbf{L}_-$ defined through

$$\mathbf{L}_+ = \mathbf{L}_x + i\mathbf{L}_y$$

$$\mathbf{L}_- = \mathbf{L}_x - i\mathbf{L}_y$$

is quite similar to that of the spin escalator operators with L replacing S, and M_L replacing M_S.

Appendix 6
Coupling of Two Spins S_A and S_B, and Clebsch–Gordon (or Wigner) Coefficients

Let us consider two local spins S_A and S_B with the local kets $|S_A,M_A\rangle$ and $|S_B,M_B\rangle$ (M_A and M_B stand for M_{S_A} and M_{S_B}, respectively). The product of these kets is noted $|S_A,S_B,M_A,M_B\rangle$. The coupling between these local spins gives rise to pair states with the pair functions $|S_A,S_B,S,M\rangle$ (M stands for M_S). S varies by an integer value from $|S_A-S_B|$ to S_A+S_B, and M varies by an integer value from $-S$ to $+S$. The $|S_A,S_B,S,M\rangle$ functions may be constructed as linear combinations of the $|S_A,S_B,M_A,M_B\rangle$ products:

$$|S_A,S_B,S,M\rangle = \sum_{M_A,M_B} |S_A,S_B,M_A,M_B\rangle\langle S_A,S_B,M_A,M_B|S_A,S_B,S,M\rangle,$$

where the $\langle S_A,S_B,M_A,M_B|S_A,S_B,S,M\rangle$ quantities are known as Clebsch–Gordon coefficients. The general expression of these coefficients has been first derived as

$$\langle S_A,S_B,M_A,M_B|S_A,S_B,S,M\rangle =$$

$$\delta(M,M_A+M_B)\{[(S_A+S_B-S)!(S+S_A-S_B)!(S+S_B-S_A)!$$

$$(2S+1)]/(S+S_A+S_B+1)!\}^{1/2}\,\Sigma_k K$$

where $\delta(M,M_A+M_B)$ is the Kronecker symbol. K is defined as

$$K = (-1)^k[(S_A+M_A)!(S_A-M_A)!(S_B+M_B)!$$

$$(S_B-M_B)!(S+M)!(S-M)!]^{1/2}/[k!(S_A+S_B-S-k)!$$

$$(S_A-M_A-k)!(S_B+M_B-k)!(S-S_B+M_A+k)!(S-S_A-M_B+k)!]$$

The expression of $\langle S_A,S_B,M_A,M_B|S_A,S_B,S,M\rangle$ above is of course rather cumbersome to utlize and the results of some specific cases are gathered below. For the sake of simplicity, the labels S_A and S_B have been omitted in the kets of both the left- and right-hand sides of the following expressions.

$S_A = 1/2$ and $S_B = 1/2$

$$|1,1\rangle = |1/2,1/2\rangle$$
$$|1,0\rangle = \sqrt{1/2}\,|1/2,-1/2\rangle + \sqrt{1/2}\,|-1/2,1/2\rangle$$
$$|1,-1\rangle = |-1/2,-1/2\rangle$$

$$|0,0\rangle = \sqrt{1/2}\,|1/2,-1/2\rangle - \sqrt{1/2}\,|-1/2,1/2\rangle$$

$S_A = 1$ and $S_B = 1/2$

$$|3/2,3/2\rangle = |1,1/2\rangle$$
$$|3/2,1/2\rangle = \sqrt{1/3}\,|1,-1/2\rangle + \sqrt{2/3}\,|0,1/2\rangle$$
$$|3/2,-1/2\rangle = \sqrt{2/3}\,|0,-1/2\rangle + \sqrt{1/3}\,|-1,1/2\rangle$$
$$|3/2,-3/2\rangle = |-1,-1/2\rangle$$

$$|1/2,1/2\rangle = \sqrt{2/3}\,|1,-1/2\rangle - \sqrt{1/3}\,|0,1/2\rangle$$
$$|1/2,-1/2\rangle = \sqrt{1/3}\,|0,-1/2\rangle - \sqrt{2/3}\,|-1,1/2\rangle$$

$S_A = 3/2$ and $S_B = 1/2$

$$|2,2\rangle = |3/2,1/2\rangle$$
$$|2,1\rangle = \sqrt{1/4}\,|3/2,-1/2\rangle + \sqrt{3/4}\,|1/2,1/2\rangle$$
$$|2,0\rangle = \sqrt{1/2}\,|1/2,-1/2\rangle + \sqrt{1/2}\,|-1/2,1/2\rangle$$
$$|2,-1\rangle = \sqrt{3/4}\,|-1/2,-1/2\rangle + \sqrt{1/4}\,|-3/2,1/2\rangle$$
$$|2,-2\rangle = |-3/2,-1/2\rangle$$

$$|1,1\rangle = \sqrt{3/4}\,|3/2,-1/2\rangle - \sqrt{1/4}\,|1/2,1/2\rangle$$
$$|1,0\rangle = \sqrt{1/2}\,|1/2,-1/2\rangle - \sqrt{1/2}\,|-1/2,1/2\rangle$$
$$|1,-1\rangle = \sqrt{1/4}\,|-1/2,-1/2\rangle - \sqrt{3/4}\,|-3/2,1/2\rangle$$

$S_A = 2$ and $S_B = 1/2$

$$|5/2,5/2\rangle = |2,1/2\rangle$$
$$|5/2,3/2\rangle = \sqrt{1/5}\,|2,-1/2\rangle + \sqrt{4/5}\,|1,1/2\rangle$$
$$|5/2,1/2\rangle = \sqrt{2/5}\,|1,-1/2\rangle + \sqrt{3/5}\,|0,1/2\rangle$$
$$|5/2,-1/2\rangle = \sqrt{3/5}\,|0,-1/2\rangle + \sqrt{2/5}\,|-1,1/2\rangle$$
$$|5/2,-3/2\rangle = \sqrt{4/5}\,|-1,-1/2\rangle + \sqrt{1/5}\,|-2,1/2\rangle$$
$$|5/2,-5/2\rangle = |-2,-1/2\rangle$$

$$|3/2,3/2\rangle = \sqrt{4/5}\,|2,-1/2\rangle - \sqrt{1/5}\,|1,1/2\rangle$$
$$|3/2,1/2\rangle = \sqrt{3/5}\,|1,-1/2\rangle - \sqrt{2/5}\,|0,1/2\rangle$$

$$|3/2,-1/2\rangle = \sqrt{2/5}\,|0,-1/2\rangle - \sqrt{3/5}\,|-1,1/2\rangle$$
$$|3/2,-3/2\rangle = \sqrt{1/5}\,|-1,-1/2\rangle - \sqrt{4/5}\,|-2,1/2\rangle$$

$S_A = 5/2$ and $S_B = 1/2$

$$|3,3\rangle = |5/2,1/2\rangle$$
$$|3,2\rangle = \sqrt{1/6}\,|5/2,-1/2\rangle + \sqrt{5/6}\,|3/2,1/2\rangle$$
$$|3,1\rangle = \sqrt{1/3}\,|3/2,-1/2\rangle + \sqrt{2/3}\,|1/2,1/2\rangle$$
$$|3,0\rangle = \sqrt{1/2}\,|1/2,-1/2\rangle + \sqrt{1/2}\,|-1/2,1/2\rangle$$
$$|3,-1\rangle = \sqrt{2/3}\,|-1/2,-1/2\rangle + \sqrt{1/3}\,|-3/2,1/2\rangle$$
$$|3,-2\rangle = \sqrt{5/6}\,|-3/2,-1/2\rangle + \sqrt{1/6}\,|-5/2,1/2\rangle$$
$$|3,-3\rangle = |-5/2,-1/2\rangle$$

$$|2,2\rangle = \sqrt{5/6}\,|5/2,-1/2\rangle - \sqrt{1/6}\,|3/2,1/2\rangle$$
$$|2,1\rangle = \sqrt{2/3}\,|3/2,-1/2\rangle - \sqrt{1/3}\,|1/2,1/2\rangle$$
$$|2,0\rangle = \sqrt{1/2}\,|1/2,-1/2\rangle - \sqrt{1/2}\,|-1/2,1/2\rangle$$
$$|2,-1\rangle = \sqrt{1/3}\,|-1/2,-1/2\rangle - \sqrt{2/3}\,|-3/2,1/2\rangle$$
$$|2,-2\rangle = \sqrt{1/6}\,|-3/2,-1/2\rangle - \sqrt{5/6}\,|-5/2,1/2\rangle$$

$S_A = 1$ and $S_B = 1$

$$|2,2\rangle\rangle = |1,1\rangle$$
$$|2,1\rangle = \sqrt{1/2}\,|1,0\rangle + \sqrt{1/2}\,|0,1\rangle$$
$$|2,0\rangle = \sqrt{1/6}\,|1,-1\rangle + \sqrt{2/3}\,|0,0\rangle + \sqrt{1/6}\,|-1,1\rangle$$
$$|2,-1\rangle = \sqrt{1/2}\,|0,-1\rangle + \sqrt{1/2}\,|-1,0\rangle$$
$$|2,-2\rangle = |-1,-1\rangle$$

$$|1,1\rangle = \sqrt{1/2}\,|1,0\rangle - \sqrt{1/2}\,|0,1\rangle$$
$$|1,0\rangle = \sqrt{1/2}\,|1,-1\rangle - \sqrt{1/2}\,|-1,1\rangle$$
$$|1,-1\rangle = \sqrt{1/2}\,|0,-1\rangle - \sqrt{1/2}\,|-1,0\rangle$$

$$|0,0\rangle = \sqrt{1/3}\,|1,-1\rangle - \sqrt{1/3}\,|0,0\rangle + \sqrt{1/3}\,|-1,1\rangle$$

$S_A = 3/2$ and $S_B = 1$

$$|5/2,5/2\rangle = |3/2,1\rangle$$
$$|5/2,3/2\rangle = \sqrt{2/5}\,|3/2,0\rangle + \sqrt{3/5}\,|1/2,1\rangle$$
$$|5/2,1/2\rangle = \sqrt{1/10}\,|3/2,-1\rangle + \sqrt{3/5}\,|1/2,0\rangle + \sqrt{3/10}\,|-1/2,1\rangle$$
$$|5/2,-1/2\rangle = \sqrt{3/10}\,|1/2,-1\rangle + \sqrt{3/5}\,|-1/2,0\rangle + \sqrt{1/10}\,|-3/2,1\rangle$$
$$|5/2,-3/2\rangle = \sqrt{3/5}\,|-1/2,-1\rangle + \sqrt{2/5}\,|-3/2,0\rangle$$
$$|5/2,-5/2\rangle = |-3/2,-1\rangle$$

$$|3/2,3/2\rangle = \sqrt{3/5}\,|3/2,0\rangle - \sqrt{2/5}\,|1/2,1\rangle$$
$$|3/2,1/2\rangle = \sqrt{2/5}\,|3/2,-1\rangle + \sqrt{1/15}\,|1/2,0\rangle - \sqrt{8/15}\,|-1/2,1\rangle$$
$$|3/2,-1/2\rangle = \sqrt{8/15}\,|1/2,-1\rangle - \sqrt{1/15}\,|-1/2,0\rangle - \sqrt{2/5}\,|-3/2,1\rangle$$
$$|3/2,-3/2\rangle = \sqrt{2/5}\,|-1/2,-1\rangle - \sqrt{3/5}\,|-3/2,0\rangle$$

$$|1/2,1/2\rangle = \sqrt{1/2}\,|3/2,-1\rangle - \sqrt{1/3}\,|1/2,0\rangle + \sqrt{1/6}\,|-1/2,1\rangle$$
$$|1/2,-1/2\rangle = \sqrt{1/6}\,|1/2,-1\rangle - \sqrt{1/3}\,|-1/2,0\rangle + \sqrt{1/2}\,|-3/2,1\rangle$$

$S_A = 2$ and $S_B = 1$

$$|3,3\rangle = |2,1\rangle$$
$$|3,2\rangle = \sqrt{1/3}\,|2,0\rangle + \sqrt{2/3}\,|1,1\rangle$$
$$|3,1\rangle = \sqrt{1/15}\,|2,-1\rangle + \sqrt{8/15}\,|1,0\rangle + \sqrt{6/15}\,|0,1\rangle$$
$$|3,0\rangle = \sqrt{1/5}\,|1,-1\rangle + \sqrt{3/5}\,|0,0\rangle + \sqrt{1/5}\,|-1,1\rangle$$
$$|3,-1\rangle = \sqrt{6/15}\,|0,-1\rangle + \sqrt{8/15}\,|-1,0\rangle + \sqrt{1/15}\,|-2,1\rangle$$
$$|3,-2\rangle = \sqrt{2/3}\,|-1,-1\rangle + \sqrt{1/3}\,|-2,0\rangle$$
$$|3,-3\rangle = |-2,-1\rangle$$

$$|2,2\rangle = \sqrt{2/3}\,|2,0\rangle - \sqrt{1/3}\,|1,1\rangle$$
$$|2,1\rangle = \sqrt{1/3}\,|2,-1\rangle + \sqrt{1/6}\,|1,0\rangle - \sqrt{1/2}\,|0,1\rangle$$
$$|2,0\rangle = \sqrt{1/2}\,|1,-1\rangle - \sqrt{1/2}\,|-1,1\rangle$$
$$|2,-1\rangle = \sqrt{1/2}\,|0,-1\rangle - \sqrt{1/6}\,|-1,0\rangle - \sqrt{1/3}\,|-2,1\rangle$$
$$|2,-2\rangle = \sqrt{1/3}\,|-1,-1\rangle - \sqrt{2/3}\,|-2,0\rangle$$

$$|1,1\rangle = \sqrt{3/5}\,|2,-1\rangle - \sqrt{3/10}\,|1,0\rangle + \sqrt{1/10}\,|0,1\rangle$$
$$|1,0\rangle = \sqrt{3/10}\,|1,-1\rangle - \sqrt{2/5}\,|0,0\rangle + \sqrt{3/10}\,|-1,1\rangle$$
$$|1,-1\rangle = \sqrt{1/10}\,|0,-1\rangle - \sqrt{3/10}\,|-1,0\rangle + \sqrt{3/5}\,|-2,1\rangle$$

$S_A = 5/2$ and $S_B = 1$

$$|7/2,7/2\rangle = |5/2,1\rangle$$
$$|7/2,5/2\rangle = \sqrt{2/7}\,|5/2,0\rangle + \sqrt{5/7}\,|3/2,1\rangle$$
$$|7/2,3/2\rangle = \sqrt{1/21}\,|5/2,-1\rangle + \sqrt{10/21}\,|3/2,0\rangle + \sqrt{10/21}\,|1/2,1\rangle$$
$$|7/2,1/2\rangle = \sqrt{1/7}\,|3/2,-1\rangle + \sqrt{4/7}\,|1/2,0\rangle + \sqrt{2/7}\,|-1/2,1\rangle$$
$$|7/2,-1/2\rangle = \sqrt{2/7}\,|1/2,-1\rangle + \sqrt{4/7}\,|-1/2,0\rangle + \sqrt{1/7}\,|-3/2,1\rangle$$
$$|7/2,-3/2\rangle = \sqrt{10/21}\,|-1/2,-1\rangle + \sqrt{10/21}\,|-3/2,0\rangle + \sqrt{1/21}\,|-5/2,1\rangle$$
$$|7/2,-5/2\rangle = \sqrt{5/7}\,|-3/2,-1\rangle + \sqrt{2/7}\,|-5/2,0\rangle$$
$$|7/2,-7/2\rangle = |-5/2,-1\rangle$$

$$|5/2,5/2\rangle = \sqrt{5/7}\,|5/2,0\rangle + \sqrt{2/7}\,|3/2,1\rangle$$
$$|5/2,3/2\rangle = \sqrt{2/7}\,|5/2,-1\rangle + \sqrt{9/35}\,|3/2,0\rangle - \sqrt{16/35}\,|1/2,1\rangle$$
$$|5/2,1/2\rangle = \sqrt{16/35}\,|3/2,-1\rangle + \sqrt{1/35}\,|1/2,0\rangle - \sqrt{18/35}\,|-1/2,1\rangle$$
$$|5/2,-1/2\rangle = \sqrt{18/35}\,|1/2,-1\rangle - \sqrt{1/35}\,|-1/2,0\rangle - \sqrt{16/35}\,|-3/2,1\rangle$$

$$|5/2,-3/2\rangle = \sqrt{16/35}\,|-1/2,-1\rangle - \sqrt{9/35}\,|-3/2,0\rangle - \sqrt{2/7}\,|-5/2,1\rangle$$
$$|5/2,-5/2\rangle = \sqrt{2/7}\,|-3/2,-1\rangle - \sqrt{5/7}\,|-5/2,0\rangle$$

$$|3/2,3/2\rangle = \sqrt{2/3}\,|5/2,-1\rangle - \sqrt{4/15}\,|3/2,0\rangle + \sqrt{1/15}\,|1/2,1\rangle$$
$$|3/2,1/2\rangle = \sqrt{2/5}\,|3/2,-1\rangle - \sqrt{2/5}\,|1/2,0\rangle + \sqrt{1/5}\,|-1/2,1\rangle$$
$$|3/2,-1/2\rangle = \sqrt{1/5}\,|1/2,-1\rangle - \sqrt{2/5}\,|-1/2,0\rangle + \sqrt{2/5}\,|-3/2,1\rangle$$
$$|3/2,-3/2\rangle = \sqrt{1/5}\,|-1/2,-1\rangle - \sqrt{4/15}\,|-3/2,0\rangle + \sqrt{2/3}\,|-5/2,1\rangle$$

$S_A = 3/2$ and $S_B = 3/2$

$$|3,3\rangle = |3/2,3/2\rangle$$
$$|3,2\rangle = \sqrt{1/2}\,|3/2,1/2| + \sqrt{1/2}\,|1/2,3/2\rangle$$
$$|3,1\rangle = \sqrt{1/5}\,|3/2,-1/2| + \sqrt{3/5}\,|1/2,1/2\rangle + \sqrt{1/5}\,|-1/2,3/2\rangle$$
$$|3,0\rangle = \sqrt{1/20}\,|3/2,-3/2| + \sqrt{9/20}\,|1/2,-1/2| + \sqrt{9/20}\,|-1/2,1/2\rangle + \sqrt{1/20}\,|-3/2,3/2\rangle$$
$$|3,-1\rangle = \sqrt{1/5}\,|1/2,-3/2\rangle + \sqrt{3/5}\,|-1/2,-1/2\rangle + \sqrt{1/5}\,|-3/2,1/2\rangle$$
$$|3,-2\rangle = \sqrt{1/2}\,|-1/2,-3/2\rangle + \sqrt{1/2}\,|-3/2,1/2\rangle$$
$$|3,-3\rangle = |-3/2,-3/2\rangle$$

$$|2,2\rangle = \sqrt{1/2}\,|3/2,1/2\rangle - \sqrt{1/2}\,|1/2,3/2\rangle$$
$$|2,1\rangle = \sqrt{1/2}\,|3/2,-1/2\rangle - \sqrt{1/2}\,|-1/2,3/2\rangle$$
$$|2,0\rangle = (1/2)|3/2,-3/2\rangle + (1/2)|1/2,-1/2\rangle - (1/2)|-1/2,1/2\rangle - (1/2)|-3/2,3/2\rangle$$
$$|2,-1\rangle = \sqrt{1/2}\,|1/2,-3/2\rangle - \sqrt{1/2}\,|-3/2,1/2\rangle$$
$$|2,-2\rangle = \sqrt{1/2}\,|-1/2,-3/2\rangle - \sqrt{1/2}\,|-3/2,-1/2\rangle$$

$$|1,1\rangle = \sqrt{3/10}\,|3/2,-1/2\rangle - \sqrt{2/5}\,|1/2,1/2\rangle + \sqrt{3/10}\,|-1/2,3/2\rangle$$
$$|1,0\rangle = \sqrt{9/20}\,|3/2,-3/2\rangle - \sqrt{1/120}\,|1/2,-1/2\rangle - \sqrt{1/120}\,|-1/2,1/2\rangle + \sqrt{9/20}\,|-3/2,3/2\rangle$$
$$|1,-1\rangle = \sqrt{3/10}\,|1/2,-3/2\rangle - \sqrt{2/5}\,|-1/2,-1/2\rangle + \sqrt{3/10}\,|-3/2,1/2\rangle$$

$$|0,0\rangle = (1/2)|3/2,-3/2\rangle - (1/2)|1/2,-1/2\rangle + (1/2)\,|-1/2,1/2\rangle - (1/2)\,|-3/2,3/2\rangle$$

$S_A = 2$ and $S_B = 3/2$

$$|7/2,7/2\rangle = |2,3/2\rangle$$
$$|7/2,5/2\rangle = \sqrt{3/7}\,|2,1/2\rangle + \sqrt{4/7}\,|1,3/2\rangle$$
$$|7/2,3/2\rangle = \sqrt{1/7}\,|2,-1/2| + \sqrt{4/7}\,|1,1/2\rangle + \sqrt{2/7}\,|0,3/2\rangle$$
$$|7/2,1/2\rangle = \sqrt{1/35}\,|2,-3/2\rangle + \sqrt{12/35}\,|1,-1/2| + \sqrt{18/35}\,|0,1/2\rangle + \sqrt{4/35}\,|-1,3/2\rangle$$
$$|7/2,-1/2\rangle = \sqrt{4/35}\,|1,-3/2\rangle + \sqrt{18/35}\,|0,-1/2\rangle + \sqrt{12/35}\,|-1,1/2\rangle + \sqrt{1/35}\,|-2,3/2\rangle$$

$$|7/2,-3/2\rangle = \sqrt{2/7}\,|0,-3/2\rangle + \sqrt{4/7}\,|-1,-1/2\rangle + \sqrt{1/7}\,|-2,1/2\rangle$$
$$|7/2,-5/2\rangle = \sqrt{4/7}\,|-1,-3/2\rangle + \sqrt{3/7}\,|-2,-1/2\rangle$$
$$|7/2,-7/2\rangle = |-2,-3/2\rangle$$

$$|5/2,5/2\rangle = \sqrt{4/7}\,|2,1/2\rangle - \sqrt{3/7}\,|1,3/2\rangle$$
$$|5/2,3/2\rangle = \sqrt{16/35}\,|2,-1/2\rangle + \sqrt{1/35}\,|1,1/2| - \sqrt{18/35}\,|0,3/2\rangle$$
$$|5/2,1/2\rangle = \sqrt{6/35}\,|2,-3/2\rangle + \sqrt{5/14}\,|1,-1/2\rangle - \sqrt{3/35}\,|0,1/2\rangle - \sqrt{27/70}\,|-1,3/2\rangle$$
$$|5/2,-1/2\rangle = \sqrt{27/70}\,|1,-3/2\rangle + \sqrt{3/35}\,|0,-1/2\rangle - \sqrt{5/14}\,|-1,1/2\rangle - \sqrt{6/35}\,|-2,3/2\rangle$$
$$|5/2,-3/2\rangle = \sqrt{18/35}\,|0,-3/2\rangle - \sqrt{1/35}\,|-1,1/2\rangle - \sqrt{16/35}\,|-2,1/2\rangle$$
$$|5/2,-5/2\rangle = \sqrt{3/7}\,|-1,-3/2\rangle - \sqrt{4/7}\,|-2,-1/2\rangle$$

$$|3/2,3/2\rangle = \sqrt{2/5}\,|2,-1/2\rangle - \sqrt{2/5}\,|1,1/2\rangle + \sqrt{1/5}\,|0,3/2\rangle$$
$$|3/2,1/2\rangle = \sqrt{2/5}\,|2,-3/2\rangle - \sqrt{1/5}\,|0,1/2\rangle + \sqrt{2/5}\,|-1,3/2\rangle$$
$$|3/2,-1/2\rangle = \sqrt{2/5}\,|1,-3/2\rangle - \sqrt{1/5}\,|0,-1/2\rangle + \sqrt{2/5}\,|-2,3/2\rangle$$
$$|3/2,-3/2\rangle = \sqrt{1/5}\,|0,-3/2\rangle - \sqrt{2/5}\,|-1,-1/2\rangle + \sqrt{2/5}\,|-2,1/2\rangle$$

$$|1/2,1/2\rangle = \sqrt{2/5}\,|2,-3/2\rangle - \sqrt{3/10}\,|1,-1/2\rangle + \sqrt{1/5}\,|0,1/2\rangle - \sqrt{1/10}\,|-1,3/2\rangle$$
$$|1/2,-1/2\rangle = \sqrt{1/10}\,|1,-3/2\rangle - \sqrt{1/5}\,|0,-1/2\rangle + \sqrt{3/10}\,|-1,1/2\rangle - \sqrt{2/5}\,|-2,3/2\rangle$$

$S_A = 2$ and $S_B = 2$

$$|4,4\rangle = |2,2\rangle$$
$$|4,3\rangle = \sqrt{1/2}\,|2,1\rangle + \sqrt{1/2}\,|1,2\rangle$$
$$|4,2\rangle = \sqrt{3/14}\,|2,0\rangle + \sqrt{8/14}\,|1,1\rangle + \sqrt{3/14}\,|0,2\rangle$$
$$|4,1\rangle = \sqrt{1/14}\,|2,-1\rangle + \sqrt{6/14}\,|1,0\rangle + \sqrt{6/14}\,|0,1\rangle + \sqrt{1/14}\,|-1,2\rangle$$
$$|4,0\rangle = \sqrt{1/70}\,|2,-2\rangle + \sqrt{16/70}\,|1,-1\rangle + \sqrt{36/70}\,|0,0\rangle + \sqrt{16/70}\,|-1,1\rangle + \sqrt{1/70}\,|2,-2\rangle$$
$$|4,-1\rangle = \sqrt{1/14}\,|1,-2\rangle + \sqrt{6/14}\,|0,-1\rangle + \sqrt{6/14}\,|-1,0\rangle + \sqrt{1/14}\,|-2,1\rangle$$
$$|4,-2\rangle = \sqrt{3/14}\,|0,-2\rangle + \sqrt{8/14}\,|-1,-1\rangle + \sqrt{3/14}\,|-2,0\rangle$$
$$|4,-3\rangle = \sqrt{1/2}\,|-1,-2\rangle + \sqrt{1/2}\,|-2,-1\rangle$$
$$|4,-4\rangle = |-2,-2\rangle$$

$$|3,3\rangle = \sqrt{1/2}\,|2,1\rangle - \sqrt{1/2}\,|1,2\rangle$$
$$|3,2\rangle = \sqrt{1/2}\,|2,0\rangle - \sqrt{1/2}\,|0,2\rangle$$
$$|3,1\rangle = \sqrt{3/10}\,|2,-1\rangle + \sqrt{2/10}\,|1,0\rangle - \sqrt{2/10}\,|0,1\rangle - \sqrt{3/10}\,|-1,2\rangle$$
$$|3,0\rangle = \sqrt{1/10}\,|2,-2\rangle + \sqrt{4/10}\,|1,-1\rangle - \sqrt{4/10}\,|-1,1\rangle - \sqrt{1/10}\,|-2,2\rangle$$
$$|3,-1\rangle = \sqrt{3/10}\,|1,-2\rangle + \sqrt{2/10}\,|0,-1\rangle - \sqrt{2/10}\,|-1,0\rangle - \sqrt{3/10}\,|-2,1\rangle$$
$$|3,-2\rangle = \sqrt{1/2}\,|0,-2\rangle - \sqrt{1/2}\,|-2,0\rangle$$
$$|3,-3\rangle = \sqrt{1/2}\,|-1,-2\rangle - \sqrt{1/2}\,|-2,-1\rangle$$

$$|2,2\rangle = \sqrt{2/7}\,|2,0\rangle - \sqrt{3/7}\,|1,1\rangle + \sqrt{2/7}\,|0,2\rangle$$
$$|2,1\rangle = \sqrt{6/14}\,|2,-1\rangle - \sqrt{1/14}\,|1,0\rangle - \sqrt{1/14}\,|0,1\rangle + \sqrt{6/14}\,|-1,2\rangle$$
$$|2,0\rangle = \sqrt{4/14}\,|2,-2\rangle + \sqrt{1/14}\,|1,-1\rangle - \sqrt{4/14}\,|0,0\rangle + \sqrt{1/14}\,|-1,1\rangle + \sqrt{4/14}\,|-2,2\rangle$$
$$|2,-1\rangle = \sqrt{6/14}\,|1,-2\rangle - \sqrt{1/14}\,|0,-1\rangle - \sqrt{1/14}\,|-1,0\rangle + \sqrt{6/14}\,|-2,1\rangle$$
$$|2,-2\rangle = \sqrt{1/2}\,|0,-2\rangle - \sqrt{1/2}\,|-2,0\rangle$$

$$|1,1\rangle = \sqrt{2/10}\,|2,-1\rangle - \sqrt{3/10}\,|1,0\rangle + \sqrt{3/10}\,|0,1\rangle - \sqrt{2/10}\,|-1,2\rangle$$
$$|1,0\rangle = \sqrt{4/10}\,|2,-2\rangle - \sqrt{1/10}\,|1,-1\rangle + \sqrt{1/10}\,|-1,1\rangle - \sqrt{4/10}\,|-2,2\rangle$$
$$|1,-1\rangle = \sqrt{2/10}\,|1,-2\rangle - \sqrt{3/10}\,|0,-1\rangle + \sqrt{3/10}\,|-1,0\rangle - \sqrt{2/10}\,|-2,1\rangle$$

$$|0,0\rangle = \sqrt{1/5}\,|2,-2\rangle - \sqrt{1/5}\,|1,-1\rangle + \sqrt{1/5}\,|0,0\rangle - \sqrt{1/5}\,|-1,1\rangle + \sqrt{1/5}\,|-2,2\rangle$$

Index

Ab initio calculation, 163–64
Active-electron approximation, 150–54, 160–62, 164–67, 171, 175, 186–87, 198–99
Allyl radical, 305–7
Anderson theory, 155–59
Anisotropy
  in chain compounds, 257–62
  in Cu(II) dinuclear compounds, 136–38
  in Cu(II)Fe(III) compounds, 127
  in Cu(II)Fe(III)Cu(II) compounds, 217–18
  of the *g*-factor, 12–14
  and first-order orbital momentum, 31–51
  and intermediate-spin state, 88
  in Mn(II)-radical compounds, 319
  in Mn(II) phthalocyanine, 324
  and rare earth-containing compounds, 50–51
  and spin-admixed state, 91–94
  in $Ti_2Cl_9Cs_3$, 207–9
  and zero-field splitting, 14–26

Biomagnetism, 2–4
Bleaney and Bowers equation, 104
Bohr magneton, 2, 10
Boltzmann distribution law, 5, 26
Bonner and Fisher method, 252. *See also* Ring chain technique
Brillouin function, 11

Carbenes, 311–13
Chain compounds
  alternating chain compounds, 263–69, 294–300
  regular chain compounds, 251–63. *See also* Ferrimagnetic chain compounds
Charge transfer, 151–53, 155–59, 199–202, 291–94
Chromium(III)
  $Cr(III)Ni(II)_3$ tetranuclear compound, 195–96, 231–34
  dinuclear compounds, 193
  and zero-field splitting, 21–23
Classical spin, 258, 273–74
  classical-quantum spin calculation, 274, 281–84
Clebsch-Gordon coefficients, 234, 307, 337, 367–73
Cobalt(II)
  in chain compounds, 261–63
  and low-spin high-spin transitions, 56, 80–82
  mononuclear compounds, 38–43
Cobalt(III)
  and low-spin high-spin transitions, 80

Configuration interaction, 7, 148–50, 300, 305–6
Cooperativity, 59–66, 79
Copper(II)
acetate, 106–8
and anisotropy of the *g*-factor, 12–14
azido-bridged compounds, 109–11, 166–67
bibridged Cu(II) dimers, 159–64
chain compounds, 251–57, 263–67
cubane-like compounds, 237–38
Cu(II)Fe(III) compounds, 126–27, 143, 196
Cu(II)Fe(III)Cu(II) compounds, 215–20
Cu(II)Gd(III)Cu(II) compounds, 245–46
Cu(II)Mn(II) chain compounds, 313–19
Cu(II)Mn(II)Cu(II) compounds, 215–20
Cu(II)Ni(II) compounds, 121–24, 186, 198
Cu(II)-radical compounds, 177–81
Cu(II)VO(II) compound, 177–81
Cu(II)VO(II)Cu(II) compound, 237–38
dinuclear compounds, 103–12, 135–38
dinuclear compounds with extended bridges, 167–74
and orbital complementarity and countercomplementarity, 164–67
tetranuclear compounds, 234–38
trinuclear compounds, 213–15
Correlation
between magnetic and structural properties, 159–61, 194, 198
Correlation length, 271, 287–88
Critical temperature
and low-spin high-spin transitions, 56, 60–67
and magnetic ordering, 284–87
Cubane, 236–241
Curie law
derivation, 9–10
and rare earth-containing compounds, 43–48
Curie-Weiss law, 26–29, 132, 302–3
Cytochrome *c* oxidase, 219

Day and Robin classification, 336
Decamethylferrocenium. *See* Metallocenium charge-transfer salts
Diamagnetic susceptibilities, 3–4
Diamagnetism, 2–4
Dinuclear compounds, 103–32, 135–43, 145–82, 185–208
heterodinuclear compounds, 120–30
Domains
and low-spin high-spin transitions, 67–69, 73
and molecular-based magnets, 288–89, 315
Double exchange, 337. *See also* Spin-dependent delocalization
Double group, 15–16, 89–90
Dzialoshinski-Moriya Hamiltonian, 138–39

Effective magnetic moment, 10
Effective spin, 42–43
Electron delocalization, 333–52
electron delocalization parameter, 341
Endor spectroscopy
in polycarbenes, 311
EPR (Electron Paramagnetic Resonance) spectroscopy
and anisotropy of the *g*-factor, 14
of Cu(II) acetate, 109
of Cu(II) dinuclear compounds, 138, 203–4
of Cu(II)Ni(II) compounds, 130
of Cu(II)Mn(II) chain compounds, 315–16
of Fe(III) dinuclear compounds, 119
and low-spin high-spin transitions, 54–55, 73
of mixed-valence compounds, 336, 348
of organic radicals, 289, 302
of polycarbenes, 312–13
and spin-admixed state, 92–93
Escalator operators, 40, 365
EXAFS (Extended X-ray Absorption Fine Structure), 254
Exchange
kinetic exchange, 156, 163
potential exchange, 155, 163, 179
Exchange-transfer, 350–51

Ferrimagnetic chains
alternating ferrimagnetic chains, 281–84
magnetic ordering of ferrimagnetic chains, 313–20
regular ferrimagnetic chains, 269–80
Ferromagnetism, 94, 287–328
Field, 1
coercive field, 289, 296, 320, 324
critical field and Haldane gap, 259

critical field and metamagnetism, 256–57
demagnetizing field, 288–89
Fisher equation, 258
Free-ion approximation, 43–47
Fullerene, 326

Galvinoxyl radical, 301–2
g-factor, *g*-tensor, 14
in dinuclear compounds, 121–25, 136
in trinuclear compounds, 212–18
and zero-field splitting, 17–26
Goodenough-Kanamori rules, 190–91
Goodenough mechanism, 199–202
Gorun and Lippard correlation, 198–99

Haldane gap, 258–61
Hartree–Fock calculation, 155–59, 163, 171, 179–81
spin-restricted, 155, 305
spin-unrestricted, 155, 305
Hatfield and Hodgson correlation, 158–61
Hay-Thibeault-Hoffmann model, 158
Heat capacity
of Co(II) chain compounds, 262
of ferrimagnetic chain compounds, 278
of galvinoxyl, 302
and low-spin high-spin transitions, 68–69, 75
and magnetic ordering, 287
of nitronyl nitroxide compounds, 303
Heisenberg-Dirac-Van Vleck Hamiltonian, 106, 112, 148, 185–87, 204–5, 239, 268, 345
Heitler-London approach, 148, 186–90, 202, 306–7
Hemerythrin, 115
High-spin molecules, 245–48, 279–80, 309–13
Hund's rule, 53, 199, 290
Hysteresis
and low-spin high-spin transitions, 55, 60–67, 72, 78
and molecular-based magnets, 289, 296, 315, 320, 322

Infrared spectroscopy
and low-spin high-spin transitions, 54, 79
Interaction
anisotropic interaction, 135–38, 202–4, 216
antiferromagnetic interaction, 26, 103–32, 145–82,
antisymmetric interaction, 138–43, 202–4, 321–25
biquadratic interaction parameter, 119–20
competing interactions, 241–45
dipolar interaction, 135–38, 261
ferromagnetic interaction, 26, 103–32, 145–82
interaction between pairs of magnetic orbitals, 190–99
isotropic interaction, 103–132, 145–82
pseudo-dipolar interaction. *See* Anisotropic interaction
Interchain interaction, 254, 313–20
Intermediate spin, 87–89, 97–100, 323–24
Intermolecular interaction, 287–90
between dinuclear compounds, 131–32, 138, 177
between mononuclear compounds, 26–29
and low–spin high-spin transitions, 59–66
Intervalence band, 336, 348
Iron
Fe(II)Fe(III) mixed-valence compounds, 347–48
Fe-S proteins, 238–41, 347, 350–52
Iron(II)
Fe(II) imidazolato, 322–23
Fe(II) phthalocyanine, 323–24
Fe(II) porphyrins, 96
$FEPS_3$, 325
and low-spin high-spin transitions, 56, 74–80
Iron(III)
dinuclear compounds, 114–20
and low-spin high-spin transitions, 56, 71–74
Fe(III) dithiocarbamato derivatives, 71–73, 93–94, 325
Fe(III)Ni(II) compound, 129–30
Fe(III) phthalocyanines, 96
Fe(III) porphyrins, 94–96
Fe(III) salen derivatives, 97, 115
oxo-bridged Fe(III) compounds, 198–99, 243–44
and spin equilibrium involving an intermediate-spin state, 97–100
and zero-field splitting, 23–25
Ising Hamiltonian, 261–63

Isotropic interaction. *See* Interaction

Jahn-Teller distortion, 35–37, 193–94, 291, 326, 349–50

Kramers doublet, 21, 37, 38–42, 88, 90–91, 129–30, 217–18, 240, 261

Lande interval rule, 119, 340, 343
Lamellar material, 325
Langmuir-Blodgett film, 79
LIESST (Light Induced Excited Spin State Trapping), 57–58, 80
Local anisotropy, 14–26, 125–30, 216–18. *See also* Zero-field splitting

MacConnell-Breslow mechanism, 291–94, 297–99
MacConnell mechanism, 303–9. *See also* MacConnell-Breslow mechanism
Magnetic circular dichroism
  of cytochrome *c* oxidase derivative, 219
Magnetic orbital, 136, 146, 185–86
  and atomic orbital contraction method, 150, 186
  interaction between pairs of magnetic orbitals, 190–99
  natural magnetic orbital, 149–50, 152–54, 167, 175–81, 189–99, 239
  orthogonalized magnetic orbital, 148–49, 150–52, 171, 175–81, 187–88, 290
Magnetic ordering, 94, 277, 279, 287–328
Magnetic susceptibility
  average magnetic susceptibility, 17–26
  of chain compounds, 251–84
  and Curie law, 9–10
  definition, 1
  of dinuclear compounds, 103–31,
  and intermolecular interactions, 26–29, 131–32
  of rare earth-containing compounds, 43–51
  of tetranuclear compounds, 229–41
  of trinuclear compounds, 211–28
Magnetization
  of $Cr(III)Ni_3$ compound, 233
  definition, 1
  field-cooled magnetization, 315
  fundamental equations, 4, 10–11
  high-field magnetization, 224–25
  of Mn(II) phthalocyanine, 324
  remnant magnetization 289–90
  saturation magnetization, 11, 295–96, 315, 322
  zero-field-cooled magnetization, 315
Maltempo model, 90–93
Manganese
  $Mn(III)_8Mn(IV)_4$ cluster, 247–48
  Mn(III)Mn(IV) mixed-valence compounds, 349–50
Manganese(II)
  Mn(II) chain compounds, 259–61
  Mn(II)Cu(II) chain compounds, 313–19
  $Mn(II)Cu(II)_3$ compound, 229–31
  Mn(II)Cu(II)Mn(II) compound, 220–23
  Mn(II) phthalocyanine, 323–24
  $MnPS_3$, 325
  Mn(II)-radical compounds, 278–80, 319–20
  Mn(II) trinuclear compounds, 223–26
  and zero-field splitting, 23–25
Manganese(III)
  and low-spin high-spin transitions, 70–71
  Mn(III) porphyrin-TCNE compound, 320–21
  oxo-bridged Mn(III) dinuclear compounds, 201–2
Metallocenium charge transfer salts, 294–300
Metamagnetism, 256–57, 297, 303
Mixed-valence compound, 333–52
Molecular-based magnets, 288–328
Molecular electronics, 82–84
Molecular-field approximation, 26, 131
Molecular orbital approach, 7, 146–48, 304–6, 310–11
Mössbauer spectroscopy
  of Cu(II)Fe(III)Cu(II) compounds, 219
  of Fe(II)Fe(III) mixed valence compounds, 348
  of Fe(III)Ni(II) compound, 130
  of Fe(III) porphyrins, 95
  of Fe_3S_4 clusters, 347
  and low-spin high-spin transitions, 54, 57, 73–76

Neutron
  inelastic neutron scattering, 107–8, 259
  polarized neutron diffraction, 261, 304

Nickel(II)
  chain compounds, 277–78
  dinuclear compounds, 194
  Ni(I)Ni(II) mixed-valence compound, 346–47
  and zero-field splitting, 17–18
Nitronyl nitroxide, 278–80, 303, 319–20
NMR (Nuclear Magnetic Resonance)
  of metallocenium charge-transfer salts, 300
Nuclear relaxation, 333–37, 343–45

Occupation operator, 341–43
Orbital complementarity and countercomplementarity, 164–67
Orbital degeneracy, 31–51, 290–94
Orbital momentum, 31–51, 357–59
Orbital reduction factor, 32, 34
Orbital reversal, 169, 255
Orthogonality (of the magnetic orbitals)
  accidental orthogonality, 163–64, 191–99
  strict orthogonality, 174–81, 191–99, 219, 228, 233, 299, 325–26
Overlap (between magnetic orbitals), 149–50, 152–54, 162, 188
Overlap density, 163–64, 178–79

Partition function, 5, 11
Pascal Tables. *See* Diamagnetic susceptibilities
Phase transition, 74–77, 287–88
Point charge approximation, 136
Polaron, 328
Polymer, 326–27
Preferred magnetization axis, 289, 315, 319

Radical, 173, 303–5
Rare earth
  Eu(II), 47–48
  Eu(III), 48–49
  Gd(III), 47–48, 245–46
  magnetic properties of rare earth-containing compounds, 43–51
  Sm(II), 48–49
  Sm(III) 49–50
Regular solution model, 60–67
Resonant delocalization, 350. *See also* Spin-dependent delocalization
Ring chain technique, 252–53, 257–58, 272–73, 278

Slichter and Drickamer model, 60–62
Soraï and Seki model, 67–69
Spin-admixed state, 89–100, 240
Spin canting, 140–41, 257, 287, 321–25
Spin density, 136, 173, 300, 304
Spin-dependent delocalization, 337–52
Spin frustration, 241–45, 325, 352
Spin-orbit coupling, 31–51, 89–93, 135–43, 202–8
Spin-Peierls transition, 267–69
Spin polarization, 155–57, 181, 300, 303–9
Spin population, 304, 307
Spin state. *See also* Spin-admixed state
  high-spin state, 53–54
  intermediate spin state, 73, 87–100
  low-spin state, 53–54, 218–20
  transition between spin states, 53–84, 97–100
Spin state structure
  irregular spin state structure, 220–23, 270
  regular spin state structure, 120, 215–31
Spontaneous magnetization, 6, 288–328
SQUID (Superconducting Quantum Interference Device), 4

Temperature-independent paramagnetism, 4, 7–8, 37, 46–47, 140
Tetranuclear compounds, 229–41
Tetrathiafulvalene
  spin–Peierls transition in TTF^+ derivatives, 268–69
Titanium(III)
  magnetism of mononuclear compounds, 32–38
  $Ti_2Cl_9^{3-}$, 204–9
Topological degeneracy, 309–13
Topology, 244, 247
Trinuclear compounds, 211–28

Units, 1, 355–56

Valence bond approach, 310–12
Vanadium
  $V(TCNE)_2.1/2CH_2Cl_2$, 320–21
Van Vleck formula
  and Curie law, 9–10
  derivation, 5
  and dinuclear compounds, 103–30, 207
  and first-order orbital momentum, 31–51

Van Vleck formula (*continued*)
and spin-admixed state, 92
and tetranuclear compounds, 229–41
and trinuclear compounds, 212–28
and zero-field splitting, 17–26
Vectorial approach, 226, 229, 240
Vibronic coupling, 336, 343–45

Weak ferromagnetism, 140–41, 287, 322–23
Wigner-Eckardt theorem, 123
Wigner coefficients. *See* Clebsch-Gordon coefficients

XY Hamiltonian, 261–63
Xα method, 181–82

Zeeman coefficients
and Cu(II)Ni(II) compounds, 123
definition, 6–7
and dinuclear compounds, 113–41
and first-order orbital momentum, 31–51
and spin-admixed state, 91
Zeeman perturbation, 7, 35, 40–42, 46, 207
Zero-field splitting, 14–26
in chain compounds, 257–61, 274
in Cu(II) dinuclear compounds, 135–41, 203–4
in Fe(III) dinuclear compounds, 119
in heterodinuclear compounds, 120–30
in trinuclear compounds, 216–18
zero-field splitting parameters, 17–26, 94
zero-field splitting tensors, 17, 125–30

Olivier Kahn

Molecular Magnetism

Molecular magnetism occupies a crossing point between two fields of research—materials science and metal biochemistry—and plays an important role in the emerging field of molecular electronics.

Molecular Magnetism is the first book to comprehensively address both the experimental and theoretical aspects of the new field of research. A leader in the field introduces the basic concepts concerning magnetization and magnetic susceptibility, establishes the fundamental equations of molecular magnetism, and examines molecules containing a unique magnetic center, including the high-spin low-spin transition compounds.

The book highlights polymetallic species, reviews the phenomenon of interaction between spin carriers from a theoretical point of view, and includes numerous examples throughout to illustrate the topics discussed.

Professionals and graduate students working in inorganic chemistry, theoretical chemistry, physical chemistry, bioinorganic chemistry, materials and molecular electronics, physics, and biophysics will find *Molecular Magnetism* invaluable.

ISBN 1-56081-566-3 VCH Publishers, Inc.
ISBN 3-527-89566-3 VCH Verlagsgesellschaft mbH

VCH